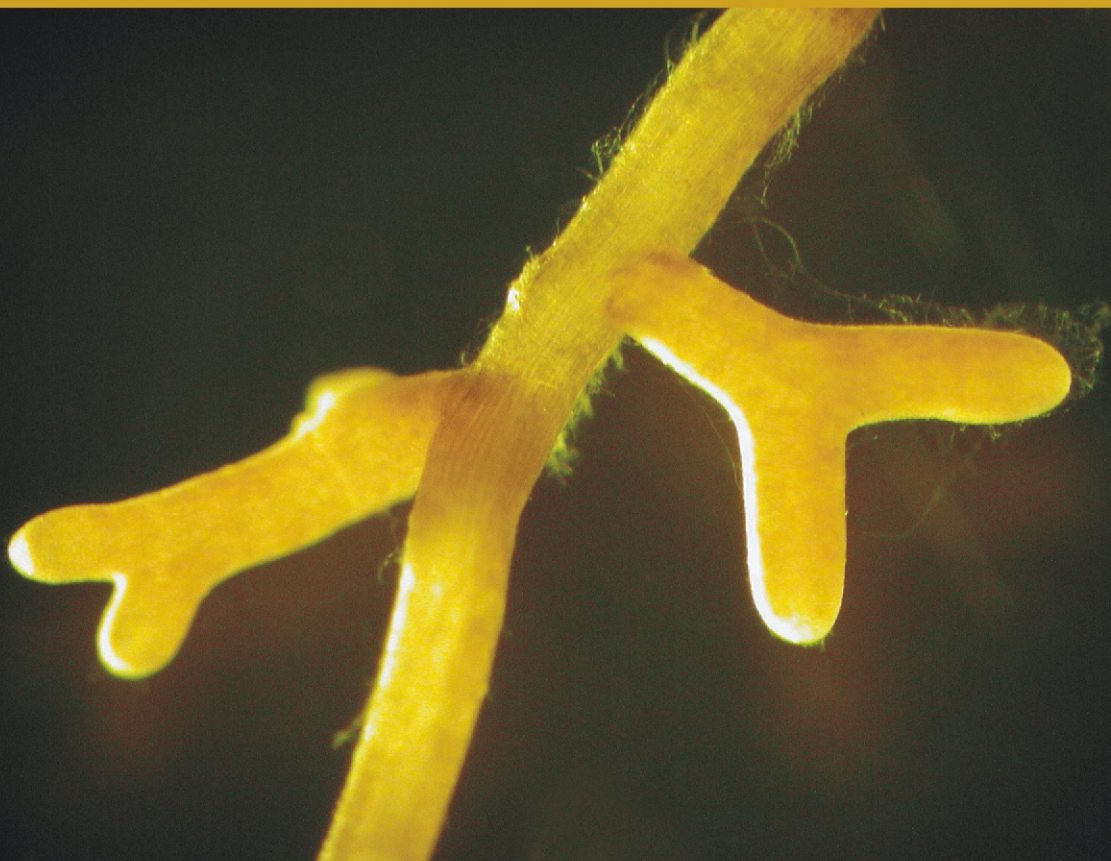


Textbook of  
**Environmental  
Microbiology**



Pradipta K. Mohapatra



# **Textbook of Environmental Microbiology**



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# **Textbook of environmental microbiology**

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*Dedicated*

*To*

*My Beloved Teacher*

*DR SATYA NARAYAN PATNAIK*

*Former Professor and Head*

*Post-Graduate Department of Botany*

*Utkal University*

*Bhubaneswar*





---

# PREFACE

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Environmental Microbiology is a branch of science which deals with the microbial diversity, interaction, association and processes operating in the environment. Historically, it was limited to the studies on microbial processes in wastewater treatment and the use of treated solid wastes to reinforce the nutrient level of crop fields. However, with the development of the understanding of environment and role of microbes in it, the scope of environmental microbiology has broadened enormously in recent years. It now virtually touches each and every aspect of the environment, where microbes have a role to play. The environmental microbiologists are now interested in discovery and application of new microbes and their products that can be used for economic development and healthcare of humankind. The drive to search novel and supermicrobes has propelled the desire of researchers to touch extreme environments like deep polar snow layer, deep sea methanogenic environment, hot-springs, hypersaline soils, etc. The discovery of the bacterium 121 and anoxygenic picoplankton are the results of such attempts.

While the nature of environmental microbiology has changed considerably in recent times, there has been limited attempts to cover the progress in a formal textbook to make the information available to the students and researchers. Plenty of literature, in the form of periodicals, books, reviews and journals are available on traditional microbiology and microbial ecology but compiled literature on environmental microbiology are truly scarce. While microbial ecology only covers the interaction of microbes among themselves, or with other organisms, traditional microbiology describes microbial processes, with respect to their growth, physiology, biochemistry and genetics. Environmental microbiology, on the other hand, is the applied aspects of microbes in the environment to remove undesirable environmental contaminants, to remineralise soil and water, to improve agricultural production and to produce value added products from the junks. Thus a book on environmental microbiology is expected to provide information on microbial diversity of various environments, and use of microbial processes for the improvement of the quality of the environment.

The studies on the microbial processes in the environment has gained momentum in recent years to remove the pollutants that man is adding continuously to his surroundings. Microbes are now applied in municipal waste treatment, wastewater purification, remediation of problem soils, improvement of agricultural productivity, immobilization and removal of recalcitrant organic and inorganic pollutants, and bioaugmentation of microbial strains for their application in human welfare. This book on environmental microbiology has been planned to cover most of the above aspects. It has been designed to provide up to

date information on the above-mentioned subjects and targets the students of Microbiology, Environmental Sciences and Biotechnology at the postgraduate level. This will also be useful to the scientists and researchers engaged in the field of Environmental Microbiology. The overall objective of this book is to describe the microbial diversity of various environments, the important microbes involved in environmental purification, and the modern aspects of environmental microbiology.

This textbook consists of 15 chapters, each with its well described content. Chapter 1 outlines the scopes and application of microbes for purification of the environment and for use of microbial processes in human welfare. Chapter 2 through 7 deal with the microbial diversity in different environments ranging from air to deep sea vents. The microbial processes of these diverse types of environments have been discussed in detail in these chapters to have a better understanding of the role of microbes in the nutrient and energy flow in these conditions. Chapter 8 contains the modern methodologies in the field of environmental microbiology. The ecological relationships among microbes as well as between microbes and other organisms have been presented in chapter 9 with special emphasis on symbiosis, which has been vividly discussed in chapter 10. The applied aspects of environmental microbiology have been discussed in the remaining five chapters. Chapter 11 covers the microbial mobilization of nutrients and toxicants in the environment while the microbial purification of wastewater has been presented in chapter 12. Chapters 13 and 14 deal with the removal as well as immobilization of recalcitrant organic and inorganic toxicants and heavy metals by microbes. The methods of biomanipulation to detoxify metal and organically polluted environments have also been discussed in these chapters. The last chapter presents the microbial indication of the environmental conditions to use them as tools in biomonitoring process. The description of each chapter has been made in simple and lucid language and has been well explained through illustrations and models. The chapters have been designed keeping in view the expectations of students.

I have been greatly encouraged by the students of the postgraduate department of Botany, Utkal University, Bhubaneswar, to write this book. I am thankful to the staff and students of the Department for their encouragement and constant support. I have also been benefited by the works of my research students and some of their finding has been presented in this book. The financial support to carryout such research has been provided by the University Grants Commission, New Delhi, under the Special Assistance Programme (UGC-SAP-DRS), which I sincerely acknowledge. I am thankful to my wife Dr Mrs Sobhana S. Panda, who has worked hard with me to develop the manuscripts. I thank The Professor and Head, P. G. Department of Botany, Utkal University, without whose help this book would not have been possible. I dedicate this book to my beloved teacher Professor Satya Narayan Patnaik, former Professor of Botany, Utkal University, who has always remained as a role model for me.

**PRADIPTA K. MOHAPATRA**

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# CONTENTS

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|                                                                    |           |
|--------------------------------------------------------------------|-----------|
| <b>1. Scope and Applications of Environmental Microbiology</b>     | <b>1</b>  |
| 1.1 Introduction                                                   | 1         |
| 1.2 Microbes and the Environment                                   | 2         |
| 1.3 Scope of Environmental Microbiology                            | 6         |
| 1.4 Microbial Application for Pollution Control                    | 10        |
| 1.5 Microbes in Human Welfare                                      | 12        |
| 1.6 Conclusion                                                     | 15        |
| <b>2. Microbial Diversity in the Environment</b>                   | <b>16</b> |
| 2.1 Introduction                                                   | 16        |
| 2.2 Classification of Microbes                                     | 16        |
| 2.3 The World of Bacteria                                          | 18        |
| 2.4 The Cyanobacteria                                              | 37        |
| 2.5 The World of Fungi                                             | 40        |
| 2.6 The World of Algae                                             | 45        |
| 2.7 The World of Protozoa                                          | 50        |
| 2.8 Conclusion                                                     | 54        |
| <b>3. Microbial Diversity and Processes in Freshwater Habitats</b> | <b>55</b> |
| 3.1 Introduction                                                   | 55        |
| 3.2 The Freshwater Environment                                     | 56        |
| 3.3 The Ecological Processes in Freshwater Habitats                | 58        |
| 3.4 The Phytoplankton Communities                                  | 62        |
| 3.5 The Zooplankton Communities                                    | 76        |
| 3.6 The Aquatic Fungi                                              | 85        |
| 3.7 The Bacterioplankton in Freshwater Ecosystems                  | 87        |
| 3.8 Microbial Activities in Sediment                               | 90        |
| 3.9 Conclusion                                                     | 96        |
| <b>4. Microbes in Marine and Estuarine Habitats</b>                | <b>97</b> |
| 4.1 Introduction                                                   | 97        |
| 4.2 The Characteristics of the Marine Environment                  | 98        |
| 4.3 Microbial Structuring of Marine Ecosystems                     | 99        |
| 4.4 Marine Bacteria                                                | 105       |
| 4.5 The Marine Fungal Diversity                                    | 112       |
| 4.6 Marine Phytoplankton                                           | 114       |

|           |                                                         |            |
|-----------|---------------------------------------------------------|------------|
| 4.7       | The Marine Protozoa                                     | 122        |
| 4.8       | The Microbial Activities in Ocean Floor                 | 123        |
| 4.9       | Conclusion                                              | 126        |
| <b>5.</b> | <b>Microbial Diversity of Terrestrial Environment</b>   | <b>128</b> |
| 5.1       | Introduction                                            | 128        |
| 5.2       | The Soil Environment                                    | 129        |
| 5.3       | Distribution of Soil Microbes                           | 137        |
| 5.4       | Microbial Diversity of Surface Soil                     | 140        |
| 5.5       | Microbial Diversity of Subsurface Soil                  | 146        |
| 5.6       | The Rhizosphere Microbes                                | 149        |
| 5.7       | The Mycorrhizae                                         | 156        |
| 5.8       | Conclusion                                              | 164        |
| <b>6.</b> | <b>Microbial Diversity of Air</b>                       | <b>165</b> |
| 6.1       | Introduction                                            | 165        |
| 6.2       | Characteristics of the Atmosphere with Viable Particles | 165        |
| 6.3       | Atmospheric Dispersal of Microbes                       | 167        |
| 6.4       | Microbial Diversity of Air                              | 173        |
| 6.5       | Microbial Survival in Air                               | 178        |
| 6.6       | The Phyllosphere Contribution to Aeromicroflora         | 180        |
| 6.7       | The Control of Bioaerosol                               | 184        |
| 6.8       | Conclusion                                              | 185        |
| <b>7.</b> | <b>Microbes of Extreme Environments</b>                 | <b>186</b> |
| 7.1       | Introduction                                            | 186        |
| 7.2       | Types of Stresses in Extreme Environments               | 187        |
| 7.3       | The Diversity of Thermophilic Microbes                  | 190        |
| 7.4       | Microbial Diversity of Deep Sea Environment             | 199        |
| 7.5       | Microbes of Cold Habitats                               | 204        |
| 7.6       | Microbes of Hypersaline Habitats                        | 209        |
| 7.7       | Microbial Diversity of Metal Contaminated Environment   | 216        |
| 7.8       | Microbes of Organically Contaminated Soil               | 220        |
| 7.9       | Conclusion                                              | 221        |
| <b>8.</b> | <b>Methods in Environmental Microbiology</b>            | <b>222</b> |
| 8.1       | Introduction                                            | 222        |
| 8.2       | Assay of Microbial Growth                               | 223        |
| 8.3       | Quantification of Carbon Utilization                    | 226        |
| 8.4       | Radiolabeling of Cellular Macromolecules                | 233        |
| 8.5       | Nucleic Acid Assay                                      | 238        |
| 8.6       | The Enzyme Assays                                       | 246        |
| 8.7       | Immunoassays                                            | 253        |
| 8.8       | The Analysis of Metagenome                              | 258        |
| 8.9       | Conclusion                                              | 260        |

|                                                                |            |
|----------------------------------------------------------------|------------|
| <b>9. Ecological Relationships Among Microorganisms</b>        | <b>262</b> |
| 9.1 Introduction                                               | 262        |
| 9.2 Relationships among Microbial Populations                  | 263        |
| 9.3 Interspecies Microbial Interactions                        | 268        |
| 9.4 Microbial Interactions in a Biofilm                        | 286        |
| 9.5 Plant-Microbe Interactions                                 | 287        |
| 9.6 Animal-Microbe Interactions                                | 292        |
| 9.7 Pathogenic Interactions of Microbes with Plants            | 294        |
| 9.8 Pathogenic Interactions of Microbes with Animals           | 297        |
| 9.9 Conclusion                                                 | 300        |
| <b>10. Symbiotic Interactions of Microbes</b>                  | <b>301</b> |
| 10.1 Introduction                                              | 301        |
| 10.2 Microbe-Microbe Symbiosis                                 | 302        |
| 10.3 Plant-Microbe Symbiosis                                   | 309        |
| 10.4 Legume-Rhizobium Interaction                              | 315        |
| 10.5 Animal-Microbe Symbiosis                                  | 324        |
| 10.6 Reconstitution of the Symbioses                           | 331        |
| 10.7 Conclusion                                                | 332        |
| <b>11. Microbial Mobilization of Nutrients</b>                 | <b>333</b> |
| 11.1 Introduction                                              | 333        |
| 11.2 Gaia Hypothesis                                           | 334        |
| 11.3 Microbial Mobilization of Carbon                          | 335        |
| 11.4 Nitrogen Cycle                                            | 343        |
| 11.5 Sulphur Cycle                                             | 353        |
| 11.6 Phosphorus Cycle                                          | 359        |
| 11.7 Interrelationship between Cycling of Nutrients            | 365        |
| 11.8 Microbial Corrosion                                       | 366        |
| 11.9 Conclusion                                                | 371        |
| <b>12. Microbial Treatment of Wastewater</b>                   | <b>373</b> |
| 12.1 Introduction                                              | 373        |
| 12.2 Microbial Processes in Wastewater Treatment               | 374        |
| 12.3 Biofilm Microbes                                          | 375        |
| 12.4 Microbial Degradation of Organic Pollutants in Ecosystems | 381        |
| 12.5 Aerobic Bacterial Degradation of Biopolymers              | 385        |
| 12.6 Anaerobic Bacterial Degradation of Biopolymers            | 390        |
| 12.7 Secondary Treatment Systems                               | 395        |
| 12.8 Microbial Removal of Nitrogen                             | 403        |
| 12.9 Microbial Removal of Phosphorus                           | 407        |
| 12.10 Conclusion                                               | 409        |
| <b>13. Microbes in Metal Pollution Control</b>                 | <b>411</b> |
| 13.1 Introduction                                              | 411        |
| 13.2 Environmental Toxicity of Heavy Metals                    | 412        |

|            |                                                                      |            |
|------------|----------------------------------------------------------------------|------------|
| 13.3       | Microbial Diversity in Metal Contaminated Habitat                    | 415        |
| 13.4       | Heavy Metal Tolerance                                                | 418        |
| 13.5       | Microbial Transformation of Metals                                   | 423        |
| 13.6       | Genetic Aspects of Heavy Metal Resistance                            | 429        |
| 13.7       | Bio-manipulation for Metal Removal                                   | 436        |
| 13.8       | Conclusion                                                           | 444        |
| <b>14.</b> | <b>Microbial Degradation of Recalcitrant Organic Pollutants</b>      | <b>446</b> |
| 14.1       | Introduction                                                         | 446        |
| 14.2       | Characteristics of Recalcitrant Xenobiotics Degrading Microorganisms | 447        |
| 14.3       | Aerobic Degradation of Hydrocarbon                                   | 451        |
| 14.4       | Anaerobic Degradation of Hydrocarbons                                | 457        |
| 14.5       | Microbial Degradation of Pesticides                                  | 464        |
| 14.6       | Cometabolic Degradation of Organopollutants                          | 473        |
| 14.7       | Fungal Degradation of Organopollutants                               | 476        |
| 14.8       | Bio-manipulation for Accelerated Degradation                         | 480        |
| 14.9       | Conclusion                                                           | 484        |
| <b>15.</b> | <b>Bioindication in the Environment</b>                              | <b>486</b> |
| 15.1       | Introduction                                                         | 486        |
| 15.2       | Standard Criteria for Indicator Organisms                            | 487        |
| 15.3       | Bacteria as Indicators                                               | 488        |
| 15.4       | Algae as Indicator                                                   | 496        |
| 15.5       | Bioindication by Bacteriophages                                      | 500        |
| 15.6       | Other Indicator Organisms                                            | 501        |
| 15.7       | Bioindication of Surface Water Quality                               | 502        |
| 15.8       | Bioindication of Groundwater Quality                                 | 508        |
| 15.9       | Conclusion                                                           | 509        |
|            | <b><i>References</i></b>                                             | <b>511</b> |
|            | <b><i>Index</i></b>                                                  | <b>519</b> |

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# SCOPE AND APPLICATIONS OF ENVIRONMENTAL MICROBIOLOGY

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## 1.1 INTRODUCTION

Environmental Microbiology, in general, refers to the study of the microbial diversity, density, interaction, association and processes of the environments—each having a definite set of environmental factors. The nature of microbial community of an environment is determined by the types and magnitude of these factors. In olden times the study of microbiology was restricted only to those microbes of mesophilic type, which have direct relevance to the human welfare. The microbial analysis of different environments started with an aim to use the microbial processes for wastewater treatment and for treatment of solid wastes to reinforce the nutrient level of crop fields. However, with the development of the understanding of the environment and the role of microbes in it, the scope of environmental microbiology has broadened enormously in recent years. It now virtually touches each and every aspect of the environment, ranging from agricultural fields to the most stressed deep sea habitats. The environmental microbiologists are now interested in discovery and application of new microbes and their products that can be used for economic development and healthcare of humankind.

Microorganisms are ubiquitous and live in most inhospitable sites across the various ecosystems, in which the existence and continuance of other organisms may not be possible. They have been associated with this planet since the origin of life and it is known that life without these tiny creatures is not feasible. The current microbiological knowledge base definitely indicates that study of microorganisms, in relation to the environment in which they live, is central to biology. This is borne out by not only their role in experimental investigations but also by their intimate association as mutualists, parasites, pathogens and similar other partnerships with plants, animals and human beings. Such millennia old partnerships have led to evolution of a staggering number of diverse microbial

cells that exist on this planet. Their major role in operation and maintenance of biogeochemical cycles and through this, soil, plant and human health is visible. While a large microbial diversity is associated with the gut and intestine across the animal world, some forms are pathogenic and are a cause of concern on account of rapid genetic change, leading to health scares in various regions of the world. However, solutions to such health issues are also being searched within the new and emerging microbial diverse gene pool. What is indeed of immense scientific interest is the cultivation of the unculturable diversity, hitherto known only from genome sequences and metagenomic studies.

## 1.2 MICROBES AND THE ENVIRONMENT

Despite the availability of a huge volume of information on these microscopic living beings, our knowledge on the distribution and abundance of microbes in different habitats is highly rudimentary. We have all now realized that there is practically no geographic limit for distribution of microbes, though the preference of species to occupy the habitat of their choice always exists. Our urge of exploring as much microbial species as possible has made us vulnerable to culture and propagate some harmful species, which can be categorized as laboratory weeds. What is the best way to characterize microbial communities without identifying every species is yet to be properly understood.

Unlike the higher plants and animals, many microbes have the ability to make their presence felt in many a diverse types of habitats. Few microbes, especially bacteria, are however highly selective in their choice for a set of conditions. Our present knowledge on the diversity of microbes of various habitats is so limited that we are now unable to correctly quantify the density and diversity of microbes of a habitat. Excepting the agricultural soil, where our research achievements have enabled us to get a fair idea of the microbial population and their association with each other and the crop plants, the data on the other habitats, especially on the extreme environments are only speculative.

With approximately only 1% culturable microbial diversity known so far, the field of environmental microbiology is wide open to investigation. The figures for undescribed microbial species have been placed by various workers at approximately 1.9 million. The figures for viruses, bacteria, and algae are only the assumptions of their densities and even those for fungi are quite variable since estimates of specialized groups such as endophytes alone are now placed at approximately 1 million. Since symbiotic associations with wild legumes are known for only 50% plants, it would be anybody's guess to provide a number of yet, undescribed forms in the context of microbial diversity. What is intriguing and interesting in the context of new searches in microbial diversity are, descriptions of novel bacteria and archaea from such common habitats as activated sludge, garden soil, gut of animals and the likes, that have otherwise been extensively surveyed. New ecosystems such as deep-hot biosphere have added not only towards new diversity spectrum but also answers to long evolutionary history of microbe-animal and environmental relationships.



Microorganisms play an unenviable role in maintenance of plant systems in nature as symbionts, pathogens, protectants, nutrient scavengers and as associative members of the root zone that is central to establishment of a species. It is known that the rhizosphere harbours 10- to 100-fold greater microbial population than the bulk soil in most agroecosystems. This is made up of plant growth promotory, deleterious and neutral populations especially in case of a soil dominant genus such as *Pseudomonas*. Contributions of this community in disease control in wheat is a well known example where antifungals comprising phenazines, phloroglucinols and pyrrolnitrin are secreted by such populations. These associative forms, currently termed, 'plant growth promoting rhizobacteria', are a major focus of diversity search since any effective isolate must also be competitive, stable and capable of root colonization. Inherent in this zone is also the component of rhizobia that nodulate legumes and are well known for fixation of nitrogen in natural crop and forest ecosystems. Their contribution in agricultural economy of the world is well known. Last few years have seen rapid search for symbiotic partners of legumes through which novel organisms have been recovered and new phylogenies established. In addition, almost 85% of plant roots are colonized by arbuscular-mycorrhizal fungi. These obligate symbionts whose role in stress tolerance, nutrient acquisition and protection towards phytopathogens are globally tested and proven. Employing spores recovered from fossilized plant specimens, it is now known that AM fungi evolved over 340 million years ago and have, therefore, been mutualists of plants for a very long time in the evolution.

Substantial effort has been devoted to understand the distribution and role of microbial biomass in soils, sediments and other important habitats in terrestrial and aquatic ecosystems. The total diversity of an identified environmental niche can be arrived using combination of culture techniques and assessment of representative population using 16S rDNA amplification and its characterization. For culturable diversity under defined growth condition part of population can be selected and characterized. However, rDNA based methodology gives the complete stock of available diversity.

### 1.2.1 The Microbial Community

Microorganisms are the largest reservoir of genetic and biochemical diversity. They have been evolving for the last 3.8 billion years and inhabit virtually every environment, often thriving under extremes of nutrient concentration, pH, salinity, pressure and temperature. Microbes drive the chemistry of life, do much of the biogeochemical cycling that keeps the world habitable, and even affect the global climate. Over billions of years, microbes have developed a wealth of functions that enable their survival in virtually every environmental niche, often where no other life forms exist. The microbial community is a black box and efforts are underway to understand the microbial wealth. The genomic information represented by such a "community genome" offers a tremendous resource for examining the extent and patterns of microbial genetic diversity and

metabolic capabilities in the natural ecosystems. Very little is known about the spatial and temporal variability of microorganisms. It is postulated that maintenance of soil organic carbon may be important for preservation of microbial diversity.

A central theme in microbial community ecology research is to understand the composition, patterns and dynamics of species diversity at various spatial and temporal scales i.e., from a micro-niche to complete site. The study of microbial diversity and community dynamics in an environment is rapidly growing in environmental microbiology. Interest in this area has been catalyzed by the rapid advancement of molecular methodologies. Through the use of culture-independent molecular techniques, new insights into the composition of uncultivated microbial communities have been gained. It is now becoming possible to define the causes of time-dependent changes in the health of microbial community on the basis of the observed molecular (16S rDNA) diversity. In addition the studies also provide basic information on mechanisms that control species coexistence, which have fundamental applications since they offer the framework that serves biologists interested in maintaining and restoring diversity in natural ecosystems. A detailed information on the microbial community of a habitat can enable make us to standardize culture conditions to isolate, maintain and use the microbial wealth of the concerned habitat.

The limitations in studying microbial diversity do not arise only from methodological limitations, but also from a lack of taxonomic knowledge. It is difficult to study the diversity of a group of microorganisms, when it is not understood how to categorize or identify the species present. Molecular techniques based on PCR have been used to overcome the limitations of culture-based methods. These techniques have been used successfully to demonstrate the presence of bacteria in extreme environments like hot springs, deep sea hydrothermal vents, acid mine drainage, heavy metal contaminated soil and on cold habitats. Microbes can withstand extreme conditions such as can survive in environment having cyanide or perhaps, the most unusual microbe reported is *Deinococcus radiodurans*, which can withstand radiation levels that usually cause massive DNA damage of many living organisms.

Microbial communities within contaminated ecosystems tend to be dominated by those organisms capable of utilizing and/ or surviving toxic contamination. As a result, these communities are typically less diverse than those in non-stressed systems, although the diversity may be influenced by the complexity of chemical mixtures present and the length of time the populations have been exposed to the stress. In these environments, communities survive with fluctuating dynamics driven by the availability and concentration of different carbon and energy sources.

There are probably millions of species in the domains of bacteria and archaea (the prokaryotes), but each follows its own set of rules at the same time governed by the community they belong. Accordingly their distribution and abundance in natural environments have been reported. One characteristic that has become clear is that prokaryote diversity in aquatic environments in order of magnitude

is less in sediments and soils. Hypotheses and models explaining such differences are under development and are beginning to offer promising insights into the mechanisms governing prokaryote diversity and ecosystem function. Despite the predominance of microbes during the course of life's history, general principles and theory of microbial evolution and ecology are not well developed. The investigators have had a fair idea on cultivation of microorganisms and on characterization of naturally occurring microbial biota without cultivation. This has facilitated the microbial community analysis of various habitats, especially of those, whose conditions are difficult to manipulate or maintain in the laboratory. A significant part of such microbial community is considered unculturable. However, we have to accept that we can upgrade our knowledge on the requirements of such microbes to rear them in the laboratory conditions.

Free from the biases of culture-based studies, molecular phylogenetic surveys have revealed a vast array of new microbial groups. Microflora from the deepest sea floor show a wide range of taxa like actinomycetes, fungi, non-extremophilic bacteria, and various extremophilic bacteria such as alkaliphiles, thermophiles and psychrophiles. These representatives are also found on surface with some diversity, indicating the versatility of this biomass. Motile bacteria could even be detected in metabolically active form from ice samples.

### 1.2.2 Microbes in Ecosystem Stability

The stability of an ecosystem is determined by the homeostasis in the rate of flow of energy and matter, which in turn, is dependent on the microbial functional diversity. Differences in microbial community structures reflect the abilities of microorganisms to respond to specific environmental factors and substrates. For example, the fluorescent pseudomonads are attracted to plant roots and show diversity between soil and plant surfaces. Species of *Penicillium* are abundant in temperate and cold climates, whereas *Aspergillus* species predominate in warmer regions. Cyanobacteria are commonly found in neutral to alkaline soils. Depending on the nature of the metabolites present in the soil, nitrogen-fixing, sulphur- and hydrogen-oxidizing and nitrifying bacteria are often found together with denitrifiers, sulphate-reducers and methanogens. Different microbial communities perform different functions in the environment which not only facilitate the matter transfer but also help in the distribution of matter in different parts of the environment. For example, though all microbes are more or less involved in degradation of organic matters in the environment, some species perform the degradation process more effectively than the other. It is more specific in nature when the organic matter is a synthetic one and there are only some specific microorganisms to metabolically or cometabolically degrade it. There are also specific species to perform other functions like nitrogen fixation, phosphorus solubilization, mineralization, etc. (Table 1.1). All these involvements of microbes in ecosystem processes bring stability to the biosphere and minimize ecosystem entropy level.

**Table 1.1:** Functional diversity of microbes in natural environment. The table contains only few examples.

| <i>Ecosystem process</i>     | <i>Common microbes</i>                                                                                                                                                                                                                                      |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organic matter decomposition | <i>Trichoderma, Fusarium, Bacillus, Streptomyces, Clostridium</i>                                                                                                                                                                                           |
| Nitrogen fixation            | <i>Rhizobium, Bradyrhizobium, Frankia, Anabaena, Azotobacter, Beijerinckia, Aerobacter, Chlorobium, Nostoc</i>                                                                                                                                              |
| Nitrogen cycles              | <i>Bacillus, Pseudomonas, Serratia, Nitrobacter, Nitrosomonas, Achromobacter, Pseudomonas</i>                                                                                                                                                               |
| Phosphate solubilization     | <i>Azotobacter, Enterobacter, Bacillus, Aspergillus, Penicillium, Rhizoctonia, Trichoderma</i>                                                                                                                                                              |
| Sulphur transformation       | <i>Desulphovibrio, Thiobacillus</i>                                                                                                                                                                                                                         |
| Iron transformation          | <i>Thiobacillus, Ferribacterium, Leptothrix</i>                                                                                                                                                                                                             |
| Siderophore production       | <i>Neurospora, Trichoderma, Agaricus, Fusarium, Penicillium, ericoid mycorrhizal fungi, Nocardia, Erwinia, Pseudomonas, Bacillus, Aeromonas</i>                                                                                                             |
| Phytohormone production      | <i>Azotobacter, Azospirillum, Pseudomonas, Rhizobium, Bacillus, Flavobacterium, Actinomyces, Nocardia, Fusarium, Gibberella, Alternaria, Penicillium</i>                                                                                                    |
| Biotic control               | <i>Pseudomonas, Bacillus, Streptomyces, Trichoderma</i>                                                                                                                                                                                                     |
| Bioremediation               | <i>Acinetobacter, Alcaligenes, Ochrobactrum, Pseudomonas, Flavimonas, Rhodococcus, Stenotrophomonas, Comamonas, Arthrobacter, Burkholderia, Ralstonia, Moraxella, Nocardia, Klebsiella, Phanerochaete, Penicillium, Aspergillus, Fusarium, Cladosporium</i> |
| Phytoremediation             | <i>Pseudomonas, Agrobacterium, Enterobacter, Rhizobium, Kluyvera, Glomus, Rhizobacteria, mycorrhizal fungi</i>                                                                                                                                              |

Adopted from Singh and Ward (2005).

## 1.3 SCOPE OF ENVIRONMENTAL MICROBIOLOGY

### 1.3.1 Microbes in Degradation of Organic Compounds

Earth is a closed system where the energy flows through the limited amount of carbon, oxygen, and nitrogen as essential elements to support life. For the most of the life on earth, the flow of energy begins with sunlight. It involves a cycle involving specialized microbes and plants to quench sunlight as a source of energy to reduce carbon dioxide to a suitable substrate. Most of the naturally occurring organics are returned back to atmosphere as a part of biogeochemical cycles and become available for remineralization by microbial activity. However,

there are organic molecules, which are chemically synthesized and once they are introduced in any environmental niche, the perturbances in the biogeochemical cycle are realized. Many of these synthetic organic chemicals are toxic to living organisms and the intensity of effects can range from abnormal cell function to complete eradication of biodiversity. Their toxicity effects on microbes range from minor growth inhibition to complete extinction of one or more, especially the sensitive species. However, microbes, with their versatile ability, provide the resilience capacity, because of which the affected ecosystem tries to attend its base line structure (homeostatic equilibrium). The assimilatory capacity to this unknown organic molecule is acquired by the ecosystem with available microbial diversity. Prolonged exposure of a microbial community triggers a modification of the gene pool of the species resulting in evolution of elite organisms with capacity to metabolically process the xenobiotics. The diverse microbes collectively reorganize the genetic pool so that an organic molecule, which was unknown to their genetic intelligence, now becomes the source for the survival of the community. Depending upon the loading of this new organic molecule, the evolved community structures itself in a way that different members play different roles to establish characteristic mineralization kinetics for this new organic molecule. It is therefore necessary to understand the strategy and dynamics of such microbial communities in specialized cases such as industrial effluent treatment plants (ETP) for making the best of their use.

In the chemical industry, the demand of the market schedules the production of different chemicals. When the series of chemicals are produced in parallel, the generated waste has organic loading due to various intermediates at different steps with traces of finished product, which is left after the recovery. It is not difficult to develop a biological solution to this pollution problem. However, there is a requirement for extensive study on the microbial processes and microbial diversity of such contaminated ecosystems in order to identify, culture and maintain the so called super bugs for their application for pollution control. The information on the genetic make-up of the microorganisms of a contaminated habitat can provide idea for molecular approach to solve the environmental contamination.

### 1.3.2 Microbial Community Analysis

Microorganisms are unique and they ‘do not follow borders’. They generally live as complex communities although extreme environments represent much simpler population structures. Until recently study of environmental microbiology (air, water, soil, extreme ecosystems) was dependent on cultivation of individual component of a community that often provided a picture different from the sum total of all the populations. This scenario has completely changed since use of newer tools permit *in situ* assessment of change in community structure as a sequel to perturbations on account of natural calamities (fires, floods) and man-made disturbances (compaction, tillage, pesticides, fertilizers, water stress, etc.). Modern tools of high-throughput genomic and post-genomic

technology, e.g., functional genomics, expression and activity analysis and proteomics are changing the manner in which environmental perspectives of microbial diversity and community structure are being investigated. Comparative genomics has added totally new dimension to ecological and environmental perspectives.

Since functions of a large number of genes are unknown, microbial community analysis, comparative genomics and associated tools will help understand complex communities of activated sludge, polluted sites and soils under stressed conditions. This information will also need being duplicated under chemostats for consortia used in waste treatment processes and clean up of aquifers so that such populations remain stable under various kinds of abiotic and biotic pressures. Metagenomics will permit greater and better understanding of the spatial and temporal heterogeneity of communities and their environments and thus help to develop better management strategies for disturbed ecosystems. Integration of genomics and ecology can help appreciate perturbations on account of enhanced levels of CO<sub>2</sub>, increased rates of N deposition, land use patterns and change, desertification, salinization of soils, and similar other natural and man-made events.

### **1.3.3 Industrial Use of Microbes**

Microorganisms are the backbone of industrial technology for long with more and more microbial species being added to the industrially important microbes list on each passing day. Biotechnological era has seen newer and novel routes to exploitative biology where wild type and modified microorganisms with improved metabolic capacities are sought for commercialization. The study of the properties of the microbes of various environments has made it possible to make use of new species in industries. However, the fact that we were dealing with only 1 % of the culturable diversity until a decade back has hastened the pace for use of new tools and techniques in search for exploitable microorganisms, especially from environments that are unique (hotsprings, deep sea, cold habitats, etc.). Such hot spots of diversity abound in nature and this is an area that has undergone a sea change with the development of new tools of proteomics and genomics. Extensive searches for enzymes and bioactive molecules have been made from microorganisms derived from activated sludge, caves, gut of animals and humans, and deep-sea sediments. It has been argued that until recently, microbiology research was focused on too few species. However, with the perception that lateral gene transfer and genetic mechanisms have had their share of impact on evolution of microbial forms, researchers have started looking for products of industrial interest from across the microbial world.

A major effort in exploitable biology relates to natural product search and discovery wherein the tools of combinatorial chemistry and miniaturized assays permits screening at a rapid pace. Almost all drugs and antibiotics are synthesized by the normal and engineered cells of bacteria and other microbes. Current estimates show that among the naturally derived drugs, over 60% of the sales



relate to antibiotics. Present biotechnological searches are, however, no longer “hitting in the dark” but based on specific targets, molecular designs and large database searches (bioinformatics). Novel and neglected taxa are often being used, e.g., endophytic microorganisms for taxol and related molecules. The other approach relates to choosing a microbial group that is known for its metabolites of a particular kind. The optimization of mass culture of microbes and down stream processing of microbial biomass and/or products has significantly reduced the cost of production. It is estimated that about one hundred thousand types of compounds are now being industrially synthesized by using the microbes of various kinds. In addition, there is a paradigm shift in expanding the scope of such application beyond drugs into sectors such as crop protection agents, food and feed ingredients, biocatalysts (extremozymes), biomaterials and others.

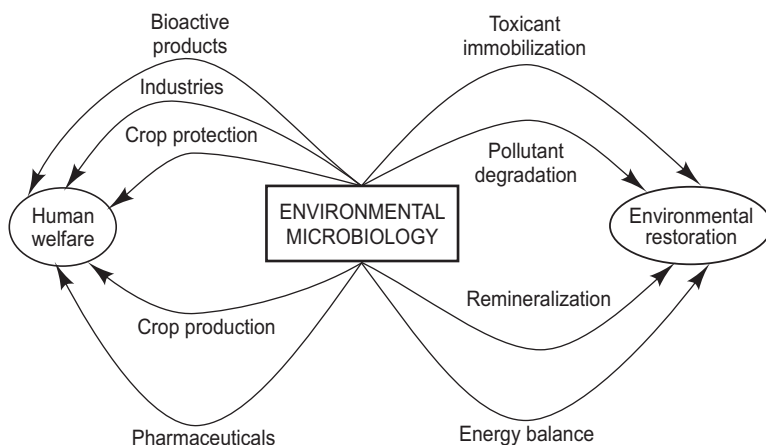
### 1.3.4 Microbes and Human Health

Microorganism-human / animal relationships are quite intimate and mostly to mutual advantage. However, a handful of microbial forms (bacteria, fungi, protozoa, viruses) are causal agents of various diseases. As it works out, the microbial-human relationships are so intense that more than 90% of the cells in human bodies are microorganisms, viz., colonization of skin, mouth and other surfaces by bacteria and fungi. They are a component of gut microflora that helps in digestion and synthesis of essential nutrients and this microbial balance can determine the physical and mental health of an individual. This is proved by the fact that animals kept under sterile conditions are much less resistant to attack by disease-causing agents than those exposed regularly to microbial components. In fact, appearance of disease is a manifestation of an ecological strategy for some microorganisms wherein human or animal body is used as a habitat for multiplication, persistence and transmission.

Despite the use of microbes for synthesis of many drugs to improve healthcare, there is also emergence of drug resistance in disease causing microbes. Since the first availability of an antimicrobial agent from a microbial source, extensive searches for a variety of molecules have continued in order to develop more effective safeguards against human / animal diseases but there has also been a continuous race on account of development of drug resistance. Modern tools of gene technology permit development of designer bioactive molecules as also recombinant vaccines for a variety of catastrophic diseases. Our knowledge on the number of microorganisms producing drugs and antibiotics can still be expanded by the study of the microbial community of different habitats and their analysis for use in human welfare.

Some of the scope of environmental microbiology have been summarized in figure 1.1. The discussed aspects are only a fraction of the hitherto known scope and many others are yet to be properly realized. Like there is no boundary of the distribution of microbes in the environment, the scope and application of microbes for human and environmental services are also not limited. Microbial applications in the environmental and industrial processes has been found to be

significantly influenced by our knowledge in understanding the potential of microorganisms. Many microbes are still beyond our use for social and environmental welfare despite our knowledge on their potential because we are not yet able to standardize their nutrient requirement and culture conditions to achieve their production at an industrial scale. However, the studies on the environmental conditions of these microbes can enable us to culture them in laboratory and further information will definitely aid to the use of more and more number of microbes for production of value added products.



**Figure 1.1** - The scopes of environmental microbiology for improving life on the earth.

## 1.4 MICROBIAL APPLICATION FOR POLLUTION CONTROL

Varieties of microhabitats with different physicochemical gradients and discontinuous environmental conditions are found in nature. Microbes adapt to these microhabitats and live together in consortia, interacting with each other and with other parts of the biota. These interactions control microbial community structure and diversity. Competitive interactions are influenced by the quality of the environment. In soil, the particle size and other factors, like pH, together with the type and amount of available organic compound may affect microbial community structure. Soil microbes are very effective in causing degradation of the environmental pollutants and regenerating the nutrients, thereby supplying them to plants. They are also subjected to considerable seasonal fluctuations in the environment, such as temperature, water content and nutrient availability, which affect their bioremediation potential.

Analysis of microbial functional diversity is important when considering the ability of microbial community to respond to changing environmental conditions, the need to conserve the microbial gene pool and utilization of the selective gene



pools for useful biotechnological applications relevant to bioremediation and phytoremediation. Fortunately, with development of advanced molecular *in situ* methods and improved cultivation procedures, better estimates of the microbial functional diversity on earth can be predicted and its role in soil ecosystem can be thoroughly evaluated. Molecular diagnostic methodologies in bioremediation have been significantly advanced during the last decade. Qualitative detection methods have been replaced with quantitative measurements of the specific microbial population present in the contaminated soils. The analysis of microbial gene pool through metagenome approach is useful to determine the metabolic efficiency of a microbial community and to use the community to get rid of the pollutants.

Microorganisms relevant to environmental biotechnology are widely distributed in the natural environment. Ecologically, hydrocarbon-metabolizing microorganisms are ubiquitous in nature. However, there is a great degree of variation among microbial groups with regard to their efficiency in metabolically degrading hydrocarbon. Nevertheless bacteria have proved themselves to be the most efficient microbial group to cause pollutant degradation in general and hydrocarbon removal in particular (Table 1.2). Numerous studies have attempted to describe microbe-microbe and microbe-hydrocarbon interactions by extrapolating from detailed laboratory studies with isolates from hydrocarbon-contaminated environments. Changes in the microbial population after petroleum spills or during bioremediation processes have been investigated by different groups of researchers. The degradation of hydrocarbon has been found both under aerobic and anaerobic conditions. There are research efforts to utilize hydrocarbon rich wastes as raw materials for microbial production of value added products.

**Table 1.2:** The common organic contaminants in the environment and their bioremediation.

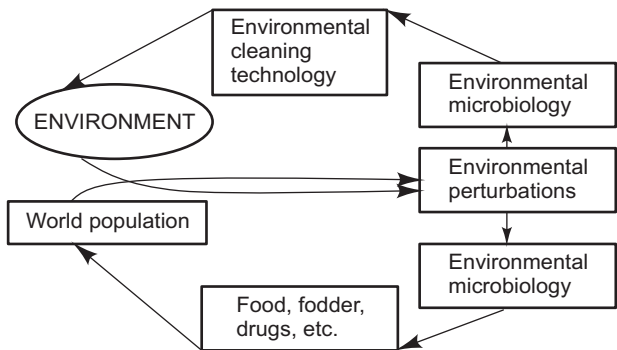
| <i>Pollutant group</i>                | <i>Frequency of occurrence</i> | <i>Dominant microbial group causing degradation</i> |
|---------------------------------------|--------------------------------|-----------------------------------------------------|
| Nutrients                             | Common                         | Algae, macrophytes                                  |
| Organic carbons                       | Common                         | Bacteria, fungi                                     |
| Metals                                | Very frequent                  | Bacteria                                            |
| Chlorinated organics                  | Very frequent                  | Bacteria, fungi                                     |
| Gasoline and fuel oils                | Common                         | Bacteria                                            |
| Polychlorinated aromatic hydrocarbons | Infrequent                     | Bacteria, fungi                                     |
| Cresol                                | Common                         | Some bacteria                                       |
| Alcohols, ketones and esters          | Common                         | Many microbial groups                               |
| Chlorinated organics                  | Very frequent                  | Few bacteria                                        |
| Polychlorinated biphenyls (PCBs)      | Infrequent                     | Few bacteria, fungi                                 |
| Nitroaromatics                        | Common                         | Few bacteria                                        |

We are beginning to understand the astonishing diversity of microbial populations and communities in the environment. However, characterization of natural microbial communities impacted upon by hydrocarbons is a daunting task

due to numerous substrate and metabolite interactions possible in soil. Biochemical/molecular tools are being developed in an attempt to better appreciate microbial abundance and distribution in natural environments to evaluate community structures with ecosystem functions. These methods are also used to determine the long-term effects of pollution and to develop appropriate remediation approaches.

1.5 MICROBES IN HUMAN WELFARE

The traditional approach of considering environmental microbiology synonymous to the study of pathogens and bioremediation has been recently changed with the input of knowledge on the potential of microbes of different environments. Environmental microbiology now ranges from the environmental quality management to the industrial application of engineered microorganisms with an aim to provide quality life to human beings (Fig. 1.2). Some of the fields, in which knowledge on environmental microbiology is a necessity, have been given in Table 1.3. Problems addressed by environmental microbiologists include the discovery and identification of new microbes and microbial products that may have practical application for protection of the environment, protection of human health, and commercial application. Examples include: a) microbial enzymes for use in detergents, food processing, and biotechnology; b) a variety of metabolites including antibiotics for use in medicine; c) surfactants that are not only used in cosmetics and food processing, but also have potential for use in remediation; and d) microbes and microbial processes for enhanced resource recovery.



**Figure 1.2** - The role of environmental microbiology for improvement of environmental quality.

Scientists in search of new microbes are eagerly examining new environments such as the deep sea thermal vents, hot springs, and deep subsurface environments. One challenge is to recover viable microbes from these unique environments. Without an understanding of growth conditions and nutrient requirements, recovery of viable microbes is problematic. Once recovered, these organisms can be characterized and tested to determine whether they produce

novel enzymes, antibiotics, or other products that can be harnessed to solve our problems. For example, pesticide application continues to increase worldwide and is an important factor in increased crop production. However, as a result of continuous pesticide use, we now find pesticide residues in many of our underground drinking water supplies. Many of our food products are contaminated to a large extent and some of them have even been rejected at the international level. The attempt now by an environmental microbiologists is to shift the focus from the traditional chemical pest eradication approach to the modern biological pest control approach. More and more microorganisms, both normal and engineered, are now being tried as biocontrol agents for pre-harvest and post-harvest control of pests in crops and many of the applications have become successful.

**Table 1.3:** The scope of environmental microbiology in relation to maintenance of environmental quality.

| <i>Field of use</i>                           | <i>Microbial issue</i>                                                                                                                    |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Air quality management                        | Collection, detection and control of pathogens or other microbes in aerosols; microbial movement in aerosols                              |
| Agricultural application                      | Biological control, nitrogen fixation, nutrient cycling                                                                                   |
| Maintenance of nutrient regime in environment | Carbon and mineral cycling, control of acid mine drainage, control of loss of fixed nitrogen                                              |
| Bioremediation                                | Degradation of organic contaminants, immobilization or removal of inorganic contaminants found in contaminated soil or water environments |
| Biotechnology                                 | Detection of pathogens or other microbes in the environment; detection of microbial activity in the environment; genetic engineering      |
| Food quality                                  | Detection of pathogens, elimination of pathogens                                                                                          |
| Resource production                           | Production of alcohol, single-cell protein                                                                                                |
| Resource recovery                             | Microbially mediated recovery of oil and metals                                                                                           |
| Solid waste degradation                       | Waste degradation, resource recovery                                                                                                      |
| Wastewater treatment                          | Degradation of waste, reduction of pathogens                                                                                              |
|                                               | Removal of organic and inorganic contaminants, detection of pathogens, elimination of pathogens                                           |
| Water quality                                 | Lowering of organic carbon load; disinfection                                                                                             |

Modified from Maier *et al.* (2006).

Microbes can also be harnessed for resource production, as for example, during recovery of metals from ore (bioleaching, e.g. production of elementary gold, silver, etc.) or recovery of oil by application of microbially produced surfactants (biosurfactants), which is also called microbially enhanced oil recovery (MEOR). Present estimate says that approximately 40% of the copper and uranium mining in the world is done with a microbial process, taking advantage of naturally occurring bacteria such as *Thiobacillus thiooxidans* and *Thiobacillus ferrooxidans*. Microbial immobilization of valuable metals from extremely

diluted solutions is still considered promising and commercially viable. However, the achievement in the microbially enhanced oil recovery is not yet noteworthy.

Bio-manipulation has recently emerged as an additional area of study for environmental microbiologists. At times, microbes are deliberately introduced into the environment for various purposes (Table 1.4). Many of these relate to enhanced crop production or improved remediation of contaminated sites. Introduced microbes can be naturally occurring or can be genetically engineered to combine and enhance metabolic capabilities. Although genetically engineered microbes (GEMs) can be very efficient, their intentional release is currently restricted because of safety concerns. The use of introduced microbes can result in significant benefits provided that they survive long enough to accomplish the desired task. Thus, environmental microbiologists have investigated many of the environmental factors, both biotic and abiotic, that affect survival. Clearly, the concept of distributing modified microbes or augmenting with naturally occurring species has profound implications for protection of the environment and human health. The option of using introduced microbes instead of more traditional approaches also appears to be attractive economically. These are such opportunities that have made the field of environmental microbiology attractive to a broad range of people, including students, researchers, society, and even politicians. However, the implications of bio-manipulations on the indigenous microflora and the autochthonous plant and animal diversity need to be properly addressed.

**Table 1.4:** Some introduced microorganisms and their contribution to environmental protection and crop production.

| <i>Introduced microbe</i>               | <i>Function</i>                                                                                  | <i>Risks</i>                                               |
|-----------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Biocontrol agents, usually bacteria     | Reduced seedling and plant disease                                                               | Sensitivity of beneficial insects                          |
| Bioremediative agents, usually bacteria | Enhanced remediation of organic- and/ or metal—contaminated sites                                | Surface water and soil contamination by leached metal ions |
| Genetically engineered microbes         | Many potential functions. An example is enhanced frost tolerance by prevention of ice nucleation | Laboratory weeds; effects on indigenous microflora         |
| Mycorrhizal fungi                       | Enhanced development of pine seedlings prior to reforestation                                    | Competitions of non-mycorrhizal plant species              |
| Rhizobia                                | Enhanced nitrogen fixation by legumes                                                            | No risk yet visualized                                     |

Modified from Maier *et al.* (2006).

## 1.6 CONCLUSION

With the increase of world population there is enhancement in the rate of resource utilization and environmental perturbations. The trend of consumerism is producing a huge quantity of wastes, some of them are truly hazardous, resulting in the pollution of air, water and soil. The energy flow, species relationship and the microbial diversity of each piece of land of the earth is now under tremendous pressure of pollution. The microbial assemblage and the processes operating therein are required to be studied in depth in order to make the best use of organisms for the productivity improvement and quality maintenance of the ecosystem. Environmental microbiology now includes the microbial diversity and processes of all possible ecosystems so as to identify the species which can be exploited in an industrial scale for human welfare. Hitherto neglected ecosystems like deep sea vents, polar ice caps, wetlands, mangroves, hypersaline habitats, peat soils, hot springs, etc. have drawn attention of environmental scientists to make the analysis of the ability of the exceptionally tolerant microbes of these habitats. The genetic potentials of these habitats have been considered to be very useful for industrial biotechnology. There are attempts to make the use of the potentials of these extremophiles to augment food production and environmental purification. Future research and extension will definitely go a long way towards application of microbes for improvement of environmental quality and ecosystem stability. Many of such aspects have been covered in this text to provide basic and applied information to the readers.

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# MICROBIAL DIVERSITY IN THE ENVIRONMENT

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## 2.1 INTRODUCTION

The microbial world in the environment is highly diverse with a lot of variation in their morphology, physiology and metabolism. The metabolic diversity and genetic elasticity of the microbial groups has enabled them to show their presence in each and every corner of the earth, ranging from a favourable mesophytic habitat to an extremely unfavourable deep sea vent. In environmental microbiology, we categorize microbes into two distinct domains: as prokaryotes and eukaryotes, both of which are intricately involved in all environmental processes. The smallest organisms are the bacteria with a diameter less than a micrometer to several micrometers. The eukaryotic microorganisms are bigger in size and have more structural complexity. However, a detailed study of all the microorganisms is beyond the scope of this book. This chapter contains only an abstract of diversity of the microbial world. Many of these microbial groups will be dealt in detail in subsequent chapters.

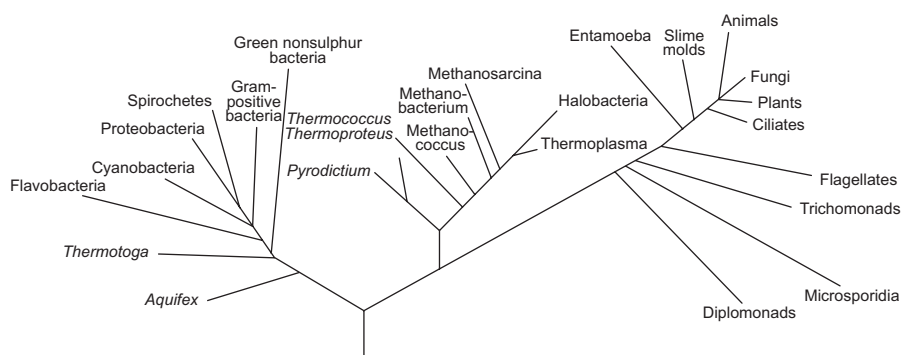
## 2.2 CLASSIFICATION OF MICROBES

Because of the bewildering diversity of microorganisms, it is desirable to classify or arrange them into groups based on their mutual similarities. People often think of microbial taxonomy as trivial and boring, simply a matter of their arrangement on the basis of some physical or metabolic similarities, which may not be required always. Actually, classification of microbes is often important for several reasons. First, it allows us to organize huge amount of knowledge about organisms because all members of a particular group share many characteristics. The more accurate the classification, the more information-rich and useful it is, especially to culture and make use of the organisms. Second, classification of microbes, basing on their metabolic requirements, allows us to make predictions and frame hypotheses for further research based on knowledge of similar organisms. It also facilitates the application of the microorganisms for efficient degradation of

specific chemicals in the environment. Third, taxonomy places microorganisms in meaningful and useful groups with precise names so that microbiologists can work with them and communicate efficiently. Fourth, taxonomy is essential for accurate identification of microorganisms. Thus classification of microbes and knowledge on the microbial diversity of different environments is a prerequisite to make proper assessment of the properties of the microbial world.

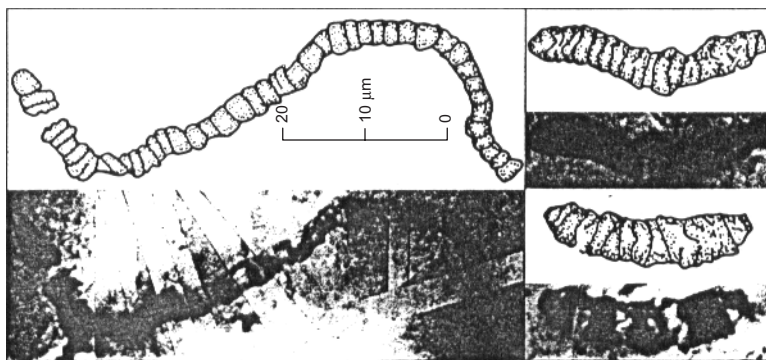
Microbial taxonomy is too broad a subject for adequate coverage in a single chapter. Therefore, this chapter emphasizes general principles and uses examples primarily from dominant microbial groups. Even though new molecular techniques are available in classifying microorganisms, more traditional approaches of using morphological and metabolic characteristics to describe the microbes of different environment are still prevalent (Table 2.1). The studies on rRNA sequences in procaryotic cells suggest that procaryotes diverged into two distinct groups very early. Figure 2.1 depicts a universal phylogenetic tree that reflects these views. The tree is divided in three major branches representing the three primary groups (domains): *Bacteria*, *Archaea* and *Eucarya*. The archaea and bacteria first diverged and then the eucaryotes developed from archaeal ancestors. These three primary groups are called domains and placed above the phylum and kingdom levels. The domains differ markedly from one another. Eucaryotic organisms with primarily glycerol fatty acyl diester membrane lipids and eucaryotic rRNA belong to the *Eucarya*. The domain *Bacteria* contains procaryotic cells with bacterial rRNA and membrane lipids that are primarily diacyl glycerol diesters. Procaryotes having isoprenoid glycerol diether or diglycerol tetraether lipids in their membranes and archaeal rRNA compose the third domain, *Archaea*. Table 2.2 summarises the characteristic features of these three primary microbial groups.

It appears likely that the prokaryotic cells appeared around 4 billion years ago and there are fossilized remains of bacterial cells considered to be of 3.5 billion years old (Fig. 2.2). The modern eucaryotic cells arose from procaryotes about



**Figure 2.1** - The universal phylogenetic tree based on three-domain classification (Adopted from Olsen and Woese, 1993).

1.4 billion years ago. There has been considerable speculation about how eucaryotes might have developed from procaryotic ancestors. It is not certain how this process occurred and two hypotheses have been proposed. According to the first, nuclei, mitochondria and chloroplasts arose by invagination of the plasma membrane to form double-membrane structures containing genetic material and capable of further development and specialization. The similarities among chloroplasts, mitochondria and modern bacteria are due to conservation of primitive procaryotic features by the slowly changing organelles. According to the more popular endosymbiotic hypothesis, the first event was nucleus formation in the pro-eucaryotic cell. The ancestral eucaryotic cell may have developed from a fusion of ancient bacteria and archaea, possibly a gram-negative one even though the conditions that compelled such fusion and the mechanism are not still clearly understood.



**Figure 2.2** - Fossilized bacteria (Adopted from Prescott *et al.*, 2005).

## 2.3 THE WORLD OF BACTERIA

Bacteria are prokaryotic microorganisms with simplest cellular organization. The cell consists of a protoplasm enclosed within a cell envelop, protected from outside by a structurally rigid cell wall. The cells are devoid of any membranous cell organelles, true nucleus and nuclear membrane. They are among the most common and ubiquitous organisms found on the earth adapting themselves metabolically to all types of environmental conditions. The abundance of bacteria can be easily visualized in soil, which can contain up to  $10^{10}$  bacteria per gram soil. The activities and importance of bacteria are evidenced throughout this book, which focuses primarily on bacterial activities in different environments and in various environmental processes. A few of their important effects are nutrient mobilization, bioremediation, pollutant degradation, plant-microbe and animal-microbe interactions, microbial symbiosis with plants and animals, bioindication of the environmental conditions and municipal waste treatment.



**Table 2.1:** The common morphological, physiological and metabolic features used in classification and identification of bacteria.

| <i>Morphological features</i>   | <i>Physiological features</i>        | <i>Metabolic features</i>                           |
|---------------------------------|--------------------------------------|-----------------------------------------------------|
| Cell shape                      | Carbon and nitrogen sources          | Energy sources                                      |
| Cell size                       | Cell wall constituents               | Fermentation products                               |
| Colonial morphology             | General nutritional type             | Luminescence                                        |
| Ultrastructural characteristics | Growth temperature optimum and range | Mechanisms of energy conversion                     |
| Staining behaviour              | Osmotic tolerance                    | Oxygen relationships                                |
| Cilia and flagella              | pH optimum and growth range          | Photosynthetic pigments                             |
| Mechanism of motility           | Salt requirements and tolerance      | Secondary metabolites formed                        |
| Endospore shape and location    |                                      | Sensitivity to metabolic inhibitors and antibiotics |
| Spore morphology and location   |                                      | Storage inclusions                                  |
| Cellular inclusions             |                                      |                                                     |
| Colour                          |                                      |                                                     |
| Motility                        |                                      |                                                     |

### 2.3.1 The Bergey's Manual

In 1923, Davis Bergey, Professor of Bacteriology at the University of Pennsylvania and his colleagues published a classification of bacteria that could be used for identification of the bacterial species. The recent edition of the *Bergey's Manual of Systematic Bacteriology* not only classifies the bacteria but also provides information of the ecological characteristics of individual taxa. Table 2.3 summarizes the organization of the manual for grouping different microbial species. The five volumes of the manual describe five different groups of bacteria having both morphological and biochemical differences. These five groups are discussed in some detail in the subsequent sections.

### 2.3.2 The Archaea

Archaea are quite diverse, both in morphology and physiology and metabolically unique group of microorganisms. They can stain either gram positive or gram negative and may be spherical, rod-shaped, spiral, lobed, plate-shaped, irregularly shaped, or pleomorphic. Some are single cells whereas others form filaments or aggregates. They range in diameter from 0.1 to over 15  $\mu\text{m}$  and some filaments can grow up to 200  $\mu\text{m}$  in length. Multiplication may be by binary fission, budding, fragmentation, or other mechanisms. Archaea can be aerobic, facultatively anaerobic or strictly anaerobic. Nutritionally they range from chemolithoautotrophs to organotrophs occupying many a different types of aquatic and terrestrial environments. They are often found in extreme habitats having various types of stresses. Some are mesophiles while others are hyperthermophiles that can grow above 100°C. Recently archaea have been discovered in cold environments (See Chapter 7). It appears that they constitute up to 34% of the procaryotic biomass in coastal Antarctic surface waters. A few are symbionts in animal digestive systems.

**Table 2.2:** A comparative account of the characteristic features of bacteria, archaea and eucarya.

| <i>Property</i>                        | <i>Bacteria</i>                                     | <i>Archaea</i>                            | <i>Eucarya</i>                         |
|----------------------------------------|-----------------------------------------------------|-------------------------------------------|----------------------------------------|
| Nucleus with nucleolus                 | Absent                                              | Absent                                    | Present                                |
| Membranous cell organelles             | Absent                                              | Absent                                    | Present                                |
| Cell wall                              | Peptidoglycan containing muramic acid               | Various types without muramic acid        | No muramic acid                        |
| Membrane lipids                        | Esterlinked straight chain fatty acids              | Esterlinked branched chain fatty acids    | Esterlinked straight chain fatty acids |
| Gas vesicles                           | Present                                             | Present                                   | Absent                                 |
| t – RNA                                | With thymine and initiation with N-formylmethionine | No thymine and initiation with methionine | Thymine and initiation with methionine |
| Polysystronic m-RNA                    | Present                                             | Present                                   | Absent                                 |
| m-RNA introns                          | Absent                                              | Absent                                    | Present                                |
| Ribosomes                              | 70S                                                 | 70S                                       | 80S                                    |
| DNA dependent RNA polymerase structure | 4 subunits                                          | 8-12 subunits                             | 12-14 subunits                         |
| ATPase similarity                      | No                                                  | Yes                                       | Yes                                    |
| Methanogenesis                         | Absent                                              | Present                                   | Absent                                 |
| Nitrogen fixation                      | Present                                             | Present                                   | Absent                                 |
| Photosynthesis                         | Present                                             | Absent                                    | Present                                |
| Chemolithotrophy                       | Present                                             | Present                                   | Absent                                 |

Source: Prescott *et al.* (2005).

Some features of archaeal genetics are similar to those of bacteria. The archaeal chromosomes that have been studied are single closed DNA circles. However, the genomes of some archaeons are significantly smaller than the normal bacterium. For example, *E. coli* DNA has a size of about  $2.5 \times 10^9$  daltons, whereas *Thermoplasma acidophilum* DNA is about  $0.8 \times 10^9$  daltons and *Methanobacterium thermoautotrophicum* DNA is  $1.1 \times 10^9$  daltons. Archaeal mRNA appears similar to that of bacteria rather than to eucaryotic mRNA. Polygenic mRNA has been discovered and there is no evidence for mRNA splicing. Archaeal promoters are similar to those in bacteria.

Archaeal metabolism varies greatly between the members of different groups. Some archaea are organotrophs, others are autotrophic. A few even carry out an unusual form of photosynthesis. Archaeal carbohydrate metabolism probably occurs without phosphorylation of glucose in Embden-Meyerhof pathway as the enzyme 6-phosphofructokinase has not been found in archaea. Extreme halophiles and thermophiles catabolize glucose using a modified form of the Entner-Doudoroff pathway. The halophiles have slightly different modifications of the pathway than do the extreme thermophiles but still produce pyruvate and NADH or NADPH. Methanogens do not catabolize glucose to any significant extent. In

contrast with glucose degradation, gluconeogenesis proceeds by a reversal of the Embden-Meyerhof pathway in halophiles and methanogens. All archaea that have been studied can oxidize pyruvate to acetyl-CoA. They lack the pyruvate dehydrogenase complex present in eucaryotes and respiratory bacteria and use the enzyme pyruvate oxidoreductase for this purpose. Halophiles and the extreme thermophile *Thermoplasma* do seem to have a functional tricarboxylic acid cycle. No methanogen has yet been found with a complete tricarboxylic acid cycle. Evidence for functional respiratory chains has been obtained in halophiles and thermophiles.

**Table 2.3:** Classification of bacteria following the *Bergey's Manual of Systematic Bacteriology*.

| <i>Taxonomic rank</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>Representative genera</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Volume 1. The Archaea and the deeply branching and phototrophic bacteria</b><br><b>Domain Archaea</b><br>Phylum Crenarchaeota<br>Phylum Euryarchaeota<br>Class I. Methanobacteria<br>Class II. Methanococci<br>Class III. Halobacteria<br>Class IV. Thermoplasmata<br>Class V. Thermococci<br>Class VI. Archaeoglobi<br>Class VII. Methanopyri<br><br><b>Domain Bacteria</b><br>Phylum Aquificae<br>Phylum Thermotogae<br>Phylum Thermodesulfobacteria<br>Phylum "Deinococcus-Thermus"<br>Phylum Chrysiogenetes<br>Phylum Chloroflexi<br>Phylum Thermomicrobia<br>Phylum Nitrospira<br>Phylum Deferribacteres<br>Phylum Cyanobacteria<br>Phylum Chlorobi | <i>Thermoproteus, Pyrodictium, Sulfolobus</i><br><br><i>Methanobacterium</i><br><i>Methanococcus</i><br><i>Halobacterium, Halococcus</i><br><i>Thermoplasma, Picrophilus</i><br><i>Thermococcus, Pyrococcus</i><br><i>Archaeoglobus</i><br><i>Methanopyrus</i><br><br><i>Aquifex, Hydrogenobacter</i><br><i>Thermotoga, Geotoga</i><br><i>Thermodesulfobacterium</i><br><i>Deinococcus, Thermus</i><br><i>Chrysogenes</i><br><i>Chloroflexus, Herpetosiphon</i><br><i>Thermomicrobium</i><br><i>Nitrospira</i><br><i>Geovibrio</i><br><i>Synechococcus, Pleurocapsa, Anabaena</i><br><i>Chlorobium, Pelodictyon</i> |
| <b>Volume 2. The Proteobacteria</b><br>Phylum Proteobacteria<br>Class I. Alphaproteobacteria<br><br>Class II. Betaproteobacteria<br><br>Class III. Gammaproteobacteria<br><br>Class IV. Deltaproteobacteria<br>Class V. Epsilonproteobacteria                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Rhodospirillum, Caulabacter, Rhizobium, Methylobacterium, Hyphomicrobium</i><br><i>Neisseria, Alcaligenes, Comamonas, Nitrosomonas, Methylophilus</i><br><i>Chromatium, Pseudomonas, Azotobacter, Escherichia, Klebsiella, Proteus, Salmonella</i><br><i>Desulfovibrio, Bdellovibrio, Myxococcus</i><br><i>Campylobacter, Helicobacter</i>                                                                                                                                                                                                                                                                       |

**Table 2.3:** (Contd.)

|                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Volume 3. The low G + C gram-positive bacteria</b><br>Phylum <i>Firmicutes</i><br>Class I. Clostridia<br>Class II. Mollicutes<br>Class III. Bacilli                                                                                                                                                                                                                                   | <i>Clostridium, Eubacterium, Heliobacterium</i><br><i>Mycoplasma, Ureaplasma, Spiroplasma</i><br><i>Bacillus, Lactobacillus, Streptococcus</i>                                                                                                                                                                           |
| <b>Volume 4. The high G + C gram-positive bacteria</b><br>Phylum <i>Actinobacteria</i><br>Class Actinobacteria                                                                                                                                                                                                                                                                           | <i>Actinomyces, Arthrobacter, Corynebacterium</i>                                                                                                                                                                                                                                                                        |
| <b>Volume 5. The Planctomycetes, Spirochaetes, Fibrobacteres, Bacteroidetes and Fusobacteria</b><br>Phylum <i>Planctomycetes</i><br>Phylum <i>Chlamydiae</i><br>Phylum <i>Spirochaetes</i><br>Phylum <i>Fibrobacteres</i><br>Phylum <i>Acidobacteria</i><br>Phylum <i>Bacteroidetes</i><br><br>Phylum <i>Fusobacteria</i><br>Phylum <i>Verrucomicrobia</i><br>Phylum <i>Dictyoglomus</i> | <i>Planctomyces, Gemmata</i><br><i>Chlamydia</i><br><i>Spirochaeta, Borrelia, Treponema</i><br><i>Fibrobacter</i><br><i>Acidobacterium</i><br><i>Bacteroides, Porphyromonas, Prevotella, Flavobacterium, Sphingobacterium</i><br><i>Fusobacterium, Streptobacillus</i><br><i>Verrucomicrobium</i><br><i>Dictyoglomus</i> |

Source: Garrity (2001).

The biosynthetic pathways for amino acids, purines and pyrimidines in the archaea are similar to those in other organisms. Some methanogens can fix atmospheric dinitrogen. Not only do many archaea use a reversal of the Embden-Meyerhof pathway to synthesize glucose, but at least some methanogens and extreme thermophiles employ glycogen as their major reserve material. Autotrophy is widespread among the methanogens and extreme thermophiles and CO<sub>2</sub> fixation occurs by the reductive tricarboxylic acid cycle (*Thermoproteus*) or by the reductive acetyl-CoA pathway (*Methanococcus*). Archaeal protein synthesis has both bacterial and eucaryotic features.

Many of the archaea constitute a significant fraction of the oceanic picoplankton. They have been divided in five major groups based on physiological and morphological differences. Table 2.4 summarizes some characteristics of these five groups and gives representatives of each. The *Bergey's Manual* divides the archaea into the phyla *Euryarchaeota* and *Crenarchaeota*. The former occupy many different ecological niches and have a variety of metabolic patterns. This phylum contains seven classes, viz., *Methanobacteria*, *Methanococci*, *Halobacteria*, *Thermoplasmata*, *Thermococci*, *Archaeglobi* and *Methanopyri*. These classes are further divided into 9 orders and 16 families. The methanogens, extreme halophiles, sulphate reducers and many extreme thermophiles with sulphur dependent

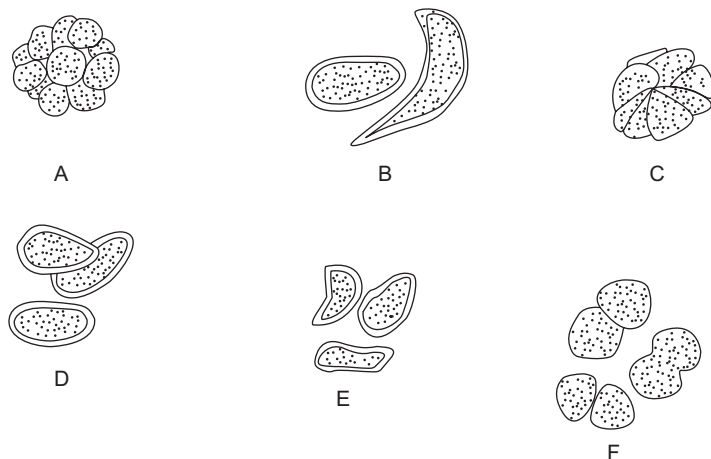
metabolism are placed in the *Euryarchaeota*. Methanogens are the dominant physiological groups (Fig. 2.3). The crenarchaeotes are thought to resemble the ancestor of the archaea and almost all the well-characterized species are thermophiles or hyperthermophiles. The phylum *Crenarchaeota* is divided into one class, *Thermoprotei*, three orders and five families. The order *Thermoproteales* contains gram-negative anaerobic to facultative, hyperthermophilic rods. They often grow chemolithoautotrophically by reducing sulphur to hydrogen sulphide. Members of the order *Sulfolobales* are coccus-shaped thermoacidophiles. The order *Desulfurococcales* contains gram-negative coccoid or disk-shaped hyperthermophiles. They grow either chemolithotrophically by hydrogen oxidation or organotrophically by either fermentation or respiration with sulphur as the electron acceptor.

**Table 2.4:** The characteristic features of the major archaeal groups.

| Group                                     | General characteristics                                                                                                                                                                                                                                                                                                  | Representative genera                                                                                                                 |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Methanogenic archaea                      | Strict anaerobes. Methane is the major metabolic end product. $S^0$ may be reduced to $H_2S$ without yielding energy. Cells possess coenzyme M, factors 420 and 430 and methanopterin                                                                                                                                    | <i>Methanobacterium</i><br><i>Methanococcus</i><br><i>Methanomicrobium</i><br><i>Methanosarcina</i>                                   |
| Archaea sulphate reducers                 | Irregular gram-negative coccoid cells. $H_2S$ formed from thiosulphate and sulphate. Autotrophic growth with thiosulphate and $H_2$ . Can grow heterotrophically. Traces of methane also formed. Extremely thermophilic and strictly anaerobic. Possess factor 420 and methanopterin but not coenzyme M or factor 430    | <i>Archaeoglobus</i>                                                                                                                  |
| Extremely halophilic archaea              | Coccoid or irregularly shaped rods. Gram-negative or gram-positive, primarily aerobic chemoorganotrophs. Require high sodium chloride concentrations for growth ( $= 1.5 M$ ). Neutrophilic or alkalophilic, mesophilic or slightly thermophilic. Some species contain bacteriorhodopsin and use light for ATP synthesis | <i>Halobacterium</i><br><i>Halococcus</i><br><i>Natronobacterium</i>                                                                  |
| Cell wall-less archaea                    | Pleomorphic cells lacking a cell wall. Thermoacidophilic and chemoorganotrophic. Facultatively anaerobic. Plasma membrane contains a mannose-rich glycoprotein and a lipoglycan                                                                                                                                          | <i>Thermoplasma</i>                                                                                                                   |
| Extremely thermophilic $S^0$ metabolizers | Gram-negative rods, filaments or cocci. Obligately thermophilic. Usually strict anaerobes but may be aerobic or facultative. Acidophilic or neutrophilic. Autotrophic or heterotrophic. Most are sulphur metabolizers. $S^0$ reduced to $H_2S$ anaerobically; $H_2S$ or $S^0$ oxidized to $H_2SO_4$ aerobically          | <i>Desulfurococcus</i><br><i>Pyrodictium</i><br><i>Pyrococcus</i><br><i>Sulfolobus</i><br><i>Thermococcus</i><br><i>Thermoproteus</i> |

Compiled from Olsen and Woese (1997); Bell and Jackson (1998); Vetriani and Reysenbach (2000).

**(a) Phylum Crenarchaeota:** Most are extremely thermophiles and many are acidophiles and sulphur dependent. The sulphur may be used either as an electron acceptor in anaerobic respiration or as an electron source by lithotrophs. Almost all are strict anaerobes growing geothermally in heated water or soils that contain elemental sulphur. Representative genera are *Thermoproteus*, *Pyrodictium* and *Sulfolobus*.



**Figure 2.3** - Morphological variation in archaeobacteria: (A) *Sulfolobus* sp., (B) *Thermoproteus* sp., (C) *Methanasarcina* sp., (D) *Methanobrevibacterium* sp., (E) *Methanogenium* sp. and (F) *Halococcus* sp.

**(b) Phylum Euryarchaeota:** It is further divided in the following groups.

**(i) The Methanogens:** This is the largest group of archaea. These are strict anaerobes that can convert  $\text{CO}_2$ ,  $\text{H}_2$ , formate, methanol, acetate and other compounds to either methane or methane and  $\text{CO}_2$ . There are five orders, viz., *Methanobacteriales*, *Methanococcales*, *Methanomicrobiales*, *Methanosarcinales* and *Methanopyrales*, which differ greatly in overall shape. One of the most unusual methanogenic groups is the class *Methanopyri*, which is extremely thermophilic rod isolated from a marine hydrothermal vent. The representative genera are *Methanococcus*, *Methanothermus*, *Methanomicrobium* and *Methanogenium*.

**(ii) The Halobacteria:** This group includes extreme halophiles having 15 genera in one family. They are aerobic chemoheterotrophs with respiratory metabolism and require complex nutrients, usually proteins and amino acids, for growth. These procaryotes have an absolute dependence on a high concentration of NaCl (at least 1.5 M). The representative genera are *Halobacterium* and *Halococcus*.

**(iii) The Thermoplasma:** These are thermoacidophiles that lack cell walls but plasma membrane is strengthened by large quantities of diglycerol tetraethers, lipopolysaccharides and glycoproteins. At present, there are three genera:

*Thermoplasma*, *Picrophilus* and *Ferroplasma*. They are sufficiently different from one another and placed in separate families. They grow in refuse piles of coal mines. These piles contain large amounts of iron pyrite (FeS), which is oxidized to sulphuric acid by chemolithotrophic bacteria. As a result the piles become very hot and acidic. This is an ideal habitat for the thermoplasmas since they grow in the temperature range between 45–65°C and pH range between 0.7 – 3.

**(iv) Sulphate-Reducing Archaea:** These are gram-negative and irregular coccoid cells with walls consisting of glycoprotein subunits. *Archaeoglobus* is extremely thermophilic (the optimum is about 83°C) and can be isolated from marine hydrothermal vents.

### 2.3.3 The Gram-Negative Nonproteobacteria

**(a) Phyla Aquificae and Thermotogae:** Two of the best-studied genera of the phylum *Aquificae* are *Aquifex* and *Hydrogenobacter*. These are gram-negative, microaerophilic rods growing in hyperthermophilic conditions with a temperature optimum of 85°C and a maximum of 95°C. *Aquifex* is an autotroph and generates energy by oxidizing donors such as hydrogen, thiosulphate and sulphur with oxygen as the acceptor. The members of the *Thermotogae* are gram-negative hyperthermophiles with a growth optimum of 80°C and a maximum of 90°C. They grow in active geothermal areas, both marine hydrothermal systems and terrestrial sulphataric springs.

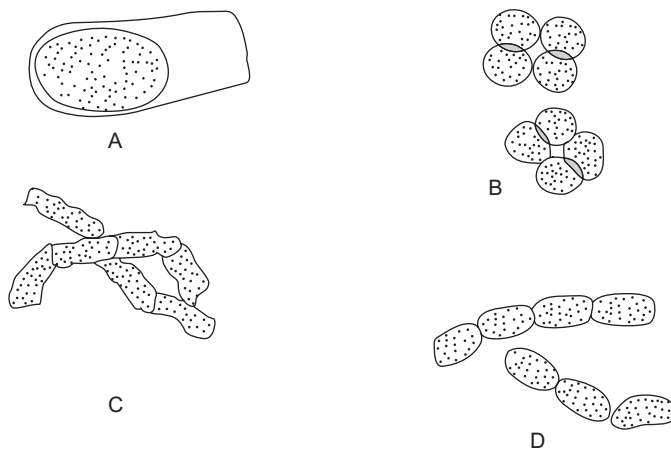
**(b) Phylum Deinococcus-Thermus:** It contains only three genera with *Deinococcus* being best studied one. Deinococci are spherical or rod-shaped aerobic and mesophilic bacteria and can produce acid from only a few sugars. Although they stain gram positive, their cell wall is layered and has an outer membrane like the gram-negative bacteria. Almost all strains are extraordinarily resistant to both desiccation and radiation and can survive as much as 3 to 5 million rad of radiation.

**(c) Photosynthetic bacteria:** There are three groups of gram-negative photosynthetic bacteria: the purple bacteria, the green bacteria and the cyanobacteria. The cyanobacteria differ most fundamentally from the green and purple photosynthetic bacteria in being able to carry out oxygenic photosynthesis. They use water as an electron donor and generate oxygen during photosynthesis. In contrast, purple and green bacteria use anoxygenic photosynthesis. Because they are unable to use water as an electron source, they employ reduced molecules such as hydrogen sulphide, sulphur, hydrogen and organic matters as their electron source for the generation of NADH and NADPH. Consequently, purple and green bacteria do not produce oxygen but many form sulphur granules. Purple sulphur bacteria accumulate granules within their cells, whereas green sulphur bacteria deposit the sulphur granules outside their cells. The purple nonsulphur bacteria normally use organic molecules as an electron source. There



are also differences in photosynthetic pigments, the organization of photosynthetic membranes, nutritional requirements and oxygen relationships among different photosynthetic bacteria (Table 2.5).

The *Bergey's Manual* places photosynthetic bacteria into seven major groups distributed among five bacterial phyla. The phylum *Chloroflexi* contains the green nonsulphur bacteria and the phylum *Chlorobi* the green sulphur bacteria. The cyanobacteria are placed in their own phylum, *Cyanobacteria*. Purple bacteria are divided in three groups. Purple sulphur bacteria are placed in the  $\gamma$ -proteobacteria. The purple nonsulphur bacteria are distributed between the  $\alpha$ -proteobacteria and  $\beta$ -proteobacteria. Finally, the gram-positive heliobacteria in the phylum *Firmicutes* are also photosynthetic having different structural peculiarities (Fig. 2.4).



**Figure 2.4** - The morphological variations in non-proteobacteria. (A) *Thermotoga maritima* showing extended loose seath, (B) *Deinococcus* sp., (C) *Pelodictyon* sp. and (D) *Chlorobium* sp.

**(d) Phylum Spirochaetes:** This phylum contains gram-negative, chemo-heterotrophic bacteria distinguished by their structure and mechanism of motility. They are slender, long bacteria (0.1 to 3.0  $\mu\text{m}$  by 5 to 250  $\mu\text{m}$ ) with a flexible, helical shape. Spirochaetes differ greatly from other bacteria with respect to motility and can move through very viscous solutions though they lack external rotating flagella. They can be anaerobic, facultatively anaerobic or aerobic. Carbohydrates, amino acids, long-chain fatty acids and long-chain fatty alcohols may serve as carbon and energy sources. This group of bacteria is exceptionally diverse ecologically and grows in habitats ranging from mud to the human mouth. Members include the genera *Spirochaeta*, *Leptospira*, *Trepollema* and *Borrelia*.



**Table 2.5:** Characteristics of the major groups of gram-negative photosynthetic bacteria.

| Characteristics                        | Green sulphur                                                                           | Green non-sulphur                                                            | Purple sulphur                                                                                                          | Purple non-sulphur                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Major photosynthetic pigments          | Bacteriochlorophylls <i>a</i> plus <i>c</i> , <i>d</i> , or <i>e</i>                    | Bacteriochlorophylls <i>a</i> and <i>c</i>                                   | Bacteriochlorophylls <i>a</i> and <i>b</i>                                                                              | Bacteriochlorophylls <i>a</i> and <i>b</i>                                                                               |
| Morphology of photosynthetic membranes | Photosynthetic system partly in chlorosomes that are independent of the plasma membrane | Chlorosomes present when grown anaerobically                                 | Photosynthetic system contains in spherical or lamellar membrane complexes that are continuous with the plasma membrane | Photosynthetic system contained in spherical or lamellar membrane complexes that are continuous with the plasma membrane |
| Photosynthetic electron donors         | H <sub>2</sub> , H <sub>2</sub> S, S                                                    | H <sub>2</sub> , H <sub>2</sub> S                                            | H <sub>2</sub> , H <sub>2</sub> S, S                                                                                    | Usually organic molecules; some times reduced sulphur compounds or H <sub>2</sub>                                        |
| Sulphur deposition                     | Outside of the cell                                                                     | None                                                                         | Inside of the cell                                                                                                      | Outside of the cell in few cases                                                                                         |
| General metabolic type                 | Obligately anaerobic                                                                    | Usually photoheterotrophic; sometimes photoautotrophic or chemoheterotrophic | Obligately anaerobic photolithoautotrophs                                                                               | Usually anaerobic photoorganoheterotrophs                                                                                |

**(e) Phylum Bacteroidetes:** The members of this phylum are anaerobic, gram-negative, nonsporing, motile or nonmotile rods of various shapes. These bacteria are chemoheterotrophic and usually produce a mixture of organic acids as fermentation end products, but they do not reduce sulphate or other sulphur compounds. These bacteria grow in habitats such as the oral cavity and intestinal tract of humans and other animals and the rumen. About 30% of the bacteria isolated from human feces are members of the genus *Bacteroides* and these organisms may provide extra nutrition by degrading cellulose, pectins and other complex carbohydrates. They are also involved in human diseases. Members of the genus *Bacteroides* are associated with diseases of major organ systems, ranging from the central nervous system to the skeletal system. Important genera are *Sphingobacterium*, *Saprospira*, *Flexibacter*, *Cytophaga*, *Sporocytophaga* and *Crenothrix*.

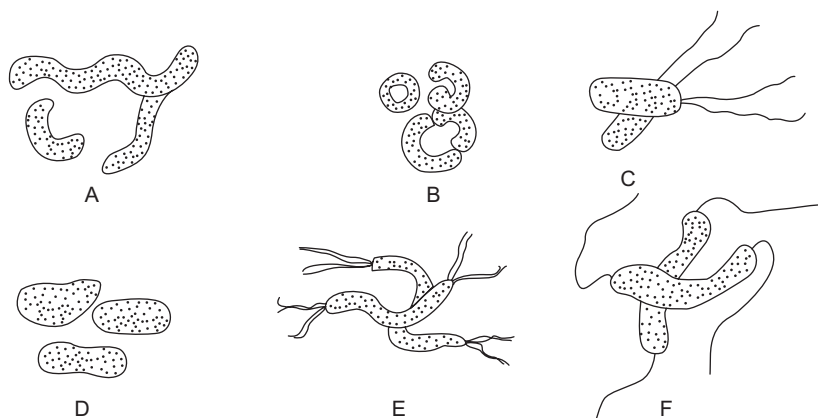
### 2.3.4 The Proteobacteria

The proteobacteria are highly diverse in their phenotypic properties, nutritional type, oxygen relationship and motility. Many of these bacteria are gram-negative with ability to accommodate themselves to a variety of habitats. They are very useful experimental tools in many laboratories. Although many of these bacteria

do not vary drastically in general appearance, they often are very diverse in their metabolism and life style, which range from obligate intracellular parasitism to a free living existence in soil and aquatic habitats (Table 2.6).

**(a)  $\alpha$ -Proteobacteria:** They include most of the oligotrophic proteobacteria. Some have unusual metabolic modes such as methylotrophy (*Methylobacterium*), chemolithotrophy (*Nitrobacter*) and the ability to fix nitrogen (*Rhizobium*). Members of genera such as *Rickettsia* and *Brucella* are important pathogens. In fact, *Rickettsia* has become an obligately intracellular parasite. Many genera are characterized by distinctive morphology such as prosthecae (Fig. 2.5).

- (i) The purple nonsulphur bacteria: All the purple bacteria use anoxygenic photosynthesis, possess bacteriochlorophylls *a* or *b* and have their photosynthetic apparatus in membrane systems that are continuous with the plasma membrane. Normally they grow anaerobically as photoorganoheterotrophs (Table 2.5). In dark, most purple nonsulphur bacteria can grow aerobically as chemoorganoheterotrophs, but some species carry out fermentation and grow anaerobically.



**Figure 2.5** - The morphological variation in proteobacteria. (A) *Rhodospirillum* sp., (B) *Rhodocyclus* sp., (C) *Rhizobium* sp., (D) *Nitrosomonas* sp., (E) *Spirillum* sp. and (F) *Thiomicrospira* sp.

- (ii) Nitrogen fixing bacteria: This group includes at least 10 families with a great variation of phenotypes. The gram-negative, aerobic genera *Rhizobium* and *Agrobacterium* are the two well studied genera. Members of the genus *Rhizobium* are motile rods, often containing poly- $\beta$ -hydroxybutyrate granules, that become pleomorphic under adverse conditions. They grow symbiotically within root nodule cells of legumes as nitrogen-fixing bacteroids. In contrast, *Azotobacter* is a free-living soil genus and fixes atmospheric nitrogen nonsymbiotically.
- (iii) Nitrifying bacteria: The taxonomy of these aerobic chemolithotrophic bacteria, those derive energy and electrons from reduced inorganic

compounds, is quite complex. Normally these bacteria use  $\text{CO}_2$  as their carbon source and thus are chemolithoautotrophs, but some can function as chemolithoheterotrophs and use reduced organic carbon sources. The nitrifying bacteria are  $\alpha$ -,  $\beta$ - and  $\gamma$ -proteobacteria type. *Nitrobacter* is  $\alpha$ -proteobacteria; *Nitrosomonas* and *Nitrospira* are  $\beta$ -proteobacteria; *Nitrococcus* and *Nitrosococcus* are  $\gamma$ -proteobacteria.

**Table 2.6:** The commonly occurring genera of gram-negative proteobacteria.

| $\alpha$ -proteobacteria                                                                                                                                    | $\beta$ -proteo-<br>bacteria                                                                                                                           | $\gamma$ -proteobacteria                                                                                                                                                                                                                          | $\delta$ -proteo-<br>bacteria                                                                                   | Nitrifying<br>bacteria                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <i>Agrobacterium</i><br><i>Caulobacter</i><br><i>Hyphomicrobium</i><br><i>Nitrobacter</i><br><i>Rhizobium</i><br><i>Rhodospirillum</i><br><i>Rickettsia</i> | <i>Bordetella</i><br><i>Burkholderia</i><br><i>Leptothrix</i><br><i>Neisseria</i><br><i>Nitrosomonas</i><br><i>Sphaerotilus</i><br><i>Thiobacillus</i> | <i>Azotobacter</i><br><i>Beggiatoa</i><br><i>Chromatium</i><br><i>Ectothiorhodospira</i><br><i>Escherichia</i><br><i>Haemophilus</i><br><i>Leucothrix</i><br><i>Methylococcus</i><br><i>Photobacterium</i><br><i>Pseudomonas</i><br><i>Vibrio</i> | <i>Bdellovibrio</i><br><i>Desulfovibrio</i><br><i>Desulfuromonas</i><br><i>Myxococcus</i><br><i>Stigmatella</i> | <i>Nitrosomonas</i><br><i>Nitrococcus</i><br><i>Nitrospira</i><br><i>Nitrosococcus</i><br><i>Nitrobacter</i> |

**(b)  $\beta$ -Proteobacteria:** The  $\beta$ -proteobacteria overlap the  $\alpha$ -proteobacteria metabolically but tend to use substances that diffuse from organic decomposition in the anaerobic zone of habitats. Some of these bacteria use hydrogen, ammonia, methane, volatile fatty acids and similar substances. The  $\beta$ -proteobacteria may be chemoheterotrophs, photolithotrophs, methylotrophs and chemolithotrophs. The subgroup contains two genera with important human pathogens: viz., *Neisseria* and *Bordetella*. The other important genera of this group are *Burkholderia*, *Nitrosomonas*, *Nitrospira*, *Gallionella* and *Spirillum*. The genus *Thiobacillus* is one of the best-studied chemolithotrophs and most prominent of the colourless sulphur bacteria. Like the nitrifying bacteria, colourless sulphur bacteria are a highly diverse group. Many are unicellular rod-shaped or spiral sulphur-oxidizing bacteria that are nonmotile or motile by flagella.

Sulphur-oxidizing  $\beta$ -proteobacteria have wide distribution and great practical importance. *Thiobacillus* grows in soil and aquatic habitats, both freshwater and marine. In marine habitats *Thiomicrospira* is more important than *Thiobacillus*. Because of their great acid tolerance, these bacteria prosper in habitats they have acidified by sulphuric acid production (pH as low as 0.5), even though most other organisms are dying. The production of large quantities of sulphuric acid and ferric iron by *T. ferrooxidans* corrodes concrete and pipe structures (See Chapter 11). Thiobacilli often cause extensive acid and metal pollution when they release metals from mine wastes. However, sulphur-oxidizing bacteria are also beneficial. They may increase soil fertility when they release elemental sulphur by oxidizing it to sulphate. Thiobacilli are used in processing low-grade metal ores because of their ability to leach metals from ore (See Chapter 13).

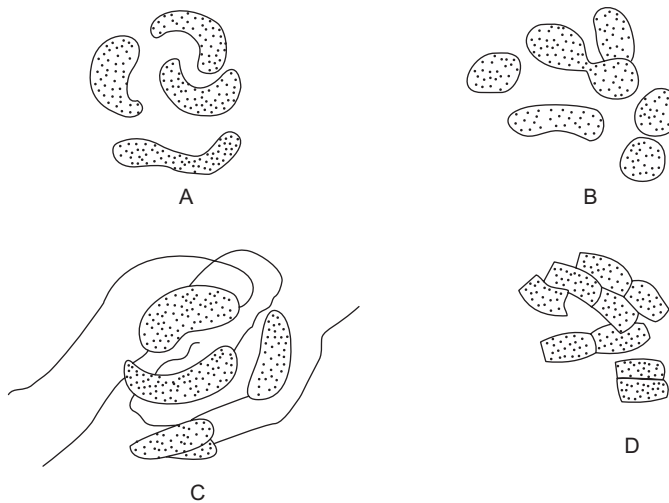
**(c)  $\gamma$ -Proteobacteria:** The  $\gamma$ -proteobacteria constitute the largest subgroup of proteobacteria with an extraordinary variety of physiological types. Many important genera are chemoorganotrophic and facultatively anaerobic. Other genera contain aerobic chemoorganotrophs, photolithotrophs, chemolithotrophs or methylotrophs. According to some DNA-rRNA hybridization studies, the  $\gamma$ -proteobacteria are composed of several deeply branching groups. One consists of the purple sulphur bacteria; a second includes the intracellular parasites *Legionella* and *Coxiella*. The two largest groups contain a wide variety of nonphotosynthetic genera. Many genera of the group are facultative anaerobes and use the glycolytic and pentose phosphate pathways to catabolize carbohydrates. The genera *Pseudomonas*, *Azotobacter*, *Moraxella*, *Xanthomonas* and *Acinetobacter* belong to this superfamily.

The family *Methylococcaceae*, belonging to this group of bacteria, contains rods, vibrios and cocci that use methane, methanol and other reduced one-carbon compounds as their sole carbon and energy sources under aerobic or microaerobic (low oxygen) conditions, i.e. they are methylotrophs. The family contains seven genera, two of which are *Methylococcus* (spherical, nonmotile cells) and *Methylomonas* (straight, curved, or branched rods with a single, polar flagellum). When oxidizing methane, the bacteria contain complex arrays of intracellular membranes. Almost all can form some types of resting stages, often a cyst, somewhat like that of the azotobacteria. Methylotrophic growth depends on the presence of methane and related compounds. Methanogenesis from substrates such as  $H_2$  and  $CO_2$  is widespread in anaerobic soil and water and methylotrophic bacteria grow above anaerobic habitats all over the world.

The pseudomonads are the most abundant and widely distributed members of  $\gamma$ -proteobacteria. *Pseudomonas* is the most well studied and well represented genus. It is straight or slightly curved gram-negative rod. These chemoheterotrophs are aerobic and carry out respiratory metabolism with  $O_2$  (and sometimes nitrate) as the electron acceptor. All pseudomonads have a functional tricarboxylic acid cycle and can oxidize substrates to  $CO_2$ . Most hexoses are degraded by the Entner-Doudoroff pathway rather than glycolytically. Many pseudomonads can degrade an exceptionally wide variety of organic molecules. Thus they are very important in the mineralization process in nature and in sewage treatment. The fluorescent pseudomonads can use approximately 80 different substances as their carbon and energy sources. Some pseudomonads are major animal and plant pathogens and also are involved in the spoilage of refrigerated milk, meat, eggs and seafood because they can grow at  $4^\circ C$  and degrade lipids and proteins.

**(d)  $\delta$ -Proteobacteria:** Although the  $\delta$ -proteobacteria are not a large assemblage of genera, they show considerable morphological and physiological diversity. These bacteria can be divided in two general groups, all of them chemoorganotrophs. Some genera are predators such as the bdellovibrios and myxobacteria. Others are anaerobes that generate sulphide from sulphate and sulphur while oxidizing organic matters. Figure 2.6 illustrates the basic structures of some representative genera.

These sulphate- or sulphur-reducing bacteria are a diverse group united by their anaerobic nature and the ability to use elemental sulphur or sulphate and other oxidized sulphur compounds as electron acceptors during anaerobic respiration. An electron transport chain generates ATP and reduces sulphur and sulphate to hydrogen sulphide. This process is very important in the cycling of sulphur within the ecosystem. Because significant amounts of sulphate are present in almost all aquatic and terrestrial habitats, sulphate reducing bacteria are widespread and active in locations made anaerobic by microbial digestion of organic materials. *Desulfovibrio* and other sulphate-reducing bacteria thrive in habitats such as muds and sediments of polluted lakes and streams, sewage lagoons and digesters and waterlogged soils. Sulphate-reducing bacteria negatively impact industry by accelerating corrosion.

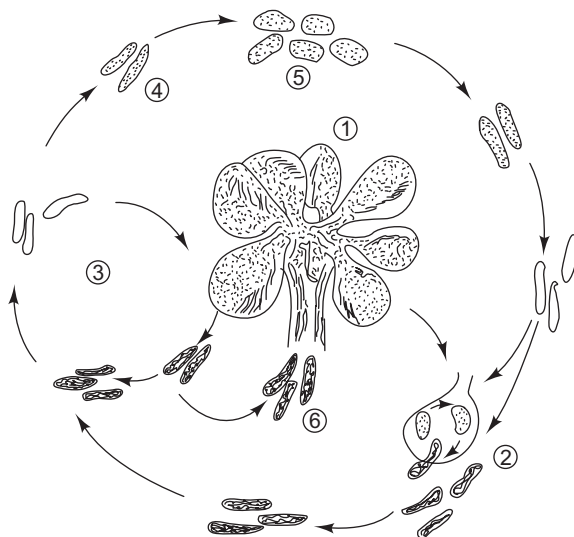


**Figure 2.6** - Morphological variations in Deltaproteobacteria: (A) *Desulfovibrio* sp., (B) *Desulfobacter* sp., (C) *Bdellovibrio* sp. and (D) *Chondromyces* sp.

*Bdellovibrio* and myxobacteria are gram-negative, aerobic soil bacteria. These are micropredators or scavengers. The latter secrete an array of digestive enzymes that lyse bacteria and also secrete antibiotics, which may kill their prey. The digestion products, primarily small peptides, are absorbed. Most myxobacteria use amino acids as their major source of carbon, nitrogen and energy. All are chemoheterotrophs with respiratory metabolism.

The myxobacterial life cycle is quite distinctive and in many ways resembles that of the cellular slime molds (Fig. 2.7). In the presence of a food supply, myxobacteria migrate along a solid surface, feeding and leaving slime trails. During this stage the cells often form a swarm and move in a coordinated fashion. Some species congregate to produce a sheet of cells that moves rhythmically to generate waves or ripples. When their nutrient supply is exhausted, the myxobacteria aggregate and differentiate into a fruiting body. This is a complex developmental process triggered by starvation and different signals given by

specific proteins and amino acids. The fruiting bodies vary in complexity from simple globular objects (*Myxococcus*) to the elaborate, branching, treelike structures (*Srigmatella*). The fruiting body develops myxospores and releases them at maturity. A mass of myxobacteria can produce enzyme concentrations sufficient to digest their prey more easily than an individual cell can.



**Figure 2.7** - The myxobacterial life cycle showing sporulation and germination processes. (1) Fruiting body from cooperative morphogenesis, (2) inversion, (3) vegetative life cycle, (4) spores from vegetative cells, (5) mixospores, (6) vegetative cells for fruiting body formation.

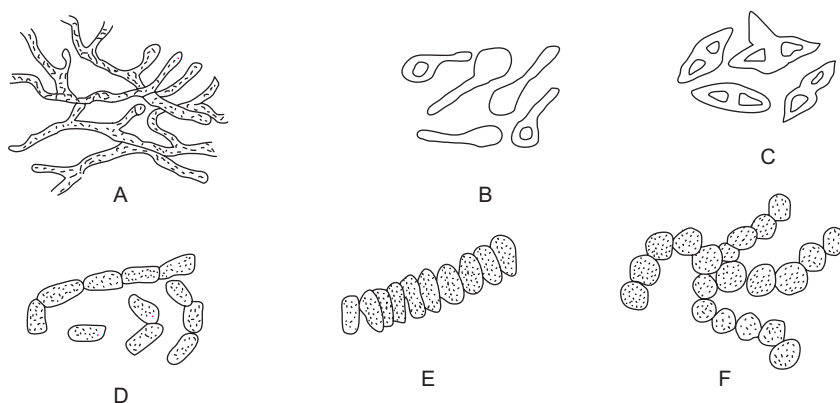
Myxobacteria are found in soils worldwide. They are most commonly isolated from neutral soils or decaying plant material such as leaves and tree bark and from animal dung. Although they grow in habitats as diverse as tropical rain forests and the arctic tundra, they are most abundant in warm areas.

**(e)  $\epsilon$ -Proteobacteria:** The  $\epsilon$ -proteobacteria are the smallest of the five proteobacterial classes. They all are slender gram-negative rods, which can be straight, curved, or helical. The two most important genera, *Campylobacter* and *Helicobacter*, are microaerophilic, motile, helical or vibrioid, gram-negative rods. The genus *Campylobacter* contains both nonpathogens and species pathogenic for humans and other animals. The species of *Helicobacter*, have been isolated from the stomachs and upper intestines of humans, dogs, cats and other mammals. In developing countries 70 to 90% of the population is infected with different species of  $\epsilon$ -proteobacteria.

### 2.3.5 The Mollicutes

Members of the class Mollicutes are commonly called mycoplasmas. These bacteria lack cell walls and cannot synthesize peptidoglycan precursors. Thus they are penicillin resistant but susceptible to lysis by osmotic shock and detergent

treatment. Because they are bounded only by a plasma membrane, these procaryotes are pleomorphic and vary in shape from spherical or pear-shaped organisms, about 0.3 to 0.8  $\mu\text{m}$  in diameter, to branched or helical filaments (Fig. 2.8A)). Some mycoplasmas (e.g., *Mycoplasma genitalium*) have a specialized terminal structure that projects from the cell and gives them a flask or pear shape. This structure aids in attachment to eucaryotic cells. They are the smallest bacteria capable of self-reproduction. Although most are nonmotile, some can glide along liquid-covered surfaces. Most species differ from the vast majority of bacteria in requiring sterols for growth. They usually are facultative anaerobes, but a few are obligate anaerobes. When growing on agar, most species form colonies with a “fried-egg” appearance because they grow into the agar surface at the centre while spreading outward on the surface at the colony edges. Their genome is one of the smallest found in prokaryotes, about 5 to  $10 \times 10^8$  daltons; the G + C content ranges from 23 to 41%. Recently the complete genome of *M. genitalium*, a parasite of the human genital and respiratory tracts, has been sequenced.



**Figure 2.8** - The morphological variations in the low G + C containing gram-positive bacteria. (A) *Mycoplasma* sp., (B) *Clostridium* sp., (C) *Desulfotomaculum* sp., (D) *Bacillus* sp., (E) *Caryophanon* sp. and (F) *Streptococcus* sp.

The metabolism of mycoplasmas is not particularly unusual, although they are deficient in several biosynthetic sequences and often require sterols, fatty acids, vitamins, amino acids, purines and pyrimidines. Mycoplasmas are usually more osmotically stable than bacterial protoplasts and their membrane sterols may be a stabilizing factor. Some produce ATP by the Embden-Meyerhof pathway and lactic acid fermentation. Others catabolize arginine or urea to generate ATP. None appears to have the complete tricarboxylic acid cycle.

Mycoplasmas are remarkably widespread and can be isolated from animals, plants, the soil and even compost piles. Indeed, about 10% of the mammalian cell cultures in use are probably contaminated with mycoplasmas, which seriously interfere with tissue culture experiments and are difficult to detect and eliminate. In animals, mycoplasmas colonize mucous membranes and joints and often are associated with diseases of the respiratory and urogenital tracts. Spiroplasmas have been isolated from insects, ticks and a variety of plants. They cause disease



in citrus plants, cabbage, broccoli, honeybees and other hosts. Presumably many more pathogenic mollicutes will be discovered as techniques for their isolation and study improve.

**2.3.6 Low G+C Gram-Positive Bacteria in Bergey’s Manual**

*Bacillus*, *Clostridium*, *Lactobacillus*, *Corynebacterium*, *Mycobacterium* and *Nocardia* are the most widely distributed genera of this group (Table 2.7). These are endospore-forming bacteria distributed widely and mostly growing as soil inhabitants. Soil conditions are often extremely variable and endospores are an obvious advantage in surviving periods of dryness or nutrient deprivation in these bacteria.

**Table 2.7:** The representative genera of some gram-positive bacteria.

| <i>Mollicutes</i>                                                                                                                                                               | <i>Clostridia and relatives</i>                                                              | <i>Bacilli</i>                                                                                                                                                                                                | <i>Actinobacteria</i>                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Acholeplasma</i><br><i>Anaeroplasma</i><br><i>Asteroleplasma</i><br><i>Entomoplasma</i><br><i>Mesoplasma</i><br><i>Mycoplasma</i><br><i>Spiroplasma</i><br><i>Ureaplasma</i> | <i>Clostridium</i><br><i>Desulfotomaculum</i><br><i>Heliobacterium</i><br><i>Veillonella</i> | <i>Bacillus</i><br><i>Caryophanon</i><br><i>Enterococcus</i><br><i>Lactobacillus</i><br><i>Lactococcus</i><br><i>Leuconostoc</i><br><i>Staphylococcus</i><br><i>Streptococcus</i><br><i>Thermoactinomyces</i> | <i>Actinoplanes</i><br><i>Arthrobacter</i><br><i>Bifidobacterium</i><br><i>Corynebacterium</i><br><i>Frankia</i><br><i>Micrococcus</i><br><i>Mycobacterium</i><br><i>Nocardia</i><br><i>Propionibacterium</i><br><i>Streptomyces</i> |

Members of the genus *Clostridium* have great practical impact. Because they are anaerobic and form heat-resistant endospores, they are responsible for many cases of food spoilage, even in canned foods. *C. botulinum* is the causative agent of botulism. Clostridia often can ferment amino acids to produce ATP by oxidizing one amino acid and using another as an electron acceptor in a process called the Stickland reaction. This reaction generates ammonia, hydrogen sulphide, fatty acids and amines during the anaerobic decomposition of proteins. These products are responsible for many unpleasant odours arising during putrefaction. Several clostridia produce toxins and are major disease causing agents.

In addition to the genus *Clostridium*, several other genera of this group are also environmentally important. For example, *Desulfotomaculum* reduces sulphate and sulphite to hydrogen sulphide during anaerobic respiration. Although it stains gram-negative, electron microscopic studies have shown that *Desulfotomaculum* has a gram-positive type cell wall despite its staining characteristics and really is a member of the low G + C gram positives. The heliobacteria, genera *Heliobacterium* and *Heliophilum*, are a group of unusual anaerobic photosynthetic bacteria characterized by the presence of bacteriochlorophyll *g*. They have a photosystem I type reaction centre like the green sulphur bacteria, but have no intracytoplasmic photosynthetic membranes. The pigments are contained in the plasma membrane.



The genus *Bacillus* contains gram-positive, endospore-forming, chemoheterotrophic rods that are usually motile and peritrichously flagellated. It is aerobic, or sometimes facultative and catalase positive (Table 2.8). Many species have been placed in new families and genera based on rRNA sequence data. Some examples of organisms that were formerly in the genus *Bacillus* are *Paenibacillus* and *Alicyclobacillus*. Many species of *Bacillus* are of considerable importance. For example, members of the genus *Bacillus* produce the antibiotics bacitracin, gramicidin and polymyxin. *B. cereus* causes some forms of food poisoning and can infect humans. *B. anthracis* is the causative agent of the disease anthrax, which can affect both farm animals and humans. Several species (*B. thuringiensis* and *B. sphaericus*) are used as insecticides.

The most important genus *Staphylococcus* is facultatively anaerobic, nonmotile, gram-positive cocci that usually form irregular clusters. Staphylococci are normally associated with the skin, skin glands and mucous membranes of warm-blooded animals and are responsible for many human diseases. *S. epidermidis* is a common skin resident that is sometimes responsible for endocarditis and infections of patients with lowered resistance (e.g., wound infections, surgical infections, urinary tract infections). *S. aureus* is the most important human staphylococcal pathogen and causes boils, abscesses, wound infections, pneumonia, toxic shock syndrome and other diseases. It is also a major cause of food poisoning.

**Table 2.8:** The list of important genera of bacteria on the basis of their choice of oxygen relationship.

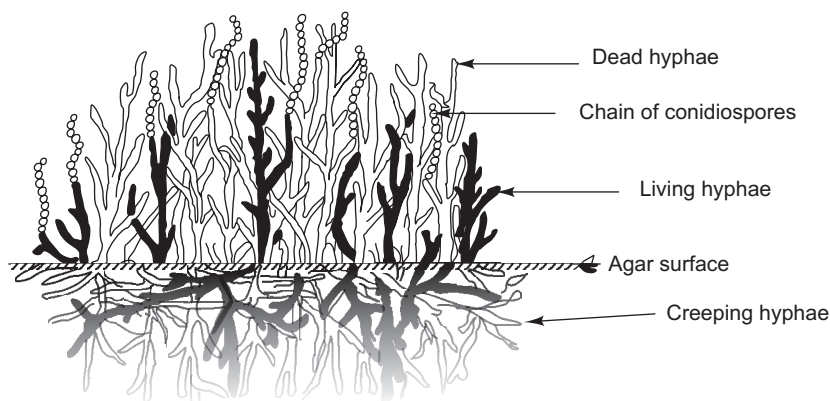
| <i>Aerobic</i>       | <i>Aerobic to microaerophilic</i> | <i>Facultative anaerobic</i> | <i>Anaerobic</i>         |
|----------------------|-----------------------------------|------------------------------|--------------------------|
| <i>Leptospira</i>    | <i>Treponema</i>                  | <i>Spirochaeta</i>           | <i>Brachyspira</i>       |
| <i>Nitrobacter</i>   | <i>Borrelia</i>                   | <i>Cristispira</i>           | <i>Serpulina</i>         |
| <i>Rhizobium</i>     | <i>Rhodospirillum</i>             | <i>Escherichia</i>           | <i>Chromatium</i>        |
| <i>Caulobacter</i>   | <i>Beggiatoa</i>                  | <i>Haemophilus</i>           | <i>Ectothiroadospira</i> |
| <i>Agrobacterium</i> | <i>Campylobacter</i>              | <i>Photobacterium</i>        | <i>Methanococcus</i>     |
| <i>Burkholderia</i>  | <i>Helicobacter</i>               | <i>Vibrio</i>                | <i>Methanospirillum</i>  |
| <i>Leptothrix</i>    | <i>Lactobacillus</i>              | <i>Bacillus</i>              | <i>Desulfovibrio</i>     |
| <i>Nitrosomonas</i>  | <i>Frankia</i>                    | <i>Enterococcus</i>          | <i>Desulfuromonas</i>    |
| <i>Thiobacillus</i>  |                                   | <i>Lactococcus</i>           | <i>Methanogenium</i>     |
| <i>Azotobacter</i>   |                                   | <i>Leuconostoc</i>           | <i>Clostridium</i>       |
| <i>Methylococcus</i> |                                   | <i>Staphylococcus</i>        | <i>Heleobacterium</i>    |
| <i>Pseudomonas</i>   |                                   | <i>Streptococcus</i>         | <i>Bifidobacterium</i>   |
| <i>Bacillus</i>      |                                   | <i>Corynebacterium</i>       |                          |
| <i>Bdellovibrio</i>  |                                   |                              |                          |
| <i>Myxococcus</i>    |                                   |                              |                          |
| <i>Stigmatella</i>   |                                   |                              |                          |
| <i>Streptomyces</i>  |                                   |                              |                          |

Compiled from Ames (1995); Takeuchi *et al.* (1995); Hanson and Hanson (1996); Radoff *et al.* (1999); Brun (2000); Hooykaas (2000); Margulis (2000).

Members of the order *Lactobacillales* produce lactic acid as their major or sole fermentation product and are sometimes collectively called lactic acid bacteria. *Streptococcus*, *Enterococcus*, *Lactococcus*, *Lactobacillus* and *Leuconostoc* are all members of this group. Lactic acid bacteria are nonsporing and usually nonmotile. They lack cytochromes and obtain energy by substrate-level phosphorylation rather than by electron transport and oxidative phosphorylation. They normally depend on sugar fermentation for energy. Nutritionally they are fastidious and many vitamins, amino acids, purines and pyrimidines must be supplied because of their limited biosynthetic capabilities. Lactic acid bacteria are usually categorized as facultative anaerobes, but some classify them as aerotolerant anaerobes (Table 2.8). Lactobacilli also create problems. They sometimes are responsible for spoilage of beer, milk and meat because their metabolic end products contribute undesirable flavours and odours. *Leuconostoc* species are involved in food spoilage and tolerate high sugar concentrations so well that they grow in syrup and are a major problem in sugar refineries.

### 2.3.7 The Actinomycetes

When growing on a solid substratum such as agar, the branching network of hyphae developed by actinomycetes grows both on the surface of the substratum and into it to form a substrate mycelium. Septa usually divide the hyphae into long cells (20  $\mu\text{m}$  and longer) containing several nucleoids. Sometimes a tissue-like mass results and may be called a thallus. Many actinomycetes also have an aerial mycelium that extends above the substratum and forms asexual, thin-walled spores called conidia (Fig. 2.9). Most actinomycetes are not motile. When motility is present, it is confined to flagellated spores.



**Figure 2.9** - A colony of actinomycetes showing the chain of conidiospores.

The taxonomically valuable properties of actinomycetes are the morphology and colour of mycelia and sporangia, the surface features and arrangement of conidiospores, the percent G + C in DNA, the phospholipid composition of cell membranes and spore heat resistance. Newer techniques are being applied to

actinomycete taxonomy. Comparison of 16S rRNA sequences has proven to be a valuable tool for classification of this group. Another useful technique is the production of large DNA fragments by restriction enzyme digestion and their separation and comparison using pulsed-field electrophoresis.

Actinomycetes have considerable practical significance. They are primarily soil inhabitants and are very widely distributed. They can degrade an enormous number and variety of organic compounds and are extremely important in the mineralization of organic matter. Actinomycetes produce most of the medically useful natural antibiotics. Although most actinomycetes are free living microorganisms, a few are pathogens of humans, other animals and some plants.

## 2.4 THE CYANOBACTERIA

The cyanobacteria are the largest and most diverse group of photosynthetic bacteria. There is little agreement about the number of cyanobacterial species. Earlier classification had as many as 2,000 or more species. In one recent system this has been reduced substantially to only few hundred genera. The second edition of *Bergey's Manual of Determinative Bacteriology* describes 57 genera in some detail. The G + C content of the group ranges from 35 to 71%. Although cyanobacteria are prokaryotes, their photosynthetic system closely resembles that of the eucaryotes because they have chlorophyll *a* and photosystem II and carry out oxygenic photosynthesis. Like the red algae, cyanobacteria use phycobiliproteins as accessory pigments. Photosynthetic pigments and electron transport chain components are located in thylakoid membranes lined with particles called phycobilisomes. These contain phycobilin pigments, particularly phycocyanin and transfer energy to photosystem II. Carbon dioxide is assimilated through the Calvin cycle and the reserve carbohydrate is glycogen. Sometimes they store extra nitrogen as polymers of arginine or aspartic acid in cyanophycin granules. Since cyanobacteria lack the enzyme 2-oxoglutarate dehydrogenase, they do not have a fully functional citric acid cycle. The pentose phosphate pathway plays a central role in their carbohydrate metabolism. Although many cyanobacteria are obligate photolithoautotrophs, some can grow slowly in the dark as chemoheterotrophs by oxidizing glucose and a few other simple sugars. Under anaerobic conditions some cyanobacteria (*e.g. Oscillatoria limnetica*) oxidize hydrogen sulphide instead of water and carry out anoxygenic photosynthesis much like the green photosynthetic bacteria. This indicates that cyanobacteria are capable of considerable metabolic flexibility adapting to many different conditions like that of photosynthetic bacteria.

Cyanobacteria also vary greatly in shape and appearance. They range in diameter from about 1 to 10  $\mu\text{m}$  and may be unicellular, exist as colonies of many shapes, or form filaments called trichomes (Fig. 2.10). A trichome is a row of bacterial cells that are in close contact with one another over a large area. In contrast, adjacent cells in a simple chain (such as those commonly found in the genus *Bacillus*) associate by only a small area of contact. Although most appear

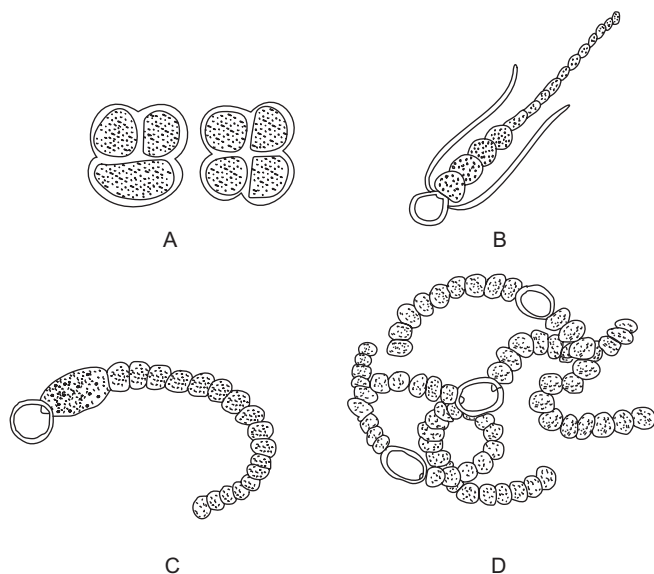
blue-green because of phycocyanin, a few cyanobacteria (e.g. *Calothrix*) are red or brown in colour because of the dominance of red pigment phycoerythrin (Table 2.9). Despite this variety, cyanobacteria have typical procaryotic cell structures and a normal gram-negative type cell wall. They often use gas vesicles to move vertically in the water and many filamentous species have gliding motility. Although cyanobacteria lack flagella, several strains of the marine genus *Synechococcus* are able to move at rates of up to 25 µm/second by means of an unknown mechanism.

**Table 2.9:** Characteristics of the cyanobacterial subsections used for their taxonomic separation.

| Subsection | General shape and growth                                                             | Reproduction                                                       | Other properties                                                                             | Common genera                                                                                                              |
|------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| I          | Unicellular rods or cocci; nonfilamentous aggregates                                 | Binary fission, budding                                            | Almost always nonmotile                                                                      | <i>Chamaesiphon</i><br><i>Chroococcus</i><br><i>Gloeotheca</i><br><i>Gleocapsa</i>                                         |
| II         | Unicellular rods or cocci; may be held together in aggregates                        | Multiple fission to form bacocytes                                 | Only some bacocytes are motile                                                               | <i>Pleurocapsa</i><br><i>Dermocarpa</i><br><i>Chroococcidiopsis</i>                                                        |
| III        | Filamentous, unbranched trichome with only vegetative cells                          | Binary fission in a single plane, fragmentation                    | Usually motile                                                                               | <i>Lyngbya</i><br><i>Oscillatoria</i><br><i>Prochlorothrix</i><br><i>Spirulina</i><br><i>Pseudoanabaena</i>                |
| IV         | Filamentous, unbranched trichome may contain specialized cells (heterocysts)         | Binary fission in a single plane, fragmentation to form hormogonia | Often motile, may produce akinetes                                                           | <i>Anabaena</i><br><i>Cylindrospermum</i><br><i>Aphanizomenon</i><br><i>Nostoc</i><br><i>Scytonema</i><br><i>Calothrix</i> |
| V          | Filamentous trichomes either with branches or composed of more than one row of cells | Binary fission in more than one plane; hormogonia formed           | May produce akenetes; greatest morphological complexity and differentiation in cyanobacteria | <i>Fischerella</i><br><i>Stigonema</i><br><i>Geitleria</i>                                                                 |

Compiled from Adams (1992); Garcia-Pichel (2000); Raymond *et al.* (2002).

Cyanobacteria show great diversity with respect to reproduction and employ a variety of mechanisms: viz., binary fission, budding, fragmentation and multiple fission. Fragmentation of filamentous cyanobacteria can generate small, motile filaments called hormogonia. Some species develop akinetes, spetialized, dormant, thick-walled resting cells that are resistant to desiccation.



**Figure 2.10** - Representatives of cyanobacteria. (A) *Chroococcus* sp., (B) *Rivularia* sp., (C) *Calothrix* sp. and (D) *Nostoc* sp.

Many filamentous cyanobacteria fix atmospheric nitrogen by means of special cells called heterocysts. Around 5 to 10% of the cells can develop into heterocysts when cyanobacteria are deprived of both nitrate and ammonia, their preferred nitrogen sources. This can go up to 20% under extreme nitrogen deficient conditions and still higher when the cyanobacteria are in symbiotic association with a host that supply metabolites to the microbial symbiont. When transforming themselves into heterocysts, cyanobacterial cells synthesize a very thick new wall, reorganize their photosynthetic membranes, discard their phycobiliproteins and photosystem II and synthesize the nitrogen-fixing enzyme nitrogenase. Photosystem I is still functional and produces ATP, but no oxygen arises from noncyclic photophosphorylation because photosystem II is absent. This inability to generate  $O_2$  is critical because the nitrogenase is extremely oxygen sensitive. The heterocyst wall slows or prevents  $O_2$  diffusion into the cell and any  $O_2$  present is consumed during respiration for its own maintenance. The structure and physiology of the heterocyst ensures that it will remain anaerobic; it is dedicated to nitrogen fixation. It obtains nutrients from adjacent vegetative cells and contributes fixed nitrogen in the form of the amino acid glutamine. Some nonheterocystous cyanobacteria also fix nitrogen when are present under dark, anoxic conditions in microbial mats. Planktonic forms such as *Trichodesmium* can fix nitrogen as well.

The classification of cyanobacteria is still in an unsettled state, partly due to the lack of pure cultures. The second edition of *Bergey's Manual* divides the cyanobacteria into five subsections and separates Prochlorophytes from the cyanobacteria. The prochlorophytes are oxygenic phototrophs that have both

chlorophyll *a* and *b* but lack phycobilins. Thus although they resemble other cyanobacteria with respect to chlorophyll *a*, they differ in also possessing chlorophyll *b*, the only procaryotes to do so. They resemble chloroplasts in their pigments and thylakoid structure, but their 5S and 16S rRNAs place them within the cyanobacteria.

The three recognized prochlorophyte genera are quite different from one another. *Prochloron* was first discovered as an extracellular symbiont growing either on the surface or within the cloacal cavity of marine colonial ascidian invertebrates. These bacteria are single-celled, spherical and from 8 to 30  $\mu\text{m}$  in diameter. *Prochlorothrix* is free living, has cylindrical cells that form filaments and has DNA with higher G + C content (53 mol%). *Prochlorococcus marinus*, a picoplankton, has recently been discovered flourishing about 100 meters below the ocean surface. It differs from other prochlorophytes in having divinyl chlorophyll *a* and  $\alpha$ -carotene instead of chlorophyll *a* and  $\beta$ -carotene. It is one of the most numerous of the marine planktons and a significant component of the marine microbial food web, especially in the nutrient deficient open ocean.

Cyanobacteria are very tolerant of environmental extremes and are present in almost all waters and soils. Thermophilic species may grow at temperatures of up to 75°C in neutral to alkaline hot springs. Because these photoautotrophs are so hard, they are primary colonizers of soils and surfaces that are devoid of plant growth. Some unicellular forms even grow in the fissures of desert rocks. In nutrient-rich warm ponds and lakes, surface cyanobacteria such as *Anacystis* and *Anabaena* can reproduce rapidly to form blooms. The release of large amounts of organic matter, upon the death of the bloom microorganisms, stimulates the growth of chemoheterotrophic bacteria that subsequently depletes the available oxygen. Some species can produce toxins that kill livestock and other animals. Other cyanobacteria, such as *Oscillatoria*, are so pollution resistant and characteristic of freshwaters with high organic matter content that they are used as water pollution indicators (see chapter 15).

Cyanobacteria are particularly successful in establishing symbiotic relationships with other organisms. They are the photosynthetic partners in most lichen associations. Cyanobacteria are symbionts with protozoa and fungi and nitrogen-fixing species form associations with a variety of plants (liverworts, mosses, gymnosperms and angiosperms) (see chapter 10).

## 2.5 THE WORLD OF FUNGI

The fungi include eucaryotic, spore-bearing organisms with absorptive nutrition, no chlorophyll and that reproduce sexually and asexually. The five-kingdom system places the fungi in the kingdom *Fungi*. According to the universal phylogenetic tree, fungi are members of the domain *Eucarya*. As presently delimited, the kingdom *Fungi* is believed to constitute a monophyletic group referred to as the true fungi or *Eumycota*.

Fungi are primarily terrestrial organisms, although a few (most probably few thousand species) are freshwater or marine. Many are pathogenic and infect

plants and animals causing numerous diseases. Fungi also form beneficial relationships with other organisms. For example, about three-fourths of all vascular plants form mycorrhizal associations between their roots and fungi (See Chapter 5). Fungi are also found in the upper portions of many plants and are abundant in the phyllosphere (See Chapter 6). These endophytic fungi affect plant reproduction and palatability to herbivores. Lichens are associations of fungi and either algae or cyanobacteria.

### 2.5.1 Environmental Importance

The world of fungi is probably the most diverse but least explored group of microbes. About 100,000 fungal species have been described out of some 1.5 million species estimated to have their existence currently on the earth (Table 2.10). Many of the fungal species are beneficial to humans but many of them are also pathogenic. With bacteria and a few other groups of heterotrophic organisms, fungi act as decomposers, a role of enormous significance. They degrade complex organic materials in the environment to simple organic compounds and inorganic molecules. In this way carbon, nitrogen, phosphorus and other critical constituents of dead organisms are released and made available for other living organisms.

**Table 2.10:** The approximate diversity of fungi in the world, which have been more or less studied.

| <i>Division</i> | <i>Common name</i> | <i>Approximate number</i> |
|-----------------|--------------------|---------------------------|
| Zygomycota      | Zygomycetes        | 600                       |
| Ascomycota      | Sac fungi          | 35,000                    |
| Basidiomycota   | Club fungi         | 30,000                    |
| Deuteromycota   | Fungi imperfecti   | 30,000                    |

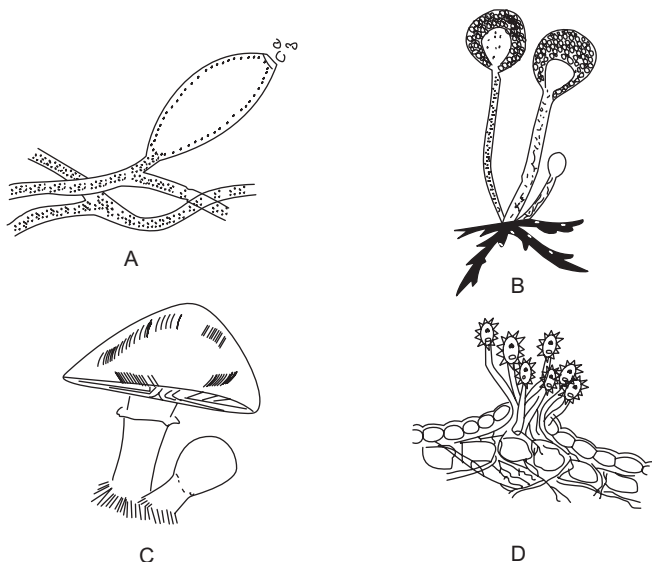
Source: Prescott *et al.* (2005).

Fungi are the major cause of plant diseases. Over 5,000 species attack economically valuable crops and garden plants and also many wild plants. In like manner many diseases of animals and humans are also caused by fungi, though this is quite low in comparison to the plant diseases. Fungi, especially the yeasts, are essential to many industrial processes involving fermentation. Examples include the making of bread, wine and beer. Fungi also play a major role in the preparation of some cheeses, soy sauce and sufu; in the commercial production of many organic acids (citric, gallic) and certain drugs (ergometrine, cortisone); and in the manufacture of many antibiotics (penicillin, griseofulvin) and the immunosuppressive drug cyclosporine. In addition, fungi are important research tools in the study of fundamental biological processes. Cytologists, geneticists, biochemists, biophysicists and microbiologists regularly use fungi as the experimental materials for their research.



### 2.5.2 Morphological Complexity

The vegetative thallus varies in complexity and size, ranging from the single-cell microscopic yeasts to multicellular molds, macroscopic puffballs and mushrooms (Fig. 2.11). The fungal cell is usually enclosed in a cell wall of chitin, which is a strong but flexible nitrogen-containing polysaccharide consisting of N-acetylglucosamine residues. A yeast is a unicellular fungus with cells that are larger than bacteria, vary considerably in size and are commonly spherical to egg shaped. They have no flagella but do possess most of the other eucaryotic organelles. A mold consists of long, branched, thread-like filamentous hyphae that form a mycelium. In some fungi, protoplasm streams through hyphae, uninterrupted by cross walls (coenocytic). The hyphae of other fungi are septate with either a single pore or multiple pores in the septa that permit cytoplasmic streaming. Many fungi, especially those that cause diseases in humans and animals, are dimorphic. The latter can change from (1) the yeast (Y) form in the animal to (2) the mold or mycelial form (M) in the external environment in response to changes in various environmental factors (nutrients, CO<sub>2</sub> tension, oxidation-reduction potentials, temperature). This shift is called the YM shift. In plant associated fungi the opposite type of dimorphism exists: the mycelial form occurs in the plant and the yeast form in the external environment.



**Figure 2.11** - Morphology of some fungi with their spore producing structures. (A) *Saprolegnia* sp., (B) *Rhizopus* sp., (C) *Agaricus* sp., (D) *Puccinia* uredial structure.

### 2.5.3 Fungal Nutrition

Fungi grow best in dark, moist habitats, but they are found wherever organic material is available. Most fungi are saprophytes, securing their nutrients from



dead organic materials. Like many bacteria, fungi release hydrolytic exoenzymes that digest external substrates. Almost all complex and lignified organic matters, which are not the metabolites for bacteria, are initially attacked by the exoenzymes of fungal origin and are converted to simple and assimilable organic matters. These simple organic carbon molecules are then processed and metabolized by both fungi and bacteria. All fungi are chemoorganoheterotrophs and use organic compounds as a source of carbon, electrons and energy.

The fungal mycelia store food and glycogen, which is the primary storage polysaccharide. Most fungi use carbohydrates (preferably glucose or maltose) and nitrogenous compounds to synthesize their own amino acids and proteins. Fungal growth is highly prolific under humid and warm conditions with sufficient supply of oxygen indicating that majority of fungi are aerobic. Some yeasts, however, are facultatively anaerobic and can obtain energy by fermentation, such as in the production of ethyl alcohol from glucose. Obligately anaerobic fungi are found in the rumen of cattle.

### 2.5.4 Classification of Fungi

The traditional taxonomic scheme used by mycologist, classifies the fungi into four divisions, based primarily on variations in sexual reproduction. Each division is equivalent to a phylum in microbial or animal classification schemes. Based on 18S rRNA studies, molecular microbiologists place the *Deuteromycota* (Fungi Imperfecti) among their closest relatives in either the *Zygomycota*, *Ascomycota* or *Basidiomycota* and add the class *Chytridiomycetes*. The distinctive characters of these fungal divisions are presented here.

**(a) Division Zygomycota:** This division contains the fungi popularly called zygomycetes. Most live on decaying plant and animal matter in the soil. A few are parasites of plants, insects, animals and humans, some becoming pathogenic. The hyphae of zygomycetes are coenocytic, with many haploid nuclei. Asexual spores, usually wind dispersed, develop in sporangia at the tips of aerial hyphae. Sexual reproduction produces tough, thick walled zygotes called zygospores that can remain dormant when the environment is too harsh for growth of the fungus. *Rhizopus* and *Mucor* are very common members of this division.

**(b) Division Ascomycota:** This division contains the fungi called ascomycetes, commonly known as the sac fungi. Many species are quite familiar and economically important. For example, most of the red, brown and blue-green molds, that cause food spoilage, are ascomycetes and are very common in moist tropical conditions. The powdery mildews that attack plant leaves and the fungi that cause chestnut blight are ascomycetes. Many yeasts as well as edible morels and truffles are ascomycetes. The pink bread mold *Neurospora crassa*, also an ascomycete, has been an extremely important research tool in genetics and biochemistry. Many ascomycetes are parasites on higher plants. *Claviceps purpurea* parasitizes rye and other grasses, causing the disease ergot. *Aspergillus*, *Penicillium*, *Erysiphe*, *Neurospora*, etc. are the common genera of the division.

**(c) Division Basidiomycota:** This division contains the basidiomycetes, commonly known as the club fungi. These are named for their characteristic structure of cell, the basidium, that is involved in sexual reproduction. A basidium is produced at the tip of hyphae and normally is club shaped. Two or more basidiospores are produced by the basidium and basidia may be held within fruiting bodies called basidiocarps. Examples of this division include the smuts, jelly fungi, rusts, shelf fungi, stinkhorns, puffballs, toadstools, mushrooms and bird's nest fungi.

Most basidiomycetes are saprophytes that decompose plant debris, especially cellulose and lignin. Many mushrooms are used as food throughout the world. Different species of *Agaricus* are edible and are cultivated widely to supplement the world food demand. Many mushrooms also produce specific alkaloids that act as either poisons or hallucinogens. One such example is the “destroying angel” mushroom, *Amanita phalloides*, which contains hepatotoxins phalloidin and  $\alpha$ -amanitin. The commonly occurring genera of the division basidiomycota are *Puccinia*, *Ustilago*, *Agaricus*, etc.

**(d) Division Deuteromycota:** The fungi, that lacks a well defined sexual phase (perfect stage), or if this phase has not yet been observed, are placed within the division *Deuteromycota*, commonly called the Fungi Imperfecti or deuteromycetes (“secondary fungi” or mitosporic fungi). Most deuteromycetes reproduce by means of conidia. Once a perfect stage is observed, the fungus is transferred to its proper division (often to the *Ascomycota*). Molecular systematics places the *Deuteromycota* among their closest relatives in the *Eumycota*. Most Fungi Imperfecti are terrestrial, with only a few being reported from freshwater and marine habitats. The majority are either saprophytes or parasites of plants. A few are parasitic on other fungi. Many of them are human pathogens, causing diseases like athlete's foot, ring worm and histoplasmosis. The common genera are *Alternaria*, *Circospora*, *Fusarium*, etc.

**(e) Division Chytridiomycota:** The simplest of the true fungi belong to the division *Chytridiomycota*. This division contains one class, *Chytridiomycetes* and its members are known as the chytrids. These are simple terrestrial and aquatic fungi that reproduce asexually by forming motile zoospores with single, posterior, whiplash flagella. The entire organism is microscopic in size and may consist of a single cell, a small multinucleate mass, or a true mycelium. Usually chitin is the major constituent of chytrid cell walls. Chytrids are thought to have been derived from a protozoan ancestor having similar flagellation. They are likely to be ancestral to the remaining groups of true fungi. Some can grow saprophytically on dead organic matter and some others are parasites of algae.

**(f) Division Myxomycota (Acellular Slime Molds):** Under appropriate conditions plasmodial (acellular) slime molds exist as streaming masses of colourful protoplasm that creep along in an amoeboid fashion over moist, rotting logs, leaves and other organic matter. Feeding is by phagocytosis. Because this streaming mass lacks cell walls, it is called a plasmodium.

**(g) Division Acrasiomycota (Cellular Slime Molds):** The vegetative stage of cellular slime molds consists of individual amoeboid cells termed myxamoebae. The myxamoebae feed phagocytically on bacteria and yeasts. When food is plenty, they divide repeatedly by mitosis and cytokinesis, producing new daughter myxamoebae. As their food supply is exhausted they form a sluglike pseudoplasmodium.

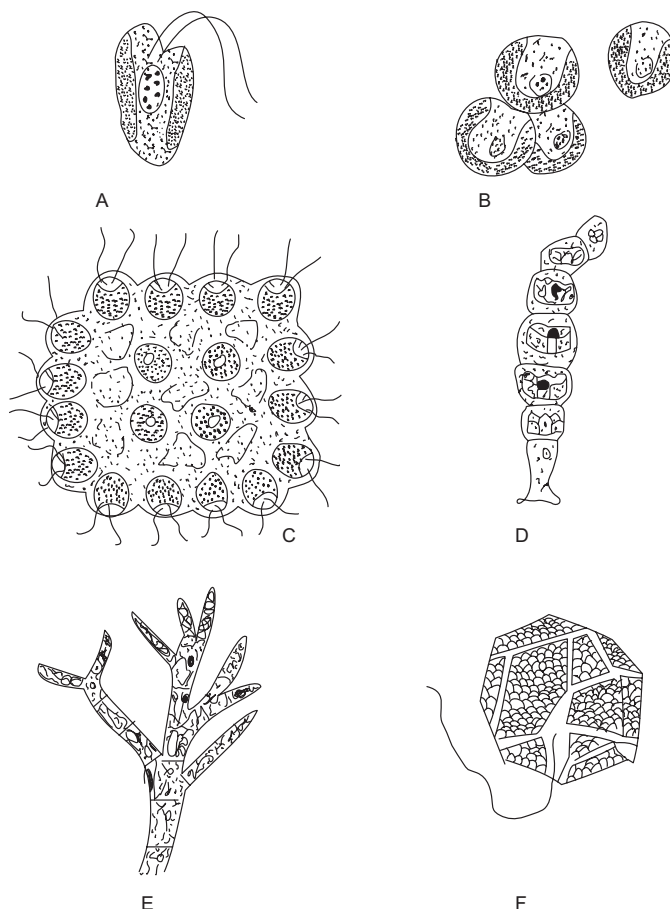
**(h) Division Oomycota:** The oomycetes (egg fungi) or water molds resemble true fungi only in appearance, consisting of finely branched filaments called hyphae. However, oomycetes have cell walls of cellulose, whereas the walls of most fungi are made of chitin. Oomycetes are also unlike the true fungi in that they have tubular mitochondrial cristae. Water molds such as *Saprolegnia* and *Achlya* are saprophytes that grow as cottony masses on dead algae and small animals, mainly in freshwater environments. They are important decomposers in aquatic ecosystems. Some water molds are parasitic on the gills of fish. Commonly occurring genera are *Peronospora*, *Phytophthora* and *Plasmopora*.

## 2.6 THE WORLD OF ALGAE

Algae are most common to aquatic systems (fresh water, marine, or brackish water) in which they may be suspended (planktonic) or attached and living on the bottom (benthic). A few algae live at the water-atmosphere interface and are termed neustonic. Some algae grow on moist rocks, wood, trees and on the surface of moist soil. Algae also live as endosymbionts in various protozoa, mollusks, worms and corals. Several algae grow as endosymbionts within plants, some are attached to the surface of various structures and a few lead a parasitic existence (See Chapter 9 and 10). Algae also associate with fungi to form lichens.

The eucaryotic algal cell is surrounded by a thin, rigid cell wall. Some algae have an outer matrix lying outside the cell wall. This usually is flexible and gelatinous, similar to bacterial capsules. Very few algae have flagella, which are the locomotor organelles. There is a true nucleus, chloroplast with pyrenoids and other membranous organelles. Mitochondrial structure varies greatly in the algae. Some algae (euglenoids) have discoid cristae; some have lamellar cristae (green and red algae) and the remaining have tubular cristae (golden-brown and yellow-green, brown and diatoms).

Algae can be either autotrophic or heterotrophic. Most are photoautotrophic requiring only light and CO<sub>2</sub> as their principal source of energy and carbon. Chemoheterotrophic algae (many planktonic green algae and cyanobacteria) require external organic compounds as carbon and energy sources. There are striking variations in the structure of algal cells and thalli and some of the structural complexities are found typical to a definite group of algae. There are also remarkable complexity of multicellular forms, such as the giant kelps. Single-celled algae may be as small as bacteria or even believed to be smaller (nanoplankton and picoplankton of open oceans). Algae are unicellular, colonial, filamentous, membranous and blade like, or tubular (Fig.2.12).



**Figure 2.12** - Representative species of algae. (A) *Cryptomonas* sp., (B) *Chlorella* sp., (C) *Gonium* sp., (D) *Ulothrix* sp., (E) *Cladophora* sp. and (F) *Ceratium* sp.

### 2.6.1 The Algal Diversity

According to the five-kingdom system of Whittaker, the algae belong to seven divisions distributed between two different kingdoms. This classical classification is based on cellular (not organismal) properties. Some more important properties include: (1) cell wall (if present) chemistry and morphology; (2) form in which food or assimilatory products of photosynthesis are stored; (3) chlorophyll molecules and accessory pigments that contribute to photosynthesis; (4) flagella number and the location of their insertion in motile cells; (5) morphology of the cells and/or body (thallus); (6) habitat; (7) reproductive structures; and (8) life history patterns. Based on these properties the algae are arranged into divisions. Table 2.11 summarizes the significant characteristics of different divisions of algae. The five kingdom classification of algae separates them into seven divisions keeping aside the cyanobacteria under the group “Photosynthetic Bacteria”. Out of these seven divisions, five (Chlorophyta, Charophyta, Euglenophyta, Chrysophyta and

Pyrrhophyta) are under the kingdom Protista and the remaining two divisions (Phaeophyta and Rhodophyta) are included in the kingdom Plantae (Table 2.12). The approximate number of species of algae placed under these divisions are believed to be about 21,000 but only about 4000 species are taxonomically known to the biologists. Many of the known species are ubiquitous in distribution.

Molecular systems of classification have placed some of the classical algae with plants (green algae), some as a separate lineage (red algae), some with the stramenopiles (golden-brown and yellow-green algae, brown algae and diatoms), some with the alveolates (dinoflagellates) and still others with some protozoa (euglenoids). Two of these groups, the alveolates and stramenopiles, have been created recently as a result of rRNA comparisons and ultrastructural studies. The alveolates have mitochondria with tubular cristae and subsurface alveoli or sacs that abut the surface. Dinoflagellates, ciliate protozoa and the apicomplexan protozoa are alveolates. The stramenopiles have mitochondria with tubular cristae and hollow hairs that give rise to a small number of fine hairs (tripartite tubular hairs). These hairs are usually on their flagella. Photosynthetic forms often have chlorophylls *a* and *c*. Some common stramenopiles are the opalinid protozoa, oomycetes, diatoms, brown algae or phaeophytes, chrysophytes and xanthophytes. Although a few groups such as the diatoms have lost their hairs, they are still considered stramenopiles based on the rRNA data, mitochondrial characteristics and other properties. However, it is beyond the scope of this book to individually discuss the characteristics of these newly created algal groups. Only salient features of dominant groups of algae is presented here.

**Table 2.11:** The pigmentation, cell constituents and habitat preference of different algal groups.

| <i>Division</i> | <i>Chlorophylls</i> | <i>Carotenoids</i>                                                        | <i>Storage products</i>   | <i>Habitat</i> |
|-----------------|---------------------|---------------------------------------------------------------------------|---------------------------|----------------|
| Chlorophyta     | <i>a, b</i>         | $\beta$ -carotene, $\pm \alpha$ -carotene, xanthophylls                   | Sugars, starch, fructosan | fw, m, bw, t   |
| Charophyta      | <i>a, b</i>         | $\alpha$ -, $\beta$ -, $\tau$ -carotene, xanthophylls                     | Starch                    | fw, bw         |
| Euglenophyta    | <i>a, b</i>         | $\beta$ -carotene, $\pm \tau$ -carotene, xanthophylls                     | Paramylon oils, sugars    | fw, m, bw, t   |
| Chrysophyta     | <i>a, c1/c2</i>     | $\alpha$ -, $\beta$ -, $\varepsilon$ -carotene, fucoxanthin, xanthophylls | Chrysolaminarin, oils     | fw, m, bw, t   |
| Pyrrhophyta     | <i>a, c1, c2</i>    | $\beta$ -carotene, fucoxanthin, dinoxanthin, peridinin                    | Starch, glucan, oils      | fw, m, bw      |
| Phaeophyta      | <i>a, c</i>         | $\beta$ -carotene, fucoxanthin, xanthophylls                              | Laminarin, mannitol, oils | m, bw          |
| Rhodophyta*     | <i>a, d</i>         | $\beta$ -carotene, $\pm \alpha$ -carotene, zeaxanthin                     | Floridian starch          | fw, m, bw      |

Note: fw: freshwater, m: marine, bw: brackish water, t: terrestrial.

\* phycobiliproteins (phycocyanin, allophycocyanin and phycoerythrin) are dominant photosynthetic pigments.

**(a) Division Chlorophyta (Green Algae):** This is an extremely varied division. These algae grow in fresh and salt water, in soil and on and within other organisms (commensal, parasitic or symbiotic). The photosynthetic pigments are chlorophylls *a* and *b* along with specific carotenoids and they store carbohydrates as starch. Many have cell walls of cellulose. They exhibit a wide diversity of body forms, ranging from unicellular to colonial, filamentous, membranous or sheet-like and tubular types. Some species have a holdfast structure that anchors them to the substratum. Both asexual and sexual reproduction occur in green algae. Most common genera are *Chlamydomonas*, *Chlorella*, *Chlorococcum*, *Pandorina*, *Eudorina*, *Selenastrum*, *Ankistrodesmus* and *Volvox*. From *Chlamydomonas*, several distinct lines of evolutionary specialization have evolved in the green algae. The first line contains nonmotile unicellular green algae, such as *Chlorella*, *Chlorococcum*, *Selenastrum* and *Ankistrodesmus*. *Volvox* represent a second major line of evolutionary specialization.

**Table 2.12:** The five kingdom system of classification of algae with their approximate diversity in the environment.

| Kingdom  | Division     | Common name     | Approximate number of species |
|----------|--------------|-----------------|-------------------------------|
| Protista | Chlorophyta  | Green algae     | 7500                          |
|          | Charophyta   | Stoneworts      | 250                           |
|          | Euglenophyta | Euglenoids      | 700                           |
|          | Chrysophyta  | Diatoms         | 6000                          |
|          | Pyrrophyta   | Dinoflagellates | 1100                          |
| Plantae  | Phaeophyta   | Brown algae     | 1500                          |
|          | Rhodophyta   | Red algae       | 3900                          |

Source: Prescott *et al.* (2005).

**(b) Division Charophyta (Stoneworts/Brittleworts):** The stoneworts are abundant in fresh to brackish waters and have a worldwide distribution. Often they appear as a dense covering on the bottom of shallow ponds. Some species precipitate calcium and magnesium carbonate from the water to form a limestone covering. *Chara* and *Nitella* are the two widely distributed members of the division.

**(c) Division Euglenophyta (Euglenoids):** The euglenoids share with the *Chlorophyta* and *Charophyta* the presence of chlorophylls *a* and *b* in their chloroplasts. The primary storage product is paramylon (a polysaccharide composed of  $\beta$ -1,3 linked glucose), which is unique to euglenoids. They occur in fresh, brackish and marine waters and on moist soils. They often form water blooms in ponds and cattle water tanks. The representative genera *Euglena* and *Phacus* are commonly found in waters with high organic load.

**(d) Division Chrysophyta (Golden-Brown and Yellow-Green Algae; Diatoms):** The division is quite diversified with respect to pigment composition, cell wall and type of flagellated cells. It is divided in three major classes: golden-brown algae, yellow-green algae and diatoms. The major



photosynthetic pigments are usually chlorophylls *a* and *c1/c2* and the carotenoid fucoxanthin. When fucoxanthin is the dominant pigment, the cells have a golden-brown colour. The major carbohydrate reserve is chrysolaminarin (a polysaccharide storage product composed principally of  $\beta$ -1,3 linked glucose residues). Some *Chrysophytes* lack cell walls while others have intricately patterned coverings external to the plasma membrane, such as scales, walls and plates. Diatoms have a distinctive two-piece wall of silica, called a frustule. Occurrence of flagella is not common. Most *Chrysophytes* are unicellular or colonial. Reproduction usually is asexual but occasionally sexual. Although some marine forms are known, most of the yellow-green and golden-brown algae live in freshwater. Some diatoms are facultative heterotrophs and can absorb carbon-containing molecules through the holes in their walls. The commonly occurring genera are *Melosira*, *Navicula*, *Nitzschia*, *Cocconeis*, *Cymbella*, *Gomphonema*, etc.

**(e) Division Pyrrophyta (Dinoflagellates):** The dinoflagellates are unicellular, photosynthetic alveolate algae. Most dinoflagellates are marine, but some live in freshwater. Along with the chrysophytes and diatoms, the dinoflagellates make up a large part of freshwater and marine plankton and are at the base of many food chains. Species of *Noctiluca*, *Pyrodinium*, *Gonyaulax* and other genera can produce light and are responsible for much of the luminescence (phosphorescence) seen in ocean waters at night. Sometimes dinoflagellate populations reach such high levels that poisonous red tides result.

The flagella, protective coats (plates) and biochemistry of the dinoflagellates are distinctive. Many dinoflagellates are armored or clad in stiff, patterned, cellulose plates or thecae, which may become encrusted with silica. Most have two flagella. Most have chlorophylls *a* and *c*, in addition to carotenoids and xanthophylls. As a result they usually have a yellowish-green to brown colour. Their mitochondria have tubular cristae. Some dinoflagellates can ingest other cells; others are colourless and heterotrophic. A few even occur as symbionts in many groups of jellyfish, sea anemones, mollusks and corals. When dinoflagellates form symbiotic relationships, they lose their cellulose plates and flagella, become spherical golden-brown globules in the host cells and are then termed zooxanthellae.

**(f) Division Phaeophyta (Brown Algae):** The division consists of multicellular organisms that occur almost exclusively in the sea. Some species have the largest linear dimensions (length) known in the eucaryotic world. Most of the conspicuous seaweeds that are brown to olive green in colour are assigned to this division. The simplest brown algae consist of small openly branched filaments; the larger, more advanced species have a complex arrangement. Some large kelps are conspicuously differentiated into flattened blades, stalks and hold fast organs that anchor them to rocks. The photosynthetic pigments are chlorophylls *a* and *c* and a comparatively high level of fucoxanthin that give the characteristic brown colour.  $\beta$ -carotene and violaxanthin are the major carotenoids. The main storage product is laminarin, which is quite similar in structure to chrysolaminarin. The common genera are *Chorda*, *Laminaria*, *Ectocarpus*, *Fucus*, *Macrocystis*, etc.

**(g) Division Rhodophyta (Red Algae):** It includes most of the seaweeds. A few members are unicellular but most are filamentous and multicellular. Some red algae are up to 1 m long. The stored food is the carbohydrate called floridean starch (composed of  $\alpha$ -1,4 and  $\alpha$ -1,6 linked glucose residues). Photosynthetic pigments are chlorophylls *a* and *d*, xanthophylls and phycobiliproteins. The characteristic colour is caused by the red pigment phycoerythrin. The presence of phycobiliproteins enables the members to live at depths of 100m or more. The cell walls of most red algae include a rigid inner part composed of microfibrils and a mucilaginous matrix. The matrix is composed of sulphated polymers of galactose called agar, funori, porphysan and carrageenan. Many red algae also deposit calcium carbonate in their cell wall and play an important role in building coral reefs. The common genera are *Batrachospermum*, *Porphyridium*, *Bangia*, *Lomentaria*, etc.

2.7 THE WORLD OF PROTOZOA

Protozoa are protists exhibiting heterotrophic nutrition and various types of locomotion. They occupy a vast array of habitats and niches and have organelles similar to those found in other eucaryotic cells and also specialized organelles. Moisture is absolutely necessary for the existence of protozoa because they are susceptible to desiccation. Most protozoa are free living and inhabit freshwater or marine environments. Many terrestrial protozoa can be found in decaying organic matter, in soil and even in beach sand; some are parasitic in plants or animals.

Protozoa play a significant role in the economy of nature. For example, they make up a large part of plankton that are an important link in many aquatic food chains and food webs of aquatic environments. They are also useful in biochemical and molecular biological studies. Some of the most important diseases of humans and animals are caused by protozoa (Table 2.13). Most protozoa are chemoheterotrophic.

**Table 2.13:** Pathogenic protozoa that cause major diseases of domestic animals.

| Protozoan group | Genus                                | Host                           | Disease                |
|-----------------|--------------------------------------|--------------------------------|------------------------|
| Amoebae         | <i>Entamoeba</i><br><i>Iodamoeba</i> | Mammals<br>Pig                 | Amebiasis<br>Enteritis |
| Sporozoa        | <i>Babesia</i>                       | Cattle                         | Sarcocystis            |
|                 | <i>Theileria</i>                     | Cattle, sheep                  | Theileriasis           |
|                 | <i>Sarcocystis</i>                   | Mammals, birds                 | Sarcosporidiosis       |
|                 | <i>Toxoplasma</i>                    | Cats                           | Toxoplasmosis          |
|                 | <i>Isospora</i>                      | Dogs                           | Coccidiosis            |
|                 | <i>Eimeria</i>                       | Cattle, cats,<br>chickens, pig | Coccidiosis            |
|                 | <i>Plasmodium</i>                    | Many animals                   | Malaria                |
|                 | <i>Leucocytozoon</i>                 | Birds                          | Leucocytozoonosis      |
|                 | <i>Cryptosporidium</i>               | Mammals                        | Cryptosporidiosis      |



Table 2.13: (Contd.)

|                         |                    |                                |                   |
|-------------------------|--------------------|--------------------------------|-------------------|
| Ciliates<br>Flagellates | <i>Balantidium</i> | Pig                            | Balantidiasis     |
|                         | <i>Leishmania</i>  | Dogs, horses,<br>sheep, cattle | Leishmaniasis     |
|                         | <i>Trypanosoma</i> | Most animals                   | Trypanosomiasis   |
|                         | <i>Trichomonas</i> | Horses, cattle                 | Trichomoniasis    |
|                         | <i>Histomonas</i>  | Birds                          | Blackhead disease |
|                         | <i>Giardia</i>     | Mammals                        | Giardiasis        |

2.7.1 Morphological Variations

Because protozoa are eucaryotic cells, in many respects their morphology and physiology are the same as the cells of multicellular animals (Fig. 2.13). However, because all of life’s various functions must be performed within the individual protozoan, some morphological and physiological features are unique to protozoan cells. In some species the cytoplasm immediately under the plasma membrane is the semisolid or gelatinous ectoplasm, giving some rigidity to the cell body. The bases of the flagella or cilia and their associated fibrillar structures are embedded in the ectoplasm. The bases of the flagella or cilia and their associated fibrillar structures are embedded in the ectoplasm. The plasma membrane and structures immediately beneath it are called the pellicle. Inside the ectoplasm is endoplasm, which is more fluid and granular in composition and contains most of the organelles. Some protozoa have one nucleus, others have two or more identical nuclei. Still other protozoa have two distinct types of nuclei - a macronucleus and

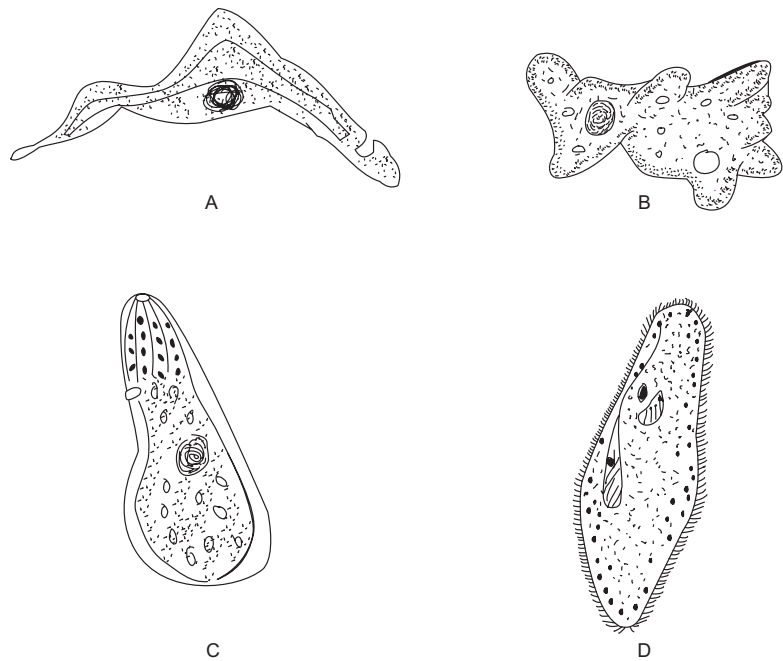


Figure 2.13 - Some members of protozoa. (A) *Trypanosoma* sp., (B) *Amoeba* sp., (C) *Apicomplexa* sp. and (D) *Paramecium* sp.

one or more micronuclei. One or more vacuoles are usually present in the cytoplasm of protozoa. Most marine protozoa and parasitic species are isotonic to their environment. Most anaerobic protozoa (such as *Trichonympha*, which lives in the gut of termites, have no mitochondria, no cytochromes and an incomplete tricarboxylic acid cycle. However, some do have small, membrane-delimited organelles termed hydrogenosomes, which are involved in hydrogenase activity.

2.7.2    **Classification of Protozoa**

Many protozoan taxonomists regard the *Protozoa* as a subkingdom, which contains seven of the 14 phyla found within the kingdom *Protista* (Table 2.14). The phylum *Sarcomastigophora* consists of flagellates and amoebae with a single type of nucleus. The phyla *Labyrinthomorpha*, *Apicomplexa*, *Microspora*, *Ascetospora* and *Myxozoa* have either saprozoic or parasitic species. The phylum *Ciliophora* has ciliated protozoa with two types of nuclei. The classification of this subkingdom into phyla is based primarily on types of nuclei, mode of reproduction and mechanism of locomotion. Some of the representative types are discussed here.

**Table 2.14:**    A brief outline of classification of protozoa.

| <i>Taxonomic group</i>           | <i>Characteristics</i>                                                                                                                           | <i>Examples</i>                                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Phylum: <i>Sarcomastigophora</i> | Locomotion by flagella, pseudopodia, or both; sexual reproduction is essentially syngamy                                                         | <i>Giardia</i><br><i>Leishmania</i><br><i>Trichomonas</i><br><i>Trichonympha</i><br><i>Trypanosoma</i>                     |
| Subphylum: <i>Mastigophora</i>   | One or more flagella; division by longitudinal binary fission; sexual reproduction in some groups                                                |                                                                                                                            |
| Class: <i>Zoomastigophorea</i>   | Parasitic without chromatophores; one to more flagella; sexuality known in some groups                                                           |                                                                                                                            |
| Subphylum: <i>Sarcodina</i>      | Locomotion by pseudopodia; shells present; flagella restricted to reproductive stages; asexual reproduction by fission                           |                                                                                                                            |
| Superclass: <i>Rhizopoda</i>     | Locomotion by pseudopodia or by protoplasmic flow; some contain tests                                                                            | <i>Labyrinthula</i>                                                                                                        |
| Phylum: <i>Labyrinthomorpha</i>  | Spindle-shaped cells capable of producing mucous tracks; trophic stage as ectoplasmic network; nonamoeboid cells parasitic on algae and seagrass |                                                                                                                            |
| Phylum: <i>Apicomplexa</i>       | Parasitic, spore-forming and have an apical complex; sexuality by syngamy; cysts often present; cilia absent                                     | <i>Cryptosporidium</i><br><i>Eimeria</i><br><i>Plasmodium</i><br><i>Toxoplasma</i><br><i>Nosema</i><br><i>Pleistophora</i> |
| Phylum: <i>Microspora</i>        | Unicellular spores with spiroplasm containing polar filaments; obligate intracellular parasites                                                  |                                                                                                                            |
|                                  |                                                                                                                                                  |                                                                                                                            |
|                                  |                                                                                                                                                  |                                                                                                                            |

**Table 2.14:** (Contd.)

|                            |                                                                                                                                                                                                                                                                 |                                                                                                                                  |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Phylum: <i>Ascetospora</i> | Spore with one or more spiroplasms; no polar capsules or filaments; parasitic in invertebrates                                                                                                                                                                  | <i>Haplosporidium</i>                                                                                                            |
| Phylum: <i>Myxozoa</i>     | Spores of multicellular origin; one or many polar capsules; all parasitic, especially in fish                                                                                                                                                                   | <i>Myxosoma</i>                                                                                                                  |
| Phylum: <i>Ciliophora</i>  | Simple cilia or compound ciliary organelles in at least one stage in the life cycle; two types of nuclei; contractile vacuole present; binary fission transverse; sexuality involving conjugation; most species free living, but many commensal, some parasitic | <i>Balantidium</i><br><i>Didinium</i><br><i>Entodinium</i><br><i>Ichthyophthirius</i><br><i>Nyctotherus</i><br><i>Paramecium</i> |

Source: Prescott *et al.* (2005).

**(a) Phylum Sarcomastigophora:** Protists that have a single type of nucleus and possess flagella or pseudopodia are included in this phylum. Both sexual and asexual reproductions are seen in this phylum. *Giardia*, *Leishmania*, *Trichomonas*, *Trichonympha* and *Trypanosoma* are the common genera. There are also many symbiotic amoebae in this phylum, most of which live in other animals. Two common genera are *Endamoeba* and *Entamoeba*.

**(b) Phylum Labyrinthomorpha:** It is a small phylum consisting of protists that have spindle-shaped or spherical nonamoeboid vegetative cells. In some genera, amoeboid cells move within a network of mucous tracks using a typical gliding motion. Most members are marine and either saprozoic or parasitic on algae. *Labyrinthula* is the most commonly occurring genus.

**(c) Phylum Apicomplexa:** The apicomplexans, often collectively called the sporozoans, have a spore-forming stage in their life cycle and lack special locomotory organelles. They are either intra- or intercellular parasites of animals and are distinguished by a unique arrangement of fibrils, microtubules, vacuoles and other organelles, collectively called the apical complex, which is located at one end of the cell. The four most important sporozoan parasites are *Plasmodium*, *Cryptosporidium*, *Toxoplasma* and *Eimeria*.

**(d) Phylum Microspora:** The small microsporans are obligatory intracellular parasites lacking mitochondria. Many of these protozoa are pathogenic. There has been an increased interest in these parasites because of their possible role as biological control agents for certain insects. Important genera are *Nosema*, *Encephalitozoon*, *Pleistophora*, *Microsporidium*, *Vittaforma*, *Trachipleistophora* and *Enterocytozoon*.

**(e) Phylum Ascetospora:** It is a relatively small phylum consisting exclusively of parasitic protists characterized by spores lacking polar caps or polar filaments. Ascetosporans, such as *Haplosporidium* are parasitic primarily in the cells, tissues and body cavities of mollusks.

**(f) Phylum Myxozoa:** The myxozoans are all parasitic, most on freshwater and marine fish. They have a resistant spore with one to six coiled polar filaments. The most economically important myxozoan is *Myxosoma*.

**(g) Phylum Ciliophora:** It is the largest of the seven protozoan phyla consisting of about 8000 species. The genera vary significantly in shape and most do not look like the slipper-shaped *Paramecium*. In some species the protozoan attaches itself to the substratum by a long stalk. Some species have tentacles for the capture of prey. Some can discharge toxic thread-like darts called toxicysts, which are used in capturing prey. A most striking feature of ciliates is their ability to capture many particles in a short time by the action of the cilia around the buccal cavity. Freshwater species possess contractile vacuoles used for osmoregulation. Common genera are *Entodinium*, *Nyctotherus*, *Balantidium*, *Vorticella*, *Stentor* and *Ichthyophthirius*.

## 2.8 CONCLUSION

The study of the diversity of microbes in the environment has become a requisite to make the use of a vast number of microorganisms in environmental purification and human welfare. Microorganisms are becoming increasingly important for pollution control and we must learn how to use them in a variety of new ways. They can be very useful to make the best use of the degraded resources, safe processing of toxic wastes, spin money from garbage and aid agriculture through their participation in nutrient and pest management. However, our current knowledge on the microbial diversity of the world is too rudimentary to make the best use of their potential. Extensive research is needed to study the growth and habitat requirements of individual microbe or microbial group. We must develop new isolation techniques and adequate system for classification of organisms, mostly on the basis of their metabolic similarities, which must include those microbes, that hitherto are considered impossible to cultivate in the laboratory. Much work is needed to be done on organisms living in extreme environments. The discovery of new microorganisms may lead to further advances in the industrial processes and enhanced environmental control.

Further developments in the study and classification of unusual microorganisms and microbial ecology will lead to a better understanding of the interactions between the microorganisms and their intimate world. Among other things this information will be very handy in solving the pollution problems of the environment and getting a sustainable solution to the management of toxic wastes. Similarly it has become increasingly clear that the microorganisms are essential partners with higher organisms in general and crop plants in particular in symbiotic relationships. Greater knowledge on growth and life cycle of microorganisms will provide new insight to the symbiotic associations which ultimately help to improve our appreciation of the living world. It will also certainly lead to improvements in the health of plants, livestock and human.

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# MICROBIAL DIVERSITY AND PROCESSES IN FRESHWATER HABITATS

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## 3.1 INTRODUCTION

The amount of freshwater on earth is very small in comparison to the water of the oceans, but the freshwaters have much more rapid renewal times making it hitherto sufficient enough to satisfy the need of all plants and animals on the earth. On a volumetric basis, freshwater is concentrated in large, deep basins of several great lakes. About 20% of surface freshwater is contained in Lake Baikal of Siberian Russia. However, the number of individual depressions of smaller lakes and reservoirs is extremely large and most of these lakes are located in temperate and sub-arctic regions of the Northern Hemisphere. Most lakes are shallow and geologically very transient.

Catastrophic events of glacial, volcanic and tectonic activities have aggregated many freshwater lakes into lake districts. The morphometry and geological substrata of the drainage basins are very important because these properties influence sediment-water interactions, the resultant productivity and the significance of littoral productivity to that of the river and entire lake ecosystem. Shallow lakes, which have more sediment area per unit of water volume, are generally more productive than deep lakes and a greater proportion of their total productivity is attributable to communities of the littoral and wetland areas of the land-water interface. The freshwater ecological processes are strongly influenced by the microbial activities, that include photosynthesis, respiration, nutrient mobilization and immobilization. The sediment characteristics and microbial processes in the sediment determine the rate of change of matter and energy in the water column and its support to the terrestrial ecosystem.

## 3.2 THE FRESHWATER ENVIRONMENT

Inland waters cover less than 2% of the earth's surface, approximately 2.5 million square kilometers. About 20 lakes are extremely deep (in excess of 400 m) and a significant portion of the world's freshwater is contained in Lake Baikal, the deepest lake of enormous volume (23,000 km<sup>3</sup>). Like Lake Baikal, nearly all other extremely deep lakes are tectonic or volcanic in origin or have formed from fjords that have subsequently become fresh. Although a few lakes of glacial origin, such as the Laurentian Great Lakes and Great Slave Lake of Canada, have basins of great depth, few exceed a depth of 300 m. Most of the very deep lakes are found in mountainous regions along the western portions of North and South America, of Europe and in the mountainous regions of central Africa and Asia. While more than 90% of the earth's freshwater remains as stagnant water in the lakes forming a characteristic ecosystem, the remaining 10% flows through the rivers and streams connecting the lakes and rivers with the terrestrial habitats.

### 3.2.1 The Lotic Ecosystems

Streams and rivers within their drainage basins are central to surface water ecosystems. A number of characteristics such as water renewal rate, sediment characteristics, sedimentation patterns, water chemistry, energy flow pattern and nature of biota, differentiate running water from lake ecosystems. The nature of a lentic system is determined by the relative residence time of water. In this respect the lotic systems differ significantly from the lentic systems. The importance of variable but continuous and rapid throughput of water and material contained in the rivers and streams is evident in the biology of most organisms inhabiting running waters. When the energy of flowing water is dissipated, as in the transition zone of reservoirs, the change to lentic characteristics is rapid.

Rivers constitute an insignificant amount (0.001 or 0.1%) of the land surface and are the most effective lotic systems. Only 0.0001% of the water of the Earth occurs in river channels. In spite of these low quantities, running waters are of enormous significance to humans. Erosion moves large amounts of dissolved and particulate matter from the land to the sea through the rivers and streams. Small streams and rivers often flow to lakes and eventually to large river systems.

### 3.2.2 The Lentic Ecosystems

The lakes constitute the most part of the lentic ecosystems. The dominant inflows to and outflows from the lakes occur in surface streams. Each lake receives the drainage water from many rivers and streams thus becoming a sink for the flow of soluble organics, nutrients and particulate matters. Because of the continuity of the rivers and lakes there is hardly any specific difference between the flora and fauna of rivers and lakes. The differences become remarkable only when there is influence of a biotic and / or abiotic factor on the biota. For example, the vegetation and the fauna of a fast flowing stream is determined by the intensity of current.

Much of our limnological understanding originates from natural lake ecosystems. Hundreds of thousands of reservoirs, however, have been created by human activities. These artificial water bodies have been created for specific purposes of water management like water storage, flood control, generation of electrical energy and recreation. Reservoirs differ in significant ways from natural lake ecosystems but still have many functional similarities with natural lakes. In order to effectively manage and utilize reservoirs, it is important to understand the structural differences between these human-made ecosystems and natural lakes while simultaneously appreciating their functional similarities. Understanding of these structural differences is mandatory for effective management and use of impounded water resources. Failure to recognize basic similarities in metabolic functioning and community interrelationships of the biota will only result in redundancies in reservoir research that can be basically understood from our existing knowledge of processes in natural lakes.

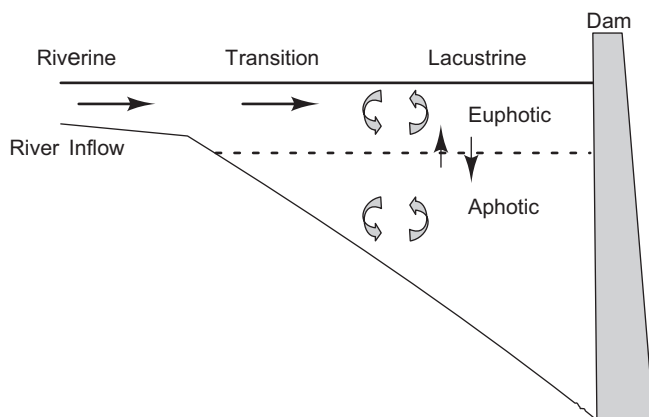
Reservoirs are most frequently constructed by damming a river valley, within which water accumulates behind the dam. As a result of the common linear morphology in a river valley, several distinct physical and biological patterns develop longitudinally along the length of reservoirs. Small reservoirs and artificial ponds are constructed also in valley depressions to which inflows occur by surface flows and/or seepage. In some cases, excavations of soil for construction or mining purposes result in reservoirs or artificial lakes as the excavations cut into water table aquifers (e.g., strip mining lakes, "barrow pit" lakes adjacent to major highways). Among many excavated reservoirs outflows may occur only by subaqueous seepage or evaporation.

Large reservoirs also bring about complex hydrodynamic characteristics with physical and chemical changes that influence aquatic biota in many ways and can be deleterious to the environment and human welfare. Large impoundments often modify the regions in their vicinities or for large distances below the dam. The reservoir serves as a source or sink for heat, sediments and solutes that can cause severe effects far downstream from the dam, sometimes for hundreds of kilometers. Diurnal and annual flooding patterns are altered and movements of anadromous fishes are impeded or inhibited. Because many reservoirs function to reduce river flows, they serve as collectors of riverine sediment loads. As a result, the longevity of reservoirs is often very short, generally less than 100 years.

Three distinct zones occur along the longitudinal gradient in a reservoir: a *riverine* zone, a zone of *transition* and a *lacustrine* zone (Fig. 3.1). Each zone possesses unique and dynamic physical, chemical and biological characteristics. The riverine zone is often relatively narrow as a result of river geomorphology. Water is usually well mixed. Although water velocities decrease as the water enters the reservoir, advective transport by currents is sufficient to move significant quantities of fine suspended particulates (silts, clays and organic particulate matter). High particulate turbidity commonly reduces light penetration and limits primary production within the water of this zone. Loading of organic matter from allochthonous sources is high in proportion to water volume in the



riverine zone. High decompositional rates often result with high consumption of dissolved oxygen, but aerobic conditions generally prevail in the shallow, well-mixed riverine zone.



**Figure 3.1** - Generalized zones along longitudinal gradients in reservoirs.

Riverine water velocities decrease as energy is dispersed over larger areas in the transitional zone. As a result an appreciable portion of the turbidity load settles out of the upper water strata. Decreased turbidity results in enhanced depth of light penetration and increased rates of photosynthetic productivity by phytoplankton. Gradually, a shift occurs to an increasing percentage of total organic matter loading from phytoplankton and, in some shallow reservoirs, from rooted vascular plants. Within the lacustrine zone, characteristics become more similar to lake ecosystems. This portion of the reservoir often stratifies thermally and assumes many of the properties of natural lakes in regard to planktonic production, limitations by nutrients, sedimentation of organic matter and decomposition in the hypolimnion.

### 3.3 THE ECOLOGICAL PROCESSES IN FRESHWATER HABITATS

Aquatic ecosystems are open and require a continual input of energy in the form of organic matter that support the entire ecosystem. This organic matter is produced almost totally by photosynthesis and is either used by consumers or stored in the ecosystem. Organic matter and energy driving the ecosystem are also imported from terrestrial or land-water interface communities to the river or lake. Within the aquatic ecosystem, organic matter is degraded physically, as in direct photolysis, utilized by decomposer organisms, or ingested by grazer (metazoan) organisms. Energy and nutrients must continually be replenished in such open systems in which available energy and other resources are constantly being utilized and respired to inorganic compounds and so lost or made unavailable.



### 3.3.1 The Drainage Basin Concept

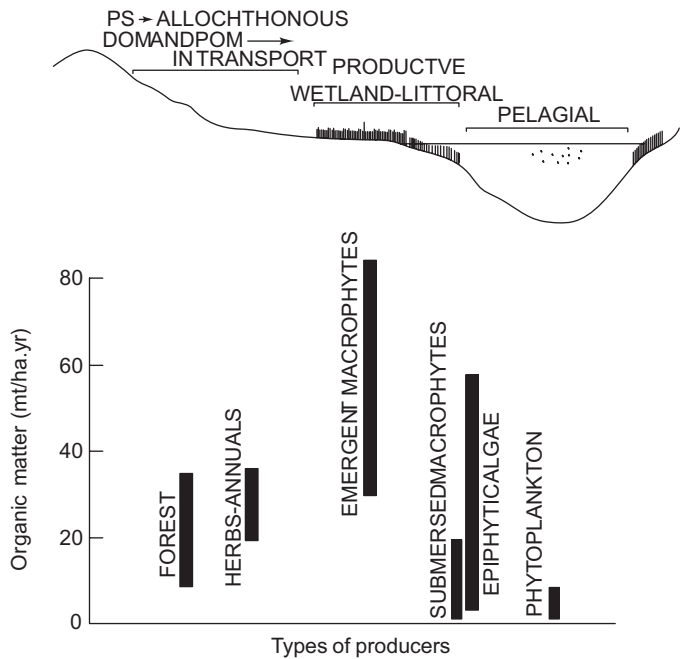
Terrestrial and aquatic ecosystems are linked by movements of water and materials through the drainage basin to recipient rivers and lakes. Chemical and biological processes modify the composition of the materials dissolved within and moved by the water. Geological features of the landscape govern the directions of movement and particularly the residence time of water during movement in surface and ground waters toward rivers and lakes. The duration of contact with soil and associated microbiota influences the content of dissolved salts and organic products conveyed in the water. Terrestrial and wetland vegetations in the drainage basin influence the chemical content of both water falling through their leaf canopy as well as water in the soil by selective assimilation and storage within their tissues.

From drainage basins that are vegetated, such as in forests, usually only about one-third of the incoming precipitation leaves as runoff and groundwater seepage to streams and rivers. Removal of vegetation greatly increases the amount and accelerates the rate of flow of runoff from the landscape. During periods of high precipitation, water in streams frequently exceeds the capacities of the river channels and inundates the flood plains, by which organic matter and many nutrients can be returned to the flooded land and result in highly productive adjacent aquatic habitats.

### 3.3.2 The Lake Ecosystem Concept

The lake ecosystem consists of the lake and its entire drainage basin, from which the lake receives the inorganic and organic nutrients. The rate of inputs of both particulate and especially dissolved organic matter to lakes from their catchments regulates the biotic activities and productivity of lakes. Extremely heterogeneous and productive wetland-littoral areas often lie at the interface between the terrestrial drainage basin and the open-water zone of the lake (Fig. 3.2). These complex wetland-littoral areas are exceedingly important in regulating lake metabolism. Since a majority of lakes of the biosphere are small and relatively shallow, the metabolically active wetland and littoral components dominate the productivity of most lakes of the world.

The effects that terrestrial, wetland and littoral biota have on the quality and quantity of inorganic and organic loading to a lake can be profound. Water laden with inorganic and organic substances flows from higher elevations to the recipient lake basin both in ground water and in surface streams. Chemical and biological reactions occur *en route* that selectively modify the quality and quantity of nutrients and organic substances entering the lake. Surface flows often pass through the wetland -littoral complex and can further selectively lose or gain inorganic and organic compounds before reaching the open water. An appreciable loading of inorganic and organic substances to the atmosphere is common from the industrial and agricultural activities of humans. These compounds reach the drainage basin and lake itself causing a significant change of the total nutrient and contaminant loading.



**Figure 3.2** - The comparison of productivity of different producers community with that of different zones of a lake.

### 3.3.3 Trophic Structure of Food Cycles

In most cases, energy required for growth and maintenance of organisms enters ecosystems as light and is converted to chemical energy by plant photosynthesis (autotrophy). Biological communities are supported by and based on the photosynthetic production of organic matter, produced either within a community (autochthonous production) or derived from an external source (allochthonous production) and transported in a dead or decomposing state to a community for utilization (heterotrophy). In self-sustaining aquatic ecosystems, the photosynthetic carbon fixation by primary producers is sufficient to support the demand of organic matters by the consumers. However, in deep lakes and saprobic streams the transport of allochthonous organic carbon is a necessity to supply the energy to the consumers inhabiting the aphotic zones.

Total biomass and numbers of individuals at each successive higher trophic level also usually decrease, although some exceptions are seen in a microbial food web. Efficiency of energy or material transfer through a trophic level is simply the ratio, expressed as a percentage of energy or material leaving a level or system to the quantity of energy or material entering the level or system. The rate of conversion of solar energy into chemical energy by photosynthesis by the first trophic level of primary producers is small, usually less than 1%, because of the great losses of light energy by non-biological scattering and absorption and dissipation as heat. The expenditure of energy for maintenance and the losses to

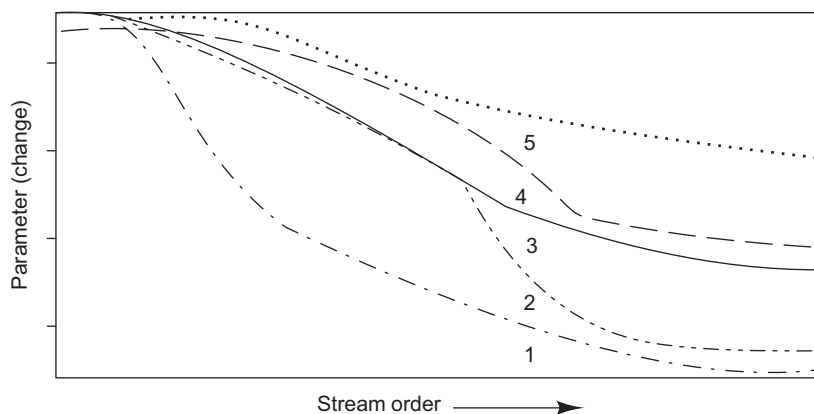
the detrital dynamic structure increase with higher trophic levels. As a result, the efficiency of energy transfer is low from one trophic level to the next (5 -15%) and usually decreases with each transfer.

### **3.3.4 The Detritus and Detrital Dynamics**

The living biota of lakes constitute only a very small portion of the total organic matter of ecosystems, even in the most productive freshwater habitats. Most of the organic matter in aquatic ecosystems is nonliving and is collectively referred to as detritus. The functioning of aquatic ecosystems centers on the cycling of organic carbon between living and nonliving components. Detritus consists of all nonliving organic matter, in both dissolved and particulate forms. In aquatic ecosystems in general, nearly all of the organic matters consist of dissolved organic carbon compounds (DOC) and particulate organic carbon compounds (POC). The ratio of DOC to POC is usually between 6:1 and 10:1 in both lake and running water ecosystems. Although living POC of the biota constitutes a very small portion of the total POC, particularly in the pelagic region, the microbial components are important in mediating the fluxes of carbon between the dissolved and particulate phases of detrital organic carbon.

Particulate organic carbon loadings from the drainage basin are relatively small; most of the organic carbon is transported and imported in dissolved form. Similarly, much of the very high net primary productivity of the wetlands and littoral land-water interface regions of lakes and river ecosystems enters the open-water regions as dissolved organic matter. This pool of dissolved organic matter is supplemented by the low phytoplanktonic primary production to varying degrees in proportion to the size and geomorphology of the lake and river basins. Clearly, most lakes are small and shallow and the ratios of organic carbon loadings from allochthonous and littoral sources to those of the pelagic are large. The pool of organic matter is variously metabolized, sedimented, or exported. Very little of either the particulate or the dissolved organic matter emanating from new photosynthetic production is consumed by animals or enters the consumer food chain but instead enters non predatory pools of dissolved and particulate detritus. Because little of this organic carbon does enter the consumer food chain, however, does not mean it is neither metabolized nor important. Not only this organic carbon is the dominant energetic pathway but also these microbial and chemical fluxes are essential to the maintenance of nutrient recycling and the metabolic stability of the ecosystem. The detritus input changes along the gradient from streams to the lakes with the greatest contribution of the detritus to the productivity seen at the land-water interface. The benthic periphyton production and the addition of autochthonous detritus from the benthic periphyton standing crop gradually decreases with the increase in the depth of a lake. Similarly the macrophyte production and the flood plain capture fine particulate detritus decreases with increase in the stream order (increase in the distance from the point of origin) (Fig. 3.3).

Metabolism of particulate and especially of dissolved organic detritus from both allochthonous and many pelagic and nonpelagic autochthonous sources dominate the material and energy fluxes. The amount of detrital organic matter is very large and it is chemically more recalcitrant to biological degradation than is living organic matter. As a result, metabolism of detritus is slow, but the quantity is large and much greater than that resulting from living organic matter that is ingested by animals. This slow decomposition provides an inherent ecosystem stability that energetically dampens the ephemeral, volatile fluctuations of higher trophic level species populations as they respond to rapidly oscillating factors governing their growth and reproduction.



**Figure 3.3** - The common pattern of changes of particulate organic matter inputs along a river continuum. Increase of stream order refers to the decrease of terrestrial influence in the river. Legends: 1- Terrestrial course particulate detritus, 2-Benthic periphyton production, 3- River phytoplankton, 4- Macrophyte production and 5- Flood plain capture fine particulate detritus (Modified from Minshall *et al.*, 1985).

### 3.4 THE PHYTOPLANKTON COMMUNITIES

The algae of the open water of lakes and large streams are dominated by phytoplankton, which consist of a diverse assemblage of nearly all major taxonomic groups. Many of these forms have different physiological requirements and vary in response to physical and chemical parameters such as light, temperature and nutrient regimen. The photosynthetic cyanobacteria constitute a major component and the primary production by this group of microbes increases with the increase of the nutrient load of the water body. Despite the diversities, both taxonomic and physiological, many algal and cyanobacterial species coexist in the same water volume. However, dominant genera in algal groupings change not only spatially (vertically and horizontally within a lake) but seasonally, as physical, chemical and biological conditions change in the water body. A general pattern of seasonal succession of

phytoplankton has been correlated with environmental factors of many lakes, although the precise reasons for many of these changes are not yet well known.

### 3.4.1 The Characteristics of Phytoplanktonic Associations

Before discussing the phytoplankton associations and population dynamics of phytoplanktonic communities, a few of the major morphological and physiological characteristics of the primary algal groups need be emphasized. The systematics of algae comprising the phytoplankton show that the latter are morphologically diverse but physiologically same. The cyanobacteria and many motile and colonial flagellates (protozoa) have similarities with true algae in certain morphological and reproductive characteristics. The important characteristic, however, is that autotrophic photosynthesis is the primary mode of nutrition and results in a major synthesis of new organic matter. Therefore, these photoautotrophic microbiota are treated collectively with the true algae.

Enumeration of organisms per unit volume of water or area is a commonly used method to evaluate microorganisms. Although number afford a great advantage in permitting qualitative differentiation between species and of organisms from detrital particles, it is not a true evaluation of phytoplankton productivity as the production is usually higher with reduction of individual cell volume. The phytoplankton differ greatly in size and the difference between the biovolume of the smallest and the largest cell/colony may be as high as five orders of magnitude (Table 3.1). The abundance of each species is better expressed as the biovolume of all individuals. The species/group of phytoplankton having the highest biovolume dominates the primary production and dictates the secondary production of the water body.

A primary algal characteristic is the presence of photosynthetic pigments—the chlorophylls, carotenoids and biliproteins. Chlorophyll *a* is the primary photosynthetic pigment of all oxygen-evolving photosynthetic organisms and is present in all algae and cyanobacteria (Table 3.2) and photosynthetic organisms other than the photosynthetic sulphur bacteria (See Chapter 2). Chlorophyll *b*, although common in higher plants, is found only in the green algae, euglenophytes and certain minor groups. Chlorophyll *c*, consisting of three spectrally distinct components (*C*, *C*<sub>1</sub>, *C*<sub>2</sub>), is probably an accessory pigment to photosystem II. Chlorophyll *d* of no known function, is a minor pigment found only in certain red algae. A fifth chlorophyll (*e*) has been isolated from a xanthophycean alga, but whether it exists *in vivo* is uncertain. Among the many carotenoids of algae, the carotenes are linear unsaturated hydrocarbons and the xanthophylls are oxygenated derivatives of carotenes.  $\beta$ -Carotene is the most widely distributed of the carotenes and is replaced by  $\alpha$ -carotene only in certain green algae and in the Cryptophyceae. The biliproteins are water-soluble, pigment-protein complexes occurring in the cyanobacteria and, to a lesser extent, in certain cryptophytes and red algae.

3.4.2    **The Algal Diversity in Freshwater**

**(a) Cyanobacteria:** The cyanobacteria have been among the most studied of all planktonic groups. Previously classified as algae in the division Cyanophyta, these organisms are true bacteria with a simple prokaryotic cell structure. Like most other bacteria, these autotrophs have murein in the cell wall, reproduce by binary fission and do not divide by mitosis as algae and higher organisms do. However, these are distinguished from other bacteria by the presence of chlorophyll *a*, which differ structurally from bacteriochlorophyll but resemble with that of algae. Cyanobacteria occur in unicellular, filamentous and colonial forms and most are enclosed in mucilaginous sheaths either individually or in colonies. Most planktonic cyanobacteria consist of members of the coccoid family Chroococcaceae (e.g., *Microcystis*, *Gomphosphaeria* and *Coccochloris*) and filamentous families (e.g., *Planktothrix*, *Limnothrix* and *Tychonema*, *Anabaena*, *Aphanizomenon*, *Nodularia* and *Gloeotrichia*).

**Table 3.1:** The calculated mean volume of the representative phytoplankton species of freshwater habitat only. Only the species common to most freshwater habitats are presented here.

| Group         | Species                                  | Volume<br>( $\mu\text{m}^3$ ) |
|---------------|------------------------------------------|-------------------------------|
| Cyanobacteria | <i>Anabaena flos-aquae</i> (col.)        | 80000                         |
|               | <i>Aphanocapsa delicatissima</i>         | 4                             |
|               | <i>Aphanothece nidulans</i>              | 5                             |
|               | <i>Chroococcus limneticus</i> (col.)     | 500                           |
|               | <i>Chroococcus turgidus</i> (col.)       | 1000                          |
|               | <i>Coelosphaerium naegelianum</i> (col.) | 15000                         |
|               | <i>Dactylococcopsis smithii</i> (col.)   | 1500                          |
|               | <i>Gloeocapsa rupestris</i>              | 18                            |
|               | <i>Gomphosphaeria lacustris</i> (col.)   | 2000                          |
|               | <i>Merismopedia tenuissima</i>           | 8                             |
|               | <i>Microcystis flos-aquae</i>            | 50                            |
|               | <i>Microcystis aeruginosa</i> (col.)     | 100000                        |
|               | <i>Oscillatoria limnetica</i> (1 mm)     | 17500                         |
|               | <i>Oscillatoria rubescens</i> (1 mm)     | 30000                         |
|               | <i>Synechococcus aeruginosus</i>         | 350                           |
| Chlorophyta   | <i>Ankistrodesmus falcatus</i>           | 250                           |
|               | <i>Botryococcus braunii</i> (col.)       | 10000                         |
|               | <i>Chlamydomonas subcompleta</i>         | 250                           |
|               | <i>Chlorella vulgaris</i>                | 200                           |
|               | <i>Closterium aciculare</i>              | 4000                          |
|               | <i>Cosmarium phaseolus</i>               | 3000                          |
|               | <i>Gloeococcus shroeteri</i> (col.)      | 5000                          |
|               | <i>Oocystis solitaria</i>                | 400                           |
|               | <i>Pandorina morum</i> (col.)            | 4000                          |
|               | <i>Scenedesmus quadricauda</i>           | 1000                          |

Table 3.1: (Contd.)

|              |                                      |        |
|--------------|--------------------------------------|--------|
| Cryptophyta  | <i>Staurostrum paradoxum</i>         | 20000  |
|              | <i>Tetraedron minimum</i>            | 40     |
|              | <i>Ulothrix zonata (1 mm)</i>        | 6000   |
|              | <i>Chroomonas nordstedtii</i>        | 35     |
|              | <i>Cryptomonas erosa</i>             | 1000   |
|              | <i>Cryptomonas ovata</i>             | 2500   |
| Chrysophyta  | <i>Rhodomonas lacustris</i>          | 135    |
|              | <i>Chromulina pyriformis</i>         | 50     |
|              | <i>Dinobryon borgei</i>              | 1500   |
|              | <i>Dinobryon divergens</i>           | 800    |
|              | <i>Mallomonas caudata</i>            | 12000  |
|              | <i>Mallomonas urnaformis</i>         | 1200   |
|              | <i>Rhizochrysis limnetica</i>        | 1200   |
|              | <i>Amphiprora ornate</i>             | 18000  |
|              | <i>Asterionella formosa</i>          | 700    |
|              | <i>Cyclotella comensis</i>           | 400    |
|              | <i>Cymatopleura solea</i>            | 80000  |
|              | <i>Diatoma vulgare</i>               | 4400   |
|              | <i>Fragilaria crotonensis (1 mm)</i> | 200000 |
|              | <i>Nitzschia gracilis</i>            | 250    |
|              | <i>Aulacoseira granulata (1 mm)</i>  | 60000  |
|              | <i>Stephanodiscus astraes</i>        | 2000   |
|              | <i>Stephanodiscus hantzschii</i>     | 200    |
|              | <i>Stephanodiscus niagarae</i>       | 5000   |
|              | <i>Synedra acus</i>                  | 250    |
|              | <i>Synedra capitata</i>              | 1000   |
|              | <i>Synedra ulna</i>                  | 50     |
|              | <i>Tabellaria fenestrata</i>         | 3000   |
| Pyrrophyta   | <i>Ceratium hirundinella</i>         | 4000   |
|              | <i>Gymnodinium fuscum</i>            | 10000  |
|              | <i>Gymnodinium helveticum</i>        | 20000  |
|              | <i>Peridinium cinctum</i>            | 40000  |
|              | <i>Peridinium willei</i>             | 40000  |
| Euglenophyta | <i>Trachelomonas hispida</i>         | 4200   |
|              | <i>Trachelomonas volvocina</i>       | 1800   |

Compiled from Lewis (1977); Hillebrand *et al.* (1999).

**(b) Green algae:** The Chlorophyta are an extremely large and morphologically diverse group of algae that is almost totally freshwater in distribution. Most planktonic green algae belong to the orders Volvocales (e.g., *Chlamydomonas*, *Sphaerocystis*, *Eudorina* and *Volvox*) and Chlorococcales (e.g., *Scenedesmus*, *Ankistrodesmus*, *Selenastrum* and *Pediastrum*). Many members are flagellated (2 or 4, rarely more) at least in the gamete stages; in the Zygnematales and the desmids (Conjugales or Desmidiaceae), the gametes are amoeboid. Whereas the planktonic Volvocales and Chlorococcales are ubiquitous in distribution among waters of differing salinity within the normal limnological range, the distribution

of most species of desmids of the Conjugales is limited to low concentrations of the divalent cations calcium and magnesium. Although not totally restricted to waters of low salinity, the desmids are most common and the species diversity is greatest in unproductive soft waters draining land forms and especially in waters with a high content of dissolved organic matter. Their abundance and diversity are often greatest in bog waters that drain through deposits of the moss *Sphagnum*. Many desmids are distributed widely, but as a whole they are less cosmopolitan than most unicellular algae.

**Table 3.2:** Distribution of photosynthetic pigments among different phytoplankton groups occurring in freshwater environments.

| <i>Pigments</i>         | <i>Cyanobacteria</i> | <i>Chlorophyta</i> | <i>Cryptophyta</i> | <i>Chrysophyta</i> | <i>Pyrrophyta</i> | <i>Euglenophyta</i> |
|-------------------------|----------------------|--------------------|--------------------|--------------------|-------------------|---------------------|
| Chlorophylls            |                      |                    |                    |                    |                   |                     |
| Chlorophyll <i>a</i>    | +                    | +                  | +                  | +                  | +                 | +                   |
| Chlorophyll <i>b</i>    | –                    | +                  | –                  | –                  | –                 | +                   |
| Chlorophyll <i>c</i>    | –                    | –                  | +                  | +                  | +                 | –                   |
| Carotenes               |                      |                    |                    |                    |                   |                     |
| $\alpha$ -Carotene      | –                    | +                  | +                  | –                  | –                 | –                   |
| $\beta$ -Carotene       | +                    | +                  | –                  | +                  | +                 | +                   |
| $\gamma$ -Carotene      | –                    | +                  | –                  | –                  | –                 | –                   |
| $\varepsilon$ -Carotene | –                    | –                  | +                  | –                  | –                 | –                   |
| Xanthophylls            |                      |                    |                    |                    |                   |                     |
| Lutein                  | +                    | +                  | –                  | +                  | –                 | –                   |
| Violaxanthin            | –                    | +                  | –                  | –                  | –                 | –                   |
| Fucoxanthin             | –                    | –                  | –                  | +                  | –                 | –                   |
| Neoxanthin              | –                    | +                  | –                  | –                  | –                 | +                   |
| Astaxanthin             | –                    | +                  | –                  | –                  | –                 | +                   |
| Diatoxanthin            | –                    | –                  | –                  | +                  | –                 | –                   |
| Diadinoxanthin          | –                    | –                  | –                  | +                  | +                 | –                   |
| Peridinin               | –                    | –                  | –                  | –                  | +                 | –                   |
| Dinoxanthin             | –                    | –                  | –                  | –                  | +                 | –                   |
| Teraxanthin             | –                    | –                  | –                  | –                  | –                 | –                   |
| Antheraxanthin          | –                    | –                  | –                  | –                  | –                 | +                   |
| Myxoxanthin             | +                    | –                  | –                  | –                  | –                 | –                   |
| Myxoxanthophyll         | +                    | –                  | –                  | –                  | –                 | –                   |
| Echinenone              | +                    | –                  | –                  | –                  | –                 | +                   |
| Biliproteins            |                      |                    |                    |                    |                   |                     |
| Phycocyanin             | +                    | –                  | +                  | –                  | –                 | –                   |
| Phycoerythrin           | +                    | –                  | +                  | –                  | –                 | –                   |

Compiled from Goodwin (1974); Hoek *et al.* (1995).



**(c) Yellow-green algae:** The Xanthophyceae are unicellular, colonial, or filamentous algae that are characterized by conspicuous amounts of carotenoids in comparison to chlorophylls that result in their predominantly yellow-green colouration (Table 3.2). Nearly all of the motile cells possess two flagella, one of which is smooth and longer than the other, “hairy” flagellum (tinsel type). Most are associated with substrata and many are epiphytic on larger aquatic plants. A few members are planktonic and include common genera such as *Chlorobotrys*, *Gloeobotrys* and *Gloeochloris*.

**(d) Golden-brown algae:** The chromatophores of the Chrysophyta often yield a distinctive golden-brown colouration because of the dominance of  $\beta$ -carotene and specific xanthophyll carotenoids in addition to chlorophyll *a*. Most are unicellular; a few are colonial; and they are rarely filamentous. Numerous chrysophycean algae form important components of the phytoplankton. The unicellular species with a single flagellum (e.g., *Chromulina*, *Chrysococcus* and *Mallomonas*) are usually very small algal representatives of the nanoplankton size range (Table 3.3). Larger colonial forms such as *Synura*, *Chrysosphaerella*, *Uroglena* and particularly *Dinobryon* are widely distributed and may become major components of the phytoplankton under certain environmental conditions, particularly in temperate, oligotrophic lakes. The conspicuous development of populations of certain species of *Dinobryon* and *Uroglena* in lakes of very low phosphorus concentrations may be related to their abilities to take up phosphate effectively at extremely low ambient concentrations (affinity adaptation to low P-level). In contrast, other species of *Dinobryon* and *Synura* have high phosphorus requirements. The phytoplankton of many oligotrophic lakes are dominated throughout the year by chrysophycean algae and secondarily by cryptophytes. The latter are particularly well adapted to low temperatures and light levels. Several species of chrysophycean algae, particularly *Dinobryon*, supplement photosynthesis with phagotrophy. These algae can obtain an appreciable fraction of their energy and nutrients by ingesting bacteria. Although the rates of water clearance are small (e.g., 0.14  $\mu\text{l/day}$ ), large populations of *Dinobryon* can collectively clear water of bacteria at rates that compete effectively with those of ciliates, rotifers and microcrustaceans.

**Table 3.3:** The size classification of plankton of freshwater ecosystems.

| Classification term | Size range ( $\mu\text{m}$ ) |
|---------------------|------------------------------|
| Femtoplankton       | 0.02-0.2                     |
| Ultraplankton       | 0.2-20                       |
| Picoplankton        | 0.2-2.0                      |
| Nanoplankton        | 2.0-20                       |
| Microplankton       | 20-200                       |
| Mesoplankton        | >200                         |

The presented values are based on the evaluations of Sieburth *et al.* (1978) and Fogg (1991).

**(e) Diatoms:** The most important group of algae of the phytoplankton is the diatoms (Bacillariophyceae), even though most species are sessile and associated with littoral substrata. Both unicellular and colonial forms are common among the diatoms. There are very few planktonic species as most prefer to grow as periphytic community on the macrophytes in the littoral zones. *Asterionella*, *Diatoma*, *Fragilaria*, *Synedra*, *Actinella*, *Eunotia*, *Achnanthes*, *Cocconies*, *Amphora*, *Cymbella*, *Gomphonema*, *Navicula*, *Nitzschia*, *Pinnularia* and *Surirella* are the common species of many oligotrophic and mesotrophic freshwater lakes and rivers. These species have distinct nutritional requirements and are therefore distributed in the water bodies with a wide range of trophic status.

**(f) Cryptomonads:** Most of the cryptophycean algae are naked, unicellular and motile. This class is very small and most of the planktonic members belong to the Cryptomonadineae (e.g., *Cryptomonas*, *Rhodomonas* and *Chroomonas*). All members are small and not a great deal is known about their ecology or physiology. Cryptomonad algae and other cryptomonad-like microflagellates occur in almost all lakes, regardless of trophic state. The cryptomonads are typically present in low numbers during most of the year but commonly increase intermittently. Such increases in abundance often follow the demise of previously dominant “bloom” algal populations. Characteristics of the cryptomonads and other microflagellates include intermittent numerical dominance, high nutritional quality, short turnover times, ability to grow and reproduce at low intensities of light and effective rapid pulse timing of growth. In combination, these characteristics implicate the cryptomonads and other photosynthetic microflagellates as an ecologically significant internal stabilizing component of planktonic communities.

**(g) Dinoflagellates:** The dinoflagellates (Dinophyceae of the Pyrrophyta) are unicellular flagellated algae, many of which are motile. Although a few species are naked or without a cell wall (e.g., *Gymnodinium*), most develop a conspicuous cell wall that often is sculptured and bears large spines and elaborate cell-wall processes (e.g., *Ceratium*, *Glenodinium*, *Peridinium*). The distribution of dinoflagellates in relation to major chemical characteristics shows that, while some species are widely tolerant and ubiquitous, especially among species of *Ceratium* and *Peridinium*, most dinoflagellate species have restricted ranges with respect to calcium, pH, dissolved organic matter and temperature. They have a high density in water bodies having high amount of dissolved and particulate organic matters. The relatively large body size (ca. 400  $\mu\text{m}$  in diameter) makes these algae relatively inedible to most grazing organisms.

**(h) Euglenoids:** Although the euglenoid algae (Euglenophyta) are a relatively large and diverse group, few species are truly planktonic. Nonetheless, when conditions are favourable, the euglenoids can develop in great profusion. Almost all euglenoids are unicellular, lack a distinct cell wall and possess one, two, or three flagella that arise from an invagination in the cell membrane. Some euglenoids are unpigmented and phagotrophic and are best treated as Protozoa,

but most are photosynthetic and facultatively heterotrophic. Their development in the phytoplankton occurs most often in seasons, strata, or lake systems in which concentrations of ammonia and especially dissolved organic matter are high. However, euglenoids are found most often in shallow water rich in organic matter, such as organically polluted lakes and farm ponds. Increase in the population of *Euglena* and *Phacus* indicates the appearance of polysaprobic conditions in the water body.

**(i) Brown and red algae:** The brown algae (Phaeophyta) are mostly filamentous or thalloid algae, which, as a group, are almost exclusively marine. Only a few genera of this large, primitive group are present in freshwater. Those found in freshwater are attached to substrata, such as rocks. No species is planktonic.

The red algae (Rhodophyta) are also very sparsely represented in freshwater and none is planktonic. A majority of the genera occurring in freshwater are found only there. The thalloid or filamentous species (e.g., *Batrachospermum*) are nearly all restricted to fast-flowing streams of well-oxygenated, cool waters.

### 3.4.3 Phytoplankton Size vs Productivity

An enormous variation in size is found among the phytoplankton (Table 3.1). Minute (0.2-1.0  $\mu\text{m}$ ) algal and cyanobacterial picoplankton, which consist mainly of small coccoid cyanobacteria of the *Synechococcus* type, are found to be abundantly and ubiquitously distributed in freshwater ecosystems. The eukaryotic *Chlorella*-like picophytoplankton tend to be larger (1.2-2.0  $\mu\text{m}$ ) and can dominate in certain extreme habitats, such as acidic or humic-stained dystrophic waters. Colonial picophytoplankton are found in oligotrophic lakes but are likely aggregations of chroococcoid cyanobacteria such as *Synechococcus*. When the size of planktonic organisms falls below about 20  $\mu\text{m}$  (the nanoplankton), the transfer of materials to and from cells is almost entirely by molecular diffusion rather than by the eddy diffusion of turbulence. These phytoplankton are simple, unicellular in form and have generation times in hours that are similar to the generation times of protistian flagellated predators (Table 3.4). The exchanges by molecular diffusion that small size makes possible, however, allows the establishment of a self-contained, dynamic community of phototrophs, viruses, heterotrophic bacteria, phagotrophic flagellates and ciliates, in which materials are rapidly cycled with little loss by sedimentation or transfer to higher trophic levels. In such an equilibrium system, population densities are determined by internal dynamics rather than by nutrient inputs.

The larger phytoplankton (microplankton) are not vulnerable to control by flagellates but are inefficient in nutrient uptake and therefore develop prolifically when nutrient concentrations are relatively high. The microphytoplankton are also vulnerable to sedimentation from the photic zone. The availability of sufficient nutrients and transport to light available zones are strongly coupled to macroturbulence (Table 3.4). Water movements are highly variable and, as a result, microphytoplankton must be able to grow and reproduce rapidly when conditions are conducive.

The microphytoplankton separate into two broad types of ecological strategies. Larger algae, such as many of the diatoms, are nonmotile and depend upon mixing regime to remain poised in the photic zone. Another type, typified by the dinoflagellates and gas vacuolate cyanobacteria, competes most effectively under relatively quiescent and stable water conditions where motility or buoyancy control allows positioning to optimize access to light and nutrients.

**Table 3.4:** A comparative account of the functional characteristics of ultra-phytoplankton (picoplankton and nanoplankton) and microphytoplankton.

| <i>Characteristics</i>                            | <i>Ultraphytoplankton</i> | <i>Microphytoplankton</i>            |
|---------------------------------------------------|---------------------------|--------------------------------------|
| Size                                              | 0.2-20 $\mu\text{m}$      | 20-200 $\mu\text{m}$                 |
| Mean physical process determining nutrient supply | Molecular diffusion       | Turbulence, motion to water relative |
| Form                                              | Simple, unicellular       | Elaborate, often colonial            |
| Gas vacuoles (in cyanos)                          | Absent                    | Present                              |
| Generation time                                   | Hours                     | Days                                 |
| Relative generation time of predators             | Equal                     | Much longer                          |
| Variation in natural population densities         | 10-100                    | 1000-100000                          |

Source: Fogg (1991, 1995).

There is a general preponderance of small algal species in the phytoplankton of oligotrophic lakes and the size of dominant algae often increases in more fertile lakes. A number of analyses of the photosynthetic productivity of phytoplankton in which the relative contributions of differently sized algae have shown that a greater proportion of the total productivity is by smaller forms in less productive waters. The contributions of the smaller algae also shift seasonally; the ultraplankton often dominate the productivity, particularly during the winter and spring. Their contribution to lake productivity is, however, less in comparison to their chlorophyll *a* content. Hence, even though the autotrophic picoplankton are ubiquitous, caution is needed to avoid overestimating their contributions to total production.

**3.4.4    The Phytoplankton Assemblage**

The coexistence of a number of algal species is a common feature of freshwater phytoplankton community. In some cases, one species is found in much greater abundance than others; more often, two or more algal species are codominant in the phytoplanktonic assemblage. A large number of rarer species can always be found among the dominant or subdominant algae. The algae co-occur even though each species has a specific niche based on its physiological requirements and the constraints of the environment. Theoretically, niche overlap can lead to

competitive exclusion and should result in dominance by a single species. This problem has been formally referred to as the “paradox of the plankton”. A multispecific equilibrium occurs in physically uniform conditions of the turbulent open water of lakes.

A number of explanations have been advanced to account for species diversity in phytoplanktonic communities. First, in order for the relatively slow process of competitive exclusion to occur, the physical conditions must be uniform for a sufficient period of time. If environmental conditions change rapidly, the advantages gained by one species that is a better competitor may not exist long enough to result in the competitive exclusion of other species. Putting otherwise, differences in the efficiencies of resource utilization among species may be too small for competitive exclusion to occur before conditions change. Secondly, competition can be negated partially if commensalism and symbiotic relations exist among species. A number of organic compounds released extracellularly by one alga can influence the metabolism of another species, although most of these compounds are inhibitory in nature. Also, most algae requiring exogenous sources of vitamins and micronutrients are motile and small (Chrysophyceae, Cryptophyceae, Euglenophyceae and Dinophyceae). The differences in organic micronutrient requirements and the release of the excess by species capable of their synthesis would serve as means of encouraging mixed equilibrium populations.

A great deal of descriptive work has been devoted to the associations of algal species among differing freshwaters. Algae are extremely diverse and many exhibit a very wide tolerance to environmental conditions found under natural limnological situations. Nonetheless, certain characteristic phytoplanktonic associations occur repeatedly in lakes of increasing nutrient enrichment. Some of the commonly observed major associations are set out in Table 3.5. Such categorizations are not very satisfactory because of the wide spectrum of intergradations often observed. Additionally, shifts occur seasonally from one type to another, especially among more productive waters. Even though such characterizations yield little insight into regulating environmental factors, they are useful from the standpoint of general correlations between qualitative and quantitative abundance of the algae and available nutrients.

### **3.4.5 The Seasonal Succession of Phytoplankton**

Seasonal successions of phytoplankton frequently include correlations of changes in dominant algal species, biomass, numbers and rates of photosynthesis with alterations in physical and biotic factors. Briefly, the temperate pattern of phytoplankton succession involves a winter minimum of small flagellates adapted to low light and temperature, a spring burst of diatom activity and biomass, followed rapidly by a smaller development of green algae and a transitional lull between spring and summer. Summer populations vary in relation to the trophic status of the lakes but can include either another diatom development in less productive lakes by late summer and early autumn or increases in nitrogen-fixing cyanobacteria in eutrophic lakes.

Distinct seasonal patterns and periodicity in the biomass of phytoplankton are observed in polar and temperate freshwaters. Many detailed descriptions of phytoplankton succession have been correlated with changes in environmental parameters, particularly in temperature, light, nutrient availability and mortality factors such as grazing and parasitism. Because seasonal succession is strongly keyed to meteorological and stratification-mixing processes, patterns in temperate ecosystems differ considerably from those of tropical waters. The seasonal periodicity of phytoplanktonic biomass is reasonably constant over the year, if the freshwater ecosystem is not perturbed by outside influences, such as human modifications of the watershed, nutrient loading, etc. The extent of change in phytoplanktonic numbers and biomass through the seasons is usually very great, up to five orders of magnitude in temperate and polar freshwaters. Seasonal variations in tropical waters are much lower, often as little as fivefold.

**Table 3.5:** The characteristics of the common phytoplankton association in relation to the trophic status of freshwater bodies.

| <i>Trophic status</i> | <i>Water characteristics</i>                                                                          | <i>Dominant algae</i>                                                                                                                    | <i>Other common algae</i>                                                                                                              |
|-----------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Oligotrophic          | Slightly acidic; very low salinity                                                                    | Desmids <i>Staurodesmus</i> , <i>Staurostrum</i>                                                                                         | <i>Sphaerocystis</i> , <i>Gleocystis</i> , <i>Rhizosolenia</i> , <i>Asterionella</i> spp., <i>Melosira</i> spp., <i>Dinobryon</i> spp. |
| Oligotrophic          | Neutral to slightly alkaline; nutrient-poor lakes                                                     | Diatoms <i>Cyclotella</i> and <i>Tabellaria</i>                                                                                          | <i>Synura</i> and <i>Uroglena</i> ; diatom <i>Tabellaria</i>                                                                           |
| Oligotrophic          | Neutral to slightly alkaline; nutrient-poor or more productive lakes at seasons of nutrient reduction | Chrysophycean algae <i>Dinobryon</i> , <i>Mallomonas</i>                                                                                 |                                                                                                                                        |
| Oligotrophic          | Neutral to slightly alkaline; nutrient-poor lakes                                                     | Chlorococcal <i>Oocystis</i> or chrysophycean <i>Botryococcus</i>                                                                        | Oligotrophic diatoms                                                                                                                   |
| Oligotrophic          | Neutral to slightly alkaline; generally nutrient poor; common in shallow Arctic lakes                 | Dinoflagellates <i>Peridinium</i> and <i>Ceratium</i> spp.                                                                               | Small chrysophytes, cryptophytes and diatoms                                                                                           |
| Mesotrophic           | Neutral to slightly alkaline; annual dominants or in eutrophic lakes at certain seasons               | Dinoflagellates, some <i>Peridinium</i> and <i>Ceratium</i> spp.                                                                         | <i>Glenodinium</i> and many other algae                                                                                                |
| Eutrophic             | Usually alkaline lakes with nutrient enrichment                                                       | Diatoms <i>Asterionella</i> spp., <i>Fragilaria crotonensis</i> , <i>Synedra</i> , <i>Stephanodiscus</i> , and <i>Melosira granulata</i> | Greens and cyanobacteria during warmer periods; desmids at fairly high DOM                                                             |
| Eutrophic             | Usually alkaline; nutrient enriched; common in warmer periods of temperate or enriched tropical lakes | Cyanobacteria <i>Microcystis</i> , <i>Aphanizomenon</i> <i>Anabaena</i>                                                                  | Other cyanobacteria; euglenophytes if organically enriched or polluted                                                                 |

Source: Wetzel (2001).

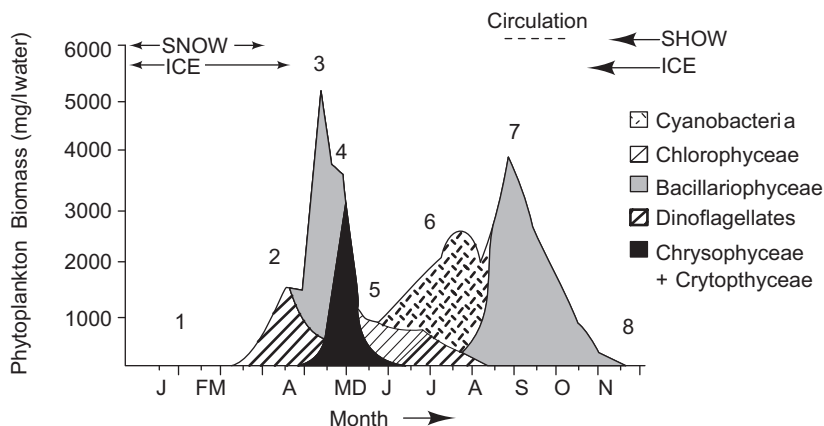


Maxima and minima of numbers and biomass of phytoplankton often are out of phase with rates of primary production. Primary productivity usually follows closely the annual cycle of incident solar radiation in temperate lakes. A periodicity of species composition continuously accompanies seasonal changes in total biomass. On an annual basis, algal communities consist of a composite of perennial species (*holoplanktonic*) that persist throughout the year in the plankton and of intermittent species (*meroplanktonic*) that periodically undergo some type of diapause in resting stages. In both cases, but much more markedly in the latter type, an interplay of environmental conditions results in fluctuations in growth and competition among other species. Among the perennial species, population numbers can decrease to extremely low levels, but cells are present in the water in sufficient numbers to re-inoculate the community when growth conditions improve and can grow to a bloom proportion if growth conditions are favourable.

**(a) Algal succession in tropical freshwaters:** Greater solar irradiance with reduced annual variations result in higher minimum water temperatures in tropical freshwater lakes and reservoirs but not in the rivers as the latter have a greater and faster water renewal. The temperature driven turbulence determines the successional sequence of phytoplankton in these water bodies. Although the successional sequence is essentially the same in tropical and temperate lakes, more successional episodes occur per year in tropical lakes because of highly distinct temperature regime. On an annual basis, the maximum of phytoplanktonic biomass in tropical lakes often is observed during the winter.

In many tropical lakes surrounded by extensive, nutrient-buffering wetlands and receiving perennial river inflow, total phytoplanktonic biomass and productivity are often both larger and more constant seasonally than those found in temperate lakes. Abrupt changes in abiotic factors can be more frequent in tropical lakes than in temperate lakes that are stratified for longer periods of time. Phytoplankton diversity in these water bodies is not markedly different from that in temperate lakes.

**(b) Algal succession in temperate freshwaters:** The seasonal succession of biomass of major phytoplankton groups in dimictic temperate and arctic lakes is generally separable into different periods. The low biomass of phytoplankton is dominated by ultraplankton, particularly by flagellates of the Chrysophyceae and Cryptophyceae. In winter Cryptophyceans, such as *Rhodomonas* and *Cryptomonas*, are particularly common, as well as pyrophyceans (*Gymnodinium*), small green algae (*Chlamydomonas*), chrysophyceans (*Dinobryon*, *Mallomonas*, *Chrysococcus* and *Synura*), some diatoms (*Synedra*, *Tabellaria* and *Fragilaria*) and euglenophyceans (*Trachelomonas*). The coldwater, low-light-adapted species are evidently quite species-specific in their tolerances to light quantity and quality and are found in narrow strata in the photic zone beneath the ice cover. A combination of good light availability and high nutrient concentrations, influxes of nutrients and other growth-promoting substances from non-frozen soils and very low grazing losses can result in algal blooms under ice in nutrient rich lakes.

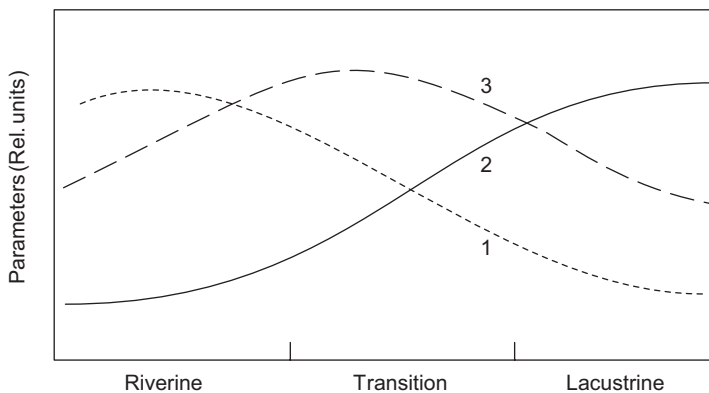


**Figure 3.4** - Model of biomass development and seasonal succession of major phytoplankton groups in a temperate lake of moderate productivity. The numbers refer to the seasonal periods (Adopted from Blomsqvist *et al.*, 1994).

Increase of temperature and disappearance of snow cover on the ice causes rapid increase of phytoplankton biomass, particularly among the dinoflagellates and small centric diatoms. Increasing light in the spring is the dominant factor contributing to the development of the spring “outburst,” during which, nutrient availability is generally high causing growth and increase in biomass, particularly by diatoms. Grazing by herbivorous zooplankton is still low, although in early stages of rapid growth, during this commonly brief period of circulation. In the early summer “clearwater” phase, algal populations decline precipitously under conditions of high temperatures, high water column stability, high light availability and markedly reduced nutrient availability. Herbivorous zooplanktons increase with the relative abundance of edible algae. Algae  $<40\ \mu\text{m}$  are particularly vulnerable and the grazing losses result in community filtration rates that exceed phytoplankton reproductive rates. This “clearwater” phase can result in accelerated microbial nutrient regeneration from zooplankton grazing and can persist for several weeks until less readily grazed and inedible algae develop to significant population sizes. Late summer stratification exhibits conditions of high temperatures, high water column stability but a progressive erosion and deepening of the metalimnion, high but declining light availability and low nutrient availability. The greatly reduced grazing pressures and increasing nutrient supply result in a more diverse phytoplanktonic community. Chrysophytes, cryptophytes and colonial green algae frequently become abundant and their growth often reduces availability of phosphorus and commonly also combined nitrogen to limiting concentrations. Dinoflagellates and cyanobacteria dominate when phosphorus and silica become limiting. Certain cyanobacterial species are recruited from the sediments, particularly during midsummer. In shallow lakes where inputs of nutrients, such as silica and/or combined nitrogen, from the sediments or external sources exceed microbial utilization, faster growing chlorophytes or diatoms can become superior competitors compared with the relatively slow-growing cyanobacteria.



**(c) Algal succession and productivity in reservoirs:** Within reservoirs the irregular dynamics of inflow and rapid, variable flushing rates markedly alter environmental conditions for biotic communities. A reservoir can be viewed as a very dynamic lake in which a significant portion of its volume possesses characteristics of running water and functions biologically as a river. Often, the riverine portion of reservoirs operates analogously to large, turbid rivers in which turbulence, sediment instability, high turbidity, reduced light penetration and other characteristics preclude extensive photosynthesis despite high nutrient availability (Fig. 3.5). Although the phytoplanktonic primary productivity of riverine sections of reservoirs can be high per unit water volume, the limited photic zone reduces areal productivity, as in large rivers. This reduction is only partially ameliorated by turbulent, intermittent recirculation of algae into the photic zone. As turbidity is reduced and the depth of the photic zone increases in the progression through the transitional to the lacustrine reservoir zones, areal primary productivity increases with the resulting greater light penetration and depth of the trophogenic zone (Fig. 3.5).



**Figure 3.5** - Commonly observed shift in the phytoplankton productivity and the sources of organic matter in the progression through the three major zones of reservoirs. Legends: 1 - Allochthonous organic carbon input, 2 - Autochthonous organic carbon input and 3 - Phytoplankton production (per  $\text{m}^3$ ) (Modified from Kimmel *et al.*, 1990).

The succession of phytoplankton communities of the lacustrine portion of reservoirs is generally similar to that found in natural lakes. Disturbance frequencies are much higher in reservoirs than in lakes, with rapid, often irregular and large changes in water level, flushing rates, turbidity and mixing. In particular, the disturbances to stratification and loadings of inflowing silt that increase turbidity are dominant factors affecting phytoplankton periodicity and succession. Green algae and planktonic diatoms are commonly dominant in summer during periods of short residence times and cyanobacteria develop more strongly when water residence times are longer, such as during dry seasons. As a result, the phytoplankton communities of reservoirs are disrupted often, which

leads to increased diversity with the development of r-strategist species whose development and successional sequences are constantly being reset.

### 3.4.6 Phytoplankton Among Aquatic Systems

Comparison of phytoplankton communities and their productivity among river, reservoir and lake ecosystems indicate a number of consistent traits. Phytoplankton diversity is very low in low-order streams but increases in high order large rivers (Table 3.6). Similarly, the diversity of phytoplankton is low in the riverine portions of reservoirs but often increases in the more stable lacustrine zones with longer retention periods. Phytoplankton diversity is distinctly higher in less productive lakes than in eutrophic waters. The biomass of phytoplankton is very low in low order streams usually because of light limitations from shading by terrestrial vegetation. Biomass can increase, despite frequent light limitation from abiotic turbidity, in larger rivers where residence times can often exceed those of the algal life cycles. The biomass of phytoplankton is usually low in riverine portions of reservoirs despite nutrient abundance because of frequent turbidity-related light limitations. Phytoplanktonic biomass fluctuates greatly in natural temperate lakes, usually by 5 or more orders of magnitude, on an annual basis, whereas biomass fluctuation in tropical lakes is much less variable (less than one order of magnitude).

The productivity of phytoplankton is low, regardless of the optimality of growing conditions, in comparison to the productivity of other photosynthetic organisms in aquatic ecosystems. The spatial distribution of phytoplankton in a dilute aqueous medium and conditions (limited light availability, sedimentation, diffusion limitations of nutrients and others) that limit productivity has been termed “aquatic desert”. Maximum rates of production of phytoplankton occur under conditions of moderate sustained nutrient availability in which the depth of the euphotic zone is large. The productivity of phytoplankton can dominate in the total productivity of some lakes where the phytoplankton themselves shade out other in-lake producer organisms. The collective productivity of phytoplankton per square meter, however, is low.

## 3.5 THE ZOOPLANKTON COMMUNITIES

Unlike a high biodiversity of large animals in freshwater ecosystems, both the phyletic representation and species diversity of planktonic animal (zooplankton) communities is very much lower in freshwater than in marine habitats. Planktonic animals are dominated by four major groups: (a) protists that include protozoa and heterotrophic flagellates, (b) the rotifers and two subclasses of the Crustacea, (c) the Cladocera and (d) the Copepoda. A few coelenterates, larval flatworms, gastrotrichs, mites and the larval stages of certain insects and fish occasionally occur among the true zooplankton, if only for a portion of their life cycles. However, excepting protists the other zooplankton are generally macroscopic and are not included in this section.

### 3.5.1 Protozoa and other Protists

Most protozoa and related protists have highly variable growth and generation times. Maximum growth rates are found when food availability is not a constraint. Under these conditions, populations generally increase directly with temperature within the limnological range and they decrease as body size increases. Many protists produce cysts under harsh environmental conditions.

The protistan zooplankton are the most important microbial consumers and have major functions in organic carbon utilization and nutrient recycling in aquatic systems. Although the instantaneous biomass of the planktonic protists is small in comparison to those of the rotifers and especially the microcrustaceans, the generation rates of protists are high. A few examples of population dynamics are presented here to evaluating the importance of these microbial communities to the utilization of organic matter and nutrient recycling in the freshwater environment.

**Table 3.6:** Comparative characteristics of phytoplankton and their productivity among river, reservoir and natural lake ecosystems.

| <i>Property</i>            | <i>Rivers</i>                                                                           | <i>Reservoirs</i>                                                                                                                                                                                             | <i>Natural lakes</i>                                                                                                                  |
|----------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Phytoplankton diversity    | Very low in low-order streams; increasing in high-order large rivers                    | Low in riverine zones; increasing in lacustrine zone                                                                                                                                                          | High diversity in oligotrophic lakes, low in eutrophic lakes; similar in tropical and temperate lakes                                 |
| Phytoplankton biomass      | Very low in low-order streams; increasing in large rivers, although often light-limited | Moderately high in response to high nutrient supply in riverine but less in lacustrine zone                                                                                                                   | Highly variable in temperate lakes seasonally; much less variable in tropical lakes                                                   |
| Phytoplankton productivity | Low, often light-limited by advective flows and turbidity                               | Highest in transitional zone; reduced in riverine zone by light availability and often in lacustrine zone by nutrients; marked horizontal gradients; volumetric productivity decreases from headwaters to dam | Low productivity; increasing with nutrient loading; declining at very high nutrient load; seasonal and vertical gradients predominate |

Compiled from Ryder and Pesendorfer (1989); Wetzel (1990).

**(a) Flagellates:** Flagellates are the most abundant component of the protozooplankton. Autotrophic flagellate species have already been discussed among the phytoplankton, although many are known to be mixotrophic and feed on bacteria as well as fix carbon photosynthetically. Common forms include the dinoflagellates (e.g., *Ceratium* and *Peridinium*), chrysomonads (*Dinobryon*,

*Mallomonas* and *Synura*), euglenids (*Euglena*), volvocids (*Volvox* and *Eudorina*), choanoflagellates (*Astrosiga*) and the diverse large group of heterotrophic flagellates. The heterotrophic groups include heterokont taxa (mainly chrysomonads and bicoseocids), choanoflagellates, kataphepharids and a number of species, collectively called *Protista incertae sedis*, that cannot be assigned to any of the major groups. The heterotrophic flagellates are differentiated into two general groups on the basis of size: (a) heterotrophic nanoflagellates (HNF) below 15  $\mu\text{m}$  and (b) large heterotrophic flagellates within a range of 15–200  $\mu\text{m}$ . Among both temperate and subtropical lakes, dinoflagellates tend to increase in dominance as the pH of the water decreases and can dominate in acidic lakes.

**(b) Ciliates:** Many major ciliate genera are represented in freshwater protozooplankton assemblages across the spectrum of trophic states. Of the three major groups, the oligotrichs, particularly *Strombidium* and *Halteria*, are found worldwide in lakes across the trophic spectrum. The tintinnid ciliates (order Choreotrichida: *Tintinnidium*, *Tintinnopsis* and *Codonella*) are also widely distributed in temperate to tropical regions. Haptorid ciliates (e.g., *Askenasia* and *Mesodinium*) are similarly distributed broadly and abundantly. Although a few ciliates are mixotrophic and supplement nutrition by photosynthesis, most are holozoic and feed on bacteria, algae, particulate detritus and other protists. A few are carnivorous and feed on small metazoans. Ciliates tend to be more significant components of the zooplankton of eutrophic lakes.

**(c) Sarcodines:** Sarcodine protozoa are poorly represented in the zooplankton of freshwaters. Even in eutrophic lakes, sarcodines occurred in modest average abundance. Heliozoans are restricted vertically largely to the epilimnion and metalimnion of stratified lakes and show great seasonal variations. Seasonal mean heliozoan production is, however, equivalent to only about 1% of the combined ciliate and flagellate production in many lakes. Testate amoebae, particularly *Diffugia*, are occasionally abundant.

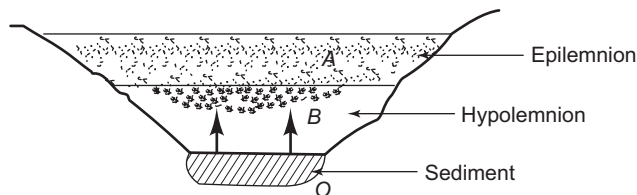
### 3.5.2 The Distribution of Microzooplankton

Although nearly all protozoa and related protists are aerobic, a few can grow very well even when oxygen concentrations are low. This microaerophilic ability is conspicuous among the planktonic and benthic ciliates, which are abundant in organic rich and organically polluted waters. Various species combinations have therefore become useful as indicators of waters of different degrees of organic stress and pollution. Populations of ciliates often develop in water strata greatly reduced in or devoid of oxygen in which bacterial populations tend to be dense, such as in the lower portions of the hypolimnia of polysaprobic lakes.

Four distinct types of ciliated protozoa can occur and be differentially distributed vertically in stratified productive lakes, viz., (1) Specialized anaerobic ciliates (e.g., *Saprodinium*) inhabit anaerobic hypolimnia, utilize particulate organic particles and have no photosynthetic symbionts, but they do have symbiotic methanogens, which probably remove  $\text{H}_2$  as  $\text{CH}_4$ . (2) Microaerophilic

ciliates (e.g., *Loxodes*) are free of symbionts, inhabit dark and microaerobic strata at the metalimnion/hypolimnion interface and move into lower anaerobic water, where they respire  $\text{NO}_3^-$ , when the microaerobic zone is illuminated. (3) A third group of ciliated protozoa (e.g., *Frontonia*) lives in the illuminated microaerobic zone of the metalimnion of eutrophic lakes where it benefits from upward diffusion of  $\text{CO}_2$  and  $\text{NH}_4^+$  that is fixed by its zoochlorellae symbionts. (4) A final group of epilimnetic ciliates (e.g., *Strombidium*) is planktonic and periodically sequesters symbiotic chloroplasts from ingested algae. Distribution of each of these ciliate types is controlled by their biochemical adaptation to anaerobic conditions, the requirements of their symbionts and their physiological responses to light, oxygen and gravity. Grazing pressure on anaerobic and microaerophilic ciliates by metazoa has been found to be very low or insignificant.

Within lakes that stratify thermally and are moderately productive, two distinct protozooplanktonic communities develop during summer stratification. An epilimnetic protistan community of obligate planktonic species overlies a hypolimnetic community of migrants from the benthic sediments (Fig. 3.6). Migration of benthic protozoan communities from the sediments into the deeper strata of the hypolimnion occur seasonally in response to reduction of dissolved oxygen. For example, two dominant species, *Loxodes* and *Spirostomum*, position their populations close to the 1 mg/l oxygen isopleth. Oxygen is toxic to *Loxodes* and its effect is exacerbated by light. There is additional evidence that *Loxodes* can respire nitrate as well as oxygen. Protozoan utilization of planktonic bacteria constitute a major pathway of energy flux among the fauna in the pelagic zone of deep water lakes. Bacterial abundances for protozoan feeding tend to be high at the metalimnetic-hypolimnetic interface.



**Figure 3.6** - The distribution pattern of protozooplankton during stratification in a eutrophic water body. (A) Obligate epilimnetic population; (B) Hyperlimnetic community of benthic migrants.

Mixotrophic ciliates and flagellates are typically distributed within the upper water column in relation to the depth distribution of light. The euphotic zone is shallower among eutrophic lakes, often  $< 5$  m, but extends much deeper in less productive lakes. Some flagellates, such as *Ceratium*, can migrate extensively to position themselves on a diurnal basis in relation to light, nutrient and oxic conditions. Heterotrophic flagellates migrate extensively and often reach very high densities ( $> 140,000/\text{ml}$ ) in anoxic hypolimnia of eutrophic lakes that contain abundant bacterial communities. Abundance of heterotrophic flagellates in most

natural plankton assemblages range from  $10^2$  to  $10^6$  per liter and occur in much greater abundance, by about 3 orders of magnitude, than the ciliates (Table 3.7).

**Table 3.7:** Representative abundance of planktonic heterotrophic flagellates and ciliates in freshwaters.

| <i>Trophic status of water body</i> | <i>Flagellates × 1000/ml</i> | <i>Ciliates / ml</i> |
|-------------------------------------|------------------------------|----------------------|
| Oligotrophic                        | 0.5 – 5.1                    | 1 – 6                |
| Mesotrophic                         | 1 – 16                       | 5 – 20               |
| Eutrophic                           | 10 – 59                      | 20 – 300             |
| Hypereutrophic                      | 30 – 400                     | 155 – 1500           |

Compiled from Bark and Watts (1984); Berninger *et al.* (1991); Hadas and Berman (1998).

**3.5.3    Seasonal Dynamics of Microzooplankton**

Seasonal variations in flagellate and ciliate abundance are variable. Cryptomonad and other microflagellates are commonly numerically dominant across a spectrum of oligotrophic through hypereutrophic lakes. Having high nutritional quality, short turnover times and ability to grow and reproduce at low intensities of light, these microorganisms function to stabilize variations in nutrient recycling in many spatial and seasonally temporal niches within lakes that are ineffectively utilized by other microbial and small metazoan organisms. Predation likely becomes increasingly important in the regulation of the abundance and community composition of protozoans and related protists as lake and river ecosystems become more productive. Where required resources for growth are abundant, the protozoan community is increasingly regulated by predation and correlated with the biomass and species composition of the metazooplankton.

Generally, ciliate abundance is coupled to maximum abundance of phytoplankton in the spring and early summer months. A secondary maximum in the autumn is frequently found in eutrophic and mesotrophic lakes. Species successions over the annual cycle indicate that tintinnids are common spring forms in eutrophic and mesotrophic temperate lakes. The tintinnids often decrease during the summer and concentrate in the metalimnion and later in the autumn increase again in a secondary maximum of lesser magnitude than in the spring. The abundance and seasonal succession patterns of microflagellate communities are controlled both by food resources, particularly in oligotrophic waters and by predatory losses. Ciliate predation on flagellates is a major regulatory mechanism, especially in more productive waters.

Among dystrophic lakes that receive very high concentrations of allochthonous dissolved organic matter, microaggregates of particulate detrital material are abundant in the plankton. These aggregates can form foci for bacteria and microflagellates and can likely serve as a significant supplemental food resource, along with bacteria and microalgae, for both microflagellates and ciliated protozoa. The highly stained waters can attenuate light severely and restrict mixotrophy among protists. Summer protozooplankton in eutrophic and

mesotrophic lakes is dominated by *Strombidium* spp., vorticellids and *Epistylis*. Mixotrophic protozoa predominate in the summer among all lake types but are particularly common in oligotrophic waters. Both heterotrophic and autotrophic flagellates commonly reach maximum abundances in midsummer.

Among tropical and subtropical monomictic lakes that are generally productive and have high average temperatures, ciliate abundance persist at high levels for longer periods of time and often exhibit a bimodal pattern (mesotrophic) or late summer-autumnal maximum. Some of the increase may be associated with increased nutrient loading and primary productivity during and following mixing in the autumn. Subtropical lakes maintain higher densities of protozooplankton than do lakes of comparable trophic status in the temperate zone. This increase in productivity is likely related to the longer growing season and higher temperatures, which allow a sustained productivity during the winter.

### 3.5.4 The Protistan Grazing on Bacterioplankton

Bacterioplankton are significantly grazed by small ( $< 5 - 8 \mu\text{m}$ ) phagotrophic nonpigmented flagellates. These flagellates often dominate among the heterotrophic planktonic protists. Larger planktonic ciliates can also be important in bacterivory and exhibit high rates of bacterial clearance from water under both natural and experimental conditions (Table 3.8). Phagotrophic phytoflagellates also ingest bacteria at rates comparable to those of nonpigmented flagellates.

**Table 3.8:** The removal rate of bacterioplankton and protozoa on grazing by protists.

| Grazing relation                | Grazing rate<br>(individuals/protist. hr) |
|---------------------------------|-------------------------------------------|
| Flagellates on bacterioplankton | 2 – 53                                    |
| Ciliates on bacterioplankton    | 6 – 375                                   |
| Ciliates on flagellates         | 1 – 42                                    |
| Rotifers on ciliates            | 32 – 259                                  |
| Rotifers on flagellates         | 0.5 – 4.8                                 |
| Copepods on flagellates         | 35 – 200                                  |

Compiled from Sommaruga and Psenner (1995); Carrias *et al.* (1996); Jack and Gilbert (1997); Hadas *et al.* (1998).

Bacterivory by flagellates and ciliates is influenced by temperature as well as the availability of bacterial prey. For example, at  $15^{\circ}\text{C}$ , ciliates filtered 10 times more body volume when bacteria were scarce, but the ingested bacteria were fewer than at high prey densities. Size and shape of prey is important and clearly prey geometry is the first-order determinant of ingestion through passive mechanical selection by protists. Among both flagellates and ciliates, but particularly among the heterotrophic microflagellates, there is a tendency to ingest larger bacterial cells at a frequency greater than they occur in the bacterial assemblage and some probability exists for chemical discrimination among prey. This size-selective grazing by microflagellates likely contributes to the frequently



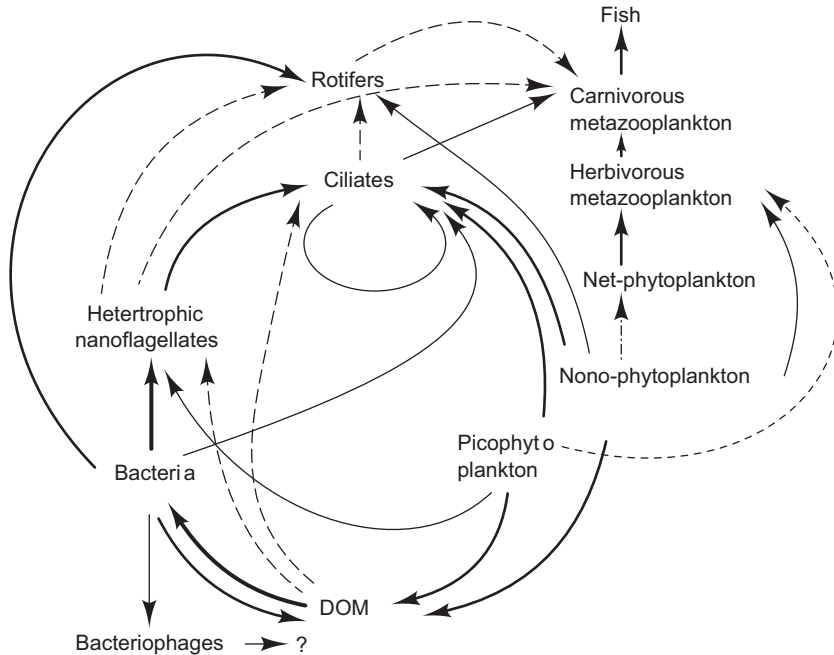
observed decrease in median bacterial cell volume during the summer period of temperate waters.

Even though the rates of bacterivory by flagellates are smaller than those by ciliates, flagellate numbers are so much greater that their collective grazing effects on bacterioplankton can be much greater than those of the ciliates. Phagotrophic flagellates are often most abundant in the epilimnion and metalimnion interface during stratification; flagellate ingestion rates tend to be greatest in the epilimnion. Ciliates are responsible for as much as 30% of the bacterivory at some depths in spring and summer, while bacterivorous rotifers are abundant only in summer. The grazing impact of cladoceran crustaceans is generally  $< 1\%$  in tropical lakes.

### 3.5.5 Trophic Relationships of Protists in Pelagic Food Webs

The functional roles of microbiota in pelagic food webs has received extraordinary attention in the past decade. This emphasis is important because clearly a major portion of the pelagic metabolism is occurring within a microbial food web. Before the importance of these microorganisms is evaluated as a support component of the metazoan food web, it is essential to examine the primary characteristics of these pelagic microbial communities.

**(a) Microbial food web:** The microbial food web concept was developed in marine studies long ago with the position that heterotrophic and phototrophic microbial metabolism constituted a significant portion of the total metabolism of the ocean. With many data from detailed studies of a lake ecosystem, Wetzel *et al.* (1972) developed a general paradigm which stated that bacteria metabolize a significant fraction of total photosynthetic production in natural waters. In particular, that work demonstrated that most of the microbial utilization of primary production occurred in littoral and benthic regions and that only a small portion of the ecosystem primary production entered the metazoan trophic levels. Many subsequent studies provided data to reinforce the importance of microbes, synthesized as the concept of the “microbial loop”. The latter is simply a model of the pathways of carbon and nutrient cycling through microbial components of pelagic aquatic communities. In addition to bacterial uptake of nonliving organic matter, many direct links exist among algae, bacteria and other heterotrophic microbes (Fig. 3.7). By these numerous pathways, fixed organic carbon can be recovered, often into larger-sized microorganisms that may be more available for consumption by larger organisms. Metazoans (metazooplankton to fish) would utilize organisms of the microbial food web by grazing phytoplankton, colourless flagellates and ciliates  $> 5 \mu\text{m}$ . In oligotrophic waters that may be dominated by phytoplankton and too small microbes ( $< 5 \mu\text{m}$ ) to be effectively consumed by many zooplankton, the phagotrophic flagellates and ciliates could function as a primary trophic link within the ecosystem.

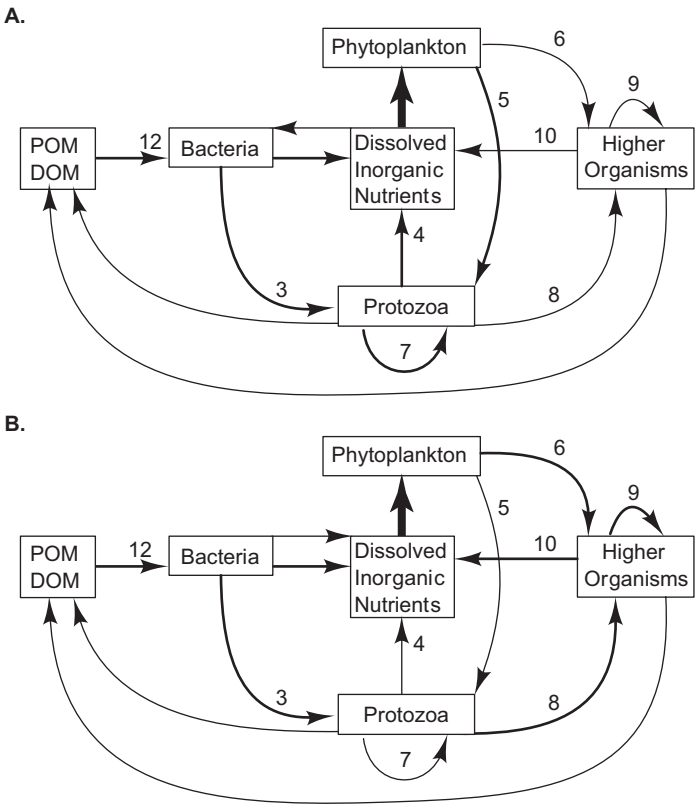


**Figure 3.7** - A microbial food web in aquatic ecosystem. The proportionate relationships and matter transfer have been presented with different thickness of arrows, which shows the relative importance of the individual process in the food web (Modified from Gunkel, 1994).

Under most prevailing conditions, significant fractions, at least 50%, of the dissolved organic matter and nonliving particulate organic materials are metabolized to gases, soluble nutrients and living bacterial biomass. This bacterial biomass may be living, dormant, senescing and utilized by other bacteria and viruses or may be consumed by bacterivores. The predominant bacterivores are protists, often microflagellates and protozoan ciliates. For example, pelagic bacterivory in eutrophic Lake Oglethorpe, Georgia, was dominated at all times by heterotrophic flagellates, which accounted for 49-81% of grazing on an areal basis and up to 98% of grazing at some depths at certain times of the year (Sanders *et al.*, 1989). Mixotrophic flagellates were major grazers during winter and spring and contributed up to a maximum of 45% of community grazing on an areal basis and to 79% at depths of maximum abundance. In late spring and early summer, ciliate bacterivory constituted as much as 30% at some depths but averaged ca. 11% over the year. Bacterivory by pelagic cladoceran crustaceans was generally < 1 % of the total and was not detected among copepods.

The bacterivores can also function as decomposers and efficient nutrient remineralizers (Fig. 3-8A) or transfer some of the fixed carbon and nutrients to higher organisms (Fig. 3-8B). In the prevailing conditions, most of the carbon and nutrients in bacterial substrates are remineralized directly by bacteria or their protistan consumers. In this case, which is likely often the case, little carbon and

nutrients are available to higher organisms as living biomass; hence, the microbial loop functions as both a carbon and a nutrient “sink,” that is, the carbon is not directly available to higher animals. However, the carbon and nutrients are not usually lost from the ecosystem but are rather displaced spatially and temporally and recycled later. The size structure, species composition and possibly the rate of production of the phytoplankton may determine the amount of the composite assemblage that is grazed by the larger zooplankton or the protozoa/protists. Much complexity within the protistan assemblage would decrease the quantitative transfer of carbon and nutrients to higher organisms.



**Figure 3.8** - A flow chart of nutrient cycling through microbial loop of pelagic communities. The thickness of arrows represents the relative importance of the pathway for nutrient flow. (A) High efficiency remineralization; (B) High efficiency of transfer to higher organisms. The numbers are given for comparison of the two pathways. (Modified from Caron, 1991).

In a stronger “link” to the metazoan hypothesis (Fig. 3-8B), carbon and nutrients in organic substrates are utilized by bacteria and incorporated into their consumers. As these organisms are utilized by higher organisms, the pathway represents a direct trophic coupling (“link”) between dissolved and particulate detrital matter and higher organisms. The large zooplankton then would dominate

carbon and nutrient cycling and remineralization by the animal microbial loop components would be less. This predation by large zooplankton can impose major regulation of carbon and nutrient fluxes to higher trophic levels.

### 3.6 THE AQUATIC FUNGI

The aquatic fungal species mostly belong to mastigomycotina, known to be present in water either as saprophytes or parasites. These fungi occur in freshwater or saline water ecosystems. They are found in streams, lakes, rivers, temporary pools, puddles and semi-aquatic habitats. They often show fluffy growth on submerged leaves, twigs, fruits, dead animals, fish eggs besides being parasitic on living fish and phytoplankton. The thallus of such fungi is either holocarpic or eucarpic. They reproduce by three kinds of zoospores; with a posterior flagellum of whiplash type, an anterior flagellum of tinsel type and biflagellate zoospores with apically or laterally attached both type of flagella. Sexual reproduction in such fungi is of planogametic copulation, gametangial contact or gametangial copulation.

The aquatic fungi can be collected as parasites on algae and also by baiting the water with cellophane, onion strips, insect exuviae, may flies, hair, midges, pine pollen, etc. Members of saprolegniales can be isolated by baiting house flies, fruit flies, ant's eggs, hemp seeds, maize grains, seeds of *Brassica*, grasses, *Crotalaria* and wheat. They also occur on submerged plant and animal debris, especially on dead fish besides being parasitic on fish and other biota. Fungi belonging to Blastocladales are also not uncommon in freshwater ecosystems.

#### 3.6.1 Aquatic Conidial Fungi

The leaves which fall into clean and well aerated streams and other water bodies are rapidly replaced by a characteristic mycoflora of fungi-imperfecti, the aquatic hyphomycetes. These fungi grow on all sorts of leaves of deciduous trees and shrubs but have not been recorded on needles of conifers. They also occur in foam or scum formed as barriers to water flow. However many of them are not taxonomically well known as our knowledge on the taxonomy and ecology of conidial fungi is far from satisfaction. More than 100 species belonging to 60 genera have been recognized so far but the diversity is expected to be 2 – 3 orders of magnitude higher. Reliable but exhaustive quantitative and qualitative data is essential for proper understanding of the taxonomy, morphology, ecology and physiology of water-borne conidial fungi. The fungi isolated from freshwater habitats by various workers have been listed in Table 3.9.

#### 3.6.2 Fungal Involvement in Remineralization Process

The aquatic fungi represent the major microbial element in the decay of leaves. Increasing attention has been given to the role of these fungi in processing of leaf material, whose degradation is probably not possible without fungal involvement. Normal brown coloured leaves associated with aerobic decay and softened tissues bore specially rich crops of fungi. The branched septate mycelium ramifies into

the tissues and separates mesophyll from the veins leading to complete degradation of leaf. There are evidences that these fungi act as intermediaries of energy flow in aquatic ecosystems. The leaves colonized and degraded by fungi are palatable to aquatic invertebrates as well as become susceptible to bacterial digestion. It is also possible that these fungi may simply ensure a supply of microbial substances by preferentially feeding on detritus, which is already colonized by fungi. It is also clear that fungi, as a group, have a definite edge over bacteria in initiating the breakdown of solid plant material because of their superior enzymatic equipment and their filamentous growth allows them to penetrate deep into solid substrates. The growth of some fungi can double the protein content of some leaf species in a short time. It can be assumed that fungi convert their substrates into their own biomass with an efficiency of 40-70%. These fungi play a key role in unlocking and distributing a major source of energy in streams.

**Table 3.9:** The commonly occurring species of fungi in freshwater habitat.

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*Actinospora megalospor*, *Alatospora acuminata*, *Anguillospora longissima*, *Articulospora tetracladia*, *Brachiosphaera* sp., *Beltrania rhombica*, *Camposporium pallucidum*, *Campylospora chaetocladia*, *C. filicladia*, *C. parvula*, *Chaetospermum haetosporum*, *Clavariopsis aquatica*, *C. brachycladia*, *Clavatospora longibrachiata*, *C. tentacula*, *Candylospora spumigena*, *Dactylella aquatica*, *Diplocladiella scalaroides*, *Emtomophthora* sp. *Flabellospora crassa*, *F. multiradiata*, *F. tetracladia*, *F. penicillioides*, *F. curvula*, *Geniculospora inflata*, *Helicomycetes* sp., *Helisporium* sp., *Helisus lugdunensis*, *H. submerses*, *Henicospora coronata*, *Ingoldiella hamata*, *Laridospora appendiculata*, *Lemonniera aquatica*, *Lunulospora curvula*, *L. cymbiformis*, *Nawawia iliformis*, *Obstipispora* sp., *Phalangispora constricta*, *Robillarda* sp., *Speiropsis* sp., *Speiropsis pedatospora*, *Tetracladium marchalianum*, *T. setigerum*, *Tetraploa aristata*, *Tricladium splendens*, *Tricladium* sp., *Tricellula aquatica*, *Tripopermium camelopardus*, *T. myrti*

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Source: Manoharachary (2005).

Fungi also represent a form of nourishment, which from the animals point of view is more than ten times as concentrated as leaf substances. Fungal growth on leaves improves their food value to the invertebrate stream fauna and facilitates the digestion. This activity is intimately connected with the amount of oxygen present in water, litter amount, pH, low temperature and other physicochemical parameters. Addition of a nitrogenous source results in an increase in the absolute quantity of nitrogen in leaves, as microbial protein. A change in the supply of phosphorus alters the quantity of immobilized nitrogen while the temperature affects the role of immobilization. Initially fungi seem more important than the bacteria in leaf breakdown, but with the progress of degradation the fungal flora becomes less important as the metabolic efficiency of bacteria in utilizing the simple organic matter is much higher than of fungi.

### 3.7 THE BACTERIOPLANKTON IN FRESHWATER ECOSYSTEMS

Biochemical transformation of particulate and dissolved detrital organic matter by bacteria and fungi are fundamental to the structure and dynamics of nutrient cycling and energy fluxes within aquatic ecosystems. The functioning and stability of aquatic ecosystems is governed by the metabolic transformation of organic matter by bacteria and fungi, who metabolize most of organic detritus. However, much of the organic matter synthesized within the lake or brought to it from the drainage basin is not decomposed in the bulk liquid, but instead is decomposed on or in the sediments. The biochemical transformations of organic matter displaced to the sediments are crucial to the operation of freshwater ecosystems and consequently deserve separate treatment (See Section 3.8).

#### 3.7.1 Distribution of Bacterioplankton

Bacterioplankton are often limited by the availability of organic substrates and nutrients, particularly phosphorus. Consequently, one would expect a general increase in bacterial biomass as an aquatic system is increasingly loaded with organic substrates. In general, the numbers and biomass of bacteria increase with increasing productivity and concentrations of inorganic and organic compounds in lakes (Table 3.10). In spite of great seasonal variations, bacterial numbers and biomass increase from oligotrophic to eutrophic inland waters. The highest levels have been observed in tropical alkaline, saline lakes and in eutrophic reservoirs. Numbers of bacteria are markedly lower in acidic dystrophic lakes, which contain high concentrations of humic matter. Similarly growth of bacteria increases with increasing phytoplanktonic productivity of lakes. The average generation time of bacteria decreases in more eutrophic lakes. A positive relationship is observed between the average rates of phytoplanktonic productivity and bacterial numbers.

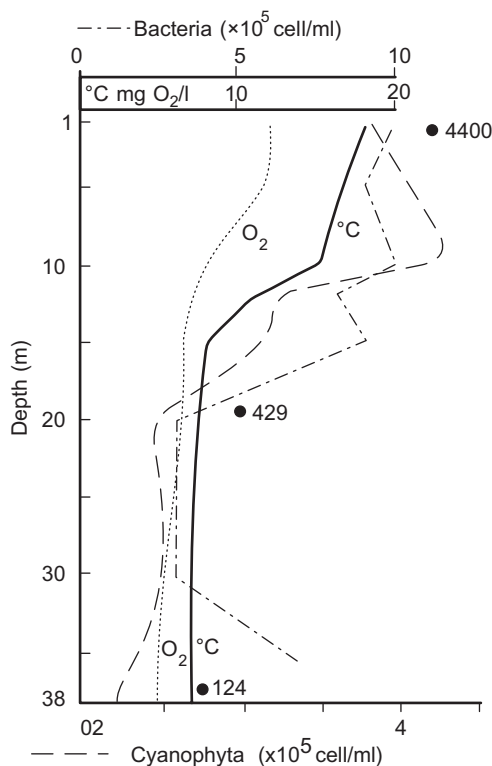
**Table 3.10:** The numbers, volumes and biomass of bacterioplankton in freshwaters of different productivity.

| <i>Trophic state</i> | <i>Number</i><br>( $10^6/\text{ml}$ ) | <i>Biovolume</i><br>( $\mu\text{m}^3$ ) | <i>Biomass</i><br>( $\text{mg/l}$ ) |
|----------------------|---------------------------------------|-----------------------------------------|-------------------------------------|
| Oligotrophic         | 0.5                                   | 0.2 – 0.4                               | 0.15                                |
| Mesotrophic          | 1.0                                   | 0.4 – 1.2                               | 0.7                                 |
| Eutrophic            | 3.7                                   | 1.5 – 2.9                               | 2.3                                 |
| Dystrophic           | 1.2                                   | 0.1 – 0.5                               | 0.4                                 |

Modified from Saunders *et al.* (1980).

The seasonal distribution of bacterial populations is highly variable from lake to lake and from year-to-year within a lake. A few generalizations, however, can be advanced even from population data determined with sampling frequencies that are far greater than the generation times of the bacteria. Although direct

correlations are often observed between the numbers of phytoplanktonic algae and heterotrophic bacteria both seasonally and vertically with depth in lakes, this does not hold true for rivers. Bacterial numbers and biomass are commonly high in the epilimnion, decrease to a minimum in the metalimnion and upper hypolimnion and increase further in the lower hypolimnion, especially if anoxic. Vertical profiles of bacterioplankton and phytoplankton are usually more strongly correlated than they are seasonally (Fig. 3.9).



**Figure 3.9** - The vertical distribution of heterotrophic bacteria in relation to planktonic algae. The solid circles are the population of saprophytes ( $10^5$  cells/ml) (Adopted from Wetzel, 2001).

The heterotrophic metabolism of the bacteria depends to a significant extent on organic substrates produced by the phytoplankton. Often, increase in bacterial populations is delayed and follow phytoplankton maxima by many days or weeks. Correlations have been found between pulses of bacteria and changes in sources of organic matter, such as death and decomposition of phytoplankton and of littoral plants, as well as changes in precipitation and runoff of organic matter from allochthonous sources, nutrients, temperature, mixing, parasitism and grazing by protists and zooplankton. Planktonic bacterial biomass and production is generally lower during winter than during summer. This relationship is correlated with low winter temperatures and reduced loading of particulate and



dissolved organic substrates from autochthonous (phytoplankton and littoral plants) and allochthonous sources. A significant portion of the planktonic bacterial community may be physiologically dormant when conditions are cold and limited organic substrates are available.

The limitation of bacterioplankton density mostly results from mortality of bacteria from viral lysis, often exceeding 50% and from grazing by protistan protozooplankton. Viral lysis and grazing activities by zooplankton on both algae and bacteria enhance the regenerated supply of dissolved organic substrates and subsequent growth of bacteria. During the summer, the same combination of biotic and abiotic factors influence bacterial growth rates and production, but usually in a more balanced manner. Gradually, as the seasons progress toward winter, declining temperatures and organic carbon substrates reduce production of bacterioplankton.

Controversy prevails concerning whether bacterioplankton are freely planktonic or attached to suspended inorganic or organic detrital particulate matter. Often most bacteria are observed to be freely planktonic, particularly in oligotrophic waters. However, bacteria are also often found to be predominantly attached to detrital particulate matter, but marked seasonal variations are observed. For example, in lakes or reservoirs with high, persistent in-lake suspended clay-organic aggregates, < 20% of the bacterioplankton may be free in the water. Microbial attachment to particles may be in response to enhanced availability of adsorbed organic substrates relative to substrate concentrations existing in the water.

Horizontal spatial variability in biomass and production of bacterioplankton is usually low in the central pelagic regions of lakes that are not morphometrically complex or disturbed by high effluent loading from point sources. The bacterial production increases markedly along a gradient from the central portions of the basins toward littoral areas. Loading of both nutrients and organic substrates from the land and littoral producers certainly enhances bacterial productivity. Patchiness in bacterial distribution is found due to differences in organic loading from human effluents. Sinking rates of individual bacteria are so low (approximately 1 mm/day) that horizontal differences are usually coupled to differences in water turbulence and variations in availability of nutrients and organic compounds (e.g., phytoplankton, littoral sources and influent sources of organic substrates).

### **3.7.2 Comparison of Bacterioplankton among Freshwater Ecosystems**

A comparative account of the bacterioplankton among rivers, lakes and reservoirs has been presented in Table 3.11. In all the systems the abundance and productivity of bacterioplankton are strongly coupled to the availability of energy-rich, readily hydrolysable organic substrates and nutrients, all of which are highly dynamic and variable among all aquatic ecosystems. Population control by protistan grazing and viral lysis is common to all freshwater bodies.

However, anthropogenic interference has been found to be more in reservoirs than in lakes resulting in some variation in the bacterioplankton distribution and productivity.

### 3.8 MICROBIAL ACTIVITIES IN SEDIMENT

A conspicuous feature of microbial populations in lakes is the great increase in numbers in the transition from the overlying water to the diffuse, uncompacted zone of the surficial sediments (Fig. 3.10). Bacteria increase about 3-5 orders of magnitude from the water to the surface sediments and decrease rapidly with increasing depth in the sediments. Expressed in numbers per gram dry weight of sediment, bacterial populations in the surficial sediments can reach 6-7 orders of magnitude greater than in an equivalent weight of overlying water. Saprophytic bacteria decrease much more rapidly than do total bacteria with increasing depth below the sediment-water interface, which suggests a reduction of readily assimilable organic substrates below the interface.

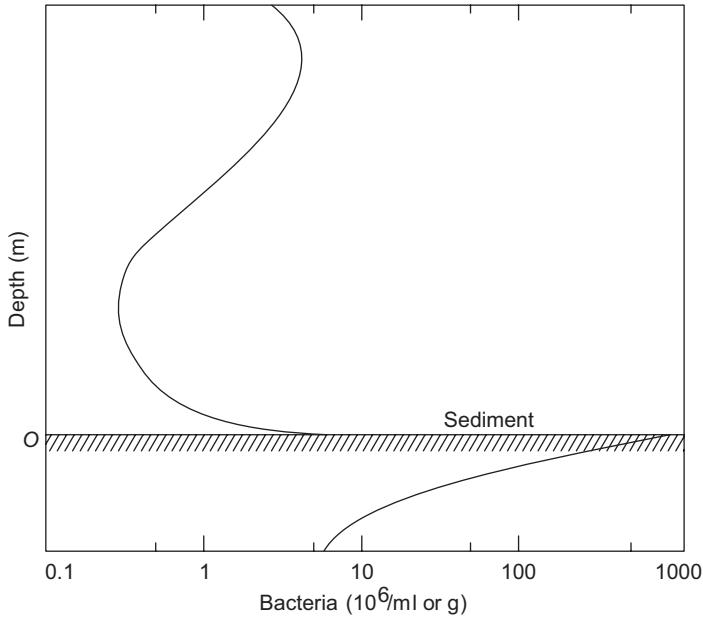
**Table 3.11:** Comparison of characteristics of bacterioplankton among rivers, lakes and reservoirs.

| <i>Characteristics of bacterioplankton</i> | <i>Rivers</i>                                                                                                                    | <i>Reservoirs</i>                                                                                            | <i>Natural lakes</i>                                                                                            |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Abundance                                  | High to very high depending on sources of organic substrates and nutrients                                                       | Moderate to high; greater proportion of bacteria attached to seston particles                                | Low; increasing with nutrient loading and productivity of phytoplankton                                         |
| Productivity                               | Low and increasing with stream order; heavy reliance on allochthonous dissolved organic matter and floodplain primary production | Low to moderate; often particle associated; commonly higher in riverine portions than in lacustrine sections | Low; greater rates in epilimnion, decreasing with depth; relatively small seasonal changes because of mortality |
| Respiration                                | High                                                                                                                             | Moderate                                                                                                     | Low                                                                                                             |
| Mortality                                  | High, from protistan predation, viral parasitism, UV photolysis and molar action                                                 | Moderate, largely from protistan predation and viral parasitism                                              | High; from protistan (nanoflagellate) predation (50 %), rest from viral parasitism, autolysis and photolysis    |

Source: Wetzel (1996).

Bacteria within the surface sediments are unevenly distributed over the lake basin as well. When the littoral zone is covered with a well-developed macrophyte community, bacterial numbers of littoral sediments are higher by

several orders of magnitude than those found in profundal sediments of deepwater areas. Numbers of bacteria in sandy, organic-poor, wave swept littoral sediments are lower than those in profundal sediments. Aerobic respiration and nitrate reduction in the sediments of the littoral are greater than those of the profundal zone, whereas methanogenesis and sulphate reduction are greater in the profundal than in sediments of the littoral zone.



**Figure 3.10** - The vertical variation of bacteria population in water and sediments of a moderately productive lake.

A more detailed analysis of bacterial concentrations in sediments within the macrovegetated littoral zone as compared with those of sediments free of vegetation showed similar relationships (Table 3.12). Nitrogen-fixing bacteria *Azotobacter* and the obligate anaerobe *Clostridium* are more abundant in sediments with macrovegetation than in sediments without vegetation or in the overlying water. Similar relationships are found among populations of hydrocarbon- and hydrogen-oxidizing bacteria, cellulose-decomposing bacteria and actinomycetes and fungi. Large numbers of methane-forming and sulphate-reducing bacteria develop under anaerobic conditions in sediments that contain large quantities of decomposition products of higher aquatic plants.

Bacterial numbers and metabolic activities of the deepwater profundal sediments show a weak correspondence with the productivity of the lakes, as one would anticipate the loading of organic matter and inorganic nutrients bound within the seston to be approximately proportional to lake productivity. Bacterial numbers in the open water tend to undergo rapid fluctuations and because of nutrient limitations and mortality from grazing and parasitism do not correlate

well with lake productivity. In extremely oligotrophic lakes, total number of bacteria of the sediments can be less than the cumulative total of the entire overlying water, which indicates that in these lakes most of the readily decomposable organic matter has been mineralized before the particulate residues reach the sediments. Hence, strong positive correlations between bacterial abundance and organic matter in pelagic sediments would not be anticipated, but certainly increases in littoral areas.

**Table 3.12:** Comparison of microbial populations ( $\times 1000/\text{g}$ ) in sediments of water bodies with and without emergent vegetation.

| <i>Microbial group</i>  | <i>With<br/>emergent<br/>vegetation</i> | <i>Without<br/>emergent<br/>vegetation</i> |
|-------------------------|-----------------------------------------|--------------------------------------------|
| Total saprophytes       | 5300                                    | 3100                                       |
| Actinomycetes           | 800                                     | 80                                         |
| Fungi                   | 300                                     | 240                                        |
| <i>Clostridium</i> spp. | 100                                     | 40                                         |
| Hydrogen oxidizers      | 10                                      | 1                                          |
| Methanogens             | 1500                                    | 140                                        |
| Sulphate reducers       | 50                                      | 1300                                       |

Modified from Wetzel (2001).

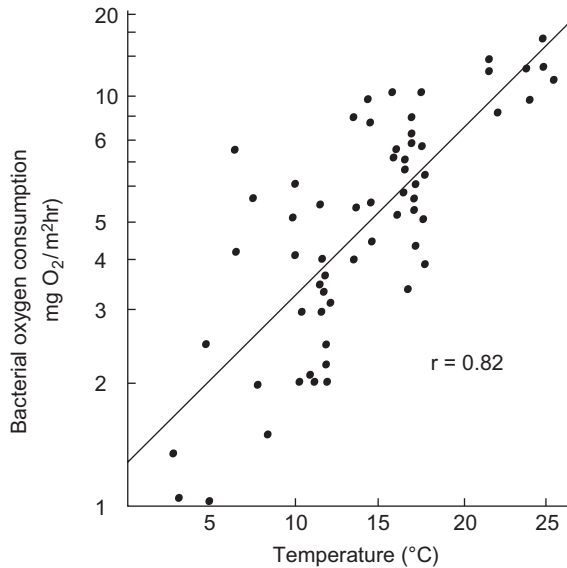
### 3.8.1 Aerobic Bacterial Activity of Sediments

The relationship of increased bacterial populations and metabolic activity to greater organic matter in sediments may appear obvious, but *in situ* measurements are very few. A significant correlation between dehydrogenase activity and organic content of surface sediments has been found in eutrophic reservoirs. These correlations, however, would not be expected to hold in organic-rich sediments in which other conditions, such as acidity as occurs in bogs, depress microbial activity or in sediments where much of the organic matter is chemically recalcitrant.

Data on oxygen consumption by the benthic community of bacteria, algae and micro- and macrofauna offer some insight into sediment respiration rates, provided that oxygen utilization by these community components can be appropriately partitioned and corrected for abiotic oxygen uptake. Under aerobic conditions, the respiratory oxygen consumption by sediment-dwelling bacteria is found to be related primarily to temperature provided the organic carbon for microbial growth is not limiting (Fig. 3.11). The proportion of community respiration resulting from bacterial respiration also varies with season. This percentage certainly increases under other conditions among lakes; in deep water sediments, for example, nearly all metabolism is by heterotrophic microbes.

Microbes associated with particulate organic detritus consume up to 3 orders of magnitude more oxygen per unit dry weight than do those of sand. Uptake rates are inversely related to particle size and directly related to particle organic

content of carbon and nitrogen. In shallow lakes, water turbulence can frequently disturb and mix sediments to appreciable depths. After brief reoxygenation by turbulence or the respiratory or feeding activities of benthic animals, community respiration and oxygen consumption rates of highly organic sediments quickly return to values similar to those of undisturbed sediments of similar organic content and composition.



**Figure 3.11** - The relationship of temperature and sediment bacterial respiration.

### 3.8.2 Anaerobic Bacterial Activity in Sediments

When oxygen is utilized more rapidly than it is generated by photosynthetic processes or intrusion from other habitats, anaerobic microbial processes utilize a variety of inorganic and organic compounds as alternative electron acceptors in sediments. Specific anaerobic processes occur largely in a fixed sequence reduction of nitrate, manganese, iron, sulphate and carbon dioxide utilizing organic substrates in methanogenesis (Table 3.13). The occurrence of specific anaerobic processes is dependent on the presence of the appropriate electron acceptor as well as competitive electron acceptors. During the succession of anaerobic oxidation processes, the redox potential of the sediments decreases as a result of the reduced products formed.

Each redox couple has a characteristic redox potential at pH 7. The potentially available maximum amount of energy generated in a redox reaction, assuming H<sub>2</sub> as the electron donor, decreases in the sequence from oxygen to carbon dioxide (Table 3.13). Generally, physiological adaptations have evolved to optimize the energy yield from the total oxidation of organic matter. Some electron-acceptor compounds, such as nitrate and ferric oxide, can be reduced by the same bacteria. These microbes, however, have a preference to utilize nitrate in the reduction of

organic compounds and are approximately mutually exclusive to utilize nitrate until the supply is exhausted before utilizing iron oxides. The reductions of ferric oxide, sulphate and carbon dioxide are also mutually exclusive but vary because the reduction is performed by physiologically different types of bacteria with different affinities for electron donors such as acetate and hydrogen. Ferric oxide reduction occurs preferentially to the reduction of sulphate and carbon dioxide owing to a stronger affinity of ferric oxide-reducing bacteria for these electron donors than the other types of anaerobic bacteria. Similarly, sulphate-reducing bacteria are better scavengers for acetate and hydrogen than are the methanogenic bacteria because methanogens have the lowest affinities for these fermentation products.

**Table 3.13:** Processes of oxidants and reductants and approximate redox potentials ( $E_o$ ) caused by reduced products of bacterial origin in sediments during a succession of anaerobic oxidation processes.

| <i>Process</i>               | <i>Substrates</i>        | <i>Products</i>    | $\Delta G^0$<br>(kJ/mol) | $E_o$<br>(mV) |
|------------------------------|--------------------------|--------------------|--------------------------|---------------|
| Disappearance of oxygen      | $O_2 + H_2$              | $2H_2O$            | -236.8                   | +330          |
| Disappearance of nitrate     | $2NO_3^- + 5H_2 + 2H^+$  | $N_2 + 6H_2O$      | -223.0                   | +220          |
|                              | $NO_3^- + 4H_2 + 2H^+$   | $NH_4^+ + 3H_2O$   | -149.8                   |               |
| Appearance of $Mn^{2+}$ ions | $MnO_2 + H_2 + 2H^+$     | $Mn^{2+} + 2H_2O$  | -156.9                   | +200          |
| Appearance of ferrous ions   | $2Fe(OH)_3 + H_2 + 4H^+$ | $2Fe^{2+} + 6H_2O$ | -43.9                    | +120          |
| Disappearance of sulphate    | $SO_4^{2-} + 4H_2$       | $S^{2-} + 4H_2O$   | -38.1                    | -150          |
| Appearance of methane        | $CO_2 + 4H_2$            | $CH_4 + 2H_2O$     | -33.1                    | -250          |

Source: Laanbroek (1990).

### 3.8.3 Littoral Decomposition and Microbial Metabolism

The quantity of total bacterioplankton in the littoral zone is usually much higher than that of the open water. Saprophytic bacteria of the littoral zone are more abundant than the other metabolic groups. The differences are the greatest during the period of active growth of the emergent macrophytes and when the littoral foliage senesce. High numbers of bacteria from the littoral zone extend out into the open water for considerable distances but they are restricted largely to the macrophyte belt.

When the macrophyte-derived dissolved organic matter (DOM) passes through attached microflora, some of the DOM remains in dissolved and colloidal form and is transported within and out of the littoral areas. Some of the DOM is mineralized by littoral bacterioplankton and some aggregates into large colloidal masses and particles. Experiments with DOM from different macrophytes showed some 23% of the dissolved organic carbon remained in solution, 58% was mineralized by bacteria and 19% formed aggregates (Alber and Valiela, 1994). The aggregates had low C:N ratios (4-12, depending on the source species) and contained large densities of active bacteria.

Relatively little, usually < 5%, of the very high total (above and below ground) annual organic productivity of wetland and littoral macrophytes is consumed by grazing animals. Most of the high net primary production senesces undisturbed and

enters the detrital pool as dissolved and particulate organic matter. Early stages of decomposition are dominated by leaching of soluble organic matter from the senescing tissues. Some of this degradation and leaching can occur while the senescing tissues are standing dead among living emergent macrophytes or after the tissues have fallen to the water. Once in the water, microbial colonization, mostly bacteria, is rapid in a succession of forms. These bacteria initially utilize primarily released soluble organic matter and the release of organic substrates influences bacterial growth and activity on leaf litter to a greater extent than does the process of fragmentation of the litter and associated increases in surface area available for bacterial colonization. Emergent macrophytic vegetation generally exhibits very slow rates of degradation in comparison to those of floating-leaved and submersed aquatic plants.

### 3.8.4 Microbial Activity in Sediments of Running Waters

Detrital particulate organic matter associated with sediments in streams and rivers is commonly separated into size classes: (a) coarse particulate organic matter (CPOM,  $> 1$  mm), which consists of large woody debris, leaves, needles, foliage of terrestrial and aquatic vegetation and animal parts and (b) fine particulate organic matter (FPOM, range,  $0.5 \mu\text{m}$ -1 mm), which consists of pieces of CPOM, imported terrestrial and wetland organic matter and aggregated dissolved organic matter. Dissolved and colloidal organic matter is of a size  $< 0.5 \mu\text{m}$ . Fungal activity causes fragmentation of CPOM making it sensitive to bacterial colonization and degradation.

Rates of decomposition of CPOM in streams vary greatly with the chemical composition of the plant material, environmental variables, particularly temperature and nutrients and the feeding activities of benthic invertebrates. Nutrient availability is critical for microbial degradation of leaf materials in flowing water bodies. For example, decomposition rates and fungal productivity of leaves in moderately productive streams are doubled by enrichments of dissolved nitrogen and phosphorus. Of the DOM leached from these materials, the more labile components are very rapidly metabolized by attached bacteria and bacterioplankton, whereas the more recalcitrant DOM compounds are metabolized more slowly.

If the POM remains senescent as leaves on plants before abscission or is standing dead aerially as is common among emergent aquatic plants or is deposited on terrestrial soils before being transported to the river, the organic matter would be exposed to long periods of leaching by precipitation. Losses of labile dissolved organic matter by leaching of aerial senescent tissue can be large. Once in the stream or river, colonization of the POM by microbes proceeds rapidly by bacteria growing on the surfaces and by fungal hyphae growing on the surface and internally. Under well-oxygenated conditions, most of the decomposition of POM in streams is by fungi, primarily hyphomycetes, during the early stages of decomposition. Bacterial degradation of these recalcitrant plant tissues is relatively minor during this period but increases and usually dominates in the later stages of the POM decomposition processes.



### 3.9 CONCLUSION

Understanding the metabolic responses of aquatic ecosystems is essential in order to confront and offset the effects of human impact and in order to achieve maximum and effective management of freshwater resources. All waters, of course, cannot be managed directly. Rather, an integration of human growth and utilization with the metabolism of freshwaters is required to minimize detrimental changes. Human induced perturbations often cause increased productivity, simplification of biotic communities and a reduction in the ability of the metabolism of the organisms to adapt to the imposed loading of nutrients. These conditions lead to reduced stability of the ecosystem. In this condition of eutrophication, excessive inputs often exceed the capacity of the ecosystem to be balanced through self purification. In reality, however, the ecosystems are out of equilibrium only with respect to the freshwater chemical and biotic characteristics that are desired by humans for specific purposes. In order to have any hope of effectively integrating humans as a component of aquatic ecosystems and of monitoring their utilization of these resources, it is mandatory that we comprehend in some detail the functional properties of freshwaters. Only then we can evaluate and predict, with reasonable certainty, the influence human activities will have on the metabolic characteristics of these ecosystems. Knowledge on the microbial density, diversity and functions in a freshwater ecosystem will certainly facilitate the effective management of the water body and maximize its potential.

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# MICROBES IN MARINE AND ESTUARINE HABITATS

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## 4.1 INTRODUCTION

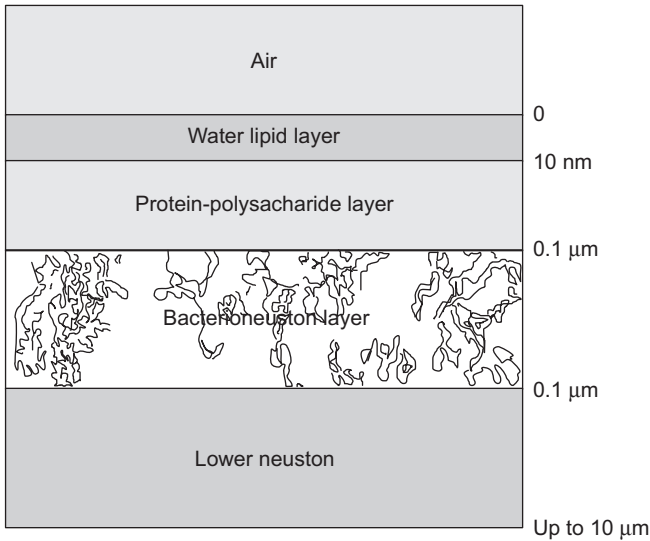
Microbes account for more than 90% of ocean biomass and constitute a hidden majority of life that flourishes in the sea. What is even more surprising is that much of this microbial life remains unknown because it is difficult to observe in nature and many of the marine microbes have failed to grow in synthetic media. Despite all the sophistication in the sampling and measurement procedures, our knowledge on the microbial diversity of the oceans is still in infancy. However, the present understanding on the microbial activities in the marine ecosystems has convinced us that marine microbes play many important roles in the Earth system. They influence our climate, are the major primary producers in the ocean, dictate much of the flow of marine energy and nutrients and provide us with a source of medicines and natural products. Microbes (which include bacteria, archaea, microbial eukaryotes and their associated viruses) are as varied as the marine environments they come from. From the ice-covered polar regions of the Arctic and Antarctic to the boiling hydrothermal vents in the depths of the sea and the calcareous oozes made up of the skeletons of single-celled eukaryotes called foraminifera that once recorded the temperature of the sea surrounding them, microbes are everywhere and they help to shape the features of our planet, past and present.

The term ‘Marine Microbes’ encompasses all microscopic organisms generally found in salt water. Most microorganisms are acellular and fall into the major categories of viruses, prokaryotes (‘bacteria’) and protists, groups which differ considerably in biological characteristics. While representatives of these groups are found in virtually everywhere in marine waters and they play nearly every ecological role imaginable, their most important function is that they form the base of the food chain in marine ecosystems.

## 4.2 THE CHARACTERISTICS OF THE MARINE ENVIRONMENT

Marine water environments, like those of lakes, are highly diverse. Marine water is characterized by salinity between 33 and 37 per thousand excepting the Baltic Sea where the salinity is comparatively low. The oceanic environment can range in depth up to 11,000 m in the deepest of ocean trenches. Oceans are not static in size or shape. Considerable mixing (especially of the surface layers) and movement are caused by the action of tides, currents, temperature upwelling and winds. Because the oceans are so expansive and their surface area is so great, the effect of sunlight is important. The ocean is divided into two zones, the upper euphotic zone, through which light can penetrate and the lower aphotic zone, whose biota is totally dependent on the organic carbon supply from the euphotic zone. Light is able up to penetrate to a depth of 200 m, depending on the turbidity of the water. In coastal areas, where the amount of suspended particulate matter in the water is high, light may, in some places, penetrate less than 1 m.

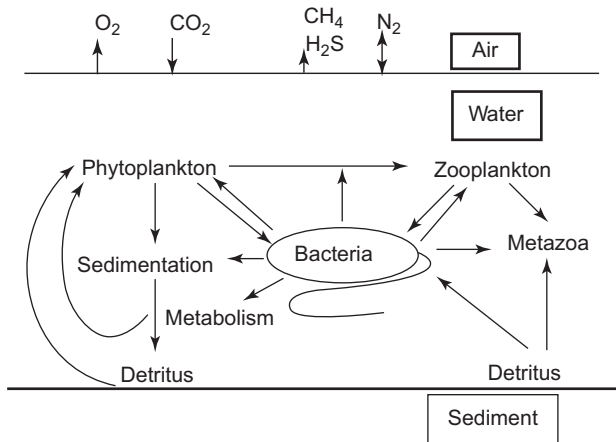
Other designations for classifying zones in marine environments are based on habitats. Four major habitats in the marine systems, are important from microbiological aspects. viz., 1) the surface of the sea (air-water interface) is the habitat referred to as the neuston (Fig. 4.1), 2) the water column consisting of the epipelagic (euphotic), mesopelagic, bathypelagic, abyssopelagic and benthopelagic (benthos) zone, 3) the epibiotic habitat, which refers to surfaces on which attached communities occur and 4) the endobiotic habitat, which pertains to organisms found within the tissues of other larger organisms such as fish. The microbes are distributed in all these habitats, always undergoing changes in their density and diversity.



**Figure 4.1** - Schematic representation of neuston that are distributed in the upper 10 µm layer of water (Adopted from Dowd *et al.*, 2006).

### 4.3 MICROBIAL STRUCTURING OF MARINE ECOSYSTEMS

One-half of global primary production occurs in the oceans and, therefore, a fundamental problem is to understand how organisms use this carbon to create spatio-temporal patterns of carbon and energy flux. Historically, it was believed that most primary production moved through a chain of small and large animals and microorganisms were largely ignored. However, several remarkable discoveries, that have been made during the past 30 years, have shown that bacteria dominate the abundance, diversity and metabolic activity of the ocean (Fig. 4.2). A large fraction of primary production becomes dissolved (dissolved organic matter; DOM) by various mechanisms in the food web and this part of the primary production is almost exclusively accessible to heterotrophic bacteria and archaea. As a result, the uptake of organic matter by bacteria is a major carbon-flow pathway and its variability can change the overall pattern of carbon flux. Further, as bacteria use behavioural and biochemical strategies to acquire organic matter, for example, by the expression of enzymes to solubilize particulate organic matter (POM), they interact with sources of organic matter and modify the ecosystem and carbon cycle in different ways.



**Figure 4.2** - Scheme of the role of bacteria in marine ecosystem processes leading to the energy and matter transfer.

Although it is impractical to measure the flux of each component of DOM into bacteria, the cumulative carbon flux (bacterial carbon demand; BCD) can be estimated as the sum of the carbon that is assimilated (growth) and respired, which can then be compared with primary production as a measure of the coupling strength ( $BCD \div \text{primary production}$ ). This parameter is useful as a global measure of bacterial performance, for example, in concepts that involve the ocean carbon flux or ecosystem-based fisheries. Measurements of coupling strength have been made on many ocean expeditions, in most oceanic locations

and in regions that differ in primary production and this has revealed important geographical and seasonal patterns. The following examples show that the coupling of bacteria with primary production is highly variable and that this variability affects ecosystem functioning.

1. In the eastern Mediterranean, bacteria take up most of the primary production, which is consistent with the poor fisheries that are present in this area (although a high BCD might be an effect rather than a cause of poor fisheries).
2. Studies carried out on a north–south oceanic transect (53°N in the Atlantic to 65°S in the Southern Ocean) showed latitudinal variation in coupling strength and, importantly, there were large net-heterotrophic regions. Bacteria took up more DOM than was present as local primary production, which indicates that the spatial or temporal import of organic matter must have occurred. Consequently, these regions have the potential for the net out-gassing of carbon dioxide. More extensive spatial and temporal coverage to account for patchy autotrophic processes might, however, reveal a metabolic imbalance.

These examples illustrate that the ability or inability of a bacterial assemblage to grow on specific types of organic matter, either owing to the constraints of community composition or environmental gene expression, can be important for the functioning of globally significant ecosystems. This underscores the need to understand how bacteria function in their natural environment to influence the flux pathways of fixed carbon. From advances in marine genomics and metagenomics it can be inferred that enormous bacterial gene diversity is available for assemblage-level bacterial interactions within the ocean.

### 4.3.1 Spatial Distribution of Microorganisms

Mostly it is assumed that bacteria take up homogeneously distributed DOM. This premise relies on the assumption that the DOM, that is released from any source, diffuses homogeneously into the bulk phase before it is taken up by an organism. This view is changing, however, with the recognition of bacterial *in situ* behavioural and physiological responses to DOM-production loci and gradients. The detection of high abundances of decomposer bacteria ( $10^6$  per ml) has led to the suggestion that the numbers and activity of primary producers (such as cyanobacteria and algae), decomposers (such as bacteria) and predators (such as viruses and protists) are similar. Typically, 1 mm<sup>3</sup> (or µl) of surface seawater (considered to be the bacterium's microenvironment) contains 10,000 viruses, 1,000 bacteria, 100 *Prochlorococcus* cells, 10 *Synechococcus* cells, 10 eukaryotic algae and 10 protists, although the numbers are highly variable. The close proximity (0–1 mm) of individual cells suggests that there is potential for many cell–cell interactions.

All microbial trophic groups are in close proximity, so that the adaptive biology of bacteria occurs in a microspatial context that integrates the interactions between all of the trophic groups. For instance, bacteria might respond behaviourally and metabolically to the DOM that is produced by a range of mechanisms, such as algal exudation and cell lysis, predation by viruses, which

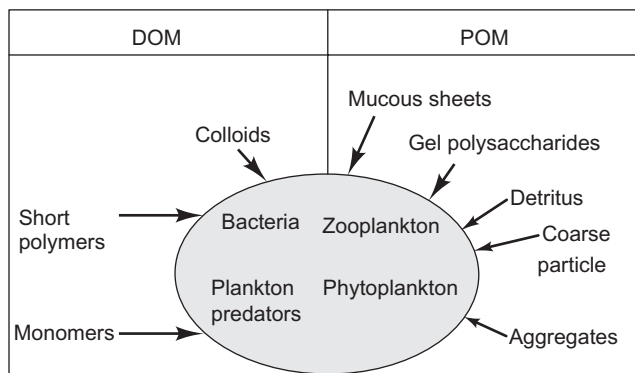
releases prey DOM and the action of protists, which ingest food vacuoles that contain DOM and regenerate N, P and  $\text{Fe}^{2+}$ . Such DOM hot spots might occur in the microenvironment of a bacterium over short timescales, perhaps even minutes, owing to the high abundance of microorganisms.

### 4.3.2 Hydrolytic Enzymes and Hydrolysis-uptake Coupling

Marine bacteria hydrolyse polymers and particles using cell-surface-bound hydrolytic enzymes or ectohydrolases (protease, glucosidase, lipase, phosphatase, nuclease and chitinase). These enzymes, together with membrane-bound transporters, make the bacterial surface reactive for organic matter transformation and uptake. This is the final step that couples bacteria to primary production in the ocean. The bacterial surface is the dominant biotic surface and it is proposed that 0.1–1 m<sup>2</sup> of the bacterial surface per m<sup>3</sup> of seawater interacts intimately with DOM. A pelagic bacterium that swims through organic matter leaves a stream of monomers and oligomers behind it naming bacteria as the “perfect swimming stomachs”.

Bacteria express multiple (multiphasic) transporters that have  $K_m$  values that range from nanomolar to millimolar, consistent with adaptation to environments that contain DOM and POM gradients (Fig. 4.3). This raises the question of whether the distinction that is made between oligotrophic and eutrophic bacteria, which is usually based on culturing studies, actually reflects the microspatial adaptations of different strains to nutrient hot spots. Perhaps, bacteria simply move to an appropriate distance from the available organic matter gradient, so that they can effectively use the carbon resources that are available.

Recent research shows that organic matter in seawater is replete with transparent gels that form tangled webs of components in the form of colloids that are approximately 10 nm long (10<sup>8</sup> colloids per ml) and mucus sheets and bundles that are up to 100 µm long (1,000 sheets or bundles per ml). Colloids,



**Figure 4.3** - The assimilation of different organic matters by the metabolic interaction of microorganisms. The thickness of arrows shows the relative rate of use of the organic matter.

sheets and bundles interact to form macromolecular networks that are 100  $\mu\text{m}$  or more long. The polymeric components of transparent gels are probably derived from microorganisms. Phytoplankton and bacteria produce cell-surface mucus, which can either be released or, alternatively, solubilized by cell-surface-acting enzymes. Phytoplankton release condensed polysaccharide particles by exocytosis, which form gels in seawater. Most (90–95%) of the DOM is refractory to degradation by bacteria and it has been proposed that some DOM components serve as stable scaffolds of a gel architecture that is inaccessible to bacterial ectohydrolases. Potentially, gels are a nutrient sink and represent a mosaic of nutrient hot spots. This nutrient pool is huge, as approximately 10% of all the DOM ( $70 \times 10^9$  tons of carbon) that is found in the ocean is present in the form of gels. This represents a carbon pool that is larger than all of the carbon that is present as biomass in the ocean. Depending on their surface properties, gels might adsorb DOM components from seawater and this could be significant given the large surface area of these gels. Bacteria can attach to gel-sand and it is possible that they hydrolyse the nutrients that are present on the gel surface using cell-surface hydrolases. This, in turn, might alter the local architecture of the gel and the nutrient dynamics at the microscale.

### 4.3.3 Bacteria–Phytoplankton Interactions

Phytoplankton are organic matter production loci. The main processes that convert up to half of all the primary production into DOM must occur before substantial amounts are transferred through multiple trophic levels and respired. The most likely mechanisms of DOM formation are direct processes, such as phytoplankton exudation and lysis (by virus attack or nutrient stress), bacterial interactions with live phytoplankton (for example, commensalism or predation) and the bacterial enzymatic degradation of recently dead phytoplankton and protists. The aggregation of phytoplankton and phytoplankton detritus facilitate bacterial access to DOM.

One way for bacteria to metabolically couple to primary production would be to swim up to phytoplankton and use cell-surface hydrolases to kill them. Bacteria have various adaptations that allow them to attach to dead phytoplankton and cause hydrolysis. If bacteria are randomly distributed, each bacterium would be a few hundred micrometres away from the nearest phytoplankton cell—a distance that most motile marine bacteria can traverse rather quickly (although many bacteria are non-motile and would not use this strategy). Bacteria can cluster around phytoplankton to create considerably higher concentrations than the average for seawater, which is  $10^6$  bacteria per ml (surface colonization). Tight spatial coupling between marine bacteria (*Pseudoalteromonas haloplanktis* and *Shewanella putrefaciens*) and a motile alga (*Pavlova lutheri*) has been filmed and referred to as the ‘pestering’ of algae, because the bacteria closely tracked motile phytoplankton. It seems that some phytoplankton-bacteria associations might even be species specific.



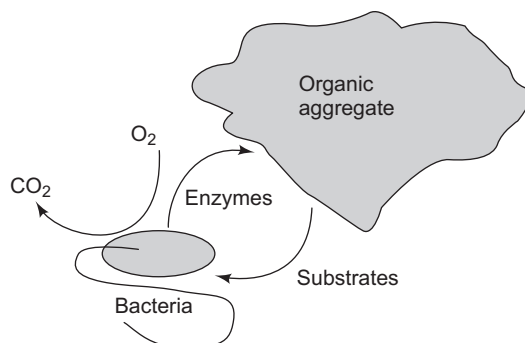
Phytoplankton produce surface mucus, polysaccharides and proteoglycans, which might serve as a protection from bacteria; this is analogous to corals, which convert substantial photosynthate into mucus for defence against microbial invasions. Some phytoplankton also produce inhibitory compounds. Mucus creates a region around the phytoplankton cell that is rich in organic matter and is known as the phycosphere. The nitrogen- and phosphorus-depleted mucus could adsorb nitrogen- and phosphorus-rich materials, such as polymers and colloids, from seawater, resulting in the development of a rich gel medium. Although much research has been done on bacteria that are attached to phytoplankton it has been technically difficult to study the *in situ* physiology of bacteria (growth, respiration and antibiotic synthesis) in the phycosphere. It is also likely that the community composition of bacteria in the phycosphere will differ from that in the local ocean environment. Significant diurnal alterations in the bacteria–phytoplankton relationship are to be expected, but these need to be addressed in a microspatial context. As the phycosphere is organically rich it has been proposed that it could support the proliferation of human pathogens or other bacteria that are adapted to high-nutrient environments.

#### 4.3.4 Bacteria-Protists Interaction

Protists consume more than half the primary production in many ecosystems. How can we reconcile the large grazing pressure on phytoplankton by protists with an equally high bacterial carbon demand? Perhaps when the BCD and protist grazing are both high, a large fraction of the organic matter that is ingested by protists is released through the lysis of these protists by viruses or other factors. Bacteria can rapidly colonize dead protists and degrade fresh protist detritus. However, it remains to be determined whether heterotrophic bacteria cause significant protist mortality in an ‘eat them before they eat you’ scenario. In this hypothetical scenario, protists are the processors of primary production, their enzymes convert the POM of prey to DOM and their newly synthesized biomass is consumed by bacteria as predators. Protists are important grazers of small, highly abundant phytoplankton, such as *Synechococcus*, *Prochlorococcus* and picoeukaryotes. Therefore, protist processing of primary production might be one mechanism that increases the coupling of bacteria to primary production.

#### 4.3.5 Interactions of Bacteria with Detritus

Dead phytoplankton cells are a source of organic matter for bacteria, which readily colonize such material to form a detritosphere (Fig. 4.4). Bacterial colonization and enzymatic hydrolysis convert detritus to DOM. However, not all bacteria attach to dead phytoplankton. In an experimental system, dense clusters of bacteria were recorded swimming around a killed dinoflagellate and these were modelled to predict bacterial interactions with DOM hot spots. Depending on the particle sinking rate, the effects of bacteria on dead phytoplankton cells might result in DOM plumes.



**Figure 4.4** - The microbial digestion of the organic matter in the detritosphere, where microbial enzymes cause the disappearance of the detritus from the surface.

Diatoms attract bacteria and freshly lysed diatoms comprise organic matter hot spots that support the robust growth of bacteria. Because diatoms require silicon for growth, the mechanisms of silicon recycling are important. Bacteria, that do not require silicon, regulate the regeneration of silicon from dead diatoms. The diatoms, that have low silicon requirement, probably do not release soluble organic matters, which is the main attraction of bacteria to diatom surface. Although bacteria use their ectohydrolases to solubilize organic matter, the action of these enzymes also removes the protective proteoglycan silaffins from the silicon frustules and it is the silaffin proteins that normally prevent the dissolution of silicon. This process requires colonization by bacteria and those bacteria with high cell-specific protease activity cause more rapid silaffin dissolution. In view of the importance of diatom production and the regeneration of silicon in the marine system, the regulation of the silicon cycle by bacteria could affect the carbon-export flux.

#### 4.3.6 Marine Snow

Under favourable conditions in coastal oceans, bloom formation imposes a temporary shortage of resources, which compels the bloom forming phytoplankton to die back. Many of these dead phytoplankton cells are grouped into aggregates by interactive activity of many bacterial species. Because the rate of production of these organic particles is many times higher than their metabolism in the water column, these particles move to the open oceanic water by current and tidal action and in the comparatively calm open water they move down as clouds called the marine snow. In general, aggregation with larger particles and agglomeration with other materials, such as phytoplankton or detritus, forms marine snow. Bacteria colonize the marine snow in population densities that often reach  $10^8 - 10^9$  per ml. The expression of several ectohydrolases, such as protease, lipase, chitinase and phosphatase, is high, but glucosidases are not well expressed. This enzyme complement digests some of the DOM in marine snow. However, the colonizing bacteria use only a fraction of the hydrolysate (weak coupling) and, therefore, the sinking marine snow leaves

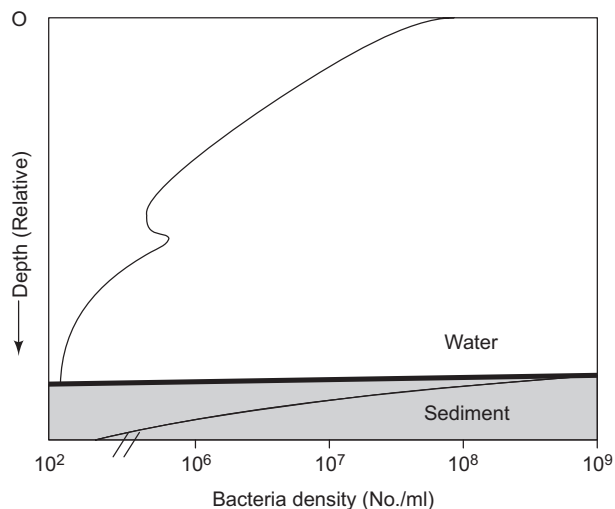
an extended plume of DOM behind them that other bacteria can use. One model predicts that approximately half of all the BCD in the ocean is satisfied by bacterial interaction with these plumes.

Marine-snow plumes help to retain nitrogen, phosphorus and iron within the upper mixed layer of the ocean and this supports primary productivity. In addition, as the glucosidase activity of bacteria in marine snow is low, a disproportionate amount of carbon sink below the upper mixed layer, thus increasing the carbon:nitrogen and carbon:phosphorus ratios in aggregates and causing carbon storage. Because marine snow is organically enriched, bacterial growth might occur with high growth efficiency. One strategy that might enable such loose organic matter–bacteria coupling to be adaptive is for the attached bacteria to release their progeny into the plume. Thus, the microscale interactions of bacteria influence fundamental biogeochemical processes, such as carbon storage and the regulation of carbon flux, as well as provide a microspatial framework for understanding the behavioural and biochemical strategies of free-living bacteria that might be adapted to using the DOM at high concentrations in the plumes. Such strategies might also be used by bacteria at depth because marine snow can sink to hundreds or even thousands of metres.

How bacteria influence the aggregation and biogeochemical fate of carbon is also relevant to concerns about ocean fertilization and carbon sequestration, in which the goal is to maximize aggregation and export carbon flux. Many bacteria are non-motile, notably pelagic Bacteroids, who are specialized for colonizing aggregates such as marine snow. Non-motile bacteria, being particle specialists, probably attach first to the highly abundant small gel particles, which are in the 10  $\mu\text{m}$  length range and are several orders-of-magnitude more abundant than marine snow (typically 1–10 aggregates per litre in the upper ocean). Attachment of gel particles to marine snow contribute diverse bacteria, including non-motile bacteria and expedite the digestion of aggregate particles.

## 4.4 MARINE BACTERIA

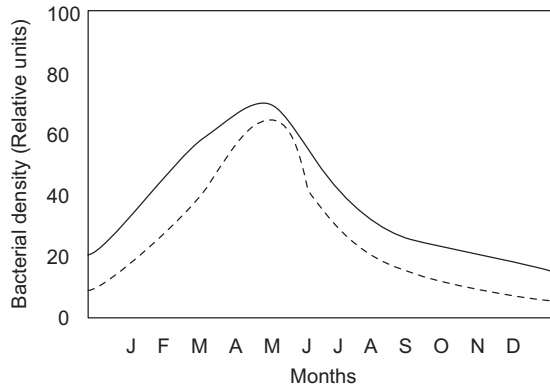
The ocean contains diverse microbial habitats. Also, depending on location (arctic to tropical zones), markedly different microbial populations predominate. As a rule, especially in deep waters, microbial concentrations are the highest within the neuston and drop markedly below this region. Immediately below the neuston, the numbers are on average close to  $10^7$  organisms/ml and decrease by more than one order at a depth of 100 m. As a result of organic input from the terrestrial environment, total bacterial numbers are on average one order of magnitude higher in coastal water than in the open ocean. This is especially true for harbours and near populated areas. The vertical distribution of the heterotrophic populations shows an increase in number of bacteria and other microbes at the thermocline because of good availability of organic carbon and oxygen. Heterotroph numbers increase again immediately above the ocean floor (Fig. 4.5).



**Figure 4.5** - The vertical variation of the density of bacteria in the open ocean well stratified to euphotic and aphotic zones. The minor peak shows the increase of number of bacteria at the thermocline.

Thermocline-induced stratification is not as dramatic in coastal waters because of the mixing of water by winds, currents and temperature. For this reason, bacterial numbers are almost uniform at all depths except when the weather is very calm for long leading to stratification of water. In addition, seasonal fluctuations in bacterial numbers, which are not observed in the open ocean, are common near the coast. In general, there are two times of the year when there is an increase in bacterial populations in coastal waters, late spring-early summer and late summer-early autumn. These are also abundant when the phytoplankton are most active. Thus, primary production is intimately tied to the overall productivity of the ocean. High phytoplankton productivity facilitates the increase of bacterial density and diversity. About 80% or more of such bacterial biomass are heterotrophic bacteria. Figure 4.6 shows the relationship between total bacteria and heterotrophic numbers. It can be seen that an increase in phototrophs is followed by an increase in heterotrophs. This tends to be true in many aquatic environments where the concentration of primary producers ranges from  $10^9$  organisms/ml in some benthic environments to  $10^8$  organisms/ml in surface zones. Heterotrophic populations generally vary considerably with depth as well, although not as much as the photosynthetic populations. The general range is from 10 organisms/ml in zones characterized as oligotrophic to  $10^8$  organisms/ml in zones where organic matter is present.

The bacterial population throughout the photosynthetic zone is closely related to the distribution of phytoplankton. The beneficial effect of the plankton may be attributed to both the organic substances they elaborate and the solid surfaces provided for bacterial aggregation. Bacterial populations differ widely with prevailing conditions. The temperature of the marine environment and the degree of salinity would be suitable for the growth of psychrophilic and halophilic physiological types. Most marine bacteria, however, are halotolerant or slightly to



**Figure 4.6** - The relationship between the relative population of total bacteria (solid line) and heterotrophic bacteria (broken line) in different parts of the year in coastal oceans of the tropics. Total bacteria include both autotrophic and heterotrophic populations.

moderately halophilic. Among the psychrophilic forms are certain luminous bacteria which can produce light in the presence of oxygen. These bacteria may exist in symbiotic association with certain species of marine animals (See Chapter 10). In general, aquatic bacteria tend to be gram-negative. It is thought that the gram-negative envelope provides a structure better suited to support life in nutritionally dilute aquatic environments than the gram-positive cell wall. Important hydrolytic enzymes are retained in the periplasmic space, rather than being excreted and lost to the aquatic environment, as would be the case for gram-positive bacteria. In addition, lipopolysaccharide (LPS) of the outer membrane affords gram-negative bacteria protection from certain toxic molecules e.g., fatty acids and antibiotics and it may serve to sequester important nutrients from water. Bacteria in the surface region of the marine environment are often pigmented, a characteristic which may afford protection from ultraviolet radiation.

#### 4.4.1 Carbon Fixation by Marine Bacteria

The two million prokaryotes, majority of whom are bacteria and Archaea, inhabit each millilitre of coastal seawater. Major advances in molecular taxonomy and ecological genomics have allowed numerous uncultured coastal bacteria to be named and, in some cases, functional roles to be assigned. It is hoped that understanding these tiny organisms better will allow us to make progress towards elucidating the main ecosystem-level processes, viz., nutrient cycling, ocean-atmosphere exchanges and productivity, that occur in the pelagic marine environment. Our knowledge will be significantly advanced by understanding the biochemistry of the individual mediators of these processes. Recently, important advances have been made in our comprehension of the intricacies of the carbon cycle, particularly the efficiency with which organic carbon is processed by marine microorganisms in the surface waters of the ocean. CO<sub>2</sub> oxidation, lithoheterotrophy and anoxygenic phototrophism are some of the well understood processes of bacteria in marine systems.

**(a) CO oxidizers:** Carboxydotrophic bacteria that oxidize CO to CO<sub>2</sub> (by the enzyme CO dehydrogenase) and that can fix a portion of the CO<sub>2</sub> into biomass (by the enzyme ribulose-1,5-bisphosphate carboxylase oxygenase) have been successfully cultured from aquatic environments and soils. However, although this phenotype might seem perfect for an oceanic CO oxidizer, physiological studies have revealed that many cultured carboxydotrophic bacteria, for example, *Oligotropha carboxidovorans* and *Hydrogenophaga pseudoflava*, require concentrations of CO that are several orders of magnitude higher than those present in ocean waters to sustain growth. For example, the  $K_m$  for CO oxidation is typically 400–1,000 nM for carboxydotrophs in the laboratory. However, the peak CO concentration in coastal environments is only ca 15 nM and even less CO is present in the open ocean (only ca 2–3 nM). Carboxydotrophs, that grow autotrophically using CO, require high concentrations of this substrate to sustain reasonable growth rates, owing to the low energy yield from the oxidation of CO. Although bacteria of this type might thrive in niches that contain high local concentrations of CO, for example, waterlogged soils, their physiology suggests that they are unlikely to be capable of growth using the CO concentrations that are present in the ocean. Some chemical oceanography studies supported this hypothesis, with field measurements of the CO oxidation rates so low that it would take more than 100 days for bacteria to consume the ambient CO pool. Thus, calculations of the ocean-atmosphere CO flux have usually indicated little or no biological participation.

By contrast, other efforts to estimate global CO budgets have indicated that microbial oxidation must be a major pathway of the CO loss from the ocean surface waters. Extensive field studies that generated system-specific values for photochemical efficiency have led to the conclusion that as much as 90% of CO that is produced in ocean surface waters was internally recycled by CO-oxidizing marine bacteria before it could escape to the atmosphere, which would represent a major shift in the CO paradigm.

**(b) Lithoheterotrophs:** Although it had been deemed unlikely that the carboxydotrophs are the main producers of oceanic CO oxidation, the microbial taxa that perform this process remained elusive. A hint at their identity came from the genome sequence of *Silicibacter pomeroyi*, which is a cultured representative of an abundant alphaproteobacteria taxon in the coastal ocean. The marine Roseobacter group, *S. pomeroyi* has two CO dehydrogenase proteins and can oxidize CO at low nanomolar concentrations that are consistent with those found in the coastal ocean. This bacterium differs from the previously characterized carboxydotrophs by one important characteristic: it has no genes for CO<sub>2</sub> fixation and, therefore, only uses CO as a supplemental energy source during heterotrophic growth on organic substrates. This metabolic strategy is termed lithoheterotrophy, which means that inorganic compounds (in this case CO) provide energy to organisms that nonetheless require organic matter as a source of carbon. Genome sequences of approximately 12 other Roseobacter group strains have indicated that genes for CO oxidation are common among cultured members of this group.

The discovery of CO lithoheterotrophy in an ecologically relevant marine bacterium suggested a resolution to the problem of how such a low concentration of CO can support bacterial growth - it does not necessarily have to. The hypothesis that lithoheterotrophs, rather than autotrophs, mediate a major fraction of the CO oxidation in ocean surface waters was supported by two findings. First, none of the CO that was oxidized by marine bacterioplankton communities seems to be incorporated into biomass and, second, diverse heterotrophic CO oxidizers from coastal seawater were successfully cultured. One possible exception to this hypothesis has been found—a *Stappia* strain (from the alphaproteobacteria) that contains genes for both CO oxidation and CO<sub>2</sub> fixation, although the two processes have not yet been shown to be physiologically coupled in this isolate. Abundance estimates indicate that CO-oxidizing bacterioplankton are not a novelty in surface waters; they have been estimated to account for 1 in every 14 cells in the oligotrophic Sargasso Sea, with similarly high frequencies estimated in the north-eastern coastal waters of the United States.

**(c) The aerobic anoxygenic phototrophs (AAPs):** Aerobic bacteria that photosynthesize without producing oxygen, are considered to be ecological oddities from specialized habitats. In 2000, a discovery was made that surprised the whole oceanography community, the primary pigment that is associated with anoxygenic phototrophy, bacteriochlorophyll *a* (Bchl *a*), was found to be ubiquitously distributed in ocean surface waters. The bacterioplankton taxa that are responsible for synthesizing the Bchl *a* and carrying out this previously unrecognized ocean process were at first completely unknown.

Early culturing attempts indicated that the marine AAPs might be Bchl *a*-containing members of the genus *Erythrobacter*, but the low representation of this taxon in 16S rRNA-based marine bacterioplankton surveys led to doubts that they were truly important ecologically. The metagenomic approaches showed that AAP genes were also present in the members of the marine Roseobacter group as well as quite distant beta- and gammaproteobacteria groups for which cultured representatives were not available. Evidence for many additional (and mostly uncultured) taxa that encode AAP genes has also been found based on a metagenomic survey of Atlantic and Pacific Ocean surface waters. Thus, many major oceanic bacterioplankton taxa have representatives with AAP genes and ca 50 genes that are required for this process seem to have been horizontally transferred with ease among the major marine groups.

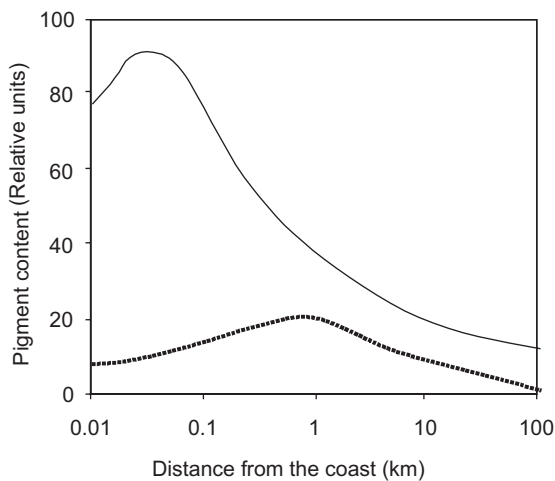
Direct microscopic counts of Bchl *a*-containing cells indicate that AAPs account for only 1–10% of the chlorophyll *a* (Chl*a*)-based plankton counts in ocean surface waters. However, in oligotrophic environments populations might be considerably larger and comprise a bigger fraction of the Chl*a*-based plankton counts. Ratios of Bchl *a* to Chl *a* in the various marine surface waters suggest that the energy that is generated by AAPs forms less than 1% of the total light-driven electron flux that is derived from standard photosynthesis (Fig. 4.7).

A direct comparison of the energy capture by the two processes does not account for the fact that AAP-derived energy is directly used by heterotrophic



cells, thereby bypassing a trophic level and the associated energy losses in the microbial food web. For example, a classical bacterial heterotroph might route only 30% of the available organic carbon substrate into macromolecule biosynthesis and cell growth, because 70% is respired to meet cellular energy needs. By contrast, an AAP that fulfils one-fifth of its cellular energy requirement from light-driven ATP synthesis could increase its cell biomass by 15% using the same amount of organic carbon, by routing 35% of the organic carbon to biosynthesis. Therefore, phototrophic electron flux by aerobic anoxygenic phototrophy might be more appropriately viewed as a supplement to heterotrophy rather than as a competing photosynthetic process.

The detection of a genome fragment from an uncultured bacterioplankton cell from Monterey Bay, California, led to the unexpected discovery of a new mechanism for light-driven energy generation in the ocean. This DNA fragment, which was derived from a member of the uncultured SAR86 lineage, contained a gene that encoded a rhodopsin-type protein. Studies established that proteorhodopsin proteins are spectrally tuned to absorb the most dominant wavelength of light in coastal versus open ocean or deep waters (green versus blue), simply by a substitution at a single amino-acid position that functions as a spectral tuning switch. The ATP-generating function of rhodopsins was already well known when the oceanic proteins were discovered, but these photoproteins were thought to be confined to highly salt-tolerant archaea from unusual environments. Proteorhodopsins are more broadly distributed taxonomically than the ‘proteo’ in their name would imply, as taxa from both within and outside the proteobacteria have now been found to contain proteorhodopsin orthologues. These include the gammaproteobacteria, the alphaproteobacteria, the bacteroidetes and the marine euryarchaeota. The recently sequenced genome of

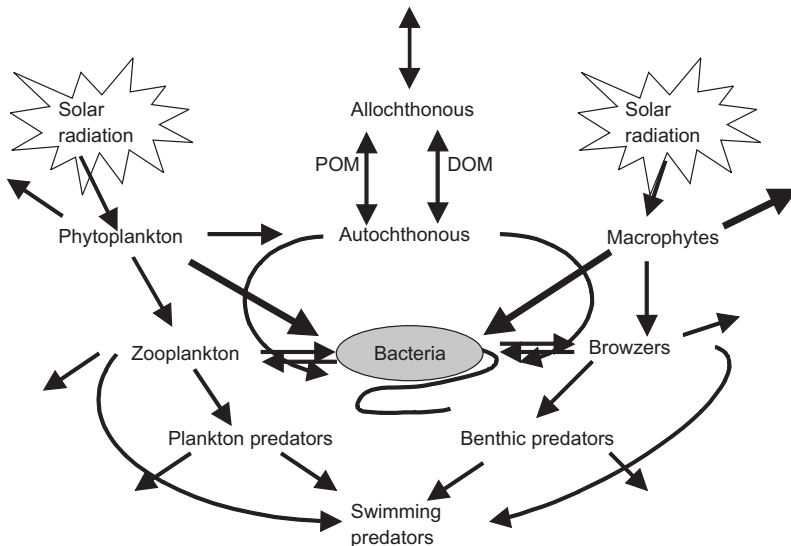


**Figure 4.7** - The relationship between the abundance of photosynthetic bacteria and phytoplankton as measured by their Chl *a* contents expressed in relative units. Legends: Solid line: Chl *a*; Broken line: Bchl *a*.

one member of a difficult-to-culture *Roseobacter* lineage (Alphaproteobacterium HTCC2255) now adds the *Roseobacter* group to the list of marine alphaproteobacteria taxa that contain proteorhodopsin genes.

#### 4.4.2 Bacterial Role in Microbial Food Web

Aquatic life exhibits a vast complex of interactions among microorganisms and between microorganisms and macroorganisms, both plant and animal. Microorganisms, particularly algae and protozoa, occupy a key role in the food chain of the aquatic environment. Numerous bacterial species perform a variety of biochemical changes in various substrates that allow the recycling of elements and nutrients. The primary producers in the marine system are cyanobacteria and eucaryotic algae, which predominate in the phytoplankton. Through photosynthesis they are capable of transforming radiant energy into chemical energy (organic compounds). The biological activity of a marine ecosystem is indeed dependent on the rate of primary production performed by these photosynthetic microorganisms. In shallow estuaries, the role of photosynthetic microbes as primary producers is considerably reduced as the main contributors of organic carbon in these systems are macrophytes. Plant growth from the shoreline contributes leaves, stems and roots of vegetation, and other organic detritus (Fig.4.8). Phytoplankton and benthic



**Figure 4.8** - The involvement of bacteria in the microbial loop operating in a coastal oceanic environment with macrophyte vegetation. The thickness of arrows indicates the rate of material transfer between the components.

and benthic algae make a small contribution to the food supply. The organic vegetation is degraded by bacteria and fungi and converted to microbial protein, which may serve as nutrient for protozoa. However, the estuary contains many detritus consumers (herbivorous and omnivorous crustaceans, mollusks, insect

larvae, nematodes, polychaetas and a few fishes). They derive their energy from vascular plant material. The food web in a deep estuary is more like that in the marine environment.

## 4.5 THE MARINE FUNGAL DIVERSITY

Fungi have recently been found to be widely distributed in the marine habitat. These are the important intermediaries of energy flow from detritus to higher trophic levels in the marine ecosystems. They require sea water for the completion of their life cycle. More than 500 species of marine fungi have been described. Higher marine fungi constitute Ascomycotina, Basidiomycotina and Deuteromycotina. Majority of them being ascomycetes, their spores show adaptation to the marine ecosystem with the production of appendages, which facilitate buoyancy in water, entrapment and adherence to substrates. Marine filamentous fungi have been reported on a variety of detritus: decaying wood, leaves, seaweeds, seagrasses, calcareous materials and chitinous substrates.

Marine fungi feed on both plants and animals in addition to bacteria and algae. A relatively small number of genera are found in the marine environments, but these can often be present at surprisingly high concentrations. Certain species of fungi can be isolated with concentrations up to  $10^3$  to  $10^4$  organisms/ml, especially in carbon-rich areas of the water column and in organic-rich benthic habitats. Another surprising fact is that fungi have been found at depths well below 1000 m.

Studies on marine fungi were initiated at the temperate parts of the world. Subsequently, tropical locations were the centres of interest to understand the abundance and diversity. Tropical regions of Atlantic and Pacific Oceans have been investigated more intensively than the Indian Ocean. A few quantitative studies on marine fungi are available from the west coast of India. Majority of the studies in the Indian Ocean are restricted to mangrove habitats, so also from the Indian Peninsula. Fungal species richness and diversity of filamentous marine fungi on woody substrates collected from the beaches, islands and harbour locations have also been studied. The evidences suggest that the abundance of fungal species is more in coastal oceans and mangrove habitats than in open oceans. The organically poor oceanic environment is highly deficient in fungal population. The temperature and seasonal cycle in the oceanic environment also play a major role in the distribution and diversity of fungi. In general the tropical oceans are more abundant in fungi in comparison to the temperate oceans.

### 4.5.1 Diversity of Fungi in Marine Environment

Marine fungi have been classified into three geographical groups by Kohlmeyer and Kohlmeyer (1979): i) cosmopolitan species; ii) temperate-water species and iii) species from tropical and subtropical waters. Based on distribution in Atlantic Ocean, Indian Ocean, South-East Asia and Pacific Ocean the geographical distribution of some representative mangrove fungi (*Halosarpheia fibrosa*, *H. marina*, *Lignincola laevis* and *Lulworthia grandispora*) has been revised.

Geographical and seasonal distribution of *Asteromyces cruciatus*, *Sigmoidea marina* and *Varicosporina ramulosa* has been correlated with their growth pattern under different temperature regimes. Geographic distribution of these species overlaps at certain coasts, which are the boundaries of temperature zones. The cosmopolitan species have a wider temperature tolerance than tropical species. There are also occurrence of terrestrial fungi in the coastal ocean, especially in the woods of terrestrial origin.

Mangrove fungal diversity is dependent on the age of the mangrove, diversity of mangrove plant species and the physicochemical features of mangrove habitat (temperature, salinity and tidal range). Mangrove fungi encompass oomycetes, thraustochytrids, ascomycetes, basidiomycetes and deuteromycetes. Detritus and live parts of mangrove vegetation (tree species, mangrove associates, ferns and palms) have been surveyed for the occurrence of higher fungi. So far about 110 species belonging to 53 genera of marine fungi are known from the peninsular India. However in recent past (up to 2000), 625 fungi encompassing 279 ascomycetes, 277 deuteromycetes, 29 basidiomycetes, 3 chitridiomycetes, 2 myxomycetes, 14 oomycetes, 9 thraustochytrids and 12 zygomycetes have been reported from mangrove forests worldwide (Schmidt and Shearer, 2003). Table 4.1 contains a consolidated list of fungi encountered in the coastal ocean and mangrove habitats. It shows that the mangrove habitats have a dense flora of fungi in comparison to that of coastal oceans. The presence of a high biomass of detritus and lignified plant tissues facilitates the abundance of fungi in the mangrove zones.

#### 4.5.2 Fungal Colonization in Marine Habitat

Organic detritus (e.g. fallen leaves, twigs, bark) of the aquatic environment, that are well aerated, support rich fungal communities. Animal remains of aquatic and terrestrial origin also support fungal flora. Senescent mangrove leaves release from plant and reach distinct habitats: a) trapped above the water level in canopy, which is not accessible for tides, reach water due to storm in dried state; b) float on the surface of water as dried or fresh; c) reach the mangrove floor under low tide; d) trapped at some depth below the water surface; and e) accumulate on or buried in sediment. The dried twigs and branches detach during monsoon and trapped in canopy, get into water with or devoid of bark, settle or entrapped in sediment. The decayed stems, prop roots and pneumatophores remain attached to plants. The above conditions of substrates influence diversity and functions of mycoflora. The lower fungi follow the 'substrate-capture' strategy, while higher fungi adapt to 'mass-accumulation' strategy. It is suspected that periodical wet and dry regime of litters in the coasts and mangrove swamps leads to higher fungal activity. Under appropriate environmental conditions (intermittent drying at the intertidal zone) the filamentous fungi compete with lower fungi. In addition, dissolved organic nutrients, hydrogen ion concentration, osmotic effects, availability of oxygen, abundance of propagules, efficiency of attachment

to substrata, hydrostatic pressure, substrate preference, temperature, tidal amplitude and light also influence the colonization and growth of fungi in marine habitats.

Marine fungi colonize natural as well as artificial structures in turbulent conditions due to their efficient adherence capacity. Fungal spore adhesion has two phases: initial attachment (impaction, chemotaxis, entrapment) followed by release of extracellular adhesive materials (germ-tubes enveloped by a hyphal sheath, development of appressoria and penetration of substratum). Many ultrastructural studies have been carried out on the nature and function of spore appendages. Majority of such studies have demonstrated that the rate of colonization on the surface of the precolonized as well as virgin particles is determined by the nature of the appendages.

Higher fungi have been recovered from several animal substrata of coral reefs, coasts, beaches, coastal sand dunes and mangroves. A wide variety of animal remains have been surveyed for the occurrence of fungi: corals, shells of snails, balanids, bivalves, foraminifers, hydrozoan exoskeletons, tunicates, shipworm tubes, cuttlefish endoskeletons, feathers, snake skin, crab exoskeletons, beetle wings, bryozoan skeleton, horse hair and teredinid tunnels. Basidiomycetian fungi are found in majority of these substrates. However, these fungi are not considered the initial colonizers of these particle surface. Rather the colonization of bacteria on these surfaces form a conducive environment for fungal colonization. *Corollospora intermedia* is usually the most frequent fungus particularly on calcareous shells and crab exoskeletons in the tropical environments.

## 4.6 MARINE PHYTOPLANKTON

There are many kinds of marine organisms that fix inorganic carbon into organic compounds using external sources of energy. There are many thousands of producer species with widely different phyletic origins, including single celled phytoplankton and bottom dwelling algae, large many celled macroalgae or seaweeds, symbiotic producers such as corals and higher plants such as seagrasses.

The highly diverse population of phytoplankton may contribute up to 95% of marine primary production. This importance of phytoplankton is largely due to the large area of the earth that is covered by the open sea. In shallow water near coasts, attached single celled algae, larger multicellular algae and vascular plants are considerably more important contributors to marine primary production than the phytoplankton. However, in the nutrient depleted open ocean, the phytoplankton productivity is the only source of carbon to the vast population of consumers in the euphotic zones. This supply is not even sufficient to satisfy the carbon demand of the food web. Therefore, the overall production of the open ocean is supported by the movement of organic detritus from coastal ocean.

**Table 4.1:** List of commonly occurring fungi in the coastal oceans and mangrove habitats.

| Group        | Coastal ocean                            | Mangrove habitats                        |
|--------------|------------------------------------------|------------------------------------------|
| Ascomycotina | <i>Amylocarpus encephaloides</i>         | <i>Aigialus parvus</i>                   |
|              | <i>Aniptodera chesapeakeensis</i>        | <i>Amylocarpus encephaloides</i>         |
|              | <i>Aniptodera longispora</i>             | <i>Aniptodera chesapeakeensis</i>        |
|              | <i>Aniptodera</i> sp.                    | <i>Aniptodera indica</i>                 |
|              | <i>Antennospora quadricornuta</i>        | <i>Aniptodera longispora</i>             |
|              | <i>Arenariomyces majusculus</i>          | <i>Aniptodera mangrovei</i>              |
|              | <i>Arenariomyces parvulus</i>            | <i>Antennospora quadricornuta</i>        |
|              | <i>Arenariomyces trifurcatus</i>         | <i>Antennospora salina</i>               |
|              | <i>Ascomycete</i> sp.                    | <i>Arenariomyces parvulus</i>            |
|              | <i>Carbosphaerella leptosphaerioides</i> | <i>Arenariomyces parvulus</i>            |
|              | <i>Caryospora rhizophorae</i>            | <i>Arenariomyces trifurcatus</i>         |
|              | <i>Ceriosporopsis halima</i>             | <i>Ascocratera manglicola</i>            |
|              | <i>Corollospora angusta</i>              | <i>Ascomycete</i> sp.                    |
|              | <i>Corollospora cinnamomea</i>           | <i>Astrosphaeriella mangroveis</i>       |
|              | <i>Corollospora colossa</i>              | <i>Astrosphaeriella nypae</i>            |
|              | <i>Corollospora filiformis</i>           | <i>Carbosphaerella leptosphaerioides</i> |
|              | <i>Corollospora indica</i>               | <i>Caryospora rhizophorae</i>            |
|              | <i>Corollospora intermedia</i>           | <i>Ceriosporopsis halima</i>             |
|              | <i>Corollospora lacera</i>               | <i>Corollospora angusta</i>              |
|              | <i>Corollospora maritima</i>             | <i>Corollospora colossa</i>              |
|              | <i>Corollospora pulchella</i>            | <i>Corollospora filiformis</i>           |
|              | <i>Corollospora quinqueseptata</i>       | <i>Corollospora indica</i>               |
|              | <i>Corollospora</i> sp.                  | <i>Corollospora intermedia</i>           |
|              | <i>Coronopapilla mangrovei</i>           | <i>Corollospora maritima</i>             |
|              | <i>Crinigera maritima</i>                | <i>Corollospora pulchella</i>            |
|              | <i>Dactylospora heliotrepha</i>          | <i>Corollospora quinqueseptata</i>       |
|              | <i>Didymosphaeria lignomaris</i>         | <i>Coronopapilla mangrovei</i>           |
|              | <i>Didymosphaeria</i> sp.                | <i>Crinigera maritima</i>                |
|              | <i>Dryosphaera novigans</i>              | <i>Dactylospora heliotrepha</i>          |
|              | <i>Dryosphaera tropicalis</i>            | <i>Didymosphaeria lignomaris</i>         |
|              | <i>Halosarpheia fibrosa</i>              | <i>Didymosphaeria</i> sp.                |
|              | <i>Halosarpheia marina</i>               | <i>Dryosphaera tropicalis</i>            |
|              | <i>Halosarpheia ratnagiriensis</i>       | <i>Eutypa bathurstensis</i>              |
|              | <i>Halosarpheia retorquens</i>           | <i>Halorosellinia oceanica</i>           |
|              | <i>Halosarpheia</i> sp.                  | <i>Halosarpheia fibrosa</i>              |
|              | <i>Halosarpheia viscosa</i>              | <i>Halosarpheia marina</i>               |
|              | <i>Halosphaeria salina</i>               | <i>Halosarpheia phragmiticola</i>        |
|              | <i>Halotithia posidoniae</i>             | <i>Halosarpheia viscosa</i>              |
|              | <i>Hypoxylon oceanicum</i>               | <i>Halosphaeria cucullata</i>            |
|              | <i>Kallichroma tethys</i>                | <i>Halosphaeria salina</i>               |
|              | <i>Leptosphaeria australiensis</i>       | <i>Halotithia posidoniae</i>             |
|              | <i>Leptosphaeria pelagica</i>            | <i>Julella avicenniae</i>                |
|              | <i>Lineolata rhizophorae</i>             | <i>Kallichroma tethys</i>                |
|              | <i>Lophiostoma mangrovei</i>             | <i>Leptosphaeria australiensis</i>       |

**Table 4.1:** (Contd.)

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | <i>Lulworthia grandispora</i><br><i>Lulworthia kniepii</i><br><i>Lulworthia sp.</i><br><i>Lulworthia sp.</i><br><i>Lulworthia sp.</i><br><i>Massarina velatospora</i><br><i>Mycosphaerella salicorniae</i><br><i>Mycosphaerella staticicola</i><br><i>Nimbospora bipolaris</i><br><i>Nimbospora effusa</i><br><i>Phaeosphaeria sp.</i><br><i>Phaeosphaeria spartinaecola</i><br><i>Pleospora avicenniae</i><br><i>Savoryella lignicola</i><br><i>Savoryella paucispora</i><br><i>Swampomyces armeniacus</i><br><i>Torpedospora radiata</i><br><i>Trematosphaeria striatispora</i><br><i>Verruculina enalia</i> | <i>Leptosphaeria australiensis</i><br><i>Leptosphaeria pelagica</i><br><i>Lineolata rhizophorae</i><br><i>Linocarpon appendiculatum</i><br><i>Linocarpon nypae</i><br><i>Littispora ratnagiriensis</i><br><i>Lophiostoma mangrovei</i><br><i>Lulworthia grandispora</i><br><i>Lulworthia sp.</i><br><i>Lulworthia sp.</i><br><i>Marinosphaera mangrovei</i><br><i>Massarina acrostichi</i><br><i>Massarina ramunculicola</i><br><i>Massarina velatospora</i><br><i>Mycosphaerella salicorniae</i><br><i>Mycosphaerella staticicola</i><br><i>Natantispora retorquens</i><br><i>Nimbospora bipolaris</i><br><i>Nimbospora effusa</i><br><i>Oxydothis nypae</i><br><i>Passeriniella mangrovei</i><br><i>Phaeosphaeria spartinaecola</i><br><i>Pleospora avicenniae</i><br><i>Rhizophila marina</i><br><i>Savoryella lignicola</i><br><i>Savoryella longispora</i><br><i>Savoryella paucispora</i><br><i>Swampomyces armeniacus</i><br><i>Swampomyces triseptatus</i><br><i>Torpedospora radiata</i><br><i>Trematosphaeria striatispora</i><br><i>Tunicatispora australiensis</i><br><i>Verruculina enalia</i> |
| Basidiomycotina | <i>Calathella mangrovei</i><br><i>Halocyphina villosa</i><br><i>Nia vibrissa</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Calathella mangrovei</i><br><i>Calathella sp.</i><br><i>Halocyphina villosa</i><br><i>Halocyphina sp.</i><br><i>Nia vibrissa</i><br><i>Nia sp.</i><br><i>Nia sp.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Deuteromycotina | <i>Alternaria sp.</i><br><i>Camarosporium sp.</i><br><i>Cirrenalia pseudomacrocephala</i><br><i>Cirrenalia sp.</i><br><i>Cladosporium algarum</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>Alternaria sp.</i><br><i>Camarosporium sp.</i><br><i>Cirrenalia basiminuta</i><br><i>Cirrenalia pseudomacrocephala</i><br><i>Cirrenalia pygmea</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



**Table 4.1:** (Contd.)

|                                     |                                       |
|-------------------------------------|---------------------------------------|
| <i>Cladosporium</i> sp.             | <i>Cirrenalia tropicalis</i>          |
| <i>Clavatospora bulbosa</i>         | <i>Cladosporium algarum</i>           |
| <i>Deuteromycete</i> sp.            | <i>Cladosporium</i> sp.               |
| <i>Deuteromycete</i> sp.            | <i>Clavatospora bulbosa</i>           |
| <i>Dictyosporium pelagicum</i>      | <i>Colletotrichum gloeosporioides</i> |
| <i>Monodictys pelagica</i>          | <i>Corynospora cassicola</i>          |
| <i>Periconia prolifica</i>          | <i>Deuteromycete</i> sp.              |
| <i>Periconia</i> sp.                | <i>Dictyosporium pelagicum</i>        |
| <i>Phoma</i> sp.                    | <i>Monodictys pelagica</i>            |
| <i>Trichocladium achrasporum</i>    | <i>Periconia prolifica</i>            |
| <i>Trichocladium alopallonellum</i> | <i>Phaeoisaria clematidis</i>         |
| <i>Trichocladium constrictum</i>    | <i>Phoma</i> sp.                      |
| <i>Trichocladium</i> sp.            | <i>Phoma suaedae</i>                  |
| <i>Zalerion maritimum</i>           | <i>Stachybotrys chartarum</i>         |
| <i>Zalerion varium</i>              | <i>Stachybotrys oenanthes</i>         |
|                                     | <i>Stachylidium bicolor</i>           |
|                                     | <i>Trichocladium achrasporum</i>      |
|                                     | <i>Trichocladium alopallonellum</i>   |
|                                     | <i>Trichocladium linderi</i>          |
|                                     | <i>Zalerion maritimum</i>             |
|                                     | <i>Zalerion varium</i>                |

Compiled from Ravikumar and Vittal (1996); Poon and Hyde (1998); Alias and Jones (2000); Patil and Borse (2001); Sarma *et al.* (2001); Ananda and Sridhar (2003, 2004); Sridhar (2005).

#### 4.6.1 The Taxonomic Groups of Phytoplankton

Similar to terrestrial vegetation, marine phytoplankton diversity is a unimodal function of phytoplankton biomass, with maximum diversity at intermediate levels of phytoplankton biomass and minimum diversity during massive blooms. Contrary to expectation, there is no relation between phytoplankton diversity and zooplankton diversity. Zooplankton diversity is a unimodal function of zooplankton biomass. Most strikingly, these marine biodiversity patterns show a worldwide consistency, despite obvious differences in environmental conditions of the various oceanographic regions.

The principal taxa of planktonic producers that are found over most of the world's oceans are diatoms, dinoflagellates, coccolithophorids, silicoflagellates and cyanobacteria. Diatoms have cell walls of silica and pectin, often with complex sculptured surfaces and range in size from 10 to 200  $\mu\text{m}$ . They occur floating in the water column or attached to surfaces, singly or as chains of cells. Diatoms are major contributors to production, especially in coastal waters. They occur everywhere in the sea, but are most abundant in colder, nutrient-rich waters and most common in spring and autumn in the coastal oceans of sub-tropics. Division occurs by fission and each fission is accompanied by a reduction in the size of the cell and silicious cell wall.

Dinoflagellates occur as single cells, either as naked cells or within cellulose cell walls and range in size from 1 to 100  $\mu\text{m}$ . They are planktonic, drifting with the water, although many species use flagella to move. Dinoflagellates are only second to diatoms in contributing to primary production and are widespread over the sea. Metabolically they are very versatile: they may engage in photosynthesis, be parasitic or symbiotic, absorb dissolved organic matter, or ingest particles of organic matter. Reproduction in dinoflagellates is by division and daughter cells grow to the size of the parent cell before dividing. The rate of division is regulated by the availability of organic carbon in the sea water.

Coccolithophorids occur as single cells and are protected by ornate calcareous plates (coccoliths) embedded in a gelatinous sheath that surrounds the cell. They are most abundant in warm, open-ocean waters, although sometimes abundant nearer shore. Coccolithophorids photosynthesize and may also absorb organic matter heterotrophically. Individuals may form cysts, from which spores develop to produce new individuals. Many species are flagellated. In organically rich coastal oceanic environments, the coccolithophytes form bloom under warmer conditions.

Silicoflagellates occur as single small (ca 60 $\mu\text{m}$ ) flagellated cells. The silicoflagellates typically secrete a silicious outer skeleton. These organisms are photosynthetic, but some may consume organic matter. They are common in cold, nutrient-rich water and reproduce by simple division.

Cyanobacteria are photosynthetic procaryotes with cell walls made of chitin. They occur as single, very small coccoid cell or as longer filaments consisting of many cells. There are also formation of colonies of single cells, which depends on the nutrient load of the water. Other groups of bacteria may be photosynthetic but are often restricted to waters with low oxygen content. The various types of bacteria may exist as single free floating cells, as single cells attached to surfaces, or as long filaments composed of many cells. Most marine bacteria are motile and gram-negative and divide by fission.

Prochlorophytes are a recently discovered group of extremely abundant producers, barely visible by traditional microscopic observations. They are most abundant in the lower layers of the illuminated regions of the water column and can reach a density of over  $10^9$  cells/ml in nutrient enriched coastal waters. However, in the upper surface of the euphotic zone these producers have low competitive advantage over the other microplankton. They have importance in the inner layers of the nutrient depleted open waters.

A new group of algae, called picobiliphytes, has been found in the nutrient poor open oceans and is believed to contribute substantially to the primary productivity due to their ability to grow in nutrient deficient open oceans. In this newly discovered algal group, the phycobiliproteins are found at the sites for photosynthesis, i.e. in the plastids.

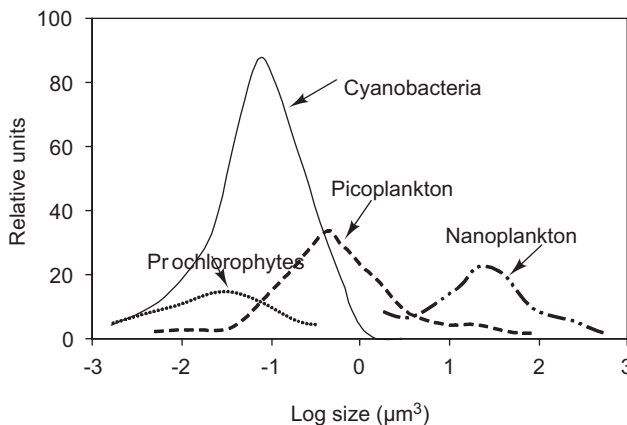
The size of the cells of bacteria and algae is an important feature of marine plankton and diameters of cells vary over several orders of magnitude. The picoplankton, mainly bacteria, prochlorophytes and cyanobacteria, dominate in the open oceans, though diatoms and desmids of smaller size classes are found in

the open oceans in certain part of the year. The nanoplankton, including diatoms, coccolithophores and silicoflagellates, are also found in open oceans, especially in the sub-tropical and temperate zones. The microplankton, green algae, desmids, diatoms and dinoflagellates, are common in coastal oceans and the mangrove vegetations contributing mostly to the coastal ocean primary productivity.

In general, phytoplankton in Indian ocean has been found to be most abundant during the upwelling period that lasts from May-June to October-November. Diatoms constitute the bulk of microplankton exhibiting rich diversity. Dinoflagellates are the next abundant group, occasionally forming blooms sometimes associated with fish kills and in rare cases resulting in paralytic shellfish poisoning. In September 2004 an incident of fish mortality, also accompanied by the release of an obnoxious gas (presumably  $H_2S$ ) from the sea that caused sickness to children, led to a public health alarm in the Kerala and Goa coast.

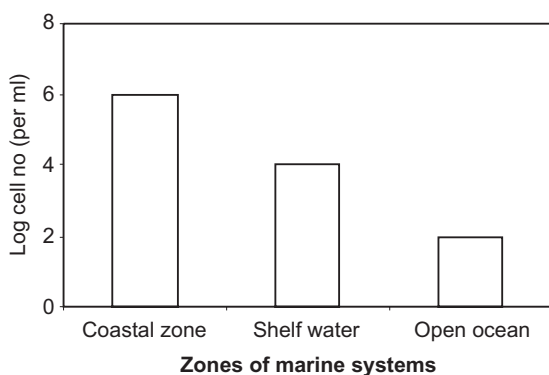
#### 4.6.2 Abundance of Various Groups of Phytoplankton

The taxonomic composition of the phytoplankton of a given body of seawater varies over time and space. There are many instances of localized short-term blooms, where one taxon may temporarily be very abundant, such as in the so-called red tides of estuaries and coastal waters, caused by rapid growth of red-pigmented dinoflagellates (*Gonyaulax*, *Gymnodinium*, etc.). Biomass of smaller producers in marine plankton is often greater than biomass of larger species. In the tropical Pacific, for example, the picoplankton makes up 39-63 % of total chlorophyll, 27-42 % is in the nanoplankton and only 9-16% is in the microplankton fraction. Picoplankton communities are generally dominated by prochlorophyte and cyanobacteria, which are common in tropical oceans and most likely represent systems associated with regenerated production. The greater abundance of picoplankton reflects the increase in macronutrient availability, which facilitate growth of microplankton (Fig. 4.9).



**Figure 4.9** - Relative abundance of phytoplanktons of different sizes in open waters of the Mediterranean Sea (Modified from Valiela, 1998).

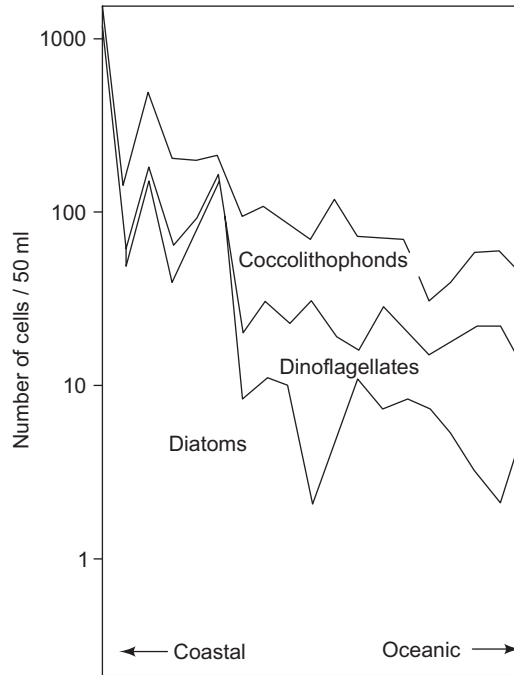
There are, however, some notable and regular regional patterns. Phytoplankton abundance decreases along coastal to oceanic gradients. Density of microplankton reaching several thousand algal cells per milliliter are common near shores and in continental shelf water while for ocean it is more typical for algal microplankton to average a few hundred cells per milliliter (Fig. 4.10). However, such pattern is not true for nano- and pico-phytoplankton. Nevertheless there is reduction in the total phytoplankton density with increase in the distance from the coast. As a result the net carbon fixed by the planktonic producers also decreases with the gradient from coast to the open ocean.



**Figure 4.10** - Abundance of phytoplankton in different zones of marine system. The representation is a general trend found in the marine system.

The fertility of sea water also affects the taxonomic composition of phytoplankton. Diatoms, for example, may dominate in the coastal regions enriched with nutrients, notably nitrogen and silica. On the other hand, the dominance of coccolithophorids and dinoflagellates increases in nutrient-poor oceanic water offshore (Fig. 4.11). The relative dominance of one or another taxon in the water column has many important consequences. For example, phytoplankton dominated by coccolithophorids provide only small particles for grazing animals or may produce a calcareous ooze in the sediment of the sea floor below. A diatom dominated phytoplankton population, in contrast, may furnish larger particles and a silicious ooze below. Such phenomena have important bearings in the energy availability to the rest of the grazing food chain of the oceanic systems.

In oceanic waters, colonial or at least aggregated phytoplankton cells can be found. Visible clumps of cells of the cyanobacterium *Trichodesmium* are common in surface waters of the Sargossa sea, Caribbean and other nutrient poor areas in the coastal oceans. Mats of the diatom *Rhizosolenia* are frequent in the tropical Atlantic. Larger multicellular algae may be locally abundant in the plankton population of the coastal regions. Species of the brown alga *Sargassum* are common in the surface waters of the tropical atlantic. In certain estuaries, free floating fronds of brown algae become dense, especially there is plenty of nutrients available.



**Figure 4.11** - Changes in composition of microplankton at a series of stations from the coast of Venezuela out to the Caribbean Sea (Adopted from Valiela, 1998).

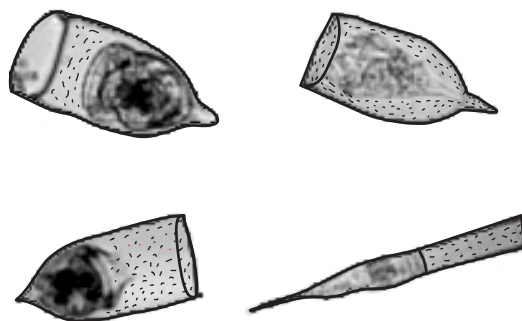
### 4.6.3 Microalgae in Coral Reefs

The symbiosis between special algal species and reef corals is the foundation of a highly productive and biologically complex ecosystem, but our understanding of how this symbiosis is established by new corals has been limited by the fact that the symbiotic algae are difficult to find and study in the ocean. But now a group of researchers has successfully identified algae of the genus known to represent coral symbionts and has gone on to show that the isolated algae are indeed capable of establishing symbioses with new corals. The findings will definitely bolster future efforts to protect and rehabilitate coral reefs. Organisms that resemble coral symbionts, dinoflagellates that are similar to those of the *Symbiodinium* genus that grow within corals, have been isolated from both sand and the water column. However, neither the locations of these populations nor their ability to establish symbioses is known. For both our understanding of reef ecosystems and their conservation, it is critical to recognize where these symbionts reside in the ocean environment. A symbiotic dinoflagellate *Symbiodinium* is found in the water column as well as on ocean-bottom substrates and can infect new coral polyps. These isolates are therefore capable of establishing symbioses with corals and thus point to environmental sources of symbionts that may prove important in the recovery of reef-building corals after bleaching events.

## 4.7 THE MARINE PROTOZOA

The marine protozoa species are structurally quite similar to those in freshwater but are perfectly adapted to a marine environment. Some of the free swimming forms eat algae and others eat smaller protozoans. They are mainly marine organisms. Individuals can retract into its *lorica* if disturbed. When it comes out, the crown of cilia is used both for locomotion and to catch food.

The marine ciliates are classified into four groups: **Prostomata** (*Litonotus*, *Coleps*)—free swimming ciliates, **Karyorelicta**—only marine; **Polyhymenophora** with large cilia (*Stentor*, *Spirostomum*, *Euplotes*, tintinnids.); and **Hypostomata** including suctoria (*Paramecium*, *Vorticella*). One very elegant family are the Tintinnids, which look like a French champagne glass (Fig. 4.12). Tintinnids are numerous in seawater only during the summer months.



**Figure 4.12** - The cell structures of some marine tintinnids. The figure shows the species of the genus *Favella*.

Unlike in freshwater, shelled amoeba, which build rigid ‘houses’ with organic debris, are not present in marine samples. But these samples are rarely taken from the sea bottom and perhaps shelled forms are living amongst suitable detritus for their ‘home building’. However, the Rhizopoda family is well represented in the sea with *Amoebae* and *Filosea*.

### 4.7.1 Protozoa in Microbial Food Web

Research during the previous two decades has shown that protozoa consume a large part of phytoplanktonic primary production. The microbial food chain starts with bacteria using dissolved organic matter excreted by phytoplankton, then passes by way of small and large protozoa to mesozooplankton. Because there can be several steps in the microbial food chain, it may be thought that the consequences of a microbially-dominated marine ecosystem is a very low ecological transfer efficiency from primary producers to zooplankton. This would indeed be the case if the only function of the microbial food web is to pass organic carbon to higher trophic levels. However, it also has another function, that of recycling mineral nutrients, as implied by the term microbial loop. The microbial food web is thought

to be most important during summer in temperate seas with active participation of protozoa. At this time, nutrients are in short supply. Although each stage in the microbial food web involves consumption of organic matter, it also releases ammonium and phosphate into the water and these recycled nutrients can be taken up by algae, stimulating additional primary production. Because small autotrophs grow best under low-nutrient conditions, but cannot easily be captured by copepods, the microbial loop provides an effective means for the exploitation of primary production under these conditions. The crustaceans can feed on the larger protozoans, which provide them with nutritious and high-energy food, having done the hard work of capturing smaller organisms and pre-processing their contents.

## 4.8 THE MICROBIAL ACTIVITIES IN OCEAN FLOOR

The global ocean is an interconnected system that has components that interact over broad spans of time and space. These scalar interactions define, in part, the biological and biogeochemical states that are found in the various oceanic regions. For example, biological connections between the surface and deep waters can be rapid, with the nearly instantaneous seasonal transport of photosynthetic biomass from the surface to the abyss. Conversely, the physical circulation of deep-ocean currents, which originate in polar surface waters and circumnavigate the seafloor to eventually emerge back at the surface, takes more than 1,000 years. The sediments collect sinking materials from above and receive geologically and biologically produced, energy-rich materials from below. Collectively, the sub-seafloor can be viewed as an enormous microbial 'bioreactor', that processes vast amounts of organic and inorganic materials. Except for the upper few millimetres, most of this enormous sub-seafloor catalytic converter is anaerobic.

The sea bed environment ranges from the desert-like deep seafloor to the rich oases that are present at seeps, vents and food falls such as whales, wood or kelp. As well as the sedimentation of organic material from above, geological processes transport chemical energy—hydrogen, methane, hydrogen sulphide and iron—to the seafloor from the subsurface below, which provides a significant proportion of the deep-sea energy. At the sites on the seafloor where chemical energy is delivered, rich and diverse microbial communities thrive. However, most subsurface microorganisms live in conditions of extreme energy limitation, with mean generation times of up to thousands of years. Even in the most remote subsurface habitats, temperature rather than energy seems to set the ultimate limit for life and in the deep biosphere, where energy is most depleted, life might even be based on the cleavage of water by natural radioisotopes. Fifty years ago, the seabed was considered to be flat, uniform and biologically inert owing to the remoteness of its location from photosynthetic ecosystems and the extreme conditions that are present at the seafloor, such as high hydrostatic pressures (up to 1,100 bar) and low temperatures (less than 4°C). In the 1950s, sediment samples that were gathered on the Danish *Galathea* Deep-Sea Expedition from



depths of more than 10,000 metres were shown to contain millions of viable bacteria per gram. Later, the Deep-Sea Drilling Project in 1968 recovered highly diverse core material from subsurface sediments and the ocean crust, which precipitated a new perception of the ocean floor as a dynamic environment. The famous dive of the deep-sea submersible ALVIN to the Pacific mid-oceanic ridge near the Galapagos Islands revealed amazingly rich benthic communities of giant clams, tube worms and microbial biofilms that were associated with the venting of hot fluids, temperatures of up to 370°C and emanating black smoke from chimneys that were composed of precipitated minerals. Since then, the development of advanced instrumentation for observing, mapping and sampling the seabed and all of its features has enabled scientists to develop a picture of the deep-sea environment as a highly dynamic geo- and biosphere. Besides the vast desert-like plains of deep-sea mud there are also diverse landscapes of canyons, cold seeps, deep-water coral reefs, mud volcanoes, carbonate mounds, brine pools, gas hydrates, seamounts, ridges, fractures and trenches that are host to rich microbial communities, which we are only beginning to map and explore.

#### 4.8.1 The Deep Seafloor

Almost 95% of the seabed (67% of the Earth's surface) lies in water depths where the light intensity is too low to sustain photosynthetic production, the temperatures are close to freezing (−1°C to 4°C) and the availability of organic matter controls benthic productivity and biomass. Depending on the productivity of the overlying waters, oxygen can be completely consumed within the upper metre of sediments. This top-metre layer of the seabed, which is mixed by burrowing animals, the order of depletion of microbial electron acceptors is usually oxygen first, then nitrate, oxidized manganese and iron minerals, then sulphate and finally bicarbonate. Anoxic ocean floors, in which sulphate and carbon dioxide are the main electron acceptors, are found in upwelling areas and oxygen-minimum zones that are present at continental margins, as well as at hydrothermal vents and cold seeps. The most oligotrophic seafloor environments are found in the central ocean gyres, particularly in the South Pacific, where oxygen fully penetrates the thin sediment cover. Two mysteries have always fascinated deep-sea researchers who are investigating life at the deep seafloor: how do benthic communities satisfy their metabolic needs and which factors account for the high biodiversity of deep-sea benthic communities, particularly in the small size classes of organisms.

Most of the  $320 \times 10^6 \text{ km}^2$  of deep-ocean seafloor is covered with fine-grained sediments that are composed of biogenic, terrigenous, vulcanogenic and authigenic particles. Therefore, compared with the water column, the ocean seafloor provides a huge area of solid surfaces and heterogeneous pore spaces, as well as a high concentration of detrital organic matter per unit volume. In the sedimentary seafloor, transport processes, such as advection and mixing, are normally limited to a few centimetres per year by sediment porosity and compaction. Benthic microbial populations are partially immobilized so that

distinct energy-rich gradients can form. Seafloor sediments contain 10–10,000-fold more cells per unit volume than productive ocean-surface waters (Fig. 4.5). This striking difference in microbial biomass between benthic and pelagic habitats is probably due to a combination of factors, such as the high energy availability at the seafloor and limited grazing pressure. Nevertheless, the largest part of the ocean seafloor receives little organic carbon from sedimentation—on average, 1 g of carbon per  $\text{m}^2$  per year. Assuming that there is an average biomass of  $10^8$  cells per  $\text{cm}^3$  in the top metre of sediment and a carbon mass of 20 kg per cell, this input has to nourish a standing stock of 2 g of carbon microbial biomass per  $\text{m}^3$ . Hence, even at an assimilation efficiency of 10% and in the absence of other detritus feeders, the average turnover of the microbial community based on fresh organic matter input must be less than 5% per year at the seafloor and considerably less in subsurface sediments. A much larger potential source of organic carbon for energy and growth could be the aged, degraded detritus material that is buried in the sediments, which comprises approximately 0.1–1% of the sediment by dry weight. This sedimentary particulate organic matter (POM) is the equivalent of approximately 1–10 kg of carbon per  $\text{m}^3$  at the seafloor, but is apparently not accessible for microbial degradation.

#### 4.8.2 Microbial Abundance and Degradation of Organic Matter at the Seafloor

The investigations of global trends in the distribution and abundance of microbial cells showed that the abundance of cells in surface sediments is usually related to the input of fresh detritus, rather than to any other oceanographic parameter. There is a global decrease in cell numbers in the surface of the seafloor (the top 0–1 cm layer) along with an increase in water depth irrespective of the water physicochemical conditions. This can be explained by the decrease in the particulate organic carbon (POC) flux that occurs with an increase in ocean depth and distance from land. The bacterial biomass and vertical POC flux show a linear relationship. Feeding and bioturbation by macrofauna have an important role in POC degradation in the sea floor. The input of phytodetritus and other labile organic materials to deep-sea sediments triggers the production of hydrolytic enzymes and, subsequently, the growth of the community, on similar timescales, to those found in shallow-water sediments. However, this labile organic matter is only a minor fraction of the aged and buried sedimentary organic material.

It remains poorly understood which fractions of the sedimentary organic material are accessible to microbial degradation and on what timescale these fractions are digested, oxidized and assimilated. Humic substances might be resistant to hydrolytic enzymes and their use as an energy and nutrient source seems to depend on a range of physicochemical and biological factors, such as the adsorption of organic matter to minerals or pre-digestion by animals. There might be unknown mechanisms by which microorganisms desorb organic matter from mineral particles and these might also have a role in the weathering of minerals in the seabed. The identification of the main microorganisms that are

present at the seafloor and the enzymes that they use for organic matter hydrolysis and fermentation is necessary to gain an understanding of the degradation and burial of POM in the seabed.

### 4.8.3 Microbial Diversity at the Seafloor

The general patterns distribution, diversity and dominance of functional and taxonomic groups of bacteria and archaea at the vast ocean seafloor are largely unknown. Considerable efforts are required to isolate and cultivate certain barophilic and psychophilic bacteria, such as some members of the *Moritella*, *Colwellia* and *Shewanella* clades. However, their environmental relevance remains undetermined. Observations have shown that the bacterial members of the heterotrophic Chloroflexi and archaeal members of the uncultivated miscellaneous Crenarcheota group dominated in the top metre of sediment cores. Although thorough studies have not yet been done, in terms of cell biomass, archaea seem to comprise a small portion of the total microbial community of the detritus-fuelled oxic seafloor, in contrast with the anoxic subsurface, where archaea might dominate. Possible shifts in benthic microbial community structure that could be associated with depth, latitude, season, composition of benthic fauna or POC fluxes have not yet well been investigated, although such shifts are evident in the community structure of benthic animals. Furthermore, little is known about the roles, host specificity or diversity of the viruses in the seabed, although their numbers and production rates indicate that they might have a significant influence on bacterial mortality. Most of the information on benthic microbial diversity has come from analyses of hot-spot ecosystems, such as hydrothermal vents, cold seeps and gas-hydrate systems (see chapter 7).

## 4.9 CONCLUSION

The past 30 years of marine microbiology and microbial oceanography have witnessed remarkable progress. Advances range from the unanticipated recognition of the large biomass of the oceanic picoplankton to the determination of the complete genome sequences of some of the most abundant inhabitants of the oceanic microbial world. The important early developments, such as the microbial-loop concept outlines the challenges that are associated with understanding cell-to-cell variability in metabolic rates and growth efficiencies. Gaining a better “bacterium’s-eye view” of pelagic microhabitats is central to deriving better models for the turnover of dissolved and particulate organic matter in the sea. In particular, the interactions between marine bacteria and archaea with live phytoplankton and particulate organic matter represent ‘hot spots’ of microbial activity in the sea. Particle consumption and dissolution, catalysed by ‘micro-local’ microbial biocatalysis, for example, have the potential to dramatically influence carbon export to the deep sea. The microscale processes have global consequences. For that reason, improved observations of microscale biogeochemistry and ecology might be crucial to better model and predict

microbial processes of global importance.

In ocean science, molecular biological and metagenomic applications have made substantial contributions to the discovery and description of several new and unsuspected metabolic features of marine microbial plankton. The balance between the amount of carbon dioxide that is fixed and organic carbon that is respired depends on the net activities of communities of autotrophic, heterotrophic and mixotrophic microorganisms. However, the metabolic properties of individuals in these communities and the balance of their activities, are still not well understood. The alternative metabolic strategies for energy acquisition, other than photosynthesis may be more important than previously supposed. New insight to the carbon pathway in the ocean will be probably the most dominant field of research in future. Modelling studies that link the Earth's history, physics and geology with microbial physiology and ecology will probably play increasingly important parts in predicting the earth-system climatic and ecological trajectories in the future.

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# MICROBIAL DIVERSITY OF TERRESTRIAL ENVIRONMENT

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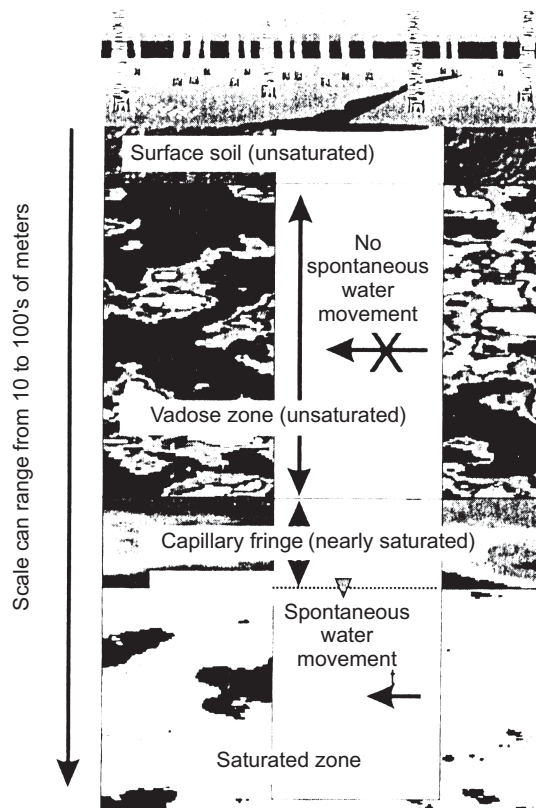
## 5.1 INTRODUCTION

Terrestrial environments, with a great degree of variation, range from the tropical rain forests in South America to the Sahara Desert in Africa. They are arguably the richest and most complex of all microbial environments. Broadly speaking the terrestrial environment, on microbiological point of view, consists of the soil environment and the active surface provided by the vegetation for microbial colonization. Irrespective of the physicochemical characteristics the soils usually teem with activity and a diversity of microorganisms. Environmental microbiologists have become more interested in this vast diversity of terrestrial microorganisms because of the potential to harness their unique activities, such as pollution abatement, sustainable agriculture and new applications in biotechnology. For example, it is increasingly apparent that terrestrial microorganisms have the capability to degrade most of the chemical pollutants that humans have added to the environment. The challenge is to stimulate the appropriate activities at a rate that is sufficient to protect human, animal and ecological health. Discoveries of microorganisms that inhabit extreme environments, such as the deep subsurface or contaminated soils, have added to the increasing interest in the unusual properties of enzymes made by these organisms for use in biotechnology. Examples include enzymes for use as detergent additives that work optimally at low temperatures, the continuing search for new antibiotics and new enzymes to help in food processing. The identification and synthesis of xenobiotic scavenging enzymes like phosphotriesterase and lindane induced proteins have added new dimension to the microbial purification of the environment. The superbugs, that can degrade recalcitrant organic pesticides and immobilize toxic metals have been identified from the contaminated soil environments. Thus, studies on the microbial

processes in the terrestrial environment in general and soil environment in particular will add new insight to the utility of microbes in human welfare.

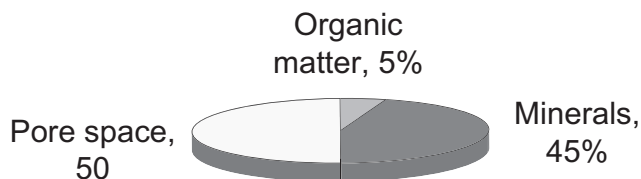
## 5.2 THE SOIL ENVIRONMENT

The key to find out new microorganisms and harness their activities is to develop a thorough understanding of the soil environment. To better understand microbial populations and their activities within these environments, it is necessary to analyse the different zones of the soil layer on the earth, viz., (1) surface soils, (2) the unsaturated or vadose zone and (3) underground aquifers or the saturated zone (Fig. 5.1). Each of these regions comprises a combination of mineral, solution and gas phases that are more commonly called a porous medium. All porous media are three-phase systems consisting of: (1) a solid or mineral inorganic phase that is often associated with organic matter, (2) a liquid or solution phase and (3) a gas phase or atmosphere. Considering the volume, almost half of the soil system is occupied by pore space under standard conditions and may be further high in highly porous and silicate rich soil. The pore space, however, is filled with soil solutions, which can be as high as 100% in flooded



**Figure 5.1** - The cross section of the subsurface showing different zones of soil.

soil (no free pore space), to as low as 0% in dry soil (Fig. 5.2). All phases interact together when the system is perturbed and then move towards an equilibrium state to create a unique environment. If left undisturbed, this environment remains fairly constant only showing minor fluctuation within a homeostatic range with change of climate and seasons. The unique properties of any porous medium are dependent on the specific composition of each of these phases.



**Figure 5.2** - The proportionate amount of three basic components of soil.

### 5.2.1 The Solid Phase

On a volume basis about 50 % of the soil medium contains solid particles, most of which (95–99 %) are mineral particles. Silicon and oxygen are the two most abundant elements found within the mineral fraction of the earth's crust. These two elements, along with lesser amounts of other elements, combine in a number of ways to form a large variety of minerals like quartz, mica, etc. These primary minerals, that are derived from the weathering of parent rock, form mineral particles of different sizes (Table 5.1). The surface area of a fine clay particle can be five orders of magnitude larger than the surface area of a 2 mm sand particle, when the surface to volume (S/V) ratio is calculated. There is also corresponding decrease in the pore space with decrease of soil particles from sand to clay types. The distribution (on a per cent by weight basis) of these minerals within a porous medium defines its texture.

On the soil surface, sand, silt and clay particles are subject to extremely active microbial processes that result in the formation of microbial gums, polysaccharides and other extracellular microbial metabolites. These materials, along with the microbial cells themselves, act as glues that bind the primary sand, silt and clay particles together into a secondary structure called soil aggregates or peds. Aggregate formation is unique for each soil and depends on the parent material, the climate and the biological activity within the soil. Soils with even modest amounts of clay usually have well-defined aggregates and hence a well-defined soil structure. Aggregates usually remain intact as long as the soil is not disturbed, for example, by ploughing and corrosion or erosion. In contrast, sandy soils with very low amounts of clay generally have less well-defined soil structure. In between the component mineral particles of a porous medium are pore spaces. These pores allow movement of air, water and microorganisms through the porous medium. Pores that exist between aggregates are called inter-aggregate pores, whereas those within the aggregates are termed as intra-aggregate pores (Fig. 5.3).



Table 5.1: Size fractionation and biological constituents of soils.

| Type of soil  | Specific surface area (m <sup>2</sup> /g) | Size (μm) | Mineral constituents                                                         | Organic constituents                       | Biological constituents                                         |
|---------------|-------------------------------------------|-----------|------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------|
| Sand          | 0.0003                                    | 2000      | Quartz, Silicates, Carbonates                                                | Organic debris                             | Poor bacterial and fungal flora                                 |
| Silt          | 0.12                                      | 50        | Quartz, Silicates, Carbonates                                                | Organic debris                             | Bacterial colonies, Fungi, Actinomycetes, Cyanobacteria         |
| Granular clay | 3                                         | 2         | Microcrystals of quartz and silicate, Illite, Mica, Smectite, etc.           | Amorphous organic matter, Humic substances | Bacteria, Fungal spores and hyphae, Large viuses, Cyanobacteria |
| Fine clay     | 30                                        | 0.2       | Swelling clay minerals, Interstratified clay minerals, Crystalline compounds | Organic matter deficient                   | Microbe deficient, Small viruses                                |

Adopted from Robert and Chenu (1992).

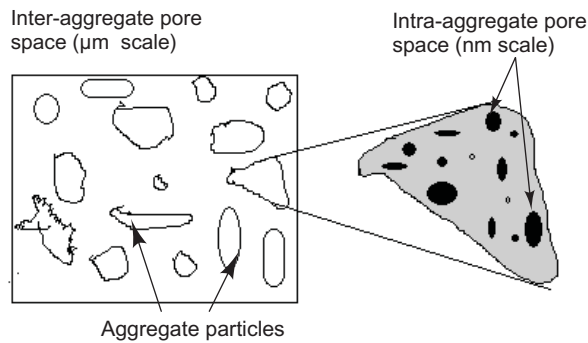


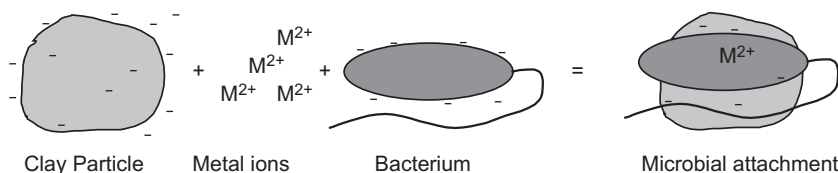
Figure 5.3 - Structure of a surface soil showing the arrangement of aggregate particles. One aggregate particle has been enlarged to show the intra-aggregate pores.

The clay particles are particularly dominant in determining the physical and chemical characteristics of a soil. For example, clays, which are often composed of aluminium silicates, add both surface area and charge to a soil. They affect not only the surface area of a porous medium but also the average pore size. Although, the average pore size is smaller in a clay soil, there are many more pores than in a sandy soil and as a result the total amount of pore space is larger in a fine-textured (clay) soil than in a coarse-textured (sandy) soil. However, because small pores do not transmit water as fast as larger pores, a fine-textured

medium is slow in the movement of any material through it, including air, water and microorganisms. At the same time, the clayey soil provides a large surface area (high S/V ratio) and sufficient substrate to microbes resulting in abundant population of bacteria, protozoa and algae. Often, fine-textured regions or layers of materials, e.g., clay lenses, can be found in sites composed primarily of coarser materials, creating very heterogeneous environments favourable for microbes to colonize. In this case, water will prefer to travel through the coarse material and flow around the finely textured lens. However, clay lenses retain water more tenaciously than sandy materials because of the smaller pore spaces. For microorganisms, which are much larger than individual water molecules, a fine-textured horizon or lens will inhibit bacterial movement either into or out of the region. Such heterogeneity poses great difficulties when trying to remove contaminants because some finely textured regions are relatively inaccessible to water flow or to microorganisms. Thus, contaminants trapped within very small pores may remain there for long periods of time, acting as a long-term “sink” of contaminant that diffuses out of the pores very slowly with time.

The total amount of negative charge in a soil, due to both clay and organic matter content, is usually measured in terms of millimoles of positive charge per kg of soil. A cation exchange capacity (CEC) of 150 to 200 millimoles positive charge per kg soil is an average value for most soils. For inorganic charged molecules, cation exchange is one of the primary mechanisms of sorption. Sorption is also a major process influencing the movement and bioavailability of essential compounds and pollutants in soil. Generally, positively charged ions, for example, calcium ( $\text{Ca}^{2+}$ ) or cadmium ( $\text{Cd}^{2+}$ ), participate in cation exchange. Since sorbed forms of these metals are in equilibrium with the soil solution, they can serve as a long-term source of essential nutrient ( $\text{Ca}^{2+}$ ) or pollutant ( $\text{Cd}^{2+}$ ) that is slowly released back into the soil solution as the soil solution concentration of the cation decreases with time.

Attachment of microorganisms can also be mediated by numerous functional groups on clays (Fig. 5.4). Although the clay surface and microbial cell surface both have net negative charges, clay surfaces are neutralized by the accumulation of positively charged counter ions such as  $\text{K}^+$ ,  $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Fe}^{3+}$  and  $\text{Al}^{3+}$ . Together, these negative and positive surface charges form an electrical double layer, to which microbes are reversibly attached. The thickness of the clay double layer depends on the valence and concentration of the cations in solution. Higher valence and increased cation concentrations shrink the electrical double layer. Because the double layers of the clay particles and microbial cells repel each



**Figure 5.4** - The attachment of bacterial cell to charged clay particles via cation bridging.

other, the thinner these layers are, the lesser is the repulsion between the clay and cell surfaces. As these repulsive forces are minimized, attractive forces such as electrostatic and van der Waals forces allow the attachment of microbial cells to the surface. Thus most microbes in terrestrial environments exist attached to soil colloids, rather than existing freely in the soil solution. However, the microbial cells are always in a dynamic state of attachment and dissociation from the clay surface, with the equilibrium more directed towards the surface. The change in surface properties determine the rate of attachment and dissociation from the active surface.

### **5.2.2 Organic Matter**

Organic matter in porous media is defined as a combination of (1) live biomass, including animals microbes and plant; (2) dead and decaying biological matter; and (3) humic substances, which are heterogeneous polymers formed during the process of decay and degradation of plant, animal and microbial biomass. Because subsurface media contain very little organic matter, the microbial density and diversity is very limited in this zone. Surface soils have a combination of these types of organic matter, whereas subsurface environments usually contain only small amounts of humic substances as well as a small amount of biomass in the form of microorganisms.

Because of their high stability and slow degradation rate, humic substances provide the stable, long-term microbial nutrient base in a soil. Humic substances have extremely complex structures that reflect the complexity and diversity of organic materials produced in a typical soil. During degradation of plant, animal and microbial biomass, a diversity of simple components such as sugars, amino acids, lipids and phenolic compounds are released. Some of these components are degraded rapidly, mostly by aerobic processes, whereas others resist degradation. Some of the more recalcitrant compounds are spontaneously or enzymatically polymerized, resulting in the formation of the humic substances. Therefore, humic substances in different soils are never exactly the same and have varying physical and chemical characteristics.

Humic substances range in molecular weight from 700 to 300,000 and can be divided in three fractions based on their chemical structures: fulvic acid, humic acid and humin. These materials are irregular polymers and their separation is based on differences in molecular weight, oxygen content and number of acidic functional groups. Each particle has a three-dimensional, sponge-like structure that contains hydrophobic and hydrophilic areas. As a result, it is generally less polar than water and therefore provides a favourable environment for solutes that are less polar than water. This means that humus can sorb nonpolar solutes from the general soil solution through a sorption process called hydrophobic binding. Humic substances also contain numerous functional groups, the most important of which are the carboxyl group and the phenolic hydroxyl group, both of which can become negatively charged in the soil solution. These groups are similar to those found on clays and can contribute to the pH dependent electrical

conductivity of soil and participate in sorption of solutes and attachment of microorganisms by cation exchange (See Section 5.6).

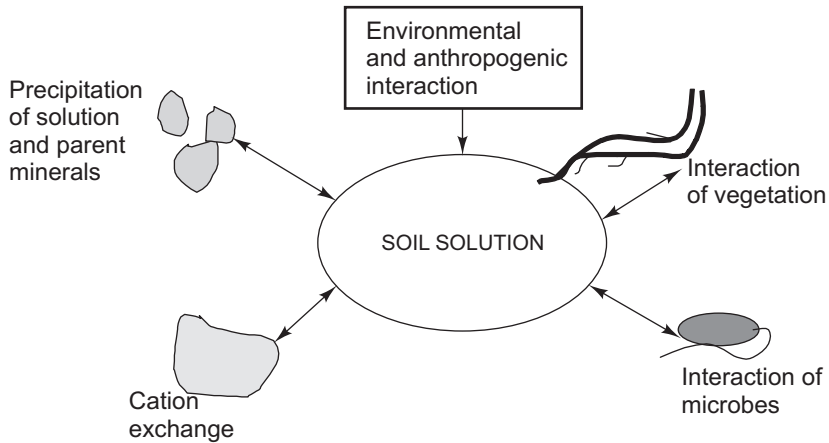
Humic substances share two properties with clay: they have a net negative charge and a large surface area. However, because humic materials are carbon based rather than silicon based, they are much more reactive in the sense that they undergo slow but constant change as a result of biological activity. Therefore, in a porous medium, organic matter and in particular humic substances exert a large influence on microbial activity determining their density, diversity and activity. Because rates of organic input, primarily from plant production, are similar to rates of organic matter degradation, undisturbed sites establish relatively constant organic matter concentrations. As a result of microbial degradation, the stable fraction of soil organic matter mineralizes at a rate of approximately 2–5% per year, depending mostly on seasonal soil temperatures. Thus, organic matter does more than just providing high surface area and charge in a soil. It also serves as a slow release source of carbon and energy for the autochthonous (indigenous), slow-growing microorganisms in the soil. Hence the number of microorganisms found in a soil often increases as the soil organic matter content increases. Numbers of microorganisms are also especially high in the soil directly surrounding plant roots and root hairs, called the rhizosphere (see section 5.6).

### 5.2.3 The Liquid Phase

The composition of the liquid phase, in soil, is extremely important for biological activity, because microorganisms are approximately 70% water and most require high levels of water activity ( $>0.95$ ) for active metabolism. Indeed, all microorganisms, even attached ones, are surrounded by a film of water from which they obtain nutrients and into which they excrete wastes. Thus, the amount and composition of the liquid phase ultimately controls both microbial and plant growth.

The soil solution composition reflects the chemistry of the soil as well as the dynamic influx and efflux of solutes in response to water movement. Water movement results from rainfall or irrigation events or from normal groundwater flow, mineral weathering and organic matter formation and decomposition (Fig. 5.5). This composition is also altered by anthropogenic activities, such as irrigation, fertilizer and pesticide addition and chemical spills. The chemistry of the soil affects not only its composition as well as of the soil solution but also modify the form and bioavailability of nutrients.

Water movement in a porous medium depends on the soil water potential, which is the work per unit quantity necessary to transfer an infinitesimal amount of water from a specified elevation and pressure to another point somewhere else in the porous medium. The soil water potential is a function of several forces acting on water, including matrix and gravitational forces. In the saturated zone, the presence of a regional hydraulic gradient usually results in a general horizontal flow of water. In contrast, flow is generally downward in the unsaturated zone. In an unsaturated zone, where the soil pores are not completely



**Figure 5.5** - The paths of mineral exchange with soil solution.

filled with water, there are several incremental forces that affect water movement. These are related to the amount of water present. In very dry soils, there is an increment of adsorbed water that exists as an extremely thin film on the order of angstroms ( $\text{\AA}$ ) in matric or capillary forces. This water exists as bridges between particle surfaces in close proximity and can actually fill small soil pores. This results in soil water potentials ranging from  $-0.1$  to  $-31$  atm. This water moves slowly from larger to smaller pores in any direction and is held against gravitational forces. The next increment of water added, free water, can be removed by gravitational forces. The soil water potential for free water ranges from  $0$  to  $-0.5$  atm. However, all categories of water actually occur as a continuum rather than with sharply defined boundaries in porous media.

The amount of water present in the pore space of a medium is another important parameter in understanding the level and type of microbial activity in an environment. The optimal environment for active aerobic microbial growth in a porous medium is one in which water is easily available but the medium is not completely saturated. Because microorganisms obtain nutrients from the water phase surrounding them, water is absolutely necessary for active microbial growth. Generally, microbial activity in the soil is greatest at  $-0.1$  atm. As the water potential becomes less negative, the soil becomes saturated. In a completely saturated environment, oxygen, which governs aerobic microbial activity, may become limiting because of its limited solubility ( $9.3$  mg/l at  $20^\circ\text{C}$  and  $1$  atm pressure) in water. Because the diffusion of oxygen through water is slow, once the available dissolved oxygen is used up, it is not replenished rapidly resulting in the formation of an anaerobic environment.

#### 5.2.4 Soil Air

There is absolutely no difference between the compositions of gases in the soil and atmospheric air. The soil air has the same basic composition: nitrogen,

oxygen and carbon dioxide as found in atmosphere. However, when aeration of soil is limited the concentration of gases in the soil become quite different from that of atmosphere. Plant and microbial activity can greatly affect the relative proportions of oxygen and carbon dioxide in soils that are not well aerated or that are removed from the soil surface. For example, in a clayey or a saturated soil, oxygen can be completely removed by the aerobic activity of respiring organisms causing elevated levels of  $\text{CO}_2$  in soil air. These changes affect the redox potential of the porous medium, which in turn changes the availability of terminal electron acceptors for aerobic and anaerobic microbes. On the other hand, in sandy or sandy loam soils, the rate of aeration is high enough to supply the required amount of oxygen for microbial respiration, thus maintaining the aerobic activity in the soil layers.

Diffusion is the prime mode of movement of gases in a soil. However, in most soils equilibrium conditions do not occur. Instead, as a result of biological activity, differences in amounts of  $\text{O}_2$  and  $\text{CO}_2$  exist between most soil environments and the atmosphere. Thus, for most soil, there is a net influx of  $\text{O}_2$  and a net efflux of  $\text{CO}_2$ . A second way in which gases can move within a soil is by mass flow, which occurs in response to a general change in pressure within a porous medium or in the atmosphere. Pressure changes can result from temperature changes, atmospheric pressure changes, or soil air pressure changes, the latter caused by a change in soil moisture.

### 5.2.5 Surface and Subsurface Environments

Surface soils are the weathered end-products of various soil forming factors, which involve action of climate and living organisms on a specific soil parent material, with a given topography over a given time period. (There are three major types of parent material: igneous rock, sedimentary rock and metamorphic rock.) The breakdown of parent materials into component minerals serves as a source of macro- and micronutrients for biological systems as they are established. It also results in the formation of physical niches for establishment and growth of microbial, plant and animal communities. As these communities become established and carry out their life cycles, organic materials are deposited in soils through the death and decay of constituent populations. This organic material is then degraded, resulting in soil organic matter.

A large proportion of the terrestrial environment contains surface soils that are unsaturated and aerobic except for brief periods of flooding during heavy rains or irrigation. Such surface soils have variable levels of productivity and organic matter content, which range from  $< 0.1\%$  (desert soil) to  $5\%$  (grassland) depending upon the microbial activity as well as biomass production. The organic matter level can even exceed  $20\%$  on a dry weight basis in wetland soil, where organic matter production is extremely high and degradation is comparatively slow due to limited supply of oxygen. Organic matter content as high as  $60\%$  or more is found in peat soil, where microbial activity is almost absent due to high acidity and temperature and nearly no availability of oxygen.

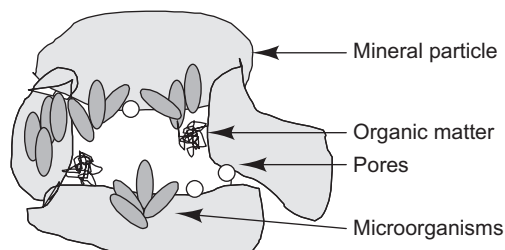
The subsurface soil is known as the vadose zone, which is an unsaturated oligotrophic environment that lies between the surface soil and the saturated zone. (The saturated and unsaturated zones are classified on the basis of water content in the pore space.) The vadose zone contains mostly unweathered parent materials and has a very low level organic carbon content (generally < 0.1 %). The thickness of the vadose zone varies considerably. When the saturated zone is shallow or near the surface, the unsaturated zone is narrow or sometimes even nonexistent, as in a wetland area. In contrast, there are many arid or semiarid areas of the world where the unsaturated zone can be up to hundreds of meters thick. These unsaturated zones, especially deep unsaturated regions, may receive little or no moisture recharge from the surface and normally have limited microbial activity because of low nutrient and/or moisture status. However, these regions are receiving more attention from a microbiological perspective, because pollutants that are present from surface contamination must pass through the unsaturated soil layers before they can reach groundwater.

The saturated zone is the most oligotrophic zone of soil with organic carbon content <0.1 %. The boundary between the vadose zone and the saturated zone is not a uniformly distinct one, because the water table can rise or fall depending on rainfall events. The area that makes up this somewhat diffuse boundary is called the capillary fringe. Saturated zones found beneath the vadose zone are commonly called aquifers and are composed of porous parent materials that are saturated with water. There are several types of aquifers, including shallow table aquifers and intermediate and deep aquifers that are separated from shallow aquifers by confining layers. The shallow aquifer systems are very active with rapid groundwater flows (meters per day) and hence usually remain aerobic. Deep aquifers, those more than 300 m in depth, are characterized by extremely slow flow rates (meters per century) and are usually anaerobic.

### 5.3 DISTRIBUTION OF SOIL MICROBES

Depending on the organic matter content and environmental conditions, soil are very diverse, qualitatively and quantitatively, in containing microorganisms. In surface soils, culturable microorganism concentrations can reach  $10^8$  per gram of dry soil, although direct counts are generally one to two orders of magnitude higher. These diverse microorganisms have been estimated to represent 10,000 species of bacteria alone. In addition, there are substantial populations of fungi (may be several thousand species), algae and protozoa. In general, microbial colonies are found in a non-uniform “patch-like” distribution on soil particle surfaces due to their tendency to maintain syntrophic activity (Fig. 5.6). Despite the large number of microorganisms found, they make up only a small fraction of the total organic carbon and a very small proportion of the soil volume (0.001 %) in most soils. Due to the presence of a large number of viable but nonculturable microbes, the culturable microbial population never exceed 2 % of the total microbial population. This indicates that a vast number of microbial processes are





**Figure 5.6** - Scheme of microbial colonization on the surface soil particles. Microbes are grouped as aggregates in patches on the soil particles to enjoy syntrophy.

controlled by hitherto unculturable microbial assemblage, opening the requirement for optimization of methods to make the best use of these microbial species.

The weathered component minerals (which serve as a source of micronutrients) and organic matter (which serves as a carbon and nitrogen source) are two of the primary differences between surface soils and subsurface materials as environments for microorganisms. These differences in nutrient content are reflected in a higher and more uniform distribution of microbial numbers and activity in surface soil environments. Further, microbial density and activity are greater in both surface and subsurface environments that have high recharge from rainfall and water flow. Microbial distribution is also dependent on soil texture and structure. In addition, in surface soils, pore space seems to be an important factor. Pore spaces in microaggregates with neck diameters less than 6  $\mu\text{m}$  have more activity than pore spaces with larger diameters, because the small pore necks protect resident bacteria from protozoal predation. Pore space also controls water content to some extent. It has been suggested that Gram-negative bacteria prefer the interior of microaggregate pore space because of the increased moisture, whereas Gram-positive bacteria, which are better adapted to withstand dry conditions, tend to occupy the microaggregate exteriors. There is also vast difference in these two groups of bacteria with regard to oxygen requirement for growth and reproduction.

Most microorganisms (80 to 90%) in soil are attached to solid surfaces in patches or colonies, which provide localized concentrations of nutrients that are contained in and recycled among the attached cells within the colony rather than being diluted into the general environment. Microbial aggregation makes it possible for many metabolites and nutrients to move from one cell to the other by direct cell to cell transfer without being released to the bulk liquid. This is especially important in oligotrophic environments. Another advantage of colony formation is that a microbial colony can alter the immediate microsite environment surrounding the cell, such as pH and gas concentration to optimize growth conditions. Genetic exchange can occur much more frequently within a colony than between isolated cells in a soil environment.

Although free-living cells are less common, dissociation of viable cells from the surface to the bulk liquid is an important mechanism of dispersion of

microorganisms. As nutrient supplies at a particular surface site are consumed, microorganisms need a mechanism by which cells can disperse to new sites that may have additional food supplies. Dissociation is also necessary when the quality of available substrates is changed due to microbial modification and succession is required to continue the degradation process. Dissociation facilitates the availability of surface to the secondary colonizers to build in their population in the biofilm. Fungi spread via spores released from fruiting bodies or via hyphal extension. Bacteria, which undergo only simple cell division, need a different mechanism of dispersal, namely release of free-living daughter cells. In fact, there is evidence that bacterial cells at the surface of a colony undergo changes in their surface properties that cause the release of a newly formed daughter cell after cell division. As these free-living daughter cells grow, their surfaces undergo a chemical change that makes attachment at a new site more favourable.

Soil is actually a discontinuous environment within which many different kinds of organisms can coexist. Viewed collectively, any given soil or subsurface material will generate a community having large numbers of organisms and great microbial diversity. For each member of the community there must be a habitat and a niche for that organism to survive. Competition for each habitat and niche ensures the “survival of the fittest” and also that all available substrates will be utilized. Thus, in “normal” soils or sediments, all available nutrients for bacteria are eventually metabolized. Readily degradable materials such as sugars and carbohydrates will exist only transiently (half-life is less than few hours), whereas substrates that are more difficult to degrade (e.g., lignin) remain present for longer periods of time. In particular, humic substances are degraded very slowly at rates acting as a constant source of nutrients for the soil microbial community (half-life can be as long as few decades).

The metabolic state of bacteria rapidly change in the soil, which occurs in response to soil physichomistry in general and soil organic matter in particular. Oxygen concentration in the pore space and the diffusion rate of evolved CO<sub>2</sub> out of soil have also some roles to play in modifying the metabolic state of the microbial assemblage. The only time that bacteria within soil exhibit high rates of metabolism is when fresh substrate is added to the medium or when a specific organism is able to exploit a previously unavailable substrate. Similarly high rates of metabolism can occur when an organism with specific degradative traits is “introduced” or added to the environment for the sole purpose of achieving biodegradation of an organic contaminant. Thus most bacteria in soil exist under conditions of limited starvation and stress as a result of competition for available nutrients. This is particularly true for populations in the subsurface, where extremely low amounts of available organic carbon can support only very low levels of microbial activity, though in surface soil nutrient depletion is ephemeral. The abiotic environmental factors like moisture and temperature also create adverse conditions for indigenous populations regulating their activity.

## 5.4 MICROBIAL DIVERSITY OF SURFACE SOIL

The indigenous populations of bacteria (including actinomycetes), fungi, algae and protozoa dominate the microflora of surface soils. In general, as the size of these organisms increases from bacteria to protozoa, the density gradually decreases. It is also known that there may be phage or viruses present on the surface soils, that can infect each class of organisms, though the virus diversity do not play any significant role in soil microbial processes. In addition, human or animal activity also introduces specific microbes into soil. This includes microbial manipulation for bioremediation of contaminated soils, irrigation by use of sewage water and use of biological agents for pest control. Animals introduce microbes through bird droppings and animal excrement. However, regardless of the source, introduced organisms rarely affect the abundance and distribution of indigenous populations in significant way. It is particularly true in habitats where the existing environmental conditions favour a high metabolic state of the indigenous assemblage and an introduced organism faces a severe competition for space (surface to colonize) and resources. Unless otherwise the indigenous microflora is stressed, the introduced microbial activity (in terms of degradation of organic matters) never exceed 5% of the total activity. The exception, however, is the conditions where intentional biomanipulation is made to facilitate the metabolism of a specific pollutant.

### 5.4.1 Soil Bacterial Diversity

On a numerical as well as diversity basis, bacteria are almost always the most abundant organism found in surface soils but most of them are difficult to culture. The culturable numbers vary depending on specific environmental conditions, particularly soil moisture and temperature but roughly lie between one ten to hundred million cells per mililitre. However, the total populations (including viable but nonculturable organisms) can exceed  $10^{10}$  cells per gram soil. In unsaturated soils, aerobic bacteria usually outnumber anaerobes by two or three orders of magnitude. Anaerobic populations increase with increasing soil depth but rarely predominate unless soils are saturated and/or clogged. Actinomycetes are an important component of bacterial populations, especially under conditions of high pH, high temperature, or water stress. Actinomycete numbers are generally one to two orders of magnitude smaller than the total bacterial population (Table 5.2). One distinguishing feature of this group of bacteria is that they are able to utilize a great variety of substrates found in soil, especially some of the less degradable insect and plant polymers such as chitin, cellulose and hemicellulose. Indigenous soil bacteria can be classified on the basis of their growth characteristics and affinity for carbon substrates.

All the soil bacteria can be grouped into two broad categories:

1. K-selected or autochthonous type that metabolize slowly released soil organic matter as a substrate and show extremely slow growth (K-strategists) and

**Table 5.2:** Characteristic features of Bacteria, Actinomycetes and Fungi found in terrestrial habitats.

| <i>Characteristic</i>                  | <i>Bacteria</i>                                      | <i>Actinomycetes</i>                                     | <i>Fungi</i>                   |
|----------------------------------------|------------------------------------------------------|----------------------------------------------------------|--------------------------------|
| Population                             | Most numerous                                        | Intermediate                                             | Least numerous                 |
| Biomass                                | As of actinomycetes<br>have similar biomass          | As of bacteria                                           | Largest biomass                |
| Degree of branching                    | Slight                                               | Filamentous, but<br>some fragment to<br>individual cells | Extensive<br>filamentous forms |
| Aerial mycelium                        | Absent                                               | Present                                                  | Present                        |
| Growth in liquid culture               | Yes-turbidity                                        | Yes-pellets                                              | Yes-pellets                    |
| Growth rate                            | Exponential                                          | Cubic                                                    | Cubic                          |
| Cell wall                              | Murein, teichoic<br>acid and lipo-<br>polysaccharide | Murein, teichoic<br>acid and lipo-<br>polysaccharide     | Chitin or cellulose            |
| Complex fruiting<br>bodies             | Absent                                               | Simple                                                   | Complex                        |
| Competitiveness for<br>simple organics | Most competitive                                     | Least competitive                                        | Intermediate                   |
| N <sub>2</sub> fixation                | Yes                                                  | Yes                                                      | No                             |
| Mode of respiration                    | Aerobic, anaerobic                                   | Mostly aerobic                                           | Aerobic except yeast           |
| Moisture stress                        | Least tolerant                                       | Intermediate                                             | Most tolerant                  |
| pH optimum                             | 6-8                                                  | 6-8                                                      | 6-8                            |
| pH competitive                         | 6-8                                                  | > 8                                                      | < 5                            |
| Competitiveness in<br>soil             | All soils                                            | Dominate dry and<br>alkaline soils                       | Dominate in acidic<br>soils    |

Source: *Pollution Science*, © 1996, Academic Press, San Diego, California.

2. r-selected or zymogenous type that metabolize easily degradable substrates and show rapid growth; they are adapted to intervals of dormancy and rapid growth, depending on substrate availability, following the addition of fresh substrate or amendment in the soil (r-strategists).

Overall estimates have indicated that surface soils can contain up to 10,000 individual species. Not all of these are culturable and of course information on the nonculturable species is limited. A conservative estimate indicates that about 60% of the soil bacteria are nonculturable making it difficult to critically analyze their exact role in soil habitat. Most studies of soil bacteria have focused on culturable organisms and a massive amount of information is available on key genera like *Arthobacter*, *Streptomyces*, *Pseudomonas* and *Bacillus* (Table 5.3). These bacterial genera mostly dominate in almost all soil habitats. The characteristics of few dominant soil bacteria have been given in Table 5.4.

**Table 5.3:** The features and functions of dominant soil bacterial genera. These genera are common to almost all soil types. Only important characteristics of the bacteria are given.

| Genus               | Characteristics                                                                                                                    | Function                                                                               |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <i>Arthrobacter</i> | Heterotrophic, aerobic, Gram variable, up to 40% of culturable soil bacteria                                                       | Nutrient cycling and biodegradation                                                    |
| <i>Streptomyces</i> | Gram positive, heterotrophic, aerobic actinomycete; 5-20% of culturable bacteria                                                   | Nutrient cycling and biodegradation; antibiotic production, e.g., <i>S. scabies</i>    |
| <i>Pseudomonas</i>  | Gram-negative heterotroph; aerobic or facultatively anaerobic; possess wide array of enzyme systems; 10-20% of culturable bacteria | Nutrient cycling and biodegradation, including recalcitrant organics; biocontrol agent |
| <i>Bacillus</i>     | Gram-positive aerobic heterotroph; produce endospores; 2-10% of culturable soil bacteria                                           | Nutrient cycling and biodegradation; biocontrol agents, e.g., <i>B.thuringiensis</i>   |

Adopted from Maier and Pepper (2006).

**5.4.2    The Soil Fungi**

Fungi other than yeasts are aerobic and are abundant in most surface soils. The population density of fungi usually range from 10<sup>5</sup> to 10<sup>6</sup> per gram of soil. Despite their lower numbers, compared with bacteria, fungi usually contribute a higher proportion of the total soil microbial biomass and contribute significantly to soil productivity. This is due to their comparatively large size, which also more or less restrict them to the inter-aggregate regions of the soil matrix. In addition the fungi have the capacity to use the organic matters (lignin and cellulose) that are most difficult to degrade and thus do not face any competition in soils rich in such organic matters. Yeasts can metabolize anaerobically (fermentation) and are less numerous than aerobic mycelium forming fungi because of their inability to compete with aerobic fungi in surface soil. Generally, yeasts are found at populations of up to 10<sup>3</sup> per gram of soil. Because of their reliance on organic sources for substrate, fungal populations are the greatest in the surface O and A horizons and numbers decrease rapidly with increasing soil depth. As with bacteria, soil fungi are normally found associated with soil particles or within plant rhizospheres.

Fungi are important components of the soil with respect to nutrient cycling and especially decomposition of organic matters, both simple (sugars) and complex (polymers such as cellulose and lignin). The role of fungi in decomposition is increasingly important as the soil pH declines because fungi tend to be more acid tolerant than bacteria. Some of the common genera of soil fungi involved in nutrient cycling are *Penicillium* and *Aspergillus*. These organisms are also important in the development of soil structure because they physically entrap soil particles with fungal hyphae. As well as being critical in the degradation of

**Table 5.4:** The features and functions of important autotrophic and heterotrophic bacteria in agricultural soil.

| Organism                                   | Characteristics                        | Function                                                                                  |
|--------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Autotrophic bacteria</b>                |                                        |                                                                                           |
| <i>Nitrosomonas</i>                        | Gram negative, aerobe                  | Converts $\text{NH}_4^+$ to $\text{NO}_2^-$ (first step of nitrification)                 |
| <i>Nitrobacter</i>                         | Gram negative, aerobe                  | Converts $\text{NO}_2^-$ to $\text{NO}_3^-$ (second step of nitrification)                |
| <i>Thiobacillus</i>                        | Gram negative, aerobe                  | Oxidizes S to $\text{SO}_4^{2-}$ (sulphur oxidation)                                      |
| <i>T. denitrificans</i>                    | Gram negative, facultative anaerobe    | Oxidizes S to $\text{SO}_4^{2-}$ (functions as a denitrifier)                             |
| <i>T. ferrooxidans</i>                     | Gram negative, aerobe                  | Oxidizes $\text{Fe}^{2+}$ to $\text{Fe}^{3+}$ (metal leaching)                            |
| <b>Heterotrophic bacteria</b>              |                                        |                                                                                           |
| Actinomycetes (e.g., <i>Streptomyces</i> ) | Gram positive, aerobic, filamentous    | Produce geosmins “earthy odour,” and antibiotics                                          |
| <i>Bacillus</i>                            | Gram positive, aerobic, spore former   | Carbon cycling, production of insecticides and antibiotics                                |
| <i>Clostridium</i>                         | Gram positive, anaerobic, spore former | Carbon cycling (fermentation), toxin production                                           |
| Methanotrophs (e.g., <i>Methylosinus</i> ) | Aerobic                                | Methane oxidizers that can cometabolize trichloroethene (TCE) using methane monooxygenase |
| <i>Alkaligenes eutrophus</i>               | Gram negative, aerobic                 | 2,4-D degradation via plasmid pJP4                                                        |
| <i>Rhizobium</i>                           | Gram negative, aerobic                 | Fixes nitrogen symbiotically with legumes                                                 |
| <i>Frankia</i>                             | Gram positive, aerobic                 | Fixes nitrogen symbiotically with non-legumes                                             |
| <i>Agrobacterium</i>                       | Gram negative, aerobic                 | Important plant pathogen, causes crown gall disease                                       |

Adopted from Maier *et al.* (2006).

complex plant polymers such as cellulose and lignin, some fungi can also degrade a variety of pollutant molecules. Fungal involvement becomes very important when cometabolism is required to initiate degradation of complex organic pollutants. In soil with high load of lignocellulosic substrates, the bacterial flora is more dependent on the fungal metabolic activities, compared to soil where there is availability of easily degradable organic matters. Many fungi in soil also thrive as plant pathogens. The biggest contribution of soil fungi to soil productivity is by the mycorrhizal fungi, which are critical for establishing plant-fungal interactions (see section 5.7).

### 5.4.3 The Soil Algae

The surface soil, where light and inorganic nutrient are not limited, is often thickly populated with filamentous and unicellular or colonial algae. High soil

moisture and good soil porosity facilitates the growth of algae but they are also recorded, though in less number and physiologically inactive forms, in comparatively dry soil. One can find algae at a depth of 1 m because some algae, including the green algae and diatoms, can grow heterotrophically as well as photoautotrophically. Many cyanobacteria also show heterotrophic growth when assimilable organic matters are available and light is limited. However, the growth and metabolic activities of cyanobacteria under heterotrophic mode is much lower than that observed in the presence of light. In general, though, algal populations are the highest in the surface 10 cm of soil, typical algal populations close to the soil surface range from 5000 to 10,000 per gram of soil. A surface soil, where a visible algal bloom has developed, can contain millions of algal cells per gram of soil. Such situation is observed when the surface soil remains barren with good availability of nutrients and light.

So far as the colonization of barren soil is concerned, algae are often the first to colonize surfaces in a soil that are devoid of preformed organic matter. Colonization by this group of microbes is important in establishing soil formation processes, especially in barren volcanic areas, desert soils and rock faces as well as to provide organic matters for subsequent occupation by a bacterial assemblage. Algae also produce and release carbonic acid, which aids in weathering the surrounding mineral particles. Further, algae produce large amounts of extracellular polysaccharides, which also aid in soil formation by causing aggregation of soil particles. Mutualistic association between soil algae and bacteria not only facilitate the growth and metabolic activity of these two groups of microbes but also hastens the soil formation process.

Populations of soil algae generally exhibit seasonal variations with numbers being the highest in rainy season in tropics but in spring and fall in subtropical and temperate region. This is because desiccation caused by water stress tends to suppress growth in the summer and cold stress affects growth in the winter. Four major groups of algae are found in soil. The green algae or the Chlorophyta, for example, *Chlamydomonas*, are the most common algae found in acidic soils. Other common green algae are the species of *Chlorella*, *Scenedesmus*, *Ankistrodesmus*, *Chlorococcum*, *Eudorina*, *Pandorina*, *Volvox* and *Selenastrum*. Also widely distributed are diatoms such as *Navicula*, which are members of the Chrysophyta. Diatoms are found primarily in neutral and alkaline soils. Less numerous are the yellow-green algae such as *Botrydiopsis*, which are also members of the Chrysophyta and the red algae (*Prophyridium*). In addition to these algal groups, there are the cyanobacteria (e.g., *Nostoc* and *Anabaena*, *Aulosira*, *Calothrix*, *Oscillatoria*, *Rivularia*, *Lyngbya*, etc.), which actually dominate the algal flora of flooded agricultural fields. The cyanobacteria participate in the soil-forming process and also they have the capacity to fix nitrogen, which is usually limiting in a barren environment as well as in rice fields. In temperate soils the relative abundance of the major algal groups follows the order green algae > diatoms > cyanobacteria > yellow-green algae. In tropical soils like ours the cyanobacteria predominate the algal flora, both in population and in metabolic performance. In a survey of rice field soil in Orissa during rainy



season in 2004, as many as 175 species of algae were reported of which 67 were cyanobacteria followed by 54 green algal species. Most of these species were reported in flooded rice fields but the diversity was significantly less in the non-flooded fields (Mohanty, 2004).

#### 5.4.4 Protozoa

Protozoa are unicellular, eukaryotic organisms that range up to 5.5 mm in length, although most are much smaller (Table 5.5). Most protozoa are heterotrophic and survive by grazing on bacteria, yeast, fungi and algae. There is evidence that they may also be involved, to some extent, in the decomposition of soil organic matter. Because of their large size and requirement for large numbers of smaller microbes as a food source, protozoa are found mainly in the top 15 to 20 cm of the soil. Protozoa are usually concentrated near root surfaces that have high densities of bacteria or other prey. Their population in the inner soil layers is limited due to poor supply of oxygen and non availability of sufficient bacteria to graze. Soil protozoa are flatter and more flexible than aquatic protozoa, which makes it easier to move around in the thin films of water that surround soil particle surfaces as well as to move into small soil pores.

**Table 5.5:** Average length and volume of soil protozoa compared with bacteria.

| Group       | Length<br>( $\mu\text{m}$ ) | Volume<br>( $\mu\text{m}^3$ ) | Shape                                        |
|-------------|-----------------------------|-------------------------------|----------------------------------------------|
| Bacteria    | <1–5                        | 2.5                           | Spherical to rod shaped                      |
| Flagellates | 2–50                        | 50                            | Spherical, pear shaped, banana shaped        |
| Amoebae     |                             |                               |                                              |
| Naked       | 2–600                       | 400                           | Protoplasmic streaming, pseudopodia          |
| Testate     | 45–200                      | 1000                          | Build oval tests or shells made of soil      |
| Giant       | 6000                        | $4 \times 10^9$               | Enormous naked amoebae                       |
| Ciliates    | 50–1500                     | 3000                          | Oval, kidney shaped, elongated and flattened |

Adopted from Maier and Pepper (2006).

There are three major categories of protozoa: the flagellates, the amoebae and the ciliates. The flagellates are the smallest of the protozoa and move by means of one to several flagella. Some flagellates (e.g., *Euglena*), contain chlorophyll, although most (e.g., *Oicomonas*) do not. The amoebae, also called rhizopods, move by protoplasmic flow, either with extensions called pseudopodia or by whole body flow. Amoebae are usually the most numerous type of protozoan found in a given soil environment. Ciliates are protozoa that move by beating short cilia that cover the surface of the cell. The protozoan population of a soil is often correlated with the bacterial population, which is the major food source present. Numbers of protozoa reported range from 30,000 per gram of soil from a nonagricultural temperate soil to  $1.6 \times 10^6$  per gram of soil from a subtropical area.

## 5.5 MICROBIAL DIVERSITY OF SUBSURFACE SOIL

### 5.5.1 Microorganisms of Upper Subsurface Layers

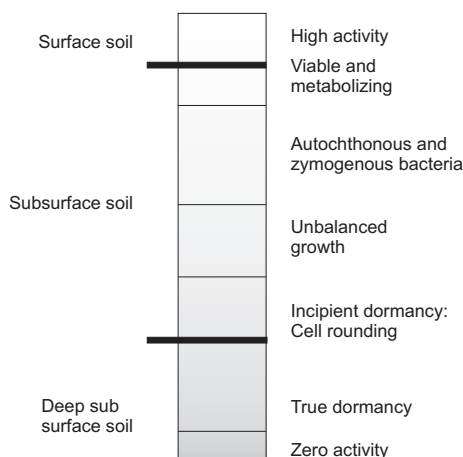
The studies on the microbial diversity of subsurface soil is relatively scanty, with most being concentrated on the numerical analysis of the microorganisms. Complicating the study of subsurface life are the facts that sterile sampling is problematic and many subsurface microorganisms are difficult to culture. Some of the initial studies evaluating subsurface populations were invalidated by contamination with surface microbes. As a result, study of subsurface organisms has required the development of new tools and approaches for sterile sampling and appropriate precautions to eliminate contamination by surface microbes. Relevant technology is also needed to be developed for microbial enumeration and identification. For example, it has been demonstrated that rich media are not suitable for culturing subsurface organisms that are adapted to oligotrophic conditions (probably the sub-surface soil microbes are K-strategists). To illustrate this point, a study by Bone and Balkwill (1988) compared viable counts from a full-strength PTGY (peptone, trypticase, glucose, yeast extract) agar medium with those from a 1: 20 diluted PTGY medium. In all cases the viable counts from the dilute PTGY medium were higher. This difference was approximately one order of magnitude for most samples, with some samples showing a difference of up to three orders of magnitude higher.

Despite limitations in collecting, enumerating and culturing the sub-surface microorganisms, there is still enough information available to know that these environments, once thought to contain very few if any microorganisms, actually have a significant and diverse population of microbes. In particular, shallow subsurface zones, specifically those with a relatively rapid rate of water recharge, have high numbers of microorganisms, most of whom are in direct exchange with the surface soil. There is evidence of the movement of microbes from surface soil to sub-surface soil and vice-versa with the movement of water. The majority of sub-surface soil microorganisms are bacteria, but protozoa and fungi, though at relatively low density and diversity, are also present. As a general rule, total numbers of bacteria, as measured by direct counts, remain fairly constant, ranging between  $10^5$  and  $10^7$  cells per gram throughout the profile of a shallow subsurface system. This is four to five order of magnitude lower than of the surface soil. This decrease in numbers is directly correlated with the low amount of inorganic nutrients and organic matter in subsurface materials. A part of the difference may also be attributed (probably) to the underestimation of the subsurface microbial density. Subsurface eukaryotic counts are also lower than surface counts by several orders of magnitude. Low eukaryotic counts are a result of low organic matter content but, perhaps more importantly, result from removal by physical straining by small soil pores as they move downward.

The soil texture significantly influences the distribution of microorganisms in the subsurface environment. Both prokaryotic and eukaryotic counts are highest in portions of the subsurface containing sandy sediments while in clayey and

muddy inner layers the count is noticeably low. This may also be due to exclusion and physical straining of microorganisms by small pores in clay-rich media. In addition, the composition of gas phase in the sub-surface environment significantly influences both prokaryotic and eukaryotic microbial diversity. The eukaryotes are mostly limited in the inner subsurface layers due to deficiency of oxygen. This is further severe if the organic load in the subsurface layers is more.

Numbers of culturable bacteria in subsurface environments show more variability than direct counts, ranging from zero to nearly equal to direct counts. Thus, in general, the difference between direct and cultural counts in the subsurface is greater than the difference in surface soils (one to two orders of magnitude). Several factors may explain the larger difference between direct and cultural counts in the subsurface. First, because nutrients are much more limiting in the subsurface, a greater proportion of the population may be in a nonculturable state. The physiological activity of the microbial assemblage gradually decreases with increase in depth of soil (Fig. 5.7). Second, the physiological and nutritional requirements of subsurface organisms are not well understood. Therefore, even though we know that a dilute nutrient medium is better than a rich medium, the type of dilute nutrient medium used may still be inappropriate.



**Figure 5.7** - The physiological states of microbial consortia in different layers of soil.

The dominant microbial population type in the subsurface soil is aerobic, heterotrophic bacteria, although there are small populations of eukaryotic organisms and small populations of anaerobic and autotrophic organisms. Subsurface microbes are diverse in type, although not as diverse as surface organisms. Predominant microorganisms vary from one depth to another within the subsurface, which, in turn, are controlled by the soil pore size and metabolite availability. Great diversity in heterotrophic activity has been found in subsurface layers and subsurface microbes have shown the capacity to degrade simple substrates such as glucose as well as more complex substrates such as aromatic

compounds, surfactants and pesticides. In general the anaerobic activity dominates in the subsurface soil layers. The physiology of subsurface microorganisms differs from that of surface microbes in that when subsurface cells are examined, they are rarely dividing and contain few ribosomes or inclusion bodies. This is not surprising considering the nutrient-limited conditions in which subsurface microbes live. What is more unexpected is that a significant portion of the viable population can quickly adapt to growth on rich nutrient media and that these organisms seem to metabolize a wide variety of organic compounds. These characteristics bode well for the promise that subsurface microorganisms can be harnessed for the *in situ* environmental cleanup of a wide variety of organic chemicals.

### 5.5.2 Microorganisms of Deep Subsurface Layers

The nutrient depleted milieu and oxygen deficiency of deep subsurface soil brings the basic conception that this environment contains a few, if any, microorganisms. However, many research findings, most notably the work of Claude Zobell, has hinted at the existence of deep subsurface microorganisms. Frank Greer, a microbiologist at the University of Chicago, was able to culture sulphate-reducing bacteria from water extracted from an oil deposit that was hundreds of meters below the earth's surface. Now a number of new techniques are available to facilitate the collection of sterile samples from deep cores in both the saturated and the unsaturated zones. Colwell (1989) collected a 70-m core from the eastern Snake River Plain, which is a semiarid, high desert area in southeastern Idaho. He observed that, though the direct counts from deep subsurface samples remained high declining by only one order of magnitude in comparison with the surface samples, the culturable counts declined by four orders of magnitude to less than 100 colony-forming units per gram of sediment (Table 5.6). The majority of the isolates from the deep subsurface in this study were Gram-positive and strictly aerobic bacteria but there is generally no trace of eukaryotes. In contrast, in surface soils Gram-negative bacteria are more numerous. The deep subsurface atmosphere was found to be similar to ambient surface air in most samples, suggesting that the deep subsurface was also aerobic. Subsequent studies have also supported the fact that the microbial numbers and activities are higher in soil layers that have had exposure to the earth's surface and plant production. These materials tend to have associated microorganisms and nutrient reserves, even very small ones, that can maintain very slow-growing populations for thousands of years.

It must be emphasized that although microbes are present in deep subsurface zones, rates of metabolic activity are much lower than rates in surface soils. The activity in deep subsurface soil is about nine orders of magnitude lower than of surface environments (Metabolic activity is expressed as the rate of CO<sub>2</sub> production and is estimated by groundwater chemical analysis and geochemical modelling.) Not only that the microorganisms of deep subsurface environment are physiologically inactive but also the limitation of growth is triggered due to unfavourable growth conditions.

**Table 5.6:** A comparison of microbial counts in surface and unsaturated subsurface environments.

| <i>Sample site</i>                            | <i>Direct counts<br/>(Counts/g)</i> | <i>Viable counts<br/>(CFU /g)</i> |
|-----------------------------------------------|-------------------------------------|-----------------------------------|
| Surface (10 cm)                               | $2.6 \times 10^6$                   | $3.5 \times 10^5$                 |
| Subsurface basalt-sediment interface (70.1 m) | $4.8 \times 10^5$                   | 50                                |
| Deep subsurface sediment layer (70.4 m)       | $1.4 \times 10^5$                   | 21                                |

Adopted from Maier and Pepper (2006).

The microbial activity of the saturated zone of soil has also been well investigated. Maximum information are available on the microbial diversity of shallow aquifers, which are just few meters below the surface soil and are not separated from the latter by an unsaturated zone. The waters from the shallow aquifers are drawn to the surface for various purpose, exposing them for interaction with the atmosphere and to the surface soil layers. There is regular input of organic matters and viable microbial cells from the surface layers, making them rich in microbes. Some of these microbes are even at the same density as in the surface layers.

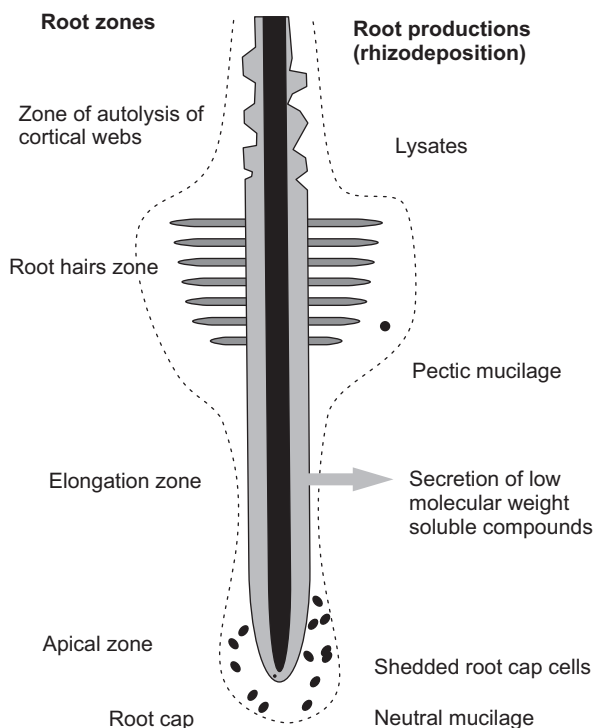
Intermediate and deep aquifers are characterized by low rates of recharge and groundwater flow that create a habitat for microorganisms different from that in shallow aquifers. Samples taken from deep cores show that there is a wide diversity of microorganisms in deep saturated zones. One of the best studied deep saturated zone sites is the U.S. Department of Energy Savannah River Site in Aiken, South Carolina. Microbial analysis of core materials of this region showed diverse and numerous populations of microbes, with total counts ranging from  $10^6$  to  $10^7$  cells per gram of sediment. The types of organisms detected included aerobic and facultatively anaerobic chemoheterotrophs, denitrifiers, methanogens, sulphate-reducers, sulphur-oxidizers, nitrifiers and nitrogen-fixing bacteria. Low numbers of unicellular cyanobacteria, fungi and protozoa were also detected in some samples (Balkwill and Boone, 1997). However, despite the high total numbers of bacteria, culturable counts were much lower.

## 5.6 THE RHIZOSPHERE MICROBES

The rhizosphere soil is the thin layer of soil adhering to a root system after shaking has removed the loose soil (Fig. 5.8). The size of the rhizosphere depends on the particular plant root structure, but generally the contact area with soil is very large. Per total plant biomass, the fibrous root structure of grassy plants provides a larger surface area than root systems characterized by a taproot. However, only 4%-10% of the actual rhizoplane comes in direct physical contact with microorganisms and also occupied by them.

An interesting and as yet insufficiently explored modification of the rhizosphere is the rhizosheath, characterized as a relatively thick soil cylinder that adheres to the plant roots, after being shaken to remove the attached soil

(Fig.5.8). The formation of rhizosheaths is typical of some desert grasses but also occurs in some grass species that grow under less extreme conditions. The sand grains in the rhizosheath are cemented together by an extracellular mucilage, apparently excreted by the root cells. The rhizosheath appears to be an adaptation for moisture conservation, but it undoubtedly also provides an environment for extensive root-microbe interactions. Increased nitrogen fixation activity has been measured in rhizosheath soil.

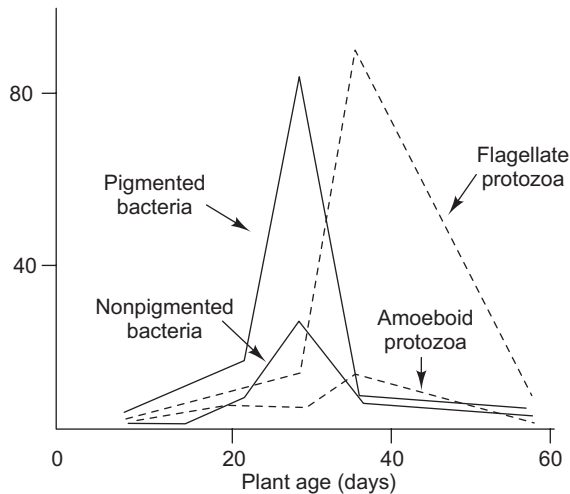


**Figure 5.8** - The schematic representation of a plant root showing the rhizoplane and rhizosphere zones (Adopted from Aragno, 2005).

### 5.6.1 The Rhizosphere Influence

The structure of the plant root system contributes to the establishment of the rhizosphere microbial population. The interactions of plant roots and rhizosphere microorganisms are based largely on interactive modification of the soil environment by processes such as water uptake by the plant, release of organic chemicals to the soil by the roots, microbial production of plant growth factors and microbially mediated availability of mineral nutrients. Within the rhizosphere, plant roots have a direct influence on the composition and density of the soil microbial community, known as the rhizosphere influence. The later can be seen by looking at the ratio of the number of microorganisms in the rhizosphere soil (R) to the number of corresponding microorganisms in soil remote from roots (S). Generally R/S

ratios range from 5 to 20, but it is common to find an R/S ratio of 100, i.e. microbial populations 100 times higher in the rhizosphere than in the surrounding root-free soil (non-rhizospheric soil) (Fig. 5.9).



**Figure 5.9** - A comparison between the microbial density of rhizosphere and the bulk soil (Adopted from Campbell, 1977).

There is a higher proportion of Gram-negative, rod-shaped bacteria and a lower proportion of Gram-positive rods, cocci and pleomorphic forms in the rhizosphere than in root-free soil. There are also a relatively higher proportion of motile, rapidly growing bacteria, such as *Pseudomonas*, in the rhizosphere than elsewhere in soil. In many cases this increase represents a direct influence of plant root exudates on soil microorganisms, which favour microorganisms with high intrinsic growth rates. Organic materials released from roots include amino acids, keto acids, vitamins, sugars, tannins, alkaloids, phosphatides and other unidentified substances. Roots surrounded by microorganisms excrete many times as much organic material as sterile roots. Although a few of these materials (mostly secondary metabolites) inhibit microorganisms, most stimulate microbial growth. The influence of materials released by the plant into the soil is evidenced by the observation that bacterial populations within the rhizosphere have markedly different nutritional properties than bacteria in root-free soil. Many rhizosphere bacteria require amino acids for maximal growth and it is likely that root exudates supply these acids.

Successional changes of microorganisms occur in the rhizosphere as the plant grows from seed germination to maturity. During plant development, a distinct rhizosphere succession results in rapidly growing, growth-factor-requiring, opportunistic microbial populations. These successional changes correspond to changes in the materials released by the plant roots to the rhizosphere during plant maturation. Initially, carbohydrate exudates and mucilaginous materials support the growth of large populations of microorganisms within the grooves of



the epidermal plant cells, on the root surface and within mucilaginous layers surrounding the roots. As the plant matures, autolysis of some of the root material takes place as part of normal root development and simple sugars and amino acids are released into the soil, stimulating the growth of *Pseudomonas* and other bacteria with high intrinsic growth rates. R/S ratios usually decline when plants stop growing and become senescent.

The microaerophilic *Azospirillum* and the aerobic *Azotobacter paspali* are nitrogen-fixing soil bacteria regularly associated with the rhizosphere of certain tropical grasses of the *Digitaria*, *Panicum* and *Paspalum* genera. These bacteria use root exudates as the energy source to support significant nitrogen fixation. However, a side effect of biochemical nitrogen fixation, especially under submerged conditions, is the evolution of hydrogen. Only some *Rhizobium* and *Bradyrhizobium* strains have the hydrogenase enzyme to utilize this side product; others wastefully evolve hydrogen gas. Nodulated root systems evolving hydrogen were found to be colonized by hydrogen-oxidizing *Acinetobacter* strains that utilized this resource not commonly available in aerobic environments (see chapter 9).

### 5.6.2 Rhizosphere Microbes and Plant Growth

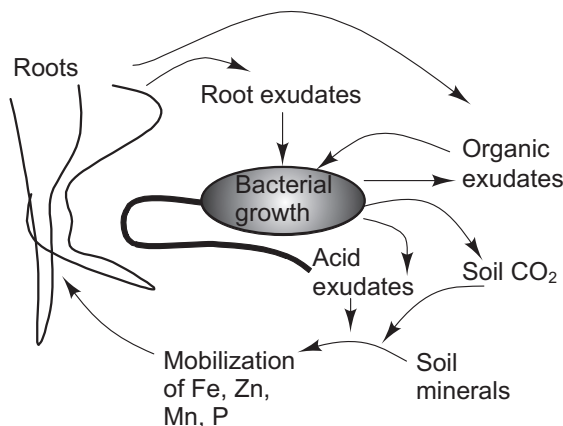
Just as plant roots have a direct effect on the surrounding microbial populations, microorganisms in the rhizosphere have a marked influence on the growth of plants. Microbial populations in the rhizosphere may benefit the plant in a variety of ways, including increased recycling and solubilization of mineral nutrients, synthesis of vitamins, amino acids, auxins, cytokinins and gibberellins and antagonism with potential plant pathogens through competition and development of amensal relationships based on production of antibiotics. Plants that grow in flooded sediments and soils have evolved adaptations for conducting oxygen from the shoot to the roots, but in such anaerobic environments the roots also have to cope with the toxic hydrogen sulphide generated by sulphate reduction. Rice and perhaps other partially submerged plants, is protected against the toxic effect of hydrogen sulphide by a mutualistic association with *Beggiatoa*. This microaerophilic, catalase-negative, sulphide-oxidizing, filamentous bacterium benefits from the oxygen and catalase enzyme provided by the rice roots and in turn, *Beggiatoa* aids the rice plant by oxidizing the toxic hydrogen sulfide to harmless sulphur or sulphate, thus protecting the cytochrome system of the rice roots from damage.

Organisms in the rhizosphere produce organic compounds and hormones that affect the proliferation of the plant root system. Microorganisms synthesize auxins and gibberellin-like compounds and these compounds increase the rate of seed germination and the development of root hairs that aid plant growth. *Arthrobacter*, *Pseudomonas* and *Agrobacterium* populations found in the rhizosphere have been reported to be capable of producing organic chemicals that stimulate growth of plants. The rhizosphere of wheat seedlings, for example, contains a significant proportion of bacteria that produce indoleacetic acid (IAA).

This may be in response to a decline in root exudate production, but it also beneficially corresponds to a decreased need for the growth hormone by the plant.

Allelopathic (antagonistic) substances released by microorganisms in the rhizosphere may allow plants to enter into amensal relationships with other plants. Such allelopathic substances surrounding some plants can prevent invasion of that habitat by other plants and this may represent a synergistic relationship between a plant and its rhizosphere microbial community. Bacterial populations in the rhizosphere of young wheat plants have been shown to inhibit the growth of pea and lettuce plants. As a wheat plant matures, the proportion of these bacteria decreases and they are replaced by a higher proportion of microorganisms capable of producing growth-promoting substances similar to gibberellic acid.

Microorganisms in the rhizosphere influence the availability of mineral nutrients to the plants, sometimes using limiting concentrations of inorganic nutrients before they can reach plant roots and in other cases increasing the availability of inorganic nutrients to the plant. Rhizosphere microorganisms increase the availability of phosphate through solubilization of materials that would otherwise be unavailable to plants. Plants have been shown to exhibit higher rates of phosphate uptake when associated with rhizosphere microorganisms than in sterile soils. The principal mechanism of increasing phosphate availability is the microbial production of acids that dissolve apatite (a common mineral group including calcium fluophosphate), releasing soluble forms of phosphorus. Iron and manganese may be more available to plants because of rhizosphere microorganisms that produce organic chelating agents, thus increasing the solubility of iron and manganese compounds. It has also been shown that microorganisms on roots significantly increase the uptake rates of calcium by the roots. This increase may be due to high concentrations in the rhizosphere of carbon dioxide produced by microorganisms, which increases the solubility and thus the availability of calcium (Fig. 5.10).



**Figure 5.10** - The metabolic interactions between the rhizospheric bacteria and the plant roots.

Even though increased uptake of minerals by plant roots in the presence of rhizosphere microorganisms is beneficial, the abundant microbial populations in the rhizosphere can sometimes create a deficiency of required minerals for the plants. For example, bacterial immobilization of zinc and oxidation of manganese cause the plant diseases “little leaf” of fruit trees and “gray speck” of oats, respectively. Microorganisms in the rhizosphere may immobilize limiting nitrogen, making it unavailable for the plant. The immobilization of nitrogen by microorganisms in the rhizosphere accounts for an appreciable loss of added nitrogen fertilizer intended for plant use. Part of the nitrogen is immobilized in the form of microbial protein, but some may also be lost to the atmosphere by denitrification. Microbial mobilization of metals in the rhizosphere may impose metal toxicity and reduce plant growth.

### 5.6.3 Rhizosphere and Bulk Soil Diversity

Before the molecular ecology methods were made available, it was often hypothesized (but without clear evidence) that bacterial diversity increased from the bulk soil to the rhizosphere, due to the higher nutrient richness of the latter. This somewhat weak argument did not take into account a well-known property of many ecosystems: for example, an oligotrophic alpine prairie may harbour a much higher plant diversity than an eutrophic field covered by a few ruderal species! Indeed, as a rule, rhizosphere bacterial communities present a decreasing (phylo)genetical diversity gradient: bulk soil → rhizosphere soil → rhizoplane/endorhizosphere. However, some phylogenetical groups are particularly favoured in the rhizosphere environment, whereas others are characteristic of bulk soil and even less abundant in the rhizosphere soil than in the distant soil. The former comprised generally proteobacteria, in particular *Pseudomonas* spp. The so-called *Holophaga-Acidobacterium* cluster (indeed a well-defined and sometimes abundant cluster distantly related to the genera *Holophaga* and *Acidobacterium* and known only from environmentally amplified DNA) and the high G + C Gram-positive bacteria (*Firmicutes*) are most abundant in the bulk soil. Therefore, there is not only a change in diversity, but also a change in specific composition, which is probably more important than the diversity itself. These changes are obviously related to the drastic modifications of the soil environment provoked by the root activities.

Probably the most important cause of the high rhizosphere microbial diversity is linked to the high nutrient flow provided by the rhizodeposition. The composition of root secretions is difficult to analyse in field conditions and is known mainly from hydroponic cultures, where exudates accumulate. In rhizosphere soil, it may be expected that most easily degradable, low molecular weight compounds such as sugars and organic or amino acids will be rapidly metabolized and so undergo a rapid turnover, maintaining their concentration quite low. For example, re-absorption of exudates by the plant is less liable to occur in natural soils than in artificial, hydroponic systems. But, one can hypothesize that there are only a few major compounds in one plant's root secretion, which will provide elective conditions for a limited number of fast-growing populations or r-strategists able to metabolize them.

Another important aspect is the selective influence of the root on the microbiota. For example, secreted flavonoids may exert a differential effect on bacterial populations, some of them being attracted, whereas others would be repelled. The selective effect is still more striking in the rhizoplane/endorhizosphere fraction. In this case, the bacteria must be adapted to adhere to the root surface or to penetrate the intercellular spaces in the rhizoderm and cortex, to become endophytes.

#### 5.6.4 The Rhizosphere Adaptation

From the preceding section it is observed that a number of bacterial populations in the rhizosphere live in mutualistic interactions with the plant. They benefit from rhizodeposition-derived nutrients and in some cases from other root-derived factors, like micro-oxic conditions (e.g. for nitrogen fixation), growth factors, attractants, or even inducers of enzyme activities. In return, these populations may exhibit properties favouring plant growth and productivity: they are termed “plant growth promoting rhizobacteria” (PGPR). The identified PGP properties are:

1. Acidolytic solubilization of minerals, particularly of calcium phosphate, making free orthophosphate available for plant nutrition.
2. Production of chelating compounds allowing the solubilization of acid-insoluble compounds, like iron or aluminium phosphates and iron (hydr)oxides. In some cases, PGP bacteria produce a plant-compatible siderophore, allowing it to share the chelated iron with the plant.
3. Production of enzymes allowing to mineralize N, S or P from organic compounds to provide the plant with the corresponding inorganic ions. Examples are uricase (urate oxidase), allantoinase and urease for the degradation of uric acid and urea into ammonium and CO<sub>2</sub> sulphatases and phosphatases, liberating sulphates and orthophosphates from the corresponding organic compounds, respectively.
4. Nitrogen fixation. The conditions existing in the rhizosphere may be particularly favourable for nitrogen-fixing bacteria, even in absence of nodule formation (associative fixation). Indeed, in a nitrogen-poor environment, the simultaneous occurrence of a low oxygen concentration and of a high energy flux would favour nitrogenase complex and high affinity ammonium assimilation (glutamine-synthase + glutamine – oxoglutarate amino-transferase pathway).
5. Production of compounds inhibiting the growth of parasites, such as diacetylphloroglucinol (DAPG) and cyanhydric acid. A good example is given by the DAPG-producing *Pseudomonas fluorescens* strain CHAO. Addition of the bacterium suppresses the effect of the co-inoculated parasite *Thielaviopsis basicola*.

The adaptations in the rhizosphere are not restricted to bilateral bacteria / root interactions. It is essential to take into account the other microbial partners which

are the fungi, essentially the mycorrhizal ones and the bacterial grazers, essentially protozoa, nematodes and rotifers. About 80% of plant species and 9 plant individuals out of 10 live in association with mycorrhizal fungi (See Section 5.7).

### 5.6.5 Microbial Food Chain in Rhizosphere

The huge nutrient and energy flow in the rhizosphere (comparable to the net plant production) would allow, if the bacteria were alone, to reach high bacteriomass densities. Due to their more efficient active transport systems, this would lead to monopolizing of soil minerals in the bacterial biomass to the detriment of the plant. Bacteria grazing is then essential to keep bacterial biomass at a level which keeps enough free minerals in soil for plant nutrition. Clarholm (1989) identified this effect on wheat cultures: the amount of nitrogen accumulated by the plant was 3-4 times higher in the presence of bacteria-grazing amoebae than in their absence. In *Arabidopsis thaliana* (a plant which does not form mycorrhizae) it has been observed that in the absence of protozoans, a complete inhibition of the growth of young shoots occur, after they reached about 1 cm in length, whereas normal growth occur in their presence. Apart from the effect on freeing minerals for plant nutrition, the triple interaction plant / bacteria / bacteria grazers may have two further advantages:

1. it favours the bacterial biomass turnover guaranteeing an active growth stage and
2. transfer of power to the plant to liberate high concentrations of ions in the root neighbourhood.

The energy transfer in the rhizosphere occurs through a microbial food chain consisting of bacteria, fungi, amoebae and small animals. The relationship is also aided by the contribution of the host plant. The heterotrophic food chain begins with the organic carbon contributed by the host plants, that facilitate bacterial and fungal growth. The bacteria are then grazed by amoebae, which in turn are grazed by small animals. The excretion of small animals and amoebae facilitates ammonia availability to the host plant. Grazing, thus, down regulates the bacterial biomass in the rhizosphere and keep it much below that which could be expected and the "cultural" R/S ratios are certainly overestimates. But the bacterial turnover will be much higher than in distant soil, favouring fast-growing r-strategists.

## 5.7 THE MYCORRHIZAE

Many plants, especially tree species, enter into a mutualistic relationship with fungal hyphae called mycorrhizal fungi, in which the later actually become integrated into the physical structure of the roots. The fungus derives nutritional benefits from the plant roots, contributes to plant nutrition and does not cause plant disease. Mycorrhizal associations differ from other plants-microbe associations by the greater specificity and organization of the plant-fungus

relationship. This, however, keeps aside the legume - rhizobium association, which is highly specific, in comparison to the mycorrhizal association. The later involves the integration of plant roots and fungal mycelia, forming integrated morphological units. The widespread existence of mycorrhizal associations between fungi and plant roots attests to the importance of this interaction.

Mycorrhizal associations exist for prolonged periods with the maintenance of a healthy physiological interaction between the plant and the fungus. The mycorrhizal associations of fungi and plant roots represent a diverse relationship, in terms of both structure and physiological function, that leads to a nutrient exchange favourable to both partners. Enhanced uptake of water and mineral nutrients, particularly phosphorus and nitrogen, has been noted in roots having mycorrhizal associations. Plants with mycorrhizal fungi are therefore able to occupy habitats they otherwise could not colonize alone. It is essential for many seedlings to have mycorrhizal association at an early stage of growth for their successful establishment in an alien habitat.

There are different types of mycorrhizae:

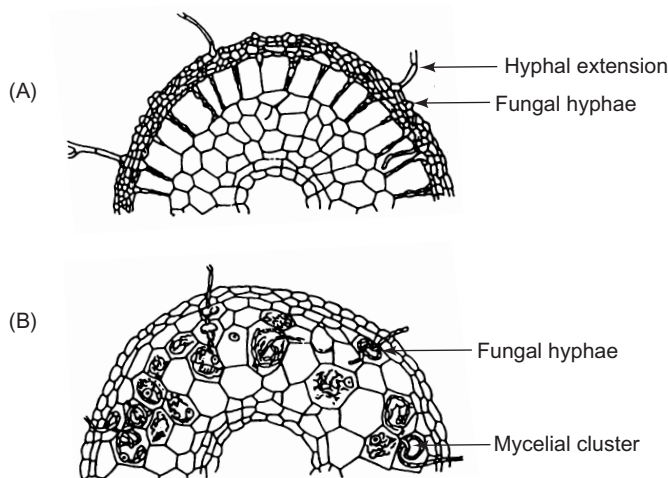
1. ectotrophic mycorrhizae (ectomycorrhizae) – mainly between ligneous plants in the sub-boreal and temperate regions and higher Basidiomycetes or some Ascomycetes (truffles);
2. arbutoid mycorrhizae – between some members of *Ericaceae* (*Arbutus*, *Arctostaphylos*, etc.) and higher Basidiomycete;
3. ericoid mycorrhizae – between most *Ericaceae* and Ascomycetes (*Hymenoscyphus*, *Oidiodendron*) exclusively mycorrhizal species;
4. orchid mycorrhizae – between members of *Orchidaceae* and Basidiomycetes or *Mycelia sterilia* affiliated to Basidiomycetes (*Rhizoctonia*); and
5. arbuscular mycorrhizae, between most plants which do not harbour other mycorrhizae and Zygomycetes belonging to the order *Glomales*, comprising exclusively mycorrhizal species.

All these mycorrhizal associations can basically be categorized into two types, basing on the position of fungal hyphae on the roots, viz., (a) ectomycorrhizae and (b) endomycorrhizae.

In ectomycorrhizae, the fungus (an ascomycete or basidiomycete) forms an external pseudoparenchymatous sheath more than 40 µm thick and constituting up to 40% of the dry weight of the combined root-fungus structure (Fig. 5.11A). The fungal hyphae penetrate the intercellular spaces of the epidermis and of the cortical region of the root but do not invade the living cells. The morphology of the root is altered, forming shorter, dichotomously branched clusters with reduced meristematic regions. The endomycorrhizae, on the other hand, invade the living cells of the root, which become filled with mycelial clusters (Fig. 5.11B). Of the various microorganisms colonizing the active root surface, mycorrhizal fungi occupy the unique ecological position of being partly inside and partly outside the host. The part of the fungus within the root does not encounter competition



with other soil microorganisms, thereby growing without being hindered by the shortage of resources. In some cases, endo- and ectomycorrhizae may be combined and are referred to as an ectendomycorrhizae.



**Figure 5.11** - A diagrammatic representation of the transverse section of plant roots showing the presence of (A) Ectomycorrhizae and (B) Endomycorrhizae (Adopted from Atlas and Bartha, 2005).

All the mycorrhizal fungi derive most of their carbon and energy from the host plant, with the exception of orchid mycorrhizae, in which the otherwise humus-inhabiting fungus brings altogether organic carbon, minerals and water to its host plant. Conversely, the fungus, with its high potential to mineralise, actively transport, accumulate and translocate minerals and water, and may provide the plant with a supplement of water and of ions often essential to its nutrition. Moreover, the volume of soil occupied by the fungus is far higher, whereas the surface in contact with soil solution is considerably increased.

Mycorrhizae are liable to alter the bacterial community around them. In the rhizosphere of wheat, Roesti (2005) has shown a significant difference in bacterial community structure in the presence and in the absence of arbuscular mycorrhizal fungi. Mycorrhizal fungi often secrete organic compounds, like oxalate or proteins, which may feed oxalotrophic and proteolytic bacteria. In return, bacteria may affect the mycorrhizal fungus either negatively or positively. Roesti (2005) showed a positive correlation between the presence of AM fungi and of calcium phosphate solubilizing bacteria: the solubilizing capability of the bacteria, combined with the translocating properties of the fungus, greatly improve phosphate transport to the plant. Bacteria may also help AM fungi root colonization by the production of cell wall softening enzymes or plant hormones. Hyphal growth may be stimulated by bacterial production of growth factors or  $\text{CO}_2$ .



### 5.7.1 The Ectomycorrhizae

Many gymnosperms and angiosperms of tropical and subtropical forests show ectomycorrhizal association at early stage of their development. Most trees in temperate forest regions have ectomycorrhizal associations. Many fungi can enter into ectomycorrhizal associations, including ascomycetes, such as truffles and basidiomycetes, such as *Boletus* and *Amanita*. Ectomycorrhizal fungi generally have optimal growth temperatures of 15-30°C and are acidophilic with optimal growth at pH 4.0 - 6.0 or even as low as pH 3.0. Most ectomycorrhizal fungi grow well on simple carbohydrates such as disaccharides and sugar alcohols. They generally utilize complex organic sources of nitrogen, amino acids and ammonium salts. Many require vitamins such as thiamine and biotin and are able to produce a variety of metabolites that they release to the plant, including auxins, gibberellins, cytokinins, vitamins, antibiotics and fatty acids. Some ectomycorrhizal fungi are capable of producing enzymes such as cellulase, but such activity is normally suppressed within the host plant and, therefore, the fungi do not digest the plant roots. The plant derives several benefits from its association with ectomycorrhizal fungi, including longevity of feeder roots, increased rates of nutrient absorption from soil, selective absorption of certain ions from soils, resistance to plant pathogens, increased tolerance to toxins and increased tolerance ranges to environmental parameters, such as temperature, drought and pH. The ectomycorrhizal fungi receive photosynthesis products from the host plant as root exudates and thus escape intense competition for organic substrates with other soil microorganisms. The nutritional benefit to the fungus is demonstrated by the fact that many mycorrhizal fungi fail to form fruiting bodies outside of a mycorrhizal association with a plant, even though vegetative saprophytic growth is usually possible under these circumstances.

An ectomycorrhizal infection has a morphogenetic effect that leads to characteristic dichotomous branching and prolonged growth and survival of plant rootlets, probably due to production of growth hormones by the ectomycorrhizal fungi. Formation of root hairs is suppressed and fungal hyphae overtake their function, thus greatly increasing the radius of nutrient availability for the plant. Ectomycorrhizal roots take up ions, such as phosphate and potassium, in excess of the rates displayed by uninfected roots. The mechanisms of uptake are dependent on fungal metabolic activity. There may be a primary accumulation of phosphate within the fungal sheath followed by transfer to the plant root. Nitrogen-containing compounds and calcium have also been found to be absorbed into the fungal mycelial sheath, followed by transfer to the plant root. Interdependence of the plant root and the ectomycorrhizal fungus is thus based in large part on their ability to supply each other with major and minor nutrients.

Arbuscular mycorrhizae occur in wheat, maize, potatoes, beans, soybeans, tomatoes, strawberries, apples, oranges, grapes, cotton, tobacco, tea, coffee, cacao, sugarcane, sugar maple, rubber trees, ash trees, hazel shrubs, honeysuckle and various herbaceous plants. They also occur in angiosperms, gymnosperms, pteridophytes and bryophytes, and in most major agricultural crop plants. Plants

with such mycorrhizae also are able to resist pathogens that otherwise would attack the plant roots. The sheaths produced by ectomycorrhizal fungi present an effective physical barrier to penetration by plant root pathogens and many basidiomycetes, that are ectomycorrhizal, have been shown to produce antibiotics. Plants with ectomycorrhizal fungi survive in soils inoculated with pathogens that enter through plant roots. Ectomycorrhizal roots also produce a variety of volatile organic acids that have fungistatic effects.

### 5.7.2 Endomycorrhizae

Endomycorrhizal associations in which the fungus penetrates into the plant root cells, which are not of arbuscular type, occur in a few orders of plants, such as the Ericales, which include heath, arbutus, azalea, rhododendron and American laurel. The endomycorrhizae of the plant genera belonging to the order Ericales are characterized by nonpathogenic penetration of the root cortex by septate fungal hyphae that often form intracellular coils. Although the fungi do not fix atmospheric nitrogen, the endomycorrhizal association may increase plant access to combined nitrogen in soil as demonstrated by better nitrogen nutrition in mycorrhizal as compared to nonmycorrhizal plants. There is greater phosphatase activity in mycorrhizal roots than in nonmycorrhizal ones and the mycorrhizal fungi can transfer phosphate from external sources to the host plant. Virtually all orchid roots are internally infected and attacked by endophytic fungal hyphae, which pass through surface cells into cortical cells to form mycorrhizae. The fungi form coils within the cells of the outer cortex and later the hyphae lose their integrity and much of their contents passes into the host cell. Orchids are obligately mycorrhizal under natural conditions, often forming associations with the fungi *Armillaria mellea* and *Rhizoctonia solani*. These endomycorrhizal associations enhance the ability of orchid seeds to germinate, but the fungi can also be parasitic to the host plant especially during its low metabolic phase. This fact and conversely the digestion of some fungal mycelium by the plant give this association the character of a precariously balanced mutual parasitism. It may also be that the association of orchids, which are pollinated at night and the bioluminescent fungus *A. mellea* is based on light production, which would attract nocturnal insects and could aid in the sexual reproduction of the orchids.

### 5.7.3 Diversity of AM Fungi

The AM fungi previously belonged to the order Glomales under the class Zygomycetes. Schüßler and his coworkers, in 2001, placed AM fungi under the phylum Glomeromycota with the class Glomeromycetes having four orders (Table 5.7). The order Glomerales was found to be ubiquitous in distribution in different ecosystems and has probably the highest number of host plants in terrestrial ecosystems. The genus *Glomus* is found in all mesophytic, xerophytic and extreme environments with ability to enter into association with numerous hosts ranging from Bryophytes to Angiosperms. In tropical forests hundreds of forest plants belonging to at least 47 families have shown AM associations. The

AM colonization level is generally higher in primary and secondary forest species than in the degraded vegetations. Fragmented forests have also poor AM association than intact forests, mostly due to disturbed soil conditions and loss of nutrient rich top soil. While *Glomus* dominate in intact vegetations, *Acaulospora* (*A. mellea*) has been reported to be more abundant in fragmented habitats. Table 5.8 summarizes the distribution of dominant AM genera in different plants and ecosystems.

Using the terminal restriction fragment length polymorphism (T-RFLP) strategy, the diversity of AM fungi has been assessed in the hosts of grassland vegetation. *Glomus* has been found to be the dominant genus in infecting a wide range of grasses, that are even taxonomically unrelated. Phylogenetic analysis revealed that at least 11 phylogenetic lines of *Glomus* clade were reported in the coastal grasslands. Arbuscular mycorrhizal associations have also been reported in boreal grasses, *Achillea ptarmica*, *Ajuga pyramidalis*, *Alchemilla glaucescens*, *Correx brunescens*, *C. pallascens*, *Crepis praemarosa*, *Hieracium lactucella*, *Rumex longifolius*, *Scorzonera humilis* and *Trifolium aureum*. The perennial xeric lands of arid tropical regions are very much infected with AM fungi of the genera *Glomus* and *Acaulospora*.

**Table 5.7:** Classification of AM fungi using molecular probe as stated by Schüßler and his coworkers.

| Taxonomic group |        | Name                 |
|-----------------|--------|----------------------|
| Phylum          |        | Glomeromycota        |
| Class           |        | Glomeromycetes       |
| 1. Order        |        | Glomeromales         |
|                 | Family | Glomeraceae          |
|                 |        | <i>Glomus</i>        |
| 2. Order        |        | Diversisporales      |
|                 | Family | Diversisporaceae     |
|                 |        | <i>Diversispora</i>  |
|                 | Family | Gigasporaceae        |
|                 |        | <i>Gigaspora</i>     |
|                 |        | <i>Scutellospora</i> |
|                 | Family | Acaulosporaceae      |
|                 |        | <i>Acaulospora</i>   |
|                 |        | <i>Entrophospora</i> |
| 3. Order        |        | Archaeosporales      |
|                 | Family | Archaeosporaceae     |
|                 |        | <i>Archaeospora</i>  |
|                 | Family | Geosiphonaceae       |
|                 |        | <i>Geosiphon</i>     |
| 4. Order        |        | Paraglomales         |
|                 | Family | Paraglomaceae        |
|                 |        | <i>Paraglomus</i>    |

Source: Khade and Adholeya (2005)

The colonization and diversity of AM fungi associated with common pteridophytes have not been so extensively investigated. Reports are also scanty on the AM investigation in gymnosperms. However, available data suggest that probably these two groups of plants have unusual preference to accommodate mycorrhizal fungi under mutualistic association. Of 34 species of ferns collected from Dujiangyan, China, 31 were found colonized by AM fungi belonging to 40 species. Hornworts, many hepatics, lycopods and psilopods also showed AM association. In all these lower plants, the structures analogous to the arbuscule of higher plants were formed in the thalli by the members of the order Glomeromales. A preponderance of association of this kind is in accordance with paleobotanical and molecular evidence indicating that Glomeraleal fungi produce the archetypal symbioses with the earliest land plants to emerge on the land.

Many epiparasitic plants depend on ectomycorrhizal fungi for carbon transfer and exhibit exceptional specificity for these fungi. Bidartondo *et al.* (2002) have observed that epiparasitism would also occur between plants linked by AM fungi. On a global scale, AM fungi form the most wide spread mycorrhizae, thus the ability of plant to cheat this symbiosis would be highly significant. Mycorrhizae between the non-green parasites *Arachnitis uniflora* and the neighbouring plants have been established. Five species of *Voyria* and one species of *Voyriella* are associated with inter-carbon transfer from the neighbouring plant through AM fungi.

#### 5.7.4 Signaling and Growth in Mycorrhizal Association

If and when the asymbiotically growing AM fungus does contact a host root, a series of signaling events occurs between the partners, which ultimately leads to the “acceptance” of AM fungus as a symbiont by the host root or rejection if the specificity is not established. Over the years, accumulated evidences indicate that roots emit a volatile signal that stimulates the directional growth of the AM fungus toward them. One of the prime candidates for this volatile signal is CO<sub>2</sub>, which can stimulate extensive hyphal growth of some AM fungi *in vitro*. A recognized hypha penetrate the host by localized production of wall-degrading enzymes (cellulase, pectinase and xyloglucanase) and by the exertion of hydrostatic pressure by the hyphal tip. The variations in colonization capacities of host tissues is related to the ability of the fungi to produce hydrolytic enzymes. Close observations have revealed that as the main hypha approaches a root, it puts out a characteristic fan-shaped complex of lateral branches. These fan-like structures called appressorium aid in fungal penetration. So far, no plant signal has been implicated in appressorium formation, but there is evidence that a shoot factor(s) may be involved in the inhibition of appressorium formation. There is every reason to doubt that there is an involvement of a secondary metabolite(s) (flavonoid ?) for the recognition of the correct fungus by a host or vice versa but neither the biochemical nor the molecular details of such signal exchange has so far been established.

**Table 5.8:** Distribution of AM fungi observed so far in different ecosystems.

| Sl. No. | Ecosystem type      | Dominant AM genera                                                                                                                               |
|---------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Tropical forests    | <i>Glomus</i> (23), <i>Acaulospora</i> (7)                                                                                                       |
| 2       | Fragmented forests  | <i>Glomus</i> (17), <i>Acaulospora</i> (8), <i>Sclerocystis</i> (1),<br><i>Scutellospora</i> (1)                                                 |
| 3       | Grass land          | <i>Glomus</i> (11), <i>Acaulospora</i> (4)                                                                                                       |
| 4       | Pteridophytes       | <i>Glomus</i> (32), <i>Acaulospora</i> (5), <i>Archaeospora</i> (1),<br><i>Entrophospora</i> (1), <i>Gigaspora</i> (1)                           |
| 5       | Bryophytes          | <i>Glomus</i> (18), <i>Acaulospora</i> (2), <i>Archaeospora</i> (1)                                                                              |
| 6       | Epiparasitic plants | <i>Glomus</i> (3), <i>Sclerocystis</i> (1)                                                                                                       |
| 7       | Saline habitats     | <i>Glomus</i> (17), <i>Acaulospora</i> (5), <i>Scutellospora</i> (3),<br><i>Entrophospora</i> (3), <i>Sclerocystis</i> (1), <i>Gigaspora</i> (1) |
| 8       | Sand dunes          | <i>Glomus</i> (14), <i>Acaulospora</i> (6), <i>Sclerocystis</i> (2),<br><i>Gigaspora</i> (2), <i>Scutellospora</i> (1)                           |

Compiled from Yamato and Iwasaki (2002); Vandenkoornhuyse *et al.* (2003); Mangel *et al.* (2004); Rosendahl and Stunkenbrock (2004); Carvalho *et al.* (2004); Khade and Adholeya (2005).

After the entry into the host tissue, the fungus develops extensively between and within root hypodermal and cortical cells and forms intraradical structures including arbuscules and lipid-rich vesicles. Arbuscules consist of hyphae that branch dichotomously and profusely within root cortical cells. Arbuscules and other fungal structures do not penetrate host cell membranes, but invaginate them. In case of arbuscules, this results in a huge increase in plant-to-fungal contact surface area, which leads to the assumption that the bidirectional transfer of nutrients in AM probably occurs at the peri-arbuscular interfaces. The arbuscules are responsible for nutrient exchange between the host and the symbiont, transporting carbohydrates from the plant to the fungus and mineral nutrients, especially phosphate and water from the fungus to the plant. Plants adapt their phosphate uptake to the interaction with the AM fungus by synthesis of specific phosphate transporters.

Colonization of root cells induces dramatic changes in the cytoplasmic organization in host: vacuole fragmentation, transformation of the plasma membrane to a periarbuscular membrane covering the arbuscule, increase of the cytoplasm volume and numbers of cell organelles, as well as movement of the nucleus into a central position. The plastids form a dense network covering the symbiotic interface. In some of these changes, microtubules are most likely involved. With regard to the molecular cross-talk between the two organisms, a number of phytohormones (cytokinins, abscisic acid, jasmonate) as well as various secondary metabolites have been examined:

- (i) Jasmonates occur at elevated level, which is accompanied by cell specific expression of genes involved in jasmonate biosynthesis that might be linked to strong carbohydrate sink function of AM roots and induced defense reactions; and

- (ii) apocarotenoids (derivatives of mycorradicin and glycosylated cyclohexenones) accumulate.

Vesicles, as globose bodies, are found within roots of the host plant and can be intercellular or intracellular. They develop to accumulate storage products in many arbuscular mycorrhizal associations. Vesicles are initiated soon after the initiation of arbuscular colonization, but continue to develop when the arbuscules senesce. Vesicles can develop thick walls in older roots and may function as propagules. Arbuscular mycorrhizal fungi belonging to family Gigasporaceae do not produce vesicles but produce auxillary cells. The latter are clusters of thin-walled cells which branch from extraradical hyphae of fungi in the genus *Gigaspora* and *Scutellospora*. Cell surfaces have spines which are reduced to knobs as in *Gigaspora* or an almost smooth surface as in *Scutellospora*.

## 5.8 CONCLUSION

There has been considerable progress in the understanding of the microbiology of terrestrial environments, which mostly deal with the microbial diversity of soil. The microbial population and the species diversity are instrumental in determining the soil productivity and the rate of remineralization. All our efforts to improve the soil productivity have failed miserably due to the chemical approach for augmentation of soil nutrient levels and use of synthetic fertilizers and chemicals. It is now an established fact that the anthropogenic addition of synthetic chemicals, petroleum products and toxicants are responsible for drastic loss of the terrestrial microflora. At the same time almost all the degradation and/ modification of environmental contaminants in terrestrial environment are known to be affected by the soil microflora. Microbes are responsible for mobilizing the soil nutrients and improving their availability to plants. There is a need to evaluate the potential of the terrestrial microflora and to make the best use of it for environmental cleaning and productivity enhancement.

The microbiology of subsurface and deep soil environments has not been fully known. Many deep subsurface microbes have been proved to be effective in degrading most difficult pollutants. These microbes need to be properly studied and methodology needs to be developed to culture the hitherto unculturable terrestrial bacteria and fungi for evaluation of their potential and their use in human welfare.

# 6

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## MICROBIAL DIVERSITY OF AIR

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### 6.1 INTRODUCTION

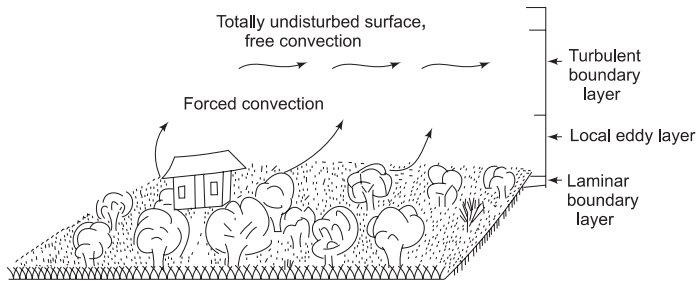
Study of microbial diversity of air or aeromicrobiology has recently become a hot topic due to widespread air pollution by viable particles and its health effects. In addition to qualitative and quantitative examination of air it also includes the study of aerosolization, aerial transmission and deposition of biological materials. From pathological point of view, many scientists have defined aeromicrobiology more specifically as the study of diseases that may be transmitted via the respiratory route. Despite the variations in definition, this relatively new field of environmental science is becoming increasingly important in many aspects of diverse scientific fields as public health, environmental science, industrial engineering, agricultural engineering, biological warfare and space exploration. Though metabolically less important, the aerially borne microbes act as the source of infection of already colonized as well as virgin habitats. Many microbes are released as spores or viable cells to air from their habitat and travel thousands of kilometers to infect fresh habitat. The microbial succession of a habitat is determined by its interaction with the aerial microflora. Some microbial processes are required to be paused by a brief period of dormancy of cells or spores, when launched from a habitat to air. Thus air acts as a medium for exchange of microbes among the habitats and aeromicrobial diversity is an amalgamation of the microbial diversity of terrestrial and aquatic habitats.

### 6.2 CHARACTERISTICS OF THE ATMOSPHERE WITH VIABLE PARTICLES

The atmospheric layers and the airflow pattern are the important forces in determining the distribution and dynamics of viable particles in air. The aeromicrobiological pathway (AMP) involves the path and pattern of movement of microbial particles in air and thereby its nature involves the atmosphere. The

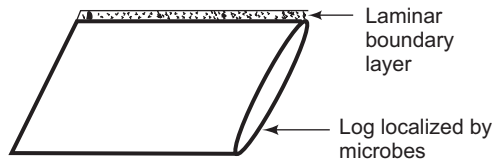


layer of most interest and significance in aeromicrobiology is the boundary layer, which extends up to 0.1 km from the earth's surface. It should be noted, however, that airborne transport of microorganisms is by no means limited to this layer and it is not uncommon to have microorganisms associated with layers of the troposphere above the turbulent boundary layer. However, it is the surface boundary layer that is largely responsible for the transport of particles over both short and long distances. The boundary layer consists of three parts: the laminar boundary layer, the turbulent boundary layer and the local eddy layer (Fig. 6.1).



**Figure 6.1** - The atmospheric layers and airflow pattern influencing bioaerosol distribution.

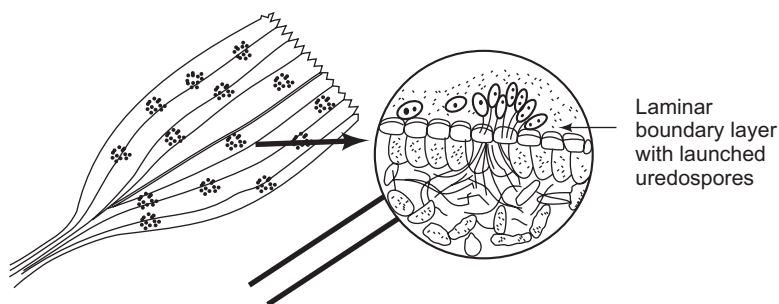
The laminar boundary layer is a layer of still air associated with the earth and all projecting surfaces, whether solid or liquid. This layer can be anywhere from 1  $\mu\text{m}$  to several meters thick, depending on weather conditions (Fig. 6.2). Still conditions cause the thickness of this layer to increase and windy conditions minimize it to a very thin layer that remains in close association with surfaces. The turbulent boundary layer is the layer that is considered to be always in motion and responsible for horizontal transport phenomena (wind dispersion), which occur whenever microorganism-associated particles are launched either indoors or outdoors. In the lower levels of the turbulent layer, the linear flow of air is interrupted by surface projections and their associated laminar boundary layers. This interaction results in the formation of friction against the airflow. This friction, which is apparent in the form of local areas of “swirling” turbulence, determines the rate of movement of the particles. The local eddy layer is the actual zone of interaction between the still laminar boundary layer of surface projections and the turbulent boundary layer.



**Figure 6.2** - The laminar boundary layer on the surface of a log colonized by microbes. The density of viable particles is the highest in this layer.

## 6.3 ATMOSPHERIC DISPERSAL OF MICROBES

The dispersal of microbes in air begins with the discharge of the microbial cells, spores or particle loaded with viable particles (aerosol) to the atmosphere. It is followed by the subsequent transport via diffusion and dispersion of these particles and finally their deposition on any surface. An example of this pathway is that of liquid aerosols containing the influenza virus launched into the air through cough, sneeze, or even through talking. These virus associated aerosols are dispersed by a cough or sneeze, transported through the air, inhaled and deposited in the lungs of a nearby person, where they may begin a new infection. Traditionally, the deposition of viable microorganisms and the resultant infection are given the most attention, but all three processes (launching, transport and deposition) are of equal importance in understanding the aerobiological pathway. While a microbial particle (hypha, cell or spore) germinate and grow, when deposited on a compatible surface, gaining the metabolic efficiency, it perishes on coming in contact with an incompatible surface. For example, when a uredospore of *Puccinia graminis* falls on a wheat leaf, it germinates and causes secondary infection. On the other hand, if it falls on a rice leaf, it fails to germinate and establish the infection (Fig. 6.3).

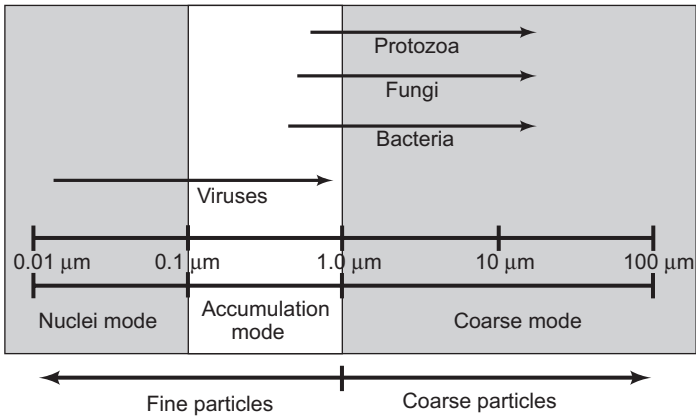


**Figure 6.3** - Launching and establishment of uredospores on the surface of a wheat leaf.

### 6.3.1 The Bioaerosols

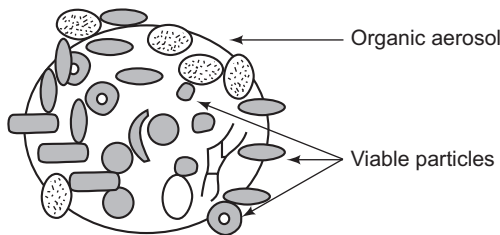
The bioaerosols are the atmospheric particles, mists or dusts of  $\mu\text{m}$  range, associated with metabolically active and/or inactive viable particles. Bioaerosols vary considerably in size and composition depends on a variety of factors including the type of microorganism or toxin, the types of particles they are associated with such as mist or dust and the gases in which the bioaerosol is suspended. Bioaerosols, in general, range from 0.02 to 100  $\mu\text{m}$  in diameter and are classified on the basis of their size. The smaller particles  $< 0.1 \mu\text{m}$  in diameter are considered to be in the nuclei mode, those ranging from 0.1 to 2  $\mu\text{m}$  are in the accumulation mode and larger particles are considered to be in the coarse mode, which undergo rapid sedimentation (Fig. 6.4). The particles in nuclei or accumulation mode are considered to be fine particles and have the capacity to move long distances. These particles have also a long residence time in the

environment. The particles in coarse mode are considered coarse particles as they settle within few meters to few kilometers from the source.



**Figure 6.4** - Diagrammatic representation of the relative sizes of bioaerosols (Dowd and Maier, 2006).

The composition of bioaerosols can be liquid or solid or a mixture of the two and should be thought of as microorganisms associated with airborne particles or as particles containing microorganisms (Fig. 6.5). This is because it is rare to have microorganisms (or toxins) that are not associated with other airborne particles such as dust or mist. This information is derived from particle size analysis experiments, which indicate that the average diameter of airborne bacterial particles is greater than 5 μm. By comparison, the average size of a soil borne bacterium, 0.3 to 1 μm, is less than one fifth this size. Similar particle size analysis experiments show the same to be true for aerosolized microorganisms of other groups (fungi, algae and viruses).



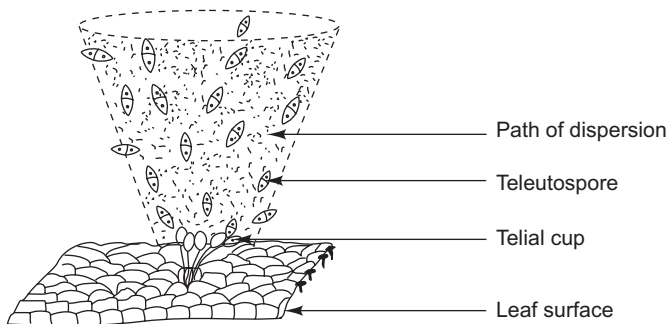
**Figure 6.5** - An organic aerosol occupied from outside with viable particles. The viable particles are attached to the aerosol surface by electrostatic attraction.

**6.3.2    Launching**

The process whereby microbes loaded particles become suspended within the earth’s atmosphere is termed launching. Because bioaerosols must be launched into the atmosphere to be transported, it is important to understand this process.

The launching of bioaerosols is mainly from terrestrial and aquatic sources, with greater airborne concentrations or atmospheric loading being associated with terrestrial sources than with aquatic sources. Some researchers speculate that there may even be atmospheric sources of bioaerosols in addition to terrestrial and aquatic ones, though the contribution from the aerial source is considered the minimum. This phenomenon is related to the limited potential for microorganisms to reproduce while airborne. This, however, is an area of aeromicrobiology for which there is little available information. In addition, a significant contribution of viable particles to the atmosphere is also made from the surfaces of plants and animals.

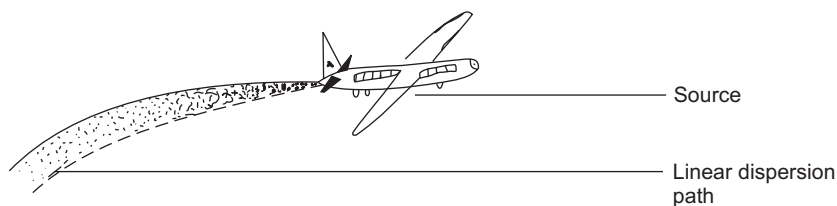
Launching into the surface boundary layers is influenced by a number of factors such as: (a) air turbulence created by the movement of humans, animals and machines; (b) the generation, storage, treatment and disposal of waste material; (c) natural mechanical processes such as the action of water and wind on contaminated solid or liquid surfaces; and (d) the release of fungal spores as a result of natural fungal life cycles. Airborne particles can be launched from either point, linear, or area sources. A point source is an isolated and well-defined site of launching such as a pile of biosolid material, before it is applied over the field or an infected leaf of a plant launching the spores of a pathogen to air. Point sources tend to display a general conical-type dispersion (Fig. 6.6). Point sources can be further defined on the basis of the type of launching phenomenon: (1) instantaneous point sources, for example, a single event such as a sneeze, or (2) continuous point sources, from which launching occurs over extended periods of time, such as the biosolid pile and the phyllosphere and phylloplane.



**Figure 6.6** - Launching of teleutospores from the surface of an infected wheat leaf showing conical type dispersion.

In contrast to point sources, linear sources and area sources involve larger, less well defined areas. When considered on the same size scale, linear and area sources display more particulate wave dispersion as opposed to the conical type of dispersion displayed by point sources. Linear and area sources can also be divided into instantaneous and continuous launching points of origin. For example, an instantaneous linear source might be a passing aircraft releasing a biological warfare agent or a passenger jet releasing the unburnt carbon particles

to air (Fig. 6.7). A continuous area source might be exemplified by release of bioaerosols from a large field that has received an application of biosolids.



**Figure 6.7** - Linear dispersion of particles in air by an aircraft.

### 6.3.3 Bioaerosol Transport

Transport or dispersion is the process by which a viable particle moves from one point to another with the speed of wind or when it is launched into air with a force. The speed of movement of an airborne particle is dependent on the kinetic energy gained by it from the force at which it is launched to the atmosphere and the wind speed. High kinetic energy of the particle can result in dissemination of airborne microorganisms over long distances. Transport of bioaerosols can be defined in terms of time and distance. Submicroscale transport involves short periods of time, under 10 minutes, as well as relatively short distances, under 100 m. This type of transport is common within buildings or other confined spaces. Microscale transport ranges from 10 minutes to 1 hour and from 100 m to 1 km and is the most common type of transport phenomenon. Mesoscale transport refers to transport in terms of days and distances up to 100 km (or even more) and in macroscale transport, the time and distance are extended even further to thousands of kilometers. Because most microorganisms have limited ability to survive when suspended in the atmosphere, the most common scales considered are the submicroscale and microscale. It should be noted, however, that some viruses, spores and spore-forming bacteria have been shown to enter into mesoscale and even macroscale transport. Some macroscale transports can be global in nature and are particularly important on pathogenic point of view. For example, the spores of wheat rust fungus can move thousands of kilometers, after being launched to air, to infect new crop fields.

As bioaerosols travel through time and space, different forces act upon them such as diffusion, inactivation and ultimately deposition. Diffusion is the scattering and/or dissipation of bioaerosols in response to a concentration gradient as well as gravity and is generally aided by airflow and atmospheric turbulence. The amount of turbulence associated with airflow and, thus, the relative amount of diffusion that may occur in association with particulates such as bioaerosols can be estimated using the method of Osbert Reynolds. Reynolds found that factors associated with mean wind velocity, the kinetic viscosity of the air and the relative dimension of the interfering structures could provide an indication of the amount of turbulence associated with linear airflow. Without turbulence, airborne particles from a point source would travel in a concentrated

stream directly downwind. The equation proposed by Reynolds quantifies the transport as Reynolds number (Re) as:

$$\text{Re} = \frac{\text{velocity} \times \text{dimension}}{\text{viscosity}}$$

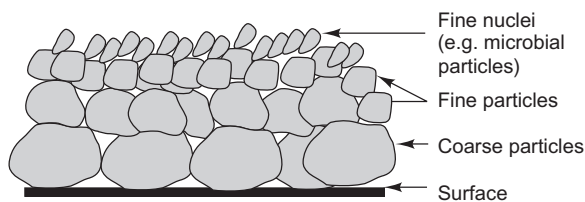
For example, at relatively high winds (36 km/hr or 10 m/sec), that are passing over a small bush (5 cm), the frictional turbulence associated with an object depends on the wind velocity being high enough and the object it is flowing over being large enough, we find that at normal air viscosity (0.14 cm<sup>2</sup>/sec) the dimensionless Re becomes 35714 [(1000 cm/sec × 5 cm)/0.14 cm<sup>2</sup>/sec].

The limiting value for the Reynolds equation is usually considered to be 2000 (for an object with 5 cm diameter, the non-turbulent wind speed is 2 km/hr), with values above this number indicating turbulent conditions. The higher the value, greater the relative turbulence of the airflow and microorganism-associated particle diffusion that occurs per unit time. Thus the rate of diffusion and transport is directly proportional to the value of Re.

### 6.3.4 Bioaerosol Deposition

Deposition is regarded as the last step in the AMB pathway. Depending on the size and the kinetic energy gained by it during launching and transport, a particle is deposited at any place beyond the source. An airborne bioaerosol will eventually leave the turbulence of the suspending gas and will ultimately be deposited on a surface by one or a combination of interrelated mechanisms. These include gravitational settling, downward molecular diffusion, surface impaction, rain deposition and electrostatic deposition. These processes are linked in many ways and even though viewed separately, they all combine to create a constant, if not steady, deposition of particles. Under standard conditions, however, the rate of deposition of a particle is directly proportional to its mass, volume and mass/volume ratio.

**(a) Gravitational settling:** The main mechanism associated with deposition is the action of gravity on particles. The force of gravity acts upon all particles heavier than air, pulling them down and essentially providing spatial and temporal limitations to the spread of airborne particles. Steady state gravitational deposition in the absence of air movement can be described in very simplistic terms by Stokes' law, which takes into account gravitational pull, particle density, particle diameter and air viscosity. The Stokes' equation is solved in terms of terminal velocity (*v*) and can be used to predict the behaviour of particles having diameters of biological significance (1 to 100 μm). Larger particles will obtain higher terminal velocities and thus settle out of the AMB pathway faster. On any surface the size of the deposited particles gradually decreases in the direction away from the surface (Fig. 6.8). It should be noted, however, that for particles of microbiological relevance that are exposed to winds above 8 × 10<sup>3</sup> m/hr, gravitational deposition may be negligible unless the particles cross out of the laminar flow via processes such as downward molecular diffusion or increase in density because of condensation reactions such as rain deposition.



**Figure 6.8** - Schematic representation of gravitational settling of particles on a surface as a function of their size. The coarse particles are settled first while the fine nuclei particles constitute the top layer.

**(b) Downward molecular diffusion:** It is a randomly occurring process caused by natural air currents and eddies that promote and enhance the downward movement of airborne particulate matters. These random movements exist even in relatively still air and tend to be in the downward direction because of gravitational effects. As a result, measured rates of gravitational deposition tend to be greater than those predicted mathematically. The increase in the rate of deposition is due to the added effects of downward molecular diffusion. Molecular diffusion is also influenced by the force of the wind and deposition rates increase with increasing wind speed and turbulence of air.

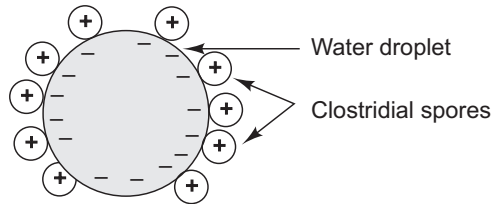
**(c) Surface impaction:** It is the process by which particles make contact with surfaces, such as leaves, trees, walls and furniture. With impaction there is an associated loss of kinetic energy. In nature, it is rare to find flat, smooth surfaces on which wind currents are unobstructed. Thus, surface impaction is a very critical factor influencing the rate of disposition of bioaerosols. The impaction potential causes the collision of a particle to a surface and facilitates its attachment to the same. However, depending on the nature of the surface a particle can bounce after collision. Bouncing off a surface causes the particle to reenter the air current at a lower rate, which can have one of the two effects: (1) it can allow subsequent downward molecular diffusion and gravitational settling to occur, resulting in deposition on another nearby surface, or (2) it can allow the particle to escape the surface and once again reenter the air current. Studies have shown that impaction is influenced by the velocity and size of the particle as well as the size, shape and nature of the surface it is approaching. Small particle has much less chance of impaction with the small object than a larger particle traveling at a greater velocity toward a much larger flat object. Thus, the larger particle has a higher impaction potential than the small particles. However, a large particle, after collision with the surface, can reenter the air only if the force of impaction is greater than the gravitational force on the particle.

**(d) Rain and electrostatic deposition:** Rainfall and electrostatic charge also can affect deposition. Rainfall deposition occurs as a condensation reaction between two particles (raindrop and bioaerosol), which combine and create a bioaerosol with a greater mass, making it to settle faster. The overall efficiency of rain deposition also depends on the spread area of the particle plume. Larger, more diffuse plumes undergo stronger impaction than smaller, more concentrated



plumes. Rain deposition is also affected by the intensity of the rainfall. The heavier the rainfall, the greater the overall rates and numbers of the condensation reactions and the greater the subsequent increase in rain deposition.

Electrostatic deposition also condenses bioaerosols, but is based on electrovalent particle attraction. All particles tend to have some type of associated charge. Microorganisms typically have an overall negative charge associated with their surfaces at neutral pH. These negatively charged particles can associate with other positively charged airborne particles, resulting in electrostatic condensation. The major phenomenon occurring may be a coagulation effect between particles (much like the condensation of the clostridial spore with the water droplet), which would increase the bioaerosol mass and enhance deposition (Fig. 6.9). It might also be assumed that as an electromagnetically charged bioaerosol comes into close proximity with an electromagnetically charged surface, electroattractive or electrorepulsive influences may be present.



**Figure 6.9** - Coagulation of clostridial spores on the surface of a water droplet. The coagulation is facilitated by the negative charges on the surface of the water droplet.

## 6.4 MICROBIAL DIVERSITY OF AIR

There is significant difference of microbial diversity and density of indoor and outdoor air. The quality of bioaerosols also differs with the regional atmospheric conditions and cultural practices. In the outdoor environment, the expanse of space and the presence of air turbulence are two controlling factors in the movement of bioaerosols. While in the outdoor conditions the bioaerosol concentration fluctuates considerably in different parts of the day and is influenced by the rate of launching and air turbulence, the fluctuation is marginal in the indoor conditions. Environmental factors such as UV radiation, temperature and relative humidity modify the effects of bioaerosols by limiting the amount of time aerosolized microorganisms will remain viable. Such limitations are, however, not observed under indoor conditions. The microbial diversity and processes of some of the outdoor and indoor conditions are given here.

### 6.4.1 The Outdoor Bioaerosols

**(a) Agriculture:** The air of agricultural areas contains the microbes related to the crops of the region. Contamination of crops and animals via bioaerosols has a huge economic impact worldwide. The list of agricultural pathogens in Table 6.1 shows the diversity of airborne microorganisms that infect plants. As the earth's

population increases, the need for a larger, more stable supply of food becomes increasingly important. Rice and wheat are the two major staple crops that are paramount to world food security. Major pathogens of such crops are the wheat rust fungi, rice smut fungi and disease causing bacterial pathogens. These spore-forming fungi cause some of the most devastating diseases of wheat and other grains. Spores of wheat rust and rice and sugarcane smut are capable of spreading hundreds of kilometers through the atmosphere. The airborne spread of rust disease has been shown to follow a predictable trend, depending upon the cultivation of wheat in the region. A rust-infected plant produces thousands of spores, which are released into the air by either natural atmospheric disturbance or mechanical disturbance during the harvesting process. Once airborne, these spores are capable of long-distance dispersal, which can cause downwind deposition onto other susceptible wheat plants.

**Table 6.1:** Important airborne fungal pathogens causing plant diseases.

| <i>Name of the disease</i>      | <i>Pathogens</i>                |
|---------------------------------|---------------------------------|
| Dutch Elm disease               | <i>Ceratocystis ulmi</i>        |
| Apple rust                      | <i>Gymnosporangium</i> spp.     |
| Potato late blight              | <i>Phytophthora infestans</i>   |
| Banana leaf spot                | <i>Mycosphaerella musicola</i>  |
| Blossom infection               | <i>Sclerotinia laxa</i>         |
| Cedar rust                      | <i>Gymnosporangium</i> spp.     |
| Leaf rust                       | <i>Puccinia recondita</i>       |
| Crown rust of oats              | <i>Puccinia coronata</i>        |
| Fusiform rust of southern pines | <i>Cronartium fusiforme</i>     |
| Loose smut of wheat             | <i>Ustilago tritici</i>         |
| Beef downy mildew               | <i>Peronospora</i> spp.         |
| Downy mildew                    | <i>Pseudoperonospora humuli</i> |
| Maize rust                      | <i>Puccinia sorghi</i>          |
| Annosus root rot                | <i>Fomes annosus</i>            |
| Powdery mildew of barley        | <i>Erysiphe graminis</i>        |
| Southern corn leaf blight       | <i>Helminthosporium maydis</i>  |
| Stem rust of wheat and rye      | <i>Puccinia graminis</i>        |
| Tobacco blue mold               | <i>Peronospora tabacina</i>     |
| Sigatoka disease of bananas     | <i>Mycosphaerella musicola</i>  |
| White pine blister rust         | <i>Cronartium ribicola</i>      |

Source Maier *et al.* (2006).

The airborne spread of pathogenic microorganisms is also highly important in the animal husbandry industry. The occurrence of foot-and-mouth disease is an example of the importance of bioaerosols in the spread of airborne disease. It has long been thought that bioaerosol spread is linked primarily to respiratory pathogens, but there is growing evidence that gastrointestinal pathogens are also important in airborne transmission of disease among animals. One example of bioaerosol spread of a gastrointestinal pathogen is transmission of *Salmonella typhimurium* among calves that are housed individually in small pens. The

pathogen can survive for long periods in an airborne state and animals exposed to aerosolized *S. typhimurium* develop disease symptoms. Aerosolized *Salmonella enteritidis* also infect laying hens. Table 6.2 contains some of the common animal pathogens found in air.

**(b) Waste disposal:** Major hazards associated with waste effluents and solid waste disposal sites are pathogenic microorganisms including bacteria, viruses, protozoa and helminths. Wastewater treatment plants utilize activated sludge and trickling filter systems and all of these treatment processes potentially create relatively large amounts of aerosols, which often include pathogenic microorganisms. Other aspects of the treatment process such as composting and land disposal are also associated with the generation of aerosols containing pathogenic microorganisms. The mixing of the solid wastes at the disposal sites and the agitation of water in wastewater treatment systems launch bioaerosols with a considerable velocity, that enables the particles to move quite a long distance and then settle. The distance travelled by the particles, however, depends on the size of the bioaerosol and air turbulence. These particles are often loaded with many pathogenic microorganisms, which cause a number of human diseases. The list of some pathogenic microbes, that are launched to the air by natural and anthropogenic processes has been given in Table 6.3.

A wide variety of airborne pathogens are present in the particles released to the atmosphere during compost application. Many pathogen indicators like *Salmonella* sp., *Clostridium* sp., *E. coli* and F+ coliphages are common viable organisms in solid and liquid aerosols at waste disposal sites. The major concern associated with the aerosolization process in relation to waste disposal operations is the exposure of waste disposal workers to pathogenic microorganisms, although nearby population centres are also at potential exposure risks. In some cases the population and the vegetation of the regions surrounding the waste disposal sites are infected with pathogenic microbes and show disease symptoms.

### 6.4.2 The Indoor Bioaerosols

The home and workplace are environments in which airborne microorganisms create major public health concerns. In comparison with the outdoor environment, indoor environments have limited circulation of external air and much less UV radiation exposure. Indoor environments also have controlled temperature and relative humidity, which are generally in the ranges that allow extended microbial survival. Thus, these conditions are suitable for the accumulation and survival of microorganisms within many enclosed environments, including office buildings, hospitals, laboratories and even spacecraft. The residence time of the bioaerosols in the indoor conditions is, however, short due to poor turbulent conditions.

In a residential building many factors such as the presence and/ or efficiency of air filtering devices, the design and operation of the air circulation systems, the health and hygiene of the occupants, the amount of clean outdoor air circulated through the building, the type of lighting used, the ambient temperature in the building and the relative humidity can influence bioaerosols and therefore the hygienic conditions of the building. Some pathogens are uniquely adapted for

survival and transmission in the indoor environment. For example, the gram-negative bacilli *Legionella* spp. are ubiquitous in the environment. They are found in association with lakes, ponds and streams and have even been found in deep terrestrial subsurface environments. *Legionella pneumophila*, the causative agent of both Legionnaires disease and Pontiac fever, has however good growth and dispersal in poorly maintained cooling tower, which provide optimal conditions for its proliferation. On aerosolization this pathogen spreads into the buildings connected to the cooling systems. Stagnant water and temperatures in the range of 35-46°C are factors that can lead to the rapid multiplication of background levels of *Legionella* spp. The pathogen can also grow intracellularly within other microbes including amoebae, cyanobacteria and protozoa. Even on the surface of any airborne particle or water droplet the pathogen can increase its number and launch the cells and spores to air.

**Table 6.2:** List of important airborne pathogenic microbes causing diseases in animals.

| <i>Microbial group</i> | <i>Disease</i>               | <i>Pathogens</i>            |
|------------------------|------------------------------|-----------------------------|
| Bacteria               | Tuberculosis                 | <i>Mycobacterium bovis</i>  |
|                        | Glanders                     | <i>Actinobacillus malei</i> |
|                        | Brucellosis                  | <i>Brucella</i> spp.        |
|                        | Salmonellosis                | <i>Salmonella</i> spp.      |
| Fungi                  | Aspergillosis                | <i>Aspergillus</i> spp.     |
|                        | Cryptococcosis               | <i>Cryptococcus</i> spp.    |
|                        | Coccidioidomycosis           | <i>Coccidioides immitis</i> |
| Viruses                | Canine herpes                | Herpesviridae               |
|                        | Eastern equine               | Alphavirus                  |
|                        | Encephalomyelitis            |                             |
|                        | Hog cholera                  | Pestivirus                  |
|                        | Influenza                    | Influenza virus             |
|                        | Feline distemper             | Morbillivirus               |
|                        | Fowl plaque                  |                             |
|                        | Rabies                       | Rhabdoviridae               |
|                        | Canine distemper             | Morbillivirus               |
|                        | Newcastle disease            |                             |
|                        | Infectious bronchitis        |                             |
|                        | Foot and mouth disease       | Influenza virus             |
|                        | Rhinderpeste                 | Morbillivirus               |
|                        | Ephemeral fever              | Alphavirus                  |
|                        | Infectious laryngotracheitis |                             |

Source: Dowd and Maier (2006).

In the highly stagnant conditions like a space flight, microbes also thrive and disperse themselves in air. However, in addition to some pathogenic microorganisms, many in the spacecraft are necessary and beneficial to the system as well as for the astronaut. Some microbes are even taken in the spacecraft to treat the wastes and facilitate the remineralisation process. The basic nutrients of the wastes are converted to biomass to support the life of the astronaut. For example,

microorganisms can be harnessed for many valuable purposes including food production, oxygen production and air purification inside the spacecraft. The use of microorganisms as components of air purification systems is a good example of a beneficial use for microorganisms in the spaceflight, in association with the AMB pathway. Biofiltration system essentially uses microorganisms to remove organic contaminants from the airstream. It is a closed two-phase system, in which the airborne organic contaminants are passed into the system and then through a membrane filter into a liquid phase. The liquid phase is made up of microorganisms and nutrients that develop into an associated biofilm on the membrane surface. Different bacterial strains or consortia of bacteria have been isolated that can completely mineralize most of the contaminants of concern to the space industry.

**Table 6.3:** List of some important airborne pathogenic microbes causing diseases in human beings.

| <i>Microbial group</i> | <i>Disease</i>                                                                                                                                                                                                                       | <i>Pathogens</i>                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bacteria               | Brucellosis<br>Pulmonary tuberculosis<br>Glanders<br>Pneumonia<br>Pulmonary anthrax<br>Respiratory infection<br>Legionellosis<br>Meningococcal infection<br>Pneumonic plague<br>Typhoid<br>Whooping cough<br>Tularemia<br>Diphtheria | <i>Brucella melitensis</i><br><i>Mycobacterium tuberculosis</i><br><i>Actinobacillus mallei</i><br><i>Klebsiella pneumoniae</i><br><i>Bacillus anthracis</i><br><i>Streptococcus pyogenes</i><br><i>Legionella</i> spp.<br><i>Neisseria meningitidis</i><br><i>Yersinia pestis</i><br><i>Salmonella typhi</i><br><i>Bordetella pertussis</i><br><i>Francisella tularensis</i><br><i>Corynebacterium diphtheriae</i> |
| Fungi                  | Aspergillosis<br>Blastomycosis<br>Coccidioidomycosis<br>Cryptococcosis<br>Histoplasmosis<br>Nocardiosis<br>Sporotrichosis                                                                                                            | <i>Aspergillus fumigatus</i><br><i>Blastomyces dermatiridi</i><br><i>Coccidioides immitis</i><br><i>Cryptococcus neoformans</i><br><i>Histoplasma capsulatum</i><br><i>Nocardia asteroides</i><br><i>Sporotrichum schenckii</i>                                                                                                                                                                                     |
| Virus                  | Influenza<br>Hemorrhagic fever<br>Hantavirus pulmonary syndrome<br>Hepatitis<br>Chicken pox<br>Common cold<br>Yellow fever<br>Dengue fever<br>Measles                                                                                | Influenza virus<br>Bunyavirus<br>Hantavirus<br>Hepatitis virus<br>Herpes virus<br>Picomavirus<br>Flavivirus<br>Flavivirus<br>Morbillivirus                                                                                                                                                                                                                                                                          |
| Protozoa               | Pneumocystosis                                                                                                                                                                                                                       | <i>Pneumocystis carinii</i>                                                                                                                                                                                                                                                                                                                                                                                         |

Source: Dowd and Maier (2006).

## 6.5 MICROBIAL SURVIVAL IN AIR

The atmosphere is an inhospitable climate for microorganisms mainly because of desiccation stress. This results in a limited time frame in which microbes can remain biologically active in air. Many microorganisms, however, have specific mechanisms that allow them to be somewhat resistant to the various environmental stresses that promote loss of biological activity. Spore forming bacteria, molds, fungi and cyst-forming protozoa all have specific mechanisms that protect them from harsh gaseous environments, increasing their ability to survive aerosolization. For organisms that have no such specific mechanisms, the survival in aerosols can often be measured in seconds. In contrast, organisms with these mechanisms can survive indefinitely. These organisms can even reproduce in the atmospheric conditions utilizing the limited amount of moisture and organic matters present on the aerosol surface. Under dry conditions, however, almost all microorganisms remain in a dormant stage in air and, if remain airborne for a long time under such dry conditions, many microorganisms lose their viability. Thus the viability is highly dependent on the environment, the length of time the organism spends in the environment and the type of microorganism. In addition, microbes may be viable but nonculturable, but when they come in contact with a suitable surface the viable and inactive cells quickly regain their activity and grow.

Many environmental factors have been shown to influence the ability of microorganisms to survive. The most important of them are relative humidity and temperature. Oxygen content, specific ions, UV radiation, various pollutants and AOFs (air associated factors) are also factors in the loss of biological activity.

### 6.5.1 Atmospheric Humidity

The relative as well as the absolute humidity content of the air play a major role in the survival of airborne microorganisms. Wells and Riley (1937) were among the first to show this phenomenon, indicating that as the relative humidity approaches 100%, the death rate of *Escherichia coli* increases. In general, it has been reported that most gram-negative bacteria associated with aerosols tend to survive for longer periods at low relative humidity by regulating their metabolic activities. The opposite tends to be true for gram-positive bacteria, which tend to remain viable longer in association with high relative humidity though, however, at 100% relative humidity, longer exposure decreases the viability vis-à-vis survival. Thus, the ability of a microorganism to remain viable in a bioaerosol is related to the organism's surface biochemistry. One mechanism that explains loss of viability in association with very low relative humidity is a structural change in the lipid bilayers of the cell membrane. As water is lost from the cell, the cell membrane bilayer changes from the typical crystalline structure to a gel phase. This structural phase transition occurs due to increase of temperature on the cell surface converting majority of the membrane lipids from rigid to fluid state i.e. the membrane becomes less integrated with more fluidity of the lipid bilayer.

This affects cell surface protein configurations, membrane integrity, active membrane transport and ultimately results in inactivation of the cell. Intracellular ionic imbalance and loss of cellular metabolites occur when the cells are exposed to unfavourable relative humidity levels.

There is information that the influenza virus is adversely affected by an increase in relative humidity. More recent work suggests that viruses possessing enveloped nucleocapsids (such as the influenza virus) have longer airborne survival when the relative humidity is below 50%, whereas viruses with naked nucleocapsids (such as the enteric viruses) are more stable at a relative humidity above 50%. It should be noted that viruses with enveloped nucleocapsids tend to have better survival in aerosols than those without. Some viruses are also stable in the AMB pathway over large ranges of relative humidity, which makes them very successful airborne pathogens. In general the survival of viruses in atmosphere is inversely proportional to the relative humidity level.

### **6.5.2 Temperature**

Temperature is a major factor in the inactivation of microorganisms. In general, high temperature promotes inactivation, mainly associated with desiccation and protein denaturation and lower temperature promotes longer survival times. When temperature approaches freezing, however, some organisms lose viability because of the formation of ice crystals on their surfaces. The effects of temperature are closely linked with many other environmental factors, including relative humidity. A comparative small change in the atmospheric temperature can cause much damage to the airborne microbes at high relative humidity than under dry conditions. The metabolic activities of microbes in air show a diurnal fluctuation in proportion to the temperature fluctuation.

### **6.5.3 Radiation**

The main sources of radiation damage to microorganisms including cyanobacteria, bacteria, viruses, algae and fungi are the shorter UV wavelengths and ionizing radiation such as X rays. The main target of UV irradiation damage/mutation in DNA, which ranges from single strand breaks to alterations in the structure of nucleic acid bases. Radiation damage to the genetic material causes inhibition of biological activity such as replication of the genome, transcription and translation and loss of expression of many vital genes. On the other hand, there may be production of some unwanted molecules in the cell due to the activity of mutated genes.

Microbes follow different strategies to protect themselves from radiation damage. These include association of microbes with larger airborne particles, possession or radiation induced synthesis of carotenoids and other pigments and efficient repair mechanism. In addition the environmental conditions like high relative humidity and cloud cover absorb or shield bioaerosols from radiation and thus provide protection to airborne microbes against radiation damage.



Many types of organisms also have mechanisms for repair of the DNA damage caused by UV radiation. An example of an organism that has a radiation resistance mechanism is *Dienococcus radiodurans*. This is a soil bacterium that is considered the most highly radiation resistant organism that has till date been isolated. An important component of its radiation resistance is its efficiency to enzymatically repair the damage to chromosomal DNA. The repair mechanism used by this bacterium and many other radiation tolerant bacteria is so highly efficient that much of the metabolic energy of the cell is dedicated exclusively to this function thereby compromising their growth for survival.

### 6.5.4 Atmospheric Oxygen and Ions

Oxygen and ions are environmental components of the atmosphere that are difficult to study at best. In general, it has been shown that these factors combine to inactivate many species of airborne microbes. Oxygen toxicity is not related to the molecular form of oxygen ( $O_2$ ), but is instead important in the inactivation of microorganisms when  $O_2$  is converted to more reactive forms. These include the reactive oxygen species like superoxide radicals, hydrogen peroxide and hydroxide radicals. These radicals arise naturally in the environment from the action of lightning, UV radiation, pollution, etc. Such reactive oxygen species cause damage to DNA by producing mutations, which can accumulate over time. There is, however, proportionate repair of such damages in the microbial cells, which controls damaging effects of reactive forms of oxygen. Similarly, the open air factors (OAF), a term coined to describe an environmental effect that cannot be replicated in laboratory experimental settings, more or less linked to oxygen toxicity and have been defined as a mixture of factors produced when ozone and hydrocarbons (generally related to ethylene) react. For example, high levels of hydrocarbons and ozone causing increased inactivation of many organisms, probably because of damaging effects on enzymes and nucleic acids. There is also irreparable damage to the microbial cells, resulting in the metabolic inhibition and death of cells in many cases.

Due to many natural and industrial processes many a different types of ions like chlorine, nitrogen and sulphur are released to the environment. Many of these ions undergo atmospheric reactions to form secondary ions and complexes, some of which are toxic to the airborne microbes. These ions have a wide range of biological activity, ranging from inactivation to physical decay of microorganisms. Positive ions cause inactivation of cell surface proteins, whereas negative ions exhibit both physical and biological effects such as internal damage to DNA.

## 6.6 THE PHYLLOSHERE CONTRIBUTION TO AEROMICROFLORA

The aerial part of plants bearing epiphytic microbial colonies and associations growing as discrete or confluent mass is called phyllosphere. In a more general

sense, the habitat adjacent to the plant surface is known as the phyllosphere and the habitat directly on the surface of the plant is called the phylloplane. Stems, leaves and fruits of plants provide suitable habitats for microbial populations. Most of the phyllosphere microbes show regular attachment and dissociation, thus making their continuous entry into atmosphere. Heterotrophic and photosynthetic bacteria, fungi (particularly yeasts), lichens and some algae, mostly cyanobacteria, regularly occur on these aerial plant surfaces. Various bacterial and fungal populations occupy the phylloplane and phyllosphere habitats. The number of microorganisms on leaf surfaces depends on the season and age of the leaf and bark. The density of the microbial population in the adjacent air also determines the number of microbial cells on the leaf surface.

Phyllosphere microbes are mostly studied by their density and physiological activity on the plant surface. Numerical taxonomic studies have also been used to identify the bacteria in the phyllosphere. The analysis of the phyllosphere diversity is used as an indirect estimate of the diversity of microbes in air. The principal populations in the phyllosphere of the green needles of some pine trees are *Pseudomonas* species, including *P. fluorescens*. Populations from the phylloplane of pine trees are relatively proficient in utilizing sugars and alcohols as carbon sources compared to bacterial populations in the litter layer underlying these trees, which exhibit more lipolytic and proteolytic activity. Bacterial populations in the phylloplane of rye exhibit seasonal changes: xanthomonads and pink chromogens have high population in May, xanthomonads and pseudomonads in July, xanthomonads in September and listeriae and staphylococci in October (Atlas and Bartha, 2005).

Yeasts frequently inhabit the leaves of plants. Populations of *Sporobolomyces roseus*, *Rhodotorula glutinis*, *R. mucilaginosa*, *Cryptococcus laurentii*, *Torulopsis ingeniosa* and *Aureobasidium pullulans* are normal inhabitants of the phyllosphere. There is an abundance of pigmented populations of yeasts and bacteria on leaf surfaces. The pigments of these microbial populations probably afford protection against exposure to direct sunlight on the leaf surface. *Sporobolomyces* is perhaps the most successful fungus that develops in the phyllosphere. It produces ballistospores that can shoot from one leaf to another, facilitating its dispersal. Many other fungi, including Ascomycetes, Basidiomycetes and Deuteromycetes, have been isolated from the phyllosphere. Some of these are undoubtedly allochthonous populations. Though the phyllosphere fungi are mostly non-pathogenic, establishing beneficial relationship with the host plant, some are associated with plant disease. Populations of *Alternaria*, *Epicoccum* and *Stemphylium* have been found to be phylloplane invaders that grow extensively only under favourable conditions. Populations of *Ascochyta*, *Leptosphaeria*, *Pleospora* and *Phoma* found in the phyllosphere are primarily saprophytes that are unable to grow extensively until the onset of senescence. Allochthonous fungal populations frequently found in the phyllosphere include *Cryptococcus*, *Myrothecium*, *Pilobolus* and various other fungal populations whose normal habitat is the soil.

Phyllosphere and phylloplane microorganisms are directly exposed to climatic changes and environmental stresses. These populations must withstand direct sunlight, periods of desiccation and periods of high and low temperatures. Most successful epiphytes are pigmented and have specialized protective cell walls, adaptive features for withstanding these adverse environmental conditions. For example, the presence of sun-screen pigments (microsporin) in cyanobacteria enable them to grow on plant surface without being photoinhibited. Epiphytic microorganisms also exhibit various spore discharge mechanisms that allow them to move from one plant surface to another. Insects play an important role in the dissemination of microorganisms on fruit surfaces. Sometimes there is a close synergistic relationship among microorganisms, insects and plants.

In many plants, flowers form a group of short-lived habitats for epiphytic microorganisms. There is easy dispersion of microbial cells and spores from flowers to the atmosphere. The high sugar content of the flower nectar makes the flower a suitable habitat for yeasts like *Candida reukaufii* and *C. pulcherrima*. Many other yeast populations, including other *Candida* species, *Torulopsis*, *Kloeckera* and *Rhodotorula*, are found in various parts of flowers. Following fertilization of flowers and during the ripening into mature fruits, the environmental conditions of this habitat change and populations of microorganisms undergo a succession, with populations of *Saccharomyces* sometimes becoming dominant. The growth of osmophilic yeasts lowers the sugar concentration, making the habitat suitable for invasion by other microbial populations. Thus flowers act as a source of inoculum of microbes to the air and also receive many airborne microbes, those settle, multiply and are dispersed to the air.

Unsaturated fatty acids produced by yeasts may inhibit development of gram-positive bacteria on fruit surfaces. Bacterial populations that develop on fruit surfaces are dependent on growth factors, such as thiamine and nicotinic acid, that are produced by yeasts. Yeast populations are also dependent on growth factors produced by bacteria on fruit surfaces. On the tree bark, lichens, various fungi, such as bracket and shelf fungi and bacteria are conspicuous colonizers. Myxomycetes (species of *Licea*, *Trichia* and *Fuligo*) form fruiting bodies on tree bark. The fungi *Colletotrichum* and *Taphrina* grow on woody stems and tree bark. A summary of the microbes growing on the plant surfaces has been given in Table 6.4.

In several interesting phyllosphere associations, fungal endophytes grow intercellularly in the wild and cultivated grasses of the Poaceae family and endow them with protection against herbivores. The best described are associations of the tall fescue (*Festuca arundinacea*) and the perennial rye grass (*Lalium perenne*) with the fungi *Acremonium coenophialum* and *A. lolii*, respectively. The endophytes cause no disease symptoms in the grasses, but while receiving photosynthate from the host plant they synthesize a range of alkaloids such as

ergopeptides, lolines, lolitrems and peramines. These act as poisons or feeding deterrents against nematodes, aphids, insects and mammalian herbivores. The lolitrems are especially potent mammalian neurotoxins and can cause livestock losses (rye grass staggers) on heavily infected pastures.

The phyllosphere has, in fact, an indigenous microflora, which complete their life cycle on the plant surface (autochthonous phyllosphere population). However, majority (> 70 %) of the phyllosphere microbes are allochthonous in nature and use the phyllosphere habitat as a transit place during their movement from soil/ water to air and vice-versa. The soil microbes, in cellular, hyphal and spore forms, are released from soil and get deposited on the phyllosphere. Many spores get appropriate conditions to germinate in the phyllosphere habitat and then get dispersed into air. The aerial microbes get settled on the plant surfaces and remained attached to it. The rate of attachment is determined by the environmental and climatic factors.

**Table 6.4:** The plant surface and the associated microorganisms most commonly growing on it.

| Surface | Microorganisms                                                                                                                 |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | Cyanobacteria/Algae                                                                                                            | Bacteria                                                                          | Fungi                                                                                                                                                                                                                                                                                                                                                                                          |
| Leaf    | <i>Anabaena</i> , <i>Nostoc</i> ,<br><i>Scytonema</i> , <i>Aulosira</i>                                                        | Pseudomonads,<br>Xanthomonads,<br>Pink chromogens,<br>Listeriae,<br>Staphylococci | <i>Sporobolomyces roseus</i> ,<br><i>Rhodotorula glutinis</i> ,<br><i>R. mucilaginis</i> ,<br><i>Cryptococcus laurentii</i> ,<br><i>Torulopsis ingeniosa</i> ,<br><i>Aureobasidium pullulans</i><br><i>Alternaria</i> , <i>Epicoccum</i> ,<br><i>Stemphylium</i> , <i>Ascochyta</i> ,<br><i>Lep-tosphaeria</i> , <i>Pleospora</i> ,<br><i>Phoma</i> , <i>Myrothecium</i> ,<br><i>Pilobolus</i> |
| Flower  | -                                                                                                                              | Pseudomonads,<br>Xanthomonads,,<br>Staphylococci                                  | <i>Candida reukaufii</i><br><i>C. pulcherrima</i> ,<br><i>Torulopsis</i> , <i>Kloeckera</i> ,<br><i>Rhodotorula</i> ,<br><i>Saccharomyces</i>                                                                                                                                                                                                                                                  |
| Fruit   | -                                                                                                                              | Pseudomonads,<br>Xanthomonads,<br>Staphylococci                                   | <i>Candida</i><br><i>Saccharomyces</i><br><i>Alternaria</i><br><i>Rhodotorula</i>                                                                                                                                                                                                                                                                                                              |
| Bark    | <i>Anabaena</i> , <i>Nostoc</i> ,<br><i>Cylindrospermum</i> ,<br><i>Chlorella</i> , <i>Selenastrum</i> ,<br><i>Scenedesmus</i> | Many gram-positive<br>bacteria,<br>Bacterial pathogens                            | <i>Licea</i> , <i>Trichia</i> , <i>Fuligo</i> ,<br><i>Colletotrichum</i> ,<br><i>Taphrina</i>                                                                                                                                                                                                                                                                                                  |

## 6.7 THE CONTROL OF BIOAEROSOL

The control of airborne microorganisms can be handled in a variety of ways. Launching, transport and deposition are all points at which the airborne spread of pathogens can be controlled. The mechanisms used to control bioaerosols include ventilation, filtration, UV treatment, biocidal agents and physical isolation. Facilitating movement of air (ventilation) is the method most commonly used to prevent the accumulation of airborne particles. This mechanism involves creating a flow of air through areas where airborne contamination occurs. However, ventilation alone may not be sufficient to significantly reduce circulating bioaerosols. It needs to be assisted with other bioaerosol control methods to make the removal of airborne viable particles effective.

Unidirectional airflow filtration is a relatively simple and yet effective method for control of airborne contamination. Some filters, for example, high efficiency particulate air (HEPA) filters are reported to remove virtually all infectious particles. These types of filters are commonly used in biological safety hoods. However, because of their high cost, they are not often used in building filtration systems. Instead, other filtration systems that rely on bag-house filtration (like a vacuum cleaner bag) are used. Typically, air filters are rated using the dust-spot percentage, which is an index of the size of the particles efficiently removed by the filter, with higher percentages representing greater filtration efficiencies. The typical rating for the filters used in most buildings is 30 to 50%. However, studies have shown that a 97% dust-spot rating is required to effectively remove virus particles from the air.

Biocidal control represents an added treatment that can be used to eradicate all airborne microorganisms, ensuring they are no longer viable and capable of causing infection. Many eradication methods are available, for example, superheating, super-dehydration, ozonation and UV irradiation. The most commonly used of these methods is ultraviolet germicidal radiation (UVGI). The latter has been shown to be able to control many types of pathogens, although some microbes show various levels of resistance. Isolation (enclosure of an environment through the use of positive or negative pressurized air gradients and airtight seals) also is used as a control mechanism for reducing the airborne microbes. It is often used to protect other people in the hospital from the pathogens (*e.g. Mycobacterium tuberculosis*) present inside the isolation area. Air from these rooms is exhausted into the atmosphere after passing through a HEPA filter and biocidal control chamber. A negative pressure isolation chamber allows fresh and filtered air into the room keeping the particle concentration of the room under control. On the other hand, positive-pressure isolation chambers work on the opposite principle by forcing air out of the room, thus protecting the occupants of the room from outside contamination.

## **6.8 CONCLUSION**

In comparison to their role in water and soil, the microbes in air are metabolically less important. But air acts as a medium for microbes to spend a brief period of inactive state and also help the microbial cells and spores to move long distances and to infect new habitats. The AMB pathway is thus important in microbially communicating one habitat with the other. This chapter has covered several areas that can be affected by the AMB pathway. There are, however, several environmental problems associated with the AMB pathway. For example, it has been suggested that airborne bacteria and fungi are causative agents in the formation of calcium oxalate films on stone monuments. The formation of this film is caused by metabolic transformations of proteins and lipids associated with the stone. These calcium oxalate films are extensive, yellow-brown in colour and resist disinfection and they diminish the beauty of the monuments. In subterranean environments, cultural relics such as tombs, prehistoric caves and underground churches are subject to biodeterioration caused by contamination with airborne microorganisms. Deposition of airborne microorganisms are also responsible for degradation of papers, parchment, leather and adhesives in libraries. This becomes an especially important cause of degradation especially if the relative humidity in the library rises above 65%. Many of these decomposers are not yet fully studied. The determination of the total microbial diversity of air is now needed to control a number of plant and animal diseases and to make the use of these microbes in human welfare.

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# MICROBES OF EXTREME ENVIRONMENTS

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## 7.1 INTRODUCTION

The condition that does not allow the normal growth and metabolic activities of widely observed mesophilic microorganisms such as *Escherichia coli* or *Bacillus subtilis* is called an extreme environment. The extreme environmental conditions are unusual habitats having one or more stress factors like high salinity, too low or too high temperature, extreme acidic or alkaline conditions, high organic or metal contaminants, etc. Extremophiles are organisms that are well adapted to these unusual conditions. They require extreme environmental conditions for normal and active growth and do not grow under what is generally referred to as “normal conditions”. The optimal growth conditions for mesophilic bacteria act as extreme and non-tolerable for extremophiles.

The microbes of extreme environments are of various types such as thermophiles (heat loving), alkaliphiles (alkaline-loving), the halophiles (salt loving), the acidophiles (acid-loving), psychrophiles (cold-loving), barophiles (pressure loving, previously called piezophiles), etc. In addition, microorganisms growing at high concentrations of heavy metal ions, high doses of UV radiation and low concentrations of nutrients (oligotrophs) are also considered extremophiles. These microorganisms are physiologically heterogeneous. They include aerobic and anaerobic chemoorganotrophs, photoautotrophs, chemolithotrophs and photoheterotrophs. They are not always limited to extreme environments but some have also been isolated from natural mesobiotic environments. For example, the halophilic alkaliphilic thermophilic microorganism, *Bacillus thermoalkaliphilus* has been isolated from soil mounds of the termite *Odontotermes obesus*, situated in a semiarid area with ambient temperature and high pH. The most alkaliphilic anaerobic thermophiles, *Clostridium paradoxum* and *C. thermoalkaliphilum* are observed in several mesobiotic sewage facilities, having even slightly acidic pH. Many of the described extremophiles exhibit only one extreme growth or physiological



condition. However, most environmental niches show combinations of different extreme conditions such as an elevated salt concentration, pH and/ or temperature. With prolonged exposure to these combinations of extreme conditions, the extremophiles develop unusual ability to tolerate them.

## **7.2 TYPES OF STRESSES IN EXTREME ENVIRONMENTS**

Almost all environments of the earth are influenced by one or more environmental stresses in most part of the year but in majority of cases such stresses are either ephemeral in nature or are mild, remaining within the tolerance limit of the existing microbial population. However, in some habitats the stresses are of high magnitude, affecting adversely the normal growth and metabolism of the mesophilic microorganisms. Such environments are defined as the extreme environments, each with its characteristic microbial diversity corresponding to the nature of the habitat. The morphology, physiology and diversity of the microbial consortium of the habitat are determined by the nature and magnitude of the stress of the concerned environment.

### **7.2.1 The High Temperature Stress**

The environment exposed to temperature that is regarded as high causing the inhibition of the cellular enzymes is regarded as thermal environment. The organisms that are able to colonise such environments and show near normal metabolic activity at such an elevated temperature are called thermophiles. In environments with temperature higher than the physiological temperature of mesophiles, growth of mesophilic bacteria is inhibited by the abnormal function of the cellular enzymes and altered permeability of the cell membrane. Increase in the environmental temperature accelerates the fluidity of the membrane lipids, thus reducing the membrane lipid packing density. This results in loss of cellular metabolites and ions and dissociation of membrane bound enzymes.

### **7.2.2 The Environment with Low Temperature Stress**

Temperatures of less than 0°C during at least a few months in the year is generally regarded as the environment imposing low temperature stress. A good portion of the Earth's biosphere (ca. 85%) experiences temperatures of less than 5°C during some part of the year and over 75% is permanently cold. The low temperature retards the growth and physiological activity of mesophilic microbes by the imposition of effect on the membrane fluidity. Lowering of temperature in the external environment first affects the membrane's physical state making the membrane rigid by converting the membrane lipids from sol to gel phase. These changes in the membrane's structural and dynamic characteristics would in turn affect the functions of membrane proteins, such as the proteins involved in respiration and transport and as a consequence either the cell adapts to the change

in its environment or it would succumb to the exposure. Increase of cytoplasmic membrane thickness and inhibition of lateral flow of ions among membrane components are caused as consequence of cold stress.

### 7.2.3 The Metal Contaminated Environment

Metal contamination of soil and water is probably the most common problem of the environment. With the growth of human civilization there is continuous increase of heavy metal concentrations (especially of Cu, Cd, Co, Hg and Zn) in the environment. Metal contaminated extreme conditions are areas receiving the acid mine drainage or having the mine overburden in the vicinity. While the organic pollutants are degraded in the environment by microbial mineralization, the metals only change their state from bioavailable form to a bio-nonavailable one by the involvement of microorganisms. Microbial activity transforms the metals from one form to the other thereby regulating their movement to the higher trophic levels and attenuating the toxic potential of heavy metals to higher organisms. However, many microbes also show sensitivity to the heavy metal toxicity and are adversely affected in the heavy metal contaminated environments. Many microbial enzymes, especially those rich in SH group, are sensitive to the heavy metals. Heavy metal toxicity induces membrane disorder, electrolyte leakage and structural integrity of cellular enzymes.

### 7.2.4 The Hypersaline Environment

Hypersaline environments are ubiquitous. Hypersaline environments usually arise due to evaporative concentration driven by sunlight. The existence of halophiles with high temperature optima is probably an adaptation to the elevated temperatures reached in these solar heated environments, even though a few may also contain hot springs such as the saline Mono Lake, California. There are two types of hypersaline environments, thalassohaline and athalassohaline. Thalassohaline environments arise due to evaporation of seawater and their ionic composition is similar to that of seawater, with sodium chloride as the predominant salt. The solar salterns that are found worldwide next to seashores are the examples of thalassohaline conditions. Athalassohaline environments arise from non-seawater sources and are usually fed by surface or river water. They are dominated by potassium, magnesium, sodium and carbonate ions. Some examples of athalassohaline environments are the alkaline soda lakes, the Dead Sea, the East African soda lakes of the Kenyan-Tanzanian Rift valley and Mono Lake, California.

In hypersaline conditions the microbes face the challenge of maintaining the cytoplasmic osmotic potential directed inward to facilitate the uptake of ions and metabolites. In addition, high concentration of salts in the cell cytosol is also achieved, which adversely affects several cellular enzymes. Thus the halophilic microbes must have the ability to facilitate salt expulsion from the cell cytoplasm but at the same time maintain the intracellular osmotic potential strong enough to support the transmembrane uptake of ions and metabolites. While some microbes redesign their biomolecules to withstand the high internal salt concentration and

act under high intracellular salt level, others have normal biomolecules, as that of mesophilic microbes, but physiological activity is maintained at normal intracellular salt concentration through the maintenance of osmotic potential by synthesis and accumulation of organic osmolites.

### 7.2.5 The Acidic and Alkaline Environments

In many hypersaline environments the presence of ions like  $\text{Cl}^-$ ,  $\text{SO}_4^{2-}$  and  $\text{CO}_3^{2-}$  increases the pH to a level from 9-11. Such hypersaline lakes are habitats to dense communities of haloalkaliphilic microorganisms. Similarly the environment with high organic load experience a comparatively high acidity, mostly due the microbial production of organic acids. Acidity or alkalinity of an environment is also caused due to release of highly acidic/alkaline domestic and industrial wastes. These extreme environments retard the growth of microorganisms of mesophilic type. In acidic and alkaline environments the microorganisms face the problem of maintaining the transmembrane electrochemical gradient, which is most essential for active transport of ions and metabolites into and out of the cells. Thus the immediate effect of increased acidity / alkalinity is an altered membrane transport and intracellular ion imbalance. In addition, the distribution of ionic groups ( $-\text{OH}$ ,  $-\text{CHO}$ ,  $=\text{CO}$ , etc.) on the cell surface is also altered due to change in medium pH, resulting in modification of the preference of a cell to ions outside and their subsequent uptake.

### 7.2.6 The Environment with Toxic Organic Pollutants

Major environmental pollutants are the products of large scale manufacturing, processing and handling of chemicals of petroleum industries and have led to serious environmental contamination. A wide variety of hazardous and toxic hydrocarbons such as polychlorinated biphenyls (PCBs), trichloroethylene (TCE), polyaromatic hydrocarbons (PAHs) and pesticides are designated as xenobiotics because of their relative recalcitrance to biodegradation and these compounds have the potential to bioaccumulate in living systems. Nitroaromatic pesticides including organochlorines (DDT,  $\alpha$ -hexachlorocyclohexane, endosulfan), organophosphates (chlorpyrifos, dimethoate, parathion and methyl parathion), nitrophenols (dinoseb, dinitrocresol) and their partially hydrolyzed products can be leached into the groundwater while their use in agriculture. On the other hand, nitrophenols, nitrobenzenes and nitrotoluenes have been primarily used as chemical intermediates in the production of dyes (azo and sulphur dyes), photographic developers, wood preservatives, hydraulics, fire retardants, pharmaceuticals, plasticizers, solvents, etc. and are often released to surface water in waste streams. Explosives such as picric acid (2,4,6-trinitrophenol), 2,4,6-trinitrotoluene (TNT), hexahydro-1,3,5-trinitro-1,3,5-triazine (RDX) and octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazine (HMX) are less biodegradable and often persist for extended periods in soil or groundwater. Several of these have been categorized as either priority pollutants or toxic and/ or hazardous chemicals.

Except for pesticides and insecticides, which are widely used in agriculture and domestic pest control operations, the other xenobiotics are spilled into the environment as waste products of different industrial processes. The pesticides and insecticides are also washed out to the surface waters and added to the ground water in small amount in comparison to the concentrations in which they are applied in the target sites. There are frequent reports of the accumulation of pesticides in the soil and the residual toxicity of these chemicals on countless non-target organisms. On the other hand, the industrial organic pollutants are spilled into the environment at considerably high concentrations and affect many organisms ranging from microbes to animals of higher trophic levels, thus changing the composition of biota. Sensitive microbes are easily eliminated by the toxicity of such pollutants but tolerant microbes have the exceptional ability to tolerate the pollutants as well as to perform their metabolic and / or cometabolic degradation.

Most of the toxic organic pollutants released to the environment are oils or lipophilic substances having high octanol-water partition coefficients. These chemicals have high affinity to the lipid rich zones of the cells and are preferably accumulated in the cell and organellar membranes. This results in the change of membrane properties and membrane energy potential. Therefore, in almost all microorganisms, the toxic organic pollutants bring anomaly in the membrane active transport processes.

## 7.3 THE DIVERSITY OF THERMOPHILIC MICROBES

The predominance of high temperatures on the ancient earth correlates with the ancient origin of thermophilic prokaryotes, representing the deepest phylogenetic branches. The ability to utilize electron acceptors other than oxygen is a unique property of prokaryotic microorganisms and is especially important for thermophilic prokaryotes. High temperatures and the presence of reducing agents of volcanic origin (hydrogen sulphide in the first place) make most of natural thermal habitats limited in oxygen. Meanwhile, many inorganic donors and acceptors of electrons occurring in hydrothermal vents are of volcanic origin and can sustain the growth of anaerobic prokaryotes. Many well studied groups of anaerobes such as methanogens and sulphur-, nitrate- and iron-reducing thermophilic prokaryotes and thermophilic microorganisms that can oxidize CO<sub>2</sub> anaerobically by forming hydrogen and water are distributed in different thermophilic environments. However, thermophilic bacterial and fungal activity is also seen under aerobic condition when the temperature dissipation from the habitat is slower than that generated by the activity of the microbes but sufficient oxygen is available to lead an aerobic life.

### 7.3.1 Aerobic Thermophilic Bacterial Diversity

*Bacillus* has been documented to be the predominant component of many compost ecosystem within the temperature range from 30°C to 65°C. *Bacillus*

*licheniformis*, *B. megaterium*, *B. subtilis* and *B. pumillus* grow considerably well in elevated temperature. In a composting system with poor dissipation of heat, there is regular succession of *Bacillus* species. Other bacterial species found in thermophilic composting systems are, *Arthrobacter arilati*, *Ochrobacterum* sp. and *Stenotrophomonas maltophilia*. Beffa *et al.* (1996) observed obligate and facultative autotrophic sulphur and hydrogen-oxidising thermophilic bacteria, like *Bacillus schlegelli* and *Hydrogenobacter* sp. at temperatures above 70°C and these also appeared within the pockets of reduced oxygen tension in a predominantly aerobic compost. *Thermus* strains have been reported to be dominant and active degraders in thermogenic composts above 70°C. Thermophilic oval spore forming bacilli are very active during composting when the temperatures are between 50°C and 65°C. However, during the thermogenic phase, temperature usually reaches between 65°C and 80°C, which drastically reduces the solubility of oxygen in the compost atmosphere driving it to the anaerobic state. No fungi and aerobic thermophilic bacilli are observed to grow at temperature above 70°C.

Thermophilic actinomycetes viz. - *Nocardia brasiliensis*, *Pseudonocardia thermophila*, *Saccharomonospora viridis*, *Saccharopolyspora rectivirgula*, *Streptomyces diastaticus*, *S. griseus*, *S. rectus*, *S. thermoviolaceus*, *S. thermovulgaris*, *S. violaceoruber*, *Thermoactinomyces thalophilus*, *T. vulgaris*, *T. chromogena*, *T. curvata* and *T. fusca* grow extensively in compost. *Thermomonospora* spp., a species with white aerial mycelium (including *T. curvata*, *T. alba* and *T. fusca* and *T. chromogena*) often dominate in the composting systems with increase of temperature but only at temperature up to 70°C. *Streptomyces diastaticus* and *Thermoactinomyces vulgaris* are the predominant members of the spent mushroom compost under mesophilic conditions. Other actinomycete isolates include, *Streptomyces albus* and *S. griseus* from such habitats.

### 7.3.2 The Haloalkalithermophiles

The haloalkalithermophiles are a special physiological group only recently recognized in the aerobic as well as anaerobic thermophilic habitats. They require, in addition to high temperature and salt concentration, alkaline pH for optimal growth. Adaptation to growth at these multiple extreme conditions represents an evolutionary interesting combination. To date, only one haloalkalithermophile has been validly described. An important factor should be taken into account during the comparison of growth optima for haloalkalithermophiles is their physiological dependence on external pH. There is a strong dependence of pH on temperature, the effect being greater with increased distance from neutral pH and varies with media composition. The effect becomes more complicated with the addition of high concentration of NaCl to the medium, as NaCl causes a shift to more acidic pH.

*Halonatronum saccharophilum* is the most recently described haloalkalithermophile. It is an anaerobic spore forming bacterium that belongs to the

order *Haloanaerobiales*. The optima for growth are 1.1-1.5M NaCl, pH 8.5 and temperature optima 36-55°C, which represents a very broad temperature range. *H. saccharophilum* exhibits a fermentative metabolism and grows on mono- and di-saccharides, starch and glycogen. Fermentation products from glucose are acetate, ethanol and H<sub>2</sub>. *Bacillus thermoalcaliphilus* is a chemoorganotrophic, facultatively anaerobic bacterium with growth optima at 1.5 M NaCl, pH 8.5-9.0 and temperature 60°C. Many other halotolerant alkalithermophilic microorganisms have been isolated but have not been characterized beyond the genus level. The examples include the facultatively aerobic, non-motile, spore-forming *Bacillus* sp. BG-11, that is capable of growth at 1.5M NaCl, pH 7.5-9.0 and temperatures greater than 55°C. *Bacillus* STS1 grows optimally at 1M NaCl, pH 9.0 and at 60°C.

No haloalkalithermophilic archaea have been isolated so far, even though the majority of the halothermophiles are archaea. All the isolated halothermophiles are neutrophiles. It is unclear whether the scarcity of haloalkalithermophilic isolates, bacteria or archaea, is due to inhibition of microbial growth at multiple extreme conditions or a limitation in the methods and techniques used to explore this group of extremophiles.

The halothermophiles show combine adaptive mechanisms of both halophiles and thermophiles. So far, none of the described isolates grows at combinations of salt concentrations and temperature greater than 4.5M and 60°C. Such extreme combinations probably require adaptive mechanisms different from those presently known and described halophiles and thermophiles. Halophilic microorganisms must maintain their cytoplasm, at least of isoosmotic level, with their surroundings in order to prevent loss of water to the environment. Maintenance of a turgor pressure requires a hyperosmotic cytoplasm. All halophilic microorganisms, with the possible exception of the halophilic archaea of the family *Halobacteriaceae*, maintain a turgor pressure.

Thermophiles are faced with the challenge of controlling cytoplasmic membrane permeability at high temperatures. Increased motion of lipid molecules at elevated temperatures results in increased permeability to protons. Due to this motion, water molecules become trapped in the lipid core of the membranes allowing protons to hop from one molecule to the other. Other ions, unlike protons, can diffuse through the membrane. Diffusion is a temperature dependent process hence membrane permeability to ions will increase as well. In addition, thermophiles must have mechanisms to preserve protein stability and structure at elevated temperatures. The halothermophiles show incredibly high tolerance to high ion concentration, high temperature, scarcity of molecular oxygen, high UV radiation and in some cases high pH. They adapt their cytoplasmic membrane content to somehow decrease permeability at high temperature. They have also a comparatively increased respiration rate enabling them to maintain an increased rate of proton/ion pumping out of the cell. This has been observed in aerobic bacteria living at temperatures around 60-65°C, e.g. *Geobacillus stearothermophilus*. However, this option is energetically unfavourable and may not be possible due to low solubility



of molecular oxygen in brines. Light energy absorbed by the retinal pigment bacteriorhodopsin (found in many haloarchaea) is used to pump protons and ions across the membrane thus generating a proton/ sodium motive force.

### 7.3.3 Thermophilic Fungi

Of more than the 100,000 species of fungi described to date, only about 40 are considered thermophilic. These fungi are generally isolated from self-heating decaying organic material, such as decaying plant and animal debris. Their optimal growth temperature varies from 35 to 50°C and their heat stable enzymes are much envisioned for biotechnological applications. *Cladosporium* sp., *Humicola fuscoatra*, *Penicillium* sp. and some yeast have been frequently isolated from moderately thermophilic compost. Bahl *et al.* (1989) reported for the first time the presence of *Acremonium persicinum*, *Papulospora byssina*, *Telachilidium brachiatum* and *Verticillum psalliotae* in compost at 50°C.

Perhaps the most employed source of industrial thermostable cellulases is the saprophyte *Humicola insolens*, whose cellulolytic complex is composed of at least five endoglucanases and two cellobiohydrolases. Another member of the *Humicola* genus, *H. grisea*, is also a potent cellulase producer, presenting a considerable potential for agricultural wastes bioconversion process. *Thermoascus aurantiacus* cellulase has a better activity at 50°C than at 30°C showing the thermal preference of the fungus. In the thermophilic fungus *Talaromyces emersonii*, cellulolytic system is mainly composed by endoglucanases and beta-glucosidases and a cellobiohydrolase. *Melanocarpus albomyces* shows thermal adaptation through expression of a heat stable cellulase, which enables the fungus to use complex substrates even at 55°C. The ascomycete *Chaetomium thermophilum* secretes two distinct cellobiohydrolases, with optimum temperature of 70°C and 75°C. A *Torula thermophila* strain, isolated from mushroom compost, produced a complex array of cellulases upon growth on microcrystalline cellulose at 50°C (Table 7.1).

The pioneer thermophilic mycoflora during composting, which has a prolonged mesophilic stage, comprises fast growing and rapidly sporulating fungi such as *Aspergillus fumigatus* and *Rhizomucor* spp. with a pH optima below 7.0 and temperature optima of about 40°C. When self-heating and ammonification starts and pH reaches 9.0, the pioneer flora disappears and paves way for *Talaromyces thermophilus* and *Thermomyces lanuginosus* during massive heat production as these fungi exhibit high thermal death point and pH tolerance. Such fungi have moderate growth rate and do not degrade cellulose. At the end of the composting process, about 50-70% of the compost biomass is constituted by thermophilic fungi. While most of the species are eliminated, *Scytalidium thermophilum* appears as near exclusive species after phase II composting and constitutes a climax species. Thus, thermophilic fungi forms the climax community in mushroom compost and contributes significantly towards the quality of compost. However, their presence throughout the course of composting is largely responsible for the maintenance of biological equilibrium. Thermophilic fungi influence growth of *Agaricus bisporus*



by decreasing the concentration of ammonia in compost and improving the apparent availability of nutrients.

**Table 7.1:** The temperature optima and thermostable enzymes of different thermophilic fungi.

| Name of the fungus                                                                                                                                                                                                | Temperature optimum (°C) | Thermostable enzyme                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------|
| <i>Cladosporium</i> sp., <i>Humicola fuscoatra</i> , <i>Penicillium</i> ,<br><i>Acremonium persicinum</i> ,<br><i>Papulospora byssina</i> ,<br><i>Telachilidium brachiatum</i> ,<br><i>Verticillum psalliotae</i> | 50                       | Cellulases                                                 |
| <i>Humicola insolens</i> , <i>H. grisea</i> ,<br><i>Thermoascus aurantiacus</i>                                                                                                                                   | 50                       | Five endoglucanases and two cellobiohydrolases             |
| <i>Talaromyces emersonii</i>                                                                                                                                                                                      | 55                       | Endoglucanases, beta-glucosidases and a cellobiohydrolase. |
| <i>Melanocarpus albomyces</i>                                                                                                                                                                                     | 50-55                    | Cellulose                                                  |
| <i>Chaetomium thermophilum</i>                                                                                                                                                                                    | 70-75                    | Cellobiohydrolases                                         |
| <i>Torula thermophila</i>                                                                                                                                                                                         | 50                       | A complex array of cellulases                              |
| <i>Aspergillus fumigatus</i> ,<br><i>Rhizomucor</i> spp.                                                                                                                                                          | 40                       | Cellulases                                                 |
| <i>Talaromyces thermophilus</i> ,<br><i>Thermomyces lanuginosus</i> ,<br><i>Scytalidium thermophilum rmophilum</i>                                                                                                | 40-45                    | Cellulases                                                 |

Thermophilic fungi of the *Torula-Humicola* complex are necessary and dominant components of the community during composting. Dominance of *S. thermophilum* and *H. grisea* has been reported in many composting systems. Other thermophilic fungal species such as, *Chaetomium thermophilum*, *Malbranchea sulfurea*, *Myriococcum thermophilum*, *Stilbella thermophila* and *Thielavia terrestris* grow profusely in partly matured compost having limited anaerobic activity. Tewari (2000) reported the presence of *Humicola lanuginosa* and *S. thermophilum* during peak heat stage of phase II compost. Antagonism also play a major role in determining the density and diversity of thermophilic fungi in composting systems. Volatiles of *Chaetomium thermophile* inhibit mycelial growth of *Humicola lanuginosa* and *Torula thermophila*. However, effect of the fungistatic volatile factors in compost ecosystem is only marginal in view of the high temperature at which they grow.

**7.3.4    Sulphur Reducing Thermophiles**

Elemental sulphur is a common component of volcanic environments, serving both as an energy substrate and an electron acceptor for thermophilic

microorganisms. Many hyperthermophilic archaea can reduce elemental sulphur in the course of anaerobic growth. Most of sulphur-reducing hyperthermophilic organotrophs use elemental sulphur as the sink for electrons produced during fermentation releasing hydrogen sulphide instead of molecular hydrogen, which otherwise is inhibitory for growth. However, many obligately and facultatively anaerobic hyperthermophilic archaea of *Euryarchaeota* and *Crenarchaeota* kingdoms are capable of sulphur respiration, using molecular hydrogen as an energy substrate and reducing elemental sulphur to  $\text{H}_2\text{S}$ . This type of metabolism is claimed to represent the most ancient type of respiration. However, the range of substrates used by sulphur-respiring hyperthermophilic archaea is rather limited. Molecular hydrogen is the most common substrate for sulphur respiration. Some organisms, for example, *Thermoproteales* are able to oxidize peptides completely, concurrently with reduction of elemental sulphur to  $\text{H}_2\text{S}$ . At present, there is no evidence on the ability of hyperthermophilic archaea to oxidize non-fermentable substrates (such as acetate, formate, etc.) via sulphur reduction.

*Desulphurella acetivorans*, isolated from an anoxygenic thermophilic cyanobacterial mat developing in a sulphide-rich hot spring of the Uzon Caldera (Kamchatka), became the first moderately thermophilic bacterium with sulphur respiration. The mat contained significant amounts of elemental sulphur deposited between layers of cyanobacteria, which could serve as a terminal electron acceptor in the course of anaerobic degradation of cyanobacterial cell mass. High concentration of *D. acetivorans* cells ( $10^4/\text{mm}^3$  of mat) indicated its active role in the cyanobacterial community with regard to remineralization. Later, three more species of the genus *Desulphurella* were isolated from different thermal environments. Differing in the range of growth substrates, all *Desulphurella* species are able to use a wide range of nonfermentable organic substrates, including volatile fatty acids, alcohols, organic acids and even saturated fatty acids, all organic substrates being completely oxidized to  $\text{CO}_2$  and  $\text{H}_2\text{S}$ . Elemental sulphur is obligately required by all except *Desulphurella propionica*, which can grow by reducing thiosulphate. The metabolic features of *Desulphurella* species suggest their participation in organic matter mineralization in organic-rich moderately hot anaerobic habitats. However, all *Desulphurella* species are capable of lithoautotrophic growth with molecular hydrogen as the energy substrate and  $\text{CO}_2$  as the carbon source. Thus, in volcanic environments, devoid of photosynthetic or allochthonous organic matter, *Desulphurella* may act as the primary producer, using molecular hydrogen and  $\text{CO}_2$  of volcanic origin.

In shallow-water submarine hot vents of New Zealand and New Guinea, moderately thermophilic sulphur-reducers have been reported, which are phenotypically very close to *Desulphurella*. Their phylogenetic analysis revealed, however, that all these marine isolates belonged to one species of the new genus named *Hipaea maritima*. Unlike *Desulphurella*, *Hipaea* lacks the capacity to grow autotrophically on the mineral medium but is similar to it in many other respects including cell morphology, the ability to use molecular hydrogen as the growth substrate, the obligate requirement of elemental sulphur and oxidation of numerous non-fermentable substrates to  $\text{CO}_2$  and  $\text{H}_2\text{S}$ .

The thermophilic sulphur-reducing bacteria is heterogenic both in the phylogenetic and metabolic respects. This also include the numerous fermentative organotrophs that use elemental sulphur as the electron acceptor for facilitated fermentation. The availability of sulphur greatly regulate the rate of growth of these organotrophs. Many new extremely thermophilic lithotrophic sulphur-respiring bacteria are found to represent the order of *Aquificales*. Obligately lithoautotrophic *Desulphurobacterium thermolithotrophum* utilizing only molecular hydrogen and elemental sulphur has been obtained from deep sea hydrothermal fields. Isolated from the same habitat is *Balnearium lithotrophicum*, another bacterium of this group with a very similar phenotype. Phylogenetically close to *Desulphurobacterium* is the genus *Thermovibrio* having exceptional ability to grow at elevated temperatures. Its members, when grown lithotrophically with molecular hydrogen, can reduce not only elemental sulphur but also nitrate, producing ammonium. The capacity to grow anaerobically with hydrogen and elemental sulphur is found in other facultatively anaerobic representatives of *Aquificales* inhabiting deep sea hydrothermal fields. *Persephonella marina* and *Hydrogenivirga caldilitoris* are obligate chemolithoautotrophs having ability to use molecular hydrogen as the substrate for sulphur or nitrate-reduction and to oxidize elemental sulphur either with nitrate as the electron acceptor, or microaerobically at low concentrations of oxygen. It is very likely that this group of thermophiles includes a large number of other extremely thermophilic obligately or facultatively anaerobic sulphur-reducing organisms.

Sulphur reduction is a characteristic feature of *Oceanithermus desulphurans* and *Deferribacter desulphuricans*. The genera of *Nautilia* and *Caminibacter* use molecular hydrogen as the substrate, although *Nautilia lithotrophica* can also grow on formate. It is obligately dependent on elemental sulphur, while *Caminibacter* species can also grow by nitrate reduction and one of them, *Caminibacter profundus*, is able to grow microaerobically at oxygen concentration being 2% or less. Recently, many new sulphur-reducing microorganisms belonging to Epsilon-proteobacteria have been isolated from deep sea hot hydrothermal environments. They include *Lebetimonas addiphila* and *Hydrogenomonas thermophilus*, which can reduce nitrate or grow under microaerobic conditions at low concentration of oxygen. These indicate that sulphur-reducing thermophilic prokaryotes are diverse and abundant in thermal habitats. While moderately thermophilic bacteria of the *Desulphurella* group occur in terrestrial and deep sea thermal environments, the extremely thermophilic *Aquificales* and moderately thermophilic Epsilon-proteobacteria are predominant in deep sea hydrothermal systems. Their metabolic properties are in good agreement with the chemistry of their natural environment, abounding in elemental sulphur.

### 7.3.5 Nitrate Reducing Thermophilic Bacteria

The ability of thermophiles to grow by means of nitrate reduction is found in several aerobic moderately or extremely thermophilic bacteria of the genera *Geobacillus*, *Thermothrix* and *Thermus*. All these organisms could grow

organotrophically by oxidizing quite a diverse organic substrates through aerobic respiration or nitrate reduction (especially when availability of  $O_2$  is limited). Several lithotrophic and organotrophic hyperthermophilic archaea are also able to grow by nitrate reduction. The first obligately anaerobic thermophilic bacterium reducing nitrate to ammonium in the course of lithotrophic growth with molecular hydrogen was *Ammonifex degensii*. As mentioned in the previous section, the ability to reduce nitrate is shared by many sulphur-reducing representatives of the order *Aquificales* using elemental sulphur as an electron acceptor and reducing it to  $H_2S$ . Like sulphur reducers, all nitrogen reducing thermophiles are hydrogen-utilizing lithotrophs and facultative anaerobes capable of aerobic or microaerophilic respiration. However recent findings have shown that many new thermophilic deep sea isolates like *Caminibacter* and *Deferribacter* utilize nitrate as an electron acceptor and reduce it to ammonium and nitrite, respectively. While *Caminibacter* spp. are obligate hydrogen-oxidizing lithoautotrophs, those of *Deferribacter* can also oxidize acetate, succinate, pyruvate and proteinaceous compounds, thus, participating in organic matter degradation.

The ability to reduce nitrate to nitrite is shared by representatives of the two new aerobic genera in the *Thermaceae* family, viz. *Oceanithermus* and *Thermus*. However, *Oceanithermus* and *Vulcanithermus* seem to be much better adapted to the deep sea thermal environment. Both these organisms are capable of lithotrophic growth with molecular hydrogen and both are microaerophiles. *O. profundus* is able to grow only at an oxygen concentration not higher than 6%. *V. mediatlanticus* can grow in the air atmosphere but only in a liquid medium. *O. profundus* is a moderate and *V. mediatlanticus* an extreme thermophile with optimum growth temperatures of 60 and 70°C, respectively. Such organisms might occur as well in terrestrial hot springs, where until now only aerobic thermophiles have been reported. Recently, a second species of the genus *Oceanithermus*, *O. desulphurans*, has been isolated from deep sea hot vents (Mori *et al.*, 2004).

Only a small number of obligate anaerobes are known among thermophilic nitrate-reducing prokaryotes, most of which are capable of aerobic or microaerophilic growth. An obligate anaerobic nitrate-reducing moderately thermophilic bacterium *Caldithrix abyssi* has been isolated from hot deposits in a Mid Atlantic Ridge hydrothermal field. Cells of the bacterium are long filaments with a Gram-negative cell wall structure. The cells can grow by nitrate reduction, using acetate or hydrogen as the growth substrates and reducing nitrate to ammonium. It can also grow by fermentation of peptides, in which nitrate is not reduced and does not stimulate growth.

### 7.3.6 Iron-reducing Thermophiles

The ability of thermophilic microorganisms to use ferric iron as an electron acceptor for anaerobic respiration was first shown for *Bacillus infernus*, isolated from a deep subsurface habitat. Thermophilic reduction of ferric iron accompanied by magnetite

formation was shown for an acetate utilizing microbial consortium obtained from a Kamchatka hot spring sample, Japan. Organotrophic growth of *Thermoterrabacterium ferrireducens* is observed in the presence of ferric iron, the reduction of which leads to the formation of magnetite. The same phenomenon is observed with *Thermoanaerobacter siderophilus* and *Thermovenabulum ferriorganovorum*. All these organisms are able to use molecular hydrogen as an electron donor for iron reduction but only *T. ferrireducens* is capable of lithoautotrophic growth with  $H_2$  as the energy substrate, Fe(III) as the electron acceptor and  $CO_2$  as the carbon source.

Many hyperthermophilic archaea and thermophilic bacteria with a fermentative type of metabolism are capable to reduce ferric iron in cell suspensions or in growing cultures, using organic substrates or molecular hydrogen as electron donors. The hyperthermophilic archaeon *Ferroglobus placidus* cause iron reduction in the presence of acetate. The same capacity, together with the ability to grow lithotrophically with molecular hydrogen and oxidize a wide range of fermentable and nonfermentable organic substrates, is found in a new hyperthermophilic archaeon *Geoglobus ahangari*. The bacterium *Thermoincola ferriacetica* oxidizes acetate in the course of Fe(III) reduction. It also shows growth on peptides, acetate, or molecular hydrogen but obligately dependent on the concurrent reduction of ferric iron to ferrous iron.

Another phylogenetic group of thermophilic iron-reducing bacteria is represented by the genus *Deferribacter*. The type strain *Deferribacter thermophilus* can grow either lithotrophically with molecular hydrogen, or organotrophically with diverse organic energy substrates, reducing Fe(III), manganese, elemental sulphur and nitrate. Recently two species of this genus *D. desulphuricans* and *D. abyssi* have been recorded in the deep sea hydrothermal environment. Two new chemolithoautotrophic, hydrogen-oxidizing, iron-reducing thermophilic bacteria, a moderately thermophilic *Geothermobacter ehrlichii* and an extremely thermophilic *Geothermobacterium ferrireducens*, have been isolated from the habitat with a temperature range of 65 to 100°C.

### 7.3.7 Hydrogen Producing Thermophiles

The ability to utilize CO in the processes of aerobic respiration, methanogenesis and acetogenesis is found in many mesophilic microorganisms. However, the information about their thermophilic analogues is rather limited. By contrast, anaerobic CO oxidation coupled with  $H_2$  formation, initially observed with mesophilic phototrophs, happens to be shared by many of new thermophilic prokaryotes. *Carboxydotherrmus hydrogenoformans* is the first representative of this metabolic group of thermophiles. It has a optimum growth at 70°C and able to grow in the atmosphere of 100% CO, oxidizing it to  $CO_2$  with the concurrent production of equimolar amounts of  $H_2$ . Similarly a marine anaerobic CO-trophic thermophile, *Carboxydobrachium pacificus*, *Thermoanaerobacter subterraneus* and *Caldanaerobacter subterraneus* of deep sea hot-spring are capable of CO oxidation although no details of this process has been reported.

In the hot spring environment *Carboxydocella thermoautotrophica* and *Thermoincola carboxydophila* are moderate thermophile, growing optimally at 58°C. These organisms are obligately dependent on CO and grows only on CO producing CO<sub>2</sub> and H<sub>2</sub>. However, *T. carboxydophila* is an alkaliphile with a pH optimum for growth of 8.5. Unlike these two microorganisms, another new moderately thermophilic isolate *Thermosinus carboxydophilus* has multiple metabolic capabilities with ability for growth with glucose or peptides. When grown on CO this organism can reduce ferric iron to ferrous iron or selenite to elemental selenium, if present and in this case less hydrogen is formed from water.

### 7.3.8 Thermal Adaptation of Microbes

Daniel *et al.* (2001) have formulated a procedure to describe genuine temperature optima of enzymes and correlated thermal stability with temperature optimum of an enzyme in cases where the temperature was a major factor in enzyme stability. In general, extreme conditions require either the adaptation of the amino acid sequence of a protein by mutations, the optimization of weak interactions within the protein at the protein-solvent boundary and the influence of extrinsic factors such as metabolites and cofactors. As regards thermal adaptations, instead of being a consequence of one dominant type of interaction or of a general stabilization strategy, adaptation to high temperatures reflects a number of subtle interactions, often characteristic for each protein species. Comparison of protein sequences of thermophilic and mesophilic *Methanococcus* spp. reveals amino acid substitutions involved in adapting proteins to higher temperatures. The main change is an increase in hydrophobic residues of the protein molecule. This adaptation minimizes surface energy and the hydration of apolar surface groups while burying hydrophobic residues and maximizing packing of the core. An increase in hydrophobic residues seems to interfere with the mechanism of adaptation to high salt concentrations, as it is seen that the borderline hydrophobic residues serine and threonine are replaced with the more hydrophobic alanine/proline and valine/ isoleucine respectively. Because of such phenomenon the almost all the obligate extreme halophiles of the family *Halobacteriaceae* are moderate thermophiles, with temperature tolerance maxima never exceeding 60°C. Above this temperature minimizing hydration and polar groups at the surface of the protein are necessary in order to maintain stability. The presence of high concentrations of intracellular ions would interfere with protein folding and decrease stability.

## 7.4. MICROBIAL DIVERSITY OF DEEP SEA ENVIRONMENT

The deep sea is a challenging extreme environment characterized by high hydrostatic pressures, low temperatures, near to total darkness and “feast or famine” nutrient regimes. The average depth of oceans is 4000 m and the hydrostatic pressure increases by 1 bar (0.1 MPa) for every 10 m depth. The deepest recorded part of the ocean at 11,033 m lies in the Mariana Trench in the



Pacific Ocean. Depths exceeding 1000 m are considered deep sea and accordingly, 88% of the oceanic environment comes under this zone. There is a constant rain of minute organic particles (marine snow) consisting of faecal matter as well as dead phyto- and zooplankton from surface to deeper waters. Carcasses of nektonic organisms, ranging from fishes to dead whales, provide large food source for deep sea scavengers. Contrary to popular belief, the bottom waters in most of the seas are well oxygenated due to the flow of nutrient and oxygen-rich deep and bottom waters by the South Equatorial Currents and the Pacific through-flow. The oxygen content of surface sea waters is 5-6 mg/l and in bottom waters is 4-5 mg/l and thus O<sub>2</sub> is rarely limiting. Moreover, solubility of oxygen increases in cold bottom waters and increasing hydrostatic pressures. The temperatures in deep seas are around 2-5°C, salinity on an average is 34-35 parts per thousand and the pH is around 8.2.

Deep-sea microorganisms that grow only under elevated hydrostatic pressure are called barophilic (also called piezophilic) and barotolerant ones grow at 1 atmospheric as well as at elevated hydrostatic pressures. These organisms are either psychrophilic (cold-loving) or psychrotolerant (cold-tolerant) and thermophilic or thermotolerant around hydrothermal vents. Isolation of deep sea microbes is mostly done from water samples collected from different depths in the sea using special water samplers such as Niskin, Nansen bottles, Rosett sampler and are cultured at relatively high atmospheric pressure.

### 7.4.1 Deep Sea Prokaryotic Diversity

Organisms growing under hydrostatic pressures of 5-100 MPa, low and high temperatures 2°–100°C, low and high organic nutrients (when dead carcass of large organisms are made available on sea floor) expected to be present in deep sea conditions. ZoBell and Morita (1957) were the first to report presence of barophiles in the deep sea. Deep-sea bacteria have been isolated even from depths of 10,500 m and cultured at > 100 MPa at 2°C and > 40 MPa above 100°C. Most of the barophilic and barotolerant bacteria belong to the Proteobacteria *g* subgroup. Sediment samples obtained from the world's deepest seafloor at 11,033 m in the Mariana Trench using unmanned submersible KAIKA belonging to JAM5TEC shows the dominance of barophilic bacteria over other groups. DNAs extracted directly from these sediments have been amplified by PCR and the products have been sequenced. The results showed coexistence of barophilic Archaea with *Pseudomonas* species in many barophilic environments. In addition, filamentous fungi and actinomycetes are also present at 1 bar pressure (0.1 MPa) with almost the same frequency as that of facultative psychrophilic bacteria. Besides, several alkalophilic, thermophilic and non-extremophiles are present at 0.1 MPa in deep sea sediments (Table 7.2).

Pradeep-Ram *et al.* (2001) isolated a chemolithotrophic nitrifying bacterium from sediments at 5000 m depth in the Central Indian Basin that grew at 1 bar pressure. This bacterial isolate exhibited dual property of oxidizing ammonia to nitrite and nitrite to nitrate under 1 bar pressure at 28°C as well as under 400 bar



pressure at 5°C. Nitrification is generally carried out by known nitrifiers, which either oxidize ammonia to nitrite (Phase 1) or nitrite to nitrate (Phase 2). However, the deep-sea bacterial isolate from the Central Indian Basin performed both the phases of nitrification. Several organic solvent-tolerant bacteria have also been isolated from deep sea sediments. Some of these can grow in the presence of toluene or benzene at concentration greater than 10–50% (v/v). The frequency of such organic solvent-tolerant bacteria in deep sea sediments is much more than in terrestrial source. Two extremely barophilic bacteria DB21MT2 and DB21MT5 isolated from sediments of the Mariana Trench at a depth of around 11,000 m were reported to contain polyunsaturated fatty acids (PUFA). This is a unique feature because it has been believed that bacteria are unable to produce PUFA. Culturable actinomycetes ranging in numbers from  $3.4 \times 10^2$  to  $1.6 \times 10^4$  CFU per gram deep sea sediments have been isolated from the Pacific Ocean.

**Table 7.2:** Barophilic and barotolerant microorganisms and their pressure adaptations.

| <i>Organism</i>                           | <i>Temperature optimum (°C)</i> | <i>Pressure optimum (Mpa)</i> |
|-------------------------------------------|---------------------------------|-------------------------------|
| <b>Bacteria</b>                           |                                 |                               |
| <i>Colwellia</i> MT41                     | 8                               | 103                           |
| <i>C. hadaliensis</i> MT199               | 10                              | 85                            |
| <i>Shewanella</i> SC2A                    | 20                              | 14                            |
| <i>S. violacea</i>                        | 10                              | 30                            |
| <i>S. benthica</i>                        | 15                              | 60                            |
| <i>Moritella</i> PE36                     | 15                              | 41                            |
| <i>M. japonica</i>                        | 15                              | 50                            |
| <i>M. yayanosi</i>                        | 10                              | 80                            |
| <i>Desulfovibrio profundus</i>            | 25                              | 15                            |
| <i>Photobacterium profundum</i>           | 10                              | 10                            |
| <b>Archaea</b>                            |                                 |                               |
| <i>Methanococcus thermolithotrophicus</i> | 65                              | 50                            |
| <i>M. jannaschii</i>                      | 86                              | 75                            |
| <i>Pyrococcus abyssii</i>                 | 95                              | 20                            |
| <i>Thermococcus barophilus</i>            | 85                              | 40                            |
| <b>Fungi</b>                              |                                 |                               |
| <i>Aspergillus restrictus</i>             | 30                              | 30                            |
| <i>A. sydowii</i>                         | 5                               | 50                            |
| <i>A. ustus</i>                           | 10                              | 10                            |
| <i>Graphium</i> sp.                       | 30                              | 10                            |

Reconstructed from Raghukumar (2005).

### 7.4.2 Deep Sea Eukaryotic Diversity

Fungal colonization of wooden panels suspended at deep sea between 500 and 3000 m depth was the first evidence of the existence of eukaryotic microbes in the deep sea. However, these fungi were not cultured for exact identification of

the species. Several culturable fungi such as *Cladosporium* spp., *Alternaria* spp., *Aspergillus sydowii*, *Nigrospora* spp. and *Penicillium* spp. from deep sea water samples up to 4450 m have been reported from different geographical locations. Several filamentous fungi have been isolated from deep sea calcareous sediments using a dilute malt extract broth prepared with seawater. Many of them are able to grow and reproduce under culturable conditions. For example, conidia of some of deep sea fungi germinate under 100 and 300 bar pressure (equivalent to 10 to 30 MPa, corresponding to 1000 and 3000 m depth, respectively) indicating their barotolerance. Most of these are common terrestrial fungi and it is possible that some of these fungi have acquired barotolerance by their prolonged exposure to the extreme habitat.

Besides fungi, heterotrophic, unicellular protist, the thraustochytrids belonging to the kingdom Straminipila, are also found in deep sea waters. Deep sea protozoan species *Bodo* from a North Atlantic sediment sample showed growth up to 45 MPa but showed no growth at atmospheric pressure. Strangely, this flagellate could be isolated without retaining the *in situ* pressure but required high pressure for its growth.

### 7.4.3 Adaptations in Deep Sea Microorganisms

Cold temperature and high pressure decrease membrane fluidity and affect a number of membrane-associated processes including ion and nutrient flux and DNA replication. A barotolerant prokaryote shows an increase in the proportion of unsaturated fatty acid composition in membrane fraction with increased pressure during its growth indicating its possible involvement in maintaining the fluidity of the cell membrane. The membrane phospholipids of two barophilic bacteria DB21MT-2 and DB21MT-5 showed wide distribution of polyunsaturated fatty acids, suggesting that adaptation of the barophiles to low temperature and high hydrostatic pressure influenced the synthesis of phospholipids containing polyunsaturated fatty acids.

Hydrostatic pressure also influences the abundance of cell surface proteins in bacteria adapted to low or high pressures. The moderately barophilic deep sea bacterium *Photobacterium* SS9 was reported to regulate the abundance of several outer membrane proteins (omp) in response to hydrostatic pressure. One outer membrane protein *ompH* was induced 10-100-fold in SS9 cells grown at 28 MPa suggesting that *ompH* probably acts as a non-specific porin protein involved in uptake of substrates larger than 400 daltons. Mutants that lacked *ompH* gene were, however, not pressure-sensitive suggesting that *ompH* is required for high pressure growth under other physiological conditions present in the deep sea such as low nutrient conditions or it may be one of the multiple proteins facilitating growth at high pressure. Barotolerant, barophilic and mesophilic bacteria show elevated levels of several pressure-inducible proteins (PIPs). Many of the identified PIPs are either cold-shock, heat shock or ribosomal proteins. High or low temperatures and high pressure are shown to destabilize tertiary and quaternary structure of proteins. Thus, pressure induced increases in the

proportion of partially unfolded proteins or dissociated protein subunits induces heat-shock proteins. High pressure also turns on both heat and cold shock proteins (hsps and csps) because of its inhibitory effects on ribosome assembly or functions.

In a barophilic strain of *Photobacterium* sp. a pressure regulated operon is expressed under elevated pressure producing the transcripts at 70 Mpa. These highly conserved pressure-regulated operons have been found in many deep sea adapted bacteria. A moderately barophilic strain of *Shewanella* sp. that grows at 30 MPa and 8°C and also at atmospheric pressure (0.1 MPa) and 8°C possessing a pressure-regulated operon (containing open reading frames ORF 1 and ORF 2) has been cloned and sequenced. The strain appears to produce different DNA-binding factors under different pressures.

Besides elevated hydrostatic pressures, the deep sea organisms are also adapted to low temperatures. Synthesis of polysaccharides is greater in cold-adapted microorganisms than in mesophiles. Some psychrotrophs show differential utilization of substrates at different temperatures. Solubility of oxygen increases at low temperatures and thus its availability also increases and therefore, psychrophilic and psychrotolerant microorganisms are more favourably affected by aeration than mesophiles. High lipid content in membranes of psychrotrophs and psychrophiles facilitates membrane transport of solutes. They also contain more unsaturated fatty acids when grown at low temperatures than mesophiles. An increase in unsaturated fatty acids leads to a decrease in lipid melting point and allows to maintain the lipid in a liquid and mobile state and helps to regulate membrane fluidity.

Cold-adapted enzymes have high specific activity and low thermostability. The thermostability is derived from high rigidity of the molecule which retards the interaction between substrate and enzyme, leading to a weak specific activity. By contrast flexibility or plasticity of a molecular structure at low temperature enables interactions at a low energy cost, which explains the high specific activity of cold-adapted enzymes.

#### 7.4.4 Adaptations of Deep Sea Eukaryotes

Yeasts have been used as model organisms for studying pressure-adaptations in eukaryotic heterotrophs. It has been hypothesized that chemical reactions, that are accompanied by an increase in volume, are inhibited by increasing pressure. In *Saccharomyces cerevisiae* it has been observed that maintenance of the cytoplasm at neutral pH might be one of the mechanisms required for piezoadaptation. Under the culture conditions the barotolerance is mainly attributed to cellular trehalose content. It has been proposed in yeast system that the effect of hydrostatic pressure is analogous to high temperatures and that a heat-shock pre-treatment is able to induce barotolerance.

Morphological variations are seen in fungi grown under elevated hydrostatic pressure. The filamentous fungus *Graphium* sp. isolated from deep sea calcareous sediment shows microconidiation, where a conidium gives rise to conidia directly

without intervening hyphal growth. In *Aspergillus ustus*, when grown under elevated pressure of 100 bar, the resting cells called hülle cells shows germination with the fungal hyphae showing thick swellings and beaded-structures. Multiple germ tube formation in yeasts *Rhodospiridium sphaerocarpum* subjected to 450 bar pressure has also been reported. Most of these deep sea fungi show better growth and germination of conidia at 30°C/200 bar than at 5°C/200 bar pressure indicating a preference of the fungi for mesophilic temperature. There are different effects of pressure on mycelium and spores of fungi. A significant population of deep sea fungi also resembles in morphology and physiology to the terrestrial fungi proving the fact that terrestrial fungi are also seen in the deep sea environment. However, the deep sea adapted terrestrial fungi show higher protease production when subjected to pressure under mesophilic temperature range.

7.5    MICROBES OF COLD HABITATS

The microbes growing more or less normally in the cold habitats are called psychrophiles. However, some mesophilic microbes also have the ability to survive and grow, though not as normally as of psychrophiles, in cold habitats and are called psychrotolerant microbes. As of now, about 133 new species of Gram-negative and Gram-positive bacteria have been identified up to the species level from various cold habitats such as from soil, sandstone, quartz stone, fresh water, marine lakes, sea ice, ocean, microbial mats, krill, etc. Among the Gram-negative bacteria species belonging to the genera *Flavobacterium*, *Psychrobacter* and *Pseudomonas* were the most predominant; and species of the genera *Arthrobacter*, *Bacillus* and *Clostridium* were prominent Gram-positive bacteria (Table 7.3). Bacteria and unicellular algae are better adapted to cold habitats than fungi and protozoa by showing superior molecular characteristics.

**Table 7.3:** The diversity of bacteria in various cold habitats.

| Sl. No. | Bacterial genera         | No. of species | Habitat                    |
|---------|--------------------------|----------------|----------------------------|
| 1       | <i>Achromobacter</i>     | 1              | Quartz stone               |
| 2       | <i>Aequorivita</i>       | 3              | Quartz stone               |
| 3       | <i>Alteromonas</i>       | 2              | Sea ice, Sea water         |
| 4       | <i>Antarctobacter</i>    | 1              | Lakes, Hills               |
| 5       | <i>Chromobacterium</i>   | 1              | Water                      |
| 6       | <i>Colwellia</i>         | 5              | Sea ice                    |
| 7       | <i>Flavobacterium</i>    | 11             | Microbial mats, Cold lakes |
| 8       | <i>Gelidibacter</i>      | 1              | Microbial mat              |
| 9       | <i>Glaciecola</i>        | 2              | Sea ice                    |
| 10      | <i>Halobacterium</i>     | 1              | Deep lake water            |
| 11      | <i>Halomonas</i>         | 3              | Sea ice                    |
| 12      | <i>Janthinobacterium</i> | 1              | Soil                       |
| 13      | <i>Loktanella</i>        | 3              | Microbial mat              |
| 14      | <i>Marinomonas</i>       | 2              | Lake water                 |
| 15      | <i>Methanococcoides</i>  | 1              | Lake water                 |

Table 7.3: (Contd.)

|    |                          |    |                                 |
|----|--------------------------|----|---------------------------------|
| 16 | <i>Methylosphaera</i>    | 1  | Lake sediment                   |
| 17 | <i>Myxococcus</i>        | 2  | Soil                            |
| 18 | <i>Octadecabacter</i>    | 1  | Sea ice                         |
| 19 | <i>Oleispira</i>         | 1  | Sea water                       |
| 20 | <i>Polaribacter</i>      | 4  | Sea water, Sea ice              |
| 21 | <i>Polaromonas</i>       | 1  | Sea water                       |
| 22 | <i>Pseudoalteromonas</i> | 4  | Sea water, Sea ice              |
| 23 | <i>Pseudomonas</i>       | 7  | Soil, Water, Cyanobacterial mat |
| 24 | <i>Psychroflexus</i>     | 2  | Soil, Hypersaline lakes         |
| 25 | <i>Psychromonas</i>      | 1  | Sediment                        |
| 26 | <i>Psychroserpens</i>    | 1  | Lake                            |
| 27 | <i>Psychrobacter</i>     | 10 | Fast ice                        |
| 28 | <i>Rhodoferrax</i>       | 1  | Microbial mat                   |
| 29 | <i>Roseobacter</i>       | 2  | Hypersaline lake                |
| 30 | <i>Roseovarians</i>      | 1  | Hypersaline lake                |
| 31 | <i>Saccharospirillum</i> | 1  | Hypersaline lake                |
| 32 | <i>Salegentibacter</i>   | 1  | Hypersaline lake                |
| 33 | <i>Shewanella</i>        | 4  | Sea ice, Marine water           |
| 34 | <i>Sphingobacterium</i>  | 1  | Soil                            |
| 35 | <i>Staleyia</i>          | 1  | Hypersaline lake                |
| 36 | <i>Sulfitobacter</i>     | 1  | Hypersaline lake                |
| 37 | <i>Arthrobacter</i>      | 7  | Sea water, Soil, Lake sediment  |
| 38 | <i>Bacillus</i>          | 5  | Soil                            |
| 39 | <i>Canobacterium</i>     | 1  | Anoxic water                    |
| 40 | <i>Clostridium</i>       | 5  | Pond sediment                   |
| 41 | <i>Cryobacterium</i>     | 1  | Soil                            |
| 42 | <i>Cyclobacterium</i>    | 1  | Sea water                       |
| 43 | <i>Exiguobacterium</i>   | 2  | Pond water                      |
| 44 | <i>Friedmanniella</i>    | 2  | Sand stone, Hypersaline lake    |
| 45 | <i>Hymenobacter</i>      | 1  | Soil                            |
| 46 | <i>Kocuria</i>           | 1  | Cyanobacterial mat              |
| 47 | <i>Leifsinia</i>         | 2  | Pond                            |
| 48 | <i>Luteococcus</i>       | 1  | Sand stone                      |
| 49 | <i>Micrococcus</i>       | 4  | Soil                            |
| 50 | <i>Modestobacter</i>     | 1  | Soil                            |
| 51 | <i>Nestenkonnia</i>      | 1  | Hypersaline lake                |
| 52 | <i>Nocardioides</i>      | 2  | Hypersaline lake                |
| 53 | <i>Paenibacillus</i>     | 2  | Soil                            |
| 54 | <i>Planococcus</i>       | 4  | Cyanobacterial mat              |
| 55 | <i>Pseudonocardia</i>    | 1  | Dry valleys                     |
| 56 | <i>Rhodococcus</i>       | 1  | Quartz stone                    |
| 57 | <i>Rhodoglobulus</i>     | 1  | Dry valleys                     |
| 58 | <i>Sporosarcina</i>      | 1  | Cyanobacterial mat              |

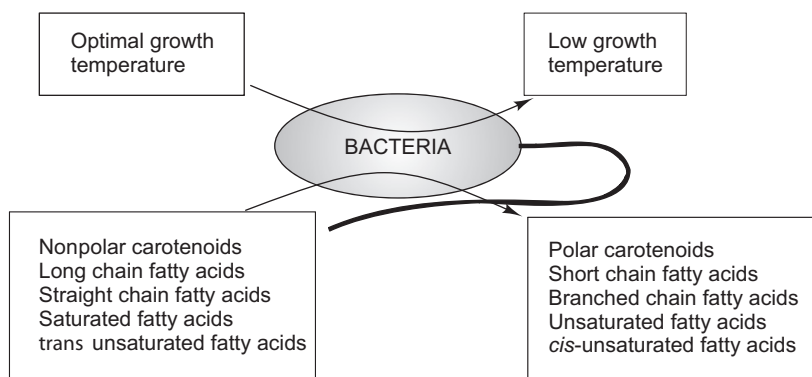
Source: Shivaji (2005)

The algal diversity of the cold habitats is mostly dominated by cyanobacteria, which are more tolerant to the environmental extremes than any other group of algae. The unicellular cyanobacteria of the genera *Synechococcus*, *Synechocystis*, *Aphanothece* and *Aphanocapsa* are found in almost all types of cold habitats. Many filamentous and nitrogen fixing cyanobacteria (*Nostoc*, *Calothrix*, *Gloeotrichia* and *Anabaena*), form cyanobacterial mats in the cold habitats during summer. In winter season there is death of the mats providing organic matters to the bacterial population inhabiting the mats. The unicellular members, however, are able to fix carbon even in the ice in winter as there is little attenuation of light up to 10 cm depth and distribution of algae is restricted in the top layer. The species of *Chlorella*, *Selenastrum*, many desmids and cold tolerant diatoms are found in the cold habitats. Some of them even form a temporary bloom in summer in nutrient rich coastal water.

### 7.5.1 Cold Adaptation in Microbes

Bacteria show subtle species-dependent differences in their response to cold-shock. But, they all appear to have a dominant common requirement to modulate membrane fluidity and membrane function predominantly by altering lipid head group, the protein content of the membrane, the type of carotenoids present, the fatty acid chain length and the proportion of *cis* to *trans* fatty acids composition (Fig.7.1). These changes in the membrane structural and dynamic characteristics would, in turn, affect the functions of membrane proteins, such as the proteins involved in respiration and transport and as a consequence the cell adapts to the change in its environment.

Polar head groups affect the packing of glycerophospholipids in the bilayer and thus influence the fluidity of the membrane. The ratios of lipid head groups may be modified by the changes in the environment temperature as observed in *Bacillus caldotenax*. Similarly certain cold shock proteins modify the fluidity of the membrane through interaction with lipid and other membrane proteins. For example, cold shock from 65°C to 45°C induces a change in the protein content of the membrane of *Bacillus caldotenax*.



**Figure 7.1-** Low temperature induced changes in bacterial membranes (Modified Chintalapati *et al.*, 2004).

A wide variety of bacteria, algae, fungi and plants modify their membrane bound carotenoids content and composition to thrive under cold shock. Carotenoids have a role in buffering membrane fluidity due to their ability to localize in the membrane: polar carotenoids stabilizing the membrane and the non-polar carotenoids increasing the fluidity of the membrane. Zeaxanthin, a polar carotenoid, was shown to confer more rigidity to a membrane than  $\alpha$ -carotene, a non-polar carotenoid. In the psychrophilic bacteria *Sphingobacterium antarcticum* and *Micrococcus roseus* the synthesis of carotenoids was dependent on the temperature for growth of the bacteria. Increase in the synthesis of polar carotenoids was observed when the bacteria were grown at low temperature so as to stabilize the membrane with a concomitant decrease in the synthesis of non-polar carotenoids.

One of the changes that is commonly observed in response to changes in temperature is a change in the proportion of short and long chain fatty acids so as to modulate the fluidity of the membrane. Fatty acids with short chains have lower melting points than those of longer chain length. Longer chains span the width of the bilayer more easily, promote acyl chain packing and thus make the membrane environment more gel-like. In contrast, shorter chains, especially those with less than 12 carbons, are unable to span the bilayer and are incapable of hydrophobic interactions with other lipids and proteins and thus maintain the fluid state of the membrane. The marine bacterium *Shewanella putrefaciens* adapts to the changes in temperature of the habitat by varying the content of short-chain fatty acids. It appears that adaptation to fluctuating temperature conditions through chain length modification is possible only in growing cells since these fatty acids are biosynthesized only in log phase cells. In addition the structural and geometrical isomers of fatty acids are known to affect membrane fluidity. Structural isomers of the straight chain fatty acids, viz., branched and cyclic fatty acids, increase the fluidity of the membrane as opposed to their straight chain isomers as the former structures disturb packing order to a great extent. The branched and cyclic fatty acids are essentially observed in Gram-positive bacteria, like *Bacillus* sp., *Micrococcus* sp., *Staphylococcus* sp., *Streptomyces* sp., *Corynebacterium* sp. and *Brevibacterium* sp. In bacteria containing *iso*- and *anteiso*- branched fatty acids, the mode of adaptation to variations in environmental temperature involves changes in the type of branching, along with changes in fatty acid chain length and degree of unsaturation. Increase in *anteiso* branched acids on lowering the growth temperature with concomitant decrease in *iso*-branched fatty acids has been observed. The other common response to a decrease in growth temperature is the observed decrease in the proportion of the highest melting straight chain fatty acid, which is replaced either by: (a) even and odd carbon *iso* branched and *anteiso*-branched fatty acids, as in the case of *Bacillus stearothermophilus*; (b) by odd-carbon *iso*-branched and *anteiso*-branched fatty acids, as in thermophilic *Bacillus*; and (c) by even carbon *iso*-branched and *anteiso*-branched acids, as in *Bacillus cereus*.

Fatty acid desaturation is the most common mechanism used by cyanobacteria, to modulate membrane fluidity in response to low temperature stress. The fatty acyl lipid desaturases are essential for such desaturation and growth at low



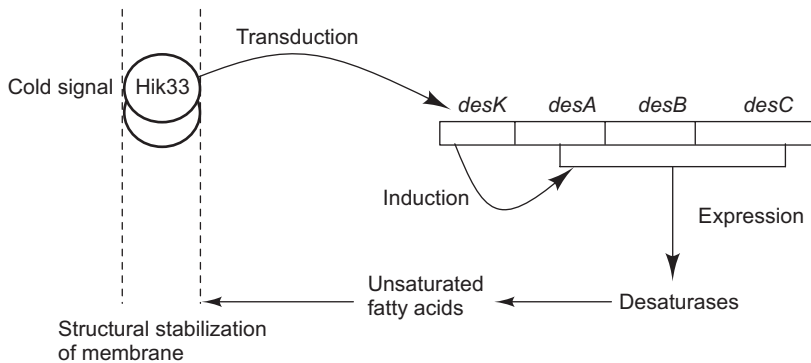
temperatures. In these cyanobacteria saturated fatty acids are converted to unsaturated fatty acids, most commonly C16:0 to C16:1(9) and C18:0 to C18:1(9), by acyl-lipid desaturases as a post-biosynthetic modification. However, different cyanobacteria show different changes in the fatty acid composition in response to cold. For instance except for the species belonging to Group I, which change both the extent of unsaturation and the chain length of fatty acids with alterations of growth temperature, all other cyanobacteria respond to a thermal shock by modulating only the extent of unsaturation of fatty acids. In species belonging to Group II levels of C18:1(9) and C18:2(9,12) unsaturated fatty acids decrease and the level of C18:3(9,12,15) increases in all major lipid classes. In some cases, C16 fatty acids show a transient change as an initial adaptive response. For example, when the growth temperature was shifted from 38°C to 22°C in *Anabaena variabilis*, a decrease in the level of C16:0 and a concomitant increase in that of C16:1(9) fatty acids occurred.

In *Synechocystis* sp. PCC 6803 mutant Fad6 and Fad12, the later is cold-sensitive and the reason resides in the fact that unlike the wild type cells, the mutant was incapable of converting C18:1(9) to C18:2(9,12), an activity associated with fatty acyl desaturase. Absence of desaturase synthesis would probably be lethal to the cells of *Synechocystis* sp. PCC 6803 as it will have no polyunsaturated fatty acids spanning the lipid core of the membrane. This implies that polyunsaturated fatty acids are absolutely essential for growth at low temperatures.

### 7.5.2 Transduction of Cold Signal

The sensing and transduction of cold signal has been partially studied in cyanobacteria but very few reports are available in other microorganisms. Murata and Wada (1995) proposed a scheme for the acclimatization of cyanobacterial cells to low temperature based on the observation that catalytic hydrogenation of fatty acids of plasma membrane lipids in *Synechocystis* sp. PCC 6803 causes a decrease in membrane fluidity and enhanced transcription of the *desA* gene. This observation led to the hypothesis that, in cyanobacterial cells, a novel type of regulatory control exists that is based on a feedback link between the physical state of the membrane and the levels of expression of the genes for desaturases. According to this model, upon a downward shift in temperature, the fluidity of the membrane decreases. This signal is detected by a membrane-bound sensor (Hik 33) for low temperature and is then transmitted to regulatory molecules that directly or indirectly interact with the regulatory regions of the genes for desaturases, leading to their activation. As a result, the levels of desaturases increases and fatty acids are desaturated. Finally, the accelerated accumulation of unsaturated fatty acids leads to the recovery of membrane fluidity to the physiologically desired level and for restoration of the activity of membrane associated enzymes (Fig. 7.2). Thus though various aspects of this scheme have been experimentally proved it was still a challenge to identify the sensor that would sense the change in fluidity and the regulator which would carry the message to the respective genes to be up-regulated.

Hik33 is assumed to span the plasma membrane twice to form a dimer whose structure may be influenced by the physical characteristics of lipids in the plasma membrane, such as their fluidity or the extent of molecular motion. This is controlled by temperature and the extent of unsaturation of fatty acids. When the temperature is decreased or fatty acids are more unsaturated, the histidine residue in the histidine kinase domain may be phosphorylated. The phosphate group is finally transferred to a response regulator. This response regulator directly interacts with the DNA molecule to regulate transcription. Microarray study of the *hik33* mutant revealed that proteins involved in translation, transcription, and DNA- and RNA unwinding are under the control of *hik33*.



**Figure 7.2** - Schematic representation of the expression of *des* (desaturase) genes in response to cold signal.

The first direct evidence for the two-component signal transduction mechanism involved in sensing cold has come from studies on *Bacillus subtilis*, which consisted of the DesK / DesR system, that recognizes and transduces low temperature signals, thus activating the transcription of C5 desaturase. It was also demonstrated by foot printing assay that DesR binds upstream to the C5 desaturase indicating its direct role in upregulation of the gene. The *des* pathway, upon decrease in constant growth temperature or temperature downshift, is activated by DesK, which senses the decrease in membrane fluidity. Recently, it was shown that only the phosphorylated form of protein DesR is able to bind to a regulatory region immediately upstream of the promoter of the C5 desaturase. The phosphate group to phosphorylate DesR protein is probably made available by the dephosphorylation of a histidine residue of Hik33 as mentioned before.

## 7.6 MICROBES OF HYPERSALINE HABITATS

Microbes inhabiting saline and hypersaline environments with sodium chloride concentrations up to saturation level are called halophilic microorganisms. These organisms are adapted to high salt concentrations and the high osmotic pressure of their environment. Halophiles thrive in hypersaline lakes, oceanic environment and other environments with large salt concentrations. Various classification

schemes have been introduced to define the salt relationship of microorganisms. A widely accepted classification according to salt tolerance/ dependence is that of Kushner (Kushner, 1985). A modified version of this classification is given in table 7.4. This classification recognizes different degrees of salt tolerance. It is important to note, however, that the classification of microorganisms based on their salt tolerance should depend on their response not only to sodium chloride (NaCl), but to other ions such as potassium, magnesium and calcium and that salt tolerance is dependent on other growth parameters such as temperature and pH.

Halophilic microorganisms are found in all three domains of life: *Bacteria*, *Archaea* and *Eucarya*. Diverse prokaryotic communities have been found over a wide range of salt concentrations from seawater (0.5M or 3.3% w/v) to lakes saturated with sodium chloride (5M or 33% w/v). Phylogenetically, halophilic microorganisms are usually interspersed between non-halophilic relatives within the same order or phylum. However, three phylogenetically related groups are composed exclusively of halophiles. In the domain *Archaea*, the family *Halobacteriaceae* (order *Halobacteriales*) comprises the most halophilic microorganisms known and its members are the dominant representatives in hypersaline environments. Among the domain *Bacteria*, the order *Haloanaerobiales* accommodates fermentative, obligate anaerobic halophilic bacteria. The family *Halomonadaceae* includes moderately halophilic aerobes. The halophilic *Archaea* belong to the family *Halobacteriaceae* (order *Halobacteriales*), which harbours the true halophiles (require at least 1.5M NaCl for growth and grow optimally at 3.5 – 4.5M NaCl) and the strictly anaerobic family *Methanosarcinaceae* (halophilic genera; *Methanohalobium* and *Methanohalophilus*).

**Table 7.4:** Classification of microorganisms according to their salt tolerance, temperature and pH optima.

| Category                | Property                                                                                                   |
|-------------------------|------------------------------------------------------------------------------------------------------------|
| Non-halophilic          | Optimal growth in media with less than 0.2 M salt                                                          |
| Halotolerant            | Optimal growth in media with 0.2–0.5 M salt but can tolerate salt concentrations up to 2.5 M               |
| Halophile               | Optimal growth in media with at least 1.5M salt                                                            |
| Extreme halophile       | Optimal growth at concentrations between 2.5–5M salt; does not tolerate salt concentrations less than 1.5M |
| Thermotolerant          | Optimal growth < 50°C, T <sub>max</sub> > 50°C                                                             |
| Thermophile             | Optimal growth above 50°, T <sub>max</sub> > 60°C                                                          |
| Extreme thermophile     | Optimal growth above > 65°C T <sub>max</sub> > 70°C                                                        |
| Hyperthermophile        | Optimal growth above 80°C T <sub>max</sub> > 85°C                                                          |
| Alkaliphile             | Optimal growth above pH 8.5, pH <sub>max</sub> 10, pH <sub>min</sub> 7.5                                   |
| Facultative alkaliphile | can tolerate pH values less than 7.5                                                                       |

### 7.6.1 The Microbes of Thalassohaline Environments

These are the obligate and extreme halophiles, halotolerant and moderate hyperthermophilic organisms. A halothermophile is defined as an organism requiring at least 1.5 M NaCl and a temperature at or above 50°C for optimal growth (Table 7.4). In addition to halothermophiles, the haloalkalithermophiles represent an additional group that tolerates or requires for growth not only elevated salt concentration and temperature but also elevated pH values. The thalassohaline halophiles show considerable physiological diversity. Three aerobic heterotrophic Bacteria, *Halomonas variabilis*, *Gracibacillus halotolerans* and *Pseudomonas halophila* are commonly observed in such environments. Several anaerobic mesophilic bacteria like the fermentative *Haloanaerobium praevalens* and *Haloanaerobium alcaliphilum* have also been isolated from thalassohaline waters. The sulphate reducing *Desulfobacter halotolerans* and *Desulfocella halophila* also grow in thalassohaline conditions, though they have salt optima far below the salt concentration of the environment from which they were isolated (0.3M and 0.9M, respectively).

Tolerance to high temperatures seems to be an adaptation by some halophilic microorganisms. *Methanohalobium evestigatum* is the only methanogenic representative of the halothermophiles. It is strictly methylotrophic, using trimethylamine, dimethylamine and methylamine as substrates for growth. It is extremely halophilic requiring 4.6M NaCl for growth. A recent study by Spear *et al.* (2003) on endoevaporites (evaporite sedimentary rock of precipitating salt deposits) revealed clear stratification and different colours due to different types of microorganisms growing in it. 16S rDNA analysis of total DNA extracted from evaporite crystals indicated that the uppermost layer contains mainly cyanobacteria. The predominant species in the deeper layers are representatives of the Cytophaga/Flexibacter/Bacteroides group. All layers contained a large number of archaeal phylotypes closely related to *Halobacterium* species and a few methanogens were found. It should be noted, however, that with the exception of *Halospirulina* and *Spirulina* spp., none of the cyanobacterial genera detected have official standing in bacterial nomenclature.

Solar salterns are artificial shallow ponds fed by seawater. They are the most widely distributed thalassohaline hypersaline environments situated next to seashores and are used for the production of salt (NaCl). Many of the halophilic archaea and bacteria found in culture collections originate from salterns. Halophilic archaea isolated from saltern ponds include *Haloterrigena thermotolerans*, *Haloarcula quadrata*, *Halobacterium salinarum*, *Halorubrum saccharovorum*, *Halorubrum coriense*, *Haloferax denitrificans* and *Haloferax mediterranei*. A great deal of variation is observed with respect to the cell morphology, carbon sources utilized, salt requirement and other properties of these isolates. Halophilic bacteria include *Halothermothrix orenii* and *Thermohalobacter berrensis*. No anaerobic, halothermophilic archaea have been isolated so far from solar saltern sediments, even though two mesophilic methanogenic isolates have been described, viz., *Methanohalophilus portucalensis* and *Halomethanococcus doli*.

Sediments and/ or benthic microbial mats often develop at the bottom of salterns favouring the growth of anaerobic microorganisms. *Halothermothrix orenii*, a fermentative bacterium of the order *Haloanaerobiales*, has been isolated from sediment of a Tunisian salt lake. *Thermohalobacter berrensensis* is also a fermentative bacterium phylogenetically related to the *Clostridiales* found in solar saltern near Berre Lagoon in France. Both of these microorganisms are characterized by being true thermophiles, growing optimally at temperatures greater than 60°C but their growth is more regulated by salt concentration than of temperature. *Halothermothrix orenii* is chemoorganotrophic and strictly halophilic anaerobe requiring 1.5M NaCl for optimal growth (NaCl range 0.6-3.3M). It oxidizes a wide variety of carbohydrates including glucose, galactose, arabinose, cellobiose and fructose. *Thermohalobacter berrensensis* has similar physiological characteristics but has smaller NaCl optima for growth, i.e. 1.2M (NaCl range 0.5-2.5M).

## 7.6.2 The Microbes of Athalassohaline Environments

The salinity in athalassohaline environments is caused by  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Cl}^-$ ,  $\text{SO}_4^{2-}$  and  $\text{CO}_3^{2-}$  with  $\text{Na}^+$  being the predominant cation. Halophilic, alkaliphilic and methylotrophic archaeon *Methanosalsum zhilinae* has been isolated from athalassohaline Bosa Lake. It grows optimally at pH 9.2 in the presence of 0.7M NaCl at 45°C and utilizes methylamines, methanol or any other methyl containing compound (but not acetate) as substrates for growth and methanogenesis. These substrates are present in hypersaline environments mainly due to microbial degradation of methylated compounds that act as osmotic solutes in many halophilic microorganisms. The other microorganisms of athalassohaline environments include the archaea *Natromonas pharonis*, *Halobacterium marismortui* and *Haloarcula marismortui*; bacteria *Ectothiorhodospira halochloris* and *E. halophila*; the primary producers *Synechococcus* spp., *Synechocystis* spp. and *Aphanothece* spp., unicellular green alga *Dunaliella* and some red algae. Halophilic archaea grow at the expense of organic material produced by these primary producers.

Other athalassohaline environments are soda lakes and soda deserts, which are the most stable and naturally occurring alkaline hypersaline environments on earth. These environments are characterized by large amounts of both  $\text{Na}_2\text{CO}_3$  and NaCl. The Kenyan-Tanzanian Rift Valley contains a number of lakes of this type, which have salt concentrations of 1M - 5M and pH values of 8.5 to > 11.5. Lake Magadi is the most intensely studied of the lakes so far as the microbial diversity is concerned. It is the site for isolation of the anaerobic haloalkalithermophile, *Halonatronum saccharophilum*. It has also been the site for isolation of a wealth of novel species, both archaea and bacteria, aerobes and anaerobes, which are adapted to grow not only to high salinity but also to enhanced temperature and alkalinity. Adaptation to grow at these multiple extreme conditions represents an evolutionary interesting combination. *Halonatronum saccharophilum* is the most recently described haloalkalithermophile. It is an anaerobic spore forming bacterium with a growth

optima at 1.1-1.5M NaCl, pH 8.5 and temperature 36-55°C. *Bacillus thermoalcaliphilus* is the second isolated species but it has not been validly published so far. The optima for growth are 1.5M NaCl, pH 8.5-9.0 and 60°C. The other isolated halotolerant alkalithermophilic microorganisms have not so far been characterized beyond the genus level. Examples include *Bacillus* sp. BG-11, isolated from an alkaline thermal environment in India and *Bacillus* sp. STS1, isolated from the same soil termite mounds as *Bacillus thermoalcaliphilus*. No haloalkalithermophilic archaea have been isolated so far, even though the majority of the halothermophiles are believed to be archaea. All the isolated halothermophiles are neutrophiles. It is unclear whether the scarcity of haloalkalithermophilic isolates, bacteria or archaea, is due to inhibition of microbial growth at multiple extreme conditions or a limitation in the methods and techniques used to explore this group of extremophiles.

### 7.6.3 The Physiological Properties of Halophiles

Halophilic microorganisms must maintain their cytoplasm, at least at isoosmotic concentration, with their surroundings, in order to prevent loss of water from cells to the environment. Maintenance of a turgor pressure, in saline water, requires a hyperosmotic cytoplasm. All halophilic microorganisms, with the possible exception of the halophilic archaea of the family *Halobacteriaceae*, maintain a turgor pressure sufficient enough to maintain the cell turgidity.

Two different strategies enable halophilic and halotolerant prokaryotes to withstand the high osmotic stress associated with their hypersaline habitats. viz., (i) Cells accumulate and maintain high concentrations of inorganic ions in the cytoplasm (the “salt-accumulation strategy” strategy). In this case, all the intracellular enzymatic machinery is adapted to the presence of high salt concentrations; (ii) Cells maintain low cytoplasmic salt concentrations but accumulate high concentrations of low molecular weight compatible solutes (the “low salt accumulation” or “organic osmolyte” strategy). No adaptation of intracellular machinery is required.

**(a) The “salt accumulation” strategy:** The salt-accumulation strategy is used by two phylogenetically unrelated groups; the extremely halophilic archaea of the family *Halobacteriaceae* and the anaerobic bacteria of the order *Haloanaerobiales*. No organic osmotic solute has been found in the members of these groups. The cytoplasm in these microorganisms is characterized by the presence of molar concentrations of  $K^+$ , several times higher than the cytoplasmic  $K^+$  concentration of a mesophilic cell. Microorganisms using the salt-accumulation strategy are obligate halophiles. High concentrations of inorganic ions, both intra- and extra-cellular, are necessary for proper maintenance of cell shape, membrane structure and enzyme activity.

The large  $K^+$  and  $Na^+$  gradients created across the membrane are maintained at the expense of energy. These concentration gradients are the result of cooperative action of a number of ion pumps and antiporters. ATP is the main source of energy used for maintenance of the ion gradients. The membranes of halophiles



investigated so far possess high activities of  $\text{Na}^+/\text{H}^+$  antiporters, which use the proton electrochemical gradient as the driving force for extrusion of  $\text{Na}^+$  from the cell. High intracellular  $\text{K}^+$  concentrations are achieved via a  $\text{K}^+$  uniport system. The negative inside membrane potential generated by respiratory electron transport promotes the diffusion of positively charged  $\text{K}^+$  ions into the cell. As  $\text{K}^+$  enters the cell,  $\text{Na}^+$  is extruded via the  $\text{Na}^+/\text{H}^+$  antiporter thus maintaining electroneutrality without any change in the transmembrane electrochemical gradient.

Molar intracellular  $\text{Cl}^-$  concentrations are maintained by the actions of two energy dependent inward chloride pumps, also a requirement to neutralize the high intracellular  $\text{K}^+$  concentration. Active uptake of  $\text{Cl}^-$  is necessary for maintenance of constant intracellular  $\text{Cl}^-$  concentration against the negative-inside membrane potential. The first  $\text{Cl}^-$  pump is light independent and couples inward  $\text{Cl}^-$  transport with  $\text{Na}^+$  expulsion. The second is the light driven halorhodopsin (protein that is probably distributed among members of the family *Halobacteriaceae*).

For microorganisms using the salt-accumulation strategy for osmotic adaptation, all structural cell components and enzymes must be adapted to the presence of molar concentrations of inorganic salts. High salt concentrations reduce the availability of free water due to hydration of ions and lead to aggregation and collapse of protein structure due to alteration of electrostatic and hydrophobic interactions on the solvent exposed surfaces of proteins. Evolutionary modifications required to engineer halophilic proteins have included the incorporation of a large number of acidic amino acids (glutamate and aspartate) into the surface of proteins. The carboxylic groups of the acidic amino acids sequester and organize a tight network of water and hydrated  $\text{K}^+$  ions around the protein and form internal salt bridges with basic amino acid residues to provide internal structural rigidity to the protein. Another characteristic feature of hydrophilic proteins is their low content of hydrophobic amino acid residues that is offset by a high content of borderline hydrophobic amino acids, serine and threonine. High salt in the surrounding environment to the proteins allows them to maintain the weak hydrophobic interactions and preserve protein structure. Most proteins of microorganisms utilizing the salt-accumulation strategy require high salt concentrations for maintenance of proper conformation and activity.

**(b) The “organic osmolite” strategy:** Organic osmolites, which are regarded as compatible solutes to build osmotic potential in cells, are polar, highly soluble molecules that are uncharged or zwitterionic at physiological pH. They include amino acids and derivatives, quaternary amines such as glycine betaine, sugars or sugar derivatives as sucrose, trehalose and 2-sulfotrehalose and polyols as glycerol and arabitol. The organic osmolite strategy is used by most halophilic bacteria with exception of the members of the order *Haloanaerobiales*. The use of organic compatible solutes, that are either synthesized by the cell or taken up from the medium when available, allows a great deal of flexibility and adaptability to a wide range of NaCl concentrations. The compatible solute strategy does not require specifically adapted proteins.



Compatible solutes are strong water structure formers and are excluded from the hydration shell of proteins. They display a general stabilizing effect by preventing unfolding and denaturation of proteins as they increase forces of cohesion between water molecules and reinforce the hydrophobic stabilization of proteins. Intracellular concentrations of compatible solutes are regulated according to the salinity of the medium allowing this strategy to provide a high degree of adaptability of cells to changes in the salt concentration of the environment. However, this strategy is only effective in environments with up to 1.5M salt. Above this concentration it is ineffective and energetically unfavourable.

It is unclear how halothermophiles exposed to high ion concentration, high temperature, in addition to scarcity of molecular oxygen, high UV radiation and in some cases high pH, are capable of adapting to multiple stress factors. It is possible that they have adapted their cytoplasmic membrane content to somehow decrease permeability at high temperature. Another mechanism is to drastically increase the respiration rate hence increase the rate of proton/ion pumping out of the cell. This has been observed in aerobic bacteria living at temperatures around 60–65°C, e.g. *Geobacillus stearothermophilus*. However, this option is energetically unfavourable and may not be possible due to the low solubility of molecular oxygen in brines. Light energy absorbed by the retinal pigment bacteriorhodopsin (found in many haloarchaea) is used to pump protons and ions across the membrane thus generating a proton / sodium motive force. But the increased permeability of the cytoplasmic membrane at high temperature and salt concentrations will make it difficult to sustain an electrochemical gradient of magnitude sufficient to support growth.

#### 7.6.4 Mechanisms of Adaptation of Haloalkalithermophiles

Haloalkalithermophiles are faced with the additional burden of elevated pH. Alkaliphiles use an electrochemical gradient of either  $H^+$  or  $Na^+$  for energy transduction. The electrochemical gradient of  $H^+$  exerts a proton motive force (pmf), which is composed of two components, a transmembrane pH gradient ( $\Delta pH$ ) and transmembrane electrical potential ( $\mu E$ ). In alkaliphiles, the  $\Delta pH$  is reversed, being alkaline on the outside and relatively acid inside. Alkalithermophiles use the  $Na^+ / H^+$  antiporter to regulate internal pH, where it drives uptake of  $H^+$  ions and extrusion of  $Na^+$  ions. Therefore, the membrane must be highly impermeable to both  $H^+$  and  $Na^+$  ions. This seems to be problematic for haloalkalithermophiles as membrane permeability to both  $H^+$  and  $Na^+$  ions is compromised at elevated temperatures and salt concentrations, because of an increase of the fluidity of the membrane lipids.

A possible mechanism for explaining difficulties encountered by haloalkalithermophiles in managing ATP synthesis could be explained by retention of protons expelled by respiration of the cytoplasmic membrane and rapidly funneling them through the ATP synthase before equilibrating with the bulk phase. This is termed “Lateral Proton Transfer” and has been observed in

*Halobacterium salinarum*. In this experiment, proton movement was monitored by linking the dye fluorescein to different cysteine residues on the cytoplasmic surface of the membrane together with pyranine (a water soluble pH indicator) to monitor  $H^+$  movement to the bulk phase. This showed that  $H^+$  is transferred laterally from the respiratory chain to ATP synthase. Lateral  $H^+$  transfer retained a constant rate over a temperature range of 4–40°C. This finding can account for the coupling of ATP synthase with protons and could also retain protons diffusing outward due to increased membrane proton permeability. Another possible explanation is that the cytoplasmic membranes of haloalkalithermophiles have developed specific fatty acid profiles and/ or other components that decrease their permeability to extraneous ions at elevated temperature/salt concentration. In any case, it is apparent that the haloalkalithermophiles have developed unique adaptations in order to cope with these three extreme environmental conditions.

## 7.7 MICROBIAL DIVERSITY OF METAL CONTAMINATED ENVIRONMENT

In contrast to other ‘extreme’ environments characterized by a relatively specific, restricted microflora, a wide range of organisms from all the major groups may be found in metal-polluted habitats and the ability to survive and grow in the presence of potentially toxic metal concentrations is frequently encountered. Thus, generalizations about a so-called ‘specific microflora’ of metal contaminated habitats cannot be made with any certainty. In general terms, heavy metals are believed to affect microbial populations in nature by reducing abundance and species diversity and selecting for a resistant population. In most of the metal contaminated environments, there are some dominant species of microbes, which successfully tolerate the metal stress and grow more or less normally or even better due to easy availability of resources.

### 7.7.1 Bacterial and Fungal Diversity

The effect of heavy metals on microbial abundance in natural habitats varies depending on the metals and microorganisms and the physicochemical attributes of that environment. General reductions in the numbers of bacteria, including actinomycetes and fungi have often been recorded in soils polluted with Cu, Cd, Pb, As, Hg and Zn. Along a steep gradient of Cu and Zn in soil towards a brass mill, fungal biomass decreased by about 75% whereas the total number of fungi in a glucose-supplemented soil was reduced by the addition of Cd or Zn, the former metal having the greatest toxic effect (Bewley and Campbell 1980). However, numerical estimates, if viewed in isolation, provide no indication as to whether the microbial populations have changed in any way as a result of metal exposure. For example, a Zn- and Cd-polluted soil located near a smelter had lower numbers of actinomycetes and fungi but not total bacteria as compared with unpolluted control soil. However, the latter contained greater number of *Nitrosomonas* sp. Combinations of Cd and Zn reduces the number of bacteria

and actinomycetes to a greater extent than numbers of fungi in glucose supplemented soil. In a study, where Cu was added to a simulated marine ecosystem, an increase in heterotrophic bacteria occurred which was thought to be the result of bacteria which survived the Cu exposure using the organic substrates released from Cu-sensitive organisms. Thus, numerical estimates may provide little meaningful information without consideration of changes in microbial groups and species and it seems evident that such changes can occur in response to metal exposure.

As in other so-called extreme environments, there can be a reduction in species diversity. In relation to pollution of leaf surfaces by aerially deposited Pb, Cd or Zn, bacteria appeared more sensitive than fungi and on polluted oak leaves *Aureobasidium pullulans* and *Cladosporium* sp. were the most numerous organisms. In fact, numbers of *A. pullulans* showed a good positive correlation with lead concentrations, whether derived from industrial or vehicular sources and it often became the dominant organism, in some cases comprising up to 97% of the leaf surface population on a numerical basis. In contrast, the ballistospore producing yeast *Sporobolomyces roseus* and heterotrophic bacteria were very low or absent from polluted samples and number of *S. roseus* showed a significant negative correlation with increasing lead concentrations. In Cu- and Zn polluted soil, the fungi *Geomyces* and *Paecilomyces* sp. and some sterile forms increased with increasing pollution, whereas *Penicillium* and *Oidiodendron* sp. declined at polluted sites. *Trichocladium asperum*, *Trichoderma hamatum*, *Zygorrhynchus moelleri* and *Chrysosporium pannorum* are isolated more frequently from organomercurial-treated soils than from untreated locations, whereas *Chaetomium*, *Fusarium*, *Penicillium* and *Paecilomyces* sp. are greatly reduced in the environment contaminated with organomercurials. Several studies have indicated sensitivity of *Penicillium* sp. towards heavy metals. However, some of the best examples of microbial metal tolerance are found in the genus *Penicillium*, which underlines the fact that metal responses may be species and strain specific. *Penicillium ochrochloron* can grow in saturated CuSO<sub>4</sub> and is frequently isolated from industrial effluents, whereas *Penicillium lilacinum* comprised 23% of all fungi isolated from soil polluted by mine drainage.

## 7.7.2 Cyanobacterial and Algal Diversity

Heavy metals can also affect the composition of cyanobacterial and algal populations. Applications of Cu favour coccoid, colonial cyanobacteria and inhibit *Anabaena flos-aquae* and many other nitrogen fixing filamentous cyanobacteria. Numerous studies have documented growth, morphological, physiological and biochemical responses of different cyanobacteria to supra-optimal concentrations of metals of the copper and zinc triads as well as many other non-essential heavy metals. Morphological effects of cadmium and elevated concentrations of zinc include inhibition of daughter cell separation, production of filaments in normally unicellular strains (such as *Synechococcus* PCC 6301) and cell elongation. In contrast, elevated copper has been shown to cause

production of sub-spherical rather than coccoid shape of cells in *Synechococcus* PCC 6301. Elevated concentrations of metals influence pigment composition, for example, with increased carotenoid to chlorophyll ratios and overall decreases in phycocyanin, chlorophyll *a* and total protein in *Synechococcus* PCC 6301 exposed to elevated zinc. An analysis of the effects of elevated concentrations of copper on *Nostoc calcicola* revealed that photosynthesis was readily inhibited. Polyphosphate bodies are implicated in the sequestration of metals of the copper and zinc triads and these increases occurred in response to elevated concentrations of such metals.

In general the planktonic green algae are more tolerant to metals than the filamentous and periphytic members. Cyanobacteria show more tolerance to metals in the soil ecosystem than of green algae. At some metal polluted aquatic sites, for example zinc enriched waters of high pH, cyanobacteria are abundant and can be the dominant organisms. Some strains of cyanobacteria isolated from metal polluted sites are resistant to elevated metal concentrations in subsequent axenic culture. Strains have also been selected in laboratory studies by step-wise adaptation of cultures to gradually increasing metal concentrations. Enhanced resistance in such a copper resistant line of *Nostoc calcicola* was correlated with enhanced copper efflux while enhanced cadmium resistance in *Synechococcus* PCC 6301 was correlated with enhanced endogenous metal chelation mediated by increased expression of metallothionein. For eukaryotic algae, Cu appeared to reduce numbers of centric diatoms, except *Skeletonema costatum* and eliminated dinoflagellates. However, other flagellates increase and pennate diatoms, particularly *Amphiphora paludosa* v. *hyalina* become dominant component of the population in the aquatic conditions contaminated with Cu. In a metal-contaminated aquatic sediment only 63 taxa of benthic algae were determined whereas 111 taxa were detected over 1 km away from the main focus of contamination (Gadd 1990). Other studies have shown *Ankistrodesmus falcatus*, *Botryococcus braunii*, *Chlorella pyrenoidosa*, *Raphidium* sp. and *Schroederia setigera* to be more abundant in metal contaminated waters.

Presence of assimilable carbon sources like glucose and organic acids has provided tolerance to many green algae to heavy metals like Zn, Hg, Cd and Ni. Sublethal concentration of mercury could become ineffective to *Chlorella vulgaris* and *Chlorococcum infusionum* at elevated concentrations of glucose, glutamate and 2- oxoglutarate. Among protozoa and pigmented flagellates, *Peranema* and *Euglena* sp. are generally quite resistant to metal toxicants although, in comparison with walled microorganisms, protozoa have high sensitivity. As with other organisms, changes in species composition, numbers and diversity occur with heavy metal exposure but there is considerable variation in recorded effects. Thus, it seems that elevated concentrations of heavy metals can affect the qualitative and quantitative composition of microbial populations although, as mentioned, it may be difficult to separate heavy metal effects from those of other environmental components. For example, phylloplane microflora may be subjected to the influence of other potential toxicants, e.g. SO<sub>2</sub>

particularly in industrial locations and these may contribute to observed effects. In addition, there are well-known theoretical and practical difficulties in obtaining meaningful assessments of microbial numbers and diversity in natural habitats which should not be ignored. Nevertheless, it is generally accepted that heavy metal induced changes in species composition do occur and a commonly associated assumption is that such changes lead to the establishment of a 'resistant' or 'tolerant' population.

### 7.7.3 Microbial Tolerance to Heavy Metals

The ability of a microorganism to survive and grow in a metal contaminated habitat may depend on genetical and/or physiological adaptation. In a given population, both may occur to varying extents, e.g. when microorganisms are equipped, physiologically or genetically, to tolerate increasing concentrations of heavy metals in laboratory media by repeated subculture on metal-containing media, there may be physiological changes which may be unstable and also the selection of genetically stable variants from the original population. While physiological changes in the cells of the microbes are meant to reduce the rate of metal uptake and reduction of intracellular metal toxicity, the genetic changes cause both intracellular and extracellular reduction of the concentrations of toxic metal species. In many cases there has been adequate consideration of toxicity modification by environmental factors or media components which may lead to the isolation and identification of apparently resistant isolates. Nevertheless, selection and isolation of metal-resistant bacteria, algae and fungi has been described from heavy metal polluted soils and waters and such resistance has been attributed to the prolonged exposure of the species to a concentration of the metal, which is considered toxic to those which are new to the contaminated habitat. However, such resistance is induced and is often lost on removal of the stress.

The ability to grow in the presence of relatively high metal concentrations is found in a wide range of microbial groups and species, including those from unpolluted sites and not in all cases is any adaptation necessary. Tolerance may result from intrinsic properties of the organism, e.g. the possession of extracellular mucilage or polysaccharide or an impermeable cell wall. In addition, environmental components may have considerable influence on toxicity and therefore apparent resistance. A good example is provided by those fungi that can grow in saturated  $\text{CuSO}_4$  (about 1.3 M) and very high concentrations of other heavy metals. Such solutions are very acidic and these organisms are sensitive to submillimolar levels at pH values around neutrality. A range of other examples are available which illustrate the difficulties involved. Although fungal abundance and species diversity were apparently reduced in Zn-polluted soil, there was little difference in zinc tolerance between fungi isolated from control or polluted sites. At control sites, *Bdellospora*, *Verticillium* and *Paecilomyces* sp. were Zn-tolerant with *Aureobasidium* and *Penicillium* sp. at polluted sites. Similar findings were obtained from Cu, Ni, Fe and Co-polluted soils, where

fungus populations were not significantly different. *Penicillium*, *Trichoderma*, *Rhodotorula*, *Oidiodendron*, *Mortierella* and *Mucor* spp. show almost equal distribution in both metal contaminated and normal soil, without having any significant metabolic adjustment.

## 7.8 MICROBES OF ORGANICALLY CONTAMINATED SOIL

In comparison to the other extreme environments, the organically polluted environments are less diverse with microorganisms. Since many of these chemical compounds have been introduced relatively recently, microorganisms have not yet had enough time to evolve metabolic pathways for their degradation resulting in their low diversity. Nevertheless there are many tolerant microorganisms having the ability to degrade a variety of organic toxicants, metabolically or cometabolically and growing on them. These microorganisms and the diversity of their metabolic enzymes are typically employed to safely remove environmental pollutants, either through direct destruction or indirectly through a transformation of the contaminant to a safer intermediate, or even to a dead endproduct. However, the microbial diversity is greatly influenced by the type and concentration of the organic pollutants. Various factors such as bioavailability, diffusion and transportation of the substrates into the cells, presence of the degradative bacteria at the site to be bioremediated and the ability of the organism to perform degradation under manipulated environmental conditions determine the total diversity of microbes of contaminated environment. Hence endeavours are being made to genetically modify microbes into more efficient degraders using modern tools of molecular biology and metabolic engineering.

### 7.8.1 Microbial Diversity in Chemically Perturbed Soil

Most of the chemically contaminated sites are colonized by the microbes that produce common enzyme types (mono- and dioxygenases, dehydrogenases). The microbial groups belonging to *Pseudomonas*, *Nocardia* and *Arthrobacter* are present in different combinations and densities at various contaminated sites showing that these species have tolerance to a wide range of chemical contaminants. There are reports on utilization of cycloalkanes by *Pseudomonas citronellolis*, *Brevibacterium erythrogenes*, *Saccharomyces cerevisiae* and *Alcanivorax borkumensis*. Pentachlorophenol (PCP), a general biocide used primarily as preservative of wood, leather, textile and related commercial products has been shown to be mineralized by *Pseudomonas* sp. strain IST 103. *Acinetobacter* sp. (utilizing *p* – Chlorobiphenyls) and radiation-resistant strain *Deinococcus radiodurans* has also been isolated from contaminated sites.

Polycyclic aromatic hydrocarbons (PAHs) are degraded by many bacterial, fungal and algal strains. Most of the bacterial species such as *Acinetobacter calcoaceticus*, *Mycobacterium* sp., *Pseudomonas cepacia* and *Rhodococcus* sp.



are reported to degrade low molecular weight PAHs, whereas there are limited reports on degradation of high molecular weight PAHs with more than four benzene rings. *Mycobacterium*, *Pseudomonas* and *Alcaligenes* sp., when present as a consortium, are able to utilize complex PAHs. A chemotactic strain of *Pseudomonas putida* isolated from oil fields of Gujarat (India) could mineralize naphthalene as sole source of carbon and energy via salicylate and harbours a 83 kb mega plasmid. *Arthrobacter sulfureus*, *Acidovorax delafieldi*, *Brevibacterium* sp. and *Pseudomonas* sp. could utilize phenanthrene as the sole source of carbon and energy. These organisms can also utilize several other PAHs such as acenaphthene, anthracene, fluorine, fluoranthene, naphthalene and pyrene, making them potentially useful in cleaning up petroleum contaminated soils containing a mixture of different PAHs.

## 7.9 CONCLUSION

The extremophiles represent an interesting group of microorganisms with representatives among all microbial groups. However, bacteria, archaea and fungi are more tolerant to the environmental stresses than other microbial groups and show their presence in almost all extreme environments. Studies carried out over the years have provided sufficient knowledge about different aspects of the extreme environments and the biological processes running in these habitats. There are still several areas about which the present knowledge is highly rudimentary. These include the proper assessment of the true diversity of extremophiles as many of them are yet to be cultured and studied in the laboratory, understanding on the potential of these microbes in human welfare and industrial application of these species. Researches on the methods and skills for isolation of microorganisms from extreme habitats might yield new microorganisms having novel biomolecules. For example, deep sea adapted microorganisms (barotolerant) are better suited for manipulation and maintenance and thus they should also receive importance in industrial microbiology.

The use of small subunit ribosomal RNA (SSU rRNA) sequences to amplify environmental samples directly has revealed a whole lot of new prokaryotic lineages at phylum and division levels. Using the same tools a vast number of heterotrophic eukaryotes in the size range of prokaryotes (1-2  $\mu$ m) have been discovered from many extreme conditions. These included ciliates, fungi and thraustochytrids besides lineages not closely related to any known organism. Culturing these huge uncultured hidden microbial diversity and exploring their environmental, industrial and pharmaceutical use is going to be a challenging task.



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# METHODS IN ENVIRONMENTAL MICROBIOLOGY

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## 8.1 INTRODUCTION

In environmental samples, measurement of microbial activity provides a relative measure of the actual or potential level of activity within that environment. The measurement of microbial activity reflects the condition of the environment and make it possible to compare it for determining the stress effect. This basal activity level is dependent on a variety of biological, chemical and physical parameters, as well as the nutrient status of the environment. In fact, measurement of microbial activity can be used to evaluate the activity dynamics of the basal microbial community in response to a variety of stimuli, including naturally variable parameters such as moisture level and human imposed variables like addition of fertilizers, pesticides, heavy metals and living organisms. In an ecological sense, measurement of microbial activity in an undisturbed environment allows one to determine the microbial contribution to nutrient cycling within an ecosystem. For example, in the case of carbon cycling, the focus of microbial activity measurements is to determine accurately the production of new biomass by each microbial component and estimate the amount of energy that is being stored in a particular group of micro- or macroorganisms. This information allows determination of the transfer of energy between trophic levels in the food chain within a particular ecosystem.

The microbial activity acts as an indicator of the general health of the environment and can be used to evaluate the impact of a disturbance on the microbial community. For example, microbial activity has been used to evaluate the impact on soil quality of certain land use practices such as farming, logging and mining. Microbial activity is also an important indicator in evaluating the process of restoration of disturbed sites. Other areas, where microbial activity measurements are often used, include engineered systems, which include municipal wastewater treatment systems, compost systems and biological reactor systems.

The microbial community analysis is a necessary component for effectively understanding the processes running in an ecosystem. The diversity of microbial community indicates the rates and patterns of energy transfer and nutrient regeneration by the microbial participation. The community analysis also provides the information on the types of stresses present in that environment and helps in planning the processes to eliminate or mitigate the stress effects. In addition to the morphological and activity analysis, the microbial community studies can be made using a number of biochemical and molecular parameters. It includes the assay of metabolites, enzymes, proteins and nucleic acids. Though a detailed description of all these methodologies is not within the scope of this book, this chapter provides an outline of some of these modern methodologies that are presently in use in the field of environmental microbiology.

## 8.2 ASSAY OF MICROBIAL GROWTH

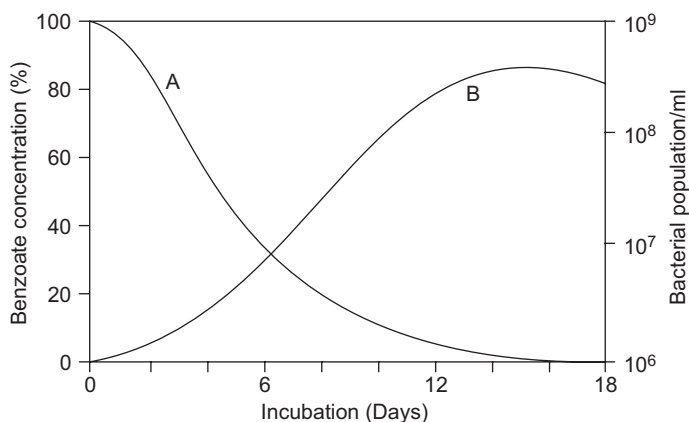
The microbial growth in natural environment or in the cultures can be measured by various methods. The most common methods used are substrate utilization, cell biomass, oxygen uptake and biomass of cellular components like carbohydrates and proteins.

### 8.2.1 Substrate Utilization

Substrate utilization can be measured in many different ways, depending on the chemistry of the substrate molecule and whether a single component or multiple components are present. For example, in a single solute system, aromatic compounds such as benzene and all benzene derivatives can be measured easily and sensitively by UV spectrophotometry at 224 nm. Analogously, a fluorimeter can be used to measure fluorescent molecules such as polyaromatic hydrocarbons (PAHs) within single-component systems. However, environmental samples contain compounds that interfere with the quantitation of the target compound. This can be considered a multiple-component system. For example, humic compounds present in soil and water samples contain many aromatic molecules that interfere strongly with measurement of aromatic substrates. In this case, a separation or chromatography step is required prior to detection of the target compound. For example, the rate of benzoate utilization by a microbial community increases with prolongation of incubation and the rate of utilization is directly proportional to the population density in the community (Fig. 8.1).

Important microbial activities, that contribute to the cycling of minerals in soil and water environments, include chemoautotrophic oxidation of ammonia (nitrification) and sulphur (sulphur oxidation). Therefore, the quantification of these substrate concentration in the microbial culture is a good parameter to estimate the microbial activity. Nitrification can be measured in several ways. A direct assay involves addition of  $^{15}\text{N}$ -labeled ammonia and measurement of the formation of  $^{15}\text{N}$ -nitrite and  $^{15}\text{N}$ -nitrate as the  $^{15}\text{N}$ -ammonia is oxidized. These  $^{15}\text{N}$ -labeled products are measured by mass spectroscopy. However, the sensitivity

of this measurement is low and thus large quantities of ammonia that far exceed those found naturally may be required in an experiment. A second approach to the measurement of nitrification is to incubate samples with  $^{14}\text{CO}_2$  to measure  $\text{CO}_2$  fixation, which is then compared in the presence and absence of the nitrification inhibitor to determine nitrification activity.



**Figure 8.1-** The relationship between (A) the benzoate concentration remaining in the growth medium and (B) bacterial population increase during 18 days of incubation.

Sulphur oxidation can be measured by addition of elemental sulphur ( $\text{S}^\circ$ ) to a sample and measuring the amount of sulphate ( $\text{SO}_4^{2-}$ ) produced during incubation. However, this is a complicated process. One such assay measures sulphate as a precipitate with barium chloride in acid solution. Following precipitation, the barium in solution is measured by atomic absorption spectroscopy (AAS). In this assay, as the amount of sulphate increases as a result of sulphur oxidation, the amount of barium in solution decreases through precipitation with the sulphate. A second approach to measurement of sulphur oxidation is to determine  $\text{CO}_2$  uptake in the presence of sulphur compounds by taking  $^{14}\text{CO}_2$ .

### 8.2.2 Terminal Electron Acceptance

A variety of terminal electron acceptors (TEAs) are available for microbial use and levels of microbial activity are reflected in the disappearance of the TEA and the appearance of the reduced TEA. For example, oxygen, unlike many other nutrients, is relatively insoluble in water and can become limiting to growth unless it is constantly replenished. Thus, aeration is an important consideration in both pure culture and environmental systems. Normal dissolved oxygen levels in an aqueous sample that is equilibrated with air range up to 10 mg/l depending on sample temperature and atmospheric pressure. Aerobic activity is maintained until the dissolved oxygen concentration in the sample falls below 1 to 2 mg/l. Because oxygen is a limiting nutrient, it is a good measure of growth and activity. Measurement of oxygen in a pure culture is relatively rapid, routine and cheap to

perform. Measurement of biological oxygen demand (BOD) of wastewater samples is a common procedure for estimation of the microbial activity. In addition the oxygen consumption can also be directly measured by the use of oxygen electrode. Colorimetric assays of oxygen consumption/ production can be made through use of reagents to produce an oxygen-dependent colour. Winkler method is the most common colorimetric method for dissolved oxygen measurement by using manganous sulphate and a potassium hydroxide-potassium iodide mixture (Wetzel and Likens, 1991). There are also several kits available for oxygen measurement that are based on a colorimetric reaction.

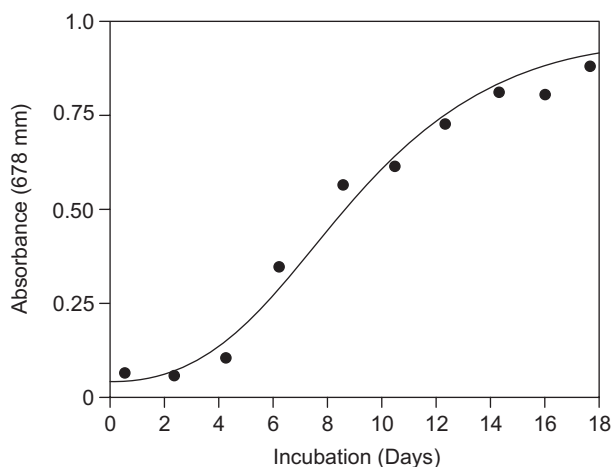
Alternate methods are also available to quantitate TEAs other than oxygen. For example, in denitrification,  $\text{NO}_3^-$  serves as a terminal electron acceptor and is reduced sequentially to  $\text{NO}_2^-$ ,  $\text{NO}$ ,  $\text{N}_2\text{O}$  and nitrogen. One common assay for denitrification involves the use of acetylene, which blocks the reduction of nitrous oxide to nitrogen. Thus, nitrous oxide accumulates in the system and can be measured using a gas chromatograph with an electron capture detector. A disadvantage of using this method in environmental samples is that acetylene also inhibits nitrification. Since nitrification and denitrification are closely coupled (e.g., the nitrate produced by nitrification is used in denitrification), this inhibition may reduce or eliminate denitrification in the sample. Therefore, in some instances, denitrification is estimated by simply measuring nitrification. A second problem with this technique is that it is not effective at low nitrate concentrations (less than 0.63 mg/l). Similarly, there are methodologies available for measuring sulphate reduction, iron and manganese reduction and methanogenic activity.

### 8.2.3 Cell Mass

The growth of microbes in a pure culture is often quantified by performing culturable plate counts or direct microscopic counts. This approach is also often used to estimate the cell mass in environmental samples. Another common approach to estimating cell mass is to measure the turbidity or the protein content in the culture being studied. In case of growth of microalgae, the growth can be spectrophotometrically measured by recording the change of culture absorbance at selected wavelengths (Fig. 8.2). Although these measurements are not suitable for environmental samples, they are easy, rapid and reproducible when used for measuring cell mass increases in pure culture. At low cell densities, there is a direct linear relationship between the number of cells and the turbidity of the suspension. At high cell densities, however, this relationship becomes nonlinear. Therefore, a standard curve must be constructed to determine the linear range for turbidity measurement for each organism being measured.

A variety of methods are available for measurement of protein to quantify the growth of the microorganisms. Currently, the most commonly used assays are based on the Folin-phenol reagent (Lowry *et al.*, 1951), the Coomassie Blue reaction (Bradford, 1976) and the bichinchoninic acid-copper reaction (Smith *et al.*, 1985). Each of these assays quantifies the amount of protein present by measuring colour produced by a reaction of the protein present with the assay

reagents. For example, the Lowry assay is based on the development of a blue colour resulting from the reaction of the Folin reagent with aromatic amino acids contained in the protein sample. The sample is measured in a spectrophotometer at 550 nm. Similarly, the Bradford assay measures the binding of Coomassie Brilliant Blue G dye to protein amino groups, which causes the development of a blue colour that can be measured at 595 nm.



**Figure 8.2** - The sigmoidal growth pattern of *Synechocystis* sp. PCC 6803 cultured in BG 11 medium. The initial inoculum density was  $10^6$  cells/ml.

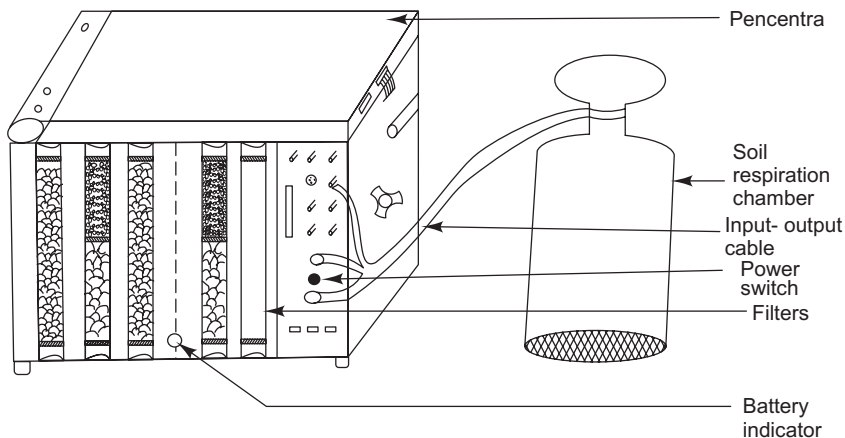
### 8.3 QUANTIFICATION OF CARBON UTILIZATION

The utilization of organic carbon by microbial community is a common process in the environment that is proportional to the density of the microbial population. Heterotrophic activity is a measure of the extent to which microbial populations utilize these organic materials as a carbon and energy source. This organic carbon degradation also involves the consumption of oxygen (under aerobic conditions) or an alternate terminal electron acceptor (under anaerobic conditions) and production of carbon dioxide. For heterotrophic activity under aerobic conditions, one can use the general assumption that 50% of the carbon goes to  $\text{CO}_2$  and 50% goes to cell mass (See Chapter 12). Under anaerobic conditions a smaller portion (ca 5%) of the carbon is used to build cell mass and a larger portion goes to  $\text{CO}_2$  and methane. Thus microbial activity within environmental samples can be monitored by measuring TEA consumption or  $\text{CO}_2$  evolution. However, for both  $\text{CO}_2$  and TEA, the relationship between the measured parameter and the amount of new cell material formed is specific to the organism, the particular substrate and environmental conditions that influence growth rate. Therefore, a conversion efficiency must be estimated to quantify bacterial production absolutely, even if an absolute value for microbial activity cannot be determined.

### 8.3.1 Measurement of Respiration

There are several approaches to measuring  $\text{CO}_2$  and TEA, in particular  $\text{O}_2$  in pure culture as described in section 8.2. Most of these approaches can also be adapted for use with environmental samples. The presence of respiration gases can also be determined by gas chromatographic analysis, typically using a thermal conductivity detector. Carbon dioxide in atmospheric samples can also be monitored using an infrared gas analyzer (IRGA), which detects  $\text{CO}_2$  from the difference of the gas between the measurement chamber and the ambient air. Some specific approaches for measurement of respiration gases in terrestrial and aquatic environments are given below. Table 8.1 summarizes some of the most commonly used methods for the measurement of microbial activities in the environmental as well as laboratory samples.

**(a) Measurement of respiration in terrestrial samples:** Microbial activity within soil or subsurface materials can be measured under controlled conditions in a laboratory or can be measured *in situ*. In a laboratory study, a sample of the porous medium is typically incubated in a sealed, airtight enclosure, usually referred to as a microcosm. The use of sealed microcosms allows the measurement of microbial activity within the sample as determined by the flux of  $\text{CO}_2$  and/or  $\text{O}_2$  within the headspace atmosphere. Headspace gas samples can be withdrawn using a gastight syringe and the  $\text{CO}_2$  or  $\text{O}_2$  concentrations can be determined using gas chromatography. One problem that can arise during long-term incubation periods is depletion of oxygen within the headspace of sealed, airtight flasks. To address this problem, flow-through incubation systems have been devised to allow headspace gases to be replenished while still allowing quantification of microbial respiration mostly by infrared gas analysis. In this case,  $\text{CO}_2$  free air is used as the flow-through gas and any  $\text{CO}_2$  in the air exiting in the culture vessel is a direct result of microbial activity, which is detected by the IRGA (Fig. 8.3).



**Figure 8.3** - Schematic diagram of an infrared gas analyzer (IRGA) used for measurement of photosynthetic carbon fixation and respiratory  $\text{CO}_2$  evolution. The cuvette shown in the diagram is used for measurement of soil respiration.

**Table 8.1:** The commonly used methods for measurement of microbial activity in environmental samples.

| <i>Test</i>                            | <i>Basis of test</i>                                                                                                                                                                                                               | <i>Application</i>                                                                                                                                                                                                                                                    | <i>Advantages</i>                                                                                                                                                                                                                                      | <i>Disadvantages</i>                                                                                                                                                                                                                                               |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Measurement of respiration gases       | Measurement of oxygen utilization or CO <sub>2</sub> production in an environmental sample. The flux of respiration gases provides an indication of overall metabolic activity, which can reflect the level of microbial activity. | Basal respiration measurements reflect microbial metabolism of organic substrates present in the environment. However, respiration by other components, such as plant roots in soil or algae in water, will also be included, depending on the environment            | Field chambers can be built and installed <i>in situ</i> to monitor flux of respiration gases in relatively undisturbed samples. The addition of an organic substrate to the sample can be used to indicate the level of potential microbial activity. | Incubation of samples in a closed chambers (microcosms) in order to monitor the flux of respiration gases can create an artificial environment. CO <sub>2</sub> production can be underestimated due to pH-dependent retention of inorganic carbon as bicarbonate. |
| Respiration of radiolabeled substrates | Metabolism of a radiolabeled substrate is monitored by measuring the evolution of labeled-CO <sub>2</sub>                                                                                                                          | This method is used to determine the potential for metabolism of a foreign substrate, such as an organic pollutant. Also, overall heterotrophic potential can be estimated by determining the turnover of organic substrates that occur naturally in the environment. | The use of radio labeling imparts a high sensitivity to the measurement, thus short incubation periods can be used. The use of specific radiolabeled compounds shows the potential for degradation of that specific substrate.                         | The concentration of substrate added is often greater than the concentration present in the environment, thus the rate of metabolism may be overestimated unless corrective procedures are used.                                                                   |
| Fixation of labeled CO <sub>2</sub>    | The photosynthesis is measured by determining the amount of <sup>14</sup> CO <sub>2</sub> incorporated autotrophically                                                                                                             | The method is used to determine the photosynthetic activity of the phytoplankton culture or assemblage under                                                                                                                                                          | The quantification of carbon fixation gives an idea about the primary production by phytoplankton in a water                                                                                                                                           | Artificial conditions are used for measurement of photosynthesis, which is significantly different from                                                                                                                                                            |

(Contd.)



**Table 8.1:** (Contd.)

| <i>Test</i>                                               | <i>Basis of test</i>                                                                                                                                           | <i>Application</i>                                                                                                                                                           | <i>Advantages</i>                                                                                                                                                                                                                                                                                                                                                                       | <i>Disadvantages</i>                                                                                                                                                                                                                  |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microelectrodes                                           | Probes with tips <20 $\mu\text{m}$ in diameter can be inserted into environmental samples to provide a continuous monitoring of activity.                      | natural or laboratory conditions<br><br>Microelectrodes have been designed to measure specific respiratory activities, including oxygen utilization and nitrate respiration. | body. The measurement can be made under controlled conditions or varied environmental conditions. It enables the investigator to determine the relative importance of different parameters.<br><br>Can monitor real-time activity at critical interfaces in biological systems. Especially useful to monitor the interdependence of aerobic and anaerobic processes in the environment. | the ambient environment. This may make the extension of the result to the field conditions difficult.<br><br>The instrumentation is delicate and relatively expensive. Extensive measurement with this device is certainly difficult. |
| Incorporation of radiolabeled thymidine into cellular DNA | Microorganisms scavenge DNA precursors, such as thymidine, from their environment. By radiolabeling thymidine, the rate of incorporation into DNA is measured. | The rate of DNA synthesis provides a reasonable estimate of the rate of cell division, thus providing an estimate of microbial biomass production.                           | When using a short incubation time, thymidine incorporation is thought to measure bacterial DNA production because the rate of bacterial incorporation of thymidine is thought to be much faster than for other organisms, which may be present in environmental samples.                                                                                                               | Not all bacteria will incorporate exogenously supplied thymidine into DNA. Also, estimating microbial activity requires the development of a conversion factor relating thymidine incorporation to biomass production.                |

(Contd.)

**Table 8.1:** (Contd.)

| <i>Test</i>                         | <i>Basis of test</i>                                                                                                                                                                      | <i>Application</i>                                                                                                                                           | <i>Advantages</i>                                                                                                                                                                                                                                                | <i>Disadvantages</i>                                                                                                                                                                                                           |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adenylate energy charge (AEC)       | AEC is a weighted ratio of ATP to total adenylates. ATP is quantified using a luciferin-luciferase substrate-enzyme system.                                                               | AEC values reflect a continuum between an active microbial community (AEC >0.8) and a community with a high proportion of dead or moribund cells (AEC <0.4). | AEC can establish the presence of a metabolically active community without the need to incubate the sample or add a surrogate substrate.                                                                                                                         | AEC is not necessarily a direct measure of microbial activity because adenylates are present in all living organisms and can also be released by aging and decaying cells.                                                     |
| Dehydrogenase assay                 | This assay measures the rate of oxidation-reduction reactions (electron transport chain activity) by monitoring the reduction of a tetrazolium salt by actively respiring microorganisms. | The rate of reduction of tetrazolium salts reflects the overall activity by all respiring microorganisms.                                                    | Actively respiring microorganisms can be visualized microscopically by the deposition of pigmented tetrazolium salts reduction products within the cell. The assay has a high sensitivity, and can be used to measure activity in low productivity environments. | The amount of tetrazolium salts reduced depends on many factors including sample incubation conditions. Direct comparison of activity between samples can be made only when the assay is performed under identical conditions. |
| Hydrolysis of fluorescein diacetate | Hydrolysis of fluorescein diacetate is performed by a variety of enzymes including esterases, proteases, and lipases                                                                      | The assay measures enzymatic activity, thus providing an estimate of total microbial activity in an environmental sample.                                    | Fluorescein diacetate hydrolysis produces a highly fluorescent product, fluorescein, which is easily detected by a fluorimeter on excitation of the pigment.                                                                                                     | Enzymes involved in hydrolysis can be intracellular or extracellular. The intracellular enzyme activity is underestimated by the use of FDA.                                                                                   |

Modified from Herman and Maier (2000).

Alternatively, field studies can be performed by placing a field chamber over a plot of surface soil and using this setup to make *in situ* measurements. In this way the soil structure is not altered by the sampling procedure and field respiration rates of the indigenous population are more reliably determined. Samples of atmosphere from within the field chamber can be withdrawn using a gastight syringe and then injected into a gas chromatograph for CO<sub>2</sub> and O<sub>2</sub> determination. By the use of a respiration measurement chamber, respiration can directly be measured from soil surface using IRGA. In some cases it may be desirable to measure respiration gases below the soil surface. In this case a gas sampling probe can be inserted beneath the soil surface and used to withdraw gas samples at discrete depths in the soil profile. However, care must be taken to avoid the creation of preferential flow paths, which could result in the introduction of above ground atmosphere into the subsurface.

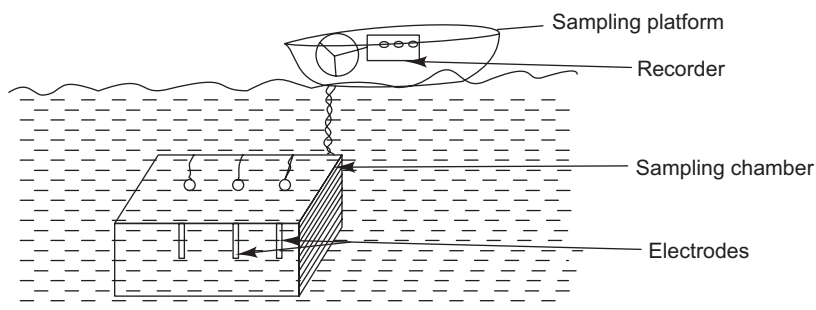
**(b) Respiration in aquatic environments:** Oxygen depletion is a common means of determining microbial activity in aquatic environments, in part because of the ease with which dissolved oxygen can be determined. However, many aquatic systems are oligotrophic, especially marine environments and thus the number and activity of heterotrophic bacteria are limited. Therefore, water samples must be incubated in sealed microcosms for extended periods, 12 hours or often longer, in order to detect significant oxygen depletion. This may, however, create an artificial environment because of alteration of several time-dependent variables in the system. These include a change in the quantity and the form of organic carbon substrate, changes in grazing pressure resulting from the presence of bacterivorous zooplankton and the exclusion of larger forms of grazing animals from the microcosm. With time, these effects can create deviations from the initial respiratory rate. To avoid this type of error, incubation time should be standardized with the existing conditions and the time required to obtain an observable effect, which will vary with the aquatic environment.

Another factor that must be considered in aquatic systems is the close association between photosynthetic and heterotrophic microorganisms. In order to prevent photosynthetic generation of oxygen, which would interfere with the measurement of heterotrophic activity using oxygen depletion, incubations should be performed in the dark in a BOD incubator. However, in this case the oxygen demand measured will include both heterotrophic activity and oxygen utilization in the dark respiration cycle of photosynthetic organisms. Because of these inherent difficulties in the measurement of respiration gases in aquatic systems, a better choice of activity measurement is usually one of the tracer methods using a passive radioisotope (<sup>14</sup>C). These techniques have the advantage of requiring much shorter incubation periods and are more specific for heterotrophic bacteria.

Direct measurement of O<sub>2</sub> consumption/evolution from a water sample can be made using an oxygen electrode. The sample is, however, required to be kept in an airtight container in the laboratory or as an airtight microcosm in the field in order to eliminate the contamination of the sample due to atmospheric interaction. Simultaneous measurement of water electrical conductivity (EC) and pH aids to the estimation of heterotrophic activity as well as the remineralization rate. There is

more or less a direct correlation between the EC and the rate of remineralization by microbial activity, though such relationship is often influenced by a number of chemical reactions in the environmental conditions.

For measuring the respiratory  $O_2$  consumption in the field condition, the sample is separated from the surrounding by an enclosure and the electrodes are inserted into it. The values are continuously recorded to determine the pattern of changes in the respiratory  $O_2$  consumption and the simultaneous changes in other physicochemical attributes of the sample. A multichambered sample cabinet can be used to record the results of several replicates together (Fig. 8.4).



**Figure 8.4** - Simultaneous measurement of electrical conductivity, pH and  $O_2$  consumption in the field by using an airtight sampling chamber.

### 8.3.2 Determination of Heterotrophic Potential by Radiolabeling

**(a) Heterotrophic activity:** The use of  $^{14}C$ -labeled carbon substrates is a common procedure of measurement of heterotrophic potential.  $^{14}C$  - labeled carbon substrate utilization reflects the biodegradation of a specific organic compound. Many radiolabeled substrates are commercially available, including sugars such as glucose, organic acids such as lactic acid, amino acids and even many representative organic pollutants. Mineralization of these substrates can be quantified by measurement of the evolution of  $^{14}CO_2$ , which can be quantified using a scintillation counter. The addition of a  $^{14}C$ -labeled substrate to an environmental sample reveals the presence of a degrading population in that sample and also indicates the rate at which the substrate can be mineralized. However, the rate of substrate mineralization is related to the amount of added substrate up to a saturation limit. Thus, depending on the level of substrate added, this measure may represent basal metabolism or may represent the potential for a microbial response to higher levels of added substrate. So the dilemma associated with the use of radiolabeled substrates is that although they can provide a very sensitive measurement of substrate utilization, the rate of utilization is usually proportional to the amount of labeled substrate added to the environmental sample. This is especially true for compounds such as glucose, which are very rapidly metabolized by microbial populations.

Measurement of heterotrophic activity has been particularly important in understanding how carbon is cycled through the aquatic environment. Heterotrophic bacterial populations have been shown to dominate the utilization of dissolved organic carbon in aquatic environments. Determination of dissolved organic carbon utilization rates revealed that the production of new biomass through heterotrophic processes can equal the amount of biomass formed through primary production by photosynthetic microorganisms. Heterotrophic activity measurements have increased our understanding of the distribution of microbial communities in different ecological zones within the aquatic environment. It has been shown that bacterial abundance and the rate of carbon cycling are greatest in freshwater sediments, as compared with the planktonic, or free-floating, environments. From this, we now understand the ecological importance of sediment-associated bacterial communities in carbon and energy cycling.

**(b) Measurement of anaerobic respiration:** Quantification of  $\text{CO}_2$  production or  $\text{O}_2$  utilization is the best way to measure soil respiration under aerobic conditions as an indicator of carbon utilization. However, many environments are completely anaerobic or contain anaerobic niches and there are situations in which monitoring the activity of anaerobic respiration is critical. For example, whereas anaerobic environments can support the biotransformation of organic pollutants, in some cases this results in the formation of more toxic metabolites. Anaerobic respiratory pathways include the use of  $\text{NO}_3^-$ ,  $\text{Fe}^{3+}$  (and other metals),  $\text{SO}_4^{2-}$  and  $\text{CO}_2$  as terminal electron acceptors. Reduction of metals (Co, Cd, Ni, Al, Pb, Zn, etc.) often produces leachable metal species (soluble and bioavailable metal species), which can become available to microbes in the aquatic environment and cause toxicity to the microbial community. Some of the terminal electron acceptors form gaseous intermediates or endproducts that can be used as a measure of anaerobic respiration. For example, nitrous oxide ( $\text{N}_2\text{O}$ ) is an intermediate in the bacterial reduction of  $\text{NO}_3^-$  to  $\text{N}_2$  (denitrification). Acetylene gas is used to inhibit the complete denitrification of  $\text{NO}_3^-$ , resulting in the accumulation of NO gas. Production of  $\text{N}_2\text{O}$  in samples treated with acetylene gas is proportional to the rate of denitrification. In terms of the soil condition, the presence of  $\text{N}_2\text{O}$  gas suggests a reducing environment in which the indigenous bacteria can utilize  $\text{NO}_3^-$  as an alternative electron acceptor. The production of  $\text{N}_2\text{O}$  also indicates that nitrification processes at some point resulted in an adequate amount of  $\text{NO}_3^-$ .

## 8.4 RADIOLABELING OF CELLULAR MACROMOLECULES

Quantification of radiolabeled tracer molecules into cellular macromolecules is a good parameter for the measurement of microbial growth. This technique is based on the fact that many species of heterotrophic bacteria can scavenge preformed molecules such as nucleotides or amino acids from their environment, incorporating them directly in cellular constituents. Monitoring the rate at which radiolabeled nucleotides or amino acids are incorporated into essential cellular macromolecules can be used as a means of determining microbial activity. Examples of specific

molecules used as tracers are the nucleoside thymidine, labeled with tritium ( $^3\text{H}$ , which is incorporated into DNA and the amino acid leucine, labeled with either tritium or  $^{14}\text{C}$ ), which is incorporated into protein. Radiolabeling of photosynthetically active cells with inorganic  $^{14}\text{C}$  is used to measure the effects of toxicants on the transport properties of the biologically active and intact plasma membrane. In addition the non-metabolizable radiolabeled substrates like 2-deoxy glucose and  $\alpha$ -aminoisobutyric acid (AIBA) are also used to study the membrane properties of bacteria and cyanobacteria.

### 8.4.1 Incorporation of Thymidine into DNA

The scavenging of the thymidine nucleoside and its incorporation into DNA by heterotrophic bacteria have been used as an indicator of the bacterial growth. Once transported across the cell membrane, the thymidine is converted into thymine monophosphate by the action of the enzyme thymidine kinase. Further phosphorylation results in incorporation into DNA as the thymine base. In very general terms, the procedure involves adding  $^3\text{H}$ -thymidine to a water sample containing planktonic bacteria or to a slurry prepared by mixing soil or sediment with water. The incubation time is usually limited to several hours and the incorporation of labeled thymidine by the cells is terminated by the addition of a chemical inhibitor or by placing the sample in an ice bath. DNA is then extracted from the cells and the radioactivity quantified to determine the amount of labeled thymidine incorporated. One problem with this technique is that during the labeling and uptake procedure, nonspecific labeling of other macromolecules, such as proteins and RNA, can occur. In this case, purification of the DNA may be required to improve the accuracy of the activity measurement. However, the rate of incorporation of thymidine to other biomolecules is much slower than that into the cellular DNA. Therefore, thymidine incorporation is still a dependable method for measurement of microbial growth. The advantage of taking thymidine for measurement of bacterial heterotrophic activity is that the rate of thymidine incorporation into bacterial DNA is much higher than the rate for other organisms tested, such as algae, fungi, or protozoa. Therefore, if the uptake study is limited to a short incubation periods (up to several hours), it is generally believed that the incorporation of these precursors into cellular macromolecules will reflect the activity of growing heterotrophic bacteria. However, not all growing bacteria will incorporate exogenously supplied thymidine into DNA. Exceptions have been noted, particularly within the genus *Pseudomonas*, which may limit the usefulness of this technique in certain environments.

A further complication in the interpretation of the results of this assay is that there are both external and internal pools of thymidine or thymidine metabolites that can compete with the added  $^3\text{H}$ -thymidine for incorporation into DNA. The internal pool is created by cells in a *de novo* process in which the nucleotides that form DNA are synthesized from cellular components. If this internal nucleotide pool is large, it may compete with scavenged labeled thymidine for incorporation into DNA. There can also be an external pool of thymidine available for uptake.

This external pool is composed of extracellular thymidine probably released from dying organisms. This is especially true for sediments, where thymidine becomes sorbed to sediment particles. The existence of the external and internal pools of thymidine metabolites will result in dilution of the added  $^3\text{H}$ -thymidine and cause underestimation of the rate of DNA synthesis. Isotope dilution, which refers to the size of the pool into which  $^3\text{H}$ -thymidine is diluted, can be estimated and used to correct estimates of DNA synthesis. Further, the addition of sufficient quantities of exogenous labeled thymidine may cause competitive inhibition of the *de novo* synthesis route and thereby limit the size of the internal thymidine pool.

The thymidine incorporation rate can be converted into a measure of microbial activity using a conversion factor that relates the number of new cells formed to a given amount of thymidine incorporated (i.e., cells formed per mole thymidine incorporated). This conversion factor is derived on the basis of measurements, or best possible estimates, of the amount of DNA per cell, the thymine content of a cell and the extent of dilution of the labeled thymidine. An alternative way to determine the conversion factor is to relate the  $^3\text{H}$ -thymidine incorporation rate directly to cell division based on a separate and independent measure of the increase in bacterial numbers (turbidometric or numerical estimation). Common estimates of this conversion factor range from 1.3 to  $2.0 \times 10^{18}$  cells/mole  $^3\text{H}$ -thymidine incorporated.

Based on thymidine incorporation, bacterial growth can range from less than  $1 \times 10^6$  cells/g soil/day in an oligotrophic aquifer environment to greater than  $1 \times 10^9$  cells / g soil. day in a marine sediment (Thorn and Ventullo, 1988). The rate of thymidine incorporation can also be used without conversion to estimate cell division. The thymidine incorporation rate provides a relative measure of microbial activity that can be used in controlled experiments to evaluate the impact of a specific factor, such as the toxicity of xenobiotics to microbial activity in environmental samples.

## 8.4.2 Incorporation of Leucine into Protein

The rate of leucine incorporation into cellular protein has also been used as a measure of microbial activity. Studies have indicated that a majority of bacteria scavenge leucine from their environment and that most of the assimilated radioactive label will be incorporated into proteins. Further, the *de novo* synthesis of leucine is inhibited by sufficient quantities of exogenous leucine. Radiolabeled leucine is taken in a medium containing all constituents for microbial growth and incubation is made in dark for a short period (up to few hours) to allow the incorporation of leucine to cellular proteins. The growth is then stopped, protein is extracted and the radioactivity is measured to quantify the uptake.

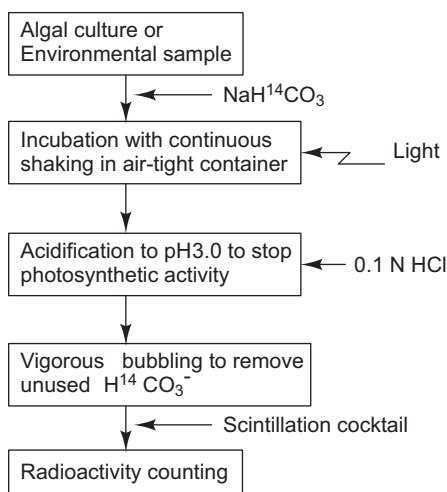
## 8.4.3 Radiolabeling for Study of Membrane Properties

**(a) Uptake of  $^{14}\text{CO}_2$ :** Uptake of inorganic  $^{14}\text{C}$  by photoautotrophic cells has always become a good method to measure the metabolic efficiency of photosynthetically active cells. The rate of uptake of inorganic  $^{14}\text{C}$  is strongly

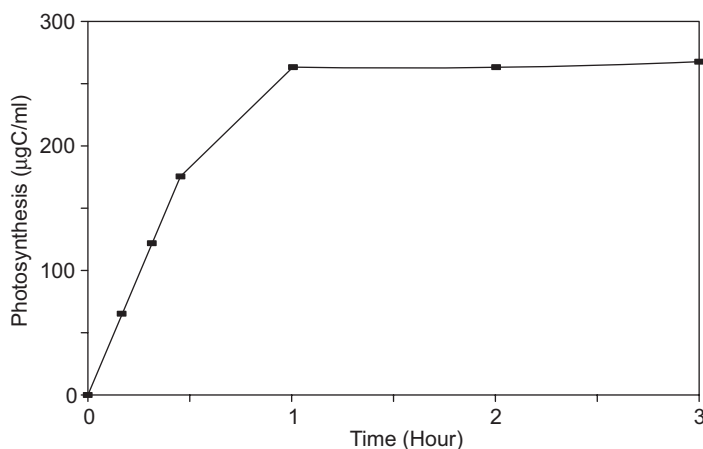


influenced by the environmental factors and stress when light is not limiting. Reduced rate of carbon fixation, in comparison to control is an indication of the stress received by the organisms. For example, the presence of growth inhibitory concentration of pesticides reduces photosynthetic activity of phytoplankton both in the environmental samples and in laboratory cultures. Similarly reduced rate of carbon fixation is also noticed with nutrient deficiency, increased acidity and high metal concentration.

For measuring the photosynthetic carbon fixation by phytoplankton, radiolabeled sodium bicarbonate ( $\text{NaH}^{14}\text{CO}_3$ ) is taken as the source of inorganic carbon. After being suspended to the required cell density ( $10^5$ – $10^7$  cells/ml), the samples are spiked with  $\text{NaH}^{14}\text{CO}_3$  (initial  $^{14}\text{C}$  of 0.1 – 0.3  $\mu\text{Ci/ml}$ ) and are incubated in light (Fig. 8.5). Care is required to be taken to provide sufficient light to drive photosynthesis but at the same time the light should not be photoinhibitory as it will reduce the rate of photosynthetic activity and underestimate the photosynthetic carbon fixation potential of the sample. It is also required to standardize the incubation period so that optimum photosynthetic activity is recorded and the sample does not come to a condition where photosynthesis is inhibited due to  $\text{CO}_2$  limitation. This is very vital especially when the sample has a very high population density. For example in a culture of the cyanobacterium *Synechocystis* sp. PCC 6803 having a cell density of  $10^7$  / ml, the  $\text{CO}_2$  limitation is felt by the culture within one hour of incubation with  $\text{NaH}^{14}\text{CO}_3$  supplementation at a concentration of about 2  $\mu\text{M}$  (Fig. 8.6). An increase in the concentration of added  $\text{NaH}^{14}\text{CO}_3$  may solve the problem but at the same time it may change the pH of the suspension, which has also an adverse impact on photosynthesis.



**Figure 8.5** - Schematic representation of the steps of uptake of inorganic  $^{14}\text{CO}_2$  by a sample containing phytoplankton. (Adopted from Mohapatra, 1995).



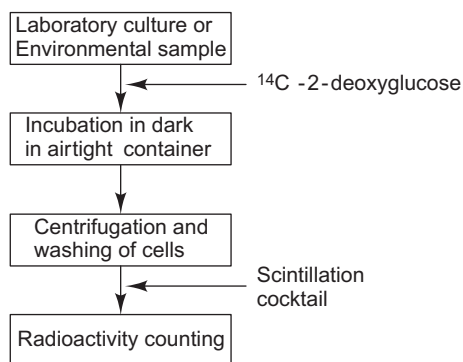
**Figure 8.6** - Photosynthetic uptake of  $\text{H}^{14}\text{CO}_3^-$  by *Synechocystis* at different times of incubation. The uptake was measured for 3 h in a medium containing 0.1 mCi ml of  $^{14}\text{C}$  by illuminating the cells with a PPFD of  $220 \mu\text{mol}/\text{m}^2 \text{ s}$ . The figure shows that almost all  $^{14}\text{C}$  was taken from the medium during first one hour of incubation (Adopted from Mohapatra, 1995).

After incubation of the sample, acidification is made with addition of 0.1 N HCl to achieve a pH of about 3.5 in the sample. The acidification not only stops photosynthesis but also facilitate the removal of  $\text{CO}_2$  from the sample solution. Removal of  $\text{CO}_2$  is needed to get rid of unused inorganic  $^{14}\text{C}$  from the sample before radioactive counting and is done by vigorously bubbling the samples for about 20 minutes. Thereafter it is mixed with a scintillant and the activity is counted. It is always required to include a standard with no acid addition to check the activity of the radioactive source used.

**(b) Incorporation of  $^{14}\text{C}$ -2 deoxyglucose:** Non-metabolizable sugars and amino acids are always good substrates to study the transport of their metabolizable forms into the microbial cells. The advantages of taking the non-metabolizable substrates are that they are taken by the same transport processes as of their metabolizable forms but get accumulated in the cell to make it possible to measure the actual uptake rate. 2-Deoxyglucose and 2-aminoinosobutyric acid (AIBA) are the two common non-metabolizable substrates taken for quantifying the rate of sugars and amino acids uptake by actively growing cells. After uptake, these substrates are accumulated in the cell cytosol at the expense of cellular energy and build a concentration gradient directed into the cell cytoplasm. This may create an energy deficient situation on prolonged exposure, due to which it is necessary to take metabolizable D-glucose along with 2-deoxyglucose to facilitate the growth of the culture. The ratio between D-glucose and  $^{14}\text{C}$ -2-deoxyglucose is usually maintained at 100:1 in the sample.

For measuring the uptake of 2-deoxyglucose, the samples are collected from the environment or reared in the laboratory and should have a comparatively high cell density ( $10^6$ – $10^8$  cells/ml). Diluted samples can also be taken but these require

longer incubation. Radioactive 2-deoxyglucose mixed with D-glucose at a ratio of 1:100 is added to the sample and then incubated in dark for a period ranging from few minutes to several hours (Fig. 8.7). After incubation the samples are acidified, filtered and the activity of the culture as well as of the supernatant are measured. Figure 8.8 shows a concentration dependent rate of uptake of glucose by actively growing *Synechocystis* cells under normal and insecticide treated conditions. This uptake has been calculated from the radioactivity counting of the harvested cells as it is expected that no loss of 2-deoxyglucose has occurred through respiratory consumption.

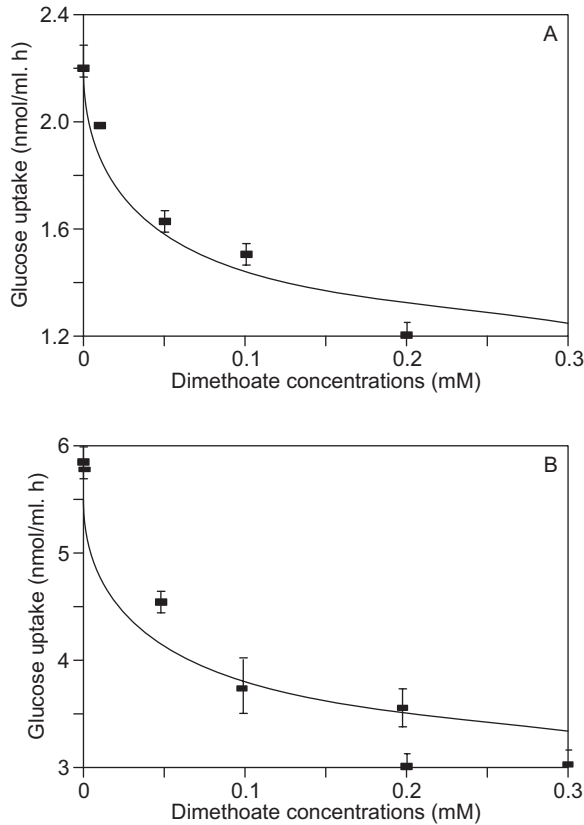


**Figure 8.7** - Schematic representation of the steps for uptake of non-metabolizable <sup>14</sup>C-2 deoxyglucose as a measurement of the active substrate transport in heterotrophic bacteria.

## 8.5 NUCLEIC ACID ASSAY

### 8.5.1 Extraction of Community Bacterial DNA

Because less than 1% of soil microflora are culturable, traditional cultural techniques tend to underestimate the true soil microbial population. Depending on the environment under investigation, several methods have been developed to obtain a more representative portion of the soil microflora. These are based on the total nucleic acid content of the community or community DNA (See Section 8.8). In the soil environment one way to obtain community DNA is to separate all cells from the soil and subsequently lyse the cells and extract the nucleic acids. Intact cells may be lysed by heat, chemical or enzymatic treatment. Following lysis, it is essential that the sample undergoes one or more purification steps for successful subsequent molecular analyses. There are problems with this approach, the major of which is the loss of DNA during extraction. In addition, the presence of metals or humic residues from environmental samples can interfere with further analysis even after careful purification. It should be noted that community DNA extracts may contain eukaryotic DNA as well as bacterial DNA. However, cell extraction followed by DNA analysis is still likely to result in a more representative estimate of the total bacterial community than a simple plate dilution on growth medium. It



**Figure 8.8** - Uptake of glucose by (A) *Synechocystis* sp. PCC 6803 and (B) *Chlorella vulgaris*. The uptake has been measured after one hour incubation in dark on treatment with different concentrations of dimethoate (Organophosphorus insecticide) (Adopted from Mohapatra *et al.*, 1997).

is generally agreed that the community based extraction methods (cells or DNA) obtain a higher proportion of the total bacterial population than dilution and plating alone. They are also likely to represent a different population of bacteria than that obtained by cultural methods.

A second approach for extraction of community DNA from soil is to lyse the bacterial cells *in situ* (within the soil matrix) and subsequently extract and purify the DNA. Several protocols for the extraction of community DNA have been published (More *et al.*, 1994) with most relying on the use of lysozyme or sodium dodecyl sulphate (SDS) as the lysing agent. Once extracted, the DNA is purified by cesium chloride density centrifugation or the use of commercial purification kits. An alternative approach is to use phenol extraction followed by ethanol precipitation as a means of DNA purification. Once the DNA has been purified, it is ready for molecular analysis.

In aquatic environments, cells are concentrated by filtering a representative portion of water through the appropriate filter. Bacteria are captured by

membrane filtration with a 0.2  $\mu\text{m}$  filter. Viruses are too small to be filtered by size exclusion and thus require an electronegative or electropositive filter for collection. In contrast, the larger parasites such as *Giardia* and *Cryptosporidium* are trapped in a simple fiber filter cartridge that captures the parasites via size exclusion. The phytoplankton and the protozoa can be easily collected by use of a plankton net or membrane filter. As with soil samples, once the organisms are collected from the water the total biomass can be lysed, purified and subjected to further molecular analysis.

### 8.5.2 Gene Probes and Probing

Gene probes are an application of nucleic acid hybridization. Typically, probes are small pieces of DNA known as oligonucleotides, which are complementary to the target sequence of interest, that are marked or labeled in some way in order to make them detectable. Probes have been used in environmental microbiology to examine soil microbial diversity, to identify a particular genotype and to test for virulence genes of suspected pathogens that have been isolated from water. Construction of gene probes and specific application methodologies are beyond the scope of this book. The readers are advised to go through the publications of Sambrook *et al.* (1989). Only the general concepts involved in the construction of gene probes along with some of their practical applications are discussed here.

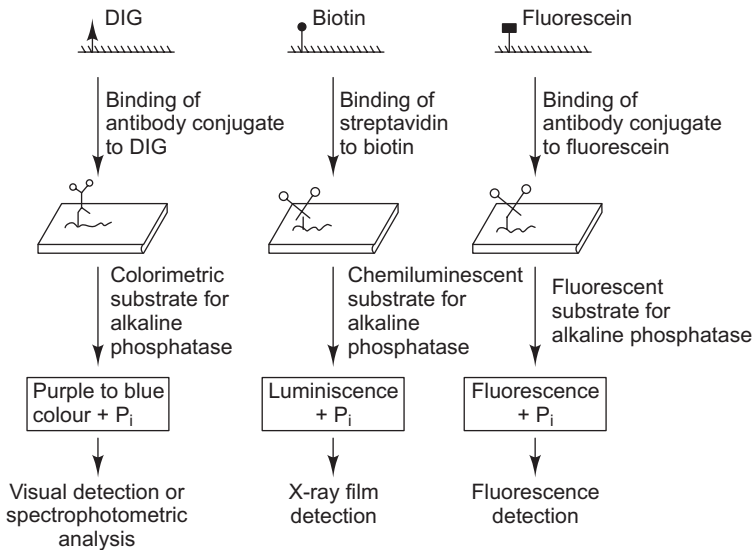
Gene probe methodology takes advantage of the fact that DNA can be denatured and reannealed. To make a gene probe, the DNA sequence of the gene of interest must be known. This gene may be unique to a particular microbial species, in which case the sequence may be useful for the specific detection of that organism. Alternatively, the gene may code for the production of an enzyme unique to some metabolic pathway and a gene probe constructed from such a sequence may indicate the potential activity of a group of bacteria in a soil or water sample. This kind of probe can be defined as a functional gene probe. For example, gene probes can be made from sequences that code for enzymes involved in degradation of phenol and such a probe could then be used to estimate whether a particular soil contained any bacteria having ability to metabolize the toxicant.

Many gene sequences are now known and are readily accessible via the gene databases. There are also computer software packages designed to search for sequences that are unique to a particular bacterium or, conversely, sequences that are conserved within particular groups of bacteria. The strategy that one uses to select a particular target sequence will depend on the intended use of the gene probe. The basic strategy in the construction of a gene probe is to obtain the sequence of the target gene and then to select a portion of this sequence for use as a probe. The size of the probe can range from 18 base pairs to as many as several hundred base pairs. The probe is then synthesized and labeled in such a way that it can be detected after it hybridizes to the target sequence.

A researcher has several options while constructing and labeling a probe. In the early days of molecular biology the only labeling option was radioactivity, but

several nonradioactive methods are now available. Radioactive labeling of a probe is typically done by labeling the sequence with a radioactive chemical, such as  $^{32}\text{P}$ , which is incorporated into the sugar-phosphate backbone of DNA. Later, when the probe or labeled sequence is hybridized to the target molecule, it can be detected by autoradiography. Non-radioactive alternatives include probes labeled with digoxigenin (DIG), biotin, or fluorescein, which can be incorporated into the sequence by chemical synthesis. The different labels are detected by binding the respective antibody or streptavidin-alkaline phosphate conjugate, which, when reacted with the appropriate substrate, will give a fluorescence signal (Fig. 8.9).

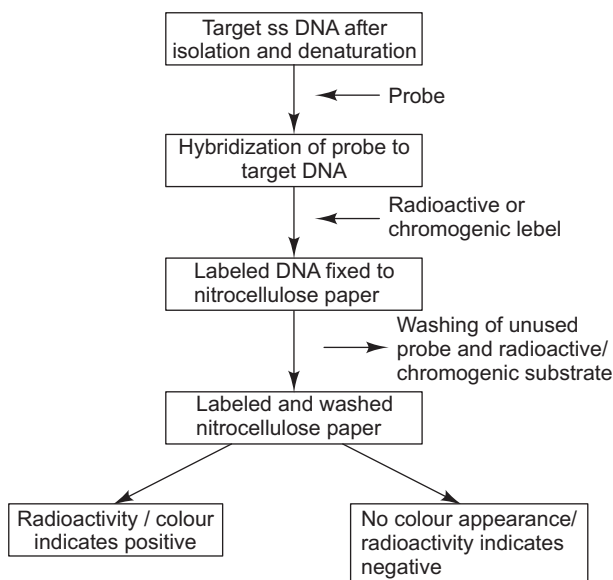
### Types of labeled DNA Probe



**Figure 8.9** - The scheme of different methods for gene probe detection of a DNA sequence.

The probing process is explained using the following example. A gene probe is constructed for a particular intestinal bacterium such as *Salmonella*. A water sample suspected to contain the pathogen is collected and then filtered onto a milipore membrane. The cells captured on the membrane are lysed and the released bacterial DNA is denatured into two single strands (Fig. 8.10). The gene probe is similarly denatured into single strands and is subsequently added to the membrane with the lysed cells. The DNA is allowed to reanneal and in some cases the single strand of the gene probe will anneal with the complementary target DNA sequence from the *Salmonella* cells. After washing the membrane free of unhybridized probe, the filter is exposed to x-ray film and the radioactivity of the  $^{32}\text{P}$  associated with the hybridized DNA results in a photographic image. A positive image indicates that the  $^{32}\text{P}$  probe annealed to target DNA on the filter,

implying that hybridization between the probe and the sample DNA has occurred. Such hybridization and hence detection, can occur only because the gene probe has annealed to the specific *Salmonella* sequence. In contrast no signal would indicate that the target *Salmonella* sequence was not present. This process can be repeated using another probe specific for an indicator organism such as *Escherichia coli*. This example illustrates how a probe can be used to detect a target organism within a sample containing a mixture of organisms. The probe can similarly be used to confirm the identity of a pure culture isolate.



**Figure 8.10** - Scheme of alternate probe labeling and detection systems.

### 8.5.3. Colony Hybridization

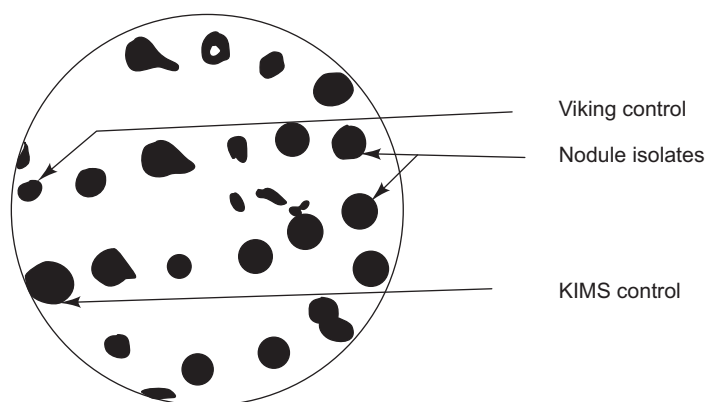
Gene probes can also be used to detect a specific gene sequence within bacterial colonies on a petri plate containing a mixed population of bacteria through use of a process termed colony hybridization or lifts. In this process a piece of filter paper is lightly pressed onto the petri plate so that some bacterial cells from each colony adhere to the paper. The cells are lysed directly on the filter paper and the DNA is fixed to the filter. The filter is then probed and detected as described above. After this procedure, only the colonies that contain the specific DNA sequence give a radioactive signal. Because the original petri plate contains all the intact colonies, the viable colony of interest can now be identified and retained for further study (Fig. 8.11 and Fig. 8.12).

### 8.5.4 Microarrays

Recent developments in high throughput screening of DNA sequences has resulted in the development of microarray technology. A microarray refers to a



set of DNA probes, usually purified PCR products from cDNA or genomic clones, deposited on a solid support (generally a glass microscope slide) with a spot density of several hundred individual spots per cm<sup>2</sup>. Arrays have been used primarily for gene expression profiling. For example, RNA is extracted from cells which have been exposed to stress and cells that have not been exposed to stress. A cDNA copy of the mRNA is labeled enzymatically by the inclusion of a fluorochrome labeled nucleotide during an amplification step. The array is hybridized with the target, the labeled cDNA copy of the mRNA and the hybridization intensity is acquired by laser scanning and CCD camera detection (Jordan, 1998). Since it is known where on the array that the individual DNA probes are located, the result is a global fingerprint or signature of the genes that are expressed as a result of a specific stress. This technology has been used in the pharmaceutical industry to look at drug responses. However, the applications of this technology seems limitless. In environmental microbiology microarrays have the potential to allow screening of a drinking water supply for several hundred pathogens with one assay. These are also helpful to screen the stress tolerant bacteria and microplankton of aquatic and terrestrial habitats.



**Figure 8.11** - A colony lift from a petri plate containing a mixed population of rhizobia isolated from root nodules. The plasmid KIM 5 has been used as a control (Reprinted from Pepper *et al.*, 1989).

### 8.5.5 RFLP Analysis of Whole Genomes

Restriction fragment length polymorphism (RFLP) is frequently used on bacterial isolates for various purposes. The method consists of the extraction, digestion and separation of DNA. The later from a pure culture of bacteria is first isolated through cell lysis followed by the DNA separation (Wilson, 1994). The genomic DNA is then cut into small fragments by using the appropriate restriction enzymes. The target sites of different restriction enzymes are different, which become useful for the fragment comparison (Table 8.2). The fragments of DNA are usually separated by gel electrophoresis and subsequently visualized by ethidium bromide staining followed by observation through UV transillumination. If the restriction enzyme

cuts at the multiple sites within a genomic DNA, several DNA fragments are formed and this provides a characteristic pattern of fragment distribution, when separated through gel electrophoresis. Such patterns are the characteristics of a bacterial isolate.

**Table 8.2:** Some restriction endonucleases<sup>a</sup> and their target sites.

| Organism                            | Restriction endonuclease | Target site                                                                   |
|-------------------------------------|--------------------------|-------------------------------------------------------------------------------|
| <i>Anabaena variabilis</i>          | <i>Ava</i> I             | C↓( <sup>C</sup> <sub>T</sub> )CG( <sup>A</sup> <sub>G</sub> )G               |
| <i>Bacillus amyloliquefaciens</i> H | <i>Bam</i> HI            | G↓GATCC                                                                       |
| <i>Bacillus globigii</i>            | <i>Bgl</i> II            | A↓GATCT                                                                       |
| <i>Escherichia coli</i> RY13        | <i>Eco</i> RI            | G↓AA <sup>*</sup> ATTC                                                        |
| <i>Escherichia coli</i> R245        | <i>Eco</i> RII           | ↓CC( <sup>A</sup> <sub>T</sub> )GG                                            |
| <i>Haemophilus aegyptius</i>        | <i>Hae</i> III           | GG↓ <sup>*</sup> CC                                                           |
| <i>Haemophilus gallinarum</i>       | <i>Hga</i> I             | GACGC                                                                         |
| <i>Haemophilus haemolyticus</i>     | <i>Hha</i> I             | G <sup>*</sup> CG↓C                                                           |
| <i>Haemophilus influenzae</i> Rd    | <i>Hind</i> II           | GT( <sup>C</sup> <sub>T</sub> )↓( <sup>A</sup> <sub>G</sub> ) <sup>*</sup> AC |
| <i>Haemophilus parainfluenzae</i>   | <i>Hpa</i> I             | GTT↓AAC                                                                       |
|                                     | <i>Hpa</i> II            | C↓ <sup>*</sup> CGG                                                           |
| <i>Klebsiella pneumoniae</i>        | <i>Kpn</i> II            | GGTAC↓C                                                                       |
| <i>Moraxella bovis</i>              | <i>Mbo</i> I             | ↓GATC                                                                         |
| <i>Providencia stuartii</i>         | <i>Pst</i> I             | CTGCA↓G                                                                       |
| <i>Serratia marcescens</i>          | <i>Sma</i> I             | CCC↓GGG                                                                       |
| <i>Streptomyces stanford</i>        | <i>Sst</i> I             | GAGCT↓C                                                                       |
| <i>Xanthomonas malvacearum</i>      | <i>Xma</i> I             | C↓CCGGG                                                                       |

Data from Old and Primrose (1989).

<sup>a</sup> Recognition sequences are written from 5' to 3', only one strand being given and the arrow indicates the point of cleavage. Bases written in parentheses signify that either base may occupy that position. Where known, bases modified by a specific methylase, are indicated by an asterisk.

<sup>\*</sup>A is N<sup>6</sup>-methyadenine; <sup>\*</sup>C is 5-methylcytosine.

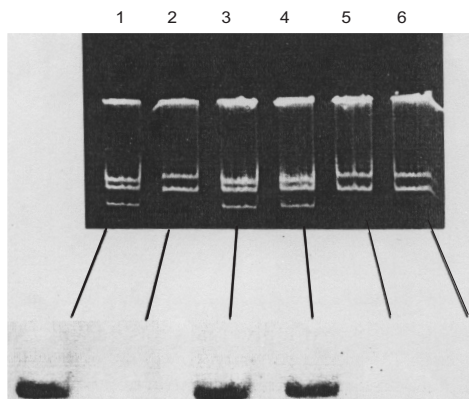
The RFLP analysis of whole genome aims at the screening of strains to understand their metabolic potential. Following cell lysis, bacterial DNA consisting of a single chromosome can be isolated and purified. Restriction enzymes can then be used to cut at specific recognition sites. Because a single base change resulting from an insertion, deletion, or substitution can create or eliminate a site, the locations of restriction sites can vary even among closely related bacteria. Distantly related bacteria would be expected to have many differently located restriction sites. These subtle changes are known as polymorphisms. DNA polymorphisms result in different recognition sites and thus different sizes of the restriction fragments from different isolates. After the enzyme digestions, polymorphisms are detected by ethidium bromide staining following gel electrophoresis.

A restriction digest of genomic bacterial DNA contains potentially hundreds of fragments of different sizes. Sometimes the observed pattern is photographed and used as the actual fingerprint. However, it is often difficult to interpret the large number of restriction fragments and in some cases there may be so many fragments that a smear is all that can be observed instead of distinct bands. To simplify the pattern, the DNA fragments can be transferred to a membrane and only fragments that contain a specific sequence detected using a gene probe. Many gram-negative enteric bacteria contain repetitive DNA sequences that are located at different points within the bacterial chromosome. Because these sequences are evolutionarily conserved, gene probes utilizing portions of these sequences can be designed. If a genomic DNA preparation of a particular isolate is cut with specific restriction enzymes, some DNA fragments will contain the conserved repetitive DNA sequence. The use of gene probes can identify these fragments and two different strains of the same species will produce different patterns of fingerprints. Insertion sequences have also been used as probes.

### 8.5.6 Plasmid Analysis

Many bacterial isolates contain one or more plasmids of variable size. In some cases, detection of these plasmids allows identification of specific bacterial isolates. The size and number of plasmids associated with a given bacterium are often unique and are usually non-repetitive. Note, however, that plasmids can be transferred to other species or in some cases lost from a bacterium if selective pressure is not present to maintain the plasmid. Curing is the process whereby a plasmid is removed from a bacterium and it can often be achieved by the application of a stress such as heat shock. For example, pJP4 is an 80-kb plasmid that encodes some of the enzymes used to degrade 2,4-dichlorophenoxyacetic acid (2,4-D) as well as sequences responsible for mercury resistance (mercuric reductase activity). This plasmid has been shown to undergo horizontal gene transfer from *Alcaligenes eutrophus* to other soil organisms and is not stable within the *Alcaligenes* isolate if grown in laboratory culture without selective pressure (Di Giovanni *et al.*, 1996). Other plasmids, particularly large megaplasmids, cannot be cured easily.

Despite the potential for plasmid transfer or loss, plasmids have been used to detect specific bacteria or genes. Specific gene sequences associated with a plasmid or homologous gene sequences among plasmids can be detected using specific gene probes. Bacteria that are associated with specific plasmids can be identified by plasmid profiles or plasmid fingerprints. Plasmid profiles are prepared by electrophoresis and staining of whole plasmids (Fig. 8.12), whereas plasmid fingerprints are produced by restriction digestion of plasmid DNA, followed by electrophoresis and staining of restriction fragments. For gene probe analysis, bacterial isolates can be lysed and probed via colony lifts using labeled plasmid DNA sequences (Fig. 8.11). Alternatively, plasmid extracts free of chromosomal DNA can be prepared, electrophoresed and probed.



**Figure 8.12** - Southern blot of a DNA sequence associated with a unique plasmid contained within the rhizobial strain KIM 5. The lanes show the original plasmid profile of two different rhizobial isolates. The bottom bands represent the Southern blot (Reprinted from Pepper *et al.*, 1989).

Crude plasmid extracts can be prepared via an alkaline lysis and used for Southern or plasmid profiles. For megaplasids larger than 20 kb, bacteria can be lysed directly in the gel wells. This *in situ* lysis reduces the shearing of megaplasids that can easily occur if other methods are employed. The large variety of plasmid functions is reflected in the variety of assays in which plasmid analyses have been utilized. For example, plasmid analyses have been used to identify *Salmonella* spp. in epidemiological studies. They have also been used to identify *Rhizobium* spp. involved in symbiotic relations with plant, in biodegradation studies and in studies of antibiotic and metal resistance.

Regardless of whether plasmid profiles or plasmid fingerprints are generated, this analysis provides a relatively quick method for identification of specific DNA sequences associated with a bacterium. Often these DNA sequences can be correlated with specific phenotypic characteristics such as resistance to antibiotics or metals or the ability to degrade specific organic contaminants (herbicides, phenol, benzene, etc.). In addition, plasmids have been used to identify specific bacterial isolates. However, in this case, the greatest disadvantage of plasmid analysis is the potential loss of the plasmid from the bacterium of interest. This can occur through a plasmid curing process, which could result in false-negative results, or the transfer of the plasmid to other isolates through horizontal gene transfer, which could result in false-positive results.

## 8.6 THE ENZYME ASSAYS

In the soil and sediment environments, enzymatic activity is essential for energy transformation and nutrient cycling reactions. For example, enzymes catalyze the hydrolysis of certain nitrogen-, phosphorus- or sulphur-containing organic compounds, releasing the ammonia, phosphate, or sulphate constituents, which are then available for assimilation by other organisms. Many enzymes are required for

hydrolysis of plant constituents, such as cellulose, starch and other polysaccharides, releasing the monomeric sugar units like glucose and fructose, which provide important energy sources for microorganisms. Extracellular enzymatic reactions in the soil environment are the determinants of soil fertility and remineralization from the sediment.

One way of monitoring enzymatic reactions is to measure the conversion of a specific substrate into a product. For example, nitrate reductase activity can be measured through the disappearance of nitrate and the formation of elemental nitrogen. More commonly, enzymatic reactions are detected using a bioassay procedure that is specific for a particular class of enzymes. The specificity refers to the assay conditions, enzyme concentration and the substrate types. A bioassay utilizes a surrogate substrate that is transformed by a specific class of enzymes, producing a product that has specific properties for detection. An example is the conversion of *p*-nitrophenol phosphate by phosphatase enzymes, producing *p*-nitrophenol and phosphate. The indicator product, *p*-nitrophenol, can be quantified spectrophotometrically. The phosphatase activity can also be fluorimetrically analyzed by taking methyl umbeliferone phosphate (MUFPP) which is metabolized by phosphatase to release the fluorescent MUF (See Section 8.6.3).

The enzymatic activity within an environmental sample can originate from a variety of sources. Enzymatic activity can be directly associated with actively growing microorganisms, including bacteria, fungi and actinomycetes. These enzymes may be contained within the cell and are often located within the cell membrane or are released from actively growing cells. Such release is intentional by the microorganisms to hydrolyze large polymers, such as the plant polymers, cellulose, hemicellulose and lignin, in order to produce assimilable monomers and facilitate their uptake by the cells for further metabolism. Enzymes may be associated for a short time with moribund cells. These enzymes may also become stabilized on clay or humic particles within the soil structure and can remain viable for a period of time. Therefore, any bioassay of enzymatic activity in environmental samples will measure activity from all sources and provide an indication of total enzymatic potential within that sample.

Quantification of enzyme activity in environmental samples can reflect the health of the environment in terms of nutrient cycling and can also reflect soil fertility parameters, such as crop yield. However, there are problems associated with use of enzyme assays to directly quantify microbial activity. First of all, enzymatic activity associated with actively growing microorganisms cannot be easily separated from the activity of extracellular enzymes stabilized in the soil environment or enzymes associated with decaying cells. In addition, enzyme assays often require the addition of a surrogate substrate and, as a consequence, the assay determines the potential enzymatic activity and not the actual level of activity in the sample. In some cases the activity is also measured by taking a non-metabolizable substrate, whose uptake becomes inhibited on prolongation of incubation due to the building up of the internal pool of the substrate. Enzyme assays are also specific for a particular substrate-enzyme combination and may not reflect the overall activity of all types of microorganisms. The simultaneous determination of a large number

of enzyme assays may be more representative of overall microbial activity, but this approach is more labour intensive and difficult to interpret.

There are many different enzyme assays to detect either specific or general microbial activity in environmental samples. Some examples are given in Table 8.3. Some of the assays are also described below.

**Table 8.3:** Some of the most common enzyme assay procedures used to measure microbial activity.

| <i>Enzyme</i>     | <i>Substrate</i>                     | <i>Assay theory and procedure</i>                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dehydrogenase     | Triphenyltetrazolium                 | Dehydrogenases convert triphenyltetrazolium chloride to triphenylformazan; the latter is extracted with methanol and quantitated spectrophotometrically.                                                                                                                                                                                                                                    |
| Phosphatase       | <i>p</i> -Nitrophenol phosphate      | Phosphatases convert the <i>p</i> -nitrophenol phosphate to <i>p</i> -nitrophenol, which is extracted in aqueous solution and quantitated spectrophotometrically.                                                                                                                                                                                                                           |
|                   | Methylumbeliferone phosphate (MUFP)  | Phosphatases convert the MUFP to MUF, which is extracted in aqueous solution and quantitated spectrofluorimetrically.                                                                                                                                                                                                                                                                       |
| Protease          | Gelatin                              | Gelatin hydrolysis, as an example of proteolytic activity, can be measured by the determination of residual protein.                                                                                                                                                                                                                                                                        |
| Amylase           | Starch                               | The amount of residual starch is quantitated spectrophotometrically by the intensity of the blue color resulting from its reaction with iodine.                                                                                                                                                                                                                                             |
| Chitinase         | Chitin                               | Production of reducing sugars is measured using anthrone reagent.                                                                                                                                                                                                                                                                                                                           |
| Cellulase         | Cellulose<br>Carboxymethyl-cellulose | Cellulases alter the viscosity of carboxymethyl-cellulose, a quantity that can be measured.                                                                                                                                                                                                                                                                                                 |
| Nitrogenase       | Acetylene                            | Nitrogenase, besides reducing dinitrogen gas (N <sub>2</sub> ) to ammonia (NH <sub>3</sub> ), is also capable of reducing acetylene (C <sub>2</sub> H <sub>2</sub> ) to ethylene (C <sub>2</sub> H <sub>4</sub> ); the rate of formation of ethylene can be monitored using a gas chromatograph and the rate of nitrogen fixation can be calculated using an appropriate conversion factor. |
| Nitrate reductase | Nitrate                              | Dissimilatory nitrate reductase can be assayed by the disappearance of nitrate or by measuring with a gas chromatograph the evolution of denitrification products, such as nitrogen gas and nitrous oxide, from samples; denitrification can be blocked at the nitrous oxide level by the addition of acetylene, permitting a simpler assay procedure.                                      |

**Table 8.3:** (Contd.)

|        |      |                                                                                                                                                                                                                          |
|--------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATPase | ATP  | ATPase activity is measured by the quantification of the phosphate produced by enzyme action. The amount of phosphate produced can be measured spectrophotometrically by using ammonium molybdate and stannous chloride. |
| Urease | Urea | Urease cause deamination of urea in the presence of a high concentration of $MgSO_4$ . The amount of ammonium produced is quantified by titration against NaOH to convert ammonium to ammonia.                           |

Modified from Atlas and Bartha (1993).

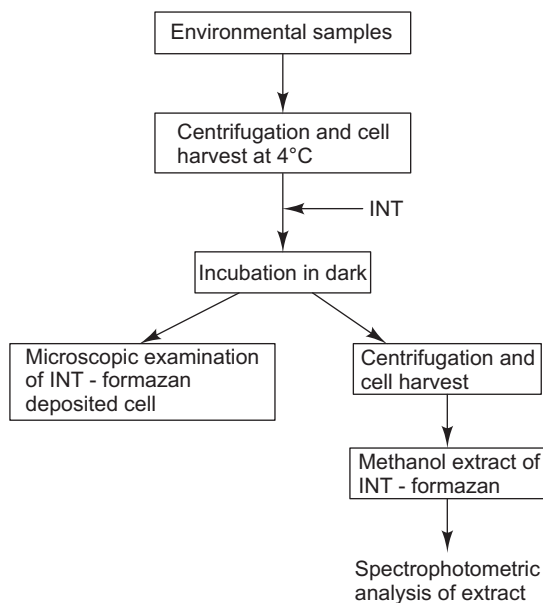
### 8.6.1 Assay of Dehydrogenase

Dehydrogenases are intracellular enzymes that catalyze oxidation-reduction reactions required for the respiration of organic compounds. Because dehydrogenases are inactive when outside the cell, this assay is considered a measure of microbial activity. Dehydrogenase reactions can be detected using a water soluble, almost colourless, tetrazolium salt, which when reduced, forms a reddish coloured formazan product that can be detected in a variety of ways. Tetrazolium salts compete with other electron acceptors for the reducing power of the electron transport chain. Thus, measurements of the reduction of tetrazolium salts will reflect electron transport chain activity. Hence, this measurement is an index of the general level of activity of a large part of the microbial community, but it is not a direct measure of microbial growth in terms of the production of new biomass.

A commonly utilized tetrazolium salt is 2-(*p*-iodophenyl)-3-(*p*-nitrophenyl)-5-phenyltetrazolium chloride (INT), which is transformed into an intensely coloured, water-insoluble formazan (INT-formazan). INT has been used to measure microbial activity in surface waters, soil and sediment samples, subsurface sediments and biofilms. Environmental samples are suspended in a solution containing INT and incubated for a matter of hours (1 – 12 hours). The production of INT-formazan can be detected by microscopic examination of the red INT-formazan deposits that form within the cells or by quantifying total INT-formazan production spectrophotometrically. Microscopic examination gives an indication of the physiological status of the microbial population by determining the percentage of total cells that are actively respiring. Total cell number can then be determined using a counterstain, such as acridine orange, which stains all cells (Fig. 8.13). The difference between the INT-formazan-containing cells and the total cell count is the proportion of the population that is metabolically active. This method is very sensitive and can be used to detect electron transport chain activity even when low numbers of microorganisms are present or when samples are incubated at low temperatures. However, when using this method, it should



be recognized that some of the cells considered inactive may in fact be respiring very slowly or not utilizing INT as an electron acceptor.



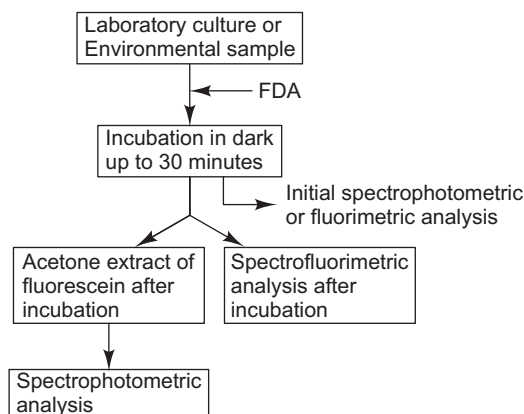
**Figure 8.13** - Schematic representation of the measurement of nitrate reductase activity using 2-(p-iodophenyl)-3-(p-nitrophenyl)-5-phenyltetrazolium chloride (INT) as chromogenic substrate.

Spectrophotometric detection of INT is done after extraction of INT-formazan from cells using methanol. The quantification of INT can be made by comparing the data with a standard curve. In terms of quantifying total INT-formazan production, this measurement provides a relative index by which electron transport chain activities in different samples can be compared. However, INT has a low efficiency as an electron acceptor and formazan production can be affected by numerous factors including the concentration of INT used, incubation time, incubation temperature, pH of the sample and whether the sample was incubated under aerobic or anaerobic conditions. Therefore, a direct comparison of activities of different samples can be made only if identical methods and experimental conditions are used.

## 8.6.2 Assay of Esterase

Another enzyme assay system used to reflect general or total microbial activity in environmental samples is based on the hydrolysis of fluorescein diacetate (FDA) by several extracellular hydrolyzing enzymes including esterases, proteases and lipases. The product of FDA hydrolysis is the highly fluorescent molecule fluorescein, which is water soluble and can be detected spectrophotometrically or

fluorimetrically. FDA hydrolysis has been used to examine microbial activity in soil, subsurface samples and activated sewage sludge. The general procedure is to incubate an environmental sample with FDA in a buffer solution for a matter of minutes to hours. The supernatant is then separated from the cells or environmental debris by centrifugation and the fluorescein production is quantified. For spectrofluimetric analysis the samples are excited with an excitation monochromator at 485 nm after incubation of the environmental sample or the assay mixture and the fluorescence is detected between 520 – 530 nm (Fig. 8.14). The fluorescence values can be calculated through a standard curve prepared by taking fluorescein disodium salt.



**Figure 8.14** - Schematic representation of the steps for measurement of esterase activity using fluorescein diacetate (FDA) as fluorogenic substrate.

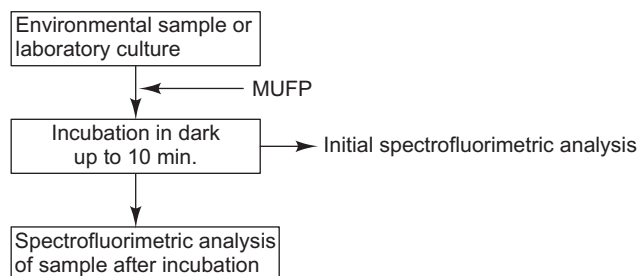
Fluorescein can also be extracted from the sample using acetone to improve detection. Interpretation of the results requires careful consideration because the enzymes involved in FDA hydrolysis can be intracellular or extracellular, which means that the assay determines enzymatic potential and not necessarily the activity of proliferating microorganisms. Several studies appear to validate the use of FDA by correlating fluorescein production with other measures of microbial activity or biomass. FDA hydrolysis usually correlates well with biomass, indicating a direct relationship with microbial activity.

### 8.6.3 Assay of Phosphatase

Phosphatases are membrane bound enzymes whose activities determine a number of membrane properties, both structural and functional. Both alkaline and acid phosphatases are used as marker enzymes to study the integrity of cellular membrane. The lipid packing density and fluidity of isolated membranes and intact cells are clearly marked by the pattern of activities of phosphatases. Because of their high stability, like other esterases, phosphatase activity is easy to monitor at room temperature as there is hardly any need to extract the enzyme in pure form for measuring its activity.

Both chromogenic and fluorogenic substrates are used for measurement of phosphatase activity. Bretauiere and Spillman (1981) have proposed a spectrophotometric method for analysis of alkaline phosphatase (EC. 3.1.3.1) activity. For measuring the activity of natural samples or of laboratory grown cultures, nitrophenylphosphate (NPP) is used as the chromogenic substrate, which on hydrolysis, produces *p*-nitrophenol. To the cell homogenate, aminomethylpropanol (AMP) is added to prepare the assay mixture. After incubation the cell free supernatant is spiked with NPP and the activity is measured spectrophotometrically for a period of up to 5 minutes by monitoring the absorbance changes at 405 nm. The absorbance values can be quantified to enzyme activity (as  $\mu\text{M}$  *p*-nitrophenol formed/ mg protein. min) using a standard curve prepared by taking *p*- nitrophenol.

Methylumbeliferone phosphate (MUFP) is a common fluorogenic substrate used for spectrofluorimetric analysis of phosphatase activity (Fig. 8.15). MUFP is added to the homogenized assay mixture at a concentration that differ with the sample and is determined by the cell density. After incubation, the production of the fluorogenic product methylumbeliferone (MUF) is measured by exciting the sample at 320 nm and then measuring the fluorescence emission at 450 nm. The excitation and emission band passes are determined by the sample type and incubation duration.

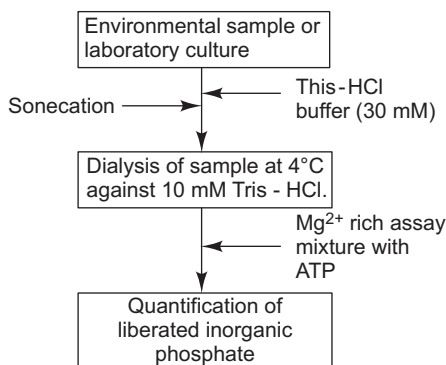


**Figure 8.15** - Schematic representation of the procedure for measurement of phosphatase activity using methyl umbeliferone phosphate (MUFP) as the fluorogenic substrate.

#### 8.6.4 Assay of ATPase

ATPase activity acts as an indicator of the metabolic state of the cells. In environmental samples, measurement of ATPase activity provides information on the metabolic efficiency of the sample to use energy rich substrates and the ability of the microbial assemblage to utilize the available energy sources. Unlike phosphatases and esterases, measurement of ATPase activity requires careful extraction and activity measurement. The cells from the culture or from the collected outdoor samples are first harvested by centrifugation and the pellet is suspended in tris-HCl buffer (30 mM) (Fig. 8.16). The sample, after a short incubation in the buffer, is sonicated for breaking the membranes and is then dialyzed against the same buffer to collect the preparation. A known amount of

ATP is added to the preparation at the presence of a high concentration of  $\text{MgCl}_2$ . A time dependent production of phosphate is measured continuously for about 10 minutes.



**Figure 8.16** - Schematic representation of steps for estimation of ATP-ase activity of samples through hydrolysis of ATP.

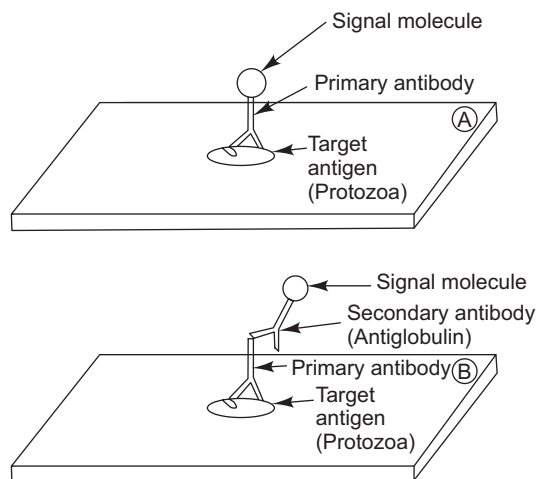
## 8.7 IMMUNOASSAYS

Immunoassays are analytical methods used for the detection and/or quantitation of the antigen-antibody interaction. For the most part, the types of immunoassays used in environmental microbiology are based on quantitation or detection of antigens as opposed to characterization of the antigens. In this case only the quantity of antigen in an environmental sample is considered without making any specific characterization. However, in order to quantitate or detect the antigen there must be a way to visualize the antigen-antibody interaction. This visual signal is produced by the attachment of specific signal molecules to the antibodies or antiglobulins used to detect the antigen within an environmental sample.

For almost all types of immunoassays, attachment of a signal molecule to the antibody and/or antigen is very important. Many types of signal molecules are used in immunoassays, including iodine, enzymes, fluorochromes and radioisotopes. These signal molecules produce a visual signal that allows quantitation of the specific antibody-antigen interaction being investigated. The signal is usually indicated by the production of some type of colour change. For example, enzymes such as horseradish peroxidase and alkaline phosphatase act by enzymatically cleaving colourless substrates to produce a coloured product. This signal is then detected qualitatively by the naked eye or quantitatively with the help of a spectrophotometer. For fluorochrome signal molecules, the antigen-antibody interaction can be detected by exciting the fluorochrome with a particular wavelength of light and then measuring the fluorescence emission. The fluorescent dye most commonly used as an antibody label is fluorescein isothiocyanate (FITC). Other examples of fluorescing chemicals used in this type of assay includes R-phycoerythrin, rhodamine and Texas red. Radioisotopes are quantitated by liquid

scintillation counting or by exposing the sample to a photographic emulsion (x-ray film), which produces a qualitative signal on the film that can be visualized as a dark spot.

The attachment of an antibody or antigen to a signal molecule is an almost universal way to allow for visual detection via immunoassays. There are also two universal formats for immunoassays. These are termed direct labeling and sandwich (often termed indirect) labeling (Fig. 8.17). Essentially with direct labeling, the primary antibody (antibody specific for the target) has the signal molecule attached to it allowing for one-step detection. Sandwich or indirect labeling involves two steps. The first is the attachment of a primary antibody to the target and the second is the attachment of a secondary (antiglobulin) antibody to the primary antibody. In indirect labeling, the secondary antibody has the signal molecule attached. Both methods work well though both have advantages and disadvantages. With direct labeling, the binding and signal are usually more specific because there is a smaller signal to background noise ratio. However, the use of sandwich labeling allows for one labeled antibody to be used with many different primary antibodies provided they are all of the same type, meaning that each primary antibody does not have to be labeled separately. For instance, if we use a mouse to produce monoclonal IgG primary antibodies against 4 different protozoa (*Microsporidia*, *Giardia*, *Cryptosporidium* and *Entamoeba*), then we can use the same labeled antiglobulin to bind to each of these. Because conjugating signal molecules to antibodies is tedious and often difficult and because a wide range of antiglobulins conjugated to various signal molecules are available commercially, this may be the format of choice. However, if we desire a one-step assay that is slightly more specific, then the use of direct labeling is often preferred.



**Figure 8.17** - The detection of target pathogens by (A) direct and (B) sandwich immunoassay.

Detailed description of various immunoassays used in the detection and examination of environmental samples has not been considered in this book. Only few important immunoassay methods are presented here.

### 8.7.1 Fluorescent Immunolabeling

Fluorescent immunolabeling (immunofluorescence) is the use of fluorescent signal molecules conjugated to antibodies to interact with and subsequently indicate the presence of a particular antigen by the production of fluorescent light. The basic procedure for immunofluorescence microscopy is to attach the sample antigen to a microscope slide. A fluorescent chemical/antibody is added to conjugate with the specific antigen and the sample is viewed under a microscope equipped with a fluorescent light source. When viewing fluorescence-labeled samples under the microscope/ the labeled antigen appears bright green against a dark background. Immunofluorescence microscopy is one of the easiest and most widely used immunoassays.

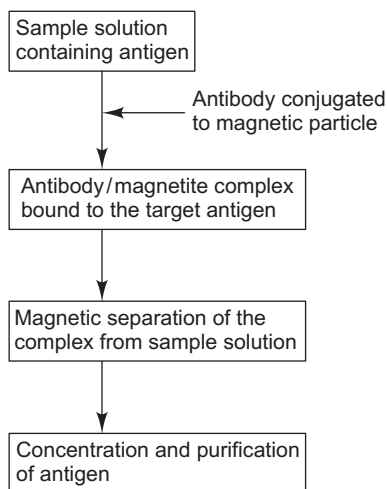
Immunofluorescence microscopy was first used in environmental microbiology for the study of Rhizobia in soil. Another important use of immunofluorescence is in the detection of protozoan parasites in water. Different fluorescein-labeled monoclonal antibodies, specific for the cyst and oocyst walls of *Giardia* sp. and *Cryptosporidium* sp., respectively are used to determine whether they passed live cysts or oocysts out in their feces. The same fluorescein-labeled monoclonal antibodies are also used to screen the feces to see whether intact and/or viable cysts or oocysts are present. Similarly fluorescein-labeled antibodies are used to stain the internal organs of the microarthropods to determine whether they have cysts or oocysts within their bodies. It may be noted that only like cysts or oocysts take and retain the labeled antibodies.

### 8.7.2 Immunomagnetic Separation Assays

Magnetic immunoseparation is an antigen capture methodology that uses antibodies conjugated to paramagnetic beads to attach to, concentrate and purify antigens. Immunomagnetic separation is rapidly becoming one of the more popular approaches to the specific manipulation of microorganisms, proteins and nucleic acids. The increase in popularity of this approach is due to its ease, mild conditions, low cost and the ability to automate the process. In its most simplistic form, an immunomagnetic separation is accomplished with antibody-coated magnetite beads and a magnet. Essentially, the antibodies coated on the magnetic beads bind with antigens in solution and are then separated from the solution using a magnet (Fig.8.18). The beads used for magnetic separation are small, ranging from 100 nm to 20  $\mu\text{m}$  and are typically made of iron oxide (magnetite). Such particles react strongly in magnetic field but do not retain any magnetism when the magnetic field is removed.

This method has been used to recover thermophilic sulphate-reducing bacteria from oil field waters below oil production platforms in the North Sea (Christensen *et al.*, 1992). These bacteria can proliferate in oil field waters and cause considerable problems for oil companies during oil recovery. So the ability to detect

their presence is of great value to the petroleum industry. In this case, immunobeads specific against cell wall antigens of the thermophilic *Thermodesulfobacterium mobile* captured several different isolates from oil-containing strata. Only one of the bacteria isolated by this method was serologically and morphologically identical to the bacterium (*T. mobile*) for which the antibodies were designed. Two others species of bacteria isolated using the immunomagnetic beads were spore-forming and similar to *Desulfotomacillum* sp., a sulphate-reducing bacterium that had previously been isolated from oil fields.



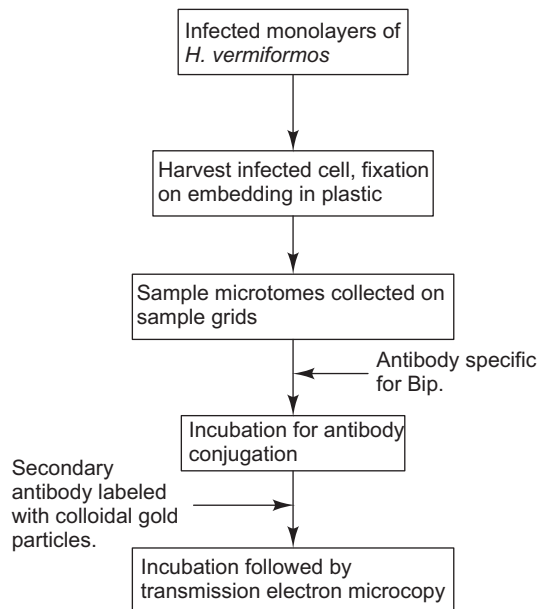
**Figure 8.18** - Schematic representation of the steps for immunomagnetic separation of antigen.

### 8.7.3 Immunocytochemical Assays

An immunocytochemical assay is used for the detection and determination of the cellular localization of target antigens. The purpose of an immunocytochemical assay is to determine where target antigens are localized within a particular cell. For instance, you can determine whether the target antigen (a particular protein for example) is localized in the cytoplasm or on the cell surface. In many cases, light microscopes are used to determine the location of antigens within a eukaryotic cell. However, immunocytochemical assays more often involve the use of electron microscopes to increase resolution and magnification of the area being studied. Many types of antibody labels can be used for electron microscopy, including heavy metals such as colloidal gold, enzymes such as horseradish peroxidase and proteins such as ferritin. These labels are visualized by the electron microscope as electron dense regions. The process involved in the preparation of a sample for immunocytological analysis is complicated but can be summarized as follows: The sample is fixed with a preservative to maintain the original localization and antigenicity of the target. The sample is then embedded in plastic, sectioned, placed on an electron microscope grid, immunolabeled, poststained and finally viewed under the electron microscope.



An example of the use of immunocytochemical techniques in environmental microbiology is in the study of the parasitism of certain amoeba by *Legionella pneumophila*. The later is an intracellular parasite and is the causative agent of legionnaires' disease. This bacterium is pervasive in aquatic environments and has been shown to parasitize and multiply within certain protozoa. The sequence of events in the intracellular infection of the amoeba *Hartmannella vermiformis* by *L. pneumophila* was examined by Kwaik (1996) using an immunocytochemical assay. The goal of his study was to compare the intracellular infection of the amoebae with the infection of human alveolar macrophages that occurs during the onset of legionnaires' pneumonia to aid in understanding the environmental life cycle of this human pathogen. Specifically, the aim was to determine whether the accumulation of ribosomes and rough endoplasmic reticulum (RER) around amoebic phagosomes containing *L. pneumophila* was similar to that observed in human phagosomes. To accomplish this, monolayers of *H. vermiformis* amoeba cells were infected with *L. pneumophila*. The infected cells were harvested, fixed and embedded in plastic. The samples were then thin sectioned, collected on sample grids and incubated with an antibody specific for Bip, a heat shock protein associated with RER. This was followed by secondary incubation with an antiglobulin labeled with colloidal gold particles (indirect immunoassay) and examination by transmission electron microscopy (Fig 8.19). The results demonstrated that there is considerable similarity in the ultrastructure of phagosomes containing *L. pneumophila* in both amoeba and humans. Using immunocytochemical assay it was also found that similar to the human system,



**Figure 8.19** - Schematic representation of immunocytochemical techniques to study the parasite infection in *Hartmannella vermiformis*.

RER-specific protein (Bip) was present in the phagosomal membrane of the amoebae. Further, the localization of the RER and ribosomes in the amoebic phagosome was identical to the localization seen in human macrophage phagosomes. This study helped to indicate a possible role of the RER in the protection and growth of the *L. pneumophila* within the phagosomes of both hosts.

The major advantage of immunocytochemical assay is the high resolution that can be achieved using electron microscopy. However, this technique is very specialized and requires a great deal of training for sample preparation and performing the microscopy. There is also a great deal of expense involved, compared to other immunology-based assays, but the information gathered is often very important in elucidation of actual cellular processes such as the example described, infection of a eukaryotic cell by a disease-causing bacterium.

## 8.8 THE ANALYSIS OF METAGENOME

To analyze the diversity of pollutant-degrading population in the environment, fragments of genes for catabolic enzymes in degradation pathways/16S rDNA have been amplified from environmental DNA samples using polymerase chain reaction (PCR). However, this approach requires prior information of the gene sequences for designing suitable primers that can indicate the presence of the gene that is searched for. Handelsman and Goodman developed a broad but still a more informative process of taking the environmental sample and extracting the entire DNA from it. The extract was then spliced by the restriction endonucleases and gel electrophoresed for the detection of the target gene. They referred to this collective genomes of soil microflora as constituting a “metagenome” and metagenomics circumvents the limitation of the requirement of known probe and sequences since it involves direct cloning of soil metagenomic DNA (Lorenz *et al.*, 2002). This approach has been successfully utilized for accessing the biosynthetic diversity of microorganisms from the environment by acquiring operons or gene(s) involved in synthesis of beneficial compounds or in biodegradation of recalcitrant pollutants. Although the genes encoding the enzymes of a degradation pathway are usually clustered in a contiguous piece of DNA, large fragments of DNA are usually cloned in bacterial artificial chromosome (BAC), fosmid or cosmid vectors such that the entire metagenomic library is represented by a finite number of screenable clones. The advantage of using BAC vectors for cloning is that they maintain very large inserts (more than 100 kb) stably in *Escherichia coli*. Moreover, the long-term goal of metagenomic analysis is to reconstruct the genomes of uncultured organisms by identifying overlapping fragments in metagenomic library clones to assemble each chromosome (Watanabe *et al.*, 2002). In case of expression-cloning approach, DNA is isolated directly from an environmental sample, digested and cloned into a high-copy-number plasmid expression vector and subsequently the clones are screened. This approach has been used to isolate novel enzymes (e.g. polyketide synthase), biotechnologically relevant gene products such as lipases, cellulases, amylases, chitinases, esterases, as well as degradative enzymes involved in degradation

**Table 8.4:** Degradative enzymes and bioactive compounds obtained from metagenomic DNA libraries.

| <i>Target enzyme</i>            | <i>Screening strategy</i>      | <i>Vector used for library construction</i> |
|---------------------------------|--------------------------------|---------------------------------------------|
| Amylase                         | Activity based, sequence based | BAC, Cosmid                                 |
| Glycerol/diol dehydratase       | Activity based, sequence based | Plasmid                                     |
| Indirubin                       | Activity based                 | BAC                                         |
| Polyketide synthase             | Sequence based                 | Cosmid                                      |
| 4-Hydroxybutyrate dehydrogenase | Activity based                 | Plasmid                                     |
| Biotin synthase                 | Activity based                 | Cosmid                                      |
| Turbomycin synthesis            | Activity based                 | BAC                                         |

Compiled from findings of Henne *et al.* (1999); Rondon *et al.* (2000); Osburne *et al.* (2000); Entcheva *et al.* (2001); Gillespie (2002); Voget *et al.* (2003); Kinetch *et al.* (2003); Courtois *et al.* (2003).

pathways, from environmental sources without prior culturing of the producing organisms (Table 8.4).

Culture-independent microbial diversity analysis methods revealed presence of vast diversity of yet uncharacterized bacteria but most of these methods fail to indicate the physiology of these “unculturable” bacteria and their role in ecosystem. Genomic information of the “unculturable” bacteria can be obtained from the clones in metagenomic libraries, which can help in understanding their physiology and also their role in ecosystem. Some success has been obtained in achieving these goals in recent years, which includes sequence snapshots of rRNA genes containing BAC clones to large scale sequencing of metagenomes of acidophilic biofilms and Sargasso Sea.

The common approaches used for the recognition of uncultivable microorganisms are the detection and analysis of 16S rDNA regions amplified by PCR with primers constructed from consensus sequences. It is known that some regions in the 16S rDNA sequence are conserved among all microorganisms characterized and the conserved sequences specific to bacterial 16S rDNA and specific to archaea are also detected. The nucleotide sequence in these conserved regions is usually used for the design of primers, which are capable of amplifying DNA fragments of 16S rDNA region from the targeted microorganisms. This approach to the microbial community in the environments has been performed by many microbiologists and researchers for other fields. The sequence information of 16S rDNA region of microorganisms, including the uncultivable microorganisms, has been obtained by PCR amplification using these primer sets. The success of this approach used by many investigators over the last 10 years has resulted in the accumulation of a large number of 16S rDNA sequences in public DNA databases.

Previous approaches in metagenomic studies were relatively limited to the environment where our access was easy. In contrast, recent interest for microbial researchers has moved on to more extreme environments where collection of usable samples is practically difficult. Such extreme environments include thermal

vents, deep sea, volcanos and hot springs. Inagaki and his coworkers showed the discovery of vertically shifted community structures of Archaea in a typical oceanic subseafloor core sample recovered from the West Philippine Basin, using a sequencing technique of 16S rDNA amplified from DNA isolated from core samples (Inagaki *et al.*, 2001). Takai *et al.* (2004) showed the spatial distribution of marine crenarchaeota group in the waters surrounding two geographically and geologically distinct hydrothermal systems by analysis of 16S rDNA clones from a microbial community in the hydrothermal environments. These studies have shown that metagenomic approach with sophisticated sampling procedures can be very handy in getting molecular and metabolic information on the microbial assemblage of extreme habitats. Since most of the extremophiles are difficult to maintain in the laboratory and most extremophilic habitats are unique with respect to their microbial community, metagenomic analysis is hitherto the best available technology to obtain molecular signature of these habitats.

Metagenomic DNA from various habitats has been exploited by PCR-based or metagenomic library-based methods in last 5-6 years for discovering novel biocatalysts or bioactive compounds. Some of these biocatalysts have also been characterized for their various industrially important properties such as substrate specificity. Many attempts have been done to prepare large insert libraries to identify novel drug molecules.

## 8.9 CONCLUSION

Different types of microbial activity measurements in both pure culture and environmental samples are useful to assess the status of the health of the microbial community and their metabolic efficiency. Although the activity measurement of pure cultures reared in the laboratory is simple, environmental community contains a wide variety of microorganisms with different rates and types of metabolic activity. However, the recent advances in the microbiological methods have made it possible to make the qualitative and quantitative estimation of the microbial community of the habitats. The measurement of respiration, rate of substrate utilization, increase of biomass and production of specific metabolites have become the basic requirement for the assessment of the microbial activity of a habitat. However, such methods are less explanatory in comparison to more accurate but sophisticated methods like the analysis of specific enzyme activity, immunoassays and the nucleic acid assay. The progress in these novel fields of microbiology and biotechnology will bring new insight to the microbial activities and health of an ecosystem, which we may not be now in a position to imagine.

In the recently progressed genomic approaches mentioned in this chapter, the DNAs utilized for construction of shotgun libraries have been obtained from the samples collected from habitats from where the sample collection was not difficult. It is relatively easy to obtain a large amount of samples from the environments where access is easy but there is still a shortage of technology and equipment to collect samples from the extreme environments. In future it is expected that the novel genes and the activities discovered from these

environment will continue to grow for their potential applications in industry. Accordingly, the collection of environmental clones and information will become more important. The genes and functions discovered from the extreme environments will be most convenient materials applicable to the industry, because of their absolute stability. Development of equipment for collection of uncharacterized microorganisms from the extreme environment will provide opportunity for expansion of environmental genomics.

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# ECOLOGICAL RELATIONSHIPS AMONG MICROORGANISMS

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## 9.1 INTRODUCTION

Microorganisms are of major importance in the environment for their role in agriculture, aquaculture, waste treatment and nutrient cycling. They reside in each and every corner of the environment, often as an assemblage and do not like to remain as a pure culture, excepting in the laboratory. For degradation of any substance in the environment, there is a need of cooperation of many microbes to effectively convert the pollutants to nutrients. Thus microbial cooperation is a very common phenomenon in nature, which occurs among individuals of a single population, among different microbial populations, between microbes and plants and between microbes and animals.

Microbes commonly interact in positive and negative ways. Neutralism is a rare and perhaps only theoretical phenomenon as the relationship of one microbe with another microbe or with a metazoan is established with a purpose. Positive interactions among microbes are often termed as cooperation or syntrophy while with plants and animals such interaction is either symbiotic or mutualistic. The negative interaction among microbial populations is termed antagonism while with plants and animals it is either a parasitism or pathogenicity. The microbial populations often draw their nutrient demands more efficiently by their cooperative activity in the environment and therefore the relationships have immense ecological importance. This chapter deals with some of such relationships. The symbiotic relationships have been separately discussed in detail in the next chapter (chapter 10).

## **9.2 RELATIONSHIPS AMONG MICROBIAL POPULATIONS**

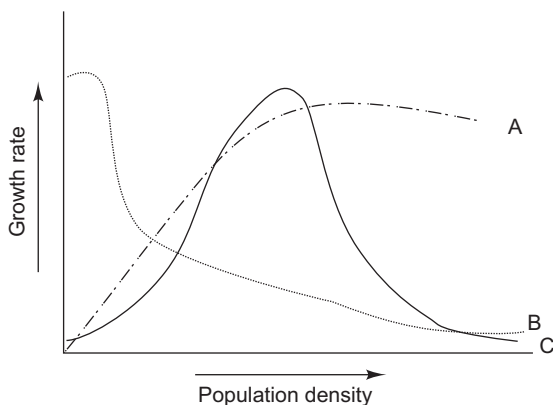
Within a single population, microbes cooperate at low (no resource limitation) and compete at high (resource limitation) cell densities. One population may benefit another one in a one-sided commensal way or two different populations may interact synergistically depending on the ecological conditions and resource availability. Such beneficial interactions are facilitated by close physical proximity, as in biofilms and flocs. Mutualism is a strong, specific, beneficial interaction essential for the survival of both partners. Competition between diverse populations for resources is keen in nature and may be sharpened by amensalism. Parasitism and predation benefit their perpetrators while damaging or destroying the host or prey, respectively. Within a community, positive interactions tend to optimize the utilization of the available resources and negative interactions act as feedback controls and prevent a complete exhaustion of resources and consequent population crashes.

### **9.2.1 Interactions within a Single Microbial Population**

Depending on the population density and growth pattern, both positive and negative interactions may occur even within a single population. Such interactions, originally observed in animals and plants, are interrelated. Under standard conditions, positive interactions increase the growth rate of a population, whereas negative interactions have the opposite effect. With increasing population density, positive interactions theoretically increase the growth rate to some asymptotic limit. In contrast, negative interactions decrease growth rate as the population density increases. An interaction can initially be positive but with the increase in the population density and limitation of resources this can be changed to a negative one. As a result, there is an optimal population density for maximal growth rate, which varies with the environmental conditions. Growth rates below the optimal density are strongly affected by positive interactions but growth rates above the optimal population density are strongly influenced by negative interactions (Fig. 9.1).

**(a) Microbial cooperation:** Microbes mostly prefer positive interactions within a population, especially when there is no competition for resources. Such interactions are positive interactions and are also called cooperation. In a single microbial population this is evidenced by an extended lag period or a complete failure of growth when a very small inoculum is used in routine culture transfer procedures. This is especially true for fastidious microorganisms (those having complex physiological requirements for growth) and is a major obstacle to all isolation procedures that require the growth of single microbial cells into colonies. Populations of intermediate density are generally more successful than individual organisms in the colonization of natural habitats. Thus there is a minimum inoculum size for each microbial species for initiating cooperation.





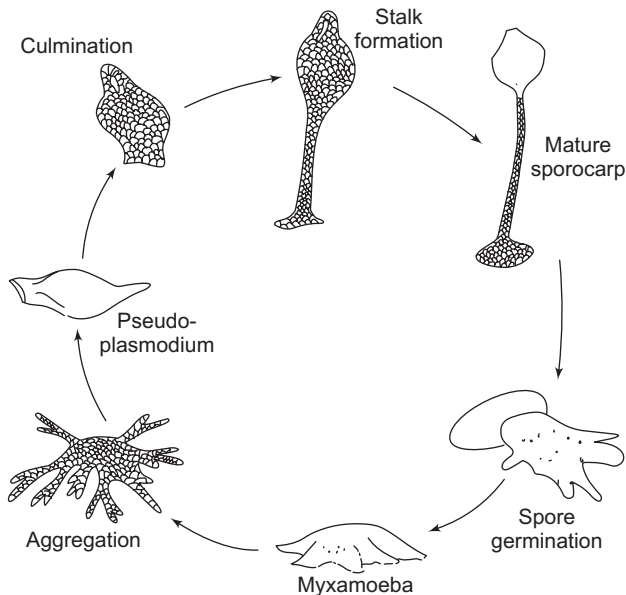
**Figure 9.1** - The growth rate of a single microbial population showing (A) positive, (B) negative and (C) combined interactions.

A well-studied example of this is the “minimum infectious dose” of pathogenic microbial populations. Usually thousands of pathogens are required to cause a disease, as a single pathogen is rarely able to overcome host defenses. Cooperation occurs within the population because the semipermeable cell membranes of microorganisms are imperfect and tend to “leak” low-molecular-weight metabolic intermediates that are essential for biosynthesis and growth. In a population, a significant extracellular concentration of these metabolites is established that counteracts further loss and facilitates reabsorption. For a single cell or at a very low density, however, population losses may exceed replacement rates and prevent growth. A substantial inoculum, i.e. a sufficiently large population, is able to adjust to an initially unfavourable culture medium redox potential, whereas a single cell or very small population may be unable to do so.

The cooperative interactions within a population facilitates colony formation in bacteria. Even motile bacteria that could move away from each other often remain in colonies only to enjoy cooperativity. Microcolonies, rather than individual microorganisms, are normally observed adhering to particles in nature. The association of a population within a colony allows more efficient utilization of available resources. Some bacterial colonies even exhibit mass movement, occasionally rotating in place or migrating in a spiral path. Such coordinated migration probably obtains beneficial results for the microorganisms within the colony as they seek new sources of nutrients.

Elaborate cooperative interactions within a population are evident in the case of the slime mold *Dictyostelium* (Fig. 9.2). When food sources become limiting, the amoeboid cells of *Dictyostelium* swarm together to a central organism. This swarming action is in response to the chemical stimulus of cyclic AMP being released and it occurs in a pulsating wave motion as the stimulus to synthesize AMP is transmitted from proximal to distal cells. The cells unite to form a fruiting body and spores that subsequently disperse. Frequently, some spores reach habitats with a more abundant food supply via this mechanism, germinate and

resume an amoeboid life stage. Communication among members of this microbial population allows for cooperative searching and utilization of resources in the habitat. Similar cooperative interactions have been described for populations of myxobacteria (Dworkin, 1996).



**Figure 9.2** - Positive interaction among the cells of slime mold *Dictyostelium* for the formation of a single structure to withstand adverse conditions (Adopted from Clark and Steck, 1979).

For degradation of insoluble substrates such as lignin and cellulose, cooperative interactions within populations is particularly important. The production of extracellular enzymes by individual members of the population makes such substrates available for all members of the population. The rate of utilization of such substrates is significantly enhanced if the population increases. At high population the cells synthesize extracellular enzymes and liberate them to the medium thereby making it possible to build a high enzyme concentration outside the cells. This cause rapid hydrolysis of the complex substrates and make the soluble products available to cells. At low population densities, the soluble products liberated by such enzymes would be rapidly lost from the population by dilution, whereas at higher population densities, the resulting soluble products can be utilized with higher efficiency. In addition, organic acids produced by individuals within a population also can solubilize inorganic elements in certain habitats, such as soils or rock surfaces, making essential elements available for all members of a population.

Cooperation in a population can also function as a protective mechanism against hostile environmental factors. It is commonly observed in the laboratory that a concentration of a metabolic inhibitor has less effect on a dense cell

suspension than on a dilute one. Microbial populations within biofilms are of several orders of magnitude and more resistant to antimicrobial agents than suspended cells of the same organisms. In natural environments, with their exposures to ultraviolet light, high densities of populations probably shield some members from direct exposure, protecting them and allowing continued growth of the population. At high population densities, microorganisms are able to depress the freezing point of water, allowing the continued growth at temperatures that otherwise would preclude availability of liquid water.

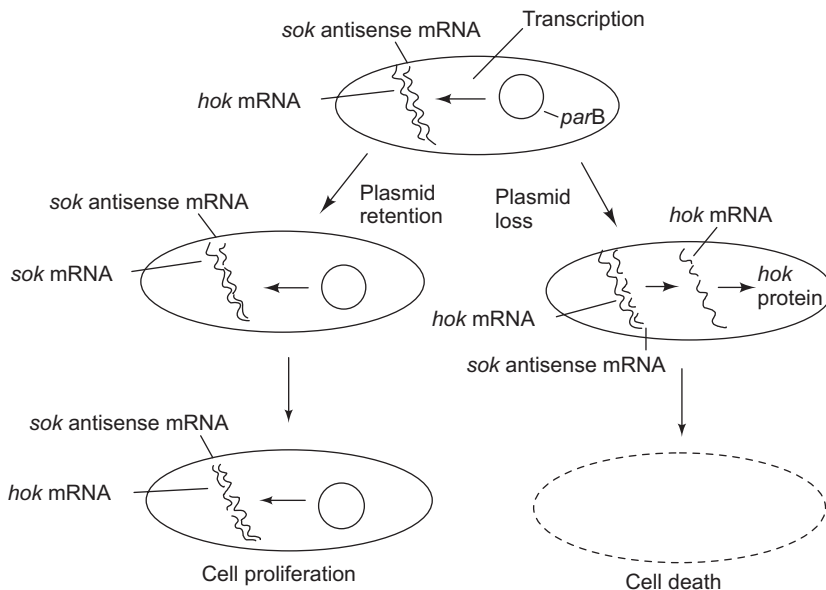
The exchange of genetic materials is another cooperative interaction that occurs within a population. Resistance to antibiotics and heavy metals and the ability to utilize unusual organic substrates are often genetically transmitted to other members of the population, thus making more and more cells able to utilize the resource or degrade the contaminant. This phenomenon allows the adaptive information that has arisen in one microorganism to be disseminated through a population. Genetic exchange is important in preventing overspecialization within a microbial population. It also facilitates the growth of the microbial assemblage on diverse substrates and in a variety of environmental conditions. Cooperative genetic exchange can occur through transformation, transduction, conjugation and sexual spore formation. Even though such genetic exchange events occur only between two members of the population, they usually require high population densities in order to increase the frequency of recombination. Bacterial populations, for example, generally require densities of greater than  $10^5/\text{ml}$  for a genetic exchange by conjugation; at lower population densities, the probability of a successful genetic exchange is very low and the process generally is not significant. Sometimes, however, aggregate formation at low population densities may facilitate genetic exchange.

**(b) Microbial competition:** Competitive interaction within a microbial population is a negative interaction, which mostly occurs under resource limited conditions. Because members of a microbial population use the same substrates and occupy the same ecological niche, resource limitation is a common phenomenon. In natural habitats with low concentrations of available substrates, increased population densities cause increased competition for the same resource. Within populations of predatory microorganisms (e.g. Protozoa) and parasites, competition occurs for prey and host cell surface, respectively.

In addition to the direct vying for good sources of food, competition in its broad sense includes actual competition for available substrates, as well as other negative interactions, such as those that result from the accumulation of toxic substances produced by members of the population. Within a high density population, metabolic products may accumulate to an inhibitory level (antagonism). For example, the accumulation of metabolic products such as low molecular weight fatty acids and hydrogen sulphide ( $\text{H}_2\text{S}$ ) constitutes a negative feedback mechanism limiting the further growth of some microbial populations even when substrate is still available. Examples of this type of competitive interaction are limitation of sulphate reduction by excessive  $\text{H}_2\text{S}$  accumulation and *Lactobacillus* growth

inhibition by lactic acid and other fatty acids accumulation. Fatty acid accumulation during hydrocarbon biodegradation, similarly can block further microbial metabolism of the hydrocarbon substrates and the accumulation of dichloroaniline from 3,4-dichloropropionanilide (propanil) catabolism can stop further growth of *Penicillium piscarium*.

An interesting genetic basis for negative interactions is the occurrence of genes that code for peptides or proteins that have lethal functions (Fig. 9.3). Strains of *Escherichia coli*, for example, have the *hok* (host-killing) gene, which codes for a peptide (Hok) that damages transmembrane potential and leads to death of cells when *hok* is expressed. To prevent *hok* expression, these *E. coli* contain a gene called *sok* (suppression of killing) that codes for an antisense mRNA to block the expression of *hok*. As long as the cell is making the mRNA encoded by *sok*, the cell is protected. However, in the absence of a more labile *sok*-encoded mRNA, the more stable *hok* mRNA is expressed and the cell is killed. Both *hok* and *sok* are encoded on the same plasmids and facilitates the survival of the population as long as the plasmids are retained. However, if the plasmids are lost, the cells of that population self-destruct. This mechanism ensures that the surviving cells retain the plasmids, i.e., it is a mechanism for plasmid survival. This may be important for the overall long-term benefit of the population since these plasmids also contain genes for antibiotic resistance that otherwise might be lost, but in the short term the population density is limited because some cells (those that lose the plasmids with *hok* and *sok*) kill themselves.



**Figure 9.3** - The negative interactions among the strains of *E. coli*. The diagram shows the expression of *hok*- and *sok*- genes. The mode of expression of the proteins has been explained in the text (Adopted with modification from Gerdes *et al.*, 1986).

### 9.3    INTERSPECIES MICROBIAL INTERACTIONS

Different microbial populations, even with different metabolite requirement, frequently interact with each other. The term *symbiosis* is used to denote any intimate relationship between two populations. In a global context, all symbiotic relationships can be viewed as beneficial because they act to maintain ecological balance. However, interactions among specific populations are often viewed in a less generic sense. The categories used to describe these interactions represent a conceptual classification system. Possible interactions between microbial populations can be recognized as negative interactions (competition and amensalism); positive interactions (commensalism, synergism and mutualism); or interactions that are positive for one but negative for the other population (parasitism and predation) (Table 9.1).

**Table 9.1:** Types of interactions between microbial populations.

| <i>Interactions</i>          | <i>Population A</i> | <i>Population B</i> |
|------------------------------|---------------------|---------------------|
| Neutralism                   | 0                   | 0                   |
| Commensalisms                | 0                   | +                   |
| Synergism (protocooperation) | +                   | +                   |
| Mutualism (symbiosis)        | +                   | +                   |
| Competition                  | –                   | –                   |
| Amensalism                   | 0 or +              | –                   |
| Predation                    | +                   | –                   |
| Parasitism                   | +                   | –                   |

Note: The effects shown are positive (+), negative (–), or neutral (0) (Adopted from Atlas and Bartha, 2005).

In communities, where the metabolic requirements of populations differ (simple communities), one or more of these interactions can be observed and studied. However, within a complex natural biological community, all these possible interactions can occur between different populations with variation of space and time. In established communities, positive interactions among indigenous populations are likely to be more developed than in newer communities. Invaders of established communities are likely to encounter severe negative interactions with the native populations. The negative interactions between populations act as feedback mechanisms that limit population densities and thereby maintain the balance in the flow of matter within the community. Positive interactions enhance the abilities of some populations to survive as part of the community within a particular habitat, where it could not have been possible without cooperation.

The establishment of positive interactions permits microorganisms to use available resources more efficiently and to occupy habitats that otherwise could not be inhabited separately by individual organism. Positive interactions between microbial populations are based on combined physical and metabolic capabilities that enhance growth and/or survival rates. Interactions between microbial populations also tend to reduce environmental stress. The negative interactions

represent self-regulation mechanisms that limit population densities. In the long run, such negative interactions benefit the species by preventing overpopulation, destruction of the habitat and extinction of species. Interactions between populations are also a driving force in the evolution of community structure.

### **9.3.1 Neutralism**

When two populations are not benefited from each other but tend to remain as a group, such interaction is termed as neutral interaction or neutralism. The latter cannot occur between populations having the same or overlapping functional roles within a community and is more likely between microbial populations with extremely different metabolic capabilities than between populations with similar capabilities. Because it is a negative proposition, it is difficult to demonstrate experimentally a total lack of interaction between two populations. Thus, examples for neutralism describe cases where interactions, if any, are of minimal importance.

Neutralism occurs between microbial populations that are spatially distant from each other. It is likely at low population densities where one microbial population does not sense the presence of another. Neutralism between microbial populations occurs in marine habitats and oligotrophic (low-nutrient) lake habitats where population densities are extremely low. In sediment or soil, it occurs when microbial populations occupy separate locations (microhabitats) on soil or sediment particles. However, physical separation alone does not ensure a relationship of neutralism. For example, a pathogenic microorganism may invade a plant root, resulting in the death of that plant and the destruction of the habitat of other microbial populations on the leaves of that plant. Even though there has been no direct contact between the two populations, the destruction of the leaf habitat by the root pathogen precludes a relationship of neutralism between the two populations.

Environmental conditions that do not permit active microbial growth favour neutralism. For example, microorganisms frozen in an ice matrix, such as in frozen food products, polar sea ice, or frozen freshwater lakes, are typically in a relationship of neutralism. Environmental conditions that favour the growth of microbial populations decrease the likelihood of neutralism. Neutralism often occurs between two microbial populations when both populations are outside their natural habitats and in environments where they cannot grow. For example, neutralism is probably the most common relationship between microbial populations in the atmosphere since all atmospheric microorganisms are allochthonous (non-indigenous) and do not naturally grow there. Resting stages of microorganisms are more likely to exhibit a relationship of neutralism with other microbial populations than with actively growing vegetative cells, mostly because of their metabolic dormancy. The low metabolic rates required for the maintenance of such resting structures do not force these organisms to competitive relationships for available resources with other microorganisms in the community. Production of spores, cysts and similar resting bodies during periods of environmental stress, such as elevated temperature or drought, allows microbial populations to maintain neutral interactions. In such situations, the only significant interaction is that with

the environmental stress, the strategy for survival is to avoid interpopulation interactions during these times. When environmental conditions become favourable, the resting stages of these organisms can germinate, forming vegetative cells, which may become cooperative or antagonistic.

Microbial populations at metabolically inactive stages are separated temporally and/or spatially within a habitat. This allows temporary coexistence of otherwise competitive populations. Some of these populations would be eliminated by competitive exclusion if such mechanisms do not exist. However, the formation of resting bodies does not ensure neutralism with other microbial populations. Some microbial populations produce enzymes that can degrade the resting stages of other microbial populations. However, many resting structures have more complex protective covering, which can withstand the enzymatic degradations.

### 9.3.2 Commensalism

A commensal relationship is a stronger positive relationship than neutralism, in which one population benefits while the other remains more or less unaffected. Such relationships are less obligatory keeping the ability of the individual microbial population to have independent existence in the absence of the other partner. Thus, commensalism is a unidirectional relationship between two populations. The unaffected population, by definition, is neither benefited from, nor is it negatively affected by, the actions of the second population. The recipient population may need the benefit provided by the unaffected population, but it may also be able to receive the necessary assistance from other populations with comparable metabolic capabilities.

Commensalism often results when the unaffected population, in the course of its normal growth and metabolism, modifies the habitat in such a way that another population benefits because the modified habitat is more suitable to its needs. For example, when a population of facultative anaerobes uses oxygen and lowers the oxygen concentration, it creates a habitat suitable for the growth of obligate anaerobes. In such a habitat, the obligate anaerobes benefit from the metabolic activities of the facultative organisms. The facultative organisms remain unaffected by the relationship as long as the two populations do not compete for the same substrates.

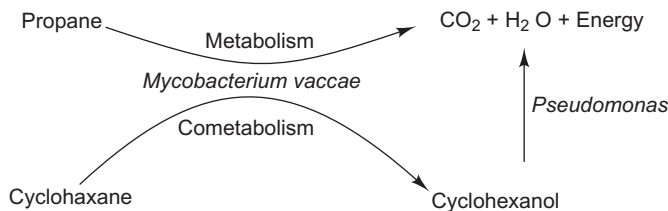
In many commensal relationships, one of the partners is benefited by the other through production of growth factors. Some microbial populations produce and excrete growth factors, such as vitamins and amino acids, which can be used by other microbial populations. For example, *Flavobacterium brevis* excretes cysteine, which *Legionella pneumophila* can use in aquatic habitats. In soil habitats, many bacteria depend upon the production of vitamins by other microbial populations to use these as growth supplements. As long as the growth factors are produced in excess and excreted from the neutral organism in the relationship, the two populations can have a commensal relationship. The production of growth factors and their excretion into the environment allow fastidious microbial populations to develop in the natural habitats.



The transformation of insoluble compounds to soluble compounds and the conversion of soluble compounds to gaseous compounds can also form the basis for a commensal relationship. The changes of state from solid to liquid and liquid to gas mobilize compounds so that they are moved to other habitats, where they can benefit other microbial populations. For example, methane produced by bacterial populations in sediment can benefit methane-oxidizing populations in the overlying water column. Methanogenesis may result from a commensal relationship for some populations but such interaction is often referred to as syntrophy. Under certain conditions, *Desulfovibrio* can supply *Methanobacterium* with acetate and hydrogen from anaerobic respiration and fermentation, using sulphate and lactate to generate these products. *Methanobacterium* can then use the products from *Desulfovibrio* to reduce carbon dioxide to methane. Other examples include the production of hydrogen sulphide in buried sediment layers that can be used by photoautotrophic sulphur bacteria on the sediment surface or within a water column and the transformation of soil-bound ammonia to nitrate by one bacterial population that allows the nitrogen to leach into the underlying soil column, where other soil microbial populations can then get their required nitrogen. The activities of one microbial population can also make a compound available to another population without actually transforming the particular compound. For example, acids produced by one microbial population may release compounds that are bound or inaccessible to the second population. Such desorption processes are probably common in soil, where many compounds can be bonded to mineral particles or humic materials.

In many commensal associations, the metabolic product of one population acts as substrates for other population, even though the population producing the product is not metabolically benefited from such activities. Some fungi enzymatically degrade complex polymeric compounds, such as cellulose, into simpler compounds, such as glucose, which are used by populations of other microorganisms that do not possess the enzymes needed to utilize the complex organic molecules. In some cases, a competitive relationship develops for the available simpler substrates. In other cases, the relationship may be truly commensal. If excess solubilized substrates would be lost for the first population by dilution in the environment, their use by a second population is not detrimental to the first one.

Commensal relationships are environmentally very meaningful for gratuitous degradation of many environmental contaminants by fungi and bacteria. Certain bacteria cometabolically oxidize a second substrate, while growing on a favoured substrate, to form the products that become sensitive to further degradation may be by the same or by a different microbe. However, according to a strict definition of cometabolism, the second substrate is not assimilated by the primary organism, but the oxidation products are available for use by other microbial populations. For example, *Mycobacterium vaccae* does not assimilate cyclohexane but cometabolize it to cyclohexanol while growing on propane. Further metabolic degradation of cyclohexanol is caused by *Pseudomonas* (Fig. 9.4).



**Figure 9.4** - Cometabolic degradation of cyclohexane in the presence of propane by commensal relationship between *Mycobacterium* and *Pseudomonas*.

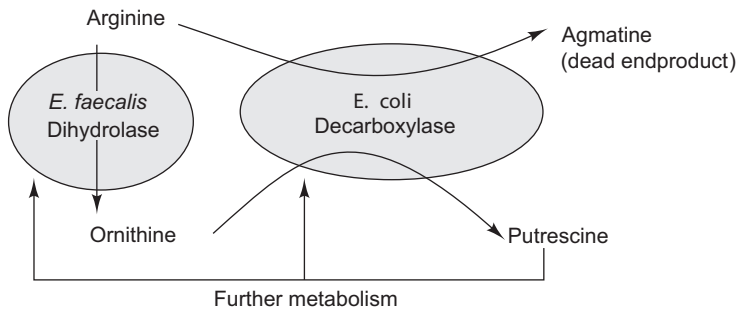
Commensal relationships are also responsible for removal or neutralization of a toxic material in the environment. The ability to destroy toxic factors is widespread in microbial communities. The oxidation of hydrogen sulphide by *Beggiatoa* is an example of detoxification that benefits  $H_2S$ -sensitive aerobic microbial populations. *Beggiatoa* is not known to benefit from its relationship with the second population. Precipitation of heavy metals, such as mercury, by sulphate reducers provides an additional example of detoxification. Production of volatile mercuric compounds by bacterial populations in aquatic habitats removes this toxic metal from the habitat. Some microbial populations are able to detoxify compounds by immobilization. For example, *Leptothrix* reduces manganese concentrations in some habitats, permitting the growth of other microbial populations, to which the higher manganese concentrations would be toxic.

In some cases, a microorganism may itself provide the suitable habitat that benefits a commensal partner. Bacteria are often observed on algal surfaces getting not only the benefit from the metabolic activities of the algae, but their association with a surface enhances intrapopulation cooperation. Such relationship may also be considered synergistic (or mutualistic) since this alga fails to grow normally in bacteria-free situations.

### 9.3.3 Synergism (Protocooperation)

Synergism is an association between two microbial populations, in which both populations benefit from the relationship, but unlike mutualism (See Chapter 10), the association is not an obligatory one. Both populations are capable of surviving in their natural environment on their own, although when initially formed, the association offers some mutual advantages. Synergistic relationships are also loose in the sense that there is less biochemical interaction between the partners and one member of the population is often easily replaced by another. In some cases, it is difficult to determine whether one of the populations is indeed benefiting and thus whether the relationship should be considered commensal or synergistic. In other cases, it is difficult to determine whether the relationship is obligatory and thus should be considered one of mutualism. Thus, protocooperation is a microbial interaction, which oscillates between commensalism and mutualism with time and metabolic relationships of participating microbes.

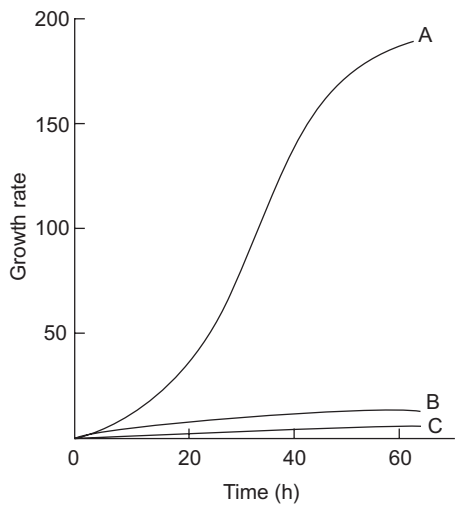
The term syntrophism is applied to the interaction of two or more populations that supply each other's nutritional needs (see chapters 11 and 12). It may allow microbial populations to perform activities, such as the synthesis of a product that neither population could perform alone. The synergistic activities of two microbial populations may allow completion of a metabolic pathway that otherwise could not be completed. A classic example of such syntrophism is exhibited by *Enterococcus faecalis* and *Escherichia coli* (Fig. 9.5). Neither organism alone is able to convert arginine to putrescine. *E. faecalis* is able to convert arginine to ornithine, which can then be utilized by the *E. coli* population to produce putrescine. *E. coli* alone is able to utilize arginine, producing agmatine, but cannot produce putrescine without assistance. Once putrescine is produced, both *E. coli* and *E. faecalis* can use it.



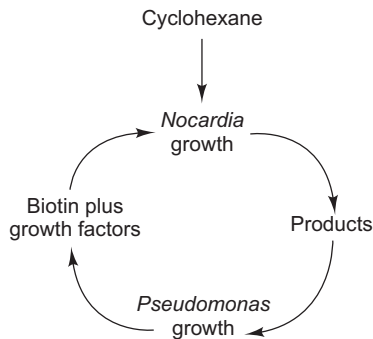
**Figure 9.5** - Synergistic relationship between *Enterococcus faecalis* and *Escherichia coli* in metabolic degradation of arginine.

Relationships of syntrophism are frequently based on the ability of one population to supply growth factors for another population. In a minimal medium, *Lactobacillus arabinosus* and *E. faecalis* are able to grow together but not alone (Fig. 9.6). The synergistic relationship is based on the fact that *E. faecalis* requires folic acid, which is produced by *Lactobacillus* and the latter requires phenylalanine, which is produced by *Enterococcus*. Together the organisms grow quite well. Similarly, cyclohexane can be degraded by a mixed population of a *Nocardia* species and a *Pseudomonas* species, but not by either population alone. The relationship is based on the ability of the *Nocardia* to metabolize cyclohexane, forming products that feed the *Pseudomonas* species. The *Pseudomonas* species produces biotin and growth factors required for the growth of the *Nocardia* species (Fig. 9.7).

Other interesting examples of synergism are based on the abilities of *Chlorobium* species to fix carbon dioxide and to oxidize hydrogen sulphide (Fig. 9.8). If there is an available source of hydrogen sulphide and carbon dioxide, *Chlorobium* is able to utilize light energy, producing organic matter. Given a supply of elemental sulphur and formate, *Spirillum* is able to produce hydrogen sulphide and carbon dioxide. Together, *Chlorobium* and *Spirillum* are able to supply each other's nutritional needs. The conversion of hydrogen sulphide to elemental sulphur by the



**Figure 9.6** - Synergistic growth of *Lactobacillus arabinosus* and *Enterococcus faecalis* in a medium lacking phenylalanine and folic acid. (A) Combined culture of both the species; (B) *E. faecalis* alone; (C) *L. arabinosus* alone (Adopted from Nurmikko, 1956).

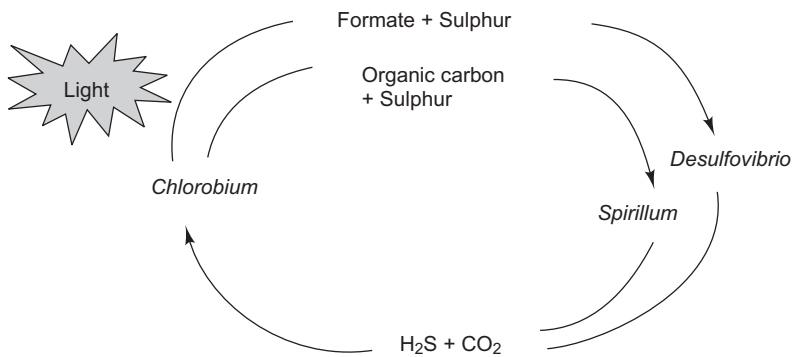


**Figure 9.7** - Synergistic degradation of cyclohexane by a *Nocardia* and a *Pseudomonas* speices (Adopted from Slater, 1978).

*Chlorobium* species plays a detoxifying step, as the hydrogen sulphide concentrations would otherwise kill the *Spirillum* species. *Chlorobium* and *Desulfovibrio* populations also exhibit such a relationship. The *Desulfovibrio* population is able to supply *Chlorobium* with hydrogen sulphide and carbon dioxide; the *Chlorobium* population, in return, is able to supply *Desulfovibrio* with sulphate and organic matter required for metabolism by the *Desulfovibrio* and to remove toxic concentrations of hydrogen sulphide.

There are similar synergistic relationships between bacterial populations involved in the cycling of nitrogen. For example, heterotrophic pseudomonads are chemotactically attracted to organic excretions formed by the heterocysts of *Anabaena spiroides* and related species. They form dense aggregates around the heterocyst, while a few bacteria are associated with the photosynthetic cells of

the filament. The pseudomonads oxidize the excreted organics and at the same time stimulate nitrogenase activity. The most likely mechanism for this stimulation is the lowering of oxygen concentration around the oxygen-sensitive nitrogenase by the use of oxygen for bacterial respiration. The exudation from the cyanobacterial cells facilitates bacterial growth. Like the heterocysts of cyanobacteria, algae also excrete organic compounds, some of which attract bacterial populations. The bacteria produce vitamins that are used by the algae. Extracellular products of algae are known to selectively stimulate certain populations of marine bacteria, enabling algae to attract and form associations with specific bacterial populations. Epiphytic bacteria are often observed on the surfaces of algae. The relationship between epiphytic bacteria and algae may be based on the ability of the alga to utilize light and produce organic compounds and oxygen for the aerobic heterotrophic metabolism of the bacteria (Fig. 9.9). The bacteria, in turn, mineralize excreted organic matter, supplying the algae with carbon dioxide and, sometimes, growth factors needed for phototrophic assimilation. Bacteria can also improve algal growth by removing oxygen.

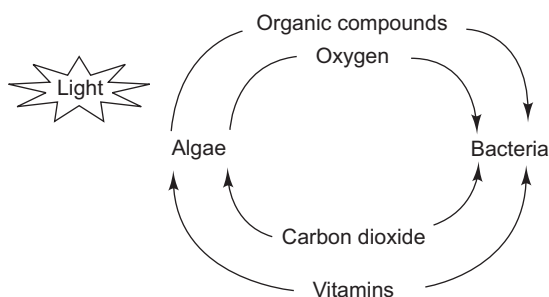


**Figure 9.8** - The cycling of C and S by the synergistic relationships between *Chlorobium* and *Desulfovibrio/Spirillum*.

Some relationships of synergism are based on the ability of a second population to accelerate the growth rate of the first one. For example, some *Pseudomonas* species can grow on orcinol but show a higher affinity for this substrate and grow more rapidly in the presence of other bacterial populations. The secondary populations, which cannot utilize orcinol alone, benefit because they are able to use the excreted organic compounds produced by the *Pseudomonas*. The *Pseudomonas* species benefits because its growth rate is accelerated, presumably by the removal of the excreted organic compounds that otherwise would act through a negative feedback mechanism to repress catabolic activity.

Some synergistic relationships allow microorganisms to produce enzymes that are not produced by either population alone. For example, populations of closely related *Pseudomonas* have been found to produce lecithinase when grown together; neither strain produces lecithinase when it is growing alone. The production of lecithinase allows both populations to benefit from the utilization of lecithin. Similarly, some mixed microbial populations produce cellulase, whereas the individual populations are unable to attack cellulose.

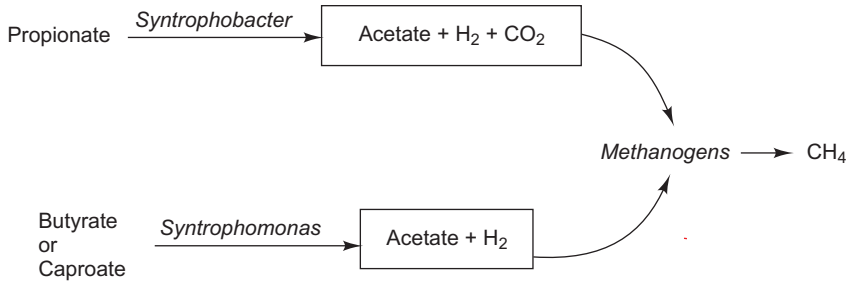
Degradation of pesticides, agrochemicals and transformation of metals in the environment occurs effectively by synergistic interactions among microbes. Some of the degradations are not even possible without microbial interaction. For example, *Arthrobacter* and *Streptomyces* strains isolated from soil are capable of completely degrading the organophosphate insecticide diazinon and can grow on this compound as the only source of carbon and energy. This is accomplished by synergistic attack; alone neither culture can mineralize the pyrimidinyl ring of diazinon or grow on this compound. In a chemostat enrichment culture, *Pseudomonas stutzeri* is capable of cleaving the organophosphate insecticide parathion to *p*-nitrophenol and diethylthiophosphate, but it is not capable of utilizing either of the resulting moieties. *Pseudomonas aeruginosa* can mineralize *p*-nitrophenol but is incapable of attacking intact parathion. Synergistically, the two-component enrichment degrades parathion with high efficiency, *P. stutzeri* apparently utilizing products excreted by *P. aeruginosa*.



**Figure 9.9** - Synergistic relationship between algae and epiphytic bacteria with the exchange of organic carbon and nutrients.

Archaeal populations involved in methane production (methanogens) have interesting synergistic relationships with bacterial and other microbial populations (See Chapter 12). The names of these bacterial genera indicate their syntrophic relationships with  $H_2$ -consuming methanogenic archaea. *Syntrophomonas* species oxidize butyric acid and caproic acid to acetate and  $H_2$ . The members of this genus also oxidize valeric acid and enanthic acid to acetate,  $CO_2$  and  $H_2$ . *Syntrophobacter* oxidizes propionic acid to acetate,  $CO_2$  and  $H_2$ . The acetate and  $H_2$  produced by these bacteria are used by methanogenic archaea to produce methane (Fig. 9.10).

Syntrophic reactions are facilitated by bacterial aggregate (floc) formations, that bring the participants into intimate contact with each other. In the anaerobic digestion of a whey effluent, lactate and ethanol are converted to methane by a three-membered syntrophic association. The bacterium *Desulfovibrio vulgaris* in the floc generates acetate and formate from ethanol and bicarbonate. The formate serves as an interspecies electron and hydrogen transfer agent to methanogens that evolves methane, for shuttling back bicarbonate to *D. vulgaris* for reduction to formate (Fig. 9.11). The acetate generated in the ethanol oxidation process is cleaved by acetoclastic methanogens to methane and  $CO_2$ . In this system, the cleavage of formate to  $CO_2$  and  $H_2$  plays only a negligible role and is performed by bacteria not associated with the floc.



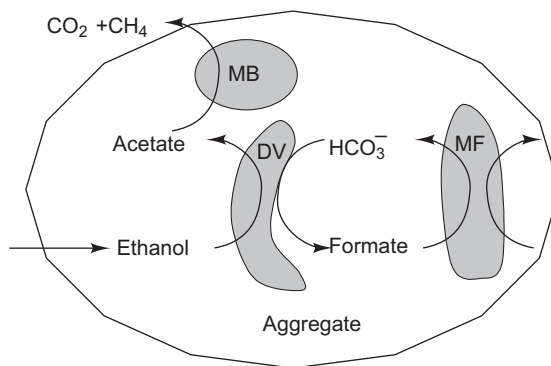
**Figure 9.10** - The syntrophic relationship between the acetogens and methanogens for the degradation of low molecular weight fatty acids like propionate, butyrate and caproate.

### 9.3.4 Mutualism

Mutualism, an intimate synergistic association between two liking partners, is an obligatory relationship that benefits both populations. This highly specific association cannot ordinarily be replaced by another related species. It requires close physical proximity between the interacting populations, between whom there is often metabolic cooperation. Relationships of mutualism allow organisms to exist in habitats that could not be occupied by either population alone. The detailed description on the physical and metabolic aspects of this microbial interaction has been given in the next chapter (See Chapter 10).

### 9.3.5 Competition

Unlike positive interactions, competition represents a negative relationship between two populations in which both populations are adversely affected with respect to their survival and growth. The populations may achieve lower maximal



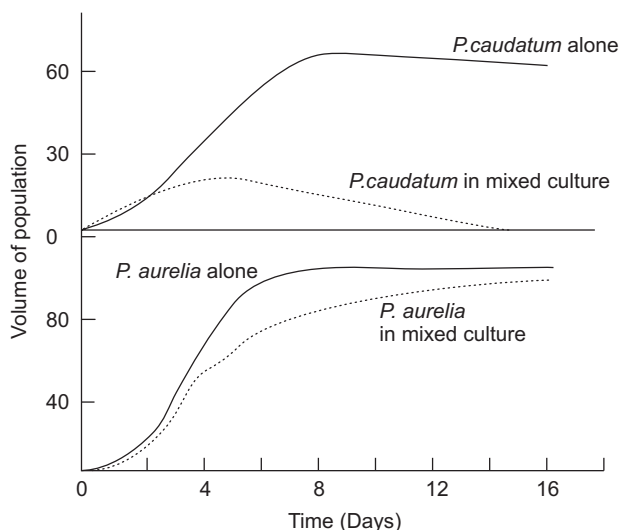
**Figure 9.11** - Syntrophic interaction of three bacterial species for anaerobic digestion of an organic aggregate. Abbreviations: MF – *Methanobacterium formicicum*; DV – *Desulfovibrio vulgaris*; MB – *Methanosarcina barkeri* (Modified from Thiele and Zeikus, 1988).



densities or lower growth rates than they would have in the absence of competition. Competition occurs when two populations use the same resource, whether space or a limiting nutrient. The microbial populations may compete for any available resources like organic carbon, nitrogen, phosphate, oxygen, iron growth factors, etc.

Competition tends to bring about ecological separation of closely related populations. This is known as the competitive exclusion principle. Competitive exclusion precludes two populations from occupying exactly the same niche, because one will win the competition and the other will be eliminated. However, populations may coexist, if the populations can avoid absolute direct competition by using different resources at different times or the size of the population is not big enough to eliminate the other. An example of competitive interaction is between *Paramecium caudatum* and *P. aurelia* (Fig.9.12). Individually, with an adequate supply of bacterial prey, both protozoan populations were able to grow and maintain a constant population level. However, when the protozoans were placed together, *P. aurelia* alone survived after 16 days. Neither organism attacked the other nor secreted toxic substances. Rather, *P. aurelia* exhibited the more rapid growth rate and thus outcompeted *P. caudatum* for available food. In contrast, a mixture of *P. caudatum* and *P. bursaria* were able to survive and reach a stable equilibrium even when grown together. They competed for the same food source, but the two populations occupied different regions of the culture flask. Thus, the habitats of *P. caudatum* and *P. bursaria* were sufficiently different to allow the two species to coexist although they had overlapping niches based on their common food source.

Intrinsic growth rates of competing populations vary under different environmental conditions, explaining the coexistence of populations in the same habitat competing for the same resources. For example, in marine habitats,



**Figure 9.12** - The competitive interactions of two species of *Paramecium*. The growth patterns are that of single and mixed cultures (Adopted from Gause, 1934).

populations of psychrophilic and psychrotrophic bacteria are found together even though both populations may be competing for the same low concentrations of organic nutrients. At low temperatures, the psychrophilic populations exhibit higher intrinsic growth rates and given sufficient time they would exclude the psychrotrophic populations. However, at higher temperatures, the psychrotrophic populations exhibit the higher intrinsic growth rate and the psychrophilic populations would be excluded. In habitats of varying temperature regimes, the advantage shifts back and forth, leading to seasonal shifts in proportions of psychrotrophic and psychrophilic populations within natural aquatic habitats. In some instances, bacterial populations are excluded from the habitat during periods when temperature or other environmental factors do not allow them to compete successfully with other populations because of their lower intrinsic growth rates. Under different environmental conditions, these same bacterial populations may compete successfully and even become dominant.

Competition within natural habitats is complex as organisms are competing for different resources, some of which may be essential and others of which may enhance growth rates. Resource-ratio theory of Tilman (1982) suggests that organisms require resources for their growth and reproduction. Different species, as well as different phenotypic strains within a species, frequently differ significantly in their efficiency of uptake and utilization of potentially growth-limiting resources. Similarly, when such physiological differences exist, they can result in variations in competitive ability among the organisms sharing these resources. When the quantitative requirements for resources and the mortality rates are known for each species, resource-ratio theory provides a framework that can be used to predict the outcome of competitive interactions between them.

Abiotic parameters, such as temperature, pH and oxygen, greatly influence the intrinsic growth rates of microbial populations and the outcome of a competitive struggle. For example, at high substrate concentrations, competition between a marine *Spirillum* and *E. coli* results in the competitive exclusion of the *Spirillum*, whereas at low concentrations the reverse occurs and the *E. coli* is excluded. Dominant microbial populations in sewage, which has a high organic substrate content, are rapidly displaced in competition with the autochthonous microbial populations of receiving streams and rivers, where the concentration of organic matter diminishes in the course of mineralization and dilution.

Tolerance of a microbial group to environmental stress is also found important in determining the outcome of competition between different populations. For example, under conditions of drought, populations that can best tolerate and survive desiccation can displace less tolerant populations. Similarly, under other stress conditions such as heavy metal leaching or high salt concentrations, the population with the greatest tolerance to that factor may have the edge in the competition. These cases represent competition for survival during nongrowth conditions. During active growth conditions, the competitive advantage returns to the population with the highest growth rate under those conditions. The efficiency to utilize the resource is also an advantage for a species to compete others in the growth-limited or nutrient-stressed conditions.

### 9.3.6 Antagonism

Mutual exclusion between microbial groups in which one group take the advantage of utilizing the resource by limiting the growth of the other group, usually through the production of a toxin, is called antagonism (also called amensalism). Microorganisms that produce substances toxic to competing populations will naturally have a competitive advantage. The first population may be unaffected by the inhibitory substance or may gain a competitive edge that is beneficial. The terms *antibiosis* and *allelopathy* have been used to describe such cases of chemical inhibition. There are cases of complex amensalism between populations in natural habitats, for example, virucidal (virus-killing) factors in seawater and fungistatic (fungi-inhibiting) factors in soil.

Antagonism may lead to the preemptive colonization of a habitat. Once an organism establishes itself within a habitat, it may prevent other populations from surviving in that habitat just to use whatever resource available in the habitat for its growth and reproduction. The production of lactic acid or similar low molecular weight fatty acids is inhibitory to many bacterial populations. Populations able to produce and tolerate high concentrations of lactic acid, as an antagonistic reaction, are able to modify the habitat so as to preclude the growth of other bacterial populations. *E. coli* is unable to grow in the rumen, probably because of the presence of volatile fatty acids produced in rumen by anaerobic heterotrophic microbial populations. Fatty acids produced by microorganisms on skin surfaces are believed to prevent the colonization of these habitats by other microorganisms, especially of those which are considered pathogenic. Populations of yeasts on skin surfaces are maintained in low numbers by microbial populations producing fatty acids. Acids produced by microbial populations in the vaginal tract are probably responsible for preventing infection by pathogens such as *Candida albicans*.

Acidification of the habitat by *Thiobacillus thiooxidans* occurs due to oxidation of sulphur to produce sulphuric acid, which facilitates the growth of the bacterium by regulating the growth of other competing species. Consumption or production of oxygen also alters the habitat in ways detrimental to some microbial populations but others are benefited from it. The production of oxygen by algae discourages the growth of obligate anaerobes. However, a few obligate anaerobes are also found in habitats where there is a significant production of oxygen by algal populations. This becomes possible by the establishment of microenvironments by the anaerobic bacteria as response to the antagonistic environment.

Production of low molecular weight alcohols such as ethanol, by some microbial populations are inhibitory to many other microbial groups. The production of ethanol by yeasts prevents the growth of most bacterial populations in habitats where the ethanol can accumulate. Thus, a few bacterial populations can grow in a habitat of fermenting grapes (wine), and therefore, such fermentation process has less problem of contamination. However, ethanol production does not eliminate all bacterial populations as some bacteria are able

to metabolize ethanol. For example, *Acetobacter* is able to convert ethanol to acetic acid under aerobic condition, which in turn, is also inhibitory to many bacterial populations. Similarly, during the decomposition of cellulose in soil, organic acids produced by microbes prevent further breakdown of cellulose metabolites in subsurface soil due to the feedback inhibition of microbial growth.

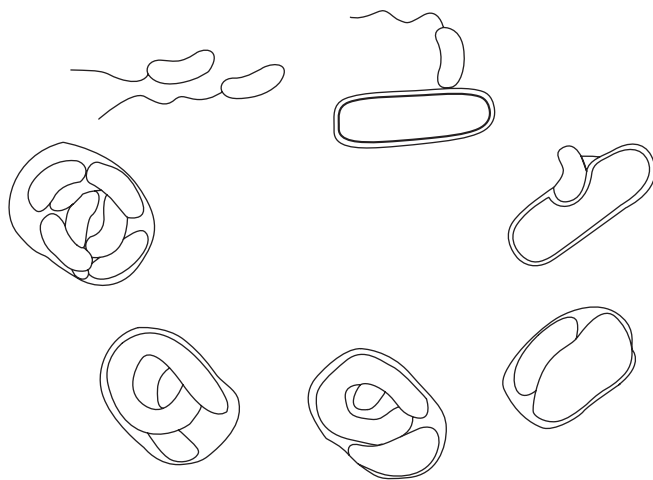
Antibiotics production by some microorganisms is a common antagonistic reaction to kill or inhibit the growth of other competing microorganisms. However, conditions that favour the production of antibiotics are not normally found in natural habitats, and therefore, efficiency of antibiotics to control unwanted microbial population in natural habitats is always questionable. Antibiotics are secondary metabolites and are usually produced when excess substrate concentrations are available to promote high levels of microbial growth. The normal condition of most soil and aquatic habitats is one of limiting organic substrates; antibiotics are not found to accumulate in natural habitats. In aquatic habitats, antibiotics probably are rapidly diluted to ineffective concentration levels. Antibiotics produced in soil may be bound to clay minerals or other particulates and thus be inactivated. Abundant *Streptomyces* and *Bacillus* populations occur in soils, but there is little evidence that they produce significant amounts of antibiotics there. Antibiotic-producing microorganisms usually do not dominate in aquatic habitats. Proportions of antibiotic-resistant strains are not exceptionally high in soils even though antibiotic-producing populations are found in soil habitats. However, if antibiotics were of no use in nature, they probably would not have been selected during evolutionary processes and would not be as widespread as they are. Rather production of antibiotics probably has a significant function in establishing amensal relationships within micro-environments and under certain environmental conditions.

### 9.3.7 Parasitism

An interactive relationship, in which one population benefits (the parasite) at the cost of the other (the host) by deriving its nutritional requirements from the latter, is called parasitism. The host-parasite relationship is characterized by a relatively long period of contact, which may be directly physical or metabolic. Usually, but not always, the parasite is smaller than the host. Some parasites remain outside the cells of the host population and are called *ectoparasites*; while other parasites penetrate the host cell and are called *endoparasites*.

Normally, the parasite-host relationship is quite specific. The available habitats for obligately parasitic populations are limited to available hosts. In some cases, host specificity depends on surface properties of the organisms that allow physical attachment of the parasite to the host cells. Viruses are obligate intracellular parasites that exhibit great host cell specificity. There are viral parasites of bacterial, fungal, algal and protozoan populations. In some habitats, viruses are suspected of being responsible for the decline and disappearance of bacterial populations. Parasitism may contribute to the disappearance of other fecal organisms in sewage that enters aquatic habitats.

The bacterium *Bdellovibrio*, like bacteriophages, is a parasite on gram-negative bacterial populations. It is highly motile, attaining speed of up to 100 cell lengths per second compared to only 10 lengths per second for *E. coli*, a potential host species. The encounter between parasite and prey appears to be random. Only a small percentage of cell contacts result in permanent attachment of *Bdellovibrio* to the outer cell membrane of its gram-negative host. Subsequently, *Bdellovibrio* penetrates the outer cell membrane of its host and enters the periplasmic space but not the cell proper; thus remaining as an ectoparasite. The parasitic interaction spans approximately one hour, during which *Bdellovibrio* modifies the cell envelope of its host in both degradative and synthetic manners. The host cell loses its original shape and becomes spherical (bdelloplast), yet it retains the cell contents for use by the parasite. The cell contents of the host are partially degraded and are used by *Bdellovibrio* with high efficiency (Fig. 9.13). When it enters the periplasmic space, *Bdellovibrio* loses its flagellum and grows into a filament without cell division. When the cell contents of the host are exhausted, the filament divides into individual cells, which develop flagella. The bdelloplast lyses and releases the progeny. The burst size (number of progeny per host cell) depends on the size of the host cell and may range from about 4 in *E. coli* to about 20 in the much larger *Spirillum serpens*.



**Figure 9.13** - The replication pattern of *Bdellovibrio* within a host cell (Adopted from Marbach *et al.*, 1976).

Many ectoparasitic microorganisms can cause lysis without direct contact. For example, myxobacteria can cause lysis of susceptible microbial strains at some distance, apparently with the aid of exoenzymes. The myxobacteria derive their nutrition from material released by the lysed microorganisms. Soil myxobacteria are able to lyse gram-negative and gram-positive bacterial populations. *Cytophaga* populations produce enzymes that cause lysis of susceptible algae. Some bacterial populations produce chitinase, degrading fungal cell walls. Other

bacterial populations produce laminarinase or cellulase enzymes that similarly attack the cell wall structures of some algal and fungal populations. In all of these cases, ectoparasitic populations cause the lysis of host cells, releasing nutrients that can be utilized by the ectoparasites, even though many of these ectoparasites are not obligate parasites and can derive nutrition by other means. Some microbial populations are more resistant to lysis than others. Microbial populations may produce resting stages, such as cysts and spores, that are more resistant to lytic activities of ectoparasites than the vegetative cells. This mechanism for escaping the pressures of ectoparasitism allows the host populations to persist.

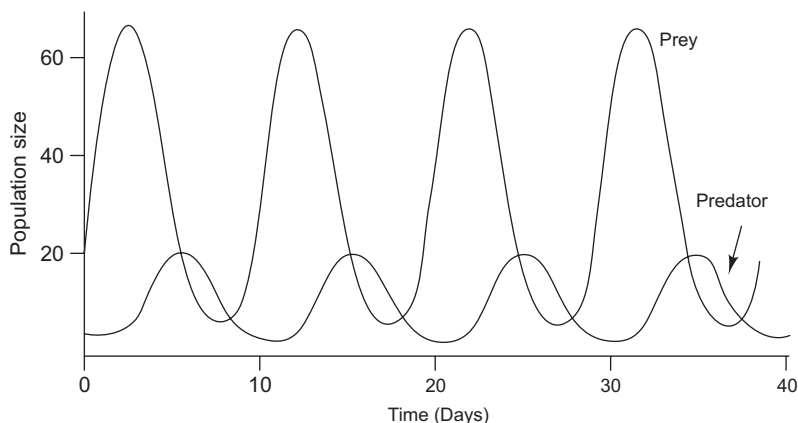
Protozoan populations are subject to parasitism by a large number of fungi, bacteria and other protozoa. The ability of the opportunistic human pathogen *Legionella pneumophila* to parasitize protozoa may play an important role in its distribution and survival in aquatic environments. Some fungal populations are parasites of protozoa. These may be endoparasites or ectoparasites. In some cases, endoparasites are maintained for long periods within the host-protozoan cells. In such cases, the line between endoparasitism and mutualism is often blurred. Algae, too, are attacked by fungi; the chitrids are notable examples. Uniflagellate chitrids are normally ectoparasites and biflagellate chitrids are normally endoparasites of algal populations. Chitrids frequently infect freshwater algae, leading to decimation of susceptible populations. Some fungal populations are parasitized by other fungal populations. Basidiomycetes, such as *Agaricus*, are frequently attacked by other fungi, such as *Trichoderma*. Such infections create difficulties in the commercial cultivation of mushrooms.

Some parasitic microorganisms may also serve as host cells for other parasitic populations, a phenomenon known as hyperparasitism. For example, *Bdellovibrio*, a bacterial parasite, may serve as a host cell for appropriate phage populations. Fungal populations that are parasitic on algae are themselves subject to bacterial and viral parasitism. Some rust fungi are parasitized by lytic bacteria.

Parasitic interactions provide a mechanism for population control. Because the intensity of parasitism is population-density dependent, parasites can thrive only as long as there are abundant host populations. Parasitism results in a reduction of host populations density facilitating the accumulation and the renewal of the environmental resources. It acts as a negative feedback control to check the unchecked growth of the host populations.

### 9.3.8 Predation

Predation typically occurs when one organism, the predator, engulfs and digests another organism, the prey. Normally, predator-prey interactions are of short duration and the predator is larger than the prey. However, in the microbial world, the distinction between parasitism and predation is not sharp. Early theoretical considerations of predator-prey relationships led to the prediction that the interactions of predator and prey species have regular cyclic fluctuations of the two populations (Fig. 9.14). In each cycle, as the prey population increases, the predator population follows, overtakes and overcomes it, producing a decline in



**Figure 9.14** - The theoretical predator-prey fluctuations (Adopted from Krebs, 1972).

the prey. The predator population then follows the declining prey supply downward until the predators reach such a low level that the prey population begins to rise again, whereupon the predator again follows it upward. Assuming that the environment is constant, the amplitude of the cycle is determined by the initial population density and remains the same. Theoretically, such predator-prey populations could cycle forever (Lotka-Volterra model).

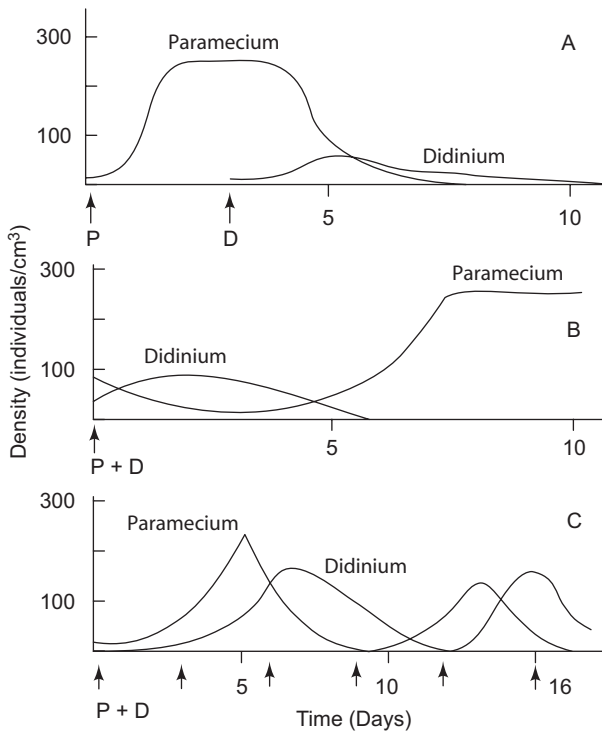
However, experimental results have rarely supported the Lotka-Volterra model. In a classic series of experiments, Gause (1934) showed that when *Didinium nasutum* was introduced with *Paramecium caudatum*, it preyed on the *Paramecium* until the *Paramecium* population became extinct. Lacking a food source, the *Didinium* population also became extinct (Fig. 9.15). If a few members of the *Paramecium* population are able to hide and escape predation by the *Didinium*, then the *Paramecium* population can recover following extinction of the *Didinium*. Only by periodically introducing *Paramecium* and *Didinium* populations, Gause was able to produce a sustained interaction with marked fluctuations in population numbers. Gause (1934) was also able to establish periodic oscillations using *P. bursaria* as a predatory species and the yeast *Schizosaccharomyces pombe* as a prey species, but was unable to establish sustained oscillations. In this heterotrophic nutrient-limited system, the amplitude of the oscillations decreases, leading to extinction of both species. The preceding experiments indicate that intense predator-prey relations may lead to the extinction of both the predator and the prey populations.

The failure of experimental models to reveal persistent population oscillations has forced a reevaluation of the underlying assumptions of the Lotka-Volterra model. The error appears to be the invalidity of the inherent assumption that predatory encounter is random in time and space and, like a bimolecular collision, is proportional to the product of predator and prey populations. Experimental evidence indicates that predator-prey populations establish nonoscillatory steady-state conditions or cycles that become limiting, such that the magnitude of



the oscillation progressively decreases. Although predation destroys individuals, the prey population as a whole may benefit from accelerated nutrient recycling. The increase in phytoplankton growth rates due to nitrogen regeneration by predaceous zooplankton has been found to fully compensate for the mortality of the individual phytoplankton caused by the zooplankton. Moderate predatory pressure keeps prey populations from exhausting the carrying capacity of their environment and maintains the prey population in a dynamic state of growth. It is important to recognize that the interactive nature of predator-prey relations is adaptive. There is an organizational control through negative feedback that regulates the population sizes of the predator and the prey. If either the predator or the prey were completely eliminated, populations of the other would be deleteriously affected.

In microbial predation, it is common to find great size differences between a predator and its prey. Populations of ciliate, flagellate and amoeboid protozoa are predatory on bacterial populations. Predation by flagellate protozoa may be



**Figure 9.15** - The predator-prey relationship between *Paramecium* and *Didinium* in a mixed culture. (A) *Didinium* introduced at high *Paramecium* levels; (B) *Didinium* introduced at low *Paramecium* levels; (C) Both species were introduced at every three days to prevent permanent extinction and to maintain the predator-prey cycle. The arrows indicate time of protozoan introduction (Adopted from Gause, 1934).

responsible for maintaining relatively stable populations of bacteria in aquatic systems. Ciliate protozoa, such as *Paramecium*, *Vorticella* and *Stentor*, have a cell mass  $10^3$  to  $10^4$  times larger than the average prey bacterium, such as *Enterobacter aerogenes*. Such relatively large predatory protozoa often employ a strategy of filter feeding so that they do not consume all their energy chasing small, low-calorie prey. Filter feeding works at an energy deficit when the prey density becomes too low. However, some microbial structures are resistant to predators. For example, soil amoebae engulf and consume vegetative cells of *Bacillus* species, but endospores of *Bacillus* are less subject to predation. The protozoan *Entodinium caudatum* is subject to predation by the larger protozoan *E. vorax*. *E. caudatum* produces spineless and spined cells; *E. vorax* preferentially consumes the spineless populations, leading to the extinction of the spineless forms while the spiny populations escape and reproduce.

## 9.4 MICROBIAL INTERACTIONS IN A BIOFILM

A complex behavioural pattern of microbes is seen during the colonization of surface environments by the pioneer species leading to the development of a film of microbial cells on a surface, called the biofilm. A wide range of species-specific behaviours like progressing through attachment, growth, microcolony formation and redistribution/dispersal are observed on the biofilm surface. There is a great degree of microbial cooperation for the selection, establishment, growth and reproduction of microbes in a biofilm. The commonest colonization strategy employed by bacteria for the formation of a biofilm is the formation of colony aggregates from a single point of bacterial colonization. In several *Pseudomonas*, *Vibrio*, *Klebsiella* and *Escherichia* species, colonies develop from an individual attached pioneering cell that goes through several division cycles, eventually forming a bacterial microcolony. The exopolysaccharides, secreted by the pioneering species during colonization on a surface, provide an interface between the cell and the external environment, influencing rates of chemical exchange and nutrient availability and facilitating the generation of microniches. Such microniches become the favourite habitats for the less competitive and less tolerant species.

Various types of interactions occur among different communities within a biofilm, viz., syntrophism, symbiosis, commensalisms and predation. These interactions are defined in terms of substrate utilization. For example, introduction of a new non-metabolizable compound into the environment can exert stress conditions, which may be relieved by mineralization of the contaminant. Thus, microbial communities may be important in developing novel degradative pathways to mineralize xenobiotic compounds during wastewater purification. While in free living state, very few species are able to mineralize xenobiotics, many species can do so when present in the biofilm. Predatory interactions between bacterial species seem to play a significant role in community development. A range of obligate and non-obligate predatory interactions can occur between bacteria and regulate cell numbers within communities.

Algal bacterial associations in bioaggregates and in biofilm communities are widespread, with algae providing sites of attachment and acting as sources of nutrients. In return, the bacteria supply nutrients for algal growth resulting in conversion of dissolved nutrients to algal biomass, which can be removed and treated separately. Algal exudates also serve as an important growth substrate for heterotrophic bacteria but their role in the bacterial mineralization of xenobiotic compounds is not known (See Chapter 12).

## 9.5 PLANT–MICROBE INTERACTIONS

Many a different types of interactions are seen between the microbes and the plants. The association ranges from symbiosis to pathogenicity. The microbes are found to be distributed on the plant surface (Phyllosphere), root surface (Rhizosphere) and inside the tissues (Endophytic symbionts and pathogens).

### 9.5.1 Rhizosphere Microbes

Interactions between soil microorganisms and plant roots satisfy important nutritional requirements for both the plant and the associated microorganisms resulting in a higher number of microbes in the vicinity of plant roots. Availability of nutrition and surface makes it possible for a large number of microorganisms to be found in the rhizoplane, which is defined as the actual surface of the plant roots within the rhizosphere. The latter includes the rhizoplane and is defined as the region of soil directly influenced by the plant roots. The thin layer of soil around the plant roots contains billions of microbes, which are in intricate metabolic association with the root zone. An interesting and as yet insufficiently explored modification of the rhizosphere is the rhizosheath, characterized as a relatively thick soil cylinder that adheres to the plant roots. The formation of rhizosheaths is typical of some desert grasses but also occurs in some grass species that grow under less extreme conditions. The sand grains in the rhizosheath are cemented together by an extracellular mucigel, apparently excreted by the root cells. The rhizosheath appears to be an adaptation for moisture conservation, but it undoubtedly also provides an environment for the extensive root-microbe interactions.

There is a higher proportion of gram-negative, rod-shaped bacteria and a lower proportion of gram-positive rods, cocci and pleomorphic forms in the rhizosphere than in root-free soil. There is also a relatively higher proportion of motile, rapidly growing bacteria, such as *Pseudomonas*, in the rhizosphere than elsewhere in soil. In many cases, this increase represents a direct influence of plant root exudates on soil microorganisms, which favour microorganisms with high intrinsic growth rates. Organic materials released from roots include amino acids, keto acids, vitamins, sugars, tannins, alkaloids, phosphatides and other unidentified substances. Roots surrounded by microorganisms excrete many times as much organic material as sterile roots. Although a few of these materials (mostly secondary metabolites) inhibit microorganisms, most stimulate microbial

growth. The influence of materials released by the plant into the soil is evidenced by the observation that bacterial populations within the rhizosphere have markedly different nutritional properties than bacteria in root-free soil. Many rhizosphere bacteria require amino acids for maximal growth and it is likely that root exudates supply these acids.

Many different types of interactions occur between plants and the microbes in the rhizosphere. Organisms in the rhizosphere produce organic compounds and hormones that affect the proliferation of the plant root system structurally modifying the root as desired by the microorganism. Microorganisms synthesize auxins and gibberellin-like compounds and these compounds increase the rate of seed germination and the development of root hairs that aid plant growth. *Arthrobacter*, *Pseudomonas* and *Agrobacterium* populations found in the rhizosphere have been reported to be capable of producing organic chemicals that stimulate growth of plants. For example, the rhizosphere of wheat seedlings contains a significant proportion of bacteria that produce indoleacetic acid (IAA). This may be in response to a decline in root exudate production, but it also beneficially corresponds to a decreased need for the growth hormone by the plant (See Chapter 5).

### 9.5.2 Mycorrhizae

Many plants, especially tree species, enter into a mutualistic relationship with fungal hyphae called mycorrhizal fungi, in which the latter actually become integrated into the physical structure of the roots. The fungus derives nutritional benefits from the plant roots, contributes to plant nutrition and does not cause plant disease. Mycorrhizal associations differ from other plants-microb associations by the greater specificity and organization of the plant-fungus relationship. It proceeds with the integration of plant roots and fungal mycelia, forming integrated morphological units. The widespread existence of mycorrhizal associations between fungi and plant roots clearly reflects the importance of this interaction for the establishment of growth of plants, especially at the early stage of development. The micorrhizal association is such an obligatory requirement of some plant species that their seedlings fail to grow if the association is not established within a first few hours of seedling germination. For example, the seedlings of Sal (*Sorea robusta*) fail to establish after germination, if the micorrhizal association does not develop with the freshly emerged root. Mycorrhizal treatment significantly enhances the germination rate and seedling establishment of Sal and Pine.

Mycorrhizal associations exist for prolonged periods with the maintenance of a healthy physiological interaction between the plant and the fungus. The mycorrhizal associations of fungi and plant roots represent a diverse relationship, in terms of both structure and physiological function, that leads to a nutrient exchange favourable to both partners. Enhanced uptake of water and mineral nutrients, particularly phosphorus and nitrogen, has been noted in many mycorrhizal associations. Therefore, plants with mycorrhizal fungi are able to occupy habitats; they otherwise could not colonize alone. It is essential for many

seedlings to have mycorrhizal association at an early stage of growth for their successful establishment in an alien habitat (See Chapter 5). Table 9.2 contains the host and microbes of some beneficial plant-microbe association.

### 9.5.3 Legume-Rhizobium Interaction

One of the most important mutualistic relationships between microorganisms and plants involves the invasion of the roots of suitable host plants by nitrogen-fixing bacteria, resulting in formation of nitrogen-fixing nodules. Within the nodule, the nitrogen-fixing bacteria are able to convert atmospheric nitrogen to ammonia, which supplies the nitrogen required for bacterial and plant growth. The fixation of nitrogen in plant nodules is of extreme importance for the maintenance of soil fertility.

The nitrogen-fixing (diazotrophic) associations of rhizobia with leguminous plants are of great importance both in global nitrogen cycling and in agriculture. Until last decade, all nodulating and nitrogen-fixing bacteria associated with leguminous plants were placed in a single genus, *Rhizobium*. Now there are four additional genera, *Azorhizobium*, *Bradyrhizobium*, *Mesorhizobium* and *Sinorhizobium* recognized and additional genera may be added as the numerous species of the legume family are investigated. The association between rhizobia and plant roots is highly specific, with mutual recognition between the two compatible partners based on chemotactic response and specific binding to the root hair prior to invasion and establishment of the root nodule. The legume plant root recognizes the right population of rhizobia, which in turn recognizes the right kind of leguminous root. Within the rhizosphere, plant roots supply the rhizobia with compounds that are transformed by them to substances involved in the initiation of the infection process and subsequent nodule development. Establishment of an adequate rhizosphere population of rhizobia is an absolute prerequisite for infection (See Chapter 10).

### 9.5.4 Non-leguminous Nitrogen-fixing Mutualistic Relationships

In addition to the mutualistic relationship between Rhizobia and leguminous plants, other symbiotic relationships between bacteria, cyanobacteria and actinomycetes with non-leguminous plants result in the formation of root nodules and the ability to fix atmospheric nitrogen. *Rhizobium*, for example, can fix nitrogen in association with *Trema*, a tree found in tropical and subtropical regions. Likewise, the actinomycete *Frankia alni* infects the roots of trees, leading to the formation of nodules. *Frankia* species also form associations with various non-leguminous plants including various woody shrubs and small trees. Such an actinomycete-type nitrogen-fixing symbiosis is especially important for angiosperms. The productivity of many forests depends on such nitrogen-fixing symbioses, especially in temperate and circumpolar regions, whereas cyanobacteria and rhizobial symbioses are more common in tropical and subtropical regions.

**Table 9.2:** The host and the microbes of some of the important plant-microbe interactions.

| <i>Non-rhizobial bacteria</i>     | <i>Host plant</i> | <i>Rhizobial bacteria</i>          | <i>Host plant</i> | <i>Mycorrhizae</i>        | <i>Host plant</i>          |
|-----------------------------------|-------------------|------------------------------------|-------------------|---------------------------|----------------------------|
| <i>Acetobacter diazotrophicus</i> | Sugarcane         | <i>Azorhizobium caulinodans</i>    | Sesbania          | <i>Glomus</i> spp.        | <i>Festuca rubra</i>       |
| <i>Herbaspirillum serapedicea</i> | Sugarcane         | <i>Bradyrhizobium elkanii</i>      | Soybean           | <i>Glomus</i> sp.         | <i>Poa pratensis</i>       |
| <i>Burkholderia brasiliensis</i>  | Sugarcane         | <i>Bradyrhizobium japonicum</i>    | Soybean           | <i>Glomus</i> sp.         | <i>Plantago lanceolata</i> |
| <i>Burkholderia cepacia</i>       | Tobacco           | <i>Bradyrhizobium liaoningense</i> | Soybean           | <i>Glomus</i> sp.         | Epiparasitic plants        |
| <i>Methylobacterium</i> sp.       | Rice              | <i>Mesorhizobium ciceri</i>        | Cicer             | <i>Glomus tenue</i>       | Lotus                      |
| <i>Curtobacterium</i> sp.         | Rice              | <i>Mesorhizobium huakuii</i>       | Astragalus        | <i>Acaulospora</i> sp.    | <i>Heiracium pilosella</i> |
| <i>Deinococcus radiophilus</i>    | Soybean           | <i>Mesorhizobium loti</i>          | Lotus             | <i>Acaulospora mellea</i> | Morisita horn community    |
| <i>Staphylococcus lentus</i>      | Lentil            | <i>Mesorhizobium plurifarum</i>    | Acacia            | <i>Acaulospora</i> sp.    | Liverworts                 |
| <i>Bacillus fastidiosus</i>       | Soybean           | <i>Rhizobium etli</i>              | Phaseolus bean    | <i>Acaulospora</i> sp.    | <i>Tamarix lanceolata</i>  |
| <i>Azospirillum brasilense</i>    | Rice              | <i>Rhizobium hainanense</i>        | Tephrosia         | <i>Entrophospora</i> sp.  | <i>Ophioglossum</i>        |
| <i>Azospirillum</i> sp.           | Tomato            | <i>Rhizobium leguminosarum</i>     | Mung bean         | <i>Gigaspora</i> sp.      | <i>Phragmites communis</i> |
| <i>Alcaligenes piechaudii</i>     | Cabbage           | <i>Rhizobium mongolense</i>        | Alfalfa           | <i>Gigaspora</i> sp.      | Ferns                      |
| <i>Khuyvera ascorbata</i>         | Cotton            | <i>Rhizobium phaseoli</i>          | Black gram        |                           |                            |
| <i>Bacillus japonicum</i>         | Soybean           | <i>Rhizobium undicola</i>          | Neptunia          |                           |                            |
|                                   |                   | <i>Sinorhizobium arboris</i>       | Acacia            |                           |                            |
|                                   |                   | <i>Sinorhizobium meliloti</i>      | Alfafa            |                           |                            |
|                                   |                   | <i>Sinorhizobium saheli</i>        | Sesbania          |                           |                            |
|                                   |                   | <i>Sinorhizobium xinjiangense</i>  | Alfalfa           |                           |                            |

Compiled from Singleton *et al.* (1991); Yamato and Iwasaki (2002); Muthukumar *et al.* (2003); Hung *et al.* (2005); Khade and Adholeya (2005); Saxena *et al.* (2005).

Some liverworts, mosses, pteridophytes, gymnosperms and angiosperms are able to establish mutualistic relationships with nitrogen-fixing cyanobacteria of the genera *Nostoc* and/or *Anabaena*. The coralloid roots of *Cycas* exhibit nodule-like structures even before invasion by the cyanobacteria. This is a modification of the neutral roots of the host plants to invite the cyanobacteria for symbiotic association. The angiosperm *Gunnera* has stem nodules structurally designed to harbour colonies of cyanobacteria. In these symbiotic relationships, the cyanobacteria are restricted to heterotrophic metabolism; the plant supplies suitable organic compounds through its photosynthetic activity and the cyanobacteria within the underground roots or deep inside the stem produce fixed forms of nitrogen. There is a relatively high rate of exudation of nitrogenous compounds by the cyanobacteria within the plant roots as compared to free-living cyanobacteria (See Chapter 10). The exuded nitrogenous compounds are effectively utilized as a source of nitrogen by the host plant. The rate of exudation can be regulated by the host plant itself by determining the rate of supply of organic compounds to the symbiont.

### 9.5.5 Microbial Interaction with Aerial Plant Surface

Colonization by heterotrophic and photosynthetic bacteria, fungi (particularly yeasts), lichens and some algae regularly occur on aerial plant surfaces (stems, leaves, fruits and flowers) called the phyllosphere. Various bacterial and fungal populations occupy the phyllosphere habitats. The number of microorganisms on leaf surfaces depend on the season and age of the leaf. The principal populations in the phyllosphere are *Pseudomonas* species, which are relatively proficient in utilizing sugars and alcohols as carbon sources compared to bacterial populations in the litter layer underlying these trees.

Populations of yeasts like *Sporobolomyces roseus*, *Rhodotorula glutinis*, *R. mucilaginosa*, *Cryptococcus laurentii*, *Torulopsis ingeniosa* and *Aureobasidium pullulans* are normal inhabitants of the phyllosphere. There is an abundance of pigmented populations of yeasts and bacteria on leaf surfaces. The pigments of these microbial populations probably afford protection against exposure to direct sunlight on the leaf surface. *Sporobolomyces* is perhaps the most successful fungus that develops in the phyllosphere with novel mechanism of spore dispersal. The fungus causes the dispersal of its balistospores from the infected surface to the distant plants by propelling them at a speed strong enough to allow the spore to reach and adhere to the new surface. Many other fungi, including Ascomycota, Basidiomycota and Deuteromycota, are present in the phyllosphere and many of them are pathogenic. Some of these allochthonous populations remain dormant and proliferate when the condition becomes favourable.

Populations of *Ascochyta*, *Leptosphaeria*, *Pleospora* and *Phoma* found in the phyllosphere are primarily saprophytes that are unable to grow extensively until the onset of senescence. Allochthonous fungal populations frequently found in the phyllosphere include *Cryptococcus*, *Myrothecium*, *Pilobolus* and various other fungal populations, which normally occur in the soil. For a temporary period, these fungi colonize the plant surface and continue their life cycle to form spores for infecting the new habitats (See Chapter 6).



## 9.6 ANIMAL–MICROBE INTERACTIONS

### 9.6.1 Predation

Predatory animals are usually unable to survive on disproportionately small prey because they would have to expend more energy capturing such prey than they would derive from their consumption. Nevertheless, many invertebrate animals can satisfy part or all of their food requirements by preying on microorganisms  $10^5$  to  $10^7$  times smaller in biomass than themselves. This feat is accomplished by two feeding strategies: grazing on microbial aggregations and filter feeding.

Aquatic Gastropoda (snails), Echinodermata (sea urchins) and Patellidae (limpets), eat microbial crust from submerged surfaces where the microbial populations are able to reach high densities because of the physical absorption of dissolved nutrients on these surfaces. The size difference between the predator and the prey becomes relatively unimportant in this feeding process because the predator pursues coherent masses of millions of microbes rather than individual prey. The microorganisms associated with fecal pellets are an important source of food for many aquatic and some terrestrial animals. That is why, the organic detritus are more nutritious and favoured food for primary consumers in aquatic systems in place of fresh organic debris. Many marine and freshwater invertebrates consume microbial populations growing on detrital particles. These animals rely on microorganisms to incorporate mineral nitrogen and to convert plant polymers in detrital particles that have a low N:C ratio into proteinaceous microbial biomass with a high N:C ratio, thereby enhancing the nutritional value of the detritus.

In the terrestrial environment, various soil microarthropods and certain rodents and lagomorphs, including rabbits, are coprophagous and regularly reingest some of their own fecal material. In aquatic environments, some invertebrates, such as snails, graze on the microbial populations developing on fecal pellets deposited by other animals. Various members of the marine meiofauna secrete slime trails, which bacterial, fungal and algal populations consume. These slime trails provide nutrients for the growth of the microbial populations. The microorganisms adhere to the slimes, which act as mucous traps. The animals then retrace their tracks and graze on the microbial populations that have grown on and are now entrapped in the mucous slime trails.

### 9.6.2 Commensal and Mutualistic Interactions

Many microorganisms maintain mutualistic and commensal relationship with animals of different structural complexity. Most warm-blooded animals contain extremely complex microbial communities within their gastrointestinal tracts. In lower intestine, the microbial flora is highly dense and diverse representing at least 400 species. In the human intestine, the strict anaerobes belonging to the genera *Bacteroides*, *Fusobacterium*, *Bifidobacterium* and *Eubacterium* are the most numerous, but no single species has a dominant role in this highly diverse

microbial community. The presence of these microbes in the intestine facilitates the digestion process and helps in degradation of many complex matters of plant and animal origin.

The rumen of animals harbours a great diversity of microorganisms (Table 9.3). Bacterial populations within the rumen include cellulose digesters, starch digesters, hemicellulose digesters, sugar fermenters, fatty acid utilizers, methanogenic bacteria, proteolytic bacteria and lipolytic bacteria. These populations include species of *Bacteroides*, *Ruminococcus*, *Succinimonas*, *Methanobacterium*, *Butyrivibrio*, *Selenomonas*, *Succinivibrio*, *Streptococcus*, *Eubacterium* and *Lactobacillus*. Many of these bacterial populations produce acetate, the predominant acid within the rumen. The bacteria also produce propionate, the only fermentation acid that can be converted into carbohydrates by the ruminant. The diverse bacterial community of the rumen possesses the broad enzymatic capabilities required for digesting the various plant components ingested by ruminants. Some nitrogen-fixation activity has also been noted in the rumen, but the amounts are too low to make a significant contribution to the nutrition of the animal.

A large population of protozoa is also found in intricate association with the processes in the rumen. Most are ciliates, but some flagellates, such as *Eutodinium*, *Diplodinium* and *Sarcodina*, are also present. Rumen ciliates are a highly specialized group that grow anaerobically, ferment plant materials for energy and tolerate the presence of dense bacterial populations. Some protozoan populations within the rumen are capable of digesting cellulose and starch; others ferment dissolved carbohydrates. Some are predators on bacterial populations. The proteins of the rumen protozoa are, in turn, digested by the ruminant's

**Table 9.3:** Fermentation products and the associated microbes in the rumen of animals.

| Microorganism                       | Major fermentation products           |
|-------------------------------------|---------------------------------------|
| <i>Bacteroides succinogenes</i>     | Acetate, succinate                    |
| <i>Bacteroides amylophilus</i>      | Acetate, succinate, formate           |
| <i>Bacteroides rumenicola</i>       | Acetate, succinate, formate           |
| <i>Ruminococcus flavefaciens</i>    | Acetate, succinate, formate, hydrogen |
| <i>Succinivibrio fibrisolvens</i>   | Acetate, succinate                    |
| <i>Succinimonas amylolytica</i>     | Succinate                             |
| <i>Ruminococcus albus</i>           | Acetate, formate, ethanol, hydrogen   |
| <i>Butyrivibrio fibrisolvens</i>    | Formate, hydrogen, butyrate           |
| <i>Eubacterium ruminantium</i>      | Formate, butyrate, lactate            |
| <i>Selenomonas ruminantium</i>      | Acetate, propionate, lactate          |
| <i>Veillonella alcalescens</i>      | Acetate, propionate, hydrogen         |
| <i>Streptococcus bovis</i>          | Lactate                               |
| <i>Lactobacillus vitulinus</i>      | Lactate                               |
| <i>Methanobacterium ruminantium</i> | Methane                               |

Adopted from Wolin (1979).

enzymes. Rumen protozoa store large amounts of carbohydrates, which the ruminant digests along with the proteins of the protozoan biomass.

The high diversity of microbial populations in the rumen allows the microbial community to respond to changes in the ruminant's diet. Some ingested plant materials contain large amounts of cellulose, others large amounts of hemicellulose materials and still others, large amounts of starches. The relative proportions of microbial populations within the rumen shift according to the nature of the plant materials ingested. Abrupt changes in diet, such as the change from dry hay in winter to pasture grass in spring, can upset the rumen fermentation system, resulting in excessive production of methane that can distend the rumen, sometimes to the extent that it compresses the lungs, suffocating the animal.

### 9.6.3 Microbe-animal Symbiosis

There exists symbiotic association of some invertebrate animals with photosynthetic microorganisms, including unicellular algae and cyanobacteria, photosynthetic bacteria, chemolithotrophs and methanotrophic microbes. These microorganisms are endozoic and are reported in various species of polychaete worms, platyhelminth worms, molluscs (including bivalves such as the giant clam), tunicates, echinoderms, hydroids, jellyfish, anemones, corals, sea fans and sponges. In these associations, the microorganism supplies the animal with organic nutrients and the animal provides a physiologically and nutritionally suitable environment for the microorganism. In some cases, morphological adaptations facilitate close contact between the microbial and animal cells, allowing efficient materials transfer between the microorganisms and the animals (See Chapter 10).

## 9.7 PATHOGENIC INTERACTIONS OF MICROBES WITH PLANTS

Microbial diseases of plants, whether caused by viruses, bacteria, fungi or protozoa, are of ecological and economic importance. The development of plant diseases due to microbial pathogens normally follows a pattern of initial contact of the microorganism with the plant, entry of the pathogen into the plant, growth of the infecting microorganisms and development of plant disease symptoms. Pathogenic microorganisms may contact the plant in the rhizoplane or the phylloplane. Because most fungal plant pathogens are dispersed through the air as spores, they often contact the leaves or stems of plants. Most viral plant diseases are transmitted by insect vectors; most of these pathogens contact the plant at the phylloplane. Some bacterial and fungal pathogens are also carried by insect vectors. Soil-borne animals, such as nematodes, transmit some plant pathogens, contacting the plant at the rhizoplane. Motile pathogens in the soil, including plant pathogenic bacteria such as *Pseudomonas* species and fungi, such as *Oidium*, are attracted to plant roots via chemotaxis and contact the plant at the rhizoplane.

Spores of plant pathogenic fungi distributed by air currents need to attach themselves to leaves or stems of susceptible plants. The conidiospores of the ascomycete *Magnaporthe grisea*, causative agent of the rice blast disease, efficiently attach themselves to hydrophobic surfaces such as the leaf cuticle or a cuticle-simulating membrane by means of a mucilage stored in the spore tip for this purpose. The adhesion of the conidiospores of the plant pathogenic fungus *Botrytis cinerea* to the cuticle of tomato leaves occurs by weak hydrophobic interactions. Plant pathogens may enter plants through wounds or natural openings, such as the stomata. Many viruses enter plants through wounds caused by the insect vector carrying the virus, although others enter through the roots with the water being taken up by the plant. Some plant pathogens are able to penetrate the cuticle of the plant directly. Such penetration involves attachment of the pathogen to the plant surface, followed by formation of a penetration peg, which passes through the cuticle and the cell wall.

Microbial pathogens, that have successfully entered the plant, disrupt normal plant function by producing degradative enzymes, toxins and growth regulators. Soil-borne plant pathogens produce pectinases, cellulases and hemicellulases that result in degeneration of the plant structure, producing soft rots and other lesions. Toxins produced or induced by pathogenic organisms in plants interfere with the normal metabolic activities of the plant. The toxin produced by *Pseudomonas tabaci*, which causes tobacco wildfire disease, has been characterized as  $\beta$ -hydroxydiaminopimelic acid. It interferes with the metabolism of methionine. Some plant pathogenic fungi produce highly selective toxins, including low molecular weight cyclic peptides and linear polyketols, which interfere with mitochondria and cell membranes.

As a result of microbial infections, plants can develop a variety of morphological or metabolic abnormalities. Invasion of plant cells by pathogenic microorganisms sometimes results in rapid death of the plant. Plant pathogens may also alter the metabolic activities of the plant. Diseased plants sometimes show changes in respiratory activity, which may be caused by electron transport uncoupling or changes in the glycolytic pathways of carbohydrate metabolism. Plant pathogens may interfere with carbon dioxide fixation, protein synthesis, nucleic acid function and activities of the enzymes.

Plants under attack of microbial pathogens may react by synthesizing polyphenolic, flavonoid, or proteinaceous antimicrobial compounds called *phytoalexins*. Their synthesis typically starts in the cells and at the sites that are under direct microbial attack, but their synthesis may spread to adjacent cells and the whole plant. Thus, the phytoalexins help to curb the original infection and also protect the plant against subsequent microbial challenges. To viral infections, plants respond typically by synthesis of antiviral proteins.

Plant pathogenic bacterial species occur in the genera *Mycoplasma*, *Spiroplasma*, *Corynebacterium*, *Agrobacterium*, *Pseudomonas*, *Xanthomonas*, *Streptomyces* and *Erwinia*. These bacteria are widely distributed and cause a number of plant diseases, including hypertrophy, wilts, rots, blights and galls (Table 9.4). Because most plant pathogenic bacteria do not form resting stages,

they must remain in intimate contact with plant tissues during all stages of their life history. Many plant pathogenic bacteria can remain viable on plant seeds and in other plant tissues during times of plant dormancy because they are resistant to desiccation. Many are obligate parasites and are unable to compete with saprophytic soil bacteria. Other opportunistic bacterial pathogens of plants are able to reproduce in soil, often causing plant diseases only after sufficient populations have developed in the soil.

**Table 9.4:** Some common bacterial diseases of plants with the concerned microorganisms.

| Organism                          | Disease                         |
|-----------------------------------|---------------------------------|
| <i>Pseudomonas tabaci</i>         | Wildfire of tobacco             |
| <i>P. angulata</i>                | Leaf spot of tobacco            |
| <i>P. phaseolicola</i>            | Halo blight of beans            |
| <i>P. pisi</i>                    | Blight of peas                  |
| <i>P. glycinea</i>                | Blight of soybeans              |
| <i>P. syringae</i>                | Blight of lilac                 |
| <i>P. solanacearum</i>            | Moko of banana                  |
| <i>P. caryophylli</i>             | Wilt of carnation               |
| <i>P. marginalis</i>              | Slippery skin of onion          |
| <i>P. sevastopol</i>              | Olive knot disease              |
| <i>P. marginata</i>               | Scab of gladiolus               |
| <i>Xanthomonas phaseoli</i>       | Blight of beans                 |
| <i>X. oryzae</i>                  | Blight of rice                  |
| <i>X. pruni</i>                   | Leaf spot of fruits             |
| <i>X. juglandis</i>               | Blight of walnut                |
| <i>X. citri</i>                   | Canker of citrus                |
| <i>X. campestris</i>              | Black rot of crucifers          |
| <i>X. vascularum</i>              | Gumming of sugarcane            |
| <i>Streptomyces scabies</i>       | Scab of potato                  |
| <i>S. ipomoeae</i>                | Pox of sweet potato             |
| <i>Erwinia amylovora</i>          | Fire blight of pears and apples |
| <i>E. tracheiphila</i>            | Wilt of cucurbits               |
| <i>E. stewartii</i>               | Wilt of corn                    |
| <i>E. carotovora</i>              | Blight of chrysanthemum         |
| <i>Corynebacterium insidiosum</i> | Wilt of alfalfa                 |
| <i>C. michiganese</i>             | Wilt of tomato                  |
| <i>C. facians</i>                 | Leafy gall of ornamentals       |
| <i>Agrobacterium tumefaciens</i>  | Crown gall of various plants    |
| <i>A. rubi</i>                    | Cane gall of raspberries        |
| <i>A. rhizogenes</i>              | Hairy root of apple             |
| <i>Mycoplasma</i> sp.             | Aster yellows                   |
| <i>Mycoplasma</i> sp.             | Elm phloem necrosis             |
| <i>Spiroplasma</i> sp.            | Citrus stubborn disease         |
| <i>Spiroplasma</i> sp.            | Bermuda grass witches' broom    |
| <i>Burkholderia cepacia</i>       | Sour skin of onion              |

Modified from Atlas and Bartha (2005).

Most plant diseases are caused by pathogenic fungi (Table 9.5). Perhaps the most important economic fungal diseases of plants are caused by the rusts and smuts. These fungi are Basidiomycota and have a complex life cycle. There are more than 20,000 species of rust fungi and more than 1000 species of smut fungi. Rust fungi require two unrelated hosts for the completion of their normal life cycle. Important diseases caused by smut fungi include loose smut of oats, corn smut, bunt or stinking smut of wheat and onion smut.

## **9.8 PATHOGENIC INTERACTIONS OF MICROBES WITH ANIMALS**

Some microorganisms, including various viruses, bacteria, fungi, protozoa and even algae, cause animal diseases. Two types of disease-causing processes are seen in animals, one in which the microorganism grows on or within the animal, causing an infection that results in a disease condition and a second in which the microorganism grows outside of the animal, producing toxic substances that cause animal diseases or altering the habitat of the animal so that the animal can no longer survive in a healthy condition.

Microorganisms growing in their natural habitats, such as water or soil, can alter environmental conditions, adversely affecting the animal populations occupying those habitats. Such conditions often arise when there are population imbalances within the microbial community. For example, when eutrophic conditions allow for large algal blooms in lakes, the growth of photoautotrophic populations adds large quantities of organic matter to the lake waters and the subsequent degradation of this organic matter by heterotrophic microorganisms depletes the dissolved oxygen, creating anoxic conditions. This depletion produces disease symptoms and death in the oxygen requiring animal populations, sometimes causing major fish kills.

Microbial populations may also produce inorganic chemicals or organic toxins that cause diseases in animals. Hydrogen sulphide produced by microbial populations in sediment can accumulate in concentrations toxic to animals burrowing in the sediment. Acid mine drainage, resulting from microbial oxidation of sulphur, can produce highly acidic conditions in drainage streams, causing disease conditions and death of aquatic animals. Toxins produced by microorganisms outside of susceptible host animals often enter the animal during ingestion of food, causing food poisoning. Many mushrooms contain toxins that are extremely poisonous to animals that ingest them, resulting in severe disease symptoms and sometimes death. Microbial toxins can remain active even under conditions that do not permit the survival and the growth of the population that produced them.

Toxins are often not produced in sufficient concentrations or are degraded or diluted so that they do not produce diseases in animal populations. Only when conditions permit extensive toxin production by microbial populations, toxins accumulate in concentrations that cause animal diseases. In some cases, toxins produced by microorganisms are concentrated within the food web, producing

**Table 9.5:** Some important fungal pathogens and the plant diseases caused by them.

| <i>Myxomycota &amp; Phycomycota</i> |                                | <i>Ascomycota</i>     |                           | <i>Basidiomycota</i> |                            | <i>Deuteromycota</i>    |                                |
|-------------------------------------|--------------------------------|-----------------------|---------------------------|----------------------|----------------------------|-------------------------|--------------------------------|
| <i>Fungus</i>                       | <i>Disease</i>                 | <i>Fungus</i>         | <i>Disease</i>            | <i>Fungus</i>        | <i>Disease</i>             | <i>Fungus</i>           | <i>Disease</i>                 |
| <i>Plasmodiophora</i>               | Clubfoot of crucifers          | <i>Taphrina</i>       | Peach leaf curl           | <i>Ustilago</i>      | Smut of corn and wheat     | <i>Phoma</i>            | Black leg of crucifers         |
| <i>Polymyxa</i>                     | Root disease of cereals        | <i>Erysiphe</i>       | Powdery mildew of grasses | <i>Tilletia</i>      | Stinking smut of wheat     | <i>Colletotrichum</i>   | Anthrachnose of crops          |
| <i>Spongospora</i>                  | Powdery scab of potatoes       | <i>Microsphaera</i>   | Powdery mildew of lilac   | <i>Sphacelotheca</i> | Loose smut of sorghum      | <i>Cylindrosporium</i>  | Leaf spot of various plants    |
| <i>Olpidium</i>                     | Root disease of various plants | <i>Podosphaera</i>    | Powdery mildew of apple   | <i>Urocystis</i>     | Smut of onion              | <i>Alternaria</i>       | Leaf spot and blight of plants |
| <i>Synchytrium</i>                  | Black wart of potato           | <i>Ceratocystis</i>   | Dutch elm disease         | <i>Puccinia</i>      | Rust of cereals            | <i>Aspergillus</i>      | Rot of seeds                   |
|                                     |                                | <i>Diaporthe</i>      | Bean pod blight           | <i>Cronartium</i>    | Pine blister rust          | <i>Botrytis</i>         | Blight of various plants       |
| <i>Urophlyctis</i>                  | Crown wart of alfalfa          | <i>Endothia</i>       | Chestnut blight           | <i>Uromyces</i>      | Rust of beans              | <i>Cladosporium</i>     | Leaf mold of tomato            |
| <i>Physoderma</i>                   | Brown spot of corn             | <i>Claviceps</i>      | Ergot of rye              | <i>Exobasidium</i>   | Stem galls of ornamentals  | <i>Fusarium</i>         | Root rot of many plants        |
| <i>Pythium</i>                      | Seed decay, root rots          | <i>Dibotryon</i>      | Black knot of cherries    | <i>Fomes</i>         | Heart rot of trees         | <i>Helminthosporium</i> | Blight of cereals              |
| <i>Phytophthora</i>                 | Blight of potato               | <i>Mycosphaerella</i> | Leaf spots of trees       | <i>Polyporus</i>     | Blight of turf grasses     | <i>Penicillium</i>      | Blue mold rot of fruits        |
| <i>Plasmopora</i>                   | Downy mildew of grapes         | <i>Ophiobolus</i>     | Take all of wheat         | <i>Armillaria</i>    | Root rots of trees         | <i>Theilaviopsis</i>    | Black root rot of tobacco      |
| <i>Rhizopus</i>                     | Soft rot of fruits             | <i>Diplocarpon</i>    | Black spot of roses       | <i>Marasmius</i>     | Fairy ring of turf grasses | <i>Verticillium</i>     | Wilt of various plants         |
|                                     |                                | <i>Lophodermium</i>   | Pine needle blight        |                      |                            | <i>Rhizoctonia</i>      | Root rot of various plants     |
|                                     |                                | <i>Sclerotinia</i>    | Soft rot of vegetables    |                      |                            |                         |                                |

Modified from Atlas and Bartha (2005).



diseases in higher consumer populations with no apparent effect on lower members of the food web. Environmental conditions that favour the growth of toxin-producing microbial populations increase the possible incidence of toxin-induced animal diseases.

In contrast to animal diseases caused by toxin-producing microorganisms, that grow outside the animal, infectious pathogens or parasitic microorganisms must be capable of growing in or on the animal's tissues. Some pathogenic microorganisms are obligate intracellular parasites and are entirely dependent, for their existence and survival, on their successful invasion and reproduction within susceptible animal cells. Although stable carrier states can occur, in which a balance between the pathogen and the host is established, disease-causing infectious microorganisms usually grow for only a limited period within susceptible animals, after which the host animal either dies, resulting in destruction of the habitat for the pathogenic microorganisms, or develops an immune response that precludes further growth of the pathogenic microorganisms. Therefore, populations of pathogenic microorganisms must be transmitted to new susceptible members of the animal population for continued growth. It is important to consider the routes of transfer, how long the pathogens can survive outside of host animals, the environmental factors that contribute to survival outside the host and the reservoirs or alternate host populations for the particular pathogenic population.

Pathogenic microorganisms usually enter animals through normal body openings, such as the respiratory and gastrointestinal tracts. Most pathogens are unable to penetrate the outer skin layers and establish infection, but there are exceptions, including a ciliate *Lchthyophthirius multifiliis* that is able to penetrate fish skin surfaces and causes ich, an often fatal disease of tropical fish characterized by the appearance of white spots on the skin, fins and eyes. Most pathogenic microorganisms are restricted to one portal of entry for establishing disease; entering another part of the body leads to a different relationship between the microbe and the animal. In some cases, a microbial population is nonpathogenic under normal conditions but can become an opportunistic pathogen under particular circumstances. For example, *Escherichia coli* is a normal inhabitant of the human intestine, where it does not produce disease, but if it enters the urinary tract, it causes a urinary tract infection.

Although the tissues of healthy animals and humans are sterile, numerous microorganisms colonize their body surfaces. The best-studied example is the human skin. The sloughing off of epidermal cells and the secretion of sebaceous and sweat glands provide keratin, lipids and fatty acids as potential growth substrates. However, generally dry conditions, salinity and inhibitory action of some fatty acids create a somewhat hostile environment that is best tolerated by some gram-positive bacteria and coryneforms and by some yeasts. Most of these microorganisms are harmless commensals, but some of them like *Staphylococcus aureus* and *Candida albicans* have the potential of becoming opportunistic pathogens in debilitated or immunocompromised individuals or if the skin's integrity is disrupted by wounds or burns. Large differences exist in the density

and species composition of the microbial populations of various skin areas, caused by variations in the dryness or moistness of different skin areas and in the density of secretory glands. *Staphylococcus* and *Corynebacterium* are the most abundant bacterial genera; gram negatives are restricted to the moister areas of the skin. The most common yeasts are *Pityrosporum ovale* and *C. albicans*.

## 9.9 CONCLUSION

Microorganisms are important contributors to the nutrition of plants and animals. They interact with each other to make the best use of the resources in the habitat in which they grow. Interactions of microorganisms and plant roots are especially important in providing nutritional requirements for the plant and its associated microorganisms. The rhizosphere microbial populations are most important to mobilize nutrients and supply them to the plant. Microorganisms within the rhizosphere are higher in numbers and physiologically different from those in root-free soil. The presence of rhizosphere microorganisms is also important to the plants, enabling some plants to grow better than otherwise would be possible.

In addition to interacting with numerous microorganisms in the rhizosphere, the roots of many plants establish a mutualistic as well as pathogenic interactions with many fungi, in which some fungi actually become an integral part of the plant. The establishment of mutualistic relationships of certain plants with nitrogen-fixing bacteria, most notably the relationship between legumes and *Rhizobium*, is extremely important to the ecology of other organisms in the biosphere. Without these nitrogen-fixing activities or human intervention through the introduction of chemical fertilizers, the abilities of all organisms to produce proteins would be highly constrained.

Plant pathogens have great ecological and economic impact as the microbial diseases are responsible for the decrease of the plant productivity. Because obligately plantpathogenic microorganisms have a limited period of viability outside of host plant tissues, it is possible to control plant pathogens of agricultural crops by appropriate management procedures. These procedures are needed to be properly standardized to improve the efficiency of disease control.

Many animals lack the enzymes necessary to digest some or all of the food resources available to them and thus require the assistance of microbial populations. Of particular importance is the ability of various microbial populations to produce the extracellular enzymes that degrade complex plant polymers, such as cellulose, which otherwise would be unavailable to the animals. A number of animal populations depend on the microbial conversion of plant polymers to fatty acids that can be used for energy production. Some microbial populations also produce vitamins, which are required but not synthesized by the animal host. The activities of microbial populations in the gastrointestinal tract thus contribute to food digestion, supply vitamins and provide considerable protection against pathogenic invaders. Some of these microbes may have important bearing on the animal nutrition which needs further research.

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# SYMBIOTIC INTERACTIONS OF MICROBES

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## 10.1 INTRODUCTION

The term symbiosis is used to denote any intimate relationship between two populations. In a global context; all symbiotic interactions can be viewed as beneficial because they act to maintain ecological balance. However, interactions among specific populations are often viewed in a less generic sense. The categories used to describe these interactions represent a conceptual classification system; many specific cases are difficult to classify without ambiguity. Symbiotic relationship between populations can be considered as an extended synergism. It is an obligatory relationship between the two populations that benefits both. A mutualistic relationship is highly specific. Generally, a member of an association can not be replaced by another related species, except in case of bacterial associations, which is not considered as true symbiosis as it is an extended mutualistic relationship.

Symbiotic interaction requires close physical proximity between the interacting populations. Relationships of symbiosis allow organisms to exist in habitats that could not be occupied by either population alone. This does not exclude the possibility that the populations may exist separately in other habitats. The metabolic activities and physiological tolerance of populations involved in mutualistic relationships are normally quite different from either of population by itself. Symbiotic relationships between microorganisms allow the microorganisms to act as if they were a single organism with a unique identity. According to the theory of serial symbiosis, some mutualistic endosymbiotic relationships had key roles in the evolution of higher organisms.

Microbes are unique in their ability to establish symbiotic associations with a remarkable range of hosts, which include both unicellular and multicellular forms. Symbiotic interactions are seen between microbial species as well as between microbes (Bacteria, Cyanobacteria, Algae and Fungi) with plants,

sponges and animals. In most of the cases, the benefit to the host is clear but many different explanations are given for the benefit enjoyed by the microbial partner from the interaction. For example, in case of symbiotic association of cyanobacteria with eukaryotic hosts, the latter is invariably benefited metabolically, mostly in the form of combined nitrogen, or occasionally carbon but the benefit to the cyanobacteria is less clear. The cyanobacteria is probably benefited through the protection from the environmental extremes and occasionally the availability of carbon from the photosynthetic hosts, though the cyanobacteria themselves are able to fix inorganic carbon.

In almost all symbiotic interactions, one of the partners is a microbe while the other may be a prokaryote, plant or animal. Broadly, the symbiosis among organisms can be grouped into three types:

1. microbe-microbe,
2. plant-microbe and
3. animal-microbe interactions.

## **10.2 MICROBE-MICROBE SYMBIOSIS**

### **10.2.1 Symbiotic Relations between Bacterial Populations**

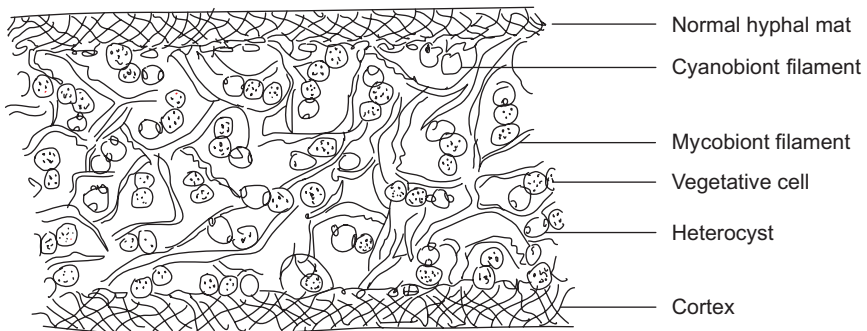
A relationship between two diverse populations of bacteria is less symbiotic and more synergistic. This interaction has been described in detail in the previous chapter (Chapter 9). Both the partners of the association are more or less benefited from the relationship. However, unlike true symbiosis, the association is not an obligatory one and can break any time even in the same habitat. Both populations are capable of surviving in their natural environment on their own, although when initially formed, the association offers some mutual advantages. Synergistic relationships are also loose in the sense that one population is often easily replaced by another giving significant flexibility to the relationship. In some cases, it is difficult to determine whether one of the populations is indeed benefiting and thus whether the relationship should be considered commensal or synergistic. In other cases, it is difficult to determine whether the relationship is obligatory and thus should be considered one of symbioses. The term syntrophism is applied to the interaction of two or more populations that supply each other's nutritional needs. Syntrophism may allow microbial populations to perform activities, such as the metabolism of a substrate or synthesis of a product, that neither population could perform alone. The synergistic activities of two microbial populations may allow completion of a metabolic pathway that otherwise could not be completed by individual population.

### **10.2.2 Algae-Fungi Interaction**

The relationships between certain algae or cyanobacteria and fungi, that result in the formation of lichens, are probably the most outstanding example of symbiotic, intermicrobial relationships. Morphologically, there are three major types of

lichens. Crustose lichens adhere closely to their substratum; foliose lichens are leafy in form and are attached to their substrata more loosely; and fruticose lichens often consist of hollow upright stalks and are the least attached to their substrata.

Lichens are composed of a primary producer (algae) and a consumer (fungi). The algal partner (the phycobiont) uses light energy and produces organic compounds that are used by the fungal partner (the mycobiont). The mycobiont provides a form of protection and mineral nutrient transport for the phycobiont. In some cases, the mycobiont also makes growth factors available to the phycobiont. The algal and fungal members of the lichen form distinct layers that function as primitive tissues (Fig. 10.1). The phycobiont may be a member of the cyanobacteria, the Chlorophyta, or the Xanthophyta. Members of at least four genera of filamentous, heterocystous cyanobacteria are known to be involved, viz., *Nostoc*, *Calothrix*, *Scytonema* and *Fischerella*, with *Nostoc* being very common phycobiont. Of the unicellular cyanobacteria, the most common are *Gloeocapsa*, *Gloeotheca* and *Synechocystis*, although their exact taxonomic status is often uncertain. A recent comparison of PCR amplified tRNA (UAA) introns from the *Nostoc* symbionts of the lichens *Nephroma arcticum*, *Peltigera aphthosa*, *P. membranacea* and *P. canina* reveals that a single symbiont is present in an individual thallus. Closely related cyanobionts are also present in the same lichen collected from different locations implying a high degree of specificity in the choice of *Nostoc* symbiont. The green algae are mostly the members of chlorophyceae; *Chlorella*, *Scenedesmus*, *Ankistrodesmus* and some colonial members.



**Figure 10.1** - Transverse section of a fruticose lichen thallus showing the distribution of the algal and fungal partners.

In the case of the unique fungal association in *Geosiphon pyriforme*, only *Nostoc* spp. seem to be cyanobionts. It is interesting that in nature *Geosiphon* occurs at the same sites as *Blasia* and *Anthoceros* and there is evidence that the *Geosiphon* cyanobionts can infect the bryophytes and vice versa.

The mycobiont is typically an ascomycete or a basidiomycete, but a few zygomycetes also form lichen associations. An analysis of DNA homologies among lichen-forming fungi indicates that they fall into at least five separate

groups. This makes it likely that lichen associations have at least five separate evolutionary origins. In contemporary lichens, a specificity in partner selection is evident but not absolute; a given algal species may be able to form a lichen association with any of several compatible fungal species and vice versa and in some lichens multiple phycobiont and/or mycobiont populations may be present. The association can be as unspecific as having taxonomically distant genera belonging both to the algal and fungal partners.

The metabolically tight association between phycobiont and mycobiont in lichens make them to survive and grow in dry and warm conditions. It is possible to dissociate the two partners and, in some cases, to reassociate the mutualistic populations but dissociation eliminates the chance of survival of the partners in that habitat though survival in their favourable habitat is possible. During reassociation of the lichen, which must be carried out under carefully controlled conditions, the flattened portions of the fungal hyphae (appressoria) make contact with the algal cells. If the reassociation is successful, all the structures typical of the lichen thallus are reestablished, including fruiting bodies and soredia. Studies on lichen association suggest that the lichen symbiosis is actually a case of controlled parasitism in which the alga has developed a degree of resistance to the parasitic fungus, such that the percentage of algal cells killed is balanced by the production of new cells. This causes very slow growth of lichens but lichens are able to colonize habitats that do not permit the growth of other microorganisms. The growth of the algal partner proceeds synchronously with the growth of lichen but the fungal partner grows at a much better rate and take active part in the formation of soredia and fruiting bodies. Under growth limiting conditions for the algal partner, the relationship is converted to somewhat parasitism, the fungal partner living successfully at the cost of the phycobiont.

Fixation of atmospheric nitrogen by some lichens is known and in some habitats they are an important source of combined nitrogen. For example, in tundra soils the lichen *Peltigera*, containing the cyanobacterium *Nostoc*, provides a major source of fixed nitrogen for the biological community. Nitrogen-fixing lichens growing in the forest canopy provide fixed forms of nitrogen for the underlying vegetation. During rainstorms, the fixed forms of nitrogen are leached from the lichens and washed to the forest floor, where the nitrogen is taken up by plant roots. However, such contribution of nitrogen by lichen is not so significant and are usually not a controlling factor of nitrogen supply to the forest vegetation.

The mutualistic association in lichens is very delicately balanced and can be disrupted by environmental changes. Although lichens are able to occupy some hostile habitats, they are particularly sensitive to industrial air pollutants and have been disappearing from industrialized areas very fast mostly due to industrial emissions. It appears that the sulphur dioxide in the atmosphere is inhibitory to the lichens, probably because it inhibits the phycobiont. The reduced efficiency of photosynthetic activity of the phycobiont allows the mycobiont to overgrow it, which leads to the elimination of the mutualistic relationship. Following destruction of the phycobiont, the fungus is unable to survive alone and is also eliminated from the habitat.



### 10.2.3 Algae-Protozoa Symbiosis

Algae, as endosymbionts, establish an interesting mutualistic relationship with protozoa. Many members of this association are unicellular, benefiting tremendously from the association. *Paramecium* can host numerous cells of the alga *Chlorella* within its cytoplasm. The alga provides the protozoan with readily assimilable organic carbon and oxygen and the protozoan presumably provides the alga with protection (photoprotection), motility, carbon dioxide and perhaps other growth factors. The presence of *Chlorella* within the ciliate allows the protozoan to move into anaerobic habitats as long as there is sufficient light. Without the alga, the protozoan can not enter and survive in such habitats. Similarly, some foraminiferans enter into mutualistic relationships with Pyrrophyta or Chrysophyta. The presence of the algae confers a red colour on the protozoa. Presumably, the symbiotic relationship is based on the ability of the alga to supply oxygen and photosynthate and on the ability of the foraminiferan to protect the algal partner from grazers. Under normal environmental conditions, such symbiotic relationships between algae and protozoa persist; but if stressed, for instance by a prolonged absence of light, the protozoan population may digest the algal population, thus converting the symbiotic relationship to parasitism.

Many other protozoa are hosts to endosymbiotic algae and cyanobacteria. Each protozoan can contain 50–100 algal cells. The algae found within freshwater protozoa normally belong to the Chlorophyta and are called zoochlorellae. The endosymbiotic algae of marine protozoa are most frequently dinoflagellates (Pyrrophyta) but can be Chrysophyta. These endozoic marine algae are called zooxanthellae. Endozoic cyanobacteria, which occur in both freshwater and marine protozoa, are called cyanellae.

### 10.2.4 Bacteria-Protozoa Symbiosis

The bacterial nature of many of the endosymbionts of protozoans has been established by electron microscopic examination, but many of the observable bacteria have yet to be isolated and cultured. Some multiply within the nucleus of the protozoan, others within the cytoplasm. The bacterial endosymbiont appears to contribute to the nutritional requirements of the protozoan. In the case of the flagellate protozoans *Blastocrithidia* and *Crithidia*, the bacterial endosymbiont provides hemin and other growth factors to the protozoa. Populations of these protozoa lacking endosymbiotic bacteria require external sources of these growth factors.

One of the most interesting mutualistic relationships between a protozoa and a bacterium is between *Paramecium aurelia* and a *Caedibacter* endosymbiont, formerly known as a kappa particle. *P. aurelia* populations occur in two forms: killers and sensitives. Killer *P. aurelia* populations contain endosymbionts but the sensitive strains lack them. The nature of the toxic substance associated with killer strains and the mechanisms by which the presence of the endosymbiont confers immunity, are not fully known. All *Caedibacter* endosymbionts that



confer the killer trait exhibit a strongly refractile inclusion body, known as an R body. The genetic information for the R body is coded on a *Caedibacter* plasmid. Although the R body itself does not appear to be the toxin, evidence indicates that the R body and the toxin are specified by the same coding region of the plasmid. It is also established that the R body is a requirement for the expression of the killer trait. The presence of *Caedibacter* endosymbionts gives an advantage to killer strains in competition with sensitive strains. The obligately endosymbiotic *Caedibacter* strains are nutritionally dependent on their protozoan host.

Many anaerobic ciliate protozoa living within the rumen, possess endosymbiotic methanogens as obligate association partner. These mutualistic methanogenic archaea belong to the genera *Methanobacterium*, *Methanocorpusculum* and *Methanoplanus*, that are different from free-living methanogens mostly in their ability to use substrates for autotrophic growth. In these relationships, intracellular methanogens appear to be much more numerous than those attached to the external cell surfaces of ciliate protozoa. It is likely that endosymbiotic methanogens can directly use molecular hydrogen produced by the ciliate protozoan (many anaerobic protozoa have organelles that generate hydrogen as a part of the protozoan cellular energy-generating metabolism). There appear to be morphological interactions between the endosymbiotic methanogens and the protozoa that facilitate material exchange. Availability of hydrogen provides opportunity to the endosymbionts to grow independently of the supply of organic carbon source.

### 10.2.5 Algae-Algae Symbiosis

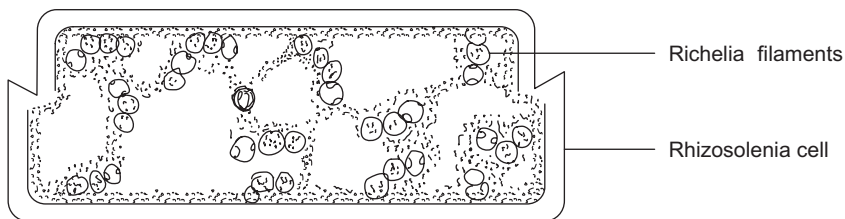
Planktonic green algae and cyanobacteria form symbiotic associations with a wide range of eukaryotic hosts and in the majority of cases the cyanobacterial symbionts (cyanobionts) are capable of nitrogen fixation, the products of which become available to the host. As a consequence of their oxygenic photosynthesis, cyanobacteria have evolved several mechanisms for protecting nitrogenase from oxygen inactivation. The most remarkable of these is the heterocyst, which develops at regular intervals within filaments of vegetative cells and provides an anaerobic environment suitable for nitrogen fixation. The specialised nature of the heterocyst means that it must rely on neighbouring vegetative cells for the provision of fixed carbon, while in turn supplying these cells with fixed nitrogen. This interdependence provides an interesting association; parallel with the cyanobacteria-plant symbioses, in which the cyanobiont provides fixed nitrogen to the host, which provides fixed carbon in return.

The infective agents in many cyanobacterial symbioses, particularly those involving plants, are specialised filaments known as hormogonia, produced by heterocystous cyanobacteria in the Nostocaceae and Stigonemataceae. Hormogonia lack heterocysts but possess gliding motility, a crucial factor in their ability to establish symbioses, because the long, immotile parent filaments are unable to gain entry to the plant structures that house the symbiotic colonies. Hormogonia formation and hormogonia themselves are responsive to plant

extracellular signals and it may be for these reasons that plant cyanobionts are almost always members of the genus *Nostoc*.

In most of the interalgal symbioses, the symbionts are capable of independent growth, although in some cases they have defied laboratory cultivation. However, there are two exceptions: the prochlorophytes and the cyanelles. The former are a group of oxygenic photosynthetic prokaryotes that closely resemble cyanobacteria, but differ in the possession of both chlorophylls *a* and *b* and the lack of phycobilin pigments. The latter is one of a range of plastids found in the cells of photosynthetic eukaryotes and was once thought to be a cyanobacterial endosymbiont. Cyanelles indeed possess many similarities with cyanobacteria, yet with genomes of only 130 to 140 kbp, they are incapable of independent growth. It is established that cyanelles evolved from a cyanobacterial ancestor and represent the extreme of adaptation to a symbiotic lifestyle in which the symbiont has lost much of its own genetic capacity and become dependent on the host.

**(a) Cyanobacteria-diatoms association:** The marine pinnate diatom *Rhizosolenia* frequently forms a symbiotic association with the filamentous, heterocystous cyanobacterium *Richelia intracellularis*, which fixes both CO<sub>2</sub> and N<sub>2</sub> for the host. Each cyanobacterial filament, of which there can be several in each diatom cell, usually contains a single basal heterocyst, although occasionally one is present at each end. The apex of *Rhizosolenia clevei* contains cytoplasm only in the region occupied by the cyanobacterium, although the cyanobiont is always external to the host cytoplasm and its cell wall remains intact (Fig. 10.2). The level of glutamine synthetase is very low in the cyanobiont and this may result in release of fixed nitrogen to the host. However, the presence of *Richelia* is not essential for the growth and survival of *Rhizosolenia*, because the diatom can be found without the cyanobiont. *Richelia intracellularis* is also found in diatoms of the genus *Hemiaulus*, which was once thought to be a relatively uncommon and unstable symbiosis.

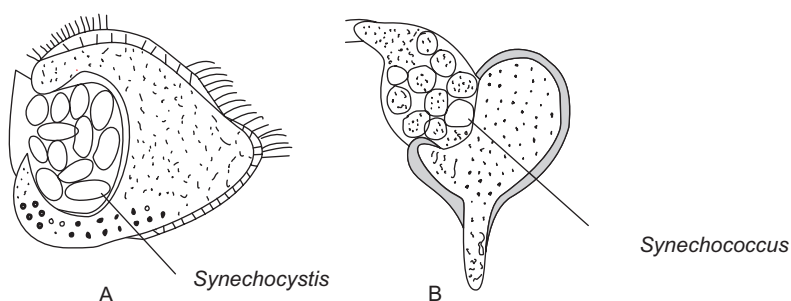


**Figure 10.2** - Distribution of *Richelia* filaments inside a *Rhizosolenia* cell.

Two diatoms of the family Epithemiaceae, *Rhopalodia gibba* and *Epithemia turgida*, possess unicellular cyanobacterial symbionts in their cytoplasm. The cyanobionts have not been isolated or identified, but appear to be capable of nitrogen fixation. Their number per diatom cell ranges from 1–8 and increases in response to decreasing external nitrate concentration. However, this increase does not occur under phosphorus limitation, possibly due to reduced cyanobiont

growth rate. This diatom-cyanobacterium symbiosis is clearly at an advantage in nitrogen-limited habitats, but probably at a disadvantage when nitrogen supply is adequate. Although the number of cyanobionts is then reduced, at least one remains in each diatom cell, presumably at a metabolic cost to the host.

**(b) Cyanobacteria-dinoflagellates association:** Cyanobacterial symbionts known as phagosomes are found in a variety of tropical marine dinoflagellates. In the subtropical waters of the Gulf of Aqaba, heterotrophic dinoflagellates of genera *Ornithocercus*, *Histioneis* and *Citharistes* have been shown to contain cyanobacterial symbionts, thought to be members of the unicellular genera *Synechococcus* and *Synechocystis*. The cyanobacteria are located between the upper and lower lists of the horizontal groove of cells of *Ornithocercus* and *Histioneis* and within special chambers inside the cells of *Citharistes* (Fig. 10.3). The cyanobacteria were thought to be benefited by the provision of a suitable microaerobic environment for the efficient functioning of nitrogenase. However, recent immunolocalisation studies with the phagosomes of *Ornithocercus magnificus*, which are cyanobacteria of the unicellular genus *Synechococcus*, failed to detect nitrogenase, even in samples collected at night, when some unicellular cyanobacteria are known to fix nitrogen to avoid the inhibitory effects of photosynthetically generated oxygen during the day. The cyanobionts seem to provide fixed carbon, rather than nitrogen, for their non-photosynthetic host, although they may also be phagotrophically consumed when they senesce.



**Figure 10.3** - Cyanobacterial symbiosis with dinoflagellates: (A) *Synechocystis*—*Histioneis* association; (B) *Synechococcus*—*Citharistes* symbiosis.

**(c) Symbiosis of cyanelles:** Cyanelles are one of a range of plastids found in the cells of photosynthetic eukaryotes and were once thought to be cyanobacterial endosymbionts of their eukaryotic hosts. They are simple carboxysome-like structures with a genome size of 130 to 140 kbp, similar in size to those of chloroplasts. Cyanelles have been found in the cells of those eukaryotes that show neither systematic nor evolutionary relatedness, although the taxonomic position of these eukaryotes is often confused. The obligately photoautotrophic alga *Cyanophora paradoxa* can be grown easily and is consequently well studied.

## 10.3 PLANT-MICROBE SYMBIOSIS

Loose associations between plants and microbes are likely to be widespread. For example, it has long been known that the presence of both free-living (including epiphytic) and symbiotic cyanobacteria in rice fields improves both growth and crop yield of rice. Epiphytic growth of *Gloeotrichia pisum* on submerged roots and stems of deepwater rice has also been observed. Similarly, colonies of the heterocystous genera *Nostoc*, *Gloeotrichia*, *Anabaena*, *Calothrix* and *Cylindrospermum* have been found growing epiphytically on the lower epidermis and the reproductive pockets of duckweed (*Lemna*) leaves and are presumably responsible for the nitrogen-fixing activity associated with duckweed blooms. Many bacteria establish loose association with plant roots and leaves. More than 90 % association of bacteria on the leaf surface is loose and easily separable. However, the intimacy of the association is somewhat high in the rhizosphere, though loose association dominates over true symbiosis in the rhizosphere.

### 10.3.1 Plant-Bacteria Symbiosis

Many rhizospheric bacteria show symbiotic relationships with leguminous as well as non-leguminous plants resulting in the formation of root nodules fixing atmospheric nitrogen. The formation of root-nodule symbiosis in legumes involves legume-*Rhizobium* symbiosis while with non-leguminous plants it occurs with *Rhizobium* populations, cyanobacteria and actinomycetes. *Rhizobium*, for example, can fix nitrogen in association with *Trema*, a tree found in tropical and subtropical regions. Likewise, the actinomycete *Frankia alni* infects the roots of trees like *Alnus*, *Myrica*, *Hippophae*, *Comptonia*, *Casuarina* and *Dryas*, leading to the formation of nodules. The productivity of many forests depends on such nitrogen-fixing symbioses. Actinomycete symbioses are more frequent in temperate and circumpolar regions, whereas cyanobacteria and rhizobial symbioses are more common in tropical and subtropical regions. In *Frankia*, a part of the hyphae becomes differentiated into specialized nitrogen fixing cells called *vesicles*. *Frankia* is also capable of forming differentiated vesicles and fixing nitrogen when it is living free of a plant. The details of the rhizospheric microbes, and their diversity and distribution have been given in Chapter 5.

In the leaf cavities of *Azolla* there are heterotrophic bacteria including abundant *Arthrobacter* spp. Some of these bacteria contain nitrogenase and are presumably capable of nitrogen fixation. The role of the bacteria in the symbiosis is unknown, but they have been considered a third member of the *Azolla*-*Anabaena* association. Some of the bacterial strains produce the plant hormone indole-3-acetic acid, although it remains to be seen what significance this may have for the symbiosis. The bacteria also produce capsular and exopolysaccharides, which may contribute to the mucilage produced by the cyanobionts. As with the cyanobiont, the bacteria appear to remain associated with *Azolla* throughout its life cycle. Indeed, bacteria are found together with cyanobacterial akinetes beneath the indusium of *Azolla* megasporocarps and they commence cell division at the same time as the akinetes germinate.

### 10.3.2 Plant-Fungi Associations

Successful and ecologically important symbiotic relationship is observed between fungal hyphae called mycorrhizal fungi and higher plants, in which the former actually become integrated into the physical structure of the roots. The fungus derives nutritional benefits from the plant roots, contributes to plant nutrition and does not cause plant disease. Mycorrhizal associations differ from other plants-microbes associations by the greater specificity and organization of the plant-fungus relationship. The widespread existence of mycorrhizal associations between fungi and plant roots attests to the importance of this interaction.

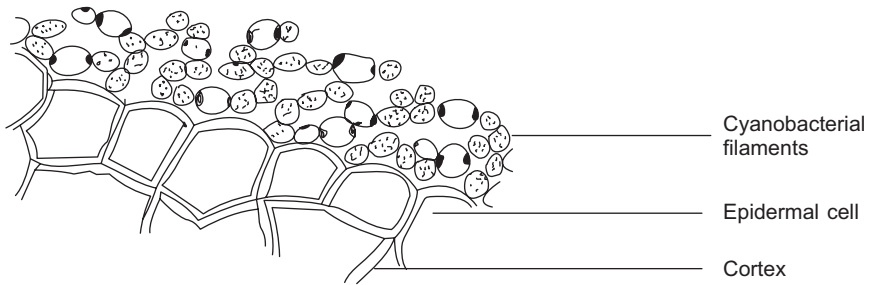
Mycorrhizal associations can basically be categorized into two types, basing on the position of fungal hyphae on the roots, viz., a) ectomycorrhizae and b) endomycorrhizae. In ectomycorrhizae, the fungal hyphae penetrate the intercellular spaces of the epidermis and of the cortical region of the root but do not invade the living cells. The endomycorrhizae, on the other hand, invade the living cells of the root, which become filled with mycelial clusters. Of the various microorganisms colonizing the active root surface, mycorrhizal fungi occupy the unique ecological position of being partly inside and partly outside the host projecting like fine hairs from root surface. The part of the fungus within the root does not encounter competition with other soil microorganisms. In a widespread form of endomycorrhizae, the microscopic appearance of intracellular hyphae clusters causes these to be called vesicular-arbuscular (VA) mycorrhizae (for details about the characteristics, diversity and distribution of micorrhizae please refer to Chapter 5).

When the hyphae of suitable actinomycetes penetrate the root, the conical cells are stimulated to divide. The hyphae penetrate the dividing cells, forming clusters within the host and vesicles form at the periphery of the infected hyphal tips. In the neighbourhood of the primary infected tissue, a root primordium is initiated that grows into the cortex. The endophytic actinomycete invades the meristem cells of the primordium and factors produced by the actinomycetes stimulate the development of the root primordium. The dichotomous division of the top meristem results in the formation of a cluster of lobes called a *rhizothamnion*, which is typical of all actinomycete nodules. Within the nodule, actinomycetes produce nitrogenase and fix atmospheric nitrogen supplying them to the host plant.

### 10.3.3 Plant-Algae Associations

Although knowledge of the effects of planktonic green algae and diatoms on crop plants is very limited, there are plenty of informations on the importance of plant-cyanobacteria root association in agriculture. The importance of cyanobacteria for rice field ecosystem has been well-studied. There are also reports of the cyanobacteria-wheat root association, which has prompted investigations in this area and has provided evidence that cyanobacteria associated with wheat roots can stimulate plant growth. For example, an ammonia-excreting mutant of *Anabaena variabilis* has been shown to enhance growth of wheat when the cyanobacterium was directly associated with the plant roots. Similarly, the roots

of hydroponically-grown wheat seedlings become colonized by certain nitrogen-fixing cyanobacteria, stimulating root length and enhancing the nitrogen content of the plants (Fig.10.4). This effect is dependent on the specific combination of plant cultivar and cyanobacterial strain, but close association of the two is not essential. The nitrogenase activity of some of the cyanobacteria can be sustained in the dark when they are associated with the roots of rice, wheat, or suitable host plants, indicating that the host supplies carbon compounds as root exudates to the symbiont.



**Figure 10.4** - A thick layer of cyanobacterial filaments on the epidermal surface of wheat roots.

In the plant-cyanobacteria symbioses, the cyanobiont is almost always *Nostoc* (Table 10.1). The ability to form hormogonia is the key to success as a symbiont, because hormogonia are the infective agents in most if not all of these symbioses. The immotile parent filaments would be unable to gain entry to the plant structures that house the symbiotic colonies. However, it is clear that a wide range of *Nostoc* species and strains are symbiotically competent and metabolically comfortable in a wide range of plants. Members of genera other than *Nostoc* can also form symbioses with plants. For example, the cycad *Encephalartos* has been reported to contain *Calothrix* spp. as cyanobionts, and has the hornwort *Phaeoceros*. It is commonly observed that *Nostoc* and *Calothrix* is heterocystous and is capable of hormogonia formation. Similarly, *Cycas* shows association with *Anabaena*, which is intracellularly located in the coralloid roots.

**(a) Cyanobacteria in bryophyte:** Cyanobacteria show epiphytic or endophytic associations with bryophytes belonging to Hepaticae (liverworts), Anthocerotae (hornworts) and even Musci (mosses). Primarily *Nostoc* enters into active and fruitful association with bryophytes but many other cyanobacteria and epiphytic attachment of green algae are also known. Association with mosses are mostly epiphytic, the exceptions being species of *Sphagnum* in which the *Nostoc* occupies hyaline cells, the function of which is unknown. *Hapalosiphon* sometimes also grows closely associated with *Sphagnum*, but critical studies on how far this can be endophytic appear to be lacking. It has been suggested that buffering within hyaline cells helps *Nostoc* to grow at pH values < 5, which is not possible for the free filaments of the cyanobacterium in the field conditions.



However, species of the nitrogen-fixing genera *Hapalosiphon* and *Tolypothrix* are sometimes frequent in pools at pH values as low as 4.1. These endophytic associations are largely limited to plants growing above the water line and have nitrogen fixation rates below those of epiphytic associations, which are usually confined to submerged plants. There is some evidence for transfer of nitrogen, fixed by the cyanobacteria, to the *Sphagnum* and for the elevation of nitrogen fixation rates of epiphytically associated *Nostoc* spp. above the same free-living strains. The nitrogen-fixing epiphytic and endophytic cyanobacterial associations with mosses such as *Sphagnum* and *Drepanocladus* may contribute to the nitrogen economy of bog and mire habitats, particularly in subarctic regions. Epiphytic associations also occur with grassland and forest mosses such as *Ceratodon* and *Funaria*, although the rates of nitrogen fixation are much lower than those of *Sphagnum* associations.

**Table 10.1:** Host, identity and location of the cyanobionts in plant-cyanobacteria symbiosis.

| <i>Plant host</i> | <i>Infected zone</i> | <i>Cyanobiont</i>                   | <i>Cyanobiont location</i> |
|-------------------|----------------------|-------------------------------------|----------------------------|
| Cycads            | Coralloid root       | <i>Nostoc</i> (+ <i>Calothrix</i> ) | Intercellular              |
| <i>Gunnera</i>    | Stem glands          | <i>Nostoc</i>                       | Intracellular              |
| Liverworts        | Auricles             | <i>Nostoc</i>                       | Intercellular              |
| Hornworts         | Slime cavities       | <i>Nostoc</i> (+ <i>Calothrix</i> ) | Intercellular              |
| <i>Azolla</i>     | Leaf cavities        | <i>Anabaena</i> ( <i>Nostoc</i> ?)  | Intercellular              |

Cyanobacterial associations with liverworts are very rare, with only 4 of over 340 genera known to be involved, two forming epiphytic and two endophytic associations. Four of the six hornwort genera form endophytic associations. The epiphytic associations are poorly understood, although they may be more common than once thought. The endophytic associations have been studied in much more detail because of their adaptability to laboratory experimentation. Hornworts, such as *Anthoceros* and *Phaeoceros* and liverworts such as *Blasia*, can be grown axenically in shaken liquid culture, with or without their symbiotic cyanobacterial partners. The symbiont-free plants can be readily reinfected with axenic cyanobacteria, making this an excellent model system for the study of cyanobacteria-plant symbiosis.

Epiphytic attachments of green algae and diatoms with liverworts and hornworts have also been reported but the biochemical relationship between the partners have not been fully established. It is assumed that the metabolite exchange occurs mostly in the form of organic carbon from one partner to the other depending on their metabolic state. Older thalli with fresh surface colonization with diatoms receive assimilable organic matters from the algal partner due to excess carbon fixation by the latter. Probably the algae receive nutrients from soil through the thalli. With extensive epiphytic associations of green algae and diatoms, the host growth is limited due to poor light availability. However, there is no evidence of carbon contribution of the epiphytic algae to the



bryophyte. Nevertheless, there is nutrient supply by the bryophyte to the epiphytic flora. Excess carbon fixed by the algal partners is exuded out on the thallus surface inviting bacteria as a third member of the association.

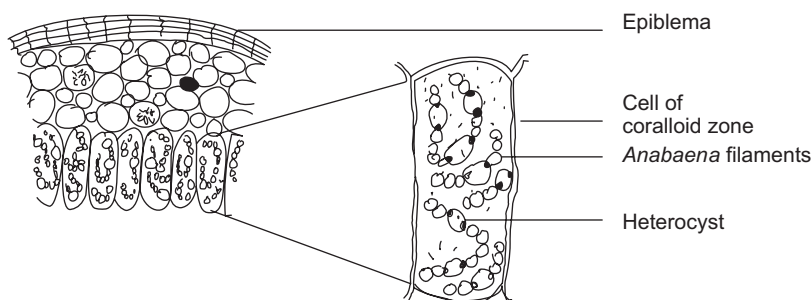
**(b) *Azolla-Anabaena* symbiosis:** The cyanobiont of *Azolla* has been traditionally referred to as *Anabaena azollae*, although this taxonomic assignment has been controversial. *Azolla* is unusual in carrying a colony of the cyanobiont within the sporocarp during sexual reproduction, making reinfection unnecessary and removing the need for the cyanobiont to produce hormogonia for infection to occur. Members of a non-hormogonium-forming genus such as *Anabaena* could therefore be considered potential symbionts in this system. However, undifferentiated filaments, resembling hormogonia, are also associated with the apical meristem of *Azolla* stem, providing the inoculum for new leaf cavities.

The *Azolla* cyanobiont was first described as *Nostoc* but later assigned it to the genus *Anabaena* and renamed as *Anabaena azollae* (Strasburger, 1884). This classification has often been questioned and the genus *Nostoc* has been proposed, based on comparisons of the arrangement of *nif* genes and the numbers of superoxide dismutase isoenzymes in symbiotic and free-living *Anabaena* and *Nostoc* strains. The results of RFLP analysis of the genomic DNA of cyanobionts from seven *Azolla* species also imply that they are more closely related to *Nostoc* than to *Anabaena*. This assertion is supported by the results of polymerase chain reaction (PCR) amplification of *Azolla* cyanobiont DNA. However, comparisons of cellular fatty acids have implied that the *Azolla* symbionts are equally distinct from both *Anabaena* and *Nostoc*. The alternative name *Trichormus azollae* has been suggested.

The true identity of the *Azolla* symbiont has remained elusive, because there appears to be more than one cyanobiont associated with the fern. The minor symbiont is readily isolated, whereas the major symbiont has proved impossible to cultivate free-living. Immunological and lectin haemagglutination studies, RFLP analyses and enzymatic comparisons have confirmed that the readily culturable cyanobionts of *Azolla* are not the same as the freshly-isolated symbiont. PCR-based techniques have been used successfully to differentiate between *Azolla* symbionts from different plant species. The genetic diversity of *Anabaena azollae* strains from different ferns has also been assessed by RFLP analysis using *nif* gene probes and the results suggest a taxonomy of *A. azollae* parallel with *Azolla* itself. Similarly, analysis of the fatty acid composition of cyanobionts from different American *Azolla* species implies that the cyanobionts form several clusters that correlate well with the taxonomic grouping of the plants, implying that the cyanobionts have coevolved with the plant, presumably as a result of the presence of *Anabaena azollae* throughout the life cycle of *Azolla*, avoiding the need for constant reinfection.

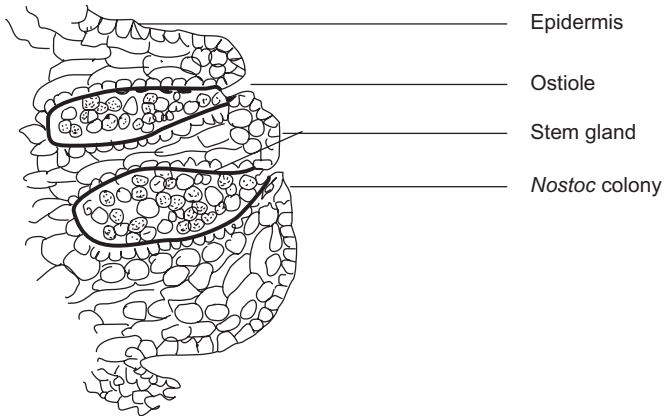
**(c) *Anabaena* in cycads:** The coralloid and apogeotropic roots of cycads are frequently infected with nitrogen-fixing cyanobacteria, forming a dark band between the inner and outer cortex of the root (Fig. 10.5). In their limited natural habitats, the cycads can make significant contributions to the local nitrogen

economy for which the credit goes to the algal partner. *Anabaena cycadoidae* occupies the algal zone in the coralloid roots of many cycads. Because these roots grow just below the soil surface, it is believed that light is available to the roots and probably the algal zone of the roots receive required light for the growth of the cyanobacterium. However, experiments have shown that the symbiont was very much dependent on the host plant for organic carbon and is almost unable to grow autotrophically in inorganic media. The cells are poor in Chlorophyll a but contain good amount of carotenoids and phycobiliproteins. This enables the cyanobacterium to use some of the trapped light energy for fixing the atmospheric nitrogen. The endosymbiont has a very high heterocyst frequency, which reaches a value that is five times higher than of the species growing independently. It enables the cyanobacterium to fix nitrogen more than that required by the host and some of this fixed nitrogen are exuded out to the soil environment.



**Figure 10.5** - The cross section of *Cycas* coralloid root showing the zone of elongated cells filled with the cyanobiont *Anabaena cycadoidae*.

**(d) Angiosperm-algae association:** Of all the angiosperms (flowering plants) only the genus *Gunnera* is known to form symbiotic associations with algae and the algal partner is always a cyanobacterium. *Gunnera* is the only one of the plant-cyanobacteria symbioses in which the cyanobiont (*Nostoc* sp.) is found intracellularly, within specific mucus-secreting glands on the plant stem at the base of each petiole (Fig. 10.6). Even the largest *Gunnera* plants colonised by *Nostoc* are able to fix sufficient nitrogen for their own needs and this becomes available to the local nitrogen economy. Since all major crop plants are angiosperms, an improved understanding of the *Nostoc-Gunnera* system may improve the chances of developing novel nitrogen-fixing symbioses with economically important plants. However, the physiological, biochemical and molecular relationships between the host and the cyanobiont have not yet been fully established making it difficult to extend the association to other angiosperms. Experimental evidences suggest that the cyanobacterium of this symbiosis is more independent than other cyanobionts (from Cycads and *Azolla*) with ability to grow independently in inorganic medium. This shows that the symbiosis is more beneficial to the host than to the cyanobiont.



**Figure 10.6** - A longitudinal section of a stem gland of the angiosperm *Gunnera chilensis* containing the colonies of *Nostoc*.

## 10.4 LEGUME-RHIZOBIUM INTERACTION

Legume-Rhizobium symbiosis involves the invasion of the roots of suitable host plants by nitrogen-fixing *Rhizobium* species, resulting in formation of a tumor-like growth called a nodule. Within the nodule, the nitrogen-fixing bacteria are able to convert atmospheric nitrogen to ammonia, which supplies the nitrogen required for bacterial and plant growth (Fig. 10.7). The fixation of nitrogen in plant nodules is of extreme importance for the maintenance of soil fertility. In agricultural practices, it is used to increase crop yields.

The fixation of atmospheric nitrogen depends on the nitrogenase enzyme system. This enzyme system is composed of dinitrogenase (MoFe protein) and dinitrogenase reductase (Fe protein). Dinitrogenase has two dissimilar polypeptides,  $\alpha_2\beta_2$ . The  $\alpha$  polypeptides are encoded by *nifD* and the  $\beta$  polypeptides by *nifK* genes. The dinitrogenase protein contains two active metalloclusters: the P cluster containing 8 iron and 7–8 sulphur atoms ( $\text{Fe}_8\text{S}_{7-8}$ ) and iron-molybdenum cofactor (FeMo-co) containing 7 iron atoms, 9 sulphur atoms, 1 molybdenum atom and 1 molecule of homocitrate ( $\text{Fe}_7\text{S}_9\text{Mo-homocitrate}$ ). The P cluster acts as an intermediate electron acceptor and probably transfers the electron to the FeMo-co cluster. The FeMo-co cluster functions as the site of nitrogen reduction. The dinitrogenase reductase protein (Fe protein) consists of two identical polypeptides,  $\gamma_2$ , encoded by the *nifH* gene. Each polypeptide contains two iron atoms. The four Fe atoms are organized into an  $\text{Fe}_4\text{S}_4$  cluster. The main function of the Fe protein is to bind and hydrolyze MgATP and to transfer electrons from the  $\text{Fe}_4\text{S}_4$  cluster to the P cluster of the MoFe protein. Both proteins are folded in such a way as to bring the active centres of each into close proximity.

Because of high sensitivity to oxygen, nitrogenase is irreversibly inactivated on exposure to even low  $\text{O}_2$  concentrations. Nitrogen fixation, therefore, often is restricted to habitats in which nitrogenase is protected from exposure to molecular oxygen. In addition to nitrogen reduction, the nitrogenase complex forms one  $\text{H}_2$

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*Bradyrhizobium* species nodulate soybeans, lupines, cowpeas and various tropical leguminous plants. *Rhizobium* species nodulate alfalfa, peas, clover and a wide variety of other leguminous plants. *Azorhizobium* is a unique member of the group that forms stem nodules on tropical leguminous trees (Table 10.2). Unlike other genera, it is capable of growing with atmospheric nitrogen in its free-living state (*azo* refers to nitrogen). *Rhizobium* and *Bradyrhizobium* are not capable of doing so, although some nitrogen fixation by free-living bacteria can be demonstrated at reduced oxygen tension. Besides its slow growth rate, *Bradyrhizobium* differs from *Rhizobium* in several characteristics (Table 10.3). However, the infection and nodulation processes are similar for both genera. Unless noted otherwise, information provided for rhizobia pertains to both *Rhizobium* and *Bradyrhizobium*.

**Table 10.2:** The rhizobia and their respective hosts for association.

| Genus                 | Species                 | Host                                            |
|-----------------------|-------------------------|-------------------------------------------------|
| <i>Azorhizobium</i>   | <i>A. caulinodans</i>   | <i>Sesbania rostrata</i>                        |
| <i>Bradyrhizobium</i> | <i>B. elkanii</i>       | <i>Glycine max</i>                              |
|                       | <i>B. japonicum</i>     | <i>Glycine max</i>                              |
|                       | <i>B. liaoningense</i>  | <i>Glycine max</i>                              |
| <i>Mesorhizobium</i>  | <i>M. amorphae</i>      | <i>Amorpha fruticosa</i>                        |
|                       | <i>M. chacoense</i>     | <i>Prosopis alba</i>                            |
|                       | <i>M. ciceri</i>        | <i>Cicer arietinum</i>                          |
|                       | <i>M. huakuii</i>       | <i>Astragalus</i>                               |
|                       | <i>M. loti</i>          | <i>Lotus corniculatus</i>                       |
|                       | <i>M. mediterraneum</i> | <i>Cicer arietinum</i>                          |
|                       | <i>M. plurifarium</i>   | <i>Acacia, Prosopis</i>                         |
|                       | <i>M. tianshanense</i>  | <i>Glycyrrhiza, Sophora and Glycine</i>         |
| <i>Rhizobium</i>      | <i>R. etli</i>          | <i>Phaseolus vulgaris</i>                       |
|                       | <i>R. galegae</i>       | <i>Galegae</i>                                  |
|                       | <i>R. gallicum</i>      | <i>Phaseolus vulgaris</i>                       |
|                       | <i>R. hainanense</i>    | <i>Centrosema, Desmodium, Stylosanthes,</i>     |
|                       | <i>Tephrosia</i>        |                                                 |
|                       | <i>R. huautlense</i>    | <i>Sesbania herbacea</i>                        |
|                       | <i>R. leguminosarum</i> | <i>Vicia, Trifolium</i>                         |
|                       | <i>R. mongolense</i>    | <i>Medicago ruthenica</i>                       |
|                       | <i>R. phaseoli</i>      | <i>P.vulgaris</i>                               |
|                       | <i>R. tropici</i>       | <i>Leucaena, P. vulgaris</i>                    |
| <i>Sinorhizobium</i>  | <i>R. undicola</i>      | <i>Neptunia natans</i>                          |
|                       | <i>R. yanglingense</i>  | <i>Amphicarpaea, Trisperma, Corollina varia</i> |
|                       | <i>S. arboris</i>       | <i>Acacia senegal, Prosopis chilensis</i>       |
|                       | <i>S. fredii</i>        | <i>G.max</i>                                    |
|                       | <i>S. kostiense</i>     | <i>A. senegal, P. chilensis</i>                 |
|                       | <i>S. medicae</i>       | <i>Medicago spp.</i>                            |
|                       | <i>S. meliloti</i>      | <i>Medicago sativa</i>                          |
|                       | <i>S. saheli</i>        | <i>Sesbania</i>                                 |
|                       | <i>S. terrangae</i>     | <i>Acacia, Sesbania</i>                         |
|                       | <i>S. xinjiangense</i>  | <i>G.max</i>                                    |

Adopted from Saxena *et al.* (2005).

Although rhizobia occur as free-living heterotrophs in soil, they are not dominant members of soil microbial communities and do not fix atmospheric nitrogen in this state. However, under appropriate conditions rhizobia can invade root hairs, initiate the formation of a nodule and develop nitrogen-fixing activity. The association between rhizobia and plant roots is specific, with mutual recognition between the two compatible partners based on chemotactic response and specific binding to the root hair prior to invasion and establishment of the root nodule. The legume plant root recognizes the right population of rhizobia, which in turn recognizes the right kind of leguminous root. Within the rhizosphere, plant roots supply the rhizobia with compounds that are transformed by them to substances involved in the initiation of the infection process and subsequent nodule development. Establishment of an adequate rhizosphere population of rhizobia is an absolute prerequisite for infection.

Soil conditions have a marked effect on rhizobia in terms of their survival and ability to infect root hairs. Rhizobia are mesophilic, but some exhibit tolerance to low temperatures, down to 5°C and others tolerate temperatures up to 40°C. Some rhizobia are sensitive to low pH and cannot establish root hair infections in acidic soils. Nitrate and nitrite ions also inhibit nodule formations at relatively low concentrations.

**Table 10.3:** Compative account of the characteristics of rhizobia (*Mesorhizobium* and *Sinorhizobium* are not much morphologically different from *Rhizobium*).

| Feature                                     | <i>Rhizobium</i> | <i>Bradyrhizobium</i> | <i>Azorhizobium</i> |
|---------------------------------------------|------------------|-----------------------|---------------------|
| Flagella                                    | Peritrichous     | Polar                 | Peritrichous        |
| Growth outside host                         | None             | None                  | All strains         |
| Growth in culture                           | Fast             | Slow                  | Fast                |
| Location of <i>nod</i> and <i>nif</i> genes | Mainly plasmid   | Chromosomal           | Chromosomal         |
| Host specificity                            | Narrow           | Broad                 | Only one host sp.   |
| Preferred host                              | Legumes          | Soybeans              | Non-legume          |

**10.4.2 Nitrogen-fixing Associations between Rhizobia and Legumes**

A complex sequence of interactions between rhizobia and plant roots takes place during nodule formation. Since plants are nonmotile, it is tempting to think that legume-bacterium association begins when bacteria advance into the legume rhizosphere. Yet chemotaxis and motility among rhizobia are clearly not essential for nodulation. Rather other more subtle factors help the partners match each other. Foremost among these are phenolic substances, especially flavonoids. Different types of *nod* gene inducing flavonoids are secreted from the roots to the rhizosphere for identifying and stimulating the infection of root hairs by the correct rhizobium (Table 10.4). Although small quantities are excreted continuously, flavonoid concentrations in the rhizosphere increase in response to compatible rhizobia. In turn, rhizobia use these plant products for their own ends.

**Table 10.4:** List of important *nod* gene inducing flavonoids secreted to the rhizosphere by legume roots.

| Flavonoid                          | Legume species                                              |
|------------------------------------|-------------------------------------------------------------|
| Apigenin                           | <i>R. pseudoacacia</i>                                      |
| Chrysoeriol                        | <i>R. pseudoacacia</i>                                      |
| Coumestrol                         | <i>G. max</i> , <i>M. sativa</i> , <i>P. vulgaris</i>       |
| Daidzein                           | <i>G. max</i> , <i>P. vulgaris</i>                          |
| 7,4'-Dihydroxy-flavone             | <i>R. pseudoacacia</i> , <i>M. sativa</i>                   |
| 7,4'-Dihydroxy-flavanone           | <i>M. sativa</i> , <i>S. rostrata</i>                       |
| 4',4'-Dihydroxy-2-methoxychalcone  | <i>M. sativa</i>                                            |
| 7,4'-Dihydroxy-3'-methoxyflavanone | <i>V. sativa</i>                                            |
| 7,3'-Dihydroxy-4'-methoxyflavanone | <i>V. sativa</i>                                            |
| Eriodictyol                        | <i>P. vulgaris</i>                                          |
| Genistein                          | <i>G. max</i> , <i>P. vulgaris</i>                          |
| Glycetein                          | <i>V. sativa</i> , <i>P. vulgaris</i>                       |
| Hesperitin                         | <i>V. sativa</i>                                            |
| Homoeriodictyol                    | <i>V. sativa</i>                                            |
| Isoliquiritigenin                  | <i>R. pseudoacacia</i> , <i>G. max</i> , <i>P. vulgaris</i> |
| Liquiritigenin                     | <i>V. sativa</i> , <i>P. vulgaris</i>                       |
| Naringenin                         | <i>R. pseudoacacia</i> , <i>P. vulgaris</i>                 |

Some flavonoids induce the expression of nodulation (*nod*, *noe* and *nol*) and related genes, while others are catabolized. Degradation leads first to the appearance of chalcones, which can be more efficient inducers of *nod* genes than the flavonoids themselves. Rhizobia respond through NodD and related proteins, which act as both sensors of the environment and activators of the transcription. Once derepressed, nodulation and other genes direct the synthesis and release of substances that profoundly affect legume roots (Fig 10.8). It is as if plants construct elaborate “bowers,” replete with flavonoids and other attractants, to entice rhizobia towards them. This is supported by the fact that legumes support considerably larger rhizosphere populations, enriched in rhizobia, than non-legumes do. Rhizobia penetrate this outer “membrane” and form an almost continuous layer 5 to 10 cells deep. Rhizobia also rapidly proliferate around the roots of developing legumes, an indication of the invitation provided by the host plant to the rhizobia to colonize. Although the evidence is not completely consistent, there are indications that production of *nod* gene-inducing flavonoids is restricted to the elongating root hair zone from which most nodules later develop.

Flavonoids or isoflavonoids secreted by the host plants induce the expression of a number of nodulation (*nod*) genes in the cognate rhizobial bacteria. Flavonoids produced via the phenylpropanoid biosynthetic pathway were identified as the strongest inducers of *nod* gene expression. Significant differences in the flavonoid compositions of seeds, roots and root exudates of *G. max*, *P. vulgaris*, *M. sativa*, *Trifolium repens* and *Vicia sativa* exist. Many flavonoids are stored and released as glycosides or related conjugates. These





### 10.4.3 Biochemical Basis of Legume-Rhizobium Association

The biochemical relationship between the rhizobia and the legume host begins with the expression of *nod* genes. Flavonoid activation of *nod* genes results in the synthesis and secretion of a family of lipo-chito-oligosaccharidic Nod factors. Simple Nod factors comprising three to six N-acetyl-D-glucosamine residues and an acyl chain from the general fatty acid metabolism have biological activity. Basic Nod factors as well as their more sophisticated cousins are mitogenic, a trait that they share with plant hormones. Yet they are unique in provoking curling of the root hairs (Hac) and it is within the folds of highly curled root hairs that rhizobia gain entry into the plant. Rhizobial mutants lacking the ability to produce Nod factor are unable to infect the legume plants. Most Nod factors are modified in species-specific ways, suggesting that each *Rhizobium* produces a different set of chemicals to recognize its correct hosts. However, Nod factors are more like master keys that are able to open many legume doors. The legumes recognize common motifs and allow the entry to the correct symbiont.

Once recognizing the host, rhizobia respond to plant root exudates and move toward localized sites on the legume roots. Both *Bradyrhizobium* and *Rhizobium* species are attracted by amino acids and dicarboxylic acids present in root exudates, as well as by very low concentrations of excreted flavonoids. Lectins, plant proteins with high affinity to carbohydrate moieties on the surface of appropriate rhizobial cells, have been identified as specific mediators of the attachment of rhizobia to susceptible root hairs. During the nodulation process, tryptophan secreted by the plant roots is metabolized to indoleacetic acid (IAA) by the rhizobia and the IAA, together with unknown cofactors probably arising from the host plant roots, initiates hair curling or branching. The root hairs grow around the bacterial cells and provide the condition to the bacterium to divide and form a colony. Polygalacturonase, secreted by the rhizobia or possibly by the plant roots, depolymerizes the cell wall and allows bacteria to invade the softened plant root tissues.

Nod factors allow rhizobia to enter the plant but do not guarantee successful symbiosis. Other inputs, from both partners, are necessary for this. Rhizobia enter roots along corridors in root hairs that are largely constructed by the plant and are called infection threads (Fig. 10.8). Although Nod factors probably initiate infection thread development, it is not unreasonable to assume that in entering a "house" of the plant's making, the rhizobia need to make certain changes for their own comfort. Failure to do so results in abortion of infection threads, a process in which phenolic substances accumulate, triggering a hypersensitive reaction. One essential modification to the corridor is insulation to protect the rhizobia from the plant cytoplasm. Part of this insulation is seen as the peribacteroid or symbiosome membrane in infected cortical cells. Histochemically, the thread wall contains the same components of pectic substances, hemicelluloses and some cellulose, as the root hair tip, and electron micrographs of infection threads in root hairs and nodules show similar structure to the primary wall of plant cells. However, it is more resistant to treatment with cell wall-degrading enzymes. Some of the

rhizobial genes that play a role in infection thread development are involved in synthesis of various polysaccharides that range from alternating b-linked residues of mannose and 3-deoxy-D-manno-2-octulosonic acid, through branched exopolysaccharides of *R. meliloti* and NGR234, to the cyclic glucans of *B. japonicum*. Perhaps the products of these genes are required for infection thread formation on certain plants, but not on others. It is also possible that extracellular carbohydrates help suppress defense reactions in the plant. Some of its' constituents are listed in Table 10.5 and many remain to be discovered. Continued infection thread development (and thus effective, nitrogen-fixing nodules) is dependent on the presence and the composition of some of these components.

**Table 10.5:** Chemical components of the infection threads of legume root hairs invaded by rhizobia.

| Partner   | Chemical compoent                            | Location                     |
|-----------|----------------------------------------------|------------------------------|
| Legume    | Arabinogalactan                              | Lumen?                       |
|           | ATPase                                       | Peribacteroid membrane (PBM) |
|           | Cellulose                                    | Wall                         |
|           | Extensin                                     | Matrix                       |
|           | Lipoxygenase                                 | Lumen                        |
|           | Esterified pectin                            | Wall                         |
|           | Non-esterified pectin                        | Wall                         |
|           | Proline-rich protein (ENOD12)                | Matrix                       |
|           | Xyloglucan                                   | Wall                         |
|           | Cyclic $\beta$ -(1,6)- $\beta$ (1,3)-glucans | ?                            |
| Rhizobium | EPS                                          | ?                            |
|           | Lipopolysaccharides                          | Lumen                        |
|           | Succinoglycan                                | ?                            |
|           | Secreted proteins                            | ?                            |
|           | NodO                                         | ?                            |

Adopted from Broughton *et al.* (2000).

The above discussion shows that at least three different sets of biochemical (symbiotic) signals are exchanged between legumes and rhizobia during nodule development. Flavonoids, the first of these, emanate from the plant and interact with rhizobial NodD proteins that serve as both environmental sensors and activators of transcription. A second set of signals is synthesized when NodD-flavonoid complexes activate transcription from *nod* boxes. Most of the genes immediately downstream of these promoters are involved in the synthesis of lipooligosaccharidic Nod factors that provoke deformation of root hairs and allow rhizobia to enter the root through infection threads. Fine-tuning of transcription of nodulation (*nod*) genes is probably related to sequence variations in individual *nod* boxes. Other rhizobial products like extracellular polysaccharides (EPS) and related compounds, as well as proteins exported from the microbiont by the type three secretion system (TTSS) seem to be necessary for continued infection thread development and these represent a third set of signals (Fig. 10.8).

Nodules have a characteristic red-brown colour owing to the presence of leghemoglobin, which is a constant and prominent feature in the central tissue of all nitrogen-fixing leguminous nodules. The leghemoglobin serves as an electron carrier, supplying oxygen to the bacteroids for the production of ATP and at the same time protecting the oxygen-sensitive nitrogenase system. The heme portion of the leghemoglobin is coded by the rhizobia and the globin portion by the plant. Leghemoglobins are unique for legume root nodules and occur nowhere else in the plant kingdom.

#### 10.4.4 Molecular Basis of Legume-Rhizobium Association

Legume-Rhizobium symbioses are marriages between two vastly different genomes but their respective contributions are probably not vastly different. In rhizobia, these include a chromosome plus zero to many plasmids, totalling 6 to 9 Mbp. In contrast, genomes of legumes are much larger, some comprising more than 20 chromosomes with total DNA contents ranging range from about 450 to 4,500 Mbp per haploid genome and at least 50 times larger than those of their microsymbionts. For example, the model legumes *Lotus japonicus* and *Medicago truncatula* have six and eight chromosomes totalling 450 and 500 Mbp per haploid genome, respectively.

Specific expression of plant and bacterial genes accompanies the development of the rhizobium-plant symbiosis. The genes involved in root nodule formation, which are collectively called *nodulin genes*, encode a series of Nod proteins that serve specific functions in the establishment of nodules that permit symbiotic nitrogen fixation. Nodulin genes essential for infection of the plant root and nodule formation by symbiotic nitrogen-fixing bacteria are divided into two classes. The first class includes genes that specify the biochemical composition of the bacterial cell surface, such as genes determining the synthesis of exopolysaccharides (*exo* genes), lipopolysaccharides (*ips* genes), capsular polysaccharides or K antigens and  $\beta$ -1,2-glucans (*ndv* genes). The *exo* and *ips* genes may play a role in determining host specificity, but this has yet to be firmly established. The second class of genes consists of the nodulation (*nod* or *nol*) genes. Inactivation of the nodulation genes can result in various plant phenotypes, such as the absence of nodulation (Nod<sup>-</sup>), a delayed but effective nodulation (Nod<sup>d</sup> Fix<sup>+</sup>), or changes in the host range. Some of the *nod* genes appear to be interchangeable for nodulation function between different species and biovars (variants) and are, therefore, designated as common *nod* genes. Other *nod* genes are involved in the nodulation of a particular host and hence are called host-specific *nod* (*hsn*) genes.

Special mechanisms that down-regulate expression of *nod* genes are apparently required for optimal nodulation of *M. sativa* by *R. meliloti* and of *Pisum sativum* by *R. leguminosarum*. In *R. meliloti*, NodR represses the transcription of the *nodD* genes as well as those genes necessary for the synthesis of the core Nod factor structures (*nodABC*). In *B. japonicum*, repression of *nod* gene expression by NodA is probably an indirect effect, perhaps mediated by

*nodD2*. *B. japonicum nolA* and *R. meliloti nolR* mutants retain the ability to nodulate their respective hosts, albeit at lower efficiency, suggesting that fine-tuning of *nod* gene expression is required for optimal nodulation.

The *nod* gene clusters coding for the infection and nodulation processes are generally located on the *Sym* plasmids, which also carry specificity genes. It is possible to transfer the *Sym* plasmids between rhizobial strains and thus to alter the range of host legumes they can infect. *Sym* plasmids may even be transferred to *Agrobacterium* and other related bacteria, which then acquire the ability to nodulate the specified plants. However, such associations usually do not fix nitrogen. The *nif* genes that code for the biochemical nitrogen-fixing mechanism may or may not be plasmid-associated in various rhizobial strains. Besides the *nod* genes, the *Sym* plasmids carry the *nif* genes and *fix* nitrogen-fixing gene clusters. The *nif* and *fix* clusters include the structural genes for nitrogenase. In *Rhizobium loti* and in *Bradyrhizobium* and *Azorhizobium* species, the symbiosis-related genes are localized on the bacterial chromosome.

The *nif* and *fix* genes of *R. meliloti*, *B. japonicum* and *A. caulinodans* are arranged in distinct patterns. The structure and clustering of these genes are unique to each species. The *nif* and *fix* gene clusters of the rhizobia are not as tightly regulated as those in *Klebsiella pneumoniae*. *R. meliloti* carries two extremely large plasmids (megaplasms) of about 1400 kbp of DNA (pSym-a or megaplasmid 1) and 1700 kbp (pSym-b or megaplasmid 2). Both cluster I (*nifHDKE*, *nifN*, *fixABCX*, *nifA*, *nifB*, *frdX*) and cluster II (*jixU*, *fixK*, *fixNOQP*, *fixGHIS*) are located on megaplasmid 1. The cluster II genes map at about 220 kbp downstream of the *nifHDKE* operon and are transcribed in opposite orientation to it. A functional duplication of the region spanning *fixK* and *fixNOQP* is about 40 kbp upstream of *nifHDKE*. A cluster of *nod* genes including the common *nod* genes (*nodABC*) is located in the 30 kbp region between *nifE* and *nifN*. Additional genes required for symbiosis are located on megaplasmid 2 and on the chromosome.

A side reaction of the nitrogen-reduction process is the evolution of hydrogen gas. The evolution of  $H_2$  wastes photosynthetic energy and leads to lowered yields. Some *Sym* plasmids also carry *hup* genes coding for hydrogenase activity. The hydrogenase oxidizes  $H_2$  to water and recovers energy by chemiosmotic coupling with ATP synthesis. This beneficial process saves photosynthetic energy that would otherwise go to waste. The genetic manipulation of the association of rhizobia with legumes is likely to increase both the range and the efficiency of the system.

## 10.5 ANIMAL-MICROBE SYMBIOSIS

All type of microbes, such as bacteria, cyanobacteria, Prochlorophytes, unicellular algae, protozoa and some fungi, enter into symbiotic associations with a diverse type of animals. Many animals are found to have metabolic dependence while others are in a less symbiotic and have more commensal relationship with the microbiont.

### 10.5.1 Microbial Symbionts in Digestive Tracts of Animals

Most warm-blooded animals contain extremely complex microbial communities within their gastrointestinal tracts. In the lower intestine, each gram of feces contains approximately  $10^{11}$  microorganisms, belonging to up to 400 different species. In the human intestine, the strict anaerobes belonging to the genera *Bacteroides*, *Fusobacterium*, *Bifidobacterium* and *Eubacterium* are most abundant, but no single species has a dominant role in this highly diverse microbial community. Recent evidences suggest that the number of bacterial cells symbiotically involved in different processes of human body is 10 times of the total number of body cells. In some animals, such as pigs, the microbial populations of the gastrointestinal tract contribute to the nutrition of the animals by fermenting carbohydrates. There is some evidence that in older pigs microorganisms digest cellulose within the intestinal tract and that the animal can utilize the products of the cellulose degradation.

In monogastric animals, the main contribution to digestion by intestinal microbial populations appears to be in the production of growth factors rather than in the production of partially degraded substrates. Although animals clearly absorb products derived from the microbial metabolism of ingested foods, it is not always clear whether any of these products are actually required. In some cases, microorganisms supply required vitamins; specific microorganisms synthesize vitamin K, e.g., germ-free animals lacking appropriate microbial populations exhibit symptoms of vitamin deficiency. Besides their contributions to digestion and nutrition, normal gastrointestinal microbial populations, by their presence and preemptive colonization, constitute an important barrier to attack by intestinal pathogens.

The relationship of the intestinal symbionts to their animal host can be characterized as largely commensal or moderately symbiotic. The animal provides the microbes with a suitable anaerobic habitat and a steady food supply. The intestinal microbes may contribute marginally to food digestion, produce some vitamins for the host and by their presence help to keep out potentially more harmful microbes. However, when animals derive most or all of their nutrition from compounds that are very difficult to digest, their intestinal symbionts become more specific and the animal's relationship to them shifts from commensal to clearly symbiotic.

Lower termites maintain a mutualistic relationship with internal populations of protozoa, which are responsible for the degradation of cellulose and production of metabolites that the termite can assimilate. The protozoan and bacterial populations found within the guts of lower termites and wood-eating cockroaches ferment cellulose anaerobically, producing carbon dioxide, hydrogen and acetate. Some of the hydrogen and carbon dioxide is converted to methane by archaea, but this pathway is largely wasteful for the insect and involves only negligible amounts. Acetogenic bacteria convert most of the  $H_2$  and  $CO_2$  to acetate. The acetate is absorbed through the wall of the termite hindgut and is oxidized aerobically, forming carbon dioxide and water. Some bacterial populations within



termite guts also fix atmospheric nitrogen. *Enterobacter agglomerans* has been isolated from the guts of some termites and shown to be capable of nitrogen fixation under conditions of reduced oxygen tension.

During their early stages of life, bloodsucking species of insects nearly always feed on microorganisms. Insects that live on such restricted diets also normally develop mutualistic relationships with microbial populations that are essential for their survival. The microbial populations, which are often maintained within mycetomes, supplement dietary deficiencies of the animal by producing growth factors. Removal of the microbial populations of mycetomes of lice, for example, results in the animal's failure to reproduce. Reproductive activity and growth can be restored if the animals are supplied with B vitamins and yeast extract.

Some plant-eating birds maintain intestinal populations of bacteria and fungi that produce cellulolytic enzymes. These microbial populations are able to degrade cellulose within the intestinal tract of the bird, providing it with nutrients. Honey guides, which consume beeswax, contain populations of the bacterium *Micrococcus cerolyticus* and the yeast *Candida albicans* that are able to utilize the beeswax, when supplied with cofactors from the bird. Birds can assimilate nutrients produced from the degradation of the beeswax.

Various fishes and aquatic invertebrates contain microbial populations within their digestive tracts that contribute to food digestion. Amphipods, for example, contain high proportions of chitinase producing *Vibrio* species. These bacterial populations partially degrade the chitin ingested by these animals, producing monomers that the animals can absorb and use. Some fish, such as catfish and carp, contain microbial populations that produce cellulase enzymes. The degradation of cellulose by bacterial populations within the fish gut produces metabolic products that the fish absorb.

### 10.5.2 Animal-Algae Symbiosis

Some invertebrate animals enter into mutualistic relationships with photosynthetic microorganisms, including unicellular algae and cyanobacteria. Classically, these microorganisms are called *endozoic algae*. They are described as zooxanthellae if the algal cells are yellow to reddish-brown, including the dinoflagellates; zoochlorellae if the algal cells are pale to bright green; and cyanellae if the dominant pigments are phycobiliproteins.

Mutualistic associations with photosynthetic microorganisms have been reported for various species of polychaete worms, platyhelminth worms, molluscs (including bivalves such as the giant clam), tunicates, echinoderms, hydroids, jellyfish, anemones, corals, sea fans and sponges. The most common occurrence of endozoic algae appears to be in the coelenterates, such as hydra, anemones and corals. The marine sponges are the most common hosts of cyanobacterial symbionts. Chlorophyta members are found primarily in freshwater invertebrates. Dinoflagellates are the most frequent algal symbionts of marine invertebrates.

Corals establish symbiotic relationships with endozoic dinoflagellates. Coral reefs also provide suitable habitats for the growth of external synergistic



populations of algae, such as calcareous red algae. The growth of the coral depends on the metabolic activities of the endozoic dinoflagellates living within the tissues of the coral polyp. Reef corals precipitate calcium from seawater mainly during periods of maximal algal photosynthesis. The assimilation of CO<sub>2</sub> shifts the equilibrium from the more soluble bicarbonate to the less soluble carbonate. The growing coral polyp benefits from algal production of organic matter and removal of ammonia that builds up within the animal. The mutualistic relationship of endozoic algae and corals appears to be based on the exchange of carbon, nitrogen, phosphorus and oxygen-containing compounds.

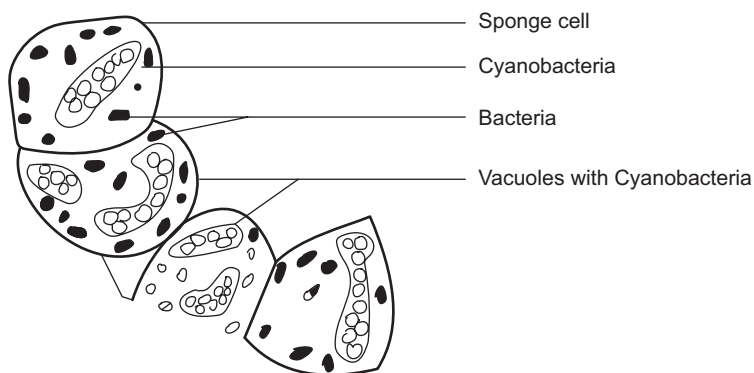
Animals harbouring endozoic algae exhibit behavioural characteristics that benefit the algal partner. Animals with endozoic algae normally exhibit phototactic responses. For example, coral polyps extend toward the light, providing the algae with accessible illumination for photosynthetic activities. Anemones with zooxanthellae also show phototactic responses. Free-swimming animals with endozoic algae move to depths with optimal light penetration for the photosynthetic activities of the endozoic algae. The mutualistic relationship between photosynthetic microorganisms and animals may be especially beneficial during some periods of the year. *Convoluta*, for example, probably requires an efficient mechanism for retaining and recycling nutrients during winter, when there is less terrestrial runoff to supply nutrients to this coastal organism.

**(a) Symbiosis of prochlorophytes:** The prochlorophytes are a group of oxygenic photosynthetic prokaryotes that resemble cyanobacteria in many ways, but differ in the possession of both chlorophylls *a* and *b* and the lack of phycobilin pigments. The first of the prochlorophytes to be discovered was *Prochloron didemni* which is found in a symbiotic relationship with marine colonial ascidians (chiefly didemnids) in (sub)tropical coastal waters. It is unicellular, with coccoid cells 6 to 30 µm in diameter and is found as a symbiont in ascidians of the Didemnidae, where it usually occurs extracellularly either on the surface, or within the cloacal cavity of the host. However, *Prochloron* has been found in the ascidian *Lissoclinum punctatum* both extracellularly and intracellularly in the tunic cells. The intracellular *Prochloron* are morphologically similar to those found extracellularly and show no signs of rejection or degeneration, implying a stable symbiosis. Cells of *Prochloron* contain carboxysomes and are bounded by a cytoplasmic membrane, periplasmic space, outer membrane and outer sheath and those cells from some didemnids contain one or more characteristic vacuoles that occupy much of the cell volume.

*Prochloron* is known to fix CO<sub>2</sub> for its host and there has been report, though limited, of acetylene reduction indicating nitrogen fixation. Chlorophyll fluorescence quenching analysis has recently been used to assess the photosynthetic performance of *Prochloron* within the host *Lissoclinum patella*. The symbiont displayed exceptionally high quantum yields under both light-limiting and light-saturated conditions. The results of oligonucleotide cataloguing and DNA-DNA reassociation studies have implied that *Prochloron* isolated from different hosts at one site, or even hosts at widely different locations, appear to be closely related, or

possibly the same species. However, an analysis of the morphological characteristics of both symbiont and host shows that different physiological states of *Prochloron* in different hosts brings in difference in morphological appearance.

**(b) Animal-cyanobacteria association:** Almost 40 genera of marine sponge are known to contain symbiotic cyanobacteria and bacteria found both intra- and inter-cellularly are distributed either throughout the sponge or just in the superficial tissues (Fig. 10.9). The cyanobionts are usually members of the unicellular genus *Aphanocapsa*, or the filamentous genus *Phormidium*, although the cyanobiont of the tropical marine sponge *Dysidea herbacea* has been identified as *Oscillatoria*. All of these genera lack heterocysts, although the cyanobionts of at least two sponges have been shown to fix nitrogen. Cyanobionts are photosynthetically active, only being found in sponges in the photic zone and transfer up to 12% of the carbon (they fix) to the host, probably in the form of glycerol.



**Figure 10.9** - Symbiotic association of cyanobacteria with the sponge *Siphonochalina tabernacula* (Redrawn from Wilkinson and Fay, 1979).

The degree of metabolic reliance of some sponges on the presence of the cyanobionts is illustrated by *Chondrilla nucula*, which cannot colonise dark sites and, if transferred to the dark, it loses its cyanobionts and dies. The cyanobionts are also thought to help protect the sponge from damage by high light intensity. The bacterial microbiont are comparatively more dependent on host and the biochemical equations of symbiosis between the two partners has not yet been fully established. However, there are reports that several bioactive natural products are produced by the heterotrophic bacteria present in the sponge tissue.

Sponges, such as *Theonella swinhoei* and *Siphonochalina labernacula*, that contain nitrogen-fixing cyanobionts, most likely benefit from the provision of combined nitrogen and this presumably benefits the local nitrogen economy. Many sponges also produce bioactive natural products, at least some of which may help protect against potential predators and bacterial invasion. *Oscillatoria spongelia*, the cyanobacterial symbiont of the marine sponge *Dysidea herbacea*, produces a group of polychlorinated and polybrominated secondary metabolites.

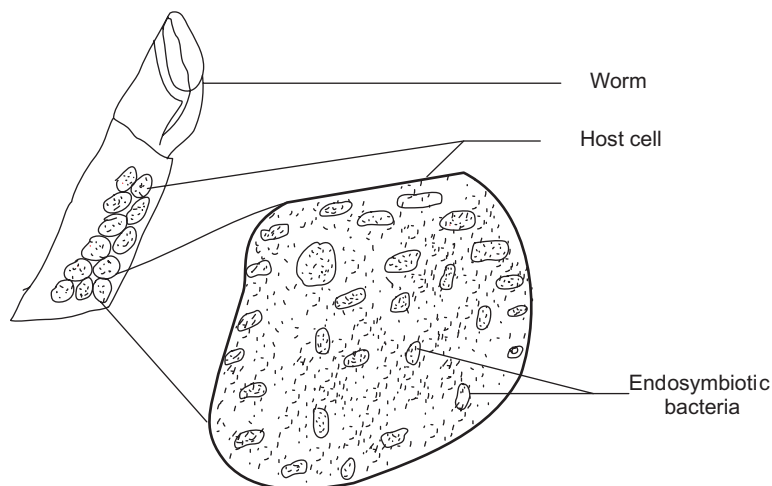
One of the latter is inhibitory to the growth of a range of Gram-positive and Gram-negative bacteria and deters feeding by generalist fishes. A novel ultraviolet absorbing mycosporine-like amino acid is also produced by the cyanobiont of *Dysidea herbacea*.

Many other cyanobacteria enter into mutualistic associations with animals. They include the species and strains of the genera *Dermocarpa*, *Xenococcus*, *Synechococcus*, *Oscillatoria*, *Anabaena* and *Nostoc*. An apparently mutualistic association occurs between the larvae of the chironomid midge *Cricotopus nostocicola* living inside colonies of *Nostoc parmelioides*. The larvae eat the *Nostoc* cells are protected from predation. In return, they supplement to the growth of *Nostoc* colony by taking  $O_2$ , which is not properly diffused out of the colony. High  $O_2$  partial pressure inside the colony inhibits the function of nitrogenase enzyme resulting in poor nitrogen fixation. Similarly, the larvae also promote inward diffusion of  $CO_2$ , which is limiting at the centre of the colony. The cyanobacterium is also benefited from the silk of the larvae, which help in the attachment of the colony to substratum. In addition, the grazing from the larvae keeps the cell density inside the colony at a low level making it more diffusible to gases ( $O_2$  and  $CO_2$ ).

### 10.5.3 Symbiosis of Invertebrates with Chemolithotrophic and Methanotrophic Bacteria

In the deep-sea hydrothermal vents, free-living chemolithotrophs oxidize reduced minerals that serve as food for various invertebrate grazers and filter feeders, but groups of invertebrates that are mutualistic with chemolithotrophic bacteria also live around the vents. These animals are unique and have not been found in any other marine environments. They are the vestimentiferan tube worms, represented by *Riftia pachyptila*, the vesicomyid clams such as *Vesicomya chordata* and *Calyptogena magnifica* and the mytilid mussels such as *Bathymodiolus thermophilus*. The giant tube worm *R. pachyptila* contains very large populations of endosymbiotic chemolithotrophic bacteria (Fig. 10.10). The worm derives its entire food and energy supply from the chemoautotrophic activity of its  $H_2S$ -oxidizing endosymbionts present in its body tube (trophosome). The animal protects these bacteria and, through its gills and circulatory system, transports favourable ratios of  $H_2S$ ,  $O_2$  and  $CO_2$  for their chemolithotrophic activity. The animal, through its respiration, also produces  $CO_2$  available for fixation and its nitrogenous wastes can be utilized by the endosymbiont as the nutrient source.

In the clam *C. magnifica*, the symbionts are located in the fleshy gills. The clam filter feeds normally and its gill symbionts appear to supplement only its nutrition. A similar association occurs in the mussel *B. thermophilus*. Although none of the endosymbionts could be cultivated, comparison of nucleic acid sequences indicate that each species of animal has its own unique chemolithotrophic endosymbiont, but these are closely related to each other. All appear to belong to the gamma subgroup of proteobacteria and the genus *Thiomicrospira* is the closest free-living relative to the chemolithotrophic endosymbionts.



**Figure 10.10** - Endosymbiotic bacteria in the tube worm *Riftia pachyptila* (Adopted from Tunncliffe, 1992).

Sulphide-oxidizing symbiotic bacteria are observed in the gills of several estuarine bivalves, such as the gutless clam *Solemya reidi* and other species of the genera *Solemya*, *Lucinoma*, *Myrtea* and *Thyasira*. Some of the symbionts are intracellular but most of them are located in separate bacteriocysts between the cuticle and the animal tissue. Some of the bivalves have retained filter-feeding capacity; others rely on their symbionts exclusively for the supply of food and nutrition. In *S. reidi*, the sulphide oxidation is mediated not only by the symbiotic bacteria in the gills but also by the mitochondria of the host clam.

#### 10.5.4 Other Symbiotic Relationships

Some marine invertebrates and fish establish mutualistic relationships with luminescent bacteria. Depending upon the particular species, the organs containing the luminescent bacteria may be localized near the eye, abdomen, rectum, or jaw. In squid, for example, luminescent bacteria occur in a pair of glands in the mantle cavity near the ink sac. To date, efforts to culture these bacteria have failed. The luminescent bacteria, *Vibrio* and *Photobacterium* species, are contained within special saclike organs that generally have external pores that allow the bacteria to enter and provide for exchange with the surrounding seawater. The fish supplies the bacteria with nutrients and protection from competing microorganisms. Luminescent bacteria, in return, emit light continuously or as flashes. Some nocturnal fishes use this light to some extent as a flashlight or headlamp. Light emission is also thought to aid schooling behaviour of fishes and perhaps repel predators.

Novel endosymbiotic relationships of bacteria and archaea in invertebrate animals are being described in rapid succession. Typically, they are based on microscopic observations in combination with the use of fluorescent

oligonucleotide probes specific for procaryotic nucleic acid sequences. The specific nature of the interaction often remains to be elucidated. This task is frequently rendered difficult by the inability to grow the endosymbiont outside its host. As an example of such still obscure mutualistic endosymbionts, all aphids have cell clusters that are called *mycetomes* and the individual cells *mycetocytes* harbouring bacteria. Upon the elimination of the endosymbionts by antibiotic treatment, the aphid stops reproducing and eventually dies. Preliminary evidence indicates that the endosymbionts may produce amino acids absent from the plant sap that the aphid feeds on.

A species of *Epulopiscium* has been identified as an endosymbiont in the intestines of surgeon fish living in the Red Sea. This bacterium is larger than the ciliate protozoan *Paramecium*; it has a cell size over 1 million times larger than a typical bacterial cell. *Epulopiscium* species have yet to be cultured and so their functional roles are yet to be known. They are related to clostridia and may play a role in cellulose digestion.

## 10.6 RECONSTITUTION OF THE SYMBIOSES

Reconstruction of symbioses by introducing axenic cultures of the two symbiotic partners has been successfully achieved with all of the plant symbioses except *Azolla*, this being the only case in which it has not been possible to grow the major cyanobiont in free culture. However, the artificial reconstruction of the *Azolla* symbiosis has been achieved by removal of a megasporocarp apical cap containing the cyanobiont and transferring this to the decapitated megasporocarp of another *Azolla* plant. In contrast, the bryophyte and legume symbioses have proved particularly easy to reconstitute, not just with the original symbiont, but with some less specific partners as well. For example, in bryophytes symbiosis could be induced by a cyanobiont isolated from *Gunnera*, cycads and even free-living strains. All of the symbiotically-competent cyanobacteria are hormogonia-forming strains of the genus *Nostoc*. Similar observations have been made with *Gunnera*, which can be infected by both symbiotic and freeliving *Nostoc* strains, but not by hormogonia-forming cyanobacteria of other genera such as *Calothrix*, *Fischerella* and *Chlorogloeopsis*.

In case of symbiotic association between microbes, the interdependence has been found to be very less with each of the partners still retaining their capacity for independent life. Therefore, the reconstitution of microbe-microbe symbioses is the easiest to perform. These also occur freely in the ecosystem. However, the environmental factors and the availability of nutrition are the strong driving forces to encourage or discourage these associations.

Reconstitution experiments are very helpful in studying the infection process and the establishment of symbiotic colonies can provide information on the specificity of the association. They confirm the broad competence of symbionts, but are likely to give a false impression of the breadth of strains capable of infecting plants (or other hosts) in the open environment, because such experiments are artificial in their use and during captive association they are

exposed to less intense environmental extremes. For example, competition experiments using several cyanobacteria at once do not mimic the natural environment in which many cyanobacterial strains will be present, not to mention the unknown influences of many other bacteria and eukaryotes. *Calothrix* and *Chlorogloeopsis*, both capable of forming hormogonia, can infect *Phaeoceros* in the laboratory, yet the latter has not been found in field samples of any bryophyte and the former only very rarely, possibly reflect both the relative abundance of these strains in the soil and their inability to compete with the natural populations of *Nostoc*.

Although it is often possible to grow separately the photobiont and mycobiont partners of lichens, resynthesis of the lichen symbiosis has proved extremely difficult. This is in part because of the very slow growth rates of the mycobionts in particular. However, there are an increasing number of examples of partial or complete resyntheses of both bipartite and tripartite lichens. In the case of *Geosiphon*, many *Nostoc* spp., other than *N. punctiforme*, can be incorporated by the fungus as endosymbionts. Indeed, it is thought that in nature *Geosiphon* and the bryophytes *Blasia* and *Anthoceros* can exchange cyanobionts.

## 10.7 CONCLUSION

The volume of symbiotic associations known to date looks extensive but there is little doubt that many more associations remain to be discovered; even our understanding of the microbial associations described in this chapter is mostly rudimentary. The best studied are the cyanobacteria-plant symbioses and legume-Rhizobium association and these have considerable potential as application systems, with regard to the augmentation of crop productivity. However, because of their flexibility and simplicity in comparison with the interactions of microbes with animals or plants, microbe-microbe symbioses have been considered as ideal systems for experimental analysis. One of the most interesting aspects of microbe-plant associations is the signalling between symbiont and host. The identity of the chemical signals and of the genes they regulate are presently unknown, but with the development of molecular and genetic techniques for symbiotic microbes, progress is now being made towards their elucidation. An understanding of the regulatory systems involved in the establishment and the maintenance of plant-microbe association may open the possibility of enhancing existing symbioses or even producing novel ones with many more higher plants. All this offers hope that the development of novel associations with agriculturally important plants may be feasible. We can hope for a better understanding and use of the microbe-microbe interaction in future.

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# MICROBIAL MOBILIZATION OF NUTRIENTS

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## 11.1 INTRODUCTION

Soil is a major reservoir of organic carbon on earth and thus constitutes an important habitat for enormous number of diverse living organisms in the soil assembled in complex and varied communities. Microbes are an essential component of the soil decomposition sub-system, in which plant and animal residues are degraded into simple and assimilable organic matters and nutrients are released into the food webs. Due to the metabolic diversity of microorganisms and their ability to derive energy from a wide variety of substrates, microbes are able to thrive well in diverse habitats and form a crucial component of the biosphere catalyzing various processes that are important for sustaining all the life on earth and are thus the engines for the biogeochemical cycles such as the carbon, nitrogen, sulphur and phosphorus cycles.

Microbial biodiversity plays a significant role in the plant nutrition because it offers countless new genes and biochemical pathways to probe for enzymes, antibiotics and other useful molecules. Bacterial species belonging to several genera, collectively known as rhizobia, are known to establish a symbiotic relationship with plants belonging to the family Fabaceae. The rhizobial symbiotic nitrogen fixation contribute the most of the biologically fixed nitrogen to the crop fields. The processes of nitrate and ammonia assimilation, ammonia oxidation, nitrification, denitrification and nitrogen transformation are responsible for nitrogen cycling in the biosphere and all these processes require numerous microbial associations. Several groups of fungi and bacteria called the phosphate solubilizing microorganisms (PSMs) assist the plants in mobilization of insoluble forms of phosphate. Soil inoculation with PSMs improves solubilization of fixed soil phosphate and applied phosphates resulting in higher crop yields and, therefore, can be effectively used as biofertilizers in enhancing crop yields in phosphate deficient soils.



Understanding biogeochemical cycles of nutrients allows scientists to decipher and predict the development of microbial communities and activities in the environment. There are many activities that can be harnessed in a beneficial way, such as for remediation of organic and metal pollutants or for recovery of precious metals such as copper or uranium from low-grade ores (See Chapter 13). There are also detrimental aspects of the cycles that can cause global environmental problems such as the formation of acid rain and acid mine drainage, metal corrosion processes and formation of corrosive nitrous oxide, which can deplete the earth's ozone layer. Thus, the knowledge of these cycles is increasingly critical as the human population continues to grow and the impact of human activity on the earth's environment becomes more significant. This chapter deals with the biogeochemical cycles of some vital nutrients, like carbon, nitrogen, sulphur and phosphorus and the role of microbes in their mobilisation.

## 11.2 GAIA HYPOTHESIS

The "Gaia Hypothesis" proposed in early 1970s by the scientist James Lovelock considers that the Earth behaves as a superorganism, with a well-integrated flow of energy and matter from one component to the other. Lovelock (1995) stated that "Living organisms and their material environment are tightly coupled. The coupled system is a superorganism and as it evolves there emerges a new property, the ability to self-regulate climate and chemistry." The basic aspect of this hypothesis is that the earth's physicochemical properties are self-regulated so that they are maintained in a favourable range facilitating the origin and the continuance of life in it. Microbial activity and later the appearance of plants, have changed the original heat-trapping carbon dioxide-rich atmosphere of the earth to the present oxidizing, carbon dioxide-poor atmosphere. This has also allowed Earth to maintain an average surface temperature of 13°C, which is favourable to the life that exists on earth.

The relationship between the Gaia Hypothesis and biogeochemical cycles can be built on the basis of similarities in nutrient flow from one component to the other in the ecosystems by the activities of numerous living organisms. The biological processes have driven the response to the slow warming of the sun resulting in the major atmospheric changes that have occurred over the last 4–5 billion years, i.e. from a reducing (anaerobic) atmosphere existed at the beginning to the present day oxidizing (aerobic) one. The initial reactions that mediated the formation of organic carbon were abiotic, driven by large influxes of ultraviolet (UV) light. The resulting reservoir of organic matter was utilized by early anaerobic heterotrophic organisms. This was followed by the development of the ability of microbes to fix carbon dioxide photosynthetically, most probably under reducing environment. Evidence from stromatolites (bacteria and/or cyanobacteria that grew in mats and entrapped or precipitated inorganic material as they grew) suggests that the ability to photosynthesize was developed at least 3.5 billion years ago. The evolution of photosynthetic organisms tapped solar energy and provided a mechanism for carbon recycling. This first carbon cycle

was maintained for approximately 1.5 billion years. Geologic evidence then suggests that approximately 2 billion years ago, photosynthetic microorganisms developed the ability to produce oxygen. This allowed oxygen to accumulate in the atmosphere, resulting in a change from reducing to oxidizing conditions.

At the same time that the carbon cycle evolved, the nitrogen cycle emerged because nitrogen was a limiting element for microbial growth. Although molecular nitrogen was abundant in the atmosphere, microbial cells could not directly utilize nitrogen as  $N_2$  gas. Cells required organic nitrogen compounds or reduced inorganic forms of nitrogen for growth. Therefore, under the reducing conditions found on early Earth, some organisms developed a mechanism for fixing nitrogen using the enzyme nitrogenase. Nitrogen fixation remains an important microbiological process but the nitrogenase enzyme is totally inhibited in the presence of oxygen. Since growth of living organisms required a wide range of nutrients, the production, conversion and degradation of organic matters caused the cycling of all major and minor nutrients. However, over the millions of years some of the nutrients (e.g., metals) have become more in concentration while some other (e.g., phosphorus) have become immobilized due to microbial activities on the earth.

When considered over this geologic time scale of several billion years, it is apparent that biogeochemical activities have been unidirectional. This means that the predominant microbial activities on earth have evolved over this long period of time to produce changes and to respond to changes that have occurred in the atmosphere, i.e., the appearance of oxygen and the decrease in carbon dioxide content. Presumably these changes will continue to occur, but they occur so slowly that we do not have the capacity to observe them. However, one can also consider biogeochemical activities on a more contemporary time scale, that of tens to hundreds of years. On this much shorter time scale, biogeochemical activities are regular and cyclic in nature. On the one hand, the presumption that earth is a superorganism and can respond to drastic environmental changes is heartening when one considers that human activity is effecting unexpected changes in the atmosphere, such as ozone depletion and buildup of carbon dioxide. However, it is important to point out that the response of a superorganism is necessarily slow.

## 11.3 MICROBIAL MOBILIZATION OF CARBON

### 11.3.1 Global Carbon Reservoirs

There are various global reservoirs of carbon, some of which are immense in size and some other are relatively small (Table 11.1). The largest carbon reservoir is carbonate rock found in the earth's sediments. This reservoir is four orders of magnitude larger than the carbonate reservoir found in the ocean and six orders of magnitude larger than the carbon reservoir found as carbon dioxide in the atmosphere. If one considers these three reservoirs, it is obvious that the atmosphere, which provide  $CO_2$  for photosynthesis, is in the smallest of the

reservoirs. However, the atmospheric CO<sub>2</sub> is most actively cycled making it sufficient to cater to the demand of the producers for photosynthesis. Industrialization, burning of fossil fuel, decrease of forest cover and biodiversity loss have affected several of the smaller carbon reservoirs, resulting in reduction of the amount of fixed organic carbon in these reservoirs and addition to the atmospheric carbon dioxide (Table 11.2). We are fortunate to have a vast oceanic surface which absorbs a significant amount of the atmospheric CO<sub>2</sub>, thus attenuating the current alarming rate of increase of atmospheric CO<sub>2</sub> concentration. The carbonate rock found in the ocean acts as a buffer between the atmospheric and sediment carbon reservoirs. However, there has still been a net efflux of carbon dioxide of approximately  $7 \times 10^9$  metric tons/year into the atmosphere. The problem with this imbalance is that because atmospheric carbon dioxide is a small carbon reservoir, the result of a continued net efflux over the past 100 years has been a 28% increase in atmospheric carbon dioxide from 0.026% to 0.033%.

**Table 11.1:** The global carbon reservoirs that exchange the nutrient among them.

| <i>Carbon reservoir</i>            | <i>Carbon content<br/>(Giga metric tons)</i> | <i>Cycling efficiency</i> |
|------------------------------------|----------------------------------------------|---------------------------|
| Atmosphere<br>CO <sub>2</sub>      | 670                                          | Active                    |
| Ocean                              |                                              |                           |
| Biomass                            | 4                                            | Less active               |
| Carbonates                         | 38000                                        | Inactive                  |
| Dissolved and particulate organics | 2100                                         | Active                    |
| Land                               |                                              |                           |
| Biota                              | 500                                          | Active                    |
| Humus                              | 1200                                         | Active                    |
| Fossil fuel                        | 10000                                        | Active                    |
| Earth's crust                      | $12 \times 10^7$                             | Inactive                  |

Source: Dobrovolsky (1994).

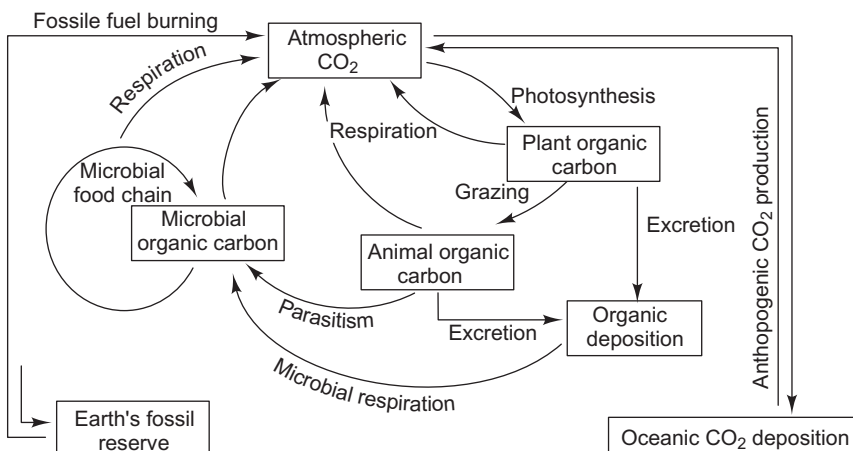
**Table 11.2:** Net rate of carbon transport between selected carbon reservoirs.

| <i>Carbon source</i>              | <i>Transport<br/>(Giga metric tons /year)</i> |
|-----------------------------------|-----------------------------------------------|
| Release by fossil fuel combustion | 7                                             |
| Land clearing                     | 3                                             |
| Forest harvest and decay          | 6                                             |
| Forest regrowth                   | −4                                            |
| Net uptake by oceans (diffusion)  | −3                                            |
| Annual flux                       | 9                                             |

Source: Maier (2000).

### 11.3.2 Energy Flow Through Carbon

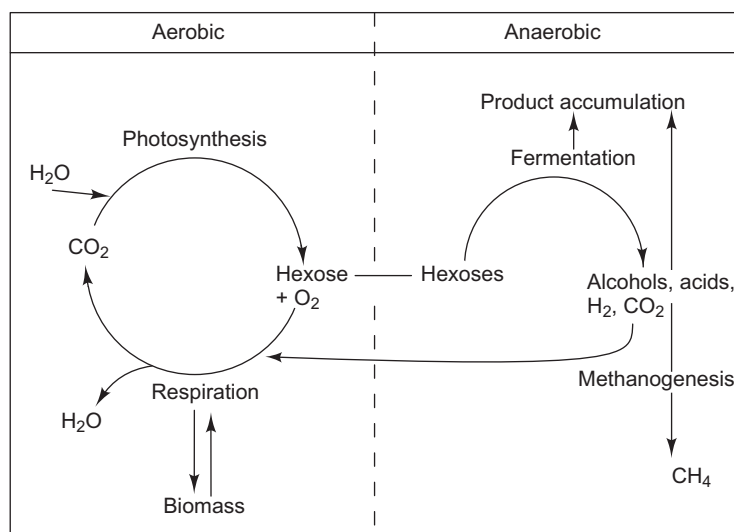
The solar energy is fixed by the autotrophs, plants and microorganisms such as algae, cyanobacteria, some bacteria and some protozoa, into organic matter that acts as a reserve of energy for the consumers and the degraders (Fig. 11.1). However, in comparison to availability, the efficiency of sunlight trapping is very low; less than 0.1 % of the sunlight that hits the earth is actually utilized. As the fixed solar energy moves up in the food web through various trophic levels and organisms of differing complexities, up to 90% of the trapped energy is lost through respiration, which is referred to as the maintenance cost. Despite this seemingly inefficient trapping, photoautotrophic primary producers support most of the considerable ecosystems found on the earth convincing us that the carbon fixation is still enough to satisfy the demand by the consumer population. Productivity varies widely among different ecosystems depending on the climate, the type of primary producer and whether the system is a managed one. For example, one of the most productive natural areas is the coral reef. Managed agricultural systems such as rice fields and sugarcane systems are also very productive, but such high productivity of these systems is due to the input of a significant amount of energy in terms of fertilizer addition, irrigation and care. The open ocean has much lower productivity due to its ultraoligotrophic nutrient status, but covers a majority of the earth's surface and so is a major contributor to primary production. In fact, the aquatic and terrestrial environments contribute almost equally to global primary production and almost 60% of the oceanic primary production occurs in the nutrient poor open ocean. However, the consumer population of open ocean is highly dependent on the detritus flux from the coastal ecosystem as the open oceanic primary production is too low to satisfy their energy demand. Plants predominate in terrestrial environments, but with the exception of immediate coastal zones, microorganisms are responsible for most primary production in aquatic environments. It follows that microorganisms are responsible for approximately one half of all primary production on the earth.



**Figure 11.1** - The scheme of carbon cycling among major components of the biosphere.

### 11.3.3 Carbon Respiration

Respiratory breakdown of organic carbon for release of energy to satisfy the energy demand of living organisms make the most use of organic matters in the ecosystem. The end-products of respiration are carbon dioxide and new cell mass. The residual organic matters that are accumulated as biomass in the ecosystems then become available for other uses like fuel and industrial raw materials. There is a balance between the carbon fixed by photosynthesis and organic carbon consumed by respiration. Many different organic compounds are found in the environment and the microbial catabolic activities have evolved in response to these organic compounds to make use of them. These include organic polymers, humus and C1 compounds such as methane ( $\text{CH}_4$ ). It is important to understand the fate of these naturally occurring organic compounds. This is because degradative activities that evolved for these compounds form the basis for degradation pathways that may be applicable to organic contaminants that are spilled in the environment. But before looking more closely at the individual carbon compounds, it should be pointed out that the carbon cycle is actually not quite as simple as depicted in Fig. 11.1. This simplified figure does not include anaerobic processes, which were predominant on the early earth and remain important in carbon cycling even today. Under anaerobic conditions, which predominated for the first several million years on earth, some cellular components were less degradable than others (Fig. 11.2). This is especially true for highly reduced molecules such as cellular lipids. These components were therefore left over and buried with sediments over time and were naturally subjected to high temperature and pressure to become the present-day fossil fuel reserves. Another carbon compound produced under anaerobic conditions is



**Figure 11.2** - The combination of aerobic and anaerobic pathways for degradation of organic carbon compounds (Modified from Maier, 2000).

methane. It is produced in soils as an end-product of anaerobic respiration. Methane is also produced under the anaerobic conditions found in ruminants such as cows.

### 11.3.4 Microbial Degradation of Organic Carbon

The organic polymers constitute the majority of organic carbons that support the heterotrophic activity on the earth. The three most common plant polymers are—cellulose, hemicellulose and lignin. There are also various other polymers including starch, chitin and peptidoglycan. These various polymers can be divided into two groups on the basis of their structures: the carbohydrate-based polymers, which include the majority of the polymers found in the environment and the phenylpropane-based polymer, lignin.

Cellulose is the most abundant polymers found on the earth. Most of the cellulose is produced by plants but there are also residual carbohydrates of animal and microbial origin. It is a linear molecule containing 1000 to 10,000  $\beta$ -1,4 linked glucose subunits and having a molecular mass of up to  $1.8 \times 10^6$ . These linear molecules are degraded by extracellular enzymes, that can begin the polymer degradation process outside the cell. The two extracellular enzymes, that initiate cellulose degradation, are  $\beta$ -1,4-endoglucanase and  $\beta$ -1,4-exoglucanase. The former hydrolyzes cellulose molecules randomly within the polymer, producing smaller and smaller cellulose molecules while the latter consecutively hydrolyzes two glucose subunits from the reducing end of the cellulose molecule, releasing the disaccharide cellobiose. A third enzyme, known as  $\beta$ -glucosidase or cellobiase, then hydrolyzes cellobiose to glucose. Both cellobiose and glucose can be taken up by many bacterial and fungal cells.

Hemicellulose is the second most common plant polymer. This molecule is more heterogeneous than cellulose, consisting of a mixture of several monosaccharides including various hexoses and pentoses as well as uronic acids. In addition, the polymer is branched instead of linear. An example of a hemicellulose polymer is the pectin molecule, which contains galacturonic acid and methylated galacturonic acid. Degradation of hemicellulose is similar to that of cellulose but many more extracellular enzymes are involved in degradation because of its heterogeneous nature. Starch, chitin and peptidoglycan are also found to be very important organic polymers on the earth requiring a variety of enzymes for their complete degradation.

Lignin is the third most common plant polymer and is strikingly different in structure from all of the carbohydrate-based polymers. The basic building blocks of lignin are the two aromatic amino acids—tyrosine and phenylalanine. These are converted to phenylpropene subunits such as coumaryl alcohol, coniferyl alcohol and sinapyl alcohol. Then 500 to 600 phenylpropene subunits are randomly polymerized, resulting in the formation of the amorphous aromatic polymer known as lignin. In plants, lignin surrounds cellulose microfibrils and strengthens the cell wall. Lignin also helps make plants more resistant to pathogens. Biodegradation of lignin is slower and less complete than degradation

of other organic polymers because it is structurally complex and has aromatic residues. Biodegradation of intact lignin polymers occurs only aerobically, which is not surprising because reactive oxygen is needed to release lignin residues. However, once residues are released, they can be degraded under anaerobic conditions. A variety of microbes including fungi, actinomycetes and bacteria degrade lignin initially through cometabolism but latter metabolic degradation occurs. The best studied organism with respect to lignin degradation is the white rot fungus *Phanerochaete chrysosporium*.

Although the majority of organic matters that are released into the environment are respired to form new cell mass and carbon dioxide, a small amount of this carbon becomes available to form humus. Humus formation is an integral part of the global carbon cycle, which adds organic carbon to soil for slow and continuous degradation for a prolonged period. Humus is formed in a two-stage process that involves the formation of reactive monomers during the degradation of organic matter, followed by the spontaneous polymerization of some of these monomers into the humus molecule. To understand this spontaneous process, consider the degradation of the common organic polymers (cellulose, hemicellulose and lignin) found in soil. Each of these polymers requires the production of extracellular enzymes that begin the polymer degradation process. In particular for lignin, these extracellular enzymes are nonspecific and produce hydrogen peroxide and oxygen radicals. Some of the reactive residues, released during polymer degradation, repolymerize and result in the production of humus. In addition, nucleic acid and protein residues, that are released from dying and decaying cells, contribute to the pool of molecules available for humus formation. Due to its high stability the turnover rate for humus ranges from 2 to 5% per year, depending on climatic conditions.

### 11.3.5 Anaerobic Microbial Degradation of Organic Carbon

Under anaerobic condition, methanogenesis is a predominant organic carbon degradation process performed by microbes, although a small amount of methane is generated naturally through volcanic activity. At present a substantial amount of methane is released to the atmosphere as a result of energy harvesting and utilization. The second way in which methane is released is through landfill gas emissions. It is estimated that landfills in India generate approximately  $8 \times 10^6$  metric tons of methane per year. Although methane makes a relatively minor carbon contribution to the global carbon cycle, global methane emission is of concern from several environmental aspects. The main reason being that methane has 21 times more global warming potential than of  $\text{CO}_2$ . In fact, it is the second most common greenhouse gas emitted to the atmosphere. Further, localized production of methane in landfills can create safety and health concerns.

The methanogens, the organisms responsible for methanogenesis, are a group of obligately anaerobic archaeobacteria capable of using a variety of simple organic matters as well as  $\text{CO}_2$  and  $\text{H}_2$  to produce methane. Thus, they can grow



autotrophically as well as heterotrophically on a limited number of other  $C_1$  and  $C_2$  substrates including acetate, methanol and formate (Table 11.3). Since there are very few carbon compounds that can be used by methanogens, these organisms are dependent on the production of these compounds by other microbes in the surrounding community. Such an interdependent community of microbes typically develops in anaerobic environments. In this community, the more complex organic molecules are catabolized by populations that ferment or respire anaerobically, generating  $C_1$  and  $C_2$  carbon substrates as well as  $CO_2$  and  $H_2$  that are then used by methanogens.

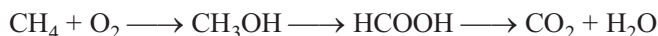
**Table 11.3:** Various substrates for methanogens under anaerobic conditions.

| <i>Substrate</i>            | <i>Products</i>           | $-\Delta G^\circ$<br>(kJ/mol $CH_4$ ) |
|-----------------------------|---------------------------|---------------------------------------|
| Acetate                     | $CH_4 + CO_2$             | 31.0                                  |
| $4H_2 + CO_2$               | $CH_4 + 2H_2O$            | 135.6                                 |
| $4HCOOH$                    | $CH_4 + 3CO_2 + 2H_2O$    | 130.1                                 |
| $4CO_2 + 2H_2O$             | $CH_4 + 3CO_2$            | 211.0                                 |
| 4Methanol                   | $3CH_4 + CO_2 + 2H_2O$    | 104.9                                 |
| Methanol + $H_2$            | $CH_4 + H_2O$             | 112.5                                 |
| 2Ethanol + $CO_2$           | $CH_4 + 2Acetate$         | 116.5                                 |
| 4Propanol + $CO_2$          | $CH_4 + 4Acetone + 2H_2O$ | 36.5                                  |
| 4Methylamine + $2H_2O$      | $3CH_4 + CO_2 + 4NH_3$    | 75.0                                  |
| 2Dimethylamine + $2H_2O$    | $3CH_4 + CO_2 + 2NH_3$    | 73.2                                  |
| 4Trimethylamine + $6H_2O$   | $9CH_4 + 3CO_2 + 4NH_3$   | 74.3                                  |
| 2Dimethylsulphide + $2H_2O$ | $3CH_4 + CO_2 + 2H_2S$    | 73.8                                  |

Source: Whitman *et al.* (1992).

### 11.3.6 Oxidation of Methane and other $C_1$ Compounds

A group of bacteria called the methanotrophs utilize methane as a source of carbon and energy for their growth and development. These are chemoheterotrophic and obligately aerobic species oxidizing methane with the help of methane monooxygenase, the first enzyme in the methane oxidation pathway. The metabolism proceeds as shown below.



Oxygenases, in general, incorporate oxygen into a substrate and are important enzymes in the initial degradation steps for hydrocarbons. However, methane monooxygenase is of particular interest because it was the first of a series of enzymes isolated that can cometabolize highly chlorinated solvents such as trichloroethylene (TCE). Until this discovery, it was believed that biodegradation of highly chlorinated solvents could occur only under anaerobic conditions as an incomplete reaction. The application of methanogens for cometabolic degradation of TCE is a strategy under development for bioremediation of groundwater contaminated with TCE.

Bacteria can also utilize many other C1 carbon compounds like CO, CH<sub>3</sub>OH, HCHO and HCOOH both under aerobic and anaerobic conditions (Table 11.4). These bacteria are called methylotrophs. Utilization of CO by methylotrophs is a very common process in the environment as the annual global production of carbon monoxide approaches  $5 \times 10^9$  metric tons/year. The two major carbon monoxide inputs are abiotic. Approximately 40% of this results from atmospheric photochemical oxidation of carbon compounds such as methane and the rest results from burning of wood, forests and fossil fuels. A small proportion (about 4%) results from biological activity in ocean and soil environments. Carbon monoxide is a highly toxic molecule because it has a strong affinity for cytochromes and on binding to cytochromes, it can completely inhibit the activity of the respiratory electron transport chain.

**Table 11.4:** The C1 carbon compounds of major environmental importance.

| <i>Carbon compound</i> | <i>Importance</i>                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------|
| Carbon dioxide         | Combustion, respiration and fermentation end-product, a major reservoir of carbon on earth.             |
| Carbon monoxide        | Combustion product, common pollutant. Product of plant, animal and microbial respiration, highly toxic. |
| Methane                | End product of anaerobic fermentation or respiration.                                                   |
| Methanol               | Generated during breakdown of hemicellulose, fermentation by-product.                                   |
| Formaldehyde           | Combustion product, intermediate metabolite.                                                            |
| Formate                | Found in plant and animal tissues, fermentation product.                                                |
| Formamide              | Formed from plant cyanides.                                                                             |
| Dimethyl ether         | Generated from methane by methanotrophs, industrial pollutant.                                          |
| Cyanide ion            | Generated by plants, fungi and bacteria. Industrial pollutant, highly toxic.                            |
| Dimethyl sulphide      | Most common organic sulphur compound found in the environment, generated by algae.                      |
| Dimethyl sulfoxide     | Generated anaerobically from dimethyl sulphide.                                                         |

*Source:* Maier (2000).

Degradation of carbon monoxide can occur abiotically by photochemical reactions in the atmosphere. Microbial processes also contribute significantly to its destruction, even though it is a highly toxic molecule. The degradation of carbon monoxide seems to be quite efficient because the level of carbon monoxide in the atmosphere has not risen significantly since industrialization began, even though carbon monoxide emissions have substantially increased. The ocean is a net producer of carbon monoxide and releases it into the atmosphere. In contrast, the terrestrial environment is a net sink for carbon monoxide and absorbs approximately 12% of the globally produced carbon monoxide.

Many methylotrophs are also methanotrophic. They are widespread in the environment because C1 compounds with methane generating potential are

ubiquitous metabolites and in response to their presence, microbes have evolved the capacity to metabolize them under aerobic as well as anaerobic conditions. Both methylotrophic and methanotrophic bacteria coexist in the environment making the metabolism of C1 compounds smooth and simple.

## 11.4 NITROGEN CYCLE

In contrast to carbon, nitrogen is taken up in the form of mineral salts and cycle oxidoreductively. It can exist in numerous oxidation states, from  $-3$  in ammonium ( $\text{NH}_4^+$ ) to  $+5$  in nitrate ( $\text{NO}_3^-$ ). It is the fourth most common element found in cells, making up approximately 12% of cell dry weight and includes the microbially catalyzed processes of nitrogen fixation, ammonium oxidation, assimilatory and dissimilatory nitrate reduction, ammonification and ammonium assimilation.

### 11.4.1 The Nitrogen Reserves in the Environment

Nitrogen in the form of the inert gas, dinitrogen ( $\text{N}_2$ ), is continually released into the atmosphere from volcanic and hydrothermal eruptions and is one of the major global reservoirs of nitrogen (Table 11.5). A second major reservoir is the nitrogen that is found in the earth's crust as bound, nonexchangeable ammonium. Neither of these reservoirs is actively cycled. The nitrogen in the earth's crust is unavailable and the nitrogen in the atmosphere must be fixed before it is available for biological use. Because nitrogen fixation is an energy-consuming process and is carried out by a limited number of microorganisms, it is a relatively slow process. Smaller reservoirs of nitrogen include the organic nitrogen found in living biomass and in dead organic matter, and the soluble inorganic nitrogen salts. These small reservoirs tend to be actively cycled, particularly because nitrogen is often a limiting nutrient in the environment. For example, soluble

**Table 11.5:** The global nitrogen reservoirs that exchange nitrogen among them.

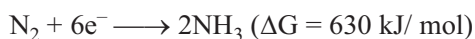
| <i>Nitrogen reservoir</i>          | <i>Nitrogen (Giga metric tons)</i> | <i>Cycling efficiency</i> |
|------------------------------------|------------------------------------|---------------------------|
| Atmosphere                         |                                    |                           |
| $\text{N}_2$                       | $39 \times 10^5$                   | Almost inactive           |
| Ocean                              |                                    |                           |
| Biomass                            | 0.52                               | Active                    |
| Dissolved and particulate organics | 300                                | Active                    |
| Soluble salts                      | 690                                | Hyperactive               |
| Dissolved $\text{N}_2$             | $2 \times 10^4$                    | Less active               |
| Land                               |                                    |                           |
| Biota                              | 25                                 | Active                    |
| Organic matter                     | 110                                | Less active               |
| Earth's crust                      | $77 \times 10^4$                   | Inactive                  |

Source: Dobrovolsky (1994).

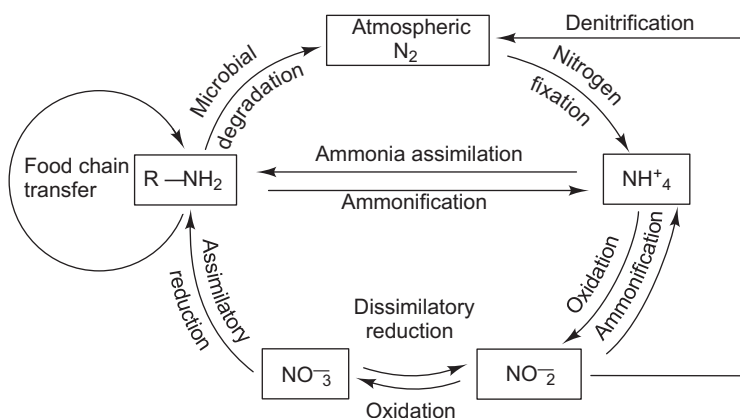
inorganic nitrogen salts in terrestrial environments can have turnover rates greater than one per day. Nitrogen in plant biomass turns over approximately once a year and nitrogen in organic matter turns over once in several decades.

### 11.4.2 Fixation of Molecular Nitrogen

The atmospheric nitrogen is fixed symbiotically as well as asymbiotically by numerous microorganisms (bacteria and cyanobacteria) possessing nitrogenase enzyme. The latter is the enzyme complex with two coproteins, one containing molybdenum plus iron and the other containing only iron. It is extremely sensitive to oxygen, requiring low oxygen tensions for activity. The fixation of nitrogen needs not only nitrogenase, but also ATP, reduced ferredoxin and perhaps other cytochromes and coenzymes. Ammonia is formed as the first detectable product of nitrogen fixation as shown in the following equation:



The ammonia is assimilated into amino acids, which are then polymerized into proteins. The energy balance of the reaction indicates that it requires high energy input. The genes responsible for nitrogen fixation, including the *nif* genes for nitrogenase production, are under stringent genetic regulation. This is an extremely complex regulatory system that controls the expression of multiple *nif* genes required for the production of active nitrogenase. The fixation of nitrogen generally occurs in response to low or limiting concentrations of ammonia (Fig. 11.3).



**Figure 11.3** - The nitrogen cycle showing chemical forms and key processes in biogeochemical cycling of nitrogen.

In terrestrial habitats, the symbiotic fixation of nitrogen by rhizobia accounts for the largest contribution of combined nitrogen. The rates of nitrogen fixation by symbiotic rhizobia are often two to three orders of magnitude higher than rates exhibited by free-living nitrogen-fixing bacteria in soil (see chapter 10). Redox potential is especially important in determining the rates of nitrogen fixation by

many bacterial species that are genetically capable of producing nitrogenase enzymes. Many aerobic nitrogen-fixing bacteria have been shown to fix nitrogen more efficiently at oxygen levels below the normal atmospheric concentration (21 %). Such conditions are frequently found in subsoil and sediment habitats. In addition to *Azotobacter* and *Beijerinckia*, the well-established genera of free-living nitrogen-fixing soil bacteria have been found over time to fix atmospheric nitrogen. They now include species of *Chromatium*, *Rhodopseudomonas*, *Rhodospirillum*, *Rhodomicrobium*, *Chlorobium*, *Azospirillum*, *Desulfovibrio*, *Desulfotomaculum*, *Klebsiella*, *Bacillus*, *Clostridium*, *Azospirillum*, *Pseudomonas*, *Vibrio* and *Thiobacillus*. Several members of Actinomycetales have also been shown to fix atmospheric nitrogen, some of which are free-living and others which fix atmospheric nitrogen in association with compatible plants (see chapter 10).

In aquatic habitats and in soils saturated with water (rice field habitats), cyanobacteria are the principal nitrogen fixers. Many of the filamentous nitrogen-fixing cyanobacteria, such as *Anabaena*, *Rivularia*, *Nostoc*, *Gloeotrichia*, *Cylindrospermum*, *Calothrix*, *Scytonema* and *Tolypothrix* have heterocysts, which are thick-walled, less pigmented and often enlarged cells occurring at intervals among vegetative cells. During their differentiation, heterocysts lose their oxygen-evolving photosystem II but retain their anoxygenic photosystem I. Nitrogen fixation is localized in the heterocysts, where the oxygen-sensitive nitrogenase is protected from inactivation by the photosynthetically produced oxygen. There is an active exchange of metabolites between the ordinary cells and the heterocysts, the former supplying disaccharides from photosynthesis and the latter supplying fixed nitrogen, mainly as glutamine, after ammonia assimilation via GS-GOGAT pathway. Some nonheterocystous cyanobacteria, like *Aphanocapsa*, *Synechocystis*, *Oscillatoria*, *Trichodesmium*, *Micrococcus* and *Lyngbya*, also fix nitrogen when present as a mat containing a high density of bacteria to filter oxygen.

Rates of nitrogen fixation by cyanobacteria are generally one to two orders of magnitude higher than by free-living non-photosynthetic soil bacteria. Under favourable conditions, cyanobacteria in a rice field may fix up to 150 kg of nitrogen per hectare per year. Cyanobacterial nitrogen fixation is observed both in freshwater and in marine habitats. Some nitrogen-fixing cyanobacteria form symbiotic associations with other microorganisms, ranging from fungi to angiosperm (see chapter 10).

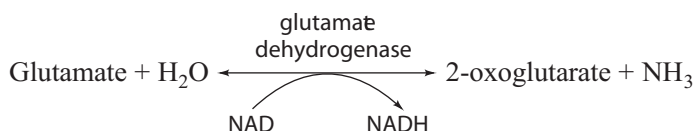
Nitrogen fixation within the phyllosphere and rhizosphere is ecologically important because some of the fixed nitrogen is likely to become available to the plants (see chapters 5 and 10). Exudates from the plant may supply some of the energy required for the nitrogen fixation process. Nitrogen fixation also occurs within the digestive tracts of some animals, such as termites. The fixation of nitrogen in organisms consuming primarily nitrogen-free cellulose is especially important for formation of proteins. Substantial nitrogen fixation has been observed in environments characterized by an abundance of carbon sources and a scarcity of nitrogen, such as decomposing logs or oil-contaminated soils.

*Azotobacter* and *Beijerinckia* can fix nitrogen at normal oxygen tensions and appear to protect their nitrogenases from oxidative inactivation by a combination of compartmentalization and complex biochemical mechanisms. Other free-living nitrogen fixers, such as *Azospirillum*, fix nitrogen only at reduced oxygen tensions. Such environments, especially on the microscale, are not rare in soil and on root surfaces. Oxygen is not a problem for nitrogenases of anoxyphototrophic bacteria such as *Chromatium*, *Rhodopseudomonas*, *Rhodospirillum*, *Rhodomicrobium* and *Chlorobium* or for the anaerobic heterotrophs such as *Clostridium*, *Desulfovibrio* and *Desulfotomaculum*. The latter are active in anaerobic sediments and in the rhizosphere of plants growing in such sediments. As in aerobic nitrogen fixation, carbon-rich, nitrogen-poor substrates, such as cellulose, favour the process. This has been known to occur indirectly as nitrogen fixers utilized low molecular weight products of cellulose fermentation.

### 11.4.3 Assimilation of Ammonium

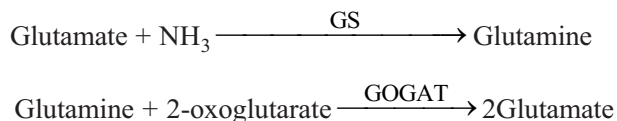
The ammonium produced by nitrogen fixation is assimilated by cells into amino acids to form proteins, cell wall components and nitrogenous bases. This process is known as ammonium assimilation or immobilization. Nitrogen can also be immobilized by the uptake and incorporation of nitrate into organic matter, a process known as assimilatory nitrate reduction. Because nitrate must be reduced to ammonium before it is incorporated into organic molecules, most organisms prefer to take up nitrogen as ammonium, if it is available. However, the bacteria and cyanobacteria show sensitivity to high ammonium concentration and feel ammonium toxicity. Ammonium also causes the inhibition of *nif* genes expression and reduces nitrogenase activity. The process that reverses immobilization, the release of ammonia from dead and decaying cells, is called ammonification or ammonium mineralization. Both immobilization and mineralization of nitrogen occur under aerobic and anaerobic conditions.

There are two pathways that microbes use to assimilate ammonium. The first is a reversible reaction that incorporates or removes ammonia from the glutamate. This reaction is driven by ammonium availability. At high ammonium concentrations and in the presence of reducing equivalents (reduced NADP), ammonium is incorporated into 2-oxoglutarate to form glutamate.



However, in most soil and in many aquatic environments, ammonium is present at low concentration. Therefore, microbes have a second ammonium uptake pathway that is energy dependent. This reaction is driven by ATP and two enzymes, glutamine synthase and glutamine oxoglutarate amino transferase (GS-GOGAT pathway). The first step in this reaction adds ammonium to glutamate to form glutamine and the second step transfers the ammonium molecule from

glutamine to 2-oxoglutarate resulting in the formation of two glutamate molecules.



#### 11.4.4 Ammonification

Ammonium mineralization or ammonification can occur intracellularly by the reversible reaction catalyzed by glutamate dehydrogenase producing metabolisable carboxylic acid and ammonia. It can also occur extracellularly catalyzed by a variety of extracellular enzymes such as proteases, lysozymes, nucleases and ureases released by microorganisms. These enzymes can initiate degradation of nitrogen-containing molecules found outside the cell such as proteins, cell walls, nucleic acids and urea. Many other plant polymers are also hydrolytically degraded extracellularly by enzymes to simple organic compounds, which subsequently become absorbed by actively growing cells. Some of the monomers are acted upon by extracellular enzymes to release ammonium directly to the environment, e.g. extracellular deamination of urea by urease.

The dominance of immobilization or mineralization, depends on whether nitrogen is the limiting nutrient. Under limiting nitrogen concentrations, immobilization becomes the more important process whereas in nitrogen enriched environment mineralization predominates. Nitrogen limitation is dictated by the carbon/nitrogen (C/N) ratio in the environment. Generally, the C/N ratio required for bacteria is 4 to 5 and for fungi is 10. So a typical average C/N ratio for soil microbial biomass is close to 8, at which there is an expected net balance between mineralization and immobilization. However, as only approximately 40% of the carbon in organic matter is actually incorporated into cell mass (the rest is lost as carbon dioxide), the C/N ratio must be increased by a factor of 2.5 to account for the carbon lost as carbon dioxide during respiration. It should also be noted that nitrogen is cycled more efficiently than carbon and there are essentially no losses in its uptake. In fact, a C/N ratio of 20 is not only the theoretical balance point but also the practically observed one. When organic amendments with C/N ratios less than 20 are added to soil, net mineralization of ammonium occurs. In contrast, when organic amendments with C/N ratios greater than 20 are added to soil, net immobilization occurs.

There are numerous possible fates for ammonium that is released into the environment as a result of ammonium mineralization. It can be taken up by plants or microorganisms and incorporated into living biomass, or it can become bound to nonliving organic matter such as soil colloids or humus. In this capacity, ammonium adds to the cation exchange capacity (CEC) of the soil. Ammonium can become fixed inside clay minerals, which essentially traps the molecule and removes the ammonium from active cycling. Some mineralized ammonium can escape into the atmosphere through volatilization. A part of the released

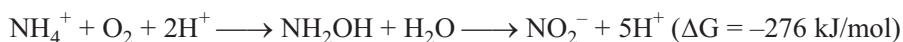


ammonium is also utilized by chemoautotrophic microbes through the nitrification process.

### 11.4.5 Nitrification

Nitrification is the microbially catalyzed conversion of ammonium to nitrate. This is predominantly an aerobic chemoautotrophic process, but some methylotrophs can use the methane monooxygenase enzyme to oxidize ammonium and a few heterotrophic fungi and bacteria can also perform this oxidation. The autotrophic nitrifiers are a closely related group of bacteria. The best studied nitrifiers are from the genus *Nitrosomonas*, which oxidizes ammonium to nitrite and *Nitrobacter*, which oxidizes nitrite to nitrate. The dominant ammonium oxidizers include *Nitrosomonas europaea*, *N. eutrophus*, *N. marina*, *Nitrosococcus nitrosus*, *N. mobilis*, *N. oceanus*, *Nitrospira briensis*, *Nitrosolobus multiformis* and *Nitrosovibrio tenuis*. In addition, nitrite oxidation is caused by *Nitrobacter winogradskyi*, *N. hamburgensis*, *N. vulgaris*, *Nitrospira gracilis*, *N. marina* and *Nitrococcus mobilis*.

Ammonium oxidation is a two-step energy producing reaction releasing energy for CO<sub>2</sub> fixation. The first step of this reaction is catalyzed by the enzyme ammonium monooxygenase, which initiates oxidation of ammonium. The enzyme has broad substrate specificity and can be used to oxidize pollutant molecules such as TCE cometabolically. In the second step, nitrite is converted to nitrate releasing energy that is much less than that of the ammonium oxidation process. The low energy yield shows that a lot of oxygen is required to be consumed to fix CO<sub>2</sub> through nitrification (There is requirement of 34 moles of ammonium or approximately 100 moles of nitrite to fix 1 mole of carbon dioxide).



These two types of nitrifiers, i.e., those that carry out the reactions shown above, are generally found together in the environment. As a result, nitrite does not normally accumulate in the environment. The optimum pH for nitrification is 6.6 to 8.0 and nitrifiers are very sensitive to pH changes. In environments with pH < 6.0, nitrification rates are slowed and below pH 4.5, nitrification seems to be completely inhibited.

Heterotrophic microbes that oxidize ammonium include some fungi and some bacteria. These organisms gain no energy from nitrification, so it is unclear why they carry out the reaction. The relative importance of autotrophic and heterotrophic nitrification in the environment has not yet been clearly determined. Although the measured rates of autotrophic nitrification in the laboratory are an order of magnitude higher than those of heterotrophic nitrification. In acidic soils, heterotrophic nitrification has a significant role to play in the nitrogen remineralization process. The rate of nitrogen utilization is so fast that there is no scope for nitrate accumulation in the natural, undisturbed ecosystems. However,

in agricultural systems that have large inputs of fertilizer, nitrification becomes an important process resulting in the production of large amounts of nitrate. Because nitrate is very mobile in soil systems, most of this excess nitrate are leached into groundwater and surface waters often causing eutrophication of water bodies.

### 11.4.6 Nitrate Reduction

Nitrate ions can be incorporated by a variety of organisms into organic matter through assimilatory nitrate reduction. A heterogeneous group of microorganisms, including many bacterial, fungal and algal species, is capable of assimilatory nitrate reduction. The process involves several enzyme systems, including nitrate and nitrite reductases, to form ammonia, which can be subsequently incorporated into amino acids. The enzyme systems appear to involve soluble metalloproteins and require reduced cofactors, including NADPH. Assimilating nitrate enzyme systems, that have been examined in bacteria, algae and fungi, are repressed by the presence of ammonia or reduced nitrogenous organic metabolites in the growth environment. Normal atmospheric concentrations of oxygen do not appear to inhibit assimilatory nitrate reductase enzyme systems. Assimilatory nitrate reduction does not result in the accumulation of high concentrations of intracellular ammonium ions, because ammonia is incorporated relatively rapidly into organic nitrogen. Excess ammonium would act through feedback inhibition to shut off nitrate reduction.

In the absence of oxygen, nitrate ions can act as terminal electron acceptors, the process is known as nitrate respiration or dissimilatory nitrate reduction. During such respiration, nitrate is converted into a variety of reduced products, while organic matter is simultaneously oxidized. In the absence of oxygen, dissimilatory nitrate reduction allows utilization of organic compounds with a much higher energy yield than fermentation processes.

There are two types of dissimilatory nitrate reduction, both of which are used by facultative chemoheterotrophic organisms under microaerophilic or anaerobic conditions. The first process, called dissimilatory nitrate reduction to ammonia (DNRA), uses nitrate as a terminal electron acceptor to produce energy to drive the oxidation of organic compounds. The end product of DNRA is ammonium. The first step in this reaction, the reduction of nitrate to nitrite, is the energy-producing step. The further reduction of nitrite to ammonium is catalyzed by an NADH-dependent reductase. This second step provides no additional energy, but it does conserve fixed nitrogen and also regenerates reducing equivalents through the reoxidation of NADH to NAD. These reducing equivalents are then used to help in the oxidation of carbon substrates. In fact, under carbon-limiting conditions nitrite accumulates (denitrification predominates), while under carbon-rich conditions ammonium is the major product (DNRA predominates). Table 11.6 lists a variety of bacteria that perform DNRA.

A variety of facultatively anaerobic bacteria, including *Alcaligenes*, *Escherichia*, *Aeromonas*, *Enterobacter*, *Bacillus*, *Flavobacterium*, *Nocardia*,

**Table 11.6:** The important genera of bacteria that convert nitrate or nitrite to ammonium.

| <i>Genus</i>          | <i>Favoured habitat</i> |
|-----------------------|-------------------------|
| Obligate anaerobes    |                         |
| <i>Clostridium</i>    | Soil, sediment          |
| <i>Desulfovibrio</i>  | Sediment                |
| <i>Selenomonas</i>    | Rumen                   |
| <i>Veillonella</i>    | Intestinal tract        |
| <i>Wolinella</i>      | Rumen                   |
| Facultative anaerobes |                         |
| <i>Citrobacter</i>    | Soil, wastewater        |
| <i>Enterobacter</i>   | Soil, wastewater        |
| <i>Erwinia</i>        | Soil                    |
| <i>Escherichia</i>    | Soil, wastewater        |
| <i>Klebsiella</i>     | Soil, wastewater        |
| <i>Photobacterium</i> | Sea water               |
| <i>Salmonella</i>     | Sewage                  |
| <i>Serratia</i>       | Intestinal tract        |
| <i>Vibrio</i>         | Sediment                |
| Microaerophiles       |                         |
| <i>Campylobacter</i>  | Oral cavity             |
| Aerobes               |                         |
| <i>Bacillus</i>       | Soil, food              |
| <i>Neisseria</i>      | Mucous membranes        |
| <i>Pseudomonas</i>    | Soil, water             |

Source: Tiedje (1988).

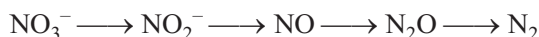
*Spirillum*, *Staphylococcus* and *Vibrio*, reduce nitrate under anaerobic conditions to nitrite. The nitrite produced by these species is excreted, or under appropriate conditions, some of these organisms reduce nitrite via hydroxylamine to ammonium (nitrate ammonification). These organisms do not produce gaseous nitrogen products, i.e. they do not denitrify. Nitrate ammonification plays an important role in stagnant water, sewage plants and some sediments. Unlike assimilatory nitrate reduction, dissimilatory nitrate reductase is not inhibited by ammonia; thus, ammonium ions can be excreted in relatively high concentrations. However, as compared to denitrification, nitrate ammonification is an environmentally less significant process for the reductive removal of nitrate and nitrite ions. Its importance appears to be limited by the number of reducing equivalents that must be consumed in the system.

### 11.4.7 Denitrification

Denitrification refers to the microbial reduction of nitrate, through various gaseous inorganic forms, to  $N_2$ . This is a second but primary type of dissimilatory nitrate reduction found in soil. It is of concern because it cycles fixed nitrogen back into  $N_2$ . When calculated in terms of energy produced for

every eight-electron transfer, denitrification provides more energy per mole of nitrate reduced than DNRA. Thus, in a carbon-limited, electron acceptor-rich environment, denitrification will be the preferred process. Further, some of the gaseous intermediates formed during denitrification, e.g., nitrous oxide ( $\text{N}_2\text{O}$ ), can cause depletion of the ozone layer and can also act as a greenhouse gas contributing to global warming.

Denitrifying nitrate reducers such as *Paracoccus denitrificans*, *Thiobacillus denitrificans* and various pseudomonads have a more complete reduction pathway, converting nitrate ( $\text{NO}_3^-$ ) through nitrite ( $\text{NO}_2^-$ ) to nitric oxide (NO) and nitrous oxide ( $\text{N}_2\text{O}$ ) to molecular nitrogen ( $\text{N}_2$ ). The denitrification process proceeds in four steps as shown below.



The first step, reduction of nitrate to nitrite, is catalyzed by the enzyme nitrate reductase. This is a membrane-bound molybdenum iron-sulphur protein that is found not only in denitrifiers but also in DNRA organisms. Both the synthesis and the activity of nitrate reductase are inhibited by oxygen. Thus, both denitrification and DNRA are inhibited by oxygen and thus predominantly occur under microaerobic conditions. The second enzyme in this pathway is nitrite reductase, which catalyzes the conversion of nitrite to nitric oxide. Nitrite reductase is unique to denitrifying organisms and is not present in the DNRA process. It is found in the periplasm and exists in two forms, a copper-containing form and a heme form, both of which are distributed widely in the environment. Synthesis of nitrite reductase is inhibited by oxygen and induced by nitrate, a feedback stimulation of activity. Nitric oxide reductase, a membrane-bound protein, is the third enzyme in the pathway, catalyzing the conversion of nitric oxide to nitrous oxide. The synthesis of this enzyme is inhibited by oxygen and induced by various nitrogen oxide forms. Nitrous oxide reductase is the last enzyme in the pathway and converts nitrous oxide to dinitrogen gas. This is a periplasmic copper-containing protein. The activity of the nitrous oxide reductase enzyme is inhibited by low pH and is even more sensitive to oxygen than the other three enzymes in the denitrification pathway. Thus, nitrous oxide is the final product of denitrification under conditions of high oxygen (in a relative sense, given a microaerophilic niche) and low pH while under complete anaerobic condition  $\text{N}_2$  is the end product.

Organisms that denitrify are found widely in the environment and display a variety of different characteristics in terms of metabolism and activities (Table 11.7). Some denitrifiers are autotrophic, some are fermentative and some are associated with other aspects of the nitrogen cycle (e.g., nitrogen fixation). In soil, the primary denitrifying genera are *Pseudomonas* and *Alcaligenes*. Many additional genera, such as *Azospirillum*, *Rhizobium*, *Rhodopseudomonas* and *Propionibacterium*, have also been reported to denitrify under some conditions. In aquatic environments, denitrification is more common in stagnant waters than in running streams and rivers.

**Table 11.7:** The genera of denitrifying bacteria involved in active removal of nitrogen compounds from soil and water.

| <i>Genus</i>             | <i>Specific characteristics</i>                                                       |
|--------------------------|---------------------------------------------------------------------------------------|
| <b>Organotrophs</b>      |                                                                                       |
| <i>Alcaligenes</i>       | Common soil bacterium                                                                 |
| <i>Agrobacterium</i>     | Some species are plant pathogens                                                      |
| <i>Aquaspirillum</i>     | Some are magnetotactic, oligotrophic                                                  |
| <i>Azospirillum</i>      | Associative N <sub>2</sub> fixer, fermentative                                        |
| <i>Bacillus</i>          | Spore former, fermentative, some species thermophilic                                 |
| <i>Blastobacter</i>      | Budding bacterium, phylogenetically related to <i>Rhizobium</i>                       |
| <i>Bradyrhizobium</i>    | Symbiotic N <sub>2</sub> fixer with legumes                                           |
| <i>Branhamella</i>       | Animal pathogen                                                                       |
| <i>Chromobacterium</i>   | Purple pigmentation                                                                   |
| <i>Cytophaga</i>         | Gliding bacterium, cellulose degrader                                                 |
| <i>Flavobacterium</i>    | Common soil bacterium                                                                 |
| <i>Flexibacter</i>       | Gliding bacterium                                                                     |
| <i>Halobacterium</i>     | Halophilic                                                                            |
| <i>Hyphomicrobium</i>    | Grows on C1 substrates, oligotrophic                                                  |
| <i>Kingella</i>          | Animal pathogen                                                                       |
| <i>Neisseria</i>         | Animal pathogen                                                                       |
| <i>Paracoccus</i>        | Halophilic, lithotrophic                                                              |
| <i>Propionibacterium</i> | Fermentative                                                                          |
| <i>Pseudomonas</i>       | Commonly isolated from soil, very diverse genus                                       |
| <i>Rhizobium</i>         | Symbiotic N <sub>2</sub> fixer with legumes                                           |
| <i>Wolinella</i>         | Animal pathogen                                                                       |
| <b>Phototrophs</b>       |                                                                                       |
| <i>Rhodopseudomonas</i>  | Anaerobic, sulphate reducer                                                           |
| <b>Lithotrophs</b>       |                                                                                       |
| <i>Alcaligenes</i>       | Uses H <sub>2</sub> , also heterotrophic                                              |
| <i>Bradyrhizobium</i>    | Uses H <sub>2</sub> , also heterotrophic, symbiotic N <sub>2</sub> fixer with legumes |
| <i>Nitrosomonas</i>      | NH <sub>3</sub> oxidizer                                                              |
| <i>Paracoccus</i>        | Uses H <sub>2</sub> , also heterotrophic, halophilic                                  |
| <i>Pseudomonas</i>       | Uses H <sub>2</sub> , also heterotrophic                                              |
| <i>Thiobacillus</i>      | S-oxidizer                                                                            |
| <i>Thiomicrospira</i>    | S-oxidizer                                                                            |
| <i>Thiosphaera</i>       | S-oxidizer, heterotrophic nitrifier, aerobic denitrifier                              |

Source: Myrold (1998).

Nitrification and denitrification in soil often occur in close proximity so that a substantial part of nitrate formed by nitrification afterward diffuses to the anaerobic denitrification zone where it is reduced to N<sub>2</sub>. Denitrification activity of some bacteria, such as *Paracoccus denitrificans*, appears to be restricted to anaerobic conditions. In contrast several strains of bacteria like *Pseudomonas*, *Aeromonas*, *Moraxella* and *Arthrobacter* are capable of respiring nitrate in the

presence of oxygen. Thus, it seems likely that the corespiration of nitrate and oxygen may indeed make a significant contribution to the flux of nitrate to nitrite in the environment.

Various steps within the nitrogen cycle are subject to repression. For example, ammonium ions normally inhibit the fixation of atmospheric nitrogen. Nitrification occurs readily in neutral, well-drained soils but is inhibited in anaerobic or highly acidic soils. Assimilatory nitrate reductase activity is repressed by ammonia; dissimilatory nitrate reductase is repressed by oxygen.

Because various environmental conditions favour specific nitrogen cycling processes, there is a spatial zonation of cycling processes. Fixation of nitrogen occurs in both surface and subsurface habitats. Nitrification occurs exclusively in aerobic habitats. Denitrification predominates in waterlogged soils and in anaerobic aquatic sediments. The cycling of nitrogen within a given habitat also exhibits seasonal fluctuations; during spring and autumn blooms of cyanobacteria, for example, rates of nitrogen fixation in aquatic habitats usually are high, reflecting population fluctuations and availability of needed energy and mineral nutrients for fixation of molecular nitrogen.

## 11.5 SULPHUR CYCLE

Sulphur is the tenth most abundant elements in the crust of Earth constituting about 1% of the mass of an actively growing microbial cell. It is a reactive element with stable valence states from  $-2$  to  $+6$  found in various compounds (Table 11.8). In living organisms, sulphur occurs mainly as sulfhydryl ( $-\text{SH}$ ) groups in amino acids and their polymers. At an average concentration of 520 ppm, it rarely becomes a limiting nutrient, though, enriched conditions are observed in some environments. Some elemental sulphur deposits and some sulphide ores appear to be of biogenic origin. Eruptive and postvolcanic activity continue to introduce additional sulphur into the ecosphere at relatively low rates. With the notable exceptions of ferric and calcium sulphates, most sulphate salts are readily soluble in water. Sulphate is the second most abundant anion in seawater and the sulphate of the marine environment represents a large, slowly cycled sulphur reservoir (Table 11.9). Living and dead organic matter compose a smaller, more rapidly cycled sulphur reservoir. Largely inert sulphur reservoirs are metal sulphides in rock, elemental sulphur deposits and fossil fuels. Human activities, including

**Table 11.8:** Oxidation states of sulphur in various compounds.

| <i>Form</i>            | <i>Example</i>        | <i>Oxidation state</i> |
|------------------------|-----------------------|------------------------|
| $\text{S}^{2-}$        | Sulphides, Mercaptans | $-2$                   |
| $\text{S}^0$           | Elemental sulphur     | $0$                    |
| $\text{S}_2\text{O}_4$ | Hyposulphite          | $+2$                   |
| $\text{SO}_3^{2-}$     | Sulphite              | $+4$                   |
| $\text{SO}_4^{2-}$     | Sulphate              | $+6$                   |

**Table 11.9:** The global sulphur reservoirs that exchange the nutrient among them.

| <i>Sulphur reservoir</i>                                           | <i>Sulphur<br/>(Giga metric tons)</i> | <i>Cycling efficiency</i> |
|--------------------------------------------------------------------|---------------------------------------|---------------------------|
| Atmosphere<br>SO <sub>2</sub> /H <sub>2</sub> S                    | 0.0014                                | Active                    |
| Ocean                                                              |                                       |                           |
| Biomass                                                            | 0.015                                 | Active                    |
| Soluble inorganic ions<br>(primary SO <sub>4</sub> <sup>2-</sup> ) | 12 × 10 <sup>5</sup>                  | Less active               |
| Land                                                               |                                       |                           |
| Living biomass                                                     | 8.5                                   | Active                    |
| Organic matter                                                     | 16                                    | Active                    |
| Earth's crust                                                      | 18 × 10 <sup>6</sup>                  | Inactive                  |

Source: Dobrovolsky (1994).

strip mining and the burning of fossil fuels, have mobilized a part of these inert sulphur reservoirs with destructive pollution consequences.

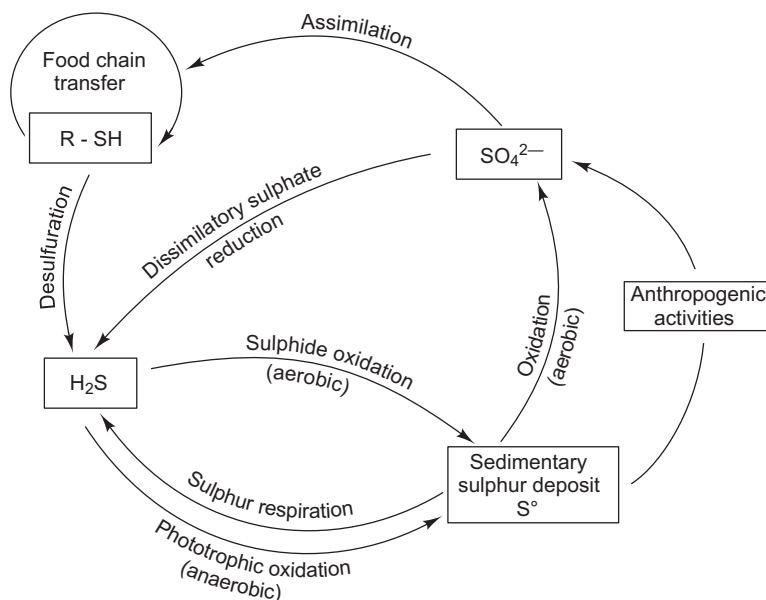
Plants, algae and many heterotrophic microorganisms assimilate sulphur in the form of sulphate. For incorporation into cysteine, methionine and coenzymes in the form of sulfhydryl (–SH) groups, sulphate needs to be reduced to the sulphide level by assimilatory sulphate reduction. A direct uptake as sulphide is not feasible for most microorganisms because of the high toxicity of H<sub>2</sub>S. In assimilatory sulphate reduction, toxicity is avoided by immediately reacting the reduced sulphur with an acceptor (e.g. serine) to yield cysteine.

Organosulphur decomposition in soils and sediments yields mercaptans and H<sub>2</sub>S by desulphuration. This reaction is analogous to ammonification forming the volatile products. In the marine environment, a major decomposition product of organosulphur is dimethylsulphide (DMS), which is highly volatile and escapes the marine environment very quickly. According to some estimates, 90% of the total sulphur flux from the marine environment to the atmosphere occurs in the form of DMS. This product originates from dimethylsulphoniopropionate (DMSP), a major metabolite of marine algae that may have a role in osmoregulation. Dimethylsulphide is released during zooplankton grazing on phytoplankton and also during decay processes. DMS, H<sub>2</sub>S and mercaptans are subject to photooxidative reactions that ultimately yield sulphate. H<sub>2</sub>S also reacts with O<sub>2</sub> in the atmosphere. If H<sub>2</sub>S does not escape to the atmosphere, it may be subject to microbial oxidation under aerobic conditions, or it may be phototrophically oxidized under anaerobic conditions. Under anaerobic conditions, sulphate as well as elemental sulphur may serve as electron acceptors and organic substrates are oxidized. Figure 11.4 summarizes the microbial cycling of sulphur.

### 11.5.1 Assimilatory Sulphate Reduction

The primary soluble form of inorganic sulphur found in soil is sulphate. Most microorganisms take up sulphur in the oxidized sulphate form and then reduce it internally (to sulphide) called assimilatory sulphate reduction. Cells assimilate





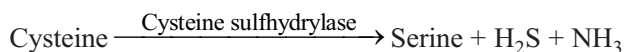
**Figure 11.4** - The biogeochemical cycle of sulphur showing the biological transformation of oxidized and reduced forms of sulphur.

sulphur in the form of sulphate because it is the most available sulphur form and because sulphide is toxic. Sulphide toxicity occurs because inside the cell sulphide reacts with metals in cytochromes to form metal-sulphide precipitates, destroying cytochrome activity. However, under the controlled conditions of sulphate reduction inside the cell, the sulphide can be removed immediately and incorporated into an organic form. Although this process does protect the cell from harmful effects of the sulphide, it is an energy-consuming reaction. After sulphate is transported inside the cell, ATP is used to convert the sulphate into the energy-rich molecule adenosine 5-phosphosulphate (APS). A second ATP molecule is used to transform APS to 3-phosphoadenosine-5-phosphosulphate (PAPS). This allows the sulphate to be reduced to sulphite and then to sulphide in two steps. First PAPS is oxidized by a reduced thioredoxin (RSH) to AMP-3-phosphate (PAP) resulting in release of a sulphite group and an oxidized thioredoxin (RSSP). In the second step, sulphite is converted to  $\text{H}_2\text{S}$  at the expense of three NADPH. Most commonly, the amino acid serine is used to remove sulphide as it is reduced, forming the sulphur-containing amino acid cysteine. The reaction steps given below summarize the reduction processes.

| Reactions                                                                                 | Enzymes              |
|-------------------------------------------------------------------------------------------|----------------------|
| ATP + Sulphate $\longrightarrow$ APS + PPi                                                | ATP sulphurylase     |
| ATP + APS $\longrightarrow$ PAPS                                                          | APS phosphokinase    |
| 2 RSH + PAPS $\longrightarrow$ Sulphite + PAP + RSSP                                      | PAPS reductase       |
| Sulphite + 3NADPH $\longrightarrow$ $\text{H}_2\text{S}$ + 3NADP                          | Sulphite reductase   |
| Serine + $\text{H}_2\text{S}$ $\longrightarrow$ Cysteine + Acetate + $\text{H}_2\text{O}$ | Serine sulphydrylase |

### 11.5.2 Sulphur Mineralization

The release of sulphur from organic forms is called sulphur mineralization. It occurs by microbial activity under both aerobic and anaerobic conditions. The enzyme serine sulphydrylase can remove sulphide from cysteine in the reverse of the reaction shown above or a second enzyme, cysteine sulphydrylase, can remove both sulphide and ammonia as shown below.



In marine environments, one of the major products of algal metabolism is the compound dimethylsulfoniopropionate (DMSP), which is used in osmoregulation of the cell. The major degradation product of DMSP is dimethylsulphide (DMS). Both  $\text{H}_2\text{S}$  and DMS are volatile compounds and therefore can be released to the atmosphere. Once in the atmosphere, these compounds are photooxidized to sulphate.

Normal biological release of reduced volatile sulphur compounds results in the formation of approximately 1 kg  $\text{SO}_4^{2-}$ /hectare year. The use of fossil fuels, which contain organic sulphur compounds, increases the amount of sulphur released to the atmosphere to up to 100 kg  $\text{SO}_4^{2-}$ /hectare year in some urban areas. Emission of sulphur compounds to the atmosphere results in the formation of sulphur acid compounds. For example, the  $\text{SO}_2$  released to the atmosphere returns to soil and surface water as sulphurous acid during rain (acid rain). It damages plant foliage, causes corrosion of stone and concrete building surfaces and can affect weakly buffered soils and lakes (See section 2.7).

### 11.5.3 Oxidative Sulphur Transformations

Under aerobic condition, reduced sulphur compounds can support the growth of a group of chemoautotrophic bacteria under strictly aerobic conditions and a group of photoautotrophic bacteria under strictly anaerobic conditions. Microaerophilic bacteria such as *Beggiatoa*, *Thioploca*, *Thiothrix* and *Thermothrix*, are capable of oxidizing  $\text{H}_2\text{S}$ , as shown below, to sulphur globules photoautotrophically, and deposit them within the cells.



In the absence of  $\text{H}_2\text{S}$ , these globules are slowly oxidized to sulphate. These typical gradient organisms position themselves on the interface of an anaerobic environment, the sediment and the partially oxygenated water in contact with the sediment.

Some species of *Thiobacillus* (*T. thioparus*, *T. novellus*) also oxidize  $\text{H}_2\text{S}$  and other reduced sulphur compounds and, because they have a low acid tolerance, they deposit elemental sulphur rather than generating sulphuric acid by further oxidation. The filamentous sulphur bacteria and the above mentioned *Thiobacillus* species are facultatively chemolithotrophic. Other members of the genus *Thiobacillus* produce sulphate from the oxidation of elemental sulphur and other inorganic sulphur compounds as shown below.



Such *Thiobacillus* species are acidophilic, grow well at pH 2–3 and are obligate chemolithotrophs, obtaining their energy exclusively from the oxidation of inorganic sulphur and their carbon from the reduction of carbon dioxide. Most *Thiobacillus* species are obligate aerobes requiring molecular oxygen for the oxidation of the inorganic sulphur compounds. However, *Thiobacillus denitrificans* can utilize nitrate ions as terminal electron acceptors in the oxidation of inorganic sulphur compounds, as shown for elemental sulphur in the following equation:



This organism, though, is not capable of assimilatory nitrate reduction and requires ammonium as a nitrogen source. Members of the archaean genus, *Sulfolobus*, oxidize elemental sulphur in hot acidic habitats to generate their required energy.

Chemoautotrophic sulphur-oxidizing bacteria are widely distributed. They are quite active in soils and aquatic habitats. A variety of other heterotrophic microorganisms oxidize inorganic sulphur to sulphate or thiosulphate but do not appear to derive energy from this transformation.

Sulphur oxidation produces substantial amount of strong mineral acid. Within soils, this can lead to solubilization and mobilization of phosphorus and other mineral nutrients with a beneficial effect on both microorganisms and plants. The activity of *Thiobacillus thiooxidans* may be used for adjusting soil pH. *T. thiooxidans* and *T. ferrooxidans* are used in microbial mining operations (See Chapter 13). When mining activities, especially strip mining, uncover large amounts of reduced sulphide rock, the activities of the same thiobacilli give rise to acid mine drainage, a destructive pollution phenomenon.

Hydrogen sulphide is also subject to phototrophic oxidation in anaerobic environments. Photosynthetic sulphur bacteria, those belonging to family Chromatiaceae, Ectothiorhodospiraceae and Chlorobiaceae, are capable of photoreducing  $\text{CO}_2$  while oxidizing  $\text{H}_2\text{S}$  to  $\text{S}^0$  (see the following equations), in striking analogy to the photosynthesis of eukaryotes.

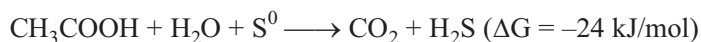


Chromatiaceae store sulphur globules intracellularly, whereas Ectothiorhodospiraceae and Chlorobiaceae excrete sulphur globules. All have a limited capacity to oxidize sulphur further to sulphate and they may contribute to biological sulphur deposition. Some cyanobacteria are capable of both oxygenic and anoxygenic photosynthesis, thus participating in the phototrophic oxidation of  $\text{H}_2\text{S}$ . Microbial oxidation of reduced sulphur is essential for continued availability of this element in nontoxic forms, but the chemoautotrophic fixation of  $\text{CO}_2$  connected with this activity contributes only minimally to the carbon cycling in most ecosystems.

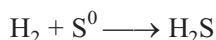
Notable exceptions are the recently discovered deep-sea hydrothermal vent ecosystems where large temperature variations cause sulphur precipitation. The vent communities, located at a depth of 800–1000 m or deeper, receive no sunlight and minimal organic nutrient input from the low-productive surface water above, yet their biomass exceeds that of the surrounding seafloor by orders of magnitude. Microbial mats cover all available surfaces on and near the vents and high densities of unique clams, mussels, vestimentiferan worms and other invertebrates cluster in the vicinity. Some of these graze or filter feed on microorganisms and others are directly symbiotic with microorganisms exhibiting chemoautotrophic activity. Energetically, the whole vent community is supported by the chemoautotrophic oxidation of reduced sulphur, primarily by *Beggiatoa*, *Thiomicrospira* and additional sulphide or sulphur oxidizers.

### 11.5.4 Reductive Sulphur Transformations

Under anaerobic or microaerobic conditions, the elemental sulphur can be used in respiratory processes. For example, the bacterium *Desulphuromonas acetoxidans* grows on acetate anaerobically reducing stoichiometric amounts of  $S^0$  to  $H_2S$  but the energy yield from this reduction is very low. However, such growth requires no other sources of carbon and energy.



*Desulphuromonas* occurs in anaerobic sediments rich in sulphide and elemental sulphur. It also lives syntrophically with the phototrophic green sulphur bacteria (Chlorobiaceae) that photooxidize  $H_2S$  to  $S^0$  and excrete elemental sulphur extracellularly. Extremely thermophilic anaerobic archaea are also capable of sulphur respiration with hydrogen gas. Several species of *Thermoproteus*, *Pyrobaculum* and *Pyrodictum* are known to carry out this chemolithotrophic reaction. Sulphur respiration also occur microbially with molecular hydrogen to produce  $H_2S$  as shown below.



When obligately anaerobic bacteria carry out dissimilatory sulphate reduction, they are referred to as sulphate reducers or sulfidogens. The sulphate-reducing genera *Desulfovibrio*, *Desulfotomaculum*, *Desulfobacter*, *Desulfobulbus*, *Desulfococcus*, *Desulfonema* and *Desulfosarcina* reduce sulphate to hydrogen sulphide through a metabolic conversion. In addition, some species of *Bacillus*, *Pseudomonas* and *Saccharomyces* have been found to liberate hydrogen sulphide from sulphate, but these additional genera do not appear to play a major role in the dissimilatory reduction of sulphate.

Sulphate reduction can occur over a wide range of pH, pressure, temperature and salinity conditions. Only relatively a few compounds can serve as electron donors for sulphate reduction. Although hydrogen and sulphate can serve as their only source of energy for growth, most of the bacterial sulphate reducers are not chemolithotrophs. Most lack the enzyme systems to assimilate  $CO_2$  and require organic carbon sources. The most common electron donors are pyruvate, lactate and molecular hydrogen. Though acetate may be utilized to support sulphate

reduction, it generally is not the preferred substrate for sulphate-reducing bacteria. Sulphate reduction is inhibited by the presence of oxygen, nitrate, or ferric ions. The rate of sulphate reduction is often carbon limited. The addition of organic compounds can result in greatly accelerated rates of dissimilatory sulphate reduction.

Biological sulphate reduction is probably also involved in the formation of some elemental sulphur deposits. Hydrogen sulphide generated by sulphate reduction may be photooxidized to elemental sulphur under anaerobic conditions by members of Chromatiaceae, Ectothiorhodospiraceae, or Chlorobiaceae. Under aerobic conditions, especially when oxygen is limiting, hydrogen sulphide may be oxidized to sulphur by the chemical reaction with oxygen or by the activities of the *Beggiatoa-Thiothrix* group. Acid-intolerant thiobacilli may also be involved in the formation of sulphur deposits.

## 11.6 PHOSPHORUS CYCLE

Phosphorus is the second major macronutrient essential for plant growth and metabolism, involved in various processes such as cell division, photosynthesis, biological oxidations and transfer of energy and nutrient uptake by crop plants. The level of phosphorus in naturally occurring soils is often below those of many 'micronutrients' in the range of 10  $\mu\text{M}$ . Phosphorus reserves are abundant in the soil, but is not found free in nature and almost always occurs in the fully oxidized state as phosphate. Due to its sedimentary characteristics around 96% of total phosphates are insoluble which are not directly available as phosphorus to the crops. Phosphorus is bound to ions of calcium, iron and aluminum, forming insoluble phosphate salts, which are unavailable to the plants. Soils of the arid and semi-arid regions have a high alkaline pH, which causes the precipitation of the poorly soluble calcium phosphates, further lowering the amount of available phosphates.

Most of the applied 'P' fertilizers are reprecipitated into insoluble mineral complexes such as calcium orthophosphate [ $\text{Ca}_3(\text{PO}_4)_2$ ], hydroxyapatite [ $\text{Ca}_5(\text{PO}_4)_3(\text{OH})$ ] and fluoroapatite [ $\text{Ca}_{10}(\text{PO}_4)_6\text{F}_2$ ] in the alkaline soils or crystalline ferric phosphates (strengite) and crystalline aluminium phosphate (variscite) in acidic soils and are not efficiently taken up by the plants. Farmers are, therefore, always advised to apply the phosphate at intervals in soil to satisfy the phosphorus requirement of a particular crop. Over-fertilization often leads to an imbalance in the soil and is one of the major environmental concerns. Biological immobilization and chemical precipitation would soon deplete the available phosphate supply leaving very little for the activity of phosphorus solubilizing microorganisms (PSMs) in the soil.

### 11.6.1 Phosphorus and Ecosystem Productivity

Productivity in many habitats is phosphate limited. In aquatic environments, phosphate concentrations exhibit seasonal fluctuations that are associated with

algal and cyanobacterial blooms. The precipitation of phosphorus, especially in marine habitats, greatly limits primary productivity. In aquatic habitats, phosphorus may exist in soluble or particulate form. These forms exhibit differential reactivity and availability to the biological community. Phosphate concentrations in freshwater habitats are closely correlated with eutrophication. Phosphate-limited lakes normally are oligotrophic. The addition of phosphates from sources such as detergent fillers cause these lakes to become eutrophic. The blooms of algae and cyanobacteria associated with eutrophication can greatly increase the concentrations of organic matter in the water bodies. During the subsequent decomposition of this organic matter, the water column can be severely depleted of oxygen, causing major fish kills.

The productivity of soil is more affected by the phosphorus deficiency than of aquatic environment. This is because of the fact that in soil, phosphorus has a low mobility and is often complexed with the soil particles and ions. The immobility is so acute that even in a phosphorus-rich soil, a plant may face phosphorus deficiency if the element is not efficiently mobilized by the existing microbial consortia. Therefore, microbial mobilization of phosphorus in agricultural soil is of prime importance to augment phosphorus availability and increase crop productivity.

### 11.6.2 Phosphorus Solubilization

Numerous soil microorganisms are reported to solubilize phosphate from inorganic phosphate compounds and make them available to the plants. Several microorganisms have been reported to solubilize mineral phosphate (Tables 11.10). Various mechanisms have been proposed to bring about the solubilization of phosphate complexes. The possibility of the involvement of organic acids in microbial phosphorus solubilization (MPS) has been studied since a long time. Organic acids released by the microorganisms act as good chelators of divalent cations of  $\text{Ca}^{2+}$  accompanying release of phosphates from insoluble phosphatic compounds. Organic acids may also form soluble complexes with metal ions associated with insoluble 'P', thus releasing phosphate. A fall in the pH during the growth of a PSM brings about the solubilization of insoluble phosphates. Many of the PSMs lower the pH of the medium either by  $\text{H}^+$  extrusion, or by secretion of organic acids (acetic, lactic, malic, succinic, tartaric, gluconic, 2-oxogluconic, oxalic and citric acids). The release of proton from the cytoplasm to the outer surface may happen in exchange for a cation (especially ammonium) uptake or with the help of translocating ATPase, which is located in the plasma-lemma and uses the energy of ATP hydrolysis. Rock phosphate, which is solubilized directly on the cell surface either as  $\text{H}_2\text{PO}_4^-$  or  $\text{HPO}_4^{2-}$  depending on the pH, is rapidly absorbed.

2-oxogluconic acid produced by bacteria plays an important role in weathering and solubilization of phosphate in soil. The non-phosphorylating oxidation pathway, which is otherwise known as the direct oxidation pathway, is one of the four known principal pathways for aldose sugar metabolism in bacteria and the solubilization is the result of acidification of the periplasmic space by the direct

**Table 11.10:** List of some microbes known to solubilize mineral phosphate.

| <i>Gram-positive bacteria</i>                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Gram-negative bacteria</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Cyanobacteria and actinomycetes</i>                                                                                                                                                                                                                                                                                                                                                             | <i>Fungi and yeast</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Bacillus brevis</i><br><i>Bacillus cereus</i><br><i>Bacillus circulans</i><br><i>Bacillus coagulans</i><br><i>Bacillus firmus</i><br><i>Bacillus licheniformis</i><br><i>Bacillus megaterium</i><br><i>Bacillus mesentericum</i><br><i>Bacillus mycoides</i><br><i>Bacillus polymyxa</i><br><i>Bacillus pumilis</i><br><i>Bacillus pulvifaciens</i><br><i>Bacillus sphaericus</i><br><i>Bacillus subtilis</i><br><i>Clostridium</i> spp. | <i>Acetobacter diazotrophicus</i><br><i>Achromobacter</i> spp.<br><i>Aerobacter aerogenes</i><br><i>Agrobacterium radiobacter</i><br><i>Alcaligenes</i> spp.<br><i>Arthobacter mysorens</i><br><i>Bradyrhizobium</i> spp.<br><i>Brevibacterium</i> spp.<br><i>Burkholderia cepacia</i><br><i>Citrobacter freundii</i><br><i>Enterobacter aerogenes</i><br><i>Escherichia freundii</i><br><i>Erwinia herbicola</i><br><i>Flavobacterium</i> spp.<br><i>Gluconobacter diazotrophicus</i><br><i>Micrococcus</i> spp.<br><i>Mycobacterium</i> spp.<br><i>Nitrosomonas</i> spp.<br><i>Nitrobacter</i> spp. | Cyanobacteria<br><br><i>Anabaena</i> spp.<br><i>Calothrix braunii</i><br><i>Haplosiphon fontinalis</i><br><i>Nostoc</i> spp.<br><i>Scytonema</i> spp.<br><i>Scytonema cinnatum</i><br><i>Tolypothrix tenuis</i><br><i>Tolypothrix ceylonica</i><br><i>Westiellopsis prolifica</i><br><br>Actinomycetes<br><br><i>Actinomyces coelicolor</i><br><i>Actinomyces</i> spp.<br><i>Streptomyces</i> spp. | <i>Achrothecium</i> spp.<br><i>Alternaria tenuis</i><br><i>Aspergillus aculeatus</i><br><i>Aspergillus carbonum</i><br><i>Aspergillus flavus</i><br><i>Aspergillus nidulans</i><br><i>Aspergillus</i> spp.<br><i>Candida</i> spp.<br><i>Cephalosporium</i> spp.<br><i>Chaetomium globosum</i><br><i>Cladosporium herbarum</i><br><i>Cunninghamella</i> spp.<br><i>Curvularia lunata</i><br><i>Fusarium oxysporum</i><br><i>Helminthosporium</i> spp.<br><i>Humicola insolens</i><br><i>Mortierella</i> spp.<br><i>Micromonospora</i> spp.<br><i>Mucor</i> spp.<br><i>Myrothecium roridum</i> |



**Table 11.10:** (Contd.)

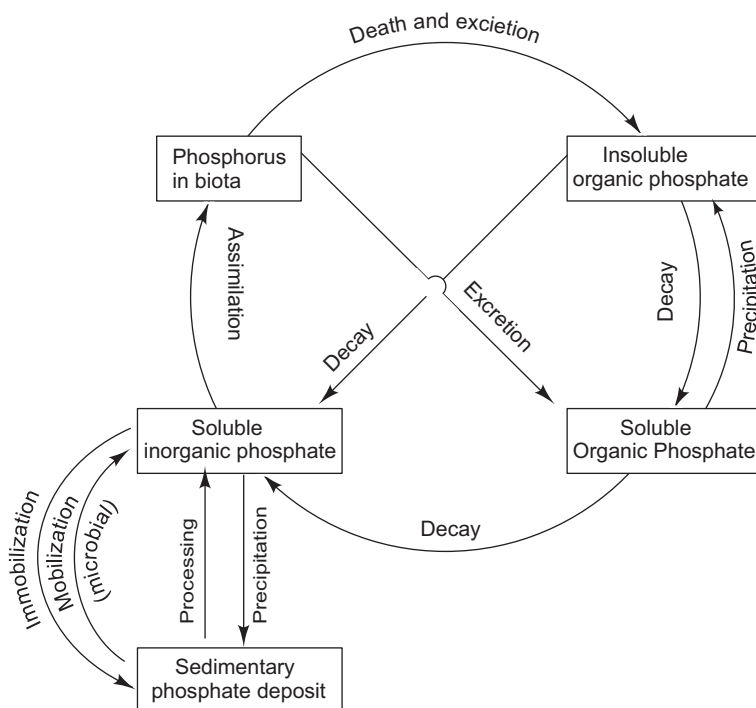
|  |                                                                                                                                                                                                                                                                                                                                               |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <i>Pseudomonas calcis</i><br><i>Pseudomonas cepacia</i><br><i>Pseudomonas putida</i><br><i>Pseudomonas rathonia</i><br><i>Pseudomonas striata</i><br><i>Serratia phosphoticum</i><br><i>Thiobacillus ferrooxidans</i><br><i>Thiobacillus thiooxidans</i><br><i>Rahnella aquatilis</i><br><i>Rhizobium meliloti</i><br><i>Xanthomonas</i> spp. |  | <i>Oideodendron</i> spp.<br><i>Paecilomyces lilacinus</i><br><i>Penicillium digitatum</i><br><i>Penicillium</i> spp.<br><i>Phoma</i> spp.<br><i>Populospora mytilina</i><br><i>Pythium</i> spp.<br><i>Rhizoctonia solani</i><br><i>Rhizopus</i> spp.<br><i>Sclerotium rolfsii</i><br><i>Torulospora globosa</i><br><i>Torula thermophila</i><br><i>Trichoderma harzianum</i><br><br>Yeast<br><br><i>Geotrichum</i> spp.<br><i>Schwanniomyces</i> spp. |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Source: Tripura *et al.* (2005).

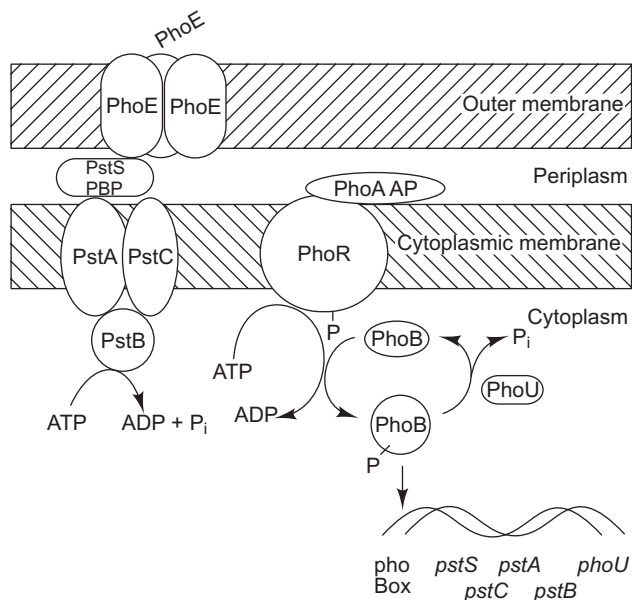
oxidation of glucose and other aldose sugars by the quinoprotein glucose dehydrogenase (PQQGDH). The direct oxidation pathway forms the metabolic basis for the strong mineral phosphate solubilizing phenotype. 2-oxogluconic acid is the major solubilizing compound released by rhizobacteria. The levels of gluconic and 2-oxogluconic acids are more in the soils adjacent to the plant roots where the availability of glucose would be higher than in the bulk soils. Gluconic and 2-oxogluconic acids are two of the strongest naturally occurring acids secreted into the extracellular medium by bacteria and are capable of acting as  $\text{Ca}^{2+}$  chelators under appropriate physico-chemical conditions. These acids provide the acidification of the external environment, necessary to dissolve the poorly soluble calcium phosphates such as tricalcium phosphate (TCP) or hydroxyapatite.

The microbial cycling of phosphorus for the most part does not alter the oxidation state of phosphorus. Most phosphorus transformations mediated by microorganisms can be viewed as transfers of inorganic to organic phosphate or as transfers of phosphate from insoluble, immobilized forms to soluble or mobile compounds (Fig. 11.5). Various microorganisms have evolved transport systems for the regulated acquisition of phosphate from the environment such as the Pst phosphate transport system of *E. coli* (Fig. 11.6).

Although phosphate is normally not reduced by microorganisms, it appears that some soil and sediment microorganisms may be capable of utilizing



**Figure 11.5** - The phosphorus cycle showing the transfer of phosphorus from organic to inorganic forms and vice versa without change in its oxidation state.



**Figure 11.6** - The Pst (phosphate transport system) expression in *E. coli* outer membrane transporters are *phoE* and *pstS* encoded outer membrane porin and periplasmic binding protein (PBP), respectively. Inner membrane transporters are PstA, PstB and PstC polypeptides. Alkaline phosphatase (AP) provide  $P_i$  from ATP to bind to PBP. PhoR and PhoB are regulatory proteins that undergo autophosphorylation—transphosphorylation cycle. The PhoU stimulates dephosphorylation of PhoB. The Pho box is the DNA binding site for the PhoB regulatory system (Adopted from Silver and Walderhaug, 1992).

phosphate as a terminal electron acceptor under appropriate environmental conditions in the absence of sulphate, nitrate and oxygen. The final product of phosphate reduction would be phosphines (PH). The latter are volatile and spontaneously ignite on contact with oxygen, producing a green glow. The production of phosphines is sometimes observed near burial sites and swamps where there is extensive decomposition of organic matter. Phosphines can also ignite the methane produced in similar environments, giving rise to “ghostly” light phenomena. A microbial involvement in phosphine production is likely but still requires experimental confirmation.

Microbial activities may also immobilize phosphorus, rendering it inaccessible to the biological community. The assimilation of phosphorus into microbial cell constituents, such as membranes, removes phosphate from the available pool. In some situations, microorganisms compete with plants for available phosphate resources.

### 11.6.3 Biomanipulation in Phosphorus Cycle

The aim to enhance agricultural output has made it necessary to manipulate the microbial diversity of soil to facilitate phosphorus mobilization and its subsequent

availability to plants. Many microbes are being used as bioinoculants to speed up the phosphorus cycle. However, these microbes have failed to perform well in varied field conditions due to their instability and inability to establish themselves in diverse ecological conditions. Hence, it is required to identify organisms native to particular soil conditions. Using transgenic approaches the agronomic efficiency of these strains has been enhanced above their natural plateau by introducing the MPS genes. There are successes in the development of strains with multiple agronomic beneficial traits. Many rhizospheric bacteria show plant growth enhancement due to mechanisms involving production of plant growth promoting substances and nitrogen fixation both symbiotically and asymbiotically. These plant growth promoting rhizobacteria (PGPR), mostly represented by the fluorescent pseudomonads such as the *Pseudomonas fluorescens*, *P. aeruginosa*, *P. cepacia*, *P. putida* and the species of *Azospirillum*, *Azotobacter*, *Bacillus*, *Bradyrhizobium*, *Enterobacter*, *Erwinia* and *Rhizobium* show a proficient colonization at the root zone. Most of the *Pseudomonas* species have ability to utilize a wide variety of organic substrates found in root exudates and thus form a very versatile group of PGPR. Some rhizobacteria, *Pseudomonas* spp. and *Burkholderia cepacia* are shown to enhance MPS activity “in vitro”. It is required to standardize the growth conditions of these microbes and to manipulate this in the field to facilitate the solubilization of immobilized phosphorus.

Agricultural microbiologists have intensively studied microbial biotransformation of poorly soluble mineral phosphates to soluble phosphates due to the potential for enhancement of crop growth and/ or the development of alternative fertilizer technologies. Majority of the phosphate solubilizing microorganisms have been isolated, based on their ability to solubilize calcium phosphate complexes *in vitro*. Efficiency of a PSM can be increased above their natural basal activity by manipulating the genes directly involved in the MPS. Broad host range non-phosphate solubilizing rhizobacteria could be modified by expressing any of the genes involved in MPS, thereby conferring upon them a MPS+ trait. Therefore, the major challenge in this area of research lies in not just identifying and isolating the MPS gene, but in modifying the gene for superior properties and express in microbes that thrive in rhizosphere conditions and solubilize mineral phosphate making it available to plants.

## 11.7 INTERRELATIONSHIP BETWEEN CYCLING OF NUTRIENTS

It is a fact that the cycle of each element is dependent on, or at least influenced by, the cycling of other elements. This is true not only for C, H and O, which are cycled by the same two processes of photosynthesis and respiration, but also for cycles that are driven by different biochemical processes and performed by distinct microorganisms. Thus, the study of biogeochemical cycle interactions is a technically challenging task for the investigators. In a complex natural system with a wide variety of microorganisms, substrates and electron sinks, it is difficult to determine which group of microorganisms degrades a certain substrate and at

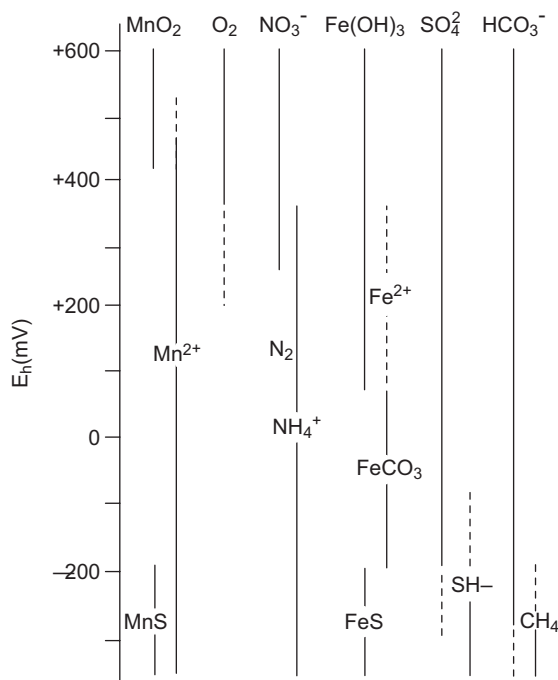
what rate. This field of investigation has gained much from the use of selective inhibitors of certain cycling activities. For examples, 2-bromoethanesulphonate can be used to selectively inhibit methanogenesis in sediments, molybdate to suppress sulphate reduction, nitrapyrin to shut down nitrification and acetylene to block denitrification on the nitrous oxide level. Although the effectivity and selectivity of these inhibitors need to be carefully established case by case, an ability to selectively inhibit a metabolic group within a complex system provides an excellent experimental tool for defining the normal contribution of that group to the system.

The reductive portions of the N, S, Fe and Mn cycles are driven by chemical energy fixed in organic substances during photosynthesis. The chemolithotrophic reoxidation of N, S, Fe and perhaps Mn are, in turn, linked to the conversion of CO<sub>2</sub> into the cell material, again involving the cycling of C, H and O. Solubilization, uptake and precipitation of Ca and Si are directly or at least energetically linked to photosynthetic and respiratory cycling of C, H and O. Acids from nitrification and sulphur oxidation help to mobilize phosphorus. Photosynthesis or respiration are required for its uptake and conversion into high energy phosphates. Sulphur is oxidized with reduction of nitrate by *Thiobacillus denitrificans* and some extremely thermophilic methanogens can transfer hydrogen not only to CO<sub>2</sub> but also to S<sup>0</sup>.

From the pool of potential electron acceptors, microorganisms within the community use the specific electron acceptor that they are genetically capable of using the maximum energy yield from the available substrate. Each electron acceptor is utilized at different redox potentials (Fig. 11.7). This seemingly intelligent decision is in part due to metabolic regulation within a single population and in part due to the inevitable outcome of competition between populations with diverse metabolic capabilities. Through various regulatory mechanisms, facultatively anaerobic microorganisms shut off their less efficient fermentative or dissimilatory nitrate reduction pathways in the presence of oxygen. In the absence of oxygen, Fe<sup>3+</sup>, Mn<sup>4+</sup> and NO<sub>3</sub><sup>-</sup> are the most oxidizing electron acceptors. From a substrate equally utilizable by nitrate and sulphate reducers, nitrate reducers obtain the higher energy yield. Therefore, they obtain the higher biomass per unit substrate utilization. They also outcompete the sulphate reducers because they have lower thresholds for carbon substrates. Nitrate and iron, usually scarce in aquatic sediments, are rapidly depleted, leaving sulphate as the most oxidizing electron acceptor. When competing for a common substrate (H<sub>2</sub>), methanogens have a lower utilization efficiency and a higher threshold for hydrogen uptake than sulphate reducers.

## 11.8 MICROBIAL CORROSION

Microbial corrosion refers to the release of elements from some surfaces, which are considered relatively stable, by the microbial communities colonized over these surfaces. The human interference with natural mineral cycles is responsible for the abnormality, in some cases accelerating or in some cases slowing part of a cycle. One example is the accelerated formation of acid leachates in metal mining



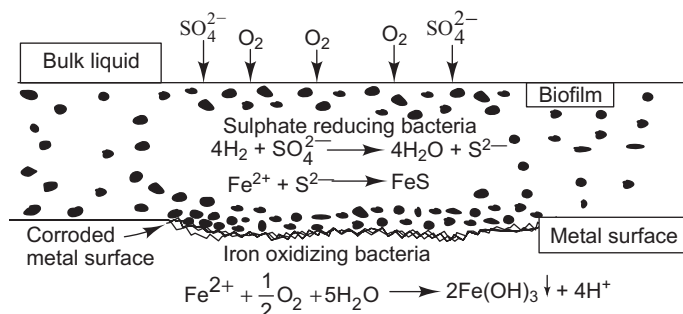
**Figure 11.7** - The redox potential ( $E_h$ ) for microbial utilization of potential electron acceptors. The figure shows the preference of a microbial community to transfer electron from an organic substrate to the most oxidizing electron acceptor available in their environment to have maximum energy yield (Adopted from Nedwell, 1984).

due to mining, which in turn results in the enhanced microbial activity of sulphur oxidation. In some instances the impact of a biogeochemical activity may be more localized but still costly in terms of damage and repair, an example being corrosion of metal, stone and concrete.

### 11.8.1 Metal Corrosion

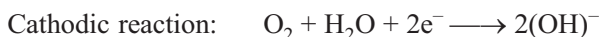
Microbially mediated corrosion damage can occur on metal structures submersed in water, such as the hulls of ships and in pipelines that carry water and oil products. It is thought that microbially influenced metal corrosion accounts for 15 to 30% of the corrosion failures in the gas and nuclear industries. It is a major cause of pipeline failures in water treatment and chemical industries and is also associated with corrosion failures and souring in gas and oil production and storage. Corrosion is mostly caused by anaerobic bacteria, when present as a multispecies assemblage in a biofilm. For example, sulphate-reducing bacteria (SRB) are active under strictly anaerobic conditions, utilizing sulphate instead of oxygen in the respiration of organic compounds. SRB-mediated corrosion requires anaerobic conditions, but oxygen is often present in the environment surrounding a metal surface. However, when biofilms form on the metal surface, anaerobic microsites can develop within the biofilm. These sites occur as a result

of rapid oxygen utilization by bacteria in the outer layers of the biofilm coupled with a slow rate of oxygen diffusion through the exopolymer matrix. These anaerobic biofilm microsites support the growth and activity of SRB, which ultimately cause the metal surface to corrode (Fig. 11.8). Because SRB are common constituents of natural environments, the use of natural water sources in cooling towers or the pumping of oil from underground reservoirs serves to inoculate metal pipeline surfaces with SRB.



**Figure 11.8** - The microbial corrosion of metal surface caused by iron oxidizing and sulphate reducing bacteria in the biofilm (Adopted from Hamilton, 1995).

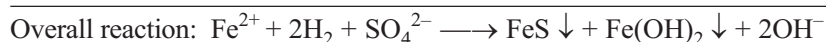
Metal corrosion is initiated by two spontaneous electrochemical reactions. In the first reaction, a differential aeration cell is set up, in which the metal surface acts as the anode to produce metal ions and in the corresponding cathodic reaction oxygen accepts the electrons produced from the oxidation of elemental iron.



In the second reaction, a concentration cell is created under anaerobic conditions in which the anodic reaction remains the same but the cathodic reaction produces  $\text{H}_2$ .



Microbial biofilms stimulate these electrochemical reactions. Biofilm formed on the metal surface facilitates the establishment of these separate electrochemical reactions by establishing an anaerobic environment at the metal surface. Able bacteria, for example, SRB, utilize  $\text{H}_2$  as an electron donor, thereby removing it from the environment and providing a driving force for the anodic reaction. The end-product of sulphate reduction is sulphide ( $\text{S}^{2-}$ ), which reacts with  $\text{Fe}^{2+}$  to form metal sulphide precipitate.





Other mechanisms also contribute to microbially influenced corrosion. For example, the biofilm matrix is thought to contribute to corrosion by trapping the corrosion products including organic acids and metal species at the metal-biofilm interface. As a result of all of these corrosion processes, pits and cracks appear on the metal surface that eventually deteriorate the integrity of the metal structure.

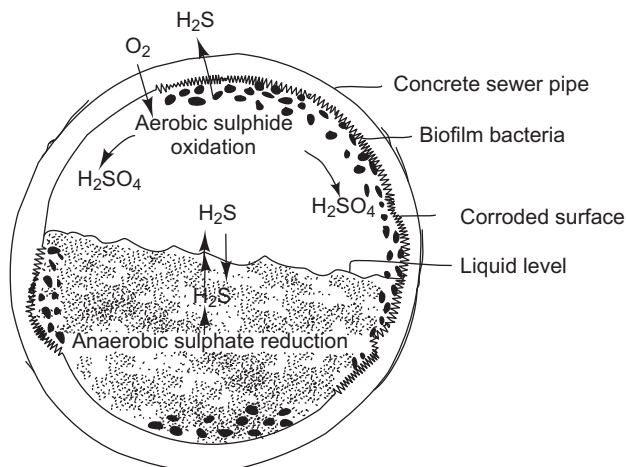
Application of quaternary ammonia compounds, phenolic compounds, surface-active substances and heavy metals such as copper on the corrosive metal surfaces check microbial metal corrosion. However, these compounds can leach into the surrounding environment, raising concern about their use. The alternate strategy is to disrupt the organization of surface biofilms, thus removing the microenvironment that supports the activity of SRB. Disinfecting chemicals, such as chlorine and addition of cationic surfactants are used to control microbial growth in pipelines, but these chemicals are more effective in killing planktonic cells than in destroying biofilms. Further, the high chlorine concentrations required to destroy biofilms may have undesirable side effects. The oil industry uses mechanical scraping (pigging) of the inside surface of pipes to prevent the prolific growth of the biofilm.

### 11.8.2 Concrete Corrosion

Microorganisms cause corrosion of concrete structures through the production of acidic metabolites. The binding components of concrete are acid sensitive and therefore microbes that produce acidic metabolites contribute to this type of corrosion as well as to corrosion of other acid-sensitive ceramic and natural stone objects. Different forms of acidic metabolites can be produced by microorganisms, including organic acids produced by fermentative bacteria. Also, the chemoautotrophic sulphur oxidizing bacteria and the nitrifying bacteria can produce sulphuric acid and nitric acid, respectively. These and other microorganisms have been found to colonize the surface of concrete structures and their level of activity is strongly influenced by the amount of available moisture.

Corrosion of concrete sanitary sewer pipes is a two-step process involving both sulphate-reducing and sulphur-oxidizing bacterial populations that cycle sulphur as described in previous sections of this chapter (see section 11.5) (Fig.11.9). Pipes carrying sewage can contain two distinct environments, namely the liquid sewage and the headspace area. In the liquid environment, anaerobic conditions are created by the high rate of microbial activity in the organic-rich sewage. Under anaerobic conditions, the SRB generate sulphide, which is converted to the volatile  $H_2S$  form and exchanged across the liquid-headspace interface. In the aerobic and moist environment of the headspace, sulphide-oxidizing bacteria colonizing the concrete wall surface oxidize  $H_2S$  to sulphuric acid. Moisture condensation along the inside of the sewer pipe walls improves the habitat for microbial activity.

The actual corrosion process occurs when the sulphuric acid produced by sulphide oxidation reacts with the calcium hydroxide binder in the concrete to form calcium sulphate, which is a soft, expansive compound with no binding capability. It effectively reduces the life expectancy of concrete sewer pipes,



**Figure 11.9** - Microbial corrosion of concrete by sulphur reducing bacteria under aerobic as well as anaerobic conditions. Aerobic activities cause sulphuric acid induced concrete corrosion (Adopted from Sydney *et al.*, 1996).

which can only be controlled by preventing the production of  $H_2S$ . The control methods include the use of caustic soda to inhibit microbial growth or the addition of alternative electron acceptors, such as iron (e.g., ferric chloride) or oxygen, to inhibit SRB activity competitively. An alternative approach is to neutralize the surface of the concrete above the sewage liquid level by spraying the concrete with a high-pH solution, such as a magnesium hydroxide slurry.

### 11.8.3 Microbial Mobilization of Halogenated Organics

Microbial reactions resulting in the production of methylated halogens are of interest in terms of global biogeochemical cycles. Halomethanes and other organohalogens are natural products of the marine environment and have been shown to be synthesized by marine algae and macroalgae, such as kelp and other seaweeds. Examples include chloromethane ( $CH_3Cl$ ), chloroform ( $CHCl_3$ ), bromoform ( $CHBr_3$ ), methyl iodide ( $CH_3I$ ) and a wide variety of other chlorinated, brominated and iodinated organic compounds. The production of halomethane metabolites is believed to be the result of reactions with haloperoxidases. These enzymes are of biochemical importance in terms of plant hormone production and other metabolic pathways. Halomethanes may be formed at random by the activity of relatively nondiscriminating haloperoxidases or other oxidase reactions. It has also been suggested that the production of toxic organohalogen metabolites may be of benefit to marine macroalgae by providing a deterrent to grazing herbivores or discouraging the colonization of their surfaces by microorganisms.

There has been increasing interest in identifying and quantifying all sources of volatile halogenated compounds in our environment and in determining the relative contribution of natural and anthropogenic sources to the global cycling

of halogenated gases. The ocean is a significant source of certain halomethanes, such as bromomethanes and chloroform. The small concentrations of halomethanes synthesized and released to the atmosphere by macroalgae are magnified to global significance by vast areas of productive macroalgae in the world's oceans.

The biological origin of organohalogen compounds also contribute to the presence of halocarbons in "unpolluted" freshwater and groundwater environments where there is no known source of anthropogenic organohalogen pollution. Some of the organohalogens present in unpolluted environments can be accounted for by long range deposition of halocarbons from industrial sources that can be located hundreds of miles away. However, there can be significant levels of organohalogens from biogenic sources. Uncertainty exists with regard to the mechanism of biological origin, although it is known that organohalogen metabolites may be formed by many types of algae, fungi and plants and that natural halogenation of humic substances produce halogenated organic compounds. One area of concern is the production of trihalomethanes in water supply reservoirs. Trihalomethanes, including chloroform, bromoform and mixed trihalomethanes, are formed as a result of chlorine application combined with algal growth or, more specifically, the release of dissolved organic compounds from algae in a water supply reservoir. Chlorination of the water in open reservoirs is used to control algal growth and is also necessary to prevent the spread of waterborne diseases in drinking water supplies. However, the small level of biogenic halocarbon production in water may be of concern because of the classification of certain trihalomethanes as possible human carcinogens.

## **11.9 CONCLUSION**

Carbon is cycled generally by two opposing processes of photosynthesis and respiration. In the absence of oxygen, organic carbon is recycled in fermentative and methanogenic processes. Nitrogen and sulphur are cycled in a complex, oxidoreductive fashion. The reduced forms support chemolithotrophic metabolism while the oxidized forms are used as electron sinks in anaerobic environments. The microbial oxidation and reduction of nitrogen and sulphur results in the transfer of these elements between the atmosphere, hydrosphere, soil and lithosphere.

Plants, animals and most microorganisms require combined forms of nitrogen for incorporation into cellular biomass, but the ability to fix nitrogen is restricted to a limited number of bacteria, archaea and symbiotic associations. Primary productivity is often dependent on biological nitrogen fixation in soil and conversion of nitrogen to assimilable forms. Soil provides a natural habitat for a diverse array of microorganisms, which coexist by interacting with one another and play an important role in the cycling of nutrients maintaining a balance in the ecosystem.

Phosphorus is present abundantly in the soil but is in a fixed form and is unavailable to the plants. In order to meet the plants phosphate requirement for

agronomically effective crop yields, annual application of soluble 'Pi' salts as fertilizers is required. But most of the applied phosphate is reprecipitated and is still unavailable to the plants. Bacteria have the ability to solubilize these insoluble phosphate complexes and bring them into solution making it available to plants. There is an increasing realization to explore the possibilities of efficiently utilizing these PSMs as biofertilizer, which is both economic, eco-friendly and serves as low cost agricultural input. The isolation, identification and manipulation of mineral phosphate solubilizing gene(s) are likely to have a significant impact on modern agriculture.

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# MICROBIAL TREATMENT OF WASTEWATER

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## 12.1 INTRODUCTION

The microbial population and metabolic activity in water is proportional to the amount of biodegradable organic matters and the availability of oxygen to maintain an aerobic environment. The amount of organic matters in water determines the rate of heterotrophic activity and the remineralization processes. While in the surface waters without much input of allochthonous organic biomass, autotrophy dominates over heterotrophy, the wastewater systems experience a dominant heterotrophic activity. Microbial activity degrades the organic matters to release bioavailable forms of nutrients, which in turn are either taken up by the producers to form autochthonous organic matter or are released to surface waters. In addition, the microbial activity in wastewater significantly reduces the biological oxygen demand and make it comparatively safe for release to the surface waters. Thus, microbial degradation of organic pollutants is the dominant process in wastewater treatment.

Various wastewater treatment processes are now available to remove the nutrients and contaminants before allowing the wastewater for release into surface water. For removal of organic and inorganic materials, a combination of physical, chemical and/or biological purification procedures can be followed. The physical and chemical purification processes are very efficient in removing the nutrients and toxic contaminants from wastewater but the treatment processes are costly and often cause secondary pollution problem. The biological treatment processes, on the other hand, are environment friendly and cost effective for removal of organic and inorganic pollutants. These processes use the catabolic reactions of bacteria, algae, fungi and other microbes for total or partial decomposition of bioorganic or organic compounds and conversion of the released minerals to usable or disposable biomass. This chapter describes some important biological treatment processes for organic and inorganic pollutants removal from wastewater.

## 12.2 MICROBIAL PROCESSES IN WASTEWATER TREATMENT

Many microbial activities are observed in a wastewater treatment system and with few exceptions, the microorganisms associated with wastewater treatment do not persist throughout the whole process or even the length of the sewerage system, being relatively fragile and therefore colonising only those areas attractive to their individual requirements. Both biofilter and activated sludge systems represent a food chain and the range and type of organism will be different in each. Frequently present in activated sludge are *Zoogloea*, *Achromobacter*, *Flavobacterium*, *Nocardia*, *Mycobacterium*, *Nitrosomonas*, *Nitrobacter*, *Beggiatoa*, *Thiothrix* and *Geotrichum*. Biofilters also host nematodes, insect larvae, microfungi, algae, small worms and freshwater leech.

Microbiological examination can identify changes to relative concentrations of the organisms at times of stress and is useful in identifying the cause of changes of effluent quality. The activity of the organisms is assessed by measuring the effect of the oxidation reactions within the system. Reductions in BOD and the oxidation of ammonia to nitrate are particularly important parameters. Two basic oxidation reactions occurring through microbial involvement in wastewater treatment systems are:

- (i) Carbonaceous oxidation of a very wide range of compounds containing C, H and O, including toxic ones such as cyanides, phenols and oils in low concentrations. The end-products are theoretically carbon dioxide and water, but complete oxidation of all components never occurs and the effluent always retains some partially oxidised material measured as BOD or COD.
- (ii) Nitrification, where the organic nitrogen present in sewage or wastewater as proteins, amines and urea are hydrolysed to ammonia, which is oxidized to nitrate. Two organisms, *Nitrosomonas* and *Nitrobacter*, are involved and the reactions proceed in two stages for converting ammonia-nitrogen to nitrate-nitrogen. In order to oxidise 1 mg of  $\text{NH}_3\text{-N}$  to  $\text{NO}_3\text{-N}$ , 4.3 mg of oxygen is required. The reaction consumes a large amount of alkaline substance. 8.64 mg of  $\text{HCO}_3^-$  is required to oxidise 1 mg of  $\text{NH}_3$ . While carbonaceous oxidation occurs first and the autotrophic heterotrophs are relatively robust, some oxidation occurs unless the system is completely poisoned or dead. Nitrifying bacteria are more demanding of the optimum conditions of a thin, mostly aerobic biomass. In particular, the pH range of the wastewater must lie between 7.3 and 8.6.

Biosystems, which are heavily loaded organically and support extensive film growth, or in the case of activated sludge or extended oxidation operated at high solids levels, will undergo effective carbonaceous oxidation but will not nitrify. The ammonia content of the effluent will then be similar to the plant influent. This state in sewage treatment indicates overloaded conditions or inadequate aeration, but may be acceptable in industrial scenarios where nitrification is

actively inhibited by some materials in the flow to treatment. The best filter medium is therefore well graded to secure good ventilation and has a rough, craggy surface with plenty of interstices to encourage the floc-forming *Zoogloea* types of rod-shaped bacteria to grow. A 'zoogloal slime' over the media surface is a healthy sign, but since the filter is a food chain, the presence of insect larvae, worms and mosses should also be expected. Only the top 1 cm of media has any appreciable slime, the lower depths of the filter being populated by nitrifying bacteria.

A number of fungi and algae also develop, the latter consuming carbon dioxide, and oxygen and utilising light and are thus restricted to the surface. Fungi perform a similar function to the bacteria, but excessive numbers in activated sludge hinder solids settlement by reducing density and causing 'bulking'. They often indicate an incorrect C:N ratio and high carbohydrate waste liquors are prone to this.

The mechanism of BOD removal in a filter or an activated sludge biomass is complex. Effluents will contain pollutants (the food source) in solution or as suspended solids and the major rate-limiting step is biophysical adsorption and absorption by the biological film. Subsequent bacterial metabolic stages are numerous and ordered by reaction rate which is temperature sensitive. Solids are broken down by extracellular enzymes and dissolved material adsorbed into the cells of the biomass. Growth and endogenous respiration is followed by excretion and grazing by numerous organisms. Rate-limiting steps in these processes are the oxygen transfer rate from the aqueous phase through the cell wall and the actual oxidation rate within the cell. Although the equilibrium saturation value for oxygen in water is between 8 and 11 mg/l over the temperature range of most liquid effluents, the rate of bio-oxidation is not adversely affected until less than 1 mg/l is available and activated sludge and extended aeration plants are often run with only 1–1.5 mg/l dissolved oxygen available.

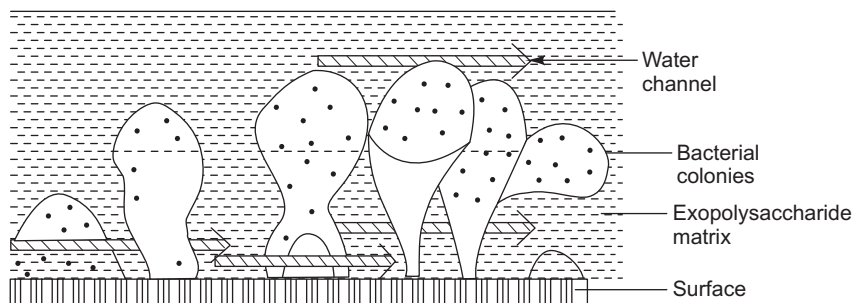
Endogenous respiration, where bacteria oxidise their own cell material, is induced by a reduction of BOD load (the food source) of the system and tends to lead to temporarily poorer effluent, as the biomass is reduced in both numbers and metabolic readiness when load is reapplied. Thus, recirculation or flow balancing to maintain reasonably stable conditions by returning viable organisms and nutrients is highly desirable. Recirculation in a filter plant also serves the function of returning fine particles to the filter to assist in floc formation by agglomeration. The polysaccharide sheath secreted by the *Zoogloea* and *Pseudomonas*-derived bacteria is the site of this activity, involving a reduction of electrostatic surface charge and the removal of much colloidal solid material as flocs.

## 12.3 BIOFILM MICROBES

There are many aquatic systems in which a macroscopic slime layer develops on the available surfaces. These slimy surfaces are called microbial biofilms with adherent populations generally containing many more cells than the planktonic



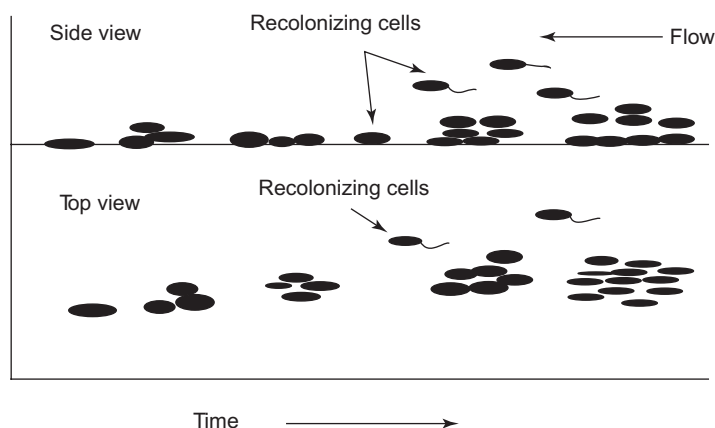
population in the bulk fluid of the ecosystem concerned. The biofilm bacteria are enveloped in large amounts of a fibrous, highly hydrated, exopolysaccharide (EP) matrix whose chemical composition is species-specific. The biofilm bacteria are distributed quite evenly and randomly throughout an abundant EP matrix that mediate their aggregation and adhesion to the colonized surface. In the 1980s, the biofilm matrix was generally conceived as being essentially homogeneous, but later it was observed that biofilm bacteria grow in matrix enclosed microcolonies interspersed with less-dense regions of the matrix that include highly permeable water channels. These water channels anastomose throughout microbial biofilms, provide direct high permeability access from the bulk fluid to the colonized surface and permit the penetration of large (2,000 kDa) organic molecules (Fig. 12.1). Individual biofilm bacteria enjoy some of the advantages of multicellular life, in that a primitive 'circulatory system' delivers nutrients from the bulk fluid to the microcolonial niche and removes metabolic products by the same process. Inhibitors and other antibacterial agents would have ready access to the water channels of the intercolonial matrix but they may still be separated from the inner cells of the microcolonies by an anionic polymeric diffusion barrier. In multispecies biofilms, mixed-species microcolonies benefit from interspecies substrate exchange and/or mutual end-product removal. The stationary biofilm mode of growth clearly makes these sustained metabolic cooperations possible and an individual within a mature multispecies biofilm typically lives in a unique microniche, where nutrients are provided by neighbouring cells by diffusion and products are removed by the same processes and where antagonists may be kept at bay by diffusion barriers.



**Figure 12.1** - The structure of biofilm bearing colonies of different bacteria within an exopolysaccharide matrix.

The formation of physiologically cooperative bacterial consortia is only possible in biofilms or in equivalent cellular aggregates. The physiology of biofilm bacteria is profoundly different from that of their planktonic counterparts. But it is in the area of ecology where biofilm bacteria differ most radically from planktonic cells. Primary surface colonisers are selected from mixed planktonic populations by their ability to adhere to a specific surface. They then act as a spatially limited stable population with the capacity to promote or to preclude

secondary colonization and succession by other organisms. While planktonic bacteria are exquisitely sensitive to antimicrobial agents, because of ready access by diffusion within the bulk fluid, biofilm bacteria are remarkably resistant to all of these agents and so they contribute stability and continuity in natural microbial ecosystems.



**Figure 12.2** - The steps of sequential cell movement and colonization on a compatible surface (Adopted from Lawrence *et al.*, 1995).

A vast majority of bacteria in most aquatic environments grow within matrix enclosed biofilms whose component bacteria grow in a large number of microniches whose properties are dictated partly by biofilm structure, diffusion and physiological activity of neighbouring organisms. Survival and reproductive success of bacteria in many systems require colonization of surface and/or integration into a biofilm community (Fig. 12.2). Though microbial behaviour at surfaces has usually been explained in terms of physicochemical interactions, several species have shown specific responses to the surface environment, including chemoadherence, morphogenesis, gene induction and variable rates of polymer production. The induction of many genetic pathways is also a surface specific phenomenon. Some behavioural strategies only function during surface colonization and growth. These strategies are essential for multicellular community growth and may manifest as specific adaptations, life cycle alternation between attached and planktonic growth, as well as formation and maintenance of microcolonies, aggregates and consortia.

Multicellular biofilm communities develop as a result of the interaction of many factors including specific cell-cell interactions and metabolic communication. Microbial interactions allow a variety of microorganisms to coexist in environments in which individual organisms cannot survive. Typically, these communities consist of various microbial species having different metabolic activities and nutritional requirements. As a result of these interdependencies among species, the range of metabolic activities carried out by microbial

communities is greater than the sum of those activities carried out by constituent members cultured in isolation. The existence of functional consortia and microcolonies within biofilms appears to be the most effective mechanism for meeting the diversity range of microbial growth requirements. Further, community membership may improve reproductive success and survival.

### 12.3.1 Regulation of Biofilm Development

Environmental conditions that influence the success of bacterial attachment to the surface have an important role in defining the population density and species diversity on newly exposed surface thereby influencing community development. Abiotic factors like substratum microroughness and chemistry, fluid dynamics and ionic strength of bulk liquid influence bacterial attachment. Surface attachment is also influenced by microbial behaviours like motility, chemotactic and sensory responses and microbe-microbe and surface-microbe interactions. Important physical factors influencing the initiation of stable microbial films include hydrodynamic conditions, shear effects, substratum roughness and physicochemical and thermodynamic forces.

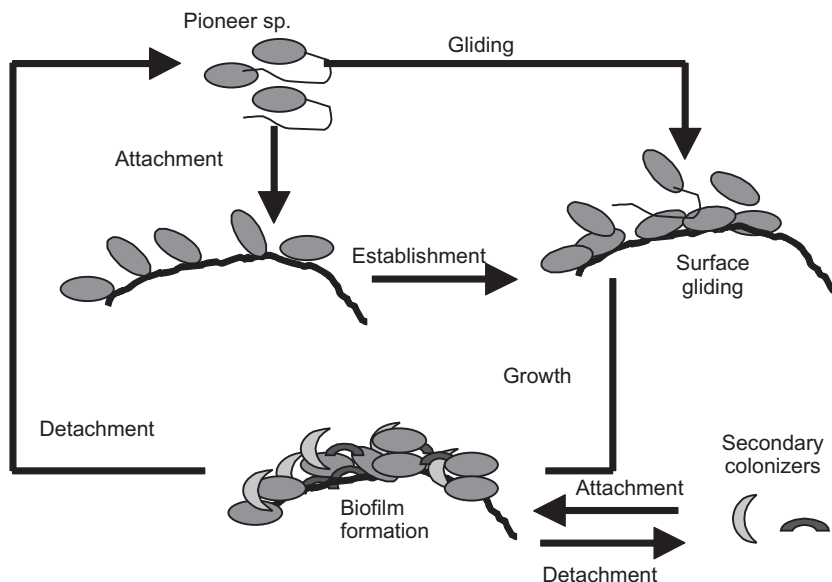
Bacteria have a variety of strategies for adapting to life in a surface habitat and also for releasing progeny from the attached to the planktonic state. Surface colonization pathways are influenced by several factors, including flow rates, viscosity, nutrient status, starvation and light intensity. An exception is the developmental pathway of some strains of *P. fluorescens*, which is unaffected by factors such as initial cell density, nutrition and laminar flow velocity.

### 12.3.2 Microbial Behaviour During Biofilm Development

A complex behavioural pattern of microbes is seen during the colonization of surface environments by the pioneer species. A wide range of species-specific behaviours like progressing through attachment, growth, microcolony formation and redistribution/dispersal are observed on the biofilm surface. Some organisms appear solitary in nature whereas others are colonial. The occurrence of 'solitary' organisms such as *Caulobacter* spp. within complex sewage bioreactors, polluted streams and intertidal zones indicate successful adaptation to community life. Solitary bacteria also sometime show some movement over surfaces. Rolling surface behaviour occurs in members of stream communities. These organisms slowly migrate over surfaces and divide without becoming firmly attached.

The commonest colonization strategy employed by bacteria may be the formation of colony aggregates from a single point of bacterial colonization. In several *Pseudomonas*, *Vibrio*, *Klebsiella* and *Escherichia* species, colonies develop from an individual attached pioneering cell that goes through several division cycles, eventually forming a bacterial microcolony (Fig. 12.3). The spatial distribution of individual microcolony members varies with cell morphology, nutrient status, cell size, mode of cell wall growth, polymer production and some unknown events at cell division. Some strains of *Pseudomonas fluorescens* follow

a kind of 'packing' microcolony formation behaviour, starting with attached individuals that grow and divide forming a tightly packed monolayer of cells on the surface. Other pseudomonads have very different patterns of microcolony formation.



**Figure 12.3** - The steps for recognition and establishment of microbes for biofilm formation.

### 12.3.3 Heterogenous Biofilm Community and Wastewater Treatment

A biofilm starts developing when a solid surface is immersed in an aqueous environment and organic molecules adsorb onto it, producing a macromolecular conditioning film. Various bacteria attach rapidly to the surface by making EP. The latter provide an interface between the cell and the external environment, influencing rates of chemical exchange and nutrient availability and facilitating the generation of micro niches. They also influence the susceptibility of bacteria to stress and facilitate cell-cell interaction. Biofilm organisms seem to benefit from the attached mode of growth through protection provided by the polymeric matrix against outside perturbations. Bacteria live in a diffusion-dominated environment and, as individuals, cannot change their environment. The close proximity of organisms to one another in a biofilm promotes interaction and genetic recombination between different species as well as significantly enhance their metabolic activity causing faster removal of the pollutants from wastewater. The larger genetic pool of the biofilm community also improves the chances of the development of new metabolic pathways and strategies to adapt to changing conditions.

A great deal of spatial heterogeneity seems to exist in biofilms, especially in arrangement of cells, EPs and space within biofilms. These arrangements create interconnected channels and pores through the biofilm matrix. Mixed species biofilms are also quite variable and extensive pore and channel networks can develop in a multispecies biofilm community.

Various types of interactions occur among different communities within biofilm, viz., syntrophism, symbiosis, commensalisms and predation. These interactions are defined in terms of substrate utilization and metabolite exchange. For example, introduction of a new non-metabolizable compound into the environment can exert stress conditions, which may be relieved by mineralization of the contaminant through microbial cooperation. Thus, microbial communities may be important in developing novel degradative pathways to mineralize xenobiotic compounds during wastewater purification. While in free living state very few species are able to mineralize xenobiotics, many species can do so when present in the biofilm. Predatory interactions between bacterial species seem to play a significant role in community development. A range of obligate and non-obligate predatory interactions can occur between bacteria and regulate cell numbers within communities. Algal bacterial associations in bioaggregates and in biofilm communities are widespread, with algae providing sites of attachment and acting as sources of assimilable organic carbon. In return, the bacteria supply nutrients for algal growth resulting in conversion of dissolved nutrients to algal biomass, which can be removed and treated separately. Algal exudates also serve as an important growth substrate for heterotrophic bacteria but their role in the bacterial mineralization of xenobiotic compounds is not known.

#### 12.3.4. Competition and Succession in Biofilm

Much work has been done in natural systems having a high degree of cellular interaction and competitive behaviour resulting from the competition of cells for available resources, whether attachment sites or growth substrates, but very little is known about these aspects in biofilms. With the change in the concentration and the type of the organic carbon in wastewater with the progress in the treatment, there is a succession of the microbes in the biofilm and the succession always lead to the establishment and dominance of those species which are adapted to use the available carbon effectively.

Adherent populations in natural habitats are in constant flux. A sequence of events occurs during surface fouling, resulting in a succession of surface colonizing species dominating at various times following immersion of clean surfaces in water. In a process termed *epibiosis*, the initial-colonizing species are latter replaced before other organisms become involved with the surface consortium. The succession involves a series of physical and biological events, starting with the adsorption of organic films and soon followed by surface colonization by bacterial species. Bacteria continue to adsorb to exposed surfaces but larger fouling organisms such as protozoa, flagellates, ciliates, diatoms and larvae subsequently colonize the surface over a period of days or weeks.

Rod-shaped bacteria are often the primary colonizers following exposure of clean surfaces to nutrient enriched aqueous systems followed by stalked bacteria such as *Caulobacter* spp., then followed by filamentous algae, diatoms and larvae.

The EPs, in the biofilm, function as an effective nutrient trap facilitating the substrate availability and their utilization by the biofilm community. Increased production of EPs by attached cells improves nutrient trapping efficiency vis-à-vis nutrient removal rate. Exopolymers have an additional role during the utilization of trapped nutrients by the localization of extracellular enzymes that hydrolyse high molecular weight organic matters into a more labile form that can be easily absorbed by the cells as well as can be diffused into the aqueous medium making their availability to the free-living bacteria.

The development of biofilm communities confers several advantages in wastewater treatment. Growth of bacteria in the biofilm may (a) strengthen adaptation by permitting a high diversity of microorganisms to develop in the presence of a broad range of micro-environmental conditions; (b) promote interactions and genetic exchange between different species so that in the presence of a narrow range of substrates, many species become able to degrade and grow on them; (c) favour sequential cometabolic transformation of compounds making the rapid degradation of complex organics possible; (d) enable ready sorption of the primary and secondary carbon and energy sources, sequestering them from the cells but releasing them for subsequent degradation as solution concentrations fall in the micro-environment; and (e) enable communities to tolerate and degrade higher concentrations of toxic organics through sequestration.

## 12.4 MICROBIAL DEGRADATION OF ORGANIC POLLUTANTS IN ECOSYSTEMS

Catabolic processes of microorganisms, algae, yeasts and lower fungi are the main pathways for total or at least partial mineralization/decomposition of bioorganic and synthetic organic compounds in wastewater. It proceeds either in the presence of molecular oxygen by respiration, under anoxic conditions by denitrification, or under anaerobic conditions by methanogenesis and/or sulphidogenesis. Respiration of soluble organic compounds or of extracellularly solubilized biopolymers such as carbohydrates, proteins, fats, or lipids in activated sludge systems leads to the formation of carbon dioxide, water and a significant amount of surplus sludge. Some ammonia and  $H_2S$  may be formed during degradation of sulphur-containing amino acids or heterocyclic compounds. Oxygen must either be supplied by aeration or by injection of pure oxygen. The two process variants for oxygen supply differ mainly in their capacity for oxygen transfer and the stripping efficiency for carbon dioxide from respiration.

Stripping of carbon dioxide is necessary to prevent a drop in pH and to remove heat energy. Respiration in the denitrification process with chemically bound

oxygen supplied in the form of nitrate or nitrite abundantly yields dinitrogen. However, some nitrates escape the reduction to dinitrogen in wastewater treatment plants and contributes about 2% of the total  $N_2O$  emissions. Denitrifiers are aerobic organisms that switch their respiratory metabolism to the utilization of nitrate or nitrite as terminal electron acceptors, if grown under anoxic conditions. Only if the nitrate in the bulk mass has been used completely, the redox potential become low enough for growth of strictly anaerobic organisms, such as methanogens or sulphate reducers. If anaerobic zones are allowed to form in sludge flocs of an activated sludge system, e.g., by limitation of the oxygen supply, methanogens and sulphate reducers may develop in the centre of sludge flocs and form traces of methane and hydrogen sulphide, found in the off-gas.

Key enzymatic reactions of aerobic biodegradation are oxidations catalyzed by oxygenases and peroxidases. Oxygenases are oxidoreductases that use  $O_2$  to incorporate it into the substrate. Degradative organisms need oxygen at two metabolic sites—the initial attack on the substrate and the end of the respiratory chain. Certain higher fungi have developed a unique oxidative system for the degradation of lignin based on extracellular ligninolytic peroxidases and laccases. This enzymatic system possesses increasing significance for the cometabolic degradation of persistent organopollutants

Under strictly anaerobic conditions, soluble carbon compounds of wastes and wastewater are degraded stepwise to  $CH_4$ ,  $CO_2$ ,  $NH_3$  and  $H_2S$  via a syntrophic interaction of fermentative and acetogenic bacteria with methanogens or sulphate reducers. The complete methanogenic degradation of biopolymers or monomers via hydrolysis/fermentation, acetogenesis and methanogenesis can proceed only at a low  $H_2$  partial pressure, which is maintained mainly by interspecies hydrogen transfer. The later is facilitated when acetogens and hydrogenolytic methanogenic bacteria are arranged in proximity in flocs or in a biofilm within short diffusion distances. The reducing equivalents for carbon dioxide reduction to methane or sulphate reduction to sulphide are derived from the fermentative metabolism, e.g., of clostridia or *Eubacterium* sp., from  $\beta$ -oxidation of fatty acids, or the oxidation of alcohol. Methane and carbon dioxide are the main products in anaerobic environments where sulphate is absent, but sulphide and carbon dioxide are the main products if sulphate is present.

The predominant degraders of organopollutants in the oxic zone of contaminated areas are chemo-organotrophic species, which are able to use a huge number of natural and xenobiotic compounds as carbon sources and as electron donors for the generation of energy. Although many bacteria are able to metabolize organic pollutants, a single bacterial species does not possess the enzymatic capability to degrade all or even most of the organic compounds in a polluted soil. Mixed microbial communities have the most powerful biodegradative potential, because the genetic information of more than one organism is necessary to degrade the complex mixtures of organic compounds present in contaminated areas.



### 12.4.1 Aerobic or Anaerobic Hydrolysis of Biopolymers: Kinetic Aspects

Hydrolysis of biopolymers and fermentation or respiration of monomers can be catalyzed by strictly anaerobic, facultative anaerobic and aerobic microorganisms. With some exceptions (e.g., small protein molecules, dextran), biopolymers are insoluble and form fibers (cellulose), grains (starch) or globules (casein after enzymatic precipitation) or can be melted or emulsified (fat) suspension. For a kinetic description of hydrolysis and fermentation, a substrate-limited first-order reaction was assumed by Buchauer (1997), who deduced that the temperature-dependent reaction rate for hydrolysis is a little lower than the reaction rate for fermentation of the hydrolysis products. The rate limiting step is therefore hydrolysis of particles and not fermentation of solubilised material. Since hydrolysis is catalyzed not only by freely soluble exoenzymes, diluted in the bulk mass of liquid, but also to a much higher extent by enzymes that are excreted in the neighborhood of bacterial colonies growing attached to the surface of the particles, the above description of complex fermentation processes is not always valid.

### 12.4.2 Aerobic or Anaerobic Hydrolysis of Biopolymers: Biological Aspects

Cellulose and lignin are the main structural compounds of plants. Both substances are the most abundant biopolymers on earth. Cellulose fibers are formed of linear chains of 100–1400 glucose units linked together by  $\beta$ -1,4-glycosidic bonds. Intermolecular and intramolecular hydrogen bonds and van der Waals interactions arrange the highly organized fibrous regions (crystalline region), which alternate with less organized amorphous regions in the cellulose fibers. The fibers are embedded in a matrix of hemicelluloses, pectin, or lignin. The hemicelluloses consist mainly of xylans or glucomannans, which have side chains of acetyl, gluconuryl, or arabinofuranosyl units. To make cellulose fibres accessible to microorganisms, the hemicellulose, pectin, or lignin matrix must be degraded microbiologically or solubilized chemically. Aerobic or anaerobic cellulose degradation is the most important step in mineralization of decaying plant materials. Cellulolytic organisms are found among aerobic soil fungi, e.g., within the genera *Trichoderma* and *Phanerochaete* and in anaerobic rumen fungi, e.g. *Neocallimastix* and *Piromyces* and among bacteria, e.g., within the genera *Cellulomonas*, *Pseudomonas* and *Thermomonospora* (aerobic cellulose degraders) and *Clostridium*, *Fibrobacter*, *Bacteroides* and *Ruminococcus* (anaerobic cellulose degraders).

Glycosyl hydrolases are involved in cellulose and hemicellulose degradation by cleaving glycosidic bonds between different carbohydrates and between carbohydrates and non-carbohydrates. Endocellulases and exocellulases, in some organisms organized in cellulosomes, must be excreted into the medium. Cellulases are complex biocatalysts and contain a catalytic site and a substrate-

binding site. The presence of a noncatalytic substrate binding site permits tight attachment to different forms of cellulose substrate and keeps the enzyme close to its cleaving sites. Substrate binding is reversible, which allows the enzyme to 'hike' along the fibers and obtain total solubilization. Many aerobic fungi and some bacteria excrete endoglucanases that hydrolyze the amorphous region of cellulose (degradation within the chain), whereas exoglucanases hydrolyze cellulose from the ends of the glucose chains. Cellobiose is cleaved off by cellobiohydrolases from the nonreducing ends in the amorphous region and finally the crystalline region is also hydrolyzed. Cleavage of cellobiose to glucose units by *p*-glucosidases is necessary to prevent cellobiose accumulation, which inhibits cellobiohydrolases.

Anaerobic bacteria such as *Clostridium thermocellum* form a stable enzyme complex, the cellulosome, at the cell surface which are active in degrading crystalline cellulose. The catalytic subunits of a cellulosome, endoglucanases and xylanases, cleave the cellulose fiber into fragments, which are simultaneously degraded further by *p*-glucosidases. Cellulosome-like proteins (cell-bound enzymes) are also found associated with the capsule or the outer membrane in *Ruminococcus* sp. and *Fibrobacter* sp. Other specific adhesions or ligand formations with the cellulose can be facilitated by fimbrial connections, glycosylated epitopes of carbohydrate-binding proteins, or the glycocalyx and carbohydrate-binding modules. Some bacteria have not developed a mechanism to adhere to cellulose fibers, but excrete cellulases into the medium. Adsorption of bacteria onto cellulose fibers via cellulosomes offers the advantage of close contact with the substrate, which is hydrolyzed mainly to glucose and then taken up for metabolism. Small amounts of cellobiose must be present initially to induce cellulase expression. The contact of bacteria with the solid substrate surface keeps them close to cellobiose and thus keeps cellulase activities high. However, accumulation of hydrolysis products such as glucose repress cellulase activity.

In nature most cellulose is degraded aerobically (about 5%–10% is thought to be degraded anaerobically). Since most ecosystems are rich in carbonaceous substances but deficient in nitrogen compounds, many cellulolytic bacteria can also fix dinitrogen. This is advantageous to them and to syntrophic or symbiotic organisms. Other examples of mutual interactions between organisms of an ecosystem are interspecies hydrogen transfer between anaerobes (anaerobic food chain), transfer of growth factors (mycorrhizae) and production of fermentable substrates for symbionts. Hydrolysis of biological structural components such as cellulose, lignin and other structural or storage polymers is difficult. The limiting step of hydrolysis seems to be liberation of the cleavage products. In contrast to the slow hydrolysis of celluloses, mainly due to lignin encrustation of naturally occurring celluloses, starch can be easily hydrolyzed. The branching and helical structure of starch facilitates hydrolysis. Where cellulose forms fibers with a large surface covered with lignin, starch forms grains with an unfavourable surface-to-volume ratio for enzymatic cleavage. Thus, although amylases may be present in high concentrations, the hydrolysis rate is limited by the limited access of the

enzymes to the substrate. Where cellulose and starch are biodegradable, other carbohydrate-derived cellular compounds are not biodegradable and, after reaction with proteins, form humic acid-like residues by the Maillard reaction.

## 12.5 AEROBIC BACTERIAL DEGRADATION OF BIOPOLYMERS

### 12.5.1 Energy Balance of Biopolymer Degradation

In ecosystems in which molecular oxygen is available, plant and animal biomass is degraded to  $\text{CO}_2$  and  $\text{H}_2\text{O}$ , catalyzed by either single species of aerobic microorganisms or the whole population of the ecosystem, in competition for the substrates. A single organism may be able to hydrolyze the polymers and oxidize the monomers to  $\text{CO}_2$  and  $\text{H}_2\text{O}$  using molecular oxygen. To make soluble and insoluble biopolymers, mainly carbohydrates, proteins and lipids accessible for respiration by bacteria, the macromolecules must be hydrolyzed by exoenzymes, which often are produced and excreted only after contact with respective inductors. The exoenzymes adsorb to the biopolymers and hydrolyze them to monomers or at least to oligomers. Only soluble, low molecular weight compounds (e.g., sugars, disaccharides, amino acids, oligopeptides, glycerol and fatty acids.) can be taken up across the membrane by microorganisms and be metabolized for energy production and cell multiplication. The two steps can be represented as:

Complex organic compounds  $\xrightarrow{\text{Exoenzymes}}$  Simple absorbable polymers

Simple polymers  $\xrightarrow{\text{Respiration}}$  Basic elements + Energy

Once taken up, degradation via glycolysis (sugars, disaccharides, glycerol), hydrolysis and deamination (amino acids, oligopeptides), or hydrolysis and  $\beta$ -oxidation (phospholipids, long-chain fatty acids) proceeds in the cells. Metabolism of almost all organic compounds leads to the formation of acetyl-CoA as the central intermediate, which is used for biosyntheses, excreted as acetate, or oxidized to  $\text{CO}_2$  and reducing equivalents in the tricarboxylic acid (TCA) cycle. The reducing equivalents are respired with molecular oxygen in the respiration chain. The energy of a maximum of only 2 mol of anhydridic phosphate bonds of ATP is conserved during glycolysis of 1 mol of glucose through substrate level phosphorylation. An additional 2 mol of ATP is formed during oxidation of 2 mol of acetate in the TCA cycle, whereas 34 mol of ATP are formed by electron transport chain phosphorylation with oxygen as the terminal electron acceptor. During oxygenic respiration, reducing equivalents react with molecular oxygen in a controlled combustion reaction to generate water and energy.

When carbohydrates are respired by aerobic bacteria, about one-third of the initial energy content is lost as heat and two-thirds are conserved biochemically

in 38 phosphoanhydride bonds of ATP. In activated sludge reactors or in wastewater treatment ponds that are not loaded with highly concentrated wastewater, wall irradiation and heat losses with the off-gas stream of aeration into the atmosphere prevent self-heating. However, in activated sludge reactors for treatment of highly concentrated wastewater, self-heating up to thermophilic temperatures may occur if the wastewater is warm in the beginning, the hydraulic retention time for biological treatment is short (short aeration time) and the air or oxygen stream for aeration is restricted so as to supply just sufficient oxygen for complete oxidation of the pollutants (small aeration volume).

The conserved energy in the terminal phosphoanhydride bond of ATP, formed during substrate chain and oxidative phosphorylation by proliferating bacteria is partially used for maintenance, metabolism and partially for cell multiplication. Partitioning between both is not constant, but depends on the nutritional state. In highly loaded activated sludge reactors with a surplus or at least a non-growth-limiting substrate supply, approximately 50% of the substrate is respired in the energy metabolism of the cells and 50% serves as a carbon source for cell growth. The biochemically conserved energy must be dissipated to be used for the maintenance, metabolism of existing cells and cell growth.

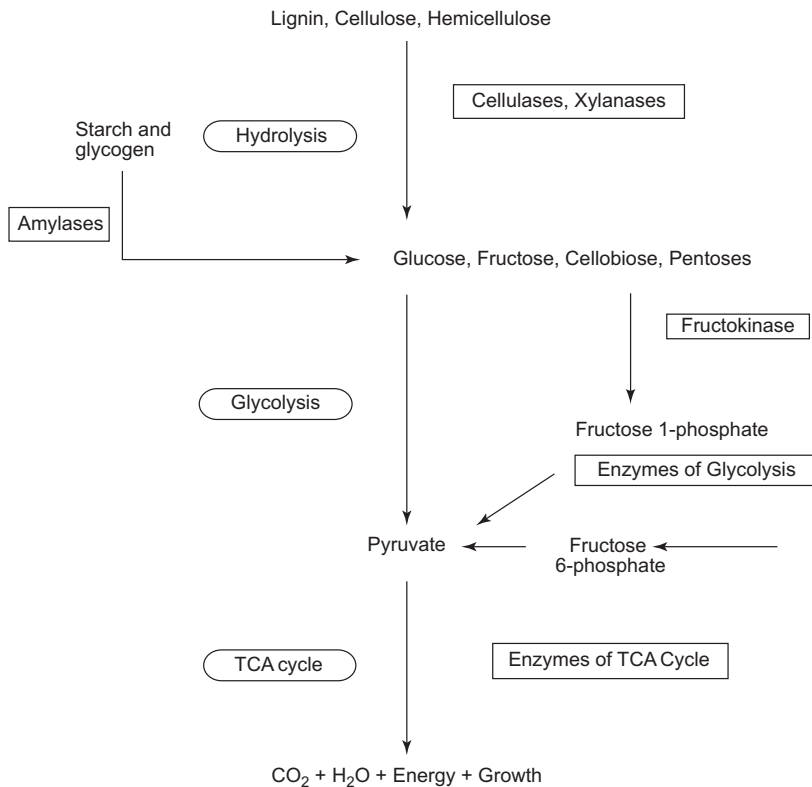
If the substrate supply is growth-limiting, e.g., in a low-loaded aerobic treatment system, a higher proportion of ATP is consumed for maintenance, representing the energy proportion that bacteria must spend for non-growth-associated cell survival metabolism and less energy is available for growth. Overall, more of the substrate carbon is respired and the ratio of respiration products to surplus sludge formed is higher, e.g., around 70% : 30% (Table 12.1). In a trickling filter system, a higher proportion of the substrate seems to be respired. This might be due to protozoa grazing off part of the biofilm. Under anaerobic condition, about 5% of the fermentable substrate is used for cell growth (surplus sludge formation) in anaerobic reactors, whereas 95% is converted to methane and CO<sub>2</sub> and most of the energy of the substrates is conserved in the fermentation products.

**Table 12.1:** Carbon flow during (A) aerobic and (B) anaerobic degradation of biopolymers.

|                                                          |                                                                               |
|----------------------------------------------------------|-------------------------------------------------------------------------------|
| (A) Aerobic degradation                                  |                                                                               |
| (a) saturating substrate supply = high-loaded conditions |                                                                               |
| 1 unit of substrate carbon                               | → 0.5 unit of CO <sub>2</sub> + 0.5 of unit cell carbon                       |
| (b) Limiting substrate supply = low-loaded systems       |                                                                               |
| 1 unit of substrate carbon                               | → 0.7 unit of CO <sub>2</sub> + 0.3 of unit cell carbon                       |
| (B) Anaerobic degradation                                |                                                                               |
| 1 unit of substrate carbon                               | → 0.95 unit of CO <sub>2</sub> and CH <sub>4</sub> + 0.05 of unit cell carbon |

### 12.5.2 Growth Associated Degradation of Carbohydrates

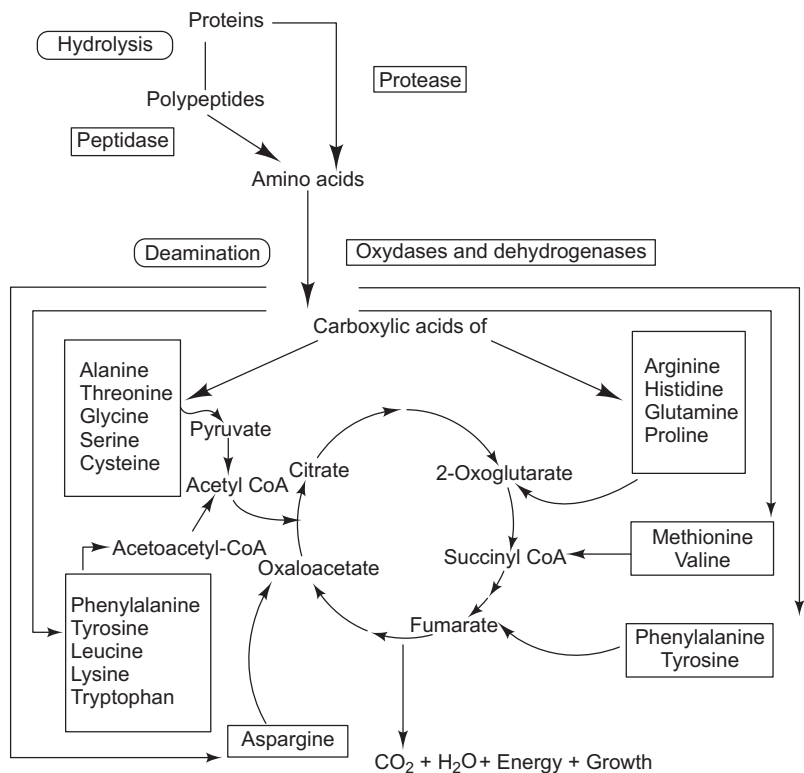
Different carbohydrates like starch, polysaccharides, lignin, cellulose and hemicellulose are microbially degraded under aerobic conditions to produce  $\text{CO}_2$  and  $\text{H}_2\text{O}$ . The energy released from the degradation is used by the microbes for their growth. The first step in the degradation process is the hydrolysis of complex polymers catalyzed by numerous auxiliary enzymes like cellulase, xylanases, amylases, phosphorylases, etc. Starch and glycogen are converted to D-glucose units by phosphorylases and phosphoglucomutase to gain energy into the glycolytic pathway. Hydrolysis of complex carbohydrates produces various types of monosaccharides other than glucose and also disaccharides like sucrose, maltose and lactose. The disaccharides are then hydrolysed to monosaccharides by the enzymes on the cell surface of the microbes and then gain entry into the cell for aerobic breakdown. The monomers are converted to pyruvate and subsequently to  $\text{CO}_2$  and  $\text{H}_2\text{O}$  in the cell through glycolysis and tricarboxylic acid cycle, respectively. A part of the carbohydrate is also utilized to produce the cell biomass as well as to be stored in the microbial cells as food reserve (Fig. 12.4).



**Figure 12.4** - Aerobic degradation of carbohydrates by microbes. The scheme shows the degradation of carbohydrates of different complexities and entry of different monosaccharides into the common aerobic degradation pathway.

12.5.3 Growth Associated Degradation of Proteins

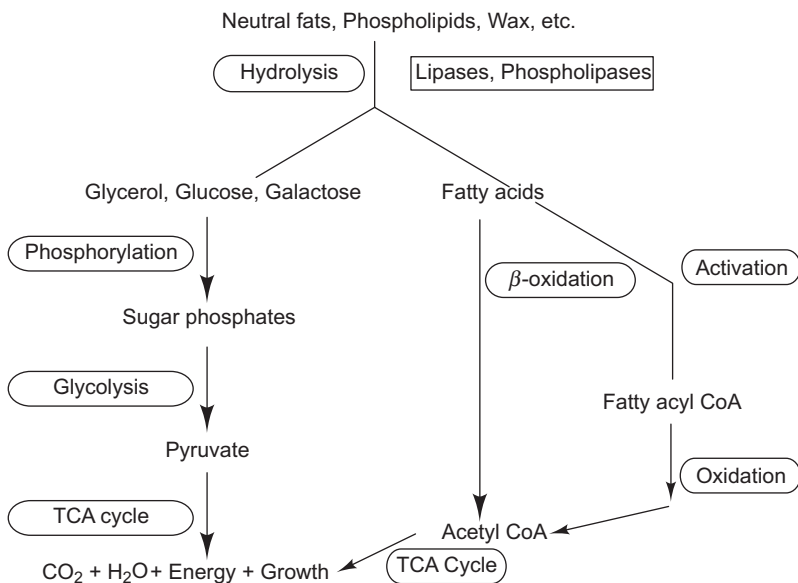
The proteins present in the wastes are aerobically degraded by microbes in three successive steps: (a) Hydrolysis of protein to amino acids by proteases and peptidases, (b) Deamination of amino acids by dehydrogenases to produce carboxylic acids and (c) Oxidative breakdown of the carboxylic acids to CO<sub>2</sub> and H<sub>2</sub>O. Proteolysis occurs on the cell surface or in the biofilm by the exoenzymes secreted from the cell assemblage. Deamination by enzymes like dehydrogenase and amino acid oxidases releases ammonia to produce carboxylic acids. Different carboxylic acids enter at different points as carbon skeleton into the tricarboxylic acid pathway. Alanine, threonine, glycine, serine and cysteine enter into the tricarboxylic acid cycle through pyruvate. Arginine, histidine, glutamine and proline get their entry through 2-oxoglutarate and isoleucine, methionine and valine enter through succinyl CoA. Certain amino acids like phenylalanine, tyrosine, leucine and tryptophan undergo fragmentation in such a way that the product enters the cycle through two different routes (Fig. 12.5). However, not all the carbon atoms of each of the 20 amino acids enter the TCA cycle since some of them are used by the cells as building blocks for growth and few others are lost en-route by decarboxylation reaction.



**Figure 12.5** - The entry of carbon skeleton of different amino acids into the aerobic oxidation pathway during microbial degradation of proteins.

### 12.5.4 Growth Associated Degradation of Lipids and Neutral Fats

Because of their hydrophobicity, fats, waxes and fatty acids are less easily degraded by microbes than carbohydrates and proteins. Different lipids and fats produce different products on hydrolysis catalyzed by the exoenzymes: lipases and phospholipases. Neutral fats, on lipolysis, produce saturated and unsaturated fatty acids (palmitic acids, linolic acids, etc.) and glycerol. Different phospholipids undergo lipolysis to generate fatty acids, glycerol and several other alcohols. Hexoses like glucose and amino alcohols are produced along with fatty acids by lipolysis of glycolipids while the sugars, alcohols and amino acids enter into aerobic degradation process at different places in the aerobic respiration pathway. The fatty acids either enter directly into the cell or undergo activation on the cell surface before entry. Long chain fatty acids are activated on the cell surface and are then transported across the plasma membrane into the cytosol. In the cytosol, the fatty acid group is transferred to CoA yielding a fatty acyl CoA thioester (Fig. 12.6). The subsequent oxidation of fatty acyl CoA occurs through dehydrogenation followed by enzymatic dehydration to form a  $\beta$ -hydroxy acyl-CoA. The later is then hydrogenated followed by a cleavage forming a molecule of acetyl CoA and fatty acyl CoA having two fewer carbon atoms than the original fatty acid. In this way, the entire fatty acyl chain is spliced to many two carbon acetyl CoA molecules, which are aerobically degraded to  $\text{CO}_2$  and  $\text{H}_2\text{O}$  through tricarboxylic acid cycle.



**Figure 12.6** - Aerobic degradation of lipids, fats and fatty acids by bacteria.



## 12.6 ANAEROBIC BACTERIAL DEGRADATION OF BIOPOLYMERS

### 12.6.1 Degradation by an Anaerobic Food Chain

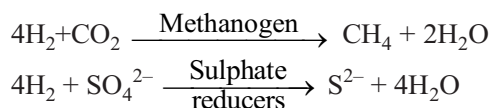
In ecosystems where molecular oxygen is deficient but organic matter content is considerably high (swamps, compost heaps, sediments, etc.), degradation occurs anoxically by reduction of electron acceptors such as nitrate and nitrite or sulphate,  $\text{Fe}^{2+}$ ,  $\text{Mn}^{4+}$  or  $\text{CO}_2$ . The oxidation of the carbon source is either complete or incomplete with acetate excretion. In the absence of inorganic electron acceptors, oxidized metabolites such as pyruvate or acetate are reduced to lactate or ethanol or biotransformed to, n-butyrate or n-butanol. In permanently anaerobic ecosystems with seasonal overfeeding, periodic accumulation of such metabolites can occur. The biopolymers decompose by extracellular enzymatic hydrolysis while the monomers are fermented and the fermentation products may be degraded further by acetogenic and methanogenic bacteria to methane. Where single cultures of aerobes can catalyze the whole mineralization process to finally form  $\text{CO}_2$  and  $\text{H}_2\text{O}$ , single cultures of strictly anaerobic bacteria are not capable of degrading biopolymers to  $\text{CH}_4$  and  $\text{CO}_2$ . Under anaerobic conditions, biopolymers must be degraded by a food chain via depolymerization (hydrolysis), fermentation (acidogenesis), oxidation of fatty acids (acetogenesis) and biogas formation (methanogenesis). The soluble mono-, di-, or oligomers are taken up by the bacteria and fermented to  $\text{CO}_2$ ,  $\text{H}_2$ , formate, acetate, propionate, butyrate, lactate, etc. If fatty acid isomers are produced, they are mainly derived from degradation of amino acids after proteolysis. Fatty acids are further oxidized by acetogenic bacteria, before the cleavage products  $\text{CO}_2$ ,  $\text{H}_2$  and acetate can be taken up by methanogens and be converted to methane and carbon dioxide. Lactate is oxidized to pyruvate, which is decarboxylated to yield acetate,  $\text{CO}_2$  and  $\text{H}_2$ . If ethanol is present, it is oxidized to acetate and hydrogen and the hydrogen is used for reduction.

In complex ecosystems, formate is formed if high concentrations of hydrogen accumulate. Syntrophic interactions are usually associated with interspecies hydrogen transfer but evidence for interspecies formate transfer has also been reported. The feasibility of the electron carrier depends on its solubility, which is much less for hydrogen than for formate and on its diffusion coefficient in water, which favours hydrogen 30 times over formate. The efficiency of the appropriate electron transfer depends mainly on the distance between the producing and consuming bacteria. It can be expected that formate transfer is favoured when the distance between communicating the bacteria is high and hydrogen transfer when the distance is small. Interspecies formate transfer is thought to play a major role in degradation of syntrophic butyrate and propionate. However, at an increased  $\text{H}_2$  partial pressure, formate is also produced by methanogens, either in pure cultures or in a sewage sludge population and this may also contribute to increasing formate concentrations. Other substrates for methanogenic bacteria, such as methyl amines and dimethyl amines, are relevant only in ecosystems

where these substances are produced during microbial decay. A few methanogens can also use reduced products such as primary, secondary and cyclic alcohols as a source of electrons for  $\text{CO}_2$  reduction.

### 12.6.2 Anaerobic Biopolymer Degradation Control by Hydrogen and Acetate

Molecular hydrogen is produced during different stages of anaerobic degradation. In the fermentative stage, organisms such as *Clostridium* sp. and *Eubacterium* sp. produce fatty acids, carbon dioxide and hydrogen from carbohydrates. In the acetogenic stage, acetogens such as *Syntrophobacter wolnii* and *Syntrophomonas wolfei* produce acetate, carbon dioxide and hydrogen or acetate and hydrogen by anaerobic oxidation of propionate and n-butyrate. Fermentative bacteria (e.g., clostridia, *Ruminococcus*, *Eubacterium* sp.) release molecular hydrogen even at a high  $\text{H}_2$  partial pressure and simultaneously excrete reduced products. However, the release of molecular hydrogen during acetogenesis of fatty acids or of other reduced metabolites may occur only when hydrogen does not accumulate, for thermodynamic reasons. Molecular hydrogen is consumed by methanogens alternatively, by sulphate reducers via interspecies hydrogen transfer.



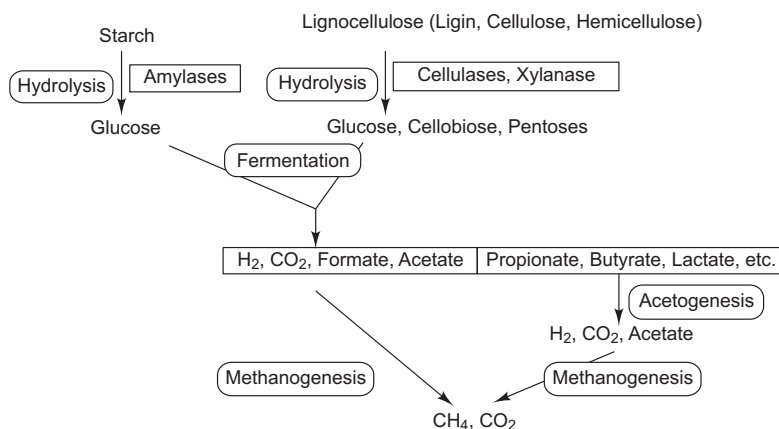
To maintain a low  $\text{H}_2$  partial pressure, a syntrophic interaction among acetogenic, hydrogen-producing, methanogenic, and hydrogen-utilizing bacteria is essential. Complete anaerobic degradation of fatty acids with hydrogen formation by obligate proton-reducing acetogenic bacteria is possible only at  $\text{H}_2$  partial pressures  $< 10^{-4}$  atm (n-butyrate) or  $10^{-5}$  atm (propionate), which cannot be maintained by methanogens or sulphate reducers. However, by reversed electron transport, electrons can be shifted to a lower redox potential suitable for proton reduction. If hydrogen accumulates beyond this threshold concentration, the anaerobic oxidation of fatty acids becomes endergonic and does not proceed, whereas hydrogen prevents  $\beta$ -oxidation of fatty acids by acetogens even at very low  $\text{H}_2$  partial pressure, much higher concentrations of acetate (in the millimolar range) are required for the same effect.

In addition to the possibility of anaerobic oxidation of organic compounds via syntrophic interactions between acetogenic bacteria and acetoclastic + hydrogenotrophic methanogens, other syntrophic associations between acetate-oxidizing bacteria and  $\text{H}_2/\text{CO}_2$ -utilizing methanogens under thermophilic and mesophilic growth are observed. Thermodynamic analysis of mesophilic syntrophic acetate oxidation revealed a hydrogen partial pressure of  $< 0.1$ – $2.6$  Pa, which is in the range found in methanogenic ecosystems. In thermophilic methanogenic reactors, acetate is degraded either by syntrophic acetate oxidizers (dominant process at low acetate concentrations) and acetate-degrading

methanogens (acetate concentration above the threshold concentration) or by acetate-utilizing methanogens of the genera *Methanosaeta* or *Methanosarcina*.

### 12.6.3 Anaerobic Degradation of Carbohydrates

The anaerobic degradation of biopolymers, in general and, of cellulose, in particular, can be divided into hydrolytic, fermentative, acetogenic and methanogenic phases (Fig. 12.7). Hydrolysis and fermentation of the hydrolysis products can be catalyzed by the same trophic group of microorganisms. The distinction of the two phases is of more theoretical than practical relevance. Concerning reaction rates in a methane fermenter that is fed with a particulate substrate, the rate-limiting step is hydrolysis rather than the subsequent fermentation of the monomers, if acetogenesis and methanogenesis proceed faster. The hydrolysis rates of polymers can be very different. Hemicellulose and pectin are hydrolyzed 10 times faster than lignin-encrusted cellulose. In the acidification reactor of a two-stage anaerobic process, hydrolysis of polymers to monomers is normally slower than fermentation of monomers to fatty acids and other fermentation products. For this reason, no sugar monomers can be detected during steady-state operation. In the methane reactor,  $\beta$ -oxidation of fatty acids, especially of propionate or n-butyrate, is the rate limiting step. Fatty acid production is the slowest reaction in a two-stage methane reactor fed with carbohydrate-containing wastewater from sugar production. Thus, the methane reactor has to be larger than the acidification reactor to permit longer hydraulic retention times.



**Figure 12.7** - Anaerobic microbial degradation pathway of simple and complex carbohydrates.

The rate of cellulose degradation depends strongly on the state of the cellulose in the wastewater. If cellulose is lignin-encrusted, lignin prevents access of cellulases to the cellulose fibres. If cellulose is mainly in a crystalline form, cellulases can easily attach to it and then hydrolysis can be a relatively fast process. At increasing

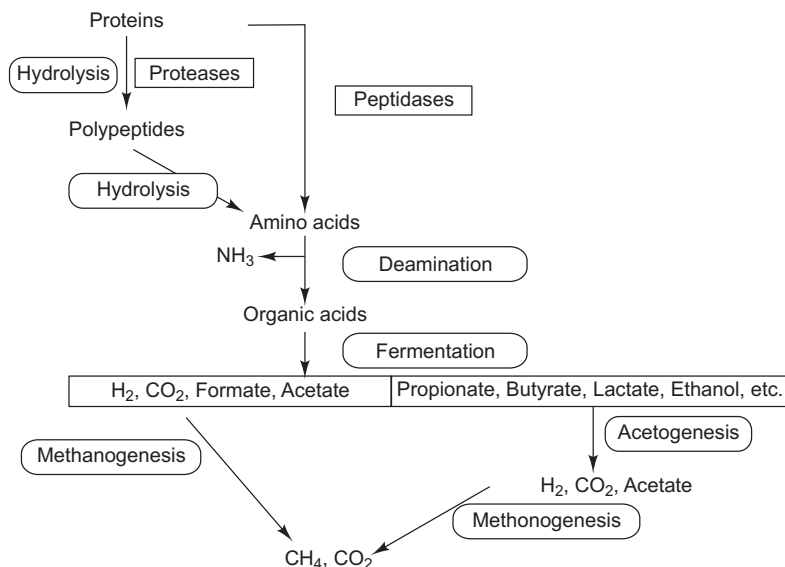
loading in an anaerobic reactor fed with crystalline cellulose, acetogenesis become the rate-limiting process, leading to propionate and butyrate formation. In decaying plant material, cellulose is very often lignin-encrusted. Due to the highly restricted access to these complexes by cellulases, hydrolysis of cellulose is the rate-limiting step in its degradation to methane and  $\text{CO}_2$ .

During anaerobic degradation of starch, hydrolysis by amylases proceeds with high velocity if good contact between starch grains and amylases is maintained whereas in an anaerobic reactor at low loading, starch degradation can proceed in the absence of acetogenic bacteria, at high loading, volatile fatty acids are formed and acetogens are essential for total degradation. Many fermentative bacteria produce only acetate, formate, carbon dioxide and hydrogen when  $\text{H}_2$ -scavenging methanogens or sulphate reducers are able to maintain a sufficiently low  $\text{H}_2$  partial pressure, but a wide spectrum of fermentation products, typical for the metabolism of the respective bacterial species in pure culture, is produced at higher  $\text{H}_2$  partial pressure.

### 12.6.4 Anaerobic Degradation of Proteins

Hydrolysis of precipitated or soluble protein is catalyzed by several types of proteases that cleave membrane-permeable amino acids, dipeptides, or oligopeptides. In contrast to the hydrolysis of carbohydrates, which proceeds favourably at a slightly acidic pH, optimal hydrolysis of proteins requires a neutral or weakly alkaline pH. Fermentation of amino acids in wastewater reactors does not lead to a significant pH change, due to acid and ammonia formation. Acetogenesis of fatty acids from deamination of amino acids requires a low  $\text{H}_2$  partial pressure for the same reasons as for carbohydrate degradation. This can be maintained by a syntrophic interaction of fermentative, protein-degrading bacteria and acetogenic and methanogenic or sulphate-reducing bacteria. Except for syntrophic interaction of amino acid-degrading bacteria with methanogens for maintenance of low  $\text{H}_2$  partial pressure, clostridia and presumably also some other sludge bacteria may couple oxidative and reductive amino acid conversion (Fig.12.8).

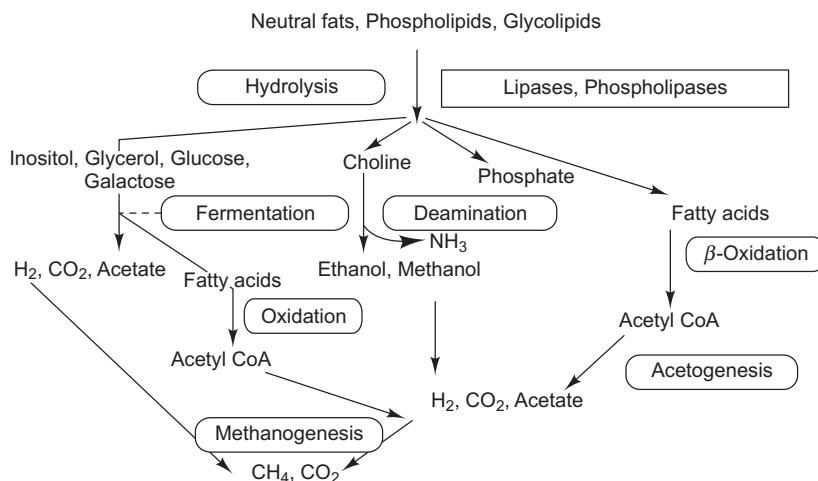
For complete degradation of amino acids in an anaerobic system, therefore, a syntrophic relationship of amino acid-fermenting anaerobic bacteria with methanogens or sulphate reducers is required. If long-chain amino acids are deaminated, fatty acids such as propionate, *i*-butyrate, or *i*-valerate are formed directly. The fatty acids require acetogenic bacteria for their degradation. During protein degradation these fatty acids are a product of deamination. In the methanogenic phase, there is no difference in methanogenic activity whether carbohydrates or proteins are fermented, except that the methanogens in a reactor fed with protein need to be more tolerant to ammonia and higher pH.



**Figure 12.8** - Anaerobic fermentation of proteins and amino acids.

### 12.6.5 Anaerobic Degradation of Neutral Fats and Lipids

Fats and lipids are another group of biopolymers that contribute significantly to the COD in sewage sludge. The first step of fat and lipid degradation is the hydrolysis to produce simpler products. Glycerol and saturated and unsaturated fatty acids (palmitic acid, linolic acid, linolenic acid, stearic acid, etc.) are formed from neutral fats. Lipolysis of phospholipids generates fatty acids, glycerol, alcohols (serine, ethanolamine, choline, inositol) and phosphate while that of sphingolipids generates fatty acids and amino alcohols (e.g., sphingosine). Sugar moieties and glycerol can be degraded to methane and carbon dioxide by interaction between fermentative and methanogenic bacteria in low-loaded systems or by cooperation between fermentative, acetogenic and methanogenic bacteria in high-loaded systems (Fig. 12.9). The long-chain fatty acids are degraded by acetogenic bacteria by  $\beta$ -oxidation to acetate and molecular hydrogen. If acetate and molecular hydrogen accumulate, the anaerobic digestion process is inhibited. Odd-numbered fatty acids are degraded to acetate, propionate and hydrogen and even-numbered fatty acids are degraded to acetate and hydrogen. Only at a very low  $\text{H}_2$  partial pressure, which can be maintained by hydrogen-utilizing methanogens or sulphate reducers,  $\beta$ -oxidation of at least n-butyrate or propionate is exergonic. After hydrolysis, fermentation and acetogenesis products of the fat components in the methanogenic phase, acetate, carbon dioxide and hydrogen are converted to biogas. All subsequent intracellular reactions can be influenced by syntrophic interaction via interspecies hydrogen transfer, except for the extracellular initial lipase reaction.



**Figure 12.9** - The anaerobic microbial metabolism of neutral fats and lipids.

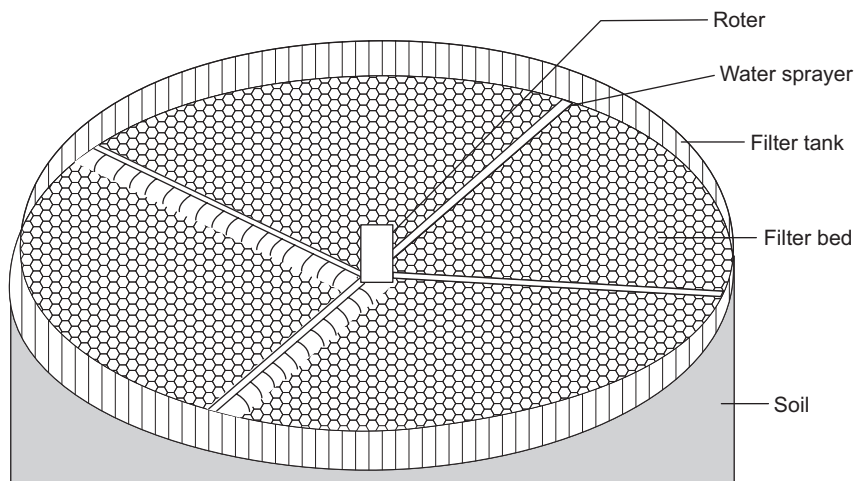
Carbohydrates, proteins, fats and biogenic oils can also be degraded anaerobically under thermophilic conditions. The overall degradation scheme is the same, but the microorganism populations are different. Thermophilic fat degradation is becoming more important in practice, since waste fat from fat separators and fat flotates of the food industry are often co-fermented in agricultural biogas plants. Since, for hygienic reasons, the input material must be autoclaved, a thermophilic process should be used, keeping the fat in a melted, soluble form for more effective metabolism. Biogas plants with co-fermentation of waste fat residues are considered waste treatment systems and must be designed to meet the hygienic demands relevant to treatment of the respective waste.

## 12.7 SECONDARY TREATMENT SYSTEMS

Many secondary treatment systems have been developed for biological degradation of the organic pollutants in industrial and municipal wastewaters. Some of the systems, that are in active use, are discussed here.

### 12.7.1 Conventional Biofilters

This system is also called percolating filters (Fig. 12.10). The usual filtration-media is blast furnace slag, although gravel, shale, coke, coal and rigid plastic shapes have also been used with success. About 2 m deep and with the media carefully graded in the size range 25–150 mm diameter with the largest material at the bottom, the whole bed is constructed on a system of underdrains. This allows a free flow of air to pass upwards through the filter by natural convection as effluent drains downwards by gravity. Retention time within the filter media is typically 20–35 minutes to achieve 80%–90% reduction. High-rate filters have



**Figure 12.10** - A circular biofilter constructed at the soil level.

rarely more than 10 minutes retention, hence the reduced removal performance. Liquid is conveyed to the media by hollow metal distributor arms with regularly spaced holes and supported by wires. On circular filters, the jetting action of the liquid leaving the arms drives the distributor round. On exposed sites, wind vanes are used to assist rotation in adverse weather. At sites where much of the flow is pumped and highly erratic, motor drives are fitted to the distributor arms principally to even out rotation speed variations.

Biofilters are only fed with settled sewage or industrial effluent after initial settlement, as a high solids load will quickly block the distributor arms and clog the top media surface. Effective preliminary treatment to remove grit and rag is also essential. The filter is either fed continuously from a distribution chamber at a slightly higher level than the distributor arms or via dosing syphons on a cyclic basis at sites where the flow is too low to maintain filter arm rotation, or much of the influent is pumped periodically. Maintenance of biofilters is confined to cleaning out the distribution holes in the arms and rodding out the whole arm—a daily task if solids levels are over 250 mg/l. Many filters can be maintained on a weekly check basis and the simplicity of design has encouraged their use in remote, unmanned areas.

Biofilter hydraulic loading rates are usually quoted at 600 l/m<sup>2</sup> day to achieve 90% nitrification, with 800 l/m<sup>2</sup> day as an upper limit for 80% removal of BOD and ammonia. However, there is no hard and fast rules for loading as biofilters can be highly individual in performance, even within the same site and fed with the same feed wastewater. There are many examples of well-graded walled filters achieving more than 90% removals at loadings approaching 1000 l/m<sup>2</sup> day where filter voidage and drainage rate and available media surface area are all high. Conversely, hydraulic underloading leading to drying out and a biomass starved to the point of endogenous respiration will not produce good effluent quality either.



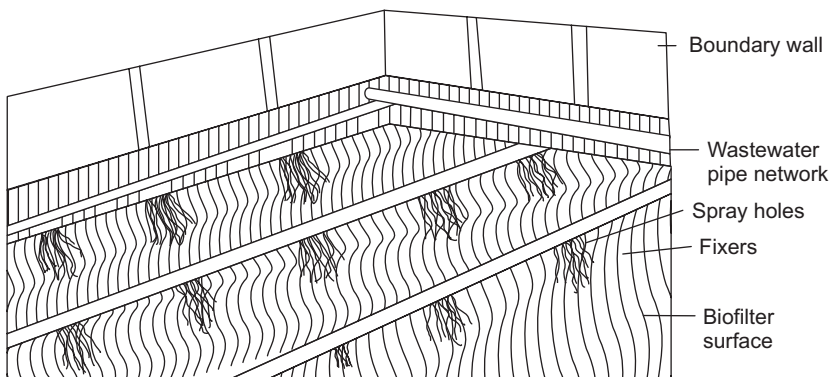
Flow balancing, continuous recirculation and double filtration (where there is more than one filter bed) are all advantageous for better performance of the biofilter. The main message is to design conservatively for hydraulic loading, but not too with a view to accommodate any further increase in wastewater volume in future. At the same time it should be kept in mind that the present feeding rate is for optimum activity of the filter.

A biological loading rate of  $0.1 \text{ kg BOD/m}^3 \text{ day}$  is frequently quoted to achieve full nitrification; 80% BOD removals may be expected at  $0.2 \text{ kg BOD/m}^3 \text{ day}$  and represents the boundary between low and high rate filtration. A poor ammonia removal of less than 50% may be expected at this rate. Effluents high in ammonia but low in BOD should be applied at the rate of  $0.05\text{--}0.08 \text{ kg NH}_3\text{N/m}^3 \text{ day}$  and conventional hydraulic rates. This can only be a rough guide, as the BOD/ $\text{NH}_3\text{N}$  ratio and the C : N : P ratio will also affect removal rates as they control the microbial growth and cooperation on the biofilm surface.

Percolating filters can adapt to the treatment of toxic materials over a period of several months provided the concentration is increased very slowly. This is required for facilitating the growth and establishment of a biofilm of tolerant species. There are thus many documented examples of the effective oxidation of phenols, cyanides, pharmaceuticals and non-chlorinated solvents and toleration of metals without significant inhibition at the microbial activity. Important factors are the maintenance of adequate flow and a reasonable nutrient balance. Once acclimatised, the biofilter system usually tends to tolerate shock loads of toxic materials and high BODs rather better than the activated sludge system.

### 12.7.2 High-rate Biofilters

High-rate biofilters are usually constructed as tower models with a height of 3–4 m, often rectangular, almost always, entirely above ground and enclosed in lightweight metal sheeting (Fig. 12.11). The packing media are mostly made up of plastics that are light weight but at the same time must be the favoured surface



**Figure 12.11** - Structure of a high rate biofilter showing the pipe network for feeding wastewater.

for microbial colonization. As plastics are not easily wetted and the voidage of high-rate filters is frequently 90%, irrigation paths and rates are crucial to good performance. Retention times are rarely more than 10 minutes. Despite this, 50% removal rates of both chemical and bacterial numbers should be achieved. The media must be randomly packed during construction to avoid shortcircuiting and spray irrigated at a rate to ensure continuous wetting. Nozzle design is important in achieving a good surface spread of liquid, whilst resisting blockage by high solids or colloidal material. Like in conventional biofilter, a preceding settlement stage is essential to remove sedimentable gross solids.

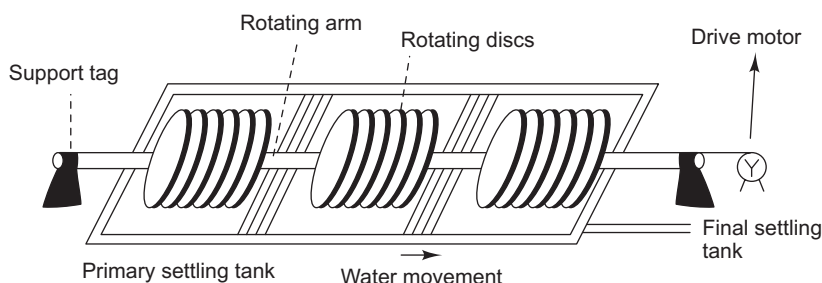
Hydraulic loading rates of 5000–10000 l/m<sup>3</sup> day are common and 25000 l/m<sup>3</sup> day can be possible. This often requires pumping the effluent onto the filter under pressure or maintaining a high recirculation/flow ratio (5–10 times) and the electricity costs of this should not be underestimated. Biological loading rates often exceed 1 kg BOD/m<sup>3</sup> day. These rates, on the other hand, are not constant and the individual circumstances will dictate performance. However, extremely heavy biological loading rate reduces the reactor performance due to increased growth of microbial assemblage leading to anaerobic pockets inside the biofilm.

Several major attractions of the high-rate filter include the low construction costs and rapid installation. Ground preparation need not be extensive. Surface loads will be low as the media is so light and excavation apart from drainage is unnecessary. The limitation of application of the high-rate filters is that these are normally constructed to perform carbonaceous oxidation only rather than nitrification and achieving the latter is often an erratic bonus at the designed hydraulic rates. Tolerance to shock loads, loss of feed and drying out are similar to conventional filters and many high-rate filters operate on a surprising cocktail of organics. Recovery is normally quicker, because of the high irrigation rate and the fact that normal performance comprises only carbonaceous oxidation, which re-establishes much quicker due to the robustness of the organisms, than the more fragile nitrifying bacteria.

### 12.7.3 Rotating Biological Contactors (RBC)

First installed in the then West Germany in 1960, this 'package' system of biotreatment has been used more for carbonaceous oxidation and as a roughing filter than in a nitrifying mode. The system consists of a series of discs or drums of plastic media that slowly rotate on a horizontal shaft, being alternately exposed to air or dipped in a trough containing the wastewater. Close-packed discs on an 8.2 m long shaft offer 9000 m<sup>2</sup> of surface area while corrugated high-density polypropylene discs offer up to 16000 m<sup>2</sup> surface for the development of active biofilm. Typically, 40% of the biofilm that develops on the discs is submerged while about 25 % of the biofilm on the surface of the shaft remain in contact with wastewater at any time. However, this varies with the systems and the portion of biofilm that remain in contact with the wastewater can go up to 70 % (Fig. 12.12). The rotation of the shaft and the discs cause regular bathing of the biofilm in wastewater and alternatively expose it to air guaranteeing the availability of the

substrate and oxygen and facilitating rapid oxidation of the organic pollutants in wastewater.



**Figure 12.12** - The structural design of a RBC system.

Typical hydraulic loadings of 80–160 l/m<sup>2</sup> of disc surface area/day and organic loadings of 10 g soluble BOD/day achieve 60% of BOD removal. A second unit is added to give nitrification where required. These loading rates normally produce a low BOD effluent after settlement. RBCs are particularly useful where a low head loss is demanded and being covered, where fly and odour nuisance precludes open percolating filters. The ‘package’ nature of the plant has led to their use for small communities or isolated factories with organic wastewater to treat, assuming there is power available. The plant occupies little room and as half is buried in the ground, and is visually unobtrusive.

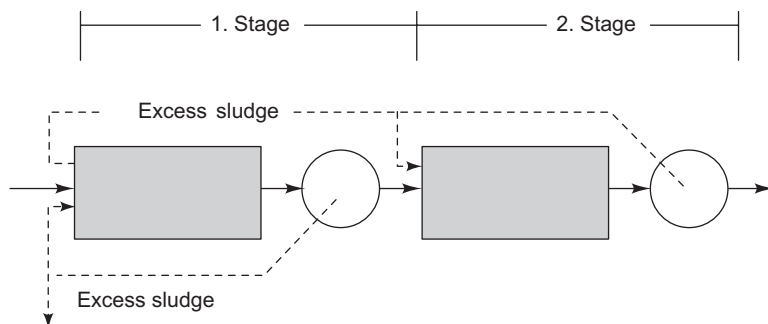
Operationally, RBCs tend to be less demanding than biofilters with rotating arms, having no dosing syphons or jets to block up and being mostly unaffected by rapid temperature fluctuation. Routine maintenance is confined to greasing, bearings and drive components. Problems with RBCs are normally centred on organic overloading. Excessive biofilm development takes place, bridging the media and generating anaerobic conditions. High shaft loadings occur as the media discs acquire a thick biofilm causing shaft and bearing collapse. After a long power failure, unequal biofilm build-up can cause the drive mechanism to stall and break the chain drive. Therefore, it is important to keep the soluble BOD load applied as low as possible by regular and thorough desludging of the settlement zones. A shear pin in the drive chain and loss-of-rotation sensing coupled to an alarm are other refinements. Time to time scrapping, collection and removal of biofilm biomass is also required to keep the RBC at operational stage.

### 12.7.4 Activated Sludge System

The first technical scale plant was a fill-and-draw activated sludge plant, which today is called the Sludge Blanket Reactor (SBR) process. An activated sludge plant is characterized by four elements: (a) An aeration tank equipped with appropriate aeration equipment, in which the biomass is mixed with wastewater and supplied with oxygen through aeration or oxygen gas under pressure, (b) A final clarifier, in which the biomass is removed from the treated wastewater by settling or other means, (c) Continuous collection of return sludge and pumping it

back into the aeration tank to maintain microbial density which facilitates efficient degradation of organic matters and (d) Withdrawal of excess sludge to maintain the appropriate concentration of mixed liquor (Fig. 12.13). Primary sedimentation is not required for activated sludge plants. However, for economic reasons, primary tanks are usually operated so as to minimize the volume of sludge for disposal.

Aeration in activated sludge system is achieved by diffused air by remotely sited, soundproofed air blowers supplying a large number of diffuser domes on the tank floor with air at sufficient pressure to overcome the head of liquid. Both fine and coarse bubble domes are available; the former are much more common. The domes are threaded into plastic supply pipe work for facilitating removal for cleaning after a period of time, usually about 5 years. Cleaning may require acid to remove scale and firing at 600–800°C to burn off organic deposits. There are also methods to clean out the grit and detritus that eluded preliminary treatment and that will have settled out, reducing treatment capacity.



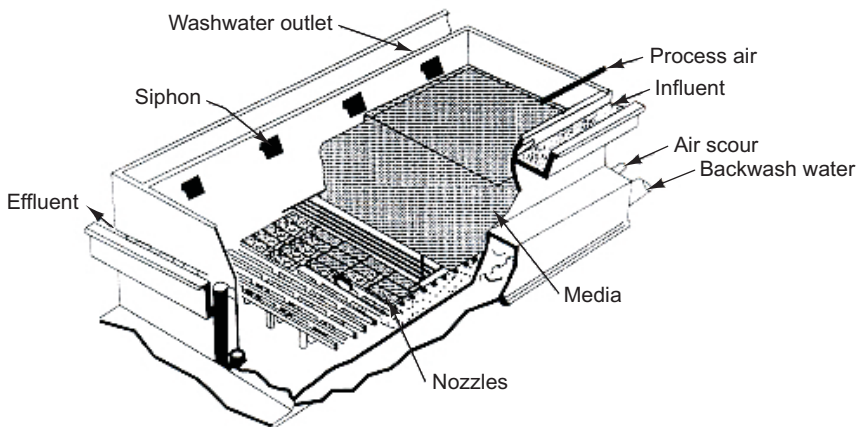
**Figure 12.13** - The flow diagram of a two-stage activated sludge treatment system.

Industrial wastewater may contain substances that precipitate on ceramic diffusers. They may also contain grease or substances that can destroy membrane material. Therefore, surface aeration or coarse bubble aeration with static mixers is a good choice. For deeper tanks, the rotating turbine aerator, which is a combination of coarse bubble aeration and a mixer to break up the large bubbles into smaller ones, has been used successfully. In a jet or ejector diffuser, mixed liquor is pumped through a venturi nozzle into which air is introduced. Very fine bubbles are released by the shear stress.

Aeration tanks are usually rectangular with evenly distributed diffusers along the tank. Wastewater and return sludge enter the tank at one end and mixed liquor leave the tank at the other end. Because of the high oxygen uptake rate, the dissolved oxygen concentration in the inlet zone is almost zero. To overcome this problem, the diffusers at the inlet zone are arranged in a higher density than at the outlet zone. This is called 'step aeration'. 'Step-feed', in which the return sludge is introduced at one end and wastewater is distributed along the tank, has been used in tanks having evenly distributed diffusers.

### 12.7.5 Submerged Aerated Filters (SAF)

The submerged or flooded aerated biological filter had seen more development in France and Britain. In this system, a combination of biological treatment and physical filtration takes place in the same reactor. Settled sewage percolates at a high rate down through a fixed bed of granular material, normally expanded shale in the size range 2–6 mm, produced by firing the natural material. Air is supplied counter-current via a grid of pipes and diffusers located 20–30 cm above the media base (Fig. 12.14). In addition, the whole media, particularly the non-aerated section under the diffusers, provides a physical filter for suspended solids. Effluent drains from the base and, in order to maintain, the filter in a flooded state is conveyed in rising pipe work to a washwater storage tank before discharge. There is no separate final settlement tank. Ancillary control, air blowers and pumps are housed in an ISO-sized container. The transportability of the units and the short installation time is a major advantage.



**Figure 12.14** - The cross-section of a biocarbon cell used as a SAF.

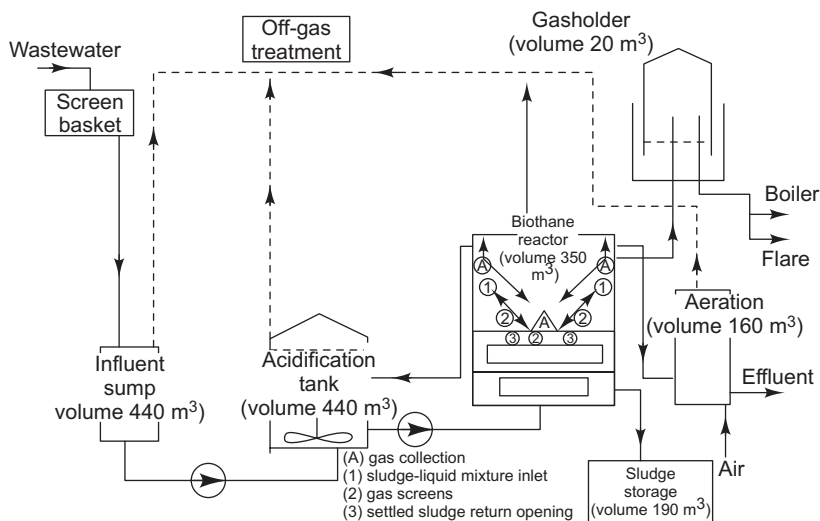
Backwashing is typically every 24 hours and is designed to remove surplus biomass. Duration and air scour rates depend on loading rates, physical characteristics of the medium and biomass growth rates and need setting after commissioning or pilot studies. Typical backwash liquor solids concentrations are 1–1.8 g/l and at sewage works installations returned to the works inlet for co-settlement with raw sludge. Biological performance is high; a 10 : 10 : 5 effluent can be reliably produced at loading rates of 2.5 kg BOD/ml media day, (six times that of conventional activated sludge) and 0.4 kg total nitrogen load/m<sup>3</sup> day. This equates to 85% of removals of BOD and SS and over 90% for NH<sub>3</sub>-N.

Power consumption is similar to a nitrifying activated sludge plant at 0.8 kWh/kg BOD and NH<sub>3</sub>-N removed. Hydraulic loading rates up to 100 m<sup>3</sup>/m<sup>2</sup> media surface day are quoted for industrial plant where applications in the USA and Japan include paper mill effluent, oil, milk products and pharmaceuticals. Sewage treatment plant installations have been hydraulically loaded at 20–80 m<sup>3</sup>/m<sup>2</sup> day. Odour and noise are low and the system can be installed enclosed in the basement

of buildings. Successful operation of the system has been observed in different European countries but its use in India is very rare.

### 12.7.6 Upward Flow Anaerobic Sludge Blanket System (UASB)

In its compactness and being a contained system, this process, developed for the sugar beet processing and related industry in the early 1970s, has similarities to SAF but operationally it is very different (Fig. 12.15) and requires pre- and post-treatment stages on either end of the main reaction vessel. By anaerobic digestion, carbonaceous oxidation of high strength wastewater is achieved with about a 75%–85% COD removal. The system is thus similar to a roughing filter but without the attendant problems of fly nuisance, odour and considerable open space requirements. Influent requirements are a warm, high strength wastewater (COD of  $> 1$  g/l) and pH control. Basic pretreatment after flow balancing is in common with anaerobic sludge digestion, the acidification vessel is heated to 30–35°C unless the effluent is very warm (this is a primary use for the biogas produced by the reactor). Other important criteria are: (a) pH is maintained in the range 6.8–7.5, (b) the COD : N : P ratio is kept at about 350 : 5 : 1 and (c) Trace elements are added if not present in the wastewater (Fe, Ni, Se, Mo, Cu, Mn, Cr and Co) to facilitate efficient growth of microbial consortium. This environment encourages acidifying bacteria to convert sugars to volatile fatty acids (acetic, propionic and butyric). Wastewater then passes to the bottom of the reactor and leaves via V-notch weirs at the top, having been separated by baffles from the methane and carbon dioxide gases produced by anaerobic digestion and the granular sludge. The bottom sludge bed is about 2 m deep and contains thick material (about 10% DM) while the top sludge blanket is about 2% DM.



**Figure 12.15** - The schematic layout of an UASB system meant for treating highly loaded industrial wastewaters.



Two distinct bacterial processes occur in the UASB. Acetogenic bacteria convert volatile acids produced in the upstream acidification tank to acetic acid, carbon dioxide and hydrogen, which the methanogenic bacteria then convert to carbon dioxide, methane and water. As the effluent is anaerobic, a post-treatment aeration tank is needed using forced air to strip out hydrogen sulphide. Excess gas can be flared off or used more productively to heat the influent; odorous waste gases are extracted and passed through a peat bed filter. Typical performance is quoted as up to 90% COD removal at organic loadings of 5–20 kg COD/m<sup>3</sup> day. Gas production of 0.35 m<sup>3</sup>/kg COD removed, retention times of 4–8 hours and a low sludge production of 0.04 kg/kg COD reduced are also the characteristics of the process.

## 12.8 MICROBIAL REMOVAL OF NITROGEN

Different forms of nitrogen and phosphorus represent the main source for eutrophication of surface water. For this reason they must be eliminated together with the organic carbon during wastewater treatment. Where phosphates form insoluble precipitates with many heavy metal ions and the precipitates can be separated by sedimentation or flotation, all nitrogen compounds, except for magnesium ammonium phosphate, are easily soluble in water and thus cannot be removed chemically by precipitation. For biological removal of amino nitrogen and of heterocyclic nitrogen compounds, their conversion to ammonia in an aerobic or anaerobic treatment process is the first step. Then ammonia must be nitrified and the nitrate denitrified to yield nitrogen. Thus, depending on the kind of nitrogen compounds present in wastewater, nitrogen removal requires up to three processes in sequence: ammonification, nitrification and denitrification.

The major portion of nitrogen compounds in municipal wastewater are reduced nitrogen compounds such as ammonia, urea, amines, amino acids and proteins. Oxidized nitrogen compounds such as nitrate and nitrite normally are not present at all or not in relevant amounts. Nitrate and nitrite may, however, represent the main nitrogen load in wastewater of certain food or metal industry branches.

The ammonia in raw municipal wastewater is mainly derived from urine and is formed in the sewer system by enzymatic cleavage of urea by ureases as given below:



Though there is possibility of release of ammonia from other sources like from proteolysis and deamination of amino acids, the residence time of the wastewater in the treatment system normally is not long enough for a significant contribution.

### 12.8.1 Ammonification

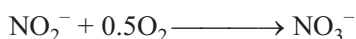
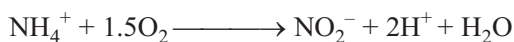
The main organic nitrogen compounds in municipal wastewater are heterocyclic compounds (like nucleic acids) and proteins. Proteolysis and degradation of amino acids leads to liberation of ammonia by various mechanisms of ammonification, including hydrolytic, oxidative, reductive and desaturative



deamination. A significant amount of ammonia from urea cleavage or from ammonification of amino acids is assimilated in aerobic treatment processes for growth of bacteria (surplus sludge formation). It can be estimated that bacteria consist of roughly 50% protein and that the nitrogen content of protein is about 16%. Thus, for synthesis of 1 g of bacterial biomass, about 80 mg of ammonia-N is required (Gallert and Winter, 2005). To eliminate ammonia that is not used for cell growth during wastewater treatment, it must first be nitrified and then denitrified to molecular nitrogen or anaerobically oxidized with nitrite.

## 12.8.2 Nitrification of Ammonia

**(a) Autotrophic nitrification:** Autotrophic nitrifiers are aerobic microorganisms oxidizing ammonia via nitrite to nitrate and the process can be given as the following:

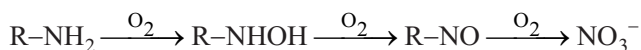


These are the energy-yielding processes which supports the autotrophic growth of the nitrifying bacteria but at the same time require a good amount of oxygen. The efficiency of the process increases with facilitated aeration, especially by streams of pure oxygen. The released energy is utilized for the assimilation of  $\text{CO}_2$  via the Calvin cycle. Since the positive redox potential of the oxidizable nitrogen compounds is not low enough to form  $\text{NADH}_2$  for  $\text{CO}_2$  reduction,  $\text{NADH}_2$  must be formed by an energy-consuming reverse electron transport. For this reason, the growth yield of nitrifying bacteria is low. For example, *Nitrosomonas* sp. must oxidize 30 g  $\text{NH}_3$  to form 1 g of cell dry mass (Sclegel, 1992).

Nitrifying bacteria belong to the genera *Nitrosomonas*, *Nitrosococcus*, *Nitrosolobus*, *Nitrospira* and *Nitrosovibrio*; organisms catalyzing nitrification include members of the genera *Nitrobacter*, *Nitrococcus* and *Nitrospira*. Due to their slow growth, autotrophic nitrifiers cannot successfully compete with heterotrophic bacteria for oxygen as the energy yield by heterotrophic bacteria is very high triggering high  $\text{O}_2$  consumption. In a highly loaded activated sludge system, the autotrophic nitrifiers are overgrown by the heterotrophic sludge flora, which consume the oxygen. Ammonia oxidation starts only if the  $\text{BOD}_5$  concentration in the wastewater is  $< 110$  mg/l. During nitrification of ammonia, the alkalinity of the wastewater increases slightly, due to  $\text{CO}_2$  consumption for autotrophic growth (pH increase), but in a counter reaction, it drastically decreases due to nitric acid formation from ammonia (pH decrease from above neutral to acidic). If the buffer capacity of the wastewater is weak, the pH drops far below 7 and thus prevents further nitrification by autotrophic nitrifiers.

**(b) Heterotrophic nitrification:** Some bacteria of the genera *Arthrobacter*, *Flavobacterium* and *Thiosphaera* are able to catalyze heterotrophic nitrification of nitrogen-containing organic substances. These microbes oxidize reduced nitrogen compounds, such as hydroxylamine and aliphatic and aromatic nitrogen containing compounds, but in contrast to autotrophic nitrification, no energy is

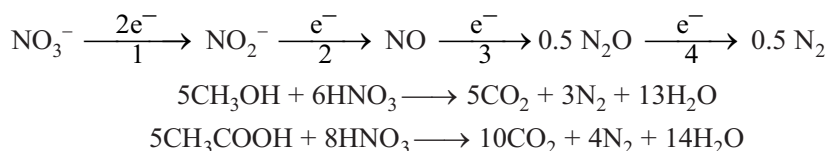
gained by nitrate formation. For this reason, an organic substrate must be respired to satisfy the energy requirement for growth and reproduction. Due to this reason, the oxygen requirement for nitrification by heterotrophic microbes is much higher than by the autotrophic nitrifiers. A general scheme of heterotrophic nitrification is given below:



Some heterotrophic nitrifiers can denitrify nitrate or nitrite under aerobic growth conditions. The nitrogen metabolism of *Thiosphaera pantotropha* and *Paracoccus denitrificans* are well documented examples (Stouthamer *et al.*, 1997). These organisms express a membrane-bound nitrate reductase under anoxic growth conditions, which works only in the absence of molecular oxygen. Under aerobic growth conditions, a periplasmic nitrate reductase is expressed, which catalyzes nitrate reduction at least to the state of nitrous oxide.

### 12.8.3 Denitrification

Many aerobic bacteria seem to be able to switch their oxidative metabolism to nitrate respiration. Similar to oxygen respiration, the nitrate respiration of heterotrophic bacteria requires a complex carbon source as an electron source for denitrification. For example, methyl alcohol and acetic acid are used as the electron source by heterotrophic bacteria. Denitrification starts with the reduction of nitrate to nitrite by membrane-bound nitrate reductase A (1). Then a membrane-bound nitrite reductase (2) catalyzes NO formation. Finally, NO reductase (3) and  $\text{N}_2\text{O}$  reductase (4) form  $\text{N}_2$ . The theoretical stoichiometry of denitrification with methanol or acetate as a carbon source is as follows:



For practical application, a surplus of carbon source must be supplied, since the wastewater is not free of oxygen and a part of the carbon source is respired until anoxic conditions are achieved.

To evaluate the stoichiometry of nitrate to organic compounds for denitrification with a complex carbon source, the oxidation-reduction state of the carbon substrates and the oxygen concentration in the wastewater should be known. In wastewater treatment plants, more than 2.85 g of COD is required for reduction of 1 g of nitrate-nitrogen (Bernet *et al.*, 1996). Except for dissimilatory nitrate reduction, many aerobic and anaerobic bacteria are capable of assimilatory nitrate reduction to supply the cells with ammonia for growth. However, the enzymes for nitrate assimilation are expressed only at concentrations of ammonia  $< 1$  mM.

Instead of nitrate, many denitrifying bacteria can use  $\text{NO}_2^-$ , NO or  $\text{N}_2\text{O}$  as terminal electron acceptors. Alternatively, they may release these intermediates during denitrification of nitrate under unfavourable conditions. If surplus nitrate

is supplied and hydrogen donors are not sufficiently available, NO and N<sub>2</sub>O can be formed. N<sub>2</sub>O formation also occurs if pH goes below 7.3, at which nitrogen oxidoreductase is inhibited.

#### 12.8.4 Combined Nitrification and Denitrification

A strict separation of the reactions participating in aerobic, autotrophic nitrification and in anoxic, heterotrophic denitrification is not required. Autotrophic ammonia oxidizers seem to be able to produce NO, N<sub>2</sub>O or N<sub>2</sub> from nitrite if oxygen is limited and ammonia as well as nitrite oxidizers can be isolated from anaerobic reactors. *Nitrosomonas europaea* can use nitrite as an electron acceptor and pyruvate as an energy source under anoxic, denitrifying growth conditions. In addition, several strains of *Nitrobacter* sp. have been reported to denitrify during anoxic heterotrophic growth.

*Thiosphaera pantotropa* respire molecular oxygen and denitrifies nitrate simultaneously, provided that suitable electron acceptors are available. The conversion rate of acetate as electron donor with nitrate as electron acceptor is twice of true aerobic state when the concentration of molecular oxygen is <30% of air saturation (Microaerobic state). Respiration, simultaneous nitrification and denitrification are observed in the presence of oxygen and ammonia. During heterotrophic denitrification by the bacterium, ammonia is first oxidized to hydroxylamine by an ammonia monooxygenase, with ubiquinone serving as electron donor. Hydroxylamine is subsequently oxidized to nitrite by a hydroxylamine oxidoreductase. During coupled nitrification-denitrification, three of the four electrons from the oxidation of hydroxylamine are used for reduction of nitrite to nitrogen and are not available to the electron transport reaction catalyzed by cytochrome oxidase. Regeneration of ubiquinone is mediated by electrons that are generated during oxidation of an organic substrate (heterotrophic nitrification) (Gallert and Winter, 2005).

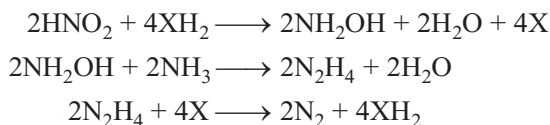
Conversion rates of ammonia by heterotrophic nitrifiers are smaller than those of autotrophic nitrifiers. If, however, the higher population density of heterotrophic is taken into consideration, the specific conversion rates are in a similar range. Heterotrophic nitrifiers can, in addition to their nitrifying capability, denitrify nitrite or nitrate to molecular nitrogen. In many wastewater treatment plants, autotrophic nitrifiers may exist, producing nitrite from ammonia under moderately aerobic conditions; the ammonia is then converted to nitrate and/or reduced to nitrogen by heterotrophic nitrifiers in the presence of a suitable carbon source. In practice, the only disadvantage of heterotrophic nitrification is more surplus sludge generation for final disposal as the carbon fixation as bacterial biomass is high under such condition.

#### 12.8.5 Anaerobic Ammonia Oxidation

The anaerobic oxidation of ammonium was once deemed biochemically impossible and non-existent in nature, but have now identified as a major pathway in global

nitrogen cycle. However, the microorganisms responsible for the process have not yet been grown in pure culture. Ammonia oxidation under anaerobic conditions proceeds through two major ways: (a) aerobic reduction of nitrite or nitrogen dioxide with hydroxylamine or ammonium as electron donor (done by *Nitrosomonas eutropha*) and (b) obligately anaerobic ammonium oxidation (Anammox). However, this conversion is still energetically weak supporting only the survival of the microorganism but is not enough to promote growth.

Under anaerobic conditions, ammonia is oxidized to nitrite in membrane-bound intracytoplasmic anammoxosomes mostly by two anaerobic ammonia-oxidizing bacteria *Brocardia anammoxidans* and *Kuenenia stuttgartiensis*. Nitrite is reduced by hydroxylamine oxidoreductase to hydroxylamine, which then reacts with ammonia to yield hydrazine ( $\text{N}_2\text{H}_4$ ), which, on oxidation to molecular nitrogen, generates four reducing equivalents required for nitrite reduction to hydroxylamine.



This process is suitable for nitrogen removal in ammonia rich effluents of anaerobic reactors that are fed with wastewater rich in TKN compounds. Nitrogenous compounds can be eliminated from this wastewater by a combination of nitrification for nitrite supply and anaerobic ammonia oxidation. The process is sensitive to high, toxic nitrite concentrations of 70–180 mg N/l (depending on the kind of biomass). Furthermore, due to the low growth rate of the ammonia oxidizing bacteria, a startup time of 100–150 days with activated sludge as inoculum seems to be necessary. In oxygen limited environments aerobic and anaerobic ammonia oxidizers are natural partners. The ammonia is oxidized to nitrite and concomitantly the oxygen level is decreased. The anammox bacteria disproportionate the nitrite and the remaining ammonia to  $\text{N}_2$ . When ammonia is limited, the affinities of aerobic and anaerobic ammonia oxidizers may lead the natural partners to change to competitors.

## 12.9 MICROBIAL REMOVAL OF PHOSPHORUS

Phosphate uptake and accumulation in microbes occurs to support the growth and to satisfy the phosphorus demand at the time of deficiency. The phosphorus accumulation more than that required for satisfying the growth demand is a common process in bacteria. Single cell accumulates polyphosphate in granules containing a few to several thousand phosphate units. Phosphate constitutes up to 12% of the cell weight of polyphosphate-accumulating bacteria, whereas bacteria that do not accumulate polyphosphate contain only 1%–3% phosphate. The accumulated polyphosphate is assumed to be an energy source for substrate assimilation during anaerobic growth conditions and poly- $\beta$ -hydroxyalkanoate (PHA) synthesis. Evidence suggests that degradation of polyphosphate may be

used to regulate intracellular pH under alkaline conditions. For biological phosphate removal by 'luxury phosphate uptake', the polyphosphate-accumulating bacteria in wastewater have to be subjected in sequence to an anaerobic and an aerobic environment. All previous attempts to isolate polyphosphate-accumulating bacteria (culture-dependent approach) led to recovery of *Acinetobacter* sp. which belongs to the  $\gamma$ -Proteobacteria and is the model organism for explanation of the polyphosphate accumulation mechanism. However, pure cultures of the polyphosphate-accumulating *Acinetobacter* sp. and wastewater communities seem to behave differently. The phosphate removal is more effective in the natural microbial assemblage of wastewater than by pure culture of *Acinetobacter*.

Polyphosphate accumulation by pure or mixed culture is interconnected with poly- $\beta$ -hydroxybutyrate (PHB) and glycogen metabolism. Under anaerobic conditions in the absence of nitrate, acetate is taken up by *Acinetobacter* cells, metabolized to  $\beta$ -hydroxybutyrate, polymerized to PHB and stored intracellularly in inclusion bodies. Reducing equivalents for PHB formation from acetate are made available by glycolysis of glucose units from stored glycogen or are alternatively formed by other microorganisms for enhanced biological phosphorus removal (EBPR) in an anaerobically operating TCA cycle. The energy for polymerization comes from hydrolysis of the anhydride bonds of ATP and polyphosphate. To supply both the maintenance metabolism and the storage metabolism of *Acinetobacter* with sufficient energy under anaerobic conditions, anhydride bonds in polyphosphate are hydrolyzed and inorganic phosphate is excreted (phosphate resolubilization). Stored glycogen in microorganisms for EBPR may also provide ATP for PHA synthesis. Under aerobic conditions, two situations may be prevalent. In wastewater that does not contain a suitable carbon source for respiration, *Acinetobacter* sp. hydrolyze PHB, which is stored in the granules and respire the  $\beta$ -hydroxybutyrate to gain energy for growth, maintenance, formation of glycogen and polymerization of phosphate taken up from the wastewater. In the presence of oxygen much more  $P_i$  is taken up by *Acinetobacter* sp. from the activated sludge flora than is released under anaerobic conditions ('luxury uptake').

In wastewater that contains suitable carbon sources for *Acinetobacter* sp., these carbon sources are partially oxidized by the TCA cycle and are partially used for PHB formation. Since reducing equivalents are needed for PHB formation, less ATPs are formed by oxidative phosphorylation. The lower ATP yield is compensated by the energy from hydrolysis of polyphosphate to  $P_i$ , which the cells use to form ADP from AMP. Two moles of ADP are disproportionated to AMP and ATP by adenylate kinase and little  $P_i$  is released into the medium.

Under the limitation of organic substrates nitrate-reducing bacteria, except *Acinetobacter*, compete with other microorganisms for EBPR. For example, most of the microbes compete for acetate as it is very easily assimilated. Nitrate-reducing bacteria also can eliminate phosphate from the aqueous environment. The simultaneous presence of obligately aerobic bacteria and nitrate-reducing bacteria

in activated sludge may explain why, in the presence of nitrate in the anaerobic phosphate resolubilization phase (oxygen absent),  $P_i$  does not accumulate in the medium. The denitrifying, polyphosphate-accumulating bacteria apparently do not inhibit  $P_i$  excretion by *Acinetobacter*, but they take up phosphate and accumulate it as polyphosphate. This explains why resolubilization of phosphate by degradation of stored polyphosphate is not always observed in wastewater treatment plants for biological phosphate removal.

Biological phosphate elimination is a highly effective process leading to final concentrations in the wastewater of  $< 0.1$  mg/l phosphate. For optimal biological phosphate removal, the COD/P ratio in the wastewater should be about 20 g COD/g phosphate to permit good growth of the polyphosphate-accumulating bacteria. If biological phosphate elimination is used in combination with chemical precipitation, the minimum COD/P ratio should be 2 g COD/g phosphate in processes with or without separation of primary sludge. If most of the particulate COD is separated in a primary sedimentation pond, ferrous ions should be added to supplement the biological phosphate removal with chemical precipitation. If no presedimentation of sludge is available or the sedimentation efficiency is low, no ferrous ions have to be added.

For biological phosphate accumulation, the ability of cells to store reserve material plays an essential role. Polymerization of substrates and storage in intracellular granules may offer the advantage that, at times of substrate shortage or in an environment with low concentrations of substrates, such organisms can survive much better than other heterotrophic bacteria that do not have effective strategies for substrate storage.

The enhanced biological phosphate removal has also some operation and stability problems. The occurrence of glycogen nonpolyphosphate-accumulating microorganisms (GAMs) in wastewater treatment system imposes a competition for the available volatile fatty acids under EBPR operation conditions (alternating anaerobic-aerobic phases during treatment). Similarly, pH also influences the performance of EBPR because at acidic pH, the growth of GAMs is favored. However, at a pH of 7–7.5, the microorganisms for EBPR outcompete the GAMs. Availability of acetate or enhanced growth of acidogenic microbes increases the efficiency of EBPR.

## 12.10 CONCLUSION

Wastewater treatment requires special attention to protect the surface waters from contamination. Many different treatment processes are available which are suitable to treat wastewater in both centralized and decentralized manner. The treatment system configuration and type can be determined by the wastewater type, production rate and the available space to make the treatment cost effective.

Establishment of active microbial biofilms in wastewater treatment systems is a prerequisite to improve the efficiency of organic matter removal from sewage. Novel systems are now available to provide surfaces for microbial colonization

and biofilm development so as to facilitate the degradation of specific chemicals. Further improvements are necessary to improve the aeration rate, organic matter distribution, reactor optimization and treatment and disposal of surplus sludge. Corrosion has always appeared as a permanent and unavoidable problem in trickling filters and sludge blanket reactors. This has limited the option of the use of concrete and metal surfaces for biofilm development. Researches on the development of non-corrosive but non-toxic materials will greatly aid to the improvement of treatment systems. Further improvement in the designing of treatment systems are needed to increase the efficiency of wastewater treatment and to make them light, potable and easy to install without massive construction.



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# MICROBES IN METAL POLLUTION CONTROL

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## 13.1 INTRODUCTION

Use of metals in domestic and industrial sectors has made the metal pollution a global issue. Anthropogenic emission of metals, especially of heavy metals at present, is approximately 100 times higher than their natural release. While the mineral deposits on the earth is decreasing day by day, the concentration of free and toxic forms of metals in all environments are showing a continuous increase. A general feature of these metal species is their well-known potential toxicity towards microbial and other life forms, which is the basis of many biocidal preparations. However, many 'trace metals' are essential for growth and metabolism in a living cell at low concentrations, e.g. Cu, Zn, Fe, Ni, Mo, Co and microorganisms possess mechanisms of varying specificity for intracellular metal accumulation from the external environment. In contrast, many other metals have no essential biological functions, e.g. Pb, Sn, Cd, Al, Hg, etc. but still may be accumulated in the biomass and are freely transferred from one organism to the other in the food chain, often resulting in bioaccumulation. These metals often impose toxicity on living organisms even at low concentrations.

Because of their toxic nature, metals are not as sensitive to biodegradation as organics. Unlike organics, metals are persistent in the environment and can not be degraded by biological, chemical or physical means. However, the microorganisms can modify the toxicity of metals by altering their bioavailability through oxidation and reduction. Thus, metals are difficult to remove from the environment but can be microbially nullified through appropriate processes. In addition, total metal concentration in an environment does not necessarily reflect the degree of biological metal toxicity, making it difficult to assess accurately the extent of risk posed by them.

Some metal-microbe interactions, where accumulation by the biomass leads to removal from the external environment, are of current biotechnological interest for the recovery of valuable metals and for detoxification of polluted effluents. An understanding of microbial responses towards heavy metals is also relevant to

the preservation and protection of natural and synthetic materials since inorganic and organometallic compounds are still widely used as microbiocidal agents in domestic and agricultural pest and pathogen control operations. Some microorganisms have mechanisms to immobilize metals whereas others actually enhance the metal bioavailability, through solubilisation, in the environment. This chapter explains the influence of metals on microbial processes and the methods for microbiological modification of metal toxicity.

## 13.2 ENVIRONMENTAL TOXICITY OF HEAVY METALS

In natural environments, average abundances of heavy metals are generally low and much of that sequestered in sediments, soils and mineral deposits may be biologically unavailable. Although elevated metal levels can occur in rather specific natural locations, e.g. deep-sea vents, hot springs and volcanic soils, it is mainly a result of industrial activities that ecosystems are increasingly subjected to heavy metal pollution (Table 13.1).

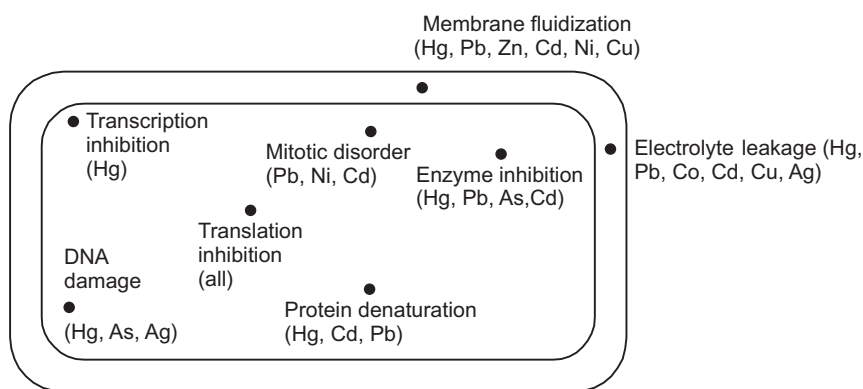
**Table 13.1:** The major natural and anthropogenic sources of heavy metals and organometallic compounds into the environment.

|                                     |                                      |
|-------------------------------------|--------------------------------------|
| Weathering processes                | Chlorine-alkali plants               |
| Deep-sea vents                      | Battery manufacture                  |
| Volcanic activities                 | Vehicular emissions                  |
| Microbial solubilization            | Nuclear waste                        |
| Metal transformations               | Sewage treatment effluents           |
| Atmospheric deposition              | Pigment manufacture                  |
| Mining and ore processing           | Seed dressing                        |
| Iron and steel production           | Animal feed additives                |
| Metal plating                       | Brewery and distillery wastes        |
| Combustion of fossil fuels          | Domestic wastes                      |
| Mine waters and leachates           | Agricultural fungicides              |
| Industrial effluents and discharges | Wood and stone preservatives         |
| Industrial accidents                | Biocides and disinfectants           |
| Fertilizer manufacture              | Pharmaceuticals                      |
| Sewage sludge fertilizers           | Industrial catalysts and stabilizers |
| Photographic waste                  | Antifouling paint biocides           |

Pollution of aquatic environments with metals of the copper and zinc triads became common knowledge in the 1950s following two incidents, both in Japan. The first resulted from the use of mercuric oxide as a catalyst in the production of acetaldehyde by a factory at Minimata Bay. Mercury discharged into the bay led to the subsequent accumulation of these metal ions in fishes and the appearance of nervous and other disorders in humans. The second resulted from cadmium containing mining effluent being released into the Jintsu river, which was used to irrigate downstream paddy fields. Contamination of many lakes, rivers and

estuaries with cadmium, mercury, zinc and/or copper, from mine drainage or industrial processes, has subsequently been documented in numerous cases. Copper pollution of aquatic environments has also resulted from its use in antifouling paint for ships' hulls (subsequently replaced by tin compounds), as an algicide and as a wood preservative. Municipal wastes often contain elevated levels of metals and discharge of municipal wastewater to the surface water lead to bioaccumulation of metals in the food chain.

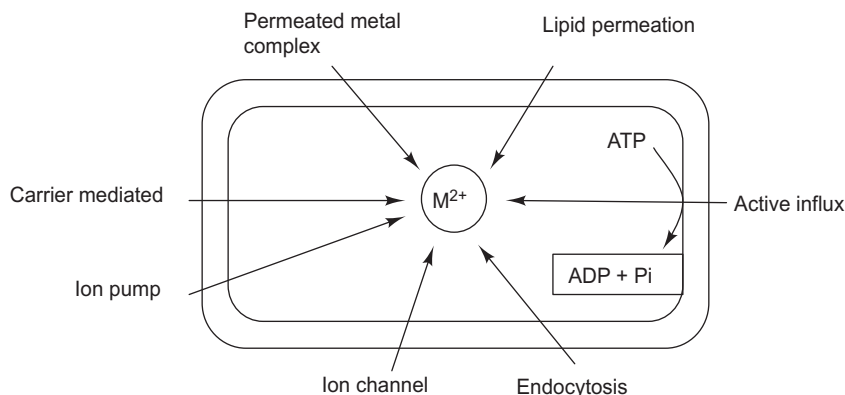
Heavy metals exert their harmful effects on microbes in many ways, although all the major mechanisms of toxicity are a consequence of the strong coordinating abilities of metal ions. The strong ionic nature of metals makes them able to bind to many cellular ligands and displace native essential metals from their normal binding sites (Fig. 13. 1). For example, arsenate can replace phosphate in the cell. Metals also disrupt proteins by binding to sulfhydryl groups and nucleic acids by binding to phosphate or hydroxyl groups. As a result, protein and DNA conformation are changed and function is disrupted. For example, cadmium competes with cellular zinc and nonspecifically binds to DNA, inducing single-strand breaks. Metals may also affect oxidative phosphorylation and membrane permeability, as seen with vanadate and mercury. Microorganisms generally use specific transport pathways to bring essential metals across the cell membrane into the cytoplasm. Toxic metals can also cross membranes, via diffusion or via pathways designed for other metals, i.e. by using ion channels, pumps and membrane carrier systems (Fig. 13.2).



**Figure 13.1** - The possible binding sites and various toxic influences of metals on a microbial cell (Modified from Roane and Pepper, 2000). The metals affecting these target sites are given in the parentheses.

For toxicity to be manifested, metals must enter microbial cells and transport systems of varying specificity may be utilized. In view of the wide variety of ligands found in living cells and the range of potentially toxic interactions that can occur, almost every index of microbial metabolism and activity in aquatic or terrestrial ecosystems can be dramatically reduced under conditions of metal pollution. These include primary productivity, nitrogen fixation, biogeochemical

cycling of C, N, S, P and other elements, litter decomposition, enzyme synthesis and activity in soils, sediments and waters. Because of the microbial involvement in such ecological processes, as well as in plant growth and productivity and symbiotic associations, it follows that heavy metal pollution can have serious and long-ranging effects, ultimately posing a threat to humans by bioaccumulation and transfer through food chains.



**Figure 13.2** - The mechanisms of metal ( $M^{2+}$ ) flux across the microbial cell membrane (Modified from Mohapatra, 2006).

The physico-chemical characteristics of a given ecosystem into which metals are deposited determine the form and the biological availability of heavy metals and therefore their toxicity. Such abiotic factors may include pH, oxidation-reduction potential ( $E'_n$ ), inorganic anions and cations, particulate and soluble organic matters, minerals, hydrous metal oxides, salinity, temperature and hydrostatic pressure. Where such factors decrease biologically available concentrations of metals, toxicity is often reduced. In soils, the availability of metals can fluctuate both spatially and temporally and these changes can operate over relatively short distances (adjacent to, or distant from, particles) and timescales (for example with rapid changes in soil moisture content). It is, of course, the potential for movement to micro niches where metal availability and the potential for swift fluctuations in metal availability that select for metal perception systems changes.

Binding interactions with organic environmental components, e.g. humic and fulvic acids, proteins, organic chelating agents, etc., can remove metals from solution or form complexes or chelates, which may not be as readily accumulated by microorganisms as free cations. A decreased pH may increase metal availability, whereas towards and above neutrality, insoluble oxides, hydroxides and carbonates may form. In addition, the pH may affect other aspects of cell physiology, e.g. nutrient transport and also the rate and extent of complexation to organic and inorganic components. Thus, pH effects may be complex and, depending on the particular system examined, toxicity may increase, decrease or be unaffected. However, in many cases, toxicity is reduced at acidic pH values,

despite the often increased metal availability and several extreme examples of microbial metal tolerance are confined to acidic conditions. Anions such as  $\text{CO}_3^{2-}$ ,  $\text{S}^{2-}$ ,  $\text{PO}_4^{3-}$ , etc., can reduce metal availability by precipitation of corresponding carbonates, sulphides and phosphates.

The toxic effects of metals can be seen on a community level as well. In response to metal toxicity, overall community numbers and diversity decrease. However, few studies have addressed community resistance. While individual microbial populations may be metal resistant, it is interesting to know the interaction of microbial populations in metal contaminated habitat. As a general response metal resistant populations interact in a way to provide resistance on the consortium of organisms. Likewise, there are also symbiotic relationships between metal resistant and metal sensitive populations, more specifically in an active biofilm, such that the metal sensitive organism receives protection from metal toxicity while providing the metal resistant organism with some essential nutrient or carbon source aimed towards syntrophic metabolism.

### 13.3 MICROBIAL DIVERSITY IN METAL CONTAMINATED HABITAT

Unlike 'extreme' environments characterized by a relatively specific, restricted microflora, a wide range of organisms from all the major groups may be found in metal-polluted habitats and the ability to survive and grow in the presence of potentially toxic metal concentrations is frequently encountered. Thus, generalizations about a so-called 'specific microflora' of metal contaminated habitats cannot be made with any certainty. In general terms, heavy metals and toxic metals, are believed to affect microbial populations in nature by reducing abundance and species diversity and selecting a resistant population to continue in the habitat. In most of the metal contaminated environments, there are some dominant species of microbes, which successfully tolerate the metal stress and grow more or less normally or even better due to easy availability of resources caused by reduced competition. Most of the species are, however, less tolerant and these only remain viable in the contaminated habitat growing extremely slowly (generation time in weeks to months).

The effect of heavy metals on microbial abundance in natural habitats varies with the metals, microorganisms and the physico-chemical attributes of that environment. General reductions in the numbers of bacteria, including actinomycetes and fungi, have often been recorded in soils polluted with Cu, Cd, Pb, As, Hg and Zn. Along a steep gradient of Cu and Zn in soil towards a brass mill, fungal biomass decreased by about 75% whereas the total number of fungi in a glucose-supplemented soil was reduced by the addition of Cd or Zn, the former metal having the greatest toxic effect. Similar reductions in numbers of aquatic cyanobacteria and algae in response to metal pollution have also been noted. However, numerical estimates, if viewed in isolation, provide no indication as to whether the microbial populations have changed in any way as a result of metal

exposure. For example, a Zn- and Cd-polluted soil located near a smelter had lower numbers of actinomycetes and fungi but bacteria as compared with unpolluted control soil. However, the latter contained greater numbers of *Nitrosomonas* sp. Combinations of Cd and Zn reduced the number of bacteria and actinomycetes to a greater extent than number of fungi in a glucose supplemented soil. In a study where Cu was added to a simulated marine ecosystem, an increase in heterotrophic bacteria occurred, which was thought to be the result of bacteria that survived the Cu exposure using the organic substrates released from Cu-sensitive organisms. Thus, numerical estimates may provide little meaningful information without consideration of changes in microbial groups and species and it seems evident that such changes can occur in response to metal exposure.

As in other so-called extreme environments, there can be a reduction in species diversity but this does not appear as a general effect of metal pollution. In relation to pollution of leaf surfaces by aerially deposited Pb, Cd or Zn, bacteria appear more sensitive than fungi and on polluted oak leaves, *Aureobasidium pullulans* and *Cladosporium* sp. are the most numerous organisms. In fact, *A. pullulans* show a good positive correlation with lead concentrations, whether derived from industrial or vehicular sources and it often becomes the dominant organism, in some cases comprising up to 97% of the leaf surface population on a numerical basis. In contrast, the ballistospore producing yeast *Sporobolomyces roseus* and heterotrophic bacteria are very low or absent from polluted samples and numbers of *S. roseus* show a significant negative correlation with increasing lead concentrations. In Cu- and Zn-polluted soil, the fungi *Geomyces* and *Paecilomyces* sp. and some sterile forms increase with increasing pollution, whereas *Penicillium* and *Oidiodendron* sp. decline at polluted sites. *Trichocladium asperum*, *Trichoderma hamatum*, *Zygorrhynchus moelleri* and *Chrysosporium pannorum* have been isolated more frequently from an organomercurial-treated golf green than from untreated locations, whereas *Chaetomium*, *Fusarium*, *Penicillium* and *Paecilomyces* sp. have been found present in greatly reduced number. Several studies have indicated sensitivity of *Penicillium* sp. towards heavy metals. However, some of the best examples of microbial metal tolerance are found in the genus *Penicillium*, which underlines the fact that metal responses may be species and strain specific. *Penicillium ochrochloron* can grow in saturated  $\text{CuSO}_4$  and is frequently isolated from industrial effluents, whereas *Penicillium lilacinum* comprised 23% of all fungi isolated from soil polluted by mine drainage

Heavy metals can also affect the composition of cyanobacterial, algal and protozoan populations. Applications of Cu favoured coccoid, colonial cyanobacteria and inhibited *Anabaena flos-aquae*. Numerous studies have documented growth, morphological, physiological and biochemical responses of different cyanobacteria to supra-optimal concentrations of metals of the copper and zinc triads as well as many other non-essential heavy metals. Morphological effects of cadmium and elevated concentrations of zinc include inhibition of

daughter cell separation, production of filaments in normally unicellular strains (such as *Synechococcus* PCC 6301) and cell elongation. In contrast, elevated copper has been shown to cause production of sub-spherical rather than coccoid *Synechococcus* PCC 6301. Elevated concentrations of metals influence pigment composition, for example, with increased carotenoid to chlorophyll ratios and overall decrease in phycocyanin, chlorophyll *a* and total protein in *Synechococcus* PCC 6301 exposed to elevated levels of zinc. An analysis of the effects of elevated concentrations of copper on *Nostoc calcicola* revealed that photosynthesis was readily inhibited. Polyphosphate bodies are implicated in the sequestration of metals of the copper and zinc triads and its increase in response to elevated concentrations of such metals.

At some metal polluted aquatic sites, for example zinc-enriched waters of high pH, cyanobacteria are abundant and can be the dominant organisms. Some strains of cyanobacteria isolated from metal polluted sites are resistant to elevated metal concentrations in subsequent axenic culture. Strains have also been selected in laboratory studies by step-wise adaptation of cultures to gradually increasing metal concentrations. Enhanced resistance in such a copper resistant *Nostoc calcicola* was correlated with enhanced copper efflux while enhanced cadmium resistance in *Synechococcus* PCC 6301 was correlated with enhanced endogenous metal chelation mediated by increased expression of metallothionein.

For eukaryotic algae, Cu appears to reduce numbers of centric diatoms, except *Skeletonema costatum* and eliminate dinoflagellates. However, other flagellates increase and pennate diatoms, particularly *Amphiphora paludosa* becomes dominant species of the population when the surface water contains an elevated concentration of copper. In metal-contaminated aquatic sediment, metal toxicity is dependent on the bioavailable species of the metal and the toxicity of metal is reflected through reduction of the total algal diversity due to elimination of sensitive species. Many studies have shown *Ankistrodesmus falcatus*, *Botryococcus braunii*, *Chlorella pyrenoidosa*, *Raphidium* sp. and *Schroederia setigera* to be more abundant in metal-contaminated waters. Presence of assimilable carbon sources like glucose and organic acids has provided tolerance to many green algae to heavy metals like Zn, Hg, Cd and Ni. Sublethal concentration of mercury could become ineffective to *Chlorella vulgaris* and *Chlorococcum infusionum* at elevated concentrations of glucose, glutamate and 2-oxoglutarate.

Among protozoa and pigmented flagellates, *Peranema* and *Euglena* sp. are generally quite resistant to metal toxicants although, in comparison with walled-microorganisms, protozoa are of high sensitivity. As with other organisms, changes in species composition, numbers and diversity occur with heavy metal exposure but there is a considerable variation in recorded effects. Thus, it seems that elevated concentrations of heavy metals can affect the qualitative and quantitative composition of microbial populations although, as mentioned, it may be difficult to separate heavy metal effects from those of other environmental components. For example, phytoplankton microflora may be subject to the influence of other potential toxicants, e.g. SO<sub>2</sub> particularly in industrial locations and these may contribute to



observed effects. In addition, there are well-known theoretical and practical difficulties in obtaining meaningful assessments of microbial numbers and diversity in natural habitats, which should not be ignored. Nevertheless, it is generally accepted that heavy metal induced changes in species composition and a commonly associated assumption is that such changes lead to the establishment of a 'resistant' or 'tolerant' population. However, there are ample evidences of the loss of resistance on removal of the metal stress from the habitat.

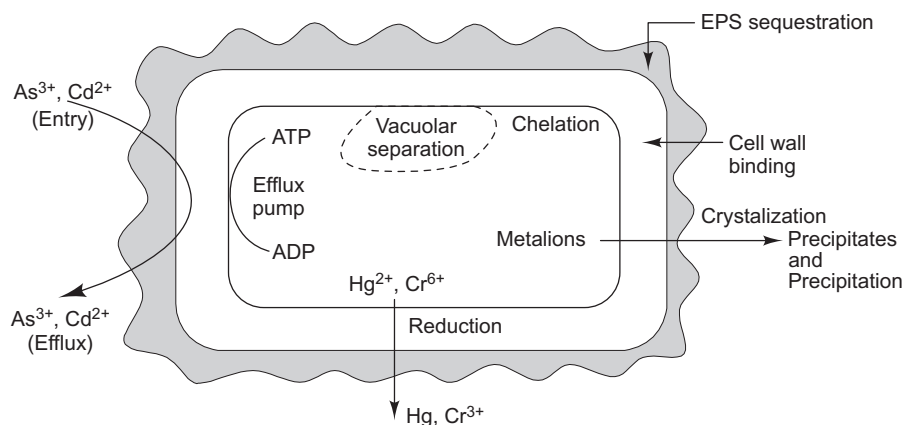
### 13.4 HEAVY METAL TOLERANCE

The ability of a microorganism to survive and grow in a metal-contaminated habitat may depend on genetical and/or physiological adaptation. In a given population, both may occur to varying extents, e.g. when microorganisms are equipped, physiologically or genetically, to tolerate increasing concentrations of heavy metals in laboratory media by repeated subculture on metal-containing media, there may be physiological changes, which may be unstable and also the selection of genetically stable variants from the original population. While physiological changes in the cells of the microbes are meant to reduce the rate of metal uptake and reduction of intracellular metal toxicity, the genetic changes cause both intracellular and extracellular reduction of the concentrations of toxic metal species. In many cases, there has been adequate consideration of toxicity modification by environmental factors or media components, which may lead to the isolation and identification of apparently resistant isolates. Nevertheless, selection and isolation of metal-resistant bacteria, algae and fungi (some mentioned previously) has been described from heavy metal polluted soils and waters and such resistance has been attributed to the prolonged exposure of the species to a concentration of the metal, which is considered toxic to those which are new to the contaminated habitat.

The ability to grow in the presence of relatively high metal concentrations is found in a wide range of microbial groups and species, including those from unpolluted sites and not in all cases, if any adaptation is necessary. Tolerance may result from intrinsic properties of the organism, e.g. the possession of extracellular mucilage or polysaccharide or an impermeable cell wall. In addition, environmental components may have considerable influence on toxicity and therefore apparent resistance. A good example is provided by those fungi that can grow in saturated  $\text{CuSO}_4$  (about 1.3 M) and very high concentrations of other heavy metals. Such solutions are very acidic and these organisms are sensitive to sub-millimolar levels at pH values around neutrality.

In response to metal toxicity, many microorganisms have developed unique mechanisms to tolerate and modify harmful metals. These mechanisms of resistance may be intracellular or extracellular and may be specific to a particular metal or a general one for a variety of metals. The important tolerance mechanisms are—(a) intracellular accumulation, (b) cell wall associated metal interaction, (c) extracellular immobilization of metals, (d) extracellular polymer-

metal interactions and (e) transformation and volatilization of metals. Fig. 13.3 represents a general scheme of metal tolerance in a bacterial cell.



**Figure 13.3** - Various metal detoxification pathways in a microbial cell. The figure shows the most common mechanisms in a prokaryotic cell.

### 13.4.1 Intracellular Accumulation

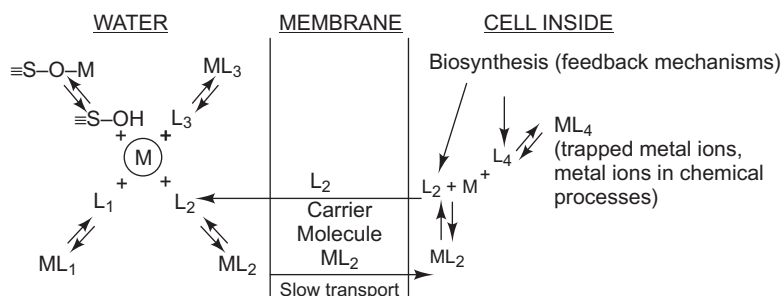
Bacterial assimilation of metals may be important in detoxification, enzyme function and physical characteristics of the cell. Potential mechanisms of toxic metal flux across a membrane are shown in Fig. 13.4. Extracellular or cell wall-attached ligands are thought to bind toxic metals. These ligands transport the complexed metals through the cell wall in a slow transport step. The metals are released inside the cell, incorporated into biochemical pathways, or trapped in an inactive form by complexation with another high-affinity ligand. Cadmium is also accumulated by a large number of organisms through the intracellular production of metal-ligand complexes. Research on *Citrobacter* suggests that it is accumulated as cell-bound cadmium phosphate. This is presumably a detoxification mechanism and is similar to the accumulation of lead as  $\text{PbHPO}_4$  by a different *Citrobacter* species. Fate and transport of the metal is, therefore, directly related to the fate and transport of the bacterial cell. When the cell dies, the metal is presumably released for further complexation. After the death of the bacterial cell, the aggregates and polymers produced by the microbial activity often produce complexes with the metals species released from the dead cells.

A number of metals are enzymatically altered intracellularly. This does not usually result in accumulation. Rather, transformation occurs to a less toxic and, in certain cases, volatile form. This will be discussed further in the next section (Section 13.5). The ability of certain bacterial cells to accumulate metals intracellularly has been exploited in mining practices, particularly in management of effluent treatment lagoons. Uranium has been shown to accumulate rapidly in cells of fungus *Saccharomyces cerevisiae*, the bacterium *Pseudomonas aeruginosa*

and many other species of bacteria and the process is being effectively used for mining of uranium from low grade deposits (See section 13.7).

### 13.4.2 Cell Wall Interactions

Binding of metals to the cell surfaces plays a dominant role in their distribution in natural waters. In addition, sorption of metals to living or dead cells is considered a practical solution to many metal contamination problems. Algal surfaces contain functional groups that have been shown to bind metals competitively with many dissolved ligands. It is suggested that carboxylic, amino, thio, hydroxo and hydroxy-carboxylic groups on the surface of phytoplankton cells interact coordinatively with metal ions and are responsible for complexation of many active metal species on the cell surface.



**Figure 13.4** - Model showing the intracellular metal accumulation through the formation of complexes of metals (M) with ligands (L).  $L_1$ ,  $L_2$  and  $L_3$  are the ligands produced by microbial cells and are released out through exudation (Adopted from Ford and Mitchell, 1992).

Bacteria possess lipopolysaccharides (LPS) in their outer membranes. These chemicals are extremely complex, consisting of a hydrophobic, phosphorylated section, known as *lipid A*; a core oligosaccharide; and variable O-specific side chains consisting of a number of unusual sugars. The side chains project out from the cell membrane and contain different functional groups capable of binding metals. Phosphoryl groups of LPS and phospholipids are the most abundant electronegative sites available for metal binding. The polyvalent toxic metals are primarily bound to LPS molecules because of the presence of closely opposed reactive sites. It has been suggested that this may provide a mechanism to immobilize toxic metals and prevent their entry into the cell thus protecting many other organisms from metal contamination.

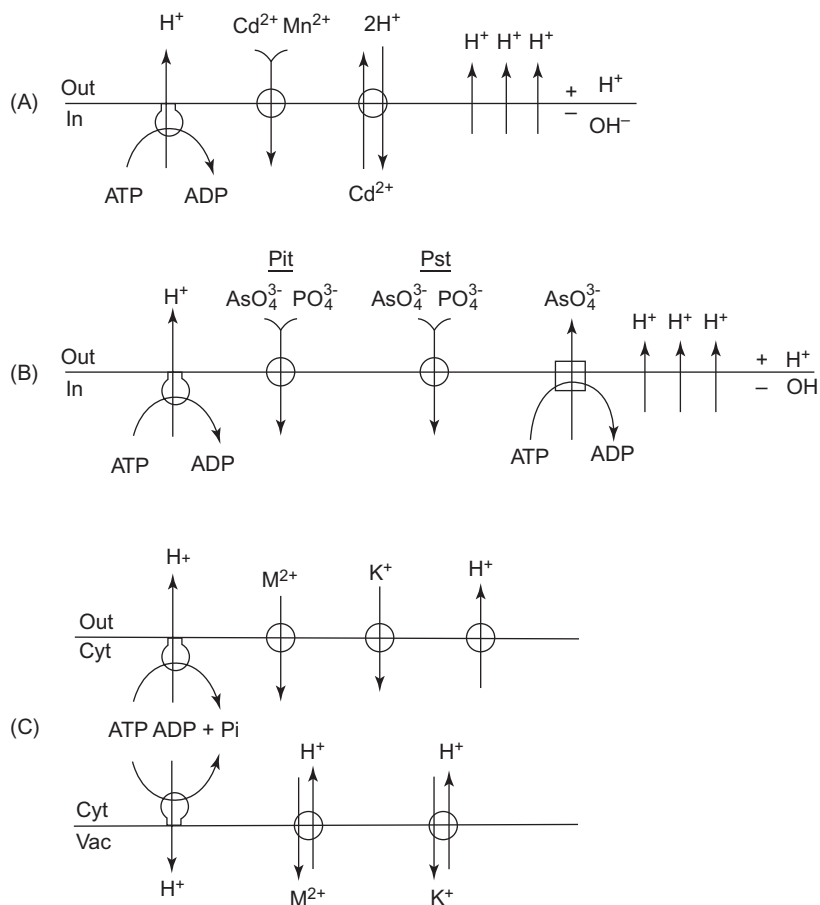
Many bacteria produce exopolysaccharides (EPS), some of which are induced by the transduction of the signal of metal toxicity and accumulate them on the cell surface. Metal binding to EPS results in metal immobilization and prevention of metal entry into the cell. For example, the immobilization of lead by exopolymers has been demonstrated in several bacterial strains, including *Staphylococcus aureus*, *Micrococcus luteus* and *Azotobacter* spp. A second type of extracellular

molecules produced microbially, that complex metals, are the siderophores. The latter are iron-complexing, low molecular weight organic compounds. Their biological function is to concentrate iron in environments where concentration is low and to facilitate its transport into the cell. Siderophores may interact with other metals that have chemistry similar to that of iron, such as aluminum, gallium and chromium (which form trivalent ions similar in size to iron). By binding metals, siderophores can reduce metal bioavailability and thereby metal toxicity. For example, siderophore complexation reduces copper toxicity in cyanobacteria.

### 13.4.3 Transport Related Metal Resistance

Metabolism-dependent intracellular transport may be a slower process than binding and is inhibited or halted by metabolic inhibitors, low temperatures, the absence of an energy source (e.g. glucose in heterotrophs, light in phototrophs) and uncouplers. In several algae, bacteria and yeasts, amounts accumulated by transport may greatly exceed amounts taken up by general biosorption. Rates of transport are also affected by the metabolic state of the cell and the nature and composition of the growth medium. Integral to heavy metal transport systems are ionic gradients across the cell membrane, e.g.  $H^+$  and/or  $K^+$  together generating the electrochemical gradient across the cell membrane and the membrane potential.

A relationship between heavy metal transport into microbial cells and toxicity is often observed, with sensitive strains taking up more metal ions than resistant strains. When transport rates do not vary, the sensitive organisms have more intracellular toxic metal species than the resistant one. It is, therefore, not surprising that decreased transport, impermeability or the occurrence of metal efflux systems constitute resistance mechanisms in many organisms. The general models of the transport of metals into the cells have been presented in Fig. 13.5. In several bacteria, including *Staphylococcus aureus* and *Bacillus subtilis*, active transport of  $Cd^{2+}$  depends on the transmembrane electrical potential and this uptake system is highly specific for  $Mn^{2+}$  as well as for  $Cd^{2+}$ . Resistance may arise from reduced  $Cd^{2+}$  transport in Gram-negative bacteria. The efflux of cadmium from the cells of actively growing *S. aureus* has been found to be an energy consuming process involving cell membrane ATPase (Fig. 13.5A). This  $Cd^{2+}/2H^+$  antiport is present in resistant strains, which results in a net reduction in  $Cd^{2+}$  uptake. Cadmium-resistant *Pseudomonas putida* strains also exhibit reduced uptake and this may also be a result of defective transport and/or the presence of a  $Cd^{2+}$  efflux system. In an *Alcaligenes* sp., adaptation to  $Cd^{2+}$  resistance has been found associated with the synthesis of a new membrane protein, which may be involved in the prevention of  $Cd^{2+}$  influx or the cause of  $Cd^{2+}$  efflux. Membrane proteins that reduce bacterial uptake of  $Hg^{2+}$  have also been described. Some  $Cu^{2+}$  resistant strains of *E. coli* exhibit reduced uptake and/or exclude  $Cu^{2+}$  due to the disappearance of outer membrane proteins, which may be involved in  $Cu^{2+}$  transport.



**Figure 13.5** - The models of metal efflux systems in bacterial cells. (A) Cd efflux system of *Staphylococcus aureus*; (B) Arsanate efflux in *E. coli* and (C) Vacuolar metal accumulation in tolerant bacterial strains (Adopted from Gadd, 1990).

The mechanism of arsenate resistance in *S. aureus* and *E. coli* is reduced accumulation by resistant strains, which is a result of an energy-dependent efflux mechanism. The efflux of arsenate from *E. coli* cells utilizes the phosphate transport system at the cell membrane and has been found to be energy dependent, the actively growing cells being able to effect more rapid metal efflux than the inactive or resource limited cells (Fig. 13.5B). Arsenate functions as a phosphate analogue and is generally accumulated by phosphate transport systems. Thus, as with  $\text{Cd}^{2+}$ , which can enter certain bacteria via a  $\text{Mn}^{2+}$  transport system, toxic anions and cations may be accumulated by means of existing transport systems for essential nutrients. However, energy-dependent efflux mechanisms, such as those for  $\text{Cd}^{2+}$  and arsenate, must be highly specific for the toxic anion or cation to prevent detrimental loss of the essential nutrient.

In *S. cerevisiae*, a connection between toxicity and energy-dependent intracellular uptake has been shown for a variety of metals including Cd, Cu, Co and Zn with decreased influx occurring in tolerant strains. Although such reduced uptake is a feature encountered in many strains of bacteria, fungi and algae, there are several examples where resistant strains take up more metal than sensitive strains, e.g.  $\text{Cu}^{2+}$ - and  $\text{Ni}^{2+}$ -tolerant strains of *Scenedesmus* sp. and  $\text{Mn}^{2+}$ -resistant strains of *S. cerevisiae*. In these strains, a significant part of the intracellular metal ions is partitioned inside the vacuoles. A metal-proton antiport removes the free metal species from the cytosol by pumping them into the vacuoles (Fig. 13.5C).

Impermeability has already been mentioned as a result of cell wall complexation, extracellular precipitation, etc. and these may be of benefit in reducing or preventing cellular entry of toxic metal ions. In the  $\text{Cu}^{2+}$ -tolerant fungus *Penicillium ochrochloron*, growth in high  $\text{Cu}^{2+}$  concentrations induced the synthesis of high concentrations of intracellular glycerol, which probably assisted in cellular exclusion of  $\text{Cu}^{2+}$ . Since a decreased influx can often determine tolerance, it follows that abiotic factors that reduce transport may also enable microbial growth and survival, e.g. ion competition and pH.  $\text{Ca}^{2+}$  can reduce  $\text{Cd}^{2+}$  toxicity to yeast by depressing rates of influx simply through the competitive reduction of Cd availability to the membrane bound carrier proteins.

### 13.5 MICROBIAL TRANSFORMATION OF METALS

Microbial transformation of toxic metal ions to less toxic or non-toxic forms is the major mechanism of their toxicity reduction in the environment. Intracellular and extracellular chelation, crystallisation, immobilization and phytochelatin mediated intracellular complexation are the major processes of toxicity reduction of metals in the environment. In addition, the metal toxicity is also reduced by conversion of bioavailable metal complexes to more mobile and highly volatile forms.

The most studied environmentally mediated transformation of toxic metals is the methylation of mercury. Pure culture experiments have shown that many bacteria and fungi, including *Clostridium*, *Neurospora*, *Pseudomonas*, *Bacillus*, *Mycobacterium*, *E. coli*, *Aerobacter aerogenes*, *Bacillus megaterium* and *Aspergillus* have the capability to methylate mercury. However, the sulphate-reducing bacteria (SRB) are the most significant and this is supported by the observations that mercury methylation is more intense in sulphate enriched habitat. However, above a certain concentration of sulphate, methylation is inhibited because of the rapid formation of the insoluble mercuric sulphide, a process that helps in immobilization of the heavy metal ions. Acidification, to certain extent, stimulates SRB activity and hence mercury methylation. The importance of SRB in mercury methylation within a particular aquatic system depends to a large extent on concentrations of methylmercury that enter from the surrounding catchment and its precipitation.

The mechanism of mercury methylation has been elucidated in detail and supported by organic matter availability. The metal is methylated intracellularly by nonenzymatic transfer of methyl groups from methyl cobalamin (vitamin B12). The methylation process is considered as a metal mobilization process as it not only increases the mercury toxicity but also the bioavailability of the metal. However, very little is known about microbial mobilization of other metals under environmentally realistic conditions. Probably the closest approximation has been the mobilization of cadmium in soil columns. Microbial mobilization of manganese by bacterial exopolymers in soil has also been reported.

Volatilization of other metals by microbial methylation has received less attention, as have other forms of transformation. The rates of other transformations, *e.g.*, ethylation and phenylation, are unlikely to be significant because of the large size of the organic groups. Methylation has, however, been shown for arsenic, lead, selenium, tellurium, tin, thallium and antimony. As a result of the use of organotins in anti-fouling paints, there is some information about biomethylation of tin by estuarine bacteria. The density of sulphate-reducing and sulphide-oxidizing bacteria significantly influences the methylation of tin in aquatic habitats. In addition, *Desulfovibrio* present in sediment also methylate tin in culture medium at similar rates to natural methylation at sediment-water interfaces.

### 13.5.1 Extracellular Complexation

Many bacteria produce large amounts of extracellular polysaccharides that have anionic properties and thus function as efficient biosorbents for metal ions. Interaction with metal ions is generally considered a direct consequence of the presence of negatively charged functional groups on these exopolymers. These groups include pyruvate, phosphate, hydroxyl, succinyl and uronic acids. A pH-dependent binding of positively charged ions can rapidly occur, with stability constants in excess of those generally measured for humic substances and other naturally occurring ligands. Such extracellular polymers, *e.g.* those produced by *Zoogloea* sp., are strongly involved in metal removal from sewage treatment processes and the extraction and removal of polymers from these and other bacterial cultures can greatly reduce biosorption capacities and also increase metal sensitivity of the concerned organism. The extracellular matrices act as an efficient barrier and prevent significant entry of metal ions into the cell. Similar interactions are also likely for those cyanobacteria, algae and fungi that produce extracellular polymers. Many cyanobacteria like *Anabaena flos-aquae*, *Microcystis* spp., *Synechocystis* spp. and *Aphanizomenon* spp. as well as green algae like species of *Chlorella*, *Ankistrodesmus*, *Oedogonium*, *Spirogyra* and Diatoms cause extracellular complexation of heavy metals in nutrient media.

Many organic metabolites can be important in metal detoxification because of their chelating or complexation properties. Organic or inorganic acids produced by microorganisms including *Thiobacillus*, *Serratia*, *Pseudomonas*, *Bacillus*, *Penicillium*, *Aspergillus*, *Anabaena*, *Microcystis* and *Aphanizomenon* are able to



extract metals from solid substrates. Citric acid is an efficient metal chelator whereas oxalic acid can precipitate metals as insoluble oxalates around cell walls and in the external medium. For example, citric acid production by a *Penicillium* spp. has been used to extract zinc selectively from industrial waste. Oxalic acid producing fungi often exhibit marked metal tolerance and this detoxification mechanism is often found in wood-rotting fungi, particularly those exposed to copper-chrome-arsenate food preservatives, e.g. *Poria* spp., *Aspergillus niger*, *Penicillium spinulosum* and *Verticillium psalliotae*.

### 13.5.2 Extracellular Precipitation and Crystalization

Certain organisms, particularly sulphate-reducing bacteria, e.g. *Desulfovibrio*, are involved in the formation of sulphide deposits, which contain large amounts of metals. Sulphide formation thus leads to metal removal from solution and this is associated with resistance in a variety of microbes. Metal-resistant strains of *Klebsiella aerogenes* precipitate Pb, Hg or Cd as insoluble sulphide granules on outer surfaces of cells and particles of  $\text{Ag}_2\text{S}$  are deposited on thiobacilli when grown in silver-containing sulphide leaching systems. Strains of the green alga *Gyanidium caldarium* can grow in acidic waters at 45°C containing high concentrations of metal ions. Iron, copper, nickel, aluminium and chromium can be removed from solution by precipitation at cell surfaces as metal sulphides. Cells can contain up to 20% metal on a dry weight basis. Yeasts can also precipitate metals as sulphides in and around cell walls and colonies may appear dark brown in the presence of copper.

Many other examples of crystallization and precipitation on microbial surfaces are known and some of these may represent resistance mechanisms. Microbes are implicated in the formation of ferromanganese nodules on ocean floors and several bacteria, e.g. *Hyphomicrobium*, algae and fungi promote  $\text{Mn}^{2+}$  oxidation in a variety of habitats and can become encrusted with manganic oxides. Other bacteria can become encrusted with oxidized iron compounds by metabolism-dependent and metabolism-independent processes. In *Thiobacillus ferrooxidans* and certain algae,  $\text{Au}^{3+}$  can be adsorbed to cell walls and the plasma membrane and be reduced to particles of elemental Au. Reducing compounds produced by silver-resistant bacteria can reduce  $\text{Ag}^+$  to metallic AgO, which results in silver deposition on glass surfaces of culture vessels or in growing colonies. Some bacterial strains have the exceptional ability to precipitate Au and Ag on their cell surface to a quantity, which can be 30 times of their cellular biomass. In metal-resistant strains of *Citrobacter*, a major mechanism of metal uptake is due to the activity of a cell-bound phosphatase, induced by growth in glycerol-2-phosphate, which can precipitate Cd, Pb, Cu and U as insoluble metal phosphates on the cell surface.

Many of the described processes are intimately dependent on metabolic processes and, in polluted environments, may be obvious determinants of tolerance and survival. However, such effects may be largely fortuitous and should not be considered true mechanisms of resistance, particularly when an

organism excretes a potential detoxifying substance in the presence or absence of potentially toxic metal concentrations.

### 13.5.3 Intracellular Compartmentation and Detoxification

Certain bacteria, yeasts, fungi and algae can actively accumulate intracellular metal ions against a gradient. Once inside the cells, metal ions may be compartmentalized and/or converted to more innocuous forms. Such processes can be effective detoxification mechanisms and microbes expressing them may be able to accumulate metals to high intracellular concentrations. It has been suggested that such mechanisms may be temporary and precede other means of expulsion of accumulated metals from the cells, possibly by means of vacuoles in eukaryotic microbes. A variety of electron dense deposits, many of unknown composition, have been recorded in cyanobacteria, bacteria, algae, protozoa and fungi after exposure to heavy metals. Polyphosphate has been implicated as a metal sequestering agent in *Pseudomonas putida*, *Anabaena cylindrica* and *Plectonema boryanum*, a variety of eukaryotic algae and certain fungi and yeasts. In eukaryotic microbes, e.g. algae, yeasts and fungi, there may be compartmentation of accumulated metals within specific organelles. In yeasts and fungi, a major proportion of accumulated  $\text{Co}^{2+}$ ,  $\text{Mn}^{2+}$ ,  $\text{Zn}^{2+}$  and  $\text{K}^{+}$  is located in the vacuole where it may be in ionic form or bound to low molecular weight polyphosphates.

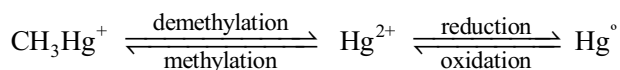
A common metal-induced response in many microorganisms is the synthesis of intracellular metal-binding proteins, which function in detoxification and also the storage and regulation of intracellular metal ion concentrations. Metallothioneins are the most widely-studied and the low molecular weight, cysteine-rich proteins that can bind metals like Cd, Zn and/or Cu. A Cd-binding prokaryotic metallothionein (molecular weight approx. 8100) was first described in the cyanobacterium *Synechococcus* sp. and others have since been reported in several bacteria including *P. putida*. Inducible Cd-binding proteins also occur in *E. coli* but these are larger than metallothioneins and may be more related to recovery from cadmium toxicity. More about the metal chelating proteins and their molecular basis of expression have been described in the next section (Section 13.6).

Recently biosurfactants have been investigated for their ability to complex metals such as cadmium, lead and zinc. Biosurfactant complexation can actually increase the apparent solubility of metals. However, the biosurfactant-complexed metal is not toxic to cells. It is not clear why some microorganisms produce biosurfactants but they may play a role in reducing metal toxicity. Recent evidence shows that biosurfactant-producing microorganisms can be isolated in greater diversity from metal-contaminated environments than from uncontaminated ones. Similarly, metal bioavailability can be influenced by common metabolic byproducts that result in metal reduction. In this case, soluble metals are reduced to less soluble metal salts, including sulphidic and phosphidic precipitates. For example, under aerobic conditions *Citrobacter* spp. can

enzymatically produce phosphate, which results in the precipitation of lead and copper. Under anaerobic conditions, high  $\text{H}_2\text{S}$  concentrations from sulphate reducing bacteria, e.g., *Desulfovibrio* spp., readily cause metal precipitation.

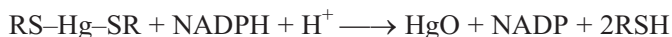
### 13.5.4 Enzymatic Transformations of Metals

Microorganisms can carry out chemical transformations of heavy metals such as oxidation, reduction, methylation and demethylation and these are relevant not only to biogeochemical cycling but may also constitute mechanisms of resistance. Most work on metal transformations has concentrated on bacterial involvement in the mercury cycle, which can simply be represented as:



Mercury resistance is a common property of Gram-positive and Gram-negative bacteria, the determinants usually being plasmid encoded, particularly in Gram-negatives. The most ubiquitous mechanism of mercury resistance in bacteria is the enzymatic reduction of  $\text{Hg}^{2+}$  by cytoplasmic mercuric reductase to metallic  $\text{Hg}^0$  which is less toxic than  $\text{Hg}^{2+}$ , volatile and rapidly lost from the environment (Fig. 13.6). This has also been recorded in certain fungi and yeasts. Organomercurial compounds are enzymatically detoxified by organomercurial lyase, which cleaves the  $\text{Hg}-\text{C}$  bond of, e.g. methyl, ethyl and phenyl mercury, to form  $\text{Hg}^{2+}$  and methane, ethane and benzene respectively. The  $\text{Hg}^{2+}$  can then be volatilized by mercuric reductase. More than one kind of organomercurial lyase may be present in a given bacterium. In organomercurial resistant *E. coli*, two enzymes are active against phenylmercuric acetate whereas only one is active against methyl and ethylmercury chloride. In a *Pseudomonas* strain there are also two enzymes of similar specificities as well as being able to cleave *p*-hydroxymercuribenzoate.

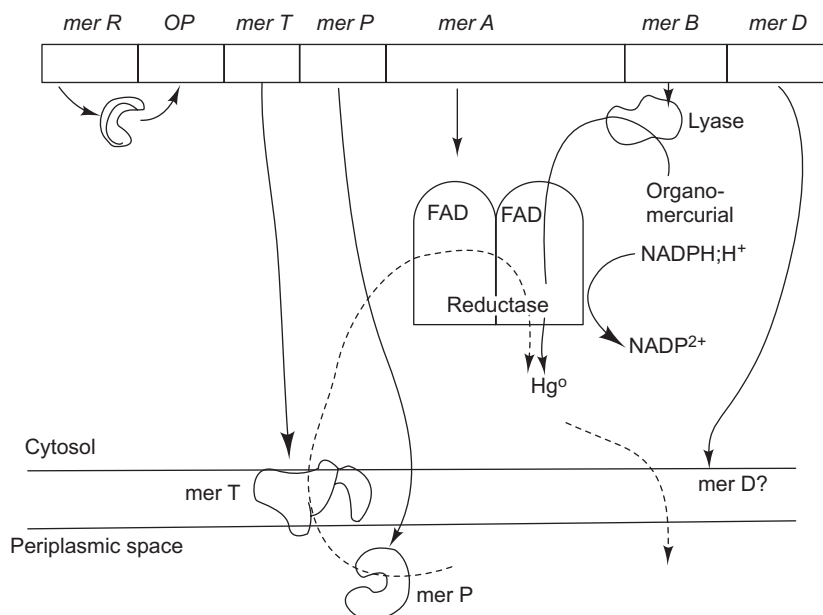
FAD-containing mercuric reductase is a flavoprotein and is the best studied metal detoxification enzyme and this and organomercurial lyase, requires an excess of thiols for activity. NADPH is the preferred reducing co-factor for mercuric reductase, as well as organomercurial lyase, but in some bacterial strains NADH is effective. The thiols prevent the formation of  $\text{NADPH}-\text{Hg}^{2+}$  complexes and ensure that the  $\text{Hg}^{2+}$  is present as a dimercaptide. The reaction catalysed by mercuric reductase *in vitro* can be represented as:



Mercuric reductase is structurally and mechanistically related to glutathione reductase and lipoamide dehydrogenase. The molecular analysis has shown that the *merR* gene product scavenges the periplasm for  $\text{Hg}^{2+}$ , which binds to a cysteine pair and releases  $2\text{H}^+$ . The  $\text{Hg}^{2+}$  is then transferred to the outermost cysteine pair of the *merT* gene product, also with release of  $2\text{H}^+$  (Fig. 13.6). Subsequent  $\text{Hg}^{2+}$  transfers are suggested to occur via such a redox-exchange mechanism. The  $\text{Hg}^{2+}$  is then transferred to the innermost cysteine pair on the inner face of *merT* gene product. Mercuric reductase may transiently associate

with the *merT* gene product in the membrane. The mechanism of mercuric reductase is probably that electrons are transferred from NADPH via FAD to reduce the active site cystine, converting it to two cysteine residues with titratable SH groups. One Cys residue forms a charge transfer complex with FAD. The active site cysteines then reduce  $\text{Hg}^{2+}$  bound to the C-terminal cysteines, forming  $\text{HgO}$ . The elemental  $\text{HgO}$  passes out through the cell envelope by simple diffusion and, since it is volatile, will be lost before it can be reoxidized. It is believed that *merD* gene product facilitates  $\text{HgO}$  diffusion out of the cell and this product is required only at the high values of mercury flux, which occur when the *mer* genes are on a high-copy-number plasmid. Mercuric reductase has been reported to exist as a monomer, dimer and a trimer with a sub-unit molecular weight of between 54000 and 69000 depending on the source, though it seems probable that the dimeric structure is active *in vivo*.

Although microbial involvement is found on the oxidation and the reduction of metals, much information are not available on these processes. A wide range of common bacteria including *E. coli*, *Pseudomonas fluorescens*, *Bacillus subtilis* and *B. megaterium* have been reported to oxidize  $\text{HgO}$  to  $\text{Hg}^{2+}$ , although such bacteria do not generally carry out subsequent methylation. There are indications that catalase is responsible for such  $\text{HgO}$  oxidation. Since this reaction results in formation of more toxic  $\text{Hg}^{2+}$ , it is regarded not as a resistance mechanism but as an activation process. In addition, the organic content of the medium greatly influences apparent  $\text{HgO}$  oxidation.



**Figure 13.6** - Proposed model of bacterial mercury resistance through mercuric reductase activity. The location of products of *mer A*, *mer T* and *mer P* are suggested (Modified from Silver *et al.*, 1986).

In mercury contaminated habitats, methylation is regarded as a very important phenomenon in determining the overall mercury toxicity to other organisms. Methylation of mercury can be accomplished by abiotic and biotic means. Microbes capable of methylation include bacteria, fungi and yeasts from a range of habitats, e.g. soil, water, sediments. Although a more toxic mercury compound results, methylation may be viewed as a detoxification mechanism since methylated species are more volatile and may be lost from the environment. In addition, the cellular elimination of organometallic compounds may be easier by diffusion. There is general agreement that methylation of  $\text{Hg}^{2+}$ , by direct and indirect microbial action, involves methylcobalamin (vitamin B12). There appears to be two mechanisms for methyl transfer from methylcobalamin to mercury and other heavy metals; first, electrophilic attack by the metal ( $\text{Hg}^{2+}$ ,  $\text{Pb}^{2+}$ ,  $\text{Cr}^{2+}$ ) on the Co–C bond of methylcobalamin and, second, methyl-radical transfer to an ion pair between the attacking metal ion and the portion of the methylcobalamin molecule that complexes to the central Co atom. The ecological significance of methylcobalamin-dependent methylation as a detoxification mechanism has been observed in many strains of bacteria. Other detoxifying enzymes include arsenite oxidase which catalyses  $\text{As}^{3+}$  to  $\text{As}^{5+}$  and chromate reductase which catalyses  $\text{Cr}^{6+}$  to  $\text{Cr}^{3+}$ . The toxicity of both these elements depends on their chemical forms and oxidation states. Thus, trivalent  $\text{As}^{3+}$  is 100–200 times more toxic than  $\text{As}^{5+}$  and so bacterial oxidation of arsenite to arsenate may be an effective detoxification mechanism. The most studied enzyme is that of a strain of *Alcaligenes faecalis*, which has a high affinity for arsenite ( $K_m = 2.2 \mu\text{M}$ ). Oxyanions of selenium and tellurium are generally of high microbial toxicity and the reduction of these may result in detoxification. Corresponding metallic compounds have been found associated with resistant bacterial cells when grown with tellurite or selenite, although the enzymes responsible have not been characterized.

### 13.6 GENETIC ASPECTS OF HEAVY METAL RESISTANCE

With a few exceptions, genetic studies of metal-microbe interactions have been limited and often confined to studies of mutant strains, mainly because knowledge of the physiology and biochemistry of resistance is fragmentary in many cases. However, certain areas of metal-microbe interactions have received precise genetical attention and this work has not only contributed to our knowledge of microbial responses to heavy metals but also to the wider contexts of molecular biology and biotechnology. The areas for which considerable genetical advances have been made to date are in heavy metal resistance in bacteria, cyanobacteria and the metallothionein of *S. cerevisiae*. In several other areas of metal studies, not only is genetic information limited but essential basic physiological and biochemical studies are also lacking.

### 13.6.1 Genetic Aspects of Heavy Metal Resistance in Bacteria

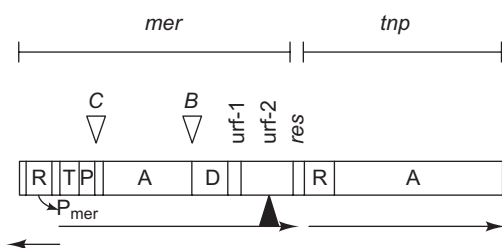
Mercury resistance is a common property of many bacteria isolated from metal contaminated environment. The mercury resistance determinants in bacteria (Gram-positive and Gram-negative) are usually plasmid encoded, particularly in Gram-negative bacteria. A common feature of the  $\text{Hg}^{2+}$ -resistance determinants of bacteria from the environment is the absence of linkage to drug resistance markers in contrast to clinical bacterial isolates where  $\text{Hg}^{2+}$  resistance is usually linked to genes for antibiotic resistance. Several mercury resistance transposons have been described that could be responsible for the widespread occurrence of  $\text{Hg}^{2+}$  resistance in Gram-negative bacteria. To date, the most recent advances have stemmed from DNA sequencing studies of the *mer* genes located on transposons *Tn501* and *Tn21* from Gram-negative bacteria and such studies have also been carried out with broad-spectrum determinants in both Gram-negative and Gram-positive bacteria (Fig. 13.7). Additional genes are also present in other *mer* operons, e.g. *merC* in *Tn21* and *merB* in the broad-spectrum operon of pDU 1358 (this does not imply that the broad-spectrum operon is part of a transposon). An insertion of a 11.2 kb gene is present in *Tn21*, which carries the streptomycin and sulphonamide resistance genes.  $P_{\text{mer}}$  in the Fig. 13.7 is the promoter regulated by the *merR* gene product. The remaining genes are involved in transposition and *res*, the internal resolution site; *tnpR*, transposition resolvase; *tnpA*, the transposase.

As described earlier, mercury resistance can be determined by the enzymatic reduction of  $\text{Hg}^{2+}$  to  $\text{Hg}^0$  by mercuric reductase and this is the system on which most of the genetical and biochemical progresses have been made. Mercuric reductase enzymes specified by plasmids in Gram-negative bacteria can be classed in two groups based on immunological studies. The prototype type I enzyme is specified by the IncP plasmid pVS1, which carries transposon *Tn501*. The prototype for the type II enzyme is specified by plasmid R100, which carries *Tn21*. Certain other plasmids may also specify these two types of enzymes. Most work has been carried out on the *Tn501*-encoded enzyme, which can constitute up to 6% of the soluble protein of *Pseudomonas aeruginosa* under conditions of  $\text{Hg}^{2+}$  induction. The enzyme encoded by plasmid R100 is very similar, the two enzymes showing 86% homology of amino acid residues and a high degree of conservative substitutions throughout their length.

Two mercury resistance phenotypes occur in Gram-negative bacteria. Narrow spectrum resistance to  $\text{Hg}^{2+}$  and one or two organomercurials, is exhibited by strains that solely express the reductase and is widespread, whereas broadspectrum resistance to  $\text{Hg}^{2+}$  and a larger number of organomercurials is found in strains specifying both reductase and lyase and is associated with plasmids of only a few incompatibility groups. In some organisms, organomercurial resistance does not involve lyase activity. All Gram-positive  $\text{Hg}^{2+}$ -resistant staphylococci express a broad-spectrum phenotype different to that of Gram-negatives. Organomercurial resistance conferred by narrow-spectrum



resistance determinants is probably due to permeability and transport processes. The archetypal *mer* operons of Gram-negative bacteria from plasmids pVS1 and R100 are located on the transposons *Tn501* and *Tn21*, respectively (Fig. 13.7). Many multiresistance transposons in Gram-negative bacteria are related to *Tn21* and may have evolved from an ancestral *mer* transposon by acquisition of DNA segments specifying resistance to different drugs. *Tn501* and *Tn21* are members of a class of transposable elements related to *Tn3*, which specifies ampicillin resistance. Several other Hg<sup>2+</sup> resistance transposons have been described and these may include multiresistance transposons *Tn1642*, *Tn1696* and *Tn2101* as well as Hg<sup>2+</sup> resistance transposons isolated from environmental bacteria, e.g. *Tn1861*, *Tn3401*, *Tn3402* and *Tn3403*. There appear to be no reports of Hg<sup>2+</sup> resistance transposons in Gram-positive bacteria.



**Figure 13.7** - Location of the *mer* and *tnp* genes in the mercury resistance transposon *Tn501*. Legends and abbreviations: ▽—Position of *merC* (C) in *Tn21* and *merB* (B) in pDU1358 operons; Δ—Position of the 11.2 kb insertion in *Tn21* which carries streptomycin and sulphonamide resistance genes; *Pmer*—Promoter regulated by the *merR* gene product; *res*—internal resolution site; *tnpR*—transposition resolvase; *tnpA*—transposase. The lower lines indicate the origin and direction of transcripts (Brown, 1985).

Initial genetic analysis of the R 100 plasmid (which contains *Tn21*) showed that at least three genes were involved in Hg<sup>2+</sup> resistance, which correlated with the biochemical processes involved. The *merR* gene is regulatory and involved in the induction of the remaining genes, the *merT* gene encodes a protein involved in Hg<sup>2+</sup> transport whereas *merA* is the structural gene for mercuric reductase (Fig. 13.6). Subsequent work with R100 from a *Shigella* has revealed additional genes: *merP*, apparently required for mercury transport and *merD* and *merG* of unknown function. This R100 system does not encode for organomercury lyase for which the gene designation *merB* is reserved.

A variety of mechanisms of cadmium resistance are found in bacteria and some have received genetical attention. Plasmid-determined cadmium resistance is found only in *Staphylococcus aureus* and two separate genes are involved, *cadA* and *cadB*, which respectively confer a large and a small increase in Cd<sup>2+</sup> resistance. The effect of *cadA* masks the *cadB* gene effect; *cadA* + *cadB* + strains are of the same resistance as *cadA* + *cadB* − strains. The mechanisms encoded are Cd<sup>2+</sup> efflux for *cadA* and increased binding for *cadB*, possibly due to a membrane component analogous to metallothionein. Cd<sup>2+</sup> and Zn<sup>2+</sup> resistance in *S. aureus*



are genetically linked, indicating that *cadA* and *cadB* also specify  $\text{Zn}^{2+}$  resistance. However, a third  $\text{Cd}^{2+}$  resistance system in *S. aureus* involves energy-dependent  $\text{Cd}^{2+}$  efflux like *cadA* but confers  $\text{Cd}^{2+}$  resistance alone. In *B. subtilis*, a chromosomal mutation is the basis of a change in the  $\text{Mn}^{2+}$  transport system, which transports both  $\text{Mn}^{2+}$  and  $\text{Cd}^{2+}$ , so that  $\text{Cd}^{2+}$  is no longer accumulated due to competitive transport regulation by  $\text{Mn}^{2+}$ .

Arsenate, arsenite and antimony (III) resistances are all coded for by an inducible operon-like system on plasmids of *S. aureus* and *E. coli*, each ion inducing all three resistances. Arsenate and arsenite resistance depend on efflux mechanisms, which result in reduced net accumulation. There is no evidence of plasmid involvement in arsenite oxidation by bacteria. Plasmid-determined resistance to chromate has been found in *Pseudomonas* strains and other bacteria, the resistance mechanism being reduced  $\text{CrO}_4^{2-}$  uptake by plasmid-bearing strains. Plasmid-determined  $\text{Cu}^{2+}$  resistance has been found on an antibiotic resistance plasmid from an *E. coli* strain, isolated from pigs fed copper feed supplements and  $\text{Cu}^{2+}$ -treated *Pseudomonas syringae*.

### 13.6.2 Genes Providing Metal Resistance in Cyanobacteria

**(a) Zinc and metallothionein—*smtA*, *mtnA*:** Metallothionein mediated metal resistance is common in many cyanobacteria. Cyanobacterial MTs are unusual in containing histidine residues, which have been implicated in the coordination of metals of the copper and zinc triads, at least in other proteins. The cyanobacterial MT coordinates three zinc ions, two exclusively to cysteine-thiolate groups and the third to a site composed of two cysteine-thiolate and two histidine-imidazole groups. MT genes have been isolated from a wide range of eukaryotes, but cyanobacterial MT genes are the only prokaryotic MT genes which have (to date) been characterised. A cyanobacterial MT-like protein was first isolated from *Anacystis nidulans* (*Synechococcus* PCC 6301) exposed to cadmium. Subsequently, MTs were purified from a marine cyanobacterium *Synechococcus* RRIMP NI and freshwater strains designated *Synechococcus* UTEX-625 and *Synechococcus* TX20. The protein from *Synechococcus* TX20 was subsequently sequenced and spectroscopic studies indicated the presence of a single metal cluster involving coordination of metal ions to thiol groups of cysteine residues. Fragments of the corresponding genes were obtained from DNA purified from *Synechococcus* PCC 7942 and *Synechococcus* PCC 6301 via PCR using degenerate primers. These PCR fragments were used as probes to isolate the corresponding genomic region and the gene designated *smtA*.

Unlike fungal MT (induced by Cu), the cyanobacterial MT induction occurs on exposure to cadmium and zinc, but not copper ions. The fungal MT, synthesised in response to copper ions, are associated with copper ions *in vivo* and mutants deficient in the corresponding genes are hypersensitive to excess copper but not to zinc. The cyanobacteria containing *smtA* accumulate more zinc than mutants deficient in *smtA*. Mutants of *Synechococcus* PCC 7942, which are deficient in a functional MT gene, are approximately 5 times more sensitive to

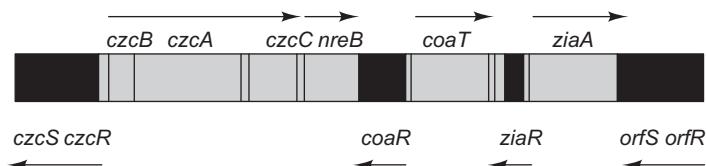
excess zinc, show some reduced tolerance to cadmium, but have 'normal' levels of tolerance to excess copper. This proves that the MT provide metal resistance in cyanobacteria most probably through intracellular metal sequestration.

**(b) Copper-transporting genes—*pacS*, *ctaA*:** Energy dependent copper efflux was proposed as a mechanism of copper resistance in *Synechococcus* sp. and in *Nostoc calcicola*. Subsequently, genes encoding copper transporting P-type ATPases, designated *pacS* and *ctaA* have been isolated, suggesting roles for this class of membrane proteins in copper acquisition, distribution and homeostasis in, at least some cyanobacteria. Kanamura and coworkers (1994) cloned two genes encoding deduced P-type ATPases, named *pacS* and *pacL*, the latter being a calcium-transporting ATPase while PacS was shown to transport copper. The *pacS* transcripts increase in abundance in response to elevated concentrations of copper and silver ions and PacS protein similarly accumulates in copper treated *Synechococcus* PCC 7942. An insertional mutant of *pacS* is hypersensitive to elevated concentrations of copper and, unexpectedly, the PacS protein was found to be localised to the thylakoid membrane, not the plasmalemma. PacS thus appears to be involved in copper homeostasis within intracellular compartments. Mutants deficient in functional *ctaA* show increased tolerance to copper compared to wild type cells suggesting a role for CtaA in the import of copper ions.

**(c) Zinc exporter *ziaA* and other homologues:** Within the genome of *Synechocystis*, PCC 6803 is a cluster of ORFs encoding deduced proteins with similarity to known proteins involved in copper and zinc homeostasis in other bacteria (Fig. 13.8). This includes sequences with similarity to *czcA* and *czcB*. These two genes were originally isolated from *Alcaligenes eutrophus* CH34 recovered from a zinc decantation tank and the plasmid encoded *czc* (for cobalt, zinc and cadmium) confers resistance to these three metal ions. The *czc* determinant is composed of three *czcCBA* structural genes. The CzcCBA proteins combine in a 1 : 1 : 1 ratio to form a cation efflux complex. Deletion of *czcC* leads to loss of cadmium and cobalt resistance but the CzcBA complex alone still confers high efficiency zinc efflux. In addition, there are two divergently transcribed (from the *czc*-like) genes designated *czcR* and *czcS* (Fig. 13.8) which show similarity to two component adaptive response regulators. Sequences encoding a histidine kinase and a response regulator are also present in sequences flanking *czc* from *Alcaligenes eutrophus* and these are known to regulate the expression of the *czc* genes. Elsewhere in the *Synechocystis* PCC 6803 genome is an ORF encoding a protein with similarity to CzcD, which is involved in the metal responsive expression of the *czc* determinant and has similarity to ZRC1 and COT1 that confer zinc/cadmium and cobalt resistance, respectively, in *Saccharomyces cerevisiae*.

The gene *ziaA* present as an ORF within this cluster encode proteins with sequence features of transition metal-transporting CPx-type ATPases and confers enhanced resistance to and reduced accumulation of zinc. The encoded protein of a second ORF has similarity to SmtB that has been shown to be a sensor of

internal zinc concentrations and redesignated ZiaR. This suggests that the cluster of genes could include two distinct zinc transporters, with *czc* responding to periplasmic zinc levels mediated by a two-component sensor and the ATPase responding to internal zinc levels mediated by an SmtB-like metal sensor ZiaR.



**Figure 13.8** - A gene cluster of *Synechocystis* sp. PCC 6803 encoding deduced proteins involved in transport and perception of metals. *orfS* and *orfR* is an additional pair of genes in the designated ORFs described in the text, which could be involved in metal perception (Robinson *et al.*, 2000).

### 13.6.3 Heavy Metal Resistance in Fungi and Other Eukaryotes

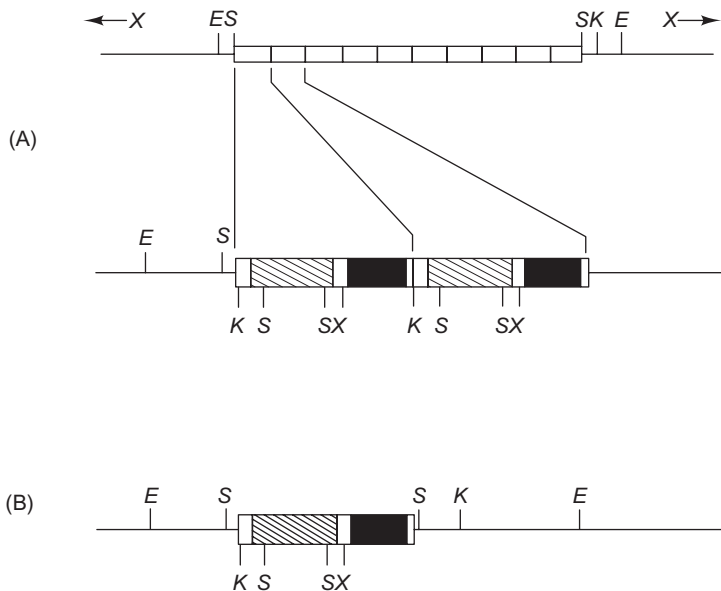
In fungi, the copper-induced metallothionein of *S. cerevisiae* (molecular weight approx. 6573) has received most attention although a copper-inducible metallothionein has also been described from *Neurospora crassa*. The yeast protein is inducible only by copper and not by cadmium or zinc and is therefore often referred to as Cu-MT or yeast MT.

Inducible Cd-binding proteins have been isolated from *Schizosaccharomyces pombe* that are structurally different to Cu-MT. A common name for these sulphur-rich metal-binding polypeptides is *phytochelatins*. The metal-binding peptides are composed of only three amino acids, L – cysteine, L – glutamic acid and glycine and are analogous to similar peptides found in plant cells exposed to heavy metals like Cd, Cu, Hg, Pb and Zn. The general structure of two peptides found in Cd-exposed *S. pombe* are ( $\gamma$ -Glu-Cys),Gly ( $n = 2$  and  $3$ ). The unit peptides are called cadystin and the phytochelatin  $n = 2$  and  $n = 3$  are called cadystin A and B, respectively. In addition to these, five homologous peptides with chain lengths from  $n = 4$  to  $n = 8$  are also observed in *S. pombe*, indicating that these are synthesized by elongation of the peptide with one ( $\gamma$ -Glu-Cys) unit at the expense of and possibly starting from glutathione. These cysteine-rich peptides of *S. pombe* can sequester several transition metals and therefore share some basic features with metallothioneins. Multiple enzymes may be involved in phytochelatins synthesis and therefore mechanisms must exist for metal-dependent induction or activation of such enzymes.

Low molecular weight cadmium-binding proteins have been detected in Cd-resistant strains of *S. cerevisiae*. In a sensitive strain, most internal Cd was bound to insoluble cytosolic material whereas Cd-binding proteins (molecular weight < 30000) were detected in the cytosol of resistant strains. Other metal binding proteins of unknown structure have been described from several other filamentous fungi and yeasts. Copper-binding proteins (molecular weight approx.

8000), similar to mammalian and yeast MT, have also been found in copper-tolerant *Scenedesmus* and metallothionein-like proteins were induced by cadmium in *Chlorella pyrenoidosa* and *Dunaliella*. A metallothionein-like protein has been identified in the ciliate *Tetrahymena pyriformis* after exposure to cadmium, which is also capable of zinc binding. The protein is rich in cysteinyl residues (32.4%) and acidic amino acids (23.7%).

Yeast contains a single Cu-MT gene, which is present in the *CUP1* locus. In genetic terms, copper resistance is ascribed to the function of the *CUP1* locus located on chromosome VIII. Copper-resistant strains of *S. cerevisiae* (*CUP1*<sup>r</sup>) contain 2 to 14 or more copies of the *CUP1* locus, which are tandemly repeated (Fig. 13.9) and can grow on media containing elevated concentrations of copper that are toxic to sensitive strains. In addition, continuous subculture of yeasts in the presence of increasing copper concentrations can result in selection of hyper-resistant yeast strains, which appear disomic for chromosome VIII that possesses the *CUP1* locus. The disomic chromosomes exhibit differential *CUP1* gene amplification patterns and this indicates that the mechanism of copper resistance involves not only amplification of the *CUP1* locus on the chromosomes but also disomy or aneuploidy of chromosome VIII.



**Figure 13.9** - Restriction maps of (A) copper resistant (*CUP1*<sup>r</sup>) and (B) copper sensitive (*cup1*<sup>s</sup>) strains of *S. cerevisiae*. The top line in (A) represents 10 tandemly repeated 2.0 kb units. The thin line represents the external flanking DNA sequences. The bottom line of (A) represents two expanded units. The black region represents coding part of the MT gene. The *cup1*<sup>s</sup> locus (B) is composed of only a single 2.0 kb unit. Abbreviations: E—*EcoRI*; K—*KpnI*, X—*XbaI*, S—*Sua3A* (Gadd, 1990).

The basis of copper resistance in *S. cerevisiae* involves amplification of the *CUP1* locus. Gene amplification is one of the few mechanisms capable of achieving increased production of certain cellular macromolecules and this can occur under selected pressure. In simple terms, the gene amplification model of *CUP1* suggests that, due to the proposed homology of DNA sequences in the *CUP1* repeats, one or multiple units are looped out. If the loop is replicated, the copy number increases (amplification) but if the loop is degraded, the copy number decreases (deamplification).

There is also genetical analysis of metal-regulated genes from other fungi although much less is known for *S. cerevisiae*. The gene for *Neurospora crassa* codes for a 26-amino acid MT protein. The metal-regulatory mechanisms of *N. crassa* and *S. cerevisiae* appear similar but no DNA sequence homology is found in upstream regulatory sequences of the gene. A strain of *Candida albicans* has been found containing DNA sequences, which hybridize with *S. cerevisiae* MT and the copper inducible proteins also show similarities with the *S. cerevisiae* MT.

### 13.7 BIOMANIPULATION FOR METAL REMOVAL

There are potentials of practical application to several areas of metal-microbe interactions. Heavy metal toxicity is the basis of many antimicrobial preparations for the control of animal and plant pathogens, algal blooms and the preservation of natural and synthetic materials. New knowledge of the mechanisms of toxicity and resistance are coming for the formulation and application of more effective compounds. Many interactions of heavy metals with microbes result in their removal from solution. These processes are of commercial relevance because the removal of potentially toxic heavy metals from industrial effluents and wastewaters can lead to detoxification and/or the recovery of valuable metals, e.g. gold. The industrial potential of microbial metal recovery systems depends on many factors, including capacities, efficiencies and selectivity, the ease of desorption, equivalence to existing physical and chemical processes in performance and economics and immunity from interference, by other effluent components or operating conditions. For full competition with existing technologies, it has been suggested that removal efficiencies must be > 99% and loading capacities > 150 mg metal/g.

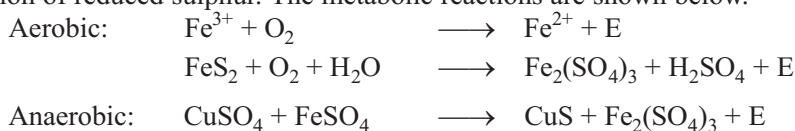
An advantage of microbial systems for commercial use is their versatility in respect of uptake mechanisms and the relative ease of morphological, physiological and genetical manipulations. Both living and dead microbes can take up heavy metals and can be products produced by or derived from microbial biomass, e.g. excreted metabolites, polysaccharides and cell wall constituents. There is no doubt that certain kinds of biomass or products can be highly efficient accumulators, even from dilute external concentrations and uptake capacities can be large by means of a variety of mechanisms ranging from purely physico-chemical to those dependent on cell metabolism. Microbe-based technologies may provide an alternative or supplementary process to conventional treatment

methods of metal removal/recovery and several are in commercial operation in the mining and metallurgical industries.

Heavy metals are usually toxic and this may hamper industrial applications of living cells. However, it may be possible to separate cell propagation from the metal contacting phase or to use strains tolerant to the metal concentrations encountered. The use of dead biomass or derived products eliminates problems of toxicity, nutrient supply and maintenance of optimal growth conditions. However, living cells can exhibit a wider variety of mechanisms for metal accumulation including transport, intracellular and extracellular precipitation, etc. although, resistance may be associated with decreased uptake.

### 13.7.1 Metal Removal by Enhanced Bioleaching

Bioleaching is the process of solubilization of metals by microbial activities to release them from their insoluble ores. The metabolic capabilities of microorganisms are being harnessed for the purpose to improve the recovery of metals from the environment. Not many microbes are known to participate in bioleaching process but still there are many species belonging to the genera *Thiobacillus*, *Sulfolobus*, *Leptospirillum*, *Burkholderia* and *Erwinia*, which efficiently perform the bioleaching process (Table. 13.2). *Thiobacillus ferrooxidans* is the best studied bioleaching bacterium known to cause the leaching of minerals, especially copper, iron and gold. It is a small rod-shaped chemolithotrophic bacterium that derives energy through the oxidation of either a reduced sulphur compound or ferrous iron. It grows best in an acidic solution both aerobically and anaerobically. Under aerobic condition, the bacterium gets energy by transferring electrons from ferric iron to oxygen while under anaerobic condition, the bacterium can use ferric iron as an alternative electron acceptor for oxidation of reduced sulphur. The metabolic reactions are shown below.



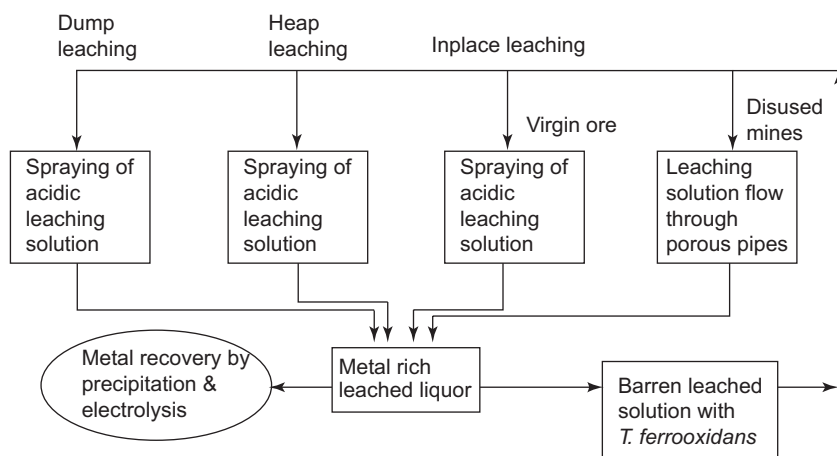
**Table 13.2:** The important microbes taking part in bioleaching process.

| Species                            | Leaching activity                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------|
| <i>Thiobacillus ferrooxidans</i>   | Leach almost all metals at temperature range 30°–50°C, pH 2.3–2.5; iron level 2–4 g/l |
| <i>T. thiooxidans</i>              | Thermophilic causing leaching of copper, arsenic and gold above 50°C                  |
| <i>Thiobacillus</i> spp.           | Metal leaching in sulphur containing environment                                      |
| <i>Sulfolobus acidocaldarius</i>   | Thermophilic copper and gold leacher                                                  |
| <i>Leptospirillum ferrooxidans</i> | Metal leaching at low iron level                                                      |
| <i>Thiobacillus</i> spp.           | Specialized molybdenum leacher                                                        |
| <i>Burkholderia cepacia</i>        | Leach phosphate from rocks                                                            |
| <i>Erwinia herbicola</i>           |                                                                                       |

*Thiobacillus thiooxidans*, a sulphur oxidizing bacterium, works in association with *T. ferrooxidans* in bioleaching process. It produces sulphuric acid and facilitates the growth and activity of other metal leaching species. Thermophilic thiobacilli have also been isolated from hot sulphur-rich environment and are known to cause metal leaching at high temperature. Members of *Sulfolobus* also oxidize ferrous iron and sulphur in a manner similar to that of thiobacilli. These are obligate thermophilic Archaea with chemoautotrophic mode of nutrition. At low ferrous iron level, leaching activities are mainly caused by *Leptospirillum* spp. and *T. thiooxidans*.

All strains of metal leachers are obligate autotrophs and use  $\text{CO}_2$  as sole source of carbon. They are even unable to grow in organic substrates. Some of them also fix nitrogen. Aeration and sufficiency of  $\text{O}_2$  is a must for active growth of these bacteria though they are able to withstand anaerobic condition getting some energy from chemolithotrophy. Aeration of the sample of iron pyrite ( $\text{FeS}$ ) in acidified water is sufficient to ensure the growth of most strains of bioleaching bacteria.

Controlled leaching of metal is done using one of several approaches including: heap leaching, vat leaching or in-place leaching (Fig. 13.10). Heap leaching usually involves excavating a long narrow ditch about 30 cm in depth and lining it with polyethylene. Air lines are evenly placed along the ditch to provide aeration. The ditch is filled with crushed ore to a height of 2 m to 3 m and then saturated with acidified water (pH ranging from 2 to 3) containing a bacterial inoculum of *Thiobacillus ferrooxidans* and *T. thiooxidans*. The leachate is collected at the bottom and recycled through the heap. Like heap leaching, dump leaching occurs on an impermeable surface. However, there is no artificial aeration so that set up costs are lower. In this case, efficiency of extraction is considerably decreased due to limitation of oxygen availability.



**Figure 13.10** - A scheme of various applications of micrometallurgy in bioleaching of metals from low grade ores and over burdens. The scheme shows only the application of *T. ferrooxidans* though many other microbes in pure and mixed cultures are used.



Vat leaching involves the continuous stirring of an ore slurry in large tanks. Again, the ore is mixed with acidified water and *T. ferrooxidans* inoculum. The advantage of vat leaching over heap leaching is the ability to control the leaching conditions efficiently as and when required. In-place leaching is carried out underground in spent mines, which contain as high as 39% of the metal of the mines. The mine is flooded with the acidic bacterial solution, resulting in an acid-rich, metal-rich leachate, from which metal is extracted and recovered after an incubation period of several months. There are several advantages to in-place leaching. In-place leaching has lower associated energy costs and has a less negative environmental impact when the production and collection of leachate is properly controlled. The in-place leaching of uranium has received considerable attention as it involves less risk of human exposure and thus better control of radiation exposure. Table 13.3 summarises some of the leaching reactions involving bacteria.

**Table 13.3:** The leaching reactions for recovery of metals from insoluble ores.

|                                                                              | Bacteria                                                                             |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $4\text{FeS}_2$ (pyrite) + $15\text{O}_2$ + $2\text{H}_2\text{O}$            | $\longrightarrow 2\text{Fe}_2(\text{SO}_4)_3 + 2\text{H}_2\text{SO}_4$               |
| $4\text{CaFeS}_2$ (chalcopyrite) + $17\text{O}_2$ + $2\text{H}_2\text{SO}_4$ | $\longrightarrow 4\text{CaSO}_4 + 2\text{Fe}_2(\text{SO}_4)_3 + 2\text{H}_2\text{O}$ |
| $2\text{FeAsS}$ (arsenopyrite) + $7\text{O}_2$ + $2\text{H}_2\text{O}$       | $\longrightarrow 2\text{FeAsO}_4 + 2\text{H}_2\text{SO}_4$                           |
| $\text{CuS}$ (covellite) + $2\text{O}_2$                                     | $\longrightarrow \text{CuSO}_4$                                                      |
| $2\text{FeS} + 2\text{Fe}_2(\text{SO}_4)_3$                                  | $\longrightarrow 6\text{FeSO}_4 + 4\text{S}$                                         |
| $\text{CuS} + \text{Fe}_2(\text{SO}_4)_3$                                    | $\longrightarrow \text{CuSO}_4 + 2\text{FeSO}_4 + \text{S}$                          |
| $\text{UO}_2 + \text{Fe}_2(\text{SO}_4)_3$                                   | $\longrightarrow \text{UO}_2\text{SO}_4 + 2\text{FeSO}_4$                            |
| $\text{UO}_3 + \text{H}_2\text{SO}_4$                                        | $\longrightarrow \text{UO}_2\text{SO}_4 + \text{H}_2\text{O}$                        |

Each of these approaches produces a metal-containing leachate. A number of methods can be used to recover the metal from the leachate, including solvent extraction, increasing the pH to precipitate the metals and electrolysis, in which an electrical current is passed through the metal solution to accumulate cationic metals at the anode and the anionic metals at the cathode. In the case of gold recovery, cyanidation is used, where free gold ( $\text{Au}^{2+}$ ) is precipitated as a cyanogold(I) species as shown below:



While uncontrolled leaching and chelation of metals can result in metal-contaminated environments, in some cases increased metal solubility can be beneficial, e.g., when essential nutrients are solubilized and delivered to plants. Metals such as zinc, copper and manganese are necessary for proper plant growth. Yet, like the toxic metals, essential metals may be present in a nonbioavailable state so that plants cannot readily absorb them. Microbial activity through acid production or the presence of microbially produced chelators can increase the solubility and hence the bioavailability of these metals.

A variety of ecological factors affect the efficiency of bioleaching. Factors that influence the activities of the most important leaching bacterium, *T. ferrooxidans*,

include temperature, generally considered optimal between 30°C and 50°C; acidity, optimally at pH 2.3–2.5; iron supply, optimally 2–4 g/l of leach liquor; oxygen and the availability of other nutrients required for growth. *T. ferrooxidans* is generally unable to initiate growth on ferrous iron above pH 3.0. Ammonium nitrogen, phosphorus, sulphate and magnesium are essential for the growth of *T. ferrooxidans*. These nutrients may be limiting in some leaching operations and need to be added. Availability of sufficient water can also be an important consideration in bioleaching processes. In open-air leaching operations, light has an inhibitory effect on some species of *Thiobacillus*, with the greatest inhibition occurring at the shorter wavelengths. Suspended particles of ferric iron offer a degree of protection against inhibitory light intensities.

Temperatures in some mineral deposits can be significantly increased by the oxidative activities of the thiobacilli and may exceed the tolerance limits of a *Thiobacillus* species, leading to decreased bioleaching activity. Because of the high temperatures associated with some leaching operations, thermophilic sulphur oxidizing microorganisms may play an important role in the bioleaching process. Members of the genus *Sulfolobus* and thermophilic *Thiobacillus* species are able to perform bioleaching process at such elevated temperature. The abilities of these acid-tolerant thermophilic bacteria to oxidize inorganic substrates make them likely candidates for use in bioleaching metal sulphides. *Sulfolobus* has been used especially in the bioleaching of molybdenite (molybdenum sulphide) as it is less sensitive to molybdenum. *Thiobacillus* species, on the other hand, are unable to leach molybdenite efficiently because of the toxicity of molybdenum to this bacterium. Still other microorganisms, perhaps in mixed populations, are potential candidates for use in bioleaching processes.

### 13.7.2 Metal Removal by Biosorption

The uptake of heavy metals by microbes can often be separated into a phase of metabolism-independent binding or adsorption to cell walls and other external surfaces followed by metabolism-dependent transport into cells. In some instances, intracellular uptake may be a result of increased cell permeability, particularly where toxicity is manifested. Metabolism-independent processes of metal uptake, including adsorption, can all be termed as ‘biosorption’, are frequently rapid and can occur in dead as well as living cells. Thus, biosorption is a passive, energy-independent process that can be mediated also by inactive cell material. Biosorption of metal ions includes mechanisms such as ion exchange, chelation, matrix entrapment and surface sorption. After biosorption or active removal of metal ions from wastewater or contaminated soil, the heavy metal-containing biomass must be separated and incinerated or regenerated by desorption or remobilization of the metals. The removal of metal ions by biologically catalyzed changes in the redox state is an alternative. The altered speciation (valence status) of metals can lead to precipitation, solubilization, or volatilization of the metal ions.

In many organisms, binding may be at least a two-stage process, first involving interactions between metal ions and reactive groups followed by inorganic

deposition of increased amounts of metal, which leads to much greater amounts of metal being accumulated than by purely ion exchange phenomena. For some metals, e.g. Pb and in certain organisms, particularly filamentous fungi, most accumulation is composed of surface phenomena. In growing cultures, biosorption and transport can be obscured or enhanced by other phenomena like precipitation, complexation, changes in the growth medium and morphology of the organism. Many potential binding sites occur in microbial cells and extracellular matrices and structural components in bacteria, algae and fungi include peptidoglycan, teichoic and teichuronic acid, polysaccharides, cellulose, uronic acid, proteins, glucans, mannans, chitin and melanin. A variety of ligands may be involved in biosorption including carboxyl, amino, phosphate, hydroxyl, carbonyl and sulphhydryl groups and the relative importance of each is difficult to resolve. Since the chemical composition of microbial cell walls can vary considerably, there are considerable differences in adsorption capacities between species, strains and even different cell types of the same organism. In general, bound metal, unless crystallized, is relatively easy to remove and binding is not generally affected over modest ranges of temperature. However, factors such as the organic and inorganic composition of the reaction mixture, pH and the biomass density, are important determinants of biosorptive processes.

An important consideration in microbe-based methods of metal accumulation is the ease of metal recovery either for reclamation or for further containment of toxic waste. Non-destructive recovery may be desired for regeneration of the biomass for reuse in multiple cycles. Destructive recovery may be by pyrometallurgical treatments or dissolution in strong acids or alkalis. If cheap, waste biomass is employed for valuable metals, destructive recovery may be economically feasible. For economic considerations, the choice of a recovery process also depends on the mechanism of accumulation. Metabolism-independent biosorption is frequently reversible and lends itself to non-destructive desorption. However, metabolism-dependent intracellular accumulation and related processes are largely irreversible and necessitate destructive recovery.

Like dissolved metal forms, particulate material can also be adsorbed onto microbial cells. Copper, lead and zinc sulphides were taken up by mycelium of *A. niger* whereas *Mucor javanus* could take up lead sulphide, zinc dust and ferric hydroxide ('ochre') from acid mine drainage. Systems using living organisms have received most use in connection with effluents containing metals at concentrations below toxic levels. Artificial stream meanders and tailing ponds, often in series, containing photosynthetic algae (e.g. *Cladophora*, *Spirogyra*, *Rhizoclonium*) and cyanobacteria (e.g. *Oscillatoria*) have been used to treat effluents from lead mining and were effective in reducing levels of Pb, Zn, Cu and Mo to those required for permissible discharge. With increasing metal concentrations and times of exposure, toxicity may result in death and settling of the biomass. Subsequent decomposition can result in H<sub>2</sub>S production by sulphate-reducing bacteria and precipitation of the metals as sulphides. This process is virtually irreversible provided reducing conditions are maintained. Some

proposed arrangements utilizing this phenomenon include single- and double-stage systems where continuous cultures of sulphate-reducing bacteria receive metal-containing inflow or  $H_2S$  from such cultures is fed into a precipitation vat that receives the metal inflow.

An example, where a living cell system acts as a 'polisher' to an existing process that is not completely efficient, is the biological removal of arsenic from copper refining effluents. After initial pyrometallurgical treatment, a two-stage algal treatment was devised with arsenic removal being accomplished in two bioreactors. Arsenic-loaded algae were subsequently harvested and incinerated.

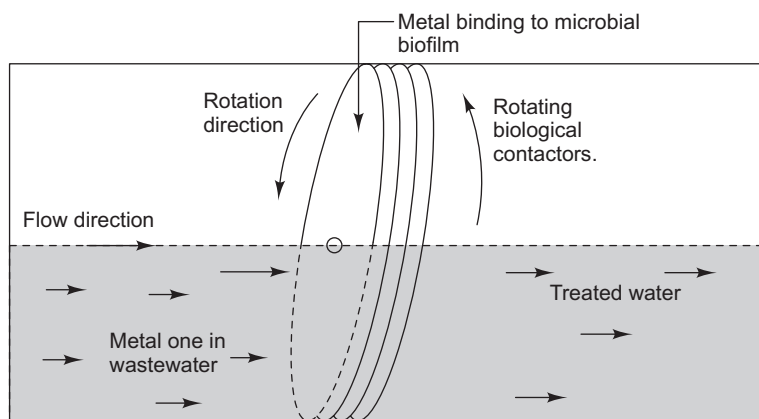
Biosorption may be important in certain survival structures, e.g. fungal chlamydospores, which have thick cell walls, frequently melanized. Such cell types exhibit higher metal tolerance than hyaline cell types and the wall appears to act as a permeability barrier preventing entry into the cell. The metal resistant strains of algae, fungi and bacteria, that are capable of accumulating metals through extracellular precipitation, are considered as the potential organisms to use them for metal removal through free living or immobilized conditions in bioreactors.

### 13.7.3 Use of Immobilized Cell Systems for Metal Removal

For rigorous industrial applications, freely suspended microbial biomass has a number of disadvantages. These include small particle size, low density and low mechanical strength, which can limit the choice of reactor systems and make biomass/effluent separation difficult and/or expensive. For use in packed bed or fluidized bed bioreactors, immobilized or pelleted biomass, living or dead, may have greater potential. The important advantages of immobilized system are easier separation of cells from the reaction mixture, better capability for biomass reuse, high biomass loadings and high flow rates with or without recirculation and minimal clogging in continuous flow systems. Polyacrylamide-immobilized cells of *Streptomyces albus* could remove uranium, copper and cobalt. Desorption could be achieved by elution with 0.1 M  $Na_2CO_3$  and uptake capacities were not significantly affected by up to five uptake-desorption cycles. A metal-resistant *Citrobacter*, immobilized in polyacrylamide, removed uranium, cadmium, copper and lead from solutions supplemented with glycerol 2-phosphate. Phosphatase-mediated cleavage of this substrate released  $HPO_4^{2-}$ , which precipitated the metals around the cells as insoluble metal phosphates, e.g.  $CdHPO_4$ . Polyacrylamide-entrapped *Chlorella* can successfully remove  $UO_2^{2+}$ ,  $Au^{3+}$ ,  $Cu^{2+}$ ,  $Hg^{2+}$ ,  $Ag^+$  and  $Zn^{2+}$  from solution and a selective elution scheme for subsequent recovery is possible.  $Cu^{2+}$  and  $Zn^{2+}$  were desorbed by lowering the pH to 2.0 whereas  $Au^{3+}$ ,  $Ag^+$  and  $Hg^{2+}$  were not. The addition of a strong ligand at differing pH values was necessary to elute these;  $Au^{3+}$  and  $Hg^{2+}$  were selectively eluted using mercaptoethanol. Other immobilization matrices include alginates and various polymeric membranes of unrevealed composition. The latter can be made to any size, contain only 10% inert material and can be used in multiple adsorption-desorption cycles. Commercial processes exist for the application of such systems in fixed bed or fluidized bed reactors, particularly for recovery of valuable metals.

A method of using immobilized cells without entrapment is to use biofilms on inert matrices such as solid supports, foams or minerals. Filamentous fungi can be grown in pellet form and therefore have some advantages in common with immobilized particles. Pellets (4 mm diameter) of *Aspergillus niger* were used for uranium removal in a fluidized bed reactor and were found to be more efficient than a commercially obtained ion-exchange resin. However, fungal pellets are prone to disintegration and similarities in density with the liquid medium makes continuous operation difficult.

Both living and dead biomass can be used in immobilized systems. If living cells are used, either entrapped or as biofilms, then it may be possible to remove other undesirable pollutants as well as heavy metals. Many biofilm microbes, including *Bacillus*, *Citrobacter*, *Arthrobacter*, *Streptomyces* and the yeasts *Saccharomyces* and *Candida*, produce exopolymers as part of their growth regime. Metals have high affinities for these anionic exopolymers. Microbial biofilms may be viable or nonviable when used in remediation. In general, the biofilm is immobilized on a support as contaminated water is passed through the support (Fig.13.11). Often, a mixture of biofilm-producing organisms grow on these supports, providing a constant supply of fresh biofilm. Live *Citrobacter* spp. and pure and mixed bacterial cultures of *Pseudomonas* biofilms are used to remove uranium from contaminated water. The bacteria are grown on anthracite coal particles in a fluidized bed or immobilized on polyvinyl chloride (PVC) or polypropylene webs. Both *Arthrobacter* spp. biofilms and biomass (non-living) are used in recovery of cadmium, lead, chromium, copper and zinc from wastewaters. Nonliving *Bacillus* biomass preparations effectively bind cadmium, chromium, copper, mercury and nickel, among other metals. A large-scale commercial process, which treats effluents from gold mining and milling, uses rotating disc, biological-contacting units to simultaneously degrade cyanide, thiocyanate and ammonia. Heavy metals are taken up by the microbial film on the disc surfaces which is periodically removed for controlled disposal.



**Figure 13.11** - Application of rotating biological contactors for metal recovery through biofilm activity (Adopted from Roane and Pepper, 2000).

To enhance metal precipitation and/or removal in activated sludge or by a biofilm, selected specialized bacteria, genetically modified bacteria, or bacteria as plasmid donors for degradative pathways can be added. Plasmid exchange from donor to recipient cells would be advantageous to the recipient bacteria, because the plasmids harbour genes whose products function in pathways for pollutant degradation, which would thus extend the substrate spectrum of the recipients. The transfer of naturally occurring mercury-resistant plasmids between *Pseudomonas* strains in biofilms was shown to occur rapidly.

Several processes are in commercial operation, which are competitive in cost and operational characteristics with standard treatments. These systems may also be used in conjunction with methods relating to the removal of other impurities. However, such systems have mainly been applied to valuable metals or radionuclides. Some of the microbe-based processes for general detoxification are relatively unsophisticated and use living cells, e.g. in ponds or streams. With increasing interest in microbe-based technologies for metal removal/recovery, further development of methodologies is inevitable and also further exploitation of so-far neglected interactions, e.g. particulate metal accumulation, extracellular precipitation and complexation and metallothioneins.

## 13.8 CONCLUSION

Despite the toxicity of heavy metals in the environment, microorganisms are commonly found in metal-polluted habitats and possess a range of morphological and physiological attributes that enable survival. Certain kinds of resistances may be genetically determined and transferable whereas, in other cases, tolerance may depend on intrinsic properties of the organism and/or detoxification due to environmental components. Many organisms can express a variety of survival strategies, providing excellent model systems for scientific endeavour in fields ranging from ecology to molecular biology. These are also of biotechnological relevance for detoxification of metal-containing environment, metal recovery, land reclamation, the preservation of natural and synthetic materials and treatment of infection.

The metal-regulated genetic system provides a powerful tool in several areas of pure and applied significance. Metal-regulated DNA sequences appear to be the most efficient elements for heterologous gene expression discovered to date and a variety of proteins have been expressed in the *CUP1* expression system for commercial and research purposes. The yeast MT gene has been successfully used to transform several brewing strains. Selection with copper is thus a useful tool for the genetic manipulation of commercial brewing strains and improvement of the product. Yeast MT may also be of potential in metal recovery since it can bind metals besides copper, including cadmium, zinc, cobalt, silver and gold, although unlike copper these elements do not induce MT synthesis. However, yeast strains with constitutive expression of MT genes may accumulate greater amounts of metals. Alternatively, systems may be engineered for the secretion of

MT proteins from yeast cells by attaching appropriate secretory signal sequences to MT genes with the related possibility of engineering a variety of MT molecules with specific affinities for different metals.

Both living and dead microbial biomass can accumulate heavy metals and some processes are already in commercial operation. It should be stressed that microbe-based processes for metal removal/recovery may not necessarily replace conventional technologies but may serve as ‘polishers’ to processes not completely efficient. Selectivity is a problem for many microbial systems and this is worthy of further attention in the future. Some degree of selectivity in both biosorption and desorption may be possible using strain selection and/or manipulation of external conditions. Living cells do have possibilities, though intracellular metal may be difficult to recover and toxicity may be a problem. Although resistant strains can be readily isolated or engineered, they often exhibit reduced accumulation. However, there are examples where resistant yeast strains take up more metal than sensitive strains, probably by means of more efficient internal sequestration and where resistant microbial consortia are also highly effective. Currently, systems that use living cells appear relatively unsophisticated and metallothioneins, particulate metal accumulation, siderophores and analogous molecules and extracellular precipitation have all received scant exploitation. With the realization that yeast metallothionein and siderophores may be able to bind precious metals and the possibility of genetic manipulation of organisms to, e.g., constitutive metallothionein expression, this may change in the future.



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# MICROBIAL DEGRADATION OF RECALCITRANT ORGANIC POLLUTANTS

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## 14.1 INTRODUCTION

The technological and industrial progresses of the present day has caused synthesis, use and disposal of a variety of industrial pollutants and agrochemicals into the environment. Majority of these chemicals are new to the environment showing resistance to microbial degradation. These chemicals are often hydrophobic in nature and are termed as recalcitrant xenobiotics. These include pesticides, aldehydes, ketones, aliphatic and aromatic hydrocarbons, amines, mineral oil constituents and halogenated products of petrochemicals. Most of these chemicals are designed for specific uses but their specificity is rarely absolute. Harmful effects are often exerted on many non-target organisms ranging from microbes to man, especially when they are present in the environment at a comparatively high concentration. Such effects are not only imposed by these pollutants in their parent forms but also by their transformation products of varying toxicity. It is, therefore, necessary to understand the pathway of degradation of these chemicals in the environment to manipulate conditions for their accelerated and efficient removal.

Microorganisms are the most effective group for removal and/or modification of majority of environmental contaminants. The microbial community within contaminated ecosystems tend to be dominated by those microorganisms capable of utilizing and/or surviving toxic contaminations. Almost all the recalcitrant contaminants in the environment are either slowly degraded or permanently modified by such adaptive group of microorganisms. However, very few microorganisms, majority of whom are bacteria, metabolically degrade these

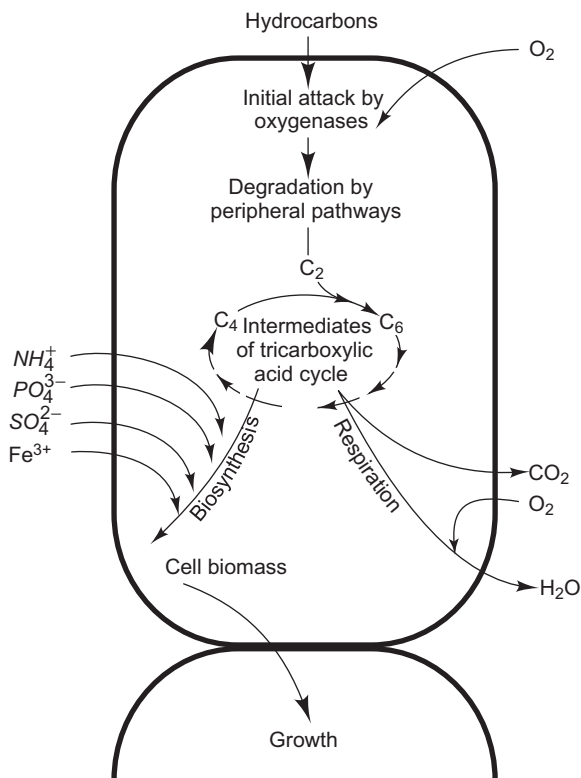
pollutants, thus requiring sufficient alternative substrates to support growth. As a result, these communities are typically less diverse than those in unstressed system, although the diversity is influenced by the complexity of chemical mixtures present and the length of time the population have been exposed. In these environments, communities survive with fluctuating dynamics driven by the availability and concentrations of different carbon and energy sources.

## **14.2 CHARACTERISTICS OF RECALCITRANT XENOBIOTICS DEGRADING MICROBES**

The capacities of aerobic microorganisms are of particular relevance for the biodegradation of such compounds and are described as examples with reference to the degradation of aliphatic and aromatic hydrocarbons as well as their chlorinated derivatives. The most rapid and complete degradation of the majority of pollutants is brought about under aerobic conditions. The initial intracellular attack on organic pollutants is an oxidative process; the activation and incorporation of oxygen is the enzymatic key reaction catalyzed by oxygenases and peroxidases. Peripheral degradation pathways convert the organic pollutants step by step into intermediates of the central intermediary metabolism, e.g., the tricarboxylic acid cycle. Biosynthesis of cell biomass from the central precursor metabolites, e.g., acetyl CoA, succinate, pyruvate and sugars required for various biosyntheses and growth must be synthesized by gluconeogenesis.

### **14.2.1 Typical Aerobic Degrading Bacteria**

The most rapid and complete degradation of the majority of pollutants in the environment is brought about under aerobic conditions (Fig. 14.1). Because of the high energy level of the activity, significant amount of carbon (approximately 50%) of the degradation reaction is used for growth. The aerobic microorganisms, degrading organic pollutants, have the characteristics as given as: (a) The metabolic processes for optimizing the contact between the microbial cells and the organic pollutants. The chemicals must be accessible to the organisms having biodegrading activities. For example, hydrocarbons are water insoluble and their degradation requires the production of biosurfactants; (b) The initial intracellular attack on organic pollutants is an oxidative process. The activation and incorporation of oxygen is the enzymatic key reaction catalyzed by oxygenases and peroxidases; (c) Peripheral degradation pathways convert the organic pollutants step by step into intermediates of the central intermediary metabolism, e.g., the tricarboxylic acid cycle; and (d) Biosynthesis of cell biomass from the central precursor metabolites, e.g., acetyl CoA, succinate and pyruvate. Sugars required for various biosyntheses and growth must be synthesized by gluconeogenesis.



**Figure 14.1** - A general metabolic degradation pathway of hydrocarbon by aerobic bacteria.

The predominant degraders of organopollutants in the oxic zone of contaminated areas are chemoorganotrophic species, which are able to use a huge number of natural and xenobiotic compounds as carbon sources and as electron donors for the generation of energy. Although many bacteria are able to metabolize organic pollutants, a single bacterial species does not possess the enzymatic capability to degrade all or even most of the organic compounds in a polluted soil. Mixed microbial communities have the most powerful biodegradative potential, because the genetic information of more than one organism is necessary to degrade the complex mixtures of organic compounds present in contaminated areas.

A huge number of bacterial and fungal genera possess the ability to degrade organic pollutants. These organisms need oxygen at two metabolic sites—the initial attack on the substrate and the end of the respiratory chain. Certain higher fungi have developed a unique oxidative system for the degradation of lignin based on extracellular ligninolytic peroxidases and laccases. This enzymatic system possesses increasing significance for the cometabolic degradation of persistent organopollutants.

The predominant bacteria in polluted soils belong to a spectrum of genera and species (Table 14.1). The table lists only bacteria that can be cultured on nutrient-

rich media. We have to consider that the majority of bacteria present in soils cannot yet be cultivated in the laboratory. The pseudomonads, aerobic gram-negative rods that never show fermentative activity, seem to have the highest degradative potential, e.g., *Pseudomonas putida*, *P. diminutum* and *P. fluorescens*. Further important degraders of organic pollutants are found within the genera *Comamonas*, *Burkholderia*, *Flavobacterium* and *Xanthomonas*. Some species utilize quite a wide variety of organic compounds as carbon sources. The immense potential of the pseudomonads does not solely depend on the catabolic enzymes, but also on their capability for metabolic regulation. A second important group of degrading bacteria are the Gram-positive rhodococci and coryneform bacteria. Many species, now classified as *Rhodococcus* spp., were originally described as *Nocardia* spp., *Mycobacterium* spp. and *Corynebacterium* spp. Rhodococci are aerobic actinomycetes that show considerable morphological diversity. A certain group of these bacteria possesses mycolic acids on the external surface of the cell. These compounds are unusual long-chain alcohols and fatty acids, esterified to the peptidoglycan of the cell wall. These lipophilic cell structures probably are important for the affinity of rhodococci to lipophilic pollutants. In general, rhodococci have high and diverse metabolic activities and can synthesize biosurfactants.

**Table 14.1:** Predominant bacteria in waste samples polluted with aliphatic and aromatic hydrocarbons, polycyclic aromatic hydrocarbons and chlorinated compounds.

| <i>Gram-negative strains</i>                                                                                                             | <i>Gram-positive strains</i>                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <i>Pseudomonas</i> spp.<br><i>Acinetobacter</i> spp.<br><i>Alcaligenes</i> spp.<br><i>Flavobacterium</i> spp.<br><i>Xanthomonas</i> spp. | <i>Nocardia</i> spp.<br><i>Mycobacterium</i> spp.<br><i>Corynebacterium</i> spp.<br><i>Arthrobacter</i> spp.<br><i>Bacillus</i> spp. |

### 14.2.2 Typical Anaerobic Degrading Bacteria

Anaerobic degradation processes have always been considered inferior to aerobic degradation in their kinetics and capacities. They are thought to be slow and inefficient, especially with certain comparably stable types of substrates. Nonetheless, in certain anoxic environments, such as the cow's rumen, the turnover of substrates (e.g. cellulose) is much faster than in the presence of oxygen. Fermentative degradation of fibers in the rumen reaches its limit with plant tissues rich in lignin which largely withstands degradation in the absence of oxygen.

Also in waste treatment, especially with high loads of easy-to-degrade organic material, anaerobic processes have proved to be efficient and far less expensive than aerobic treatment: they require only small amounts of energy input, in contrast to treatment in aeration basins and can produce a mixture of CH<sub>4</sub> and

CO<sub>2</sub> ('biogas'), which can be used efficiently for energy generation. This holds true for most waste materials that are easily accessible to degradation without the participation of oxygen, such as polysaccharides, proteins, fats, nucleic acids, etc. These polymers are hydrolyzed through specific extracellular enzymes (mostly esterases), and the oligo- and monomers can be degraded inside the cell through enzyme reactions similar to those known in aerobic metabolism, after the uptake and the assimilation of oligo- and monomers. The specific activities of such enzymes in anaerobic cultures are in the same range as those of aerobic bacteria and thus the transformation rates per unit biomass should be equivalent. Nonetheless, anaerobic bacteria obtain far less energy from substrate turnover than their aerobic counterparts. Whereas aerobic oxidation of hexose to six CO<sub>2</sub> yields 2870 kJ per mol, dismutation of hexose to three CH<sub>4</sub> and three CO<sub>2</sub> yields only 390 kJ per mol, about 15% of the aerobic process and this small amount of energy has to be shared by at least three different metabolic groups of bacteria (Acetogenic, Hemicetogenic and Methanogenic). The anaerobic bacteria can produce far less biomass per substrate molecule than aerobes can (see chapter 12). They have highly active syntrophic interaction in the anaerobic environment but the microbial communities in the anaerobic reactors still are comparably sluggish in reacting to changes in substrate composition or in their reestablishment after accidental population losses due to toxic ingredients in the feeding waste. The metabolic efficiency of the anaerobic bacteria is also greatly influenced by the substrate type.

Degradation of organic matter in the absence of oxygen can be coupled to the reduction of alternative electron acceptors following a certain sequence that appears to be determined by the respective redox potentials. Molecular oxygen (O<sub>2</sub>/H<sub>2</sub>O) is followed by nitrate (NO<sub>3</sub><sup>-</sup>/NO<sub>2</sub><sup>-</sup>), manganese(IV) oxide (MnO<sub>2</sub>/Mn<sup>2+</sup>), iron(III) hydroxides (FeOOH/Fe<sup>2+</sup>), sulphate (SO<sub>4</sub><sup>2-</sup>/HS<sup>-</sup>) and finally CO<sub>2</sub> (CO<sub>2</sub>/CH<sub>4</sub>), with the release of nitrite, ammonia, dinitrogen, manganese(II) and iron(II) carbonates, sulphides and finally methane as products. Reduction of these acceptors with electrons from organic matter (average redox potential for glucose to 6 CO<sub>2</sub> is -0.434 V) provides metabolic energy in the mentioned sequence. Thus, the energy yields of the various anaerobes mentioned differ and the availability of either high- or low-potential electron acceptors may also influence the biochemistry of anaerobic degradation processes.

Anaerobic degradation processes are often found useful for treatment of specific wastewaters rich in phenolic compounds, e.g., from the chemical industry, to avoid formation of unwanted side products such as condensed polyphenols. In other situations, aerobic treatment may cause technical problems, e.g., by extensive foam formation during aerobic treatment of surface-active compounds such as tensides. Thus, knowledge of the limits and the principles of anaerobic degradation processes under the various conditions prevailing in natural habitats might help to design suitable alternative techniques for cleanup of contaminated soils or for treatment of specific wastewaters that have so far been applied only insufficiently. On the other hand, in aerobic condition,

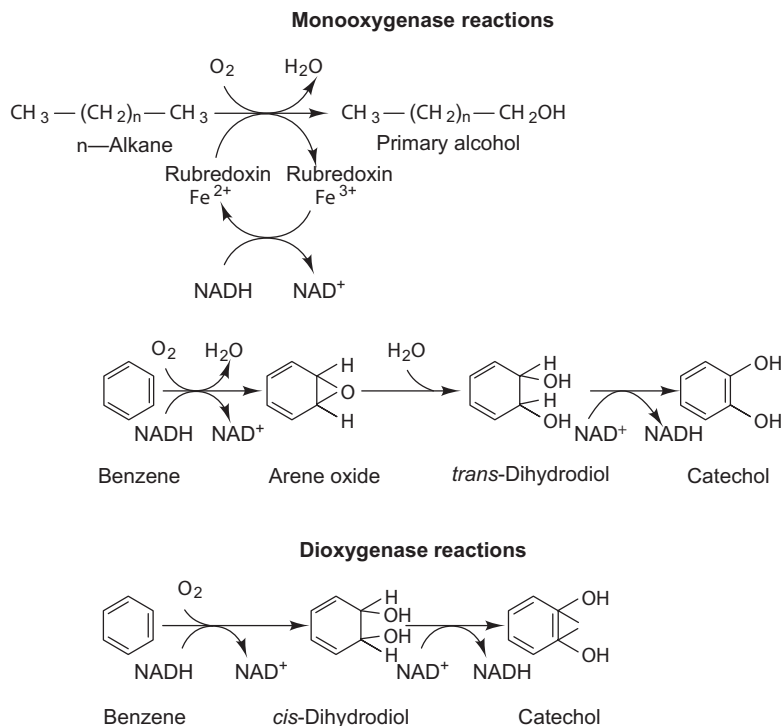
oxygenases introduce hydroxyl groups into aromatics and further oxygen may lead to the formation of phenol radicals that initiate uncontrolled polymerization and condensation to polymeric derivatives, similar to humic compounds in soil, which are very difficult to degrade further, whether anaerobically or aerobically.

## 14.3 AEROBIC DEGRADATION OF HYDROCARBON

### 14.3.1 Growth-associated Degradation of Aliphatics

The initial aerobic attack on aliphatic and cycloaliphatic hydrocarbons requires molecular oxygen. Two types of enzymatic reactions are involved in these processes (Fig. 14.2). Whether a monooxygenase or a dioxygenase reaction occurs depends on the nature of the substrate and the enzymes possessed by the microorganisms. The n-alkanes are the main constituents of mineral oil contaminations. Long-chain n-alkanes ( $C_{10}$ – $C_{24}$ ) are degraded most rapidly by monooxygenase activity (Fig. 14.3). Short-chain alkanes (less than four) are toxic to many microorganisms but they evaporate rapidly from petroleum-contaminated sites and thus their environmental toxicity is greatly reduced. Oxidation of alkanes is classified as monoterminal or diterminal. Monoterminal oxidation is the main pathway and proceeds by formation of the corresponding alcohol, aldehyde and fatty acid. Beta-oxidation of fatty acids results in the formation of acetyl-CoA. n-Alkanes having an odd number of carbon atoms are degraded to propionyl-CoA, which is in turn carboxylated to methylmalonyl-CoA and further converted to succinyl-CoA. Fatty acids having a physiological chain length may be directly incorporated into membrane lipids but most degradation products are fed into the tricarboxylic acid cycle. Subterminal oxidation occurs with lower ( $C_3$ – $C_6$ ) and longer alkanes with formation of a secondary alcohol and subsequently of a ketone. Unsaturated alkenes are oxidized at the saturated end of the chains. A minor pathway proceeds via an epoxide, which is converted to a fatty acid. Branching generally reduces the rate of biodegradation. Methyl side groups do not noticeably decrease the biodegradability, whereas complex branched chains, e.g., the tertiary butyl group, hinder the action of the degradative enzymes.

Cyclic alkanes represent minor components of mineral oil and are relatively resistant to microbial attack. The absence of an exposed terminal methyl group complicates the primary attack. A few species can use cyclohexane as a sole carbon source, but it is more commonly cometabolized by mixed cultures. The pathway of cyclohexane degradation is shown in Fig. 14.4. In general, the presence of alkyl side chains on cycloalkanes facilitates their degradation. Aliphatic hydrocarbons become less water-soluble with increasing chain length; those with a chain length of  $C_{12}$  or more are virtually water-insoluble. Two mechanisms are involved in microbial uptake of these lipophilic substrates: attachment of microbial cells to oil droplets and production of biosurfactants. The uptake mechanism linked to attachment of the cells is still unknown but the effect of biosurfactants has been studied well (Fig. 14.5).



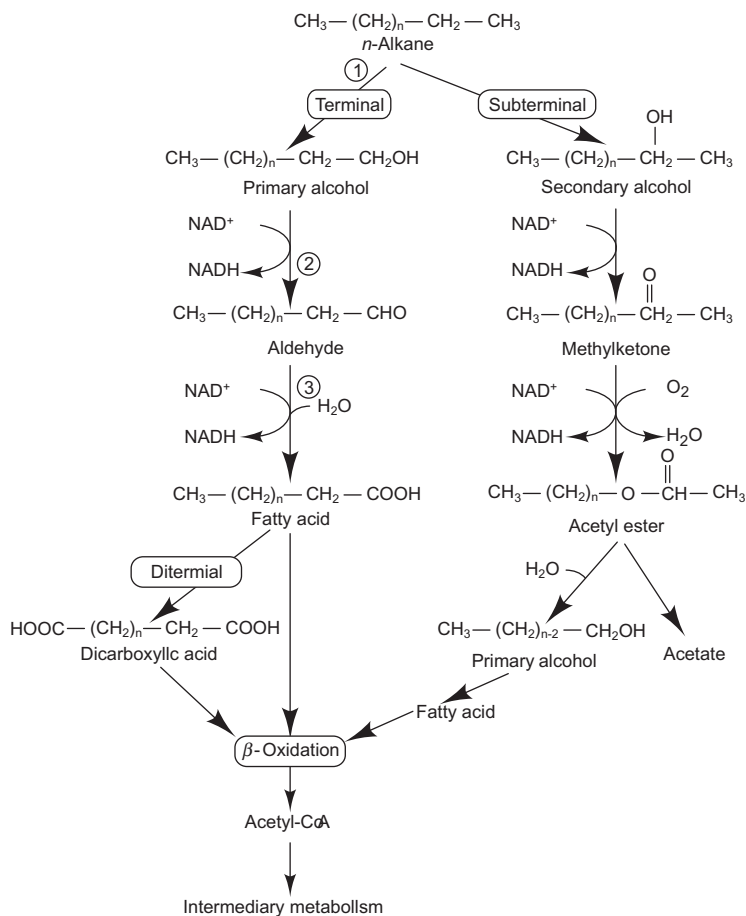
**Figure 14.2** - Oxygenase catalyzed metabolism of hydrocarbon. Monoxygenases incorporate one atom of  $\text{O}_2$  while dioxygenases incorporate one molecule of  $\text{O}_2$  into the substrate.

The products of hydrocarbon degradation that are fed into the central tricarboxylic acid cycle have a dual function: as substrates of energy metabolism and building blocks for the biosynthesis of cell biomass (Fig. 14.1). The cell biomass can be mineralized after exhaustion of degradable pollutants in a contaminated site.

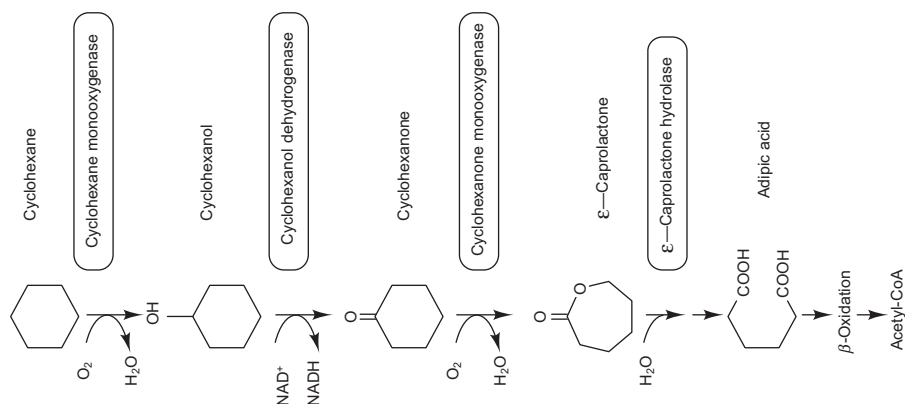
### 14.3.2 Degradation of Aromatic Compounds

Aromatic hydrocarbons, e.g., benzene, toluene, ethylbenzene and xylenes (BTEX compounds) and naphthalene belong to the large-volume petrochemicals, which are widely used as fuels and industrial solvents. Phenols and chlorophenols are released into the environment as products and waste materials from industry. Aromatic compounds are formed in large amounts by all organisms, e.g., as aromatic amino acids, phenols or hydroquinones/quinones. Thus, it is not surprising that many microorganisms have evolved catabolic pathways to degrade aromatic compounds. In general, man-made organic chemicals (xenobiotics) can be degraded by microorganisms when they are similar to natural compounds. A range of man-made aromatics are converted enzymatically by microorganisms to the natural intermediates of degradation: catechol and protocatechuate (Fig. 14.6).



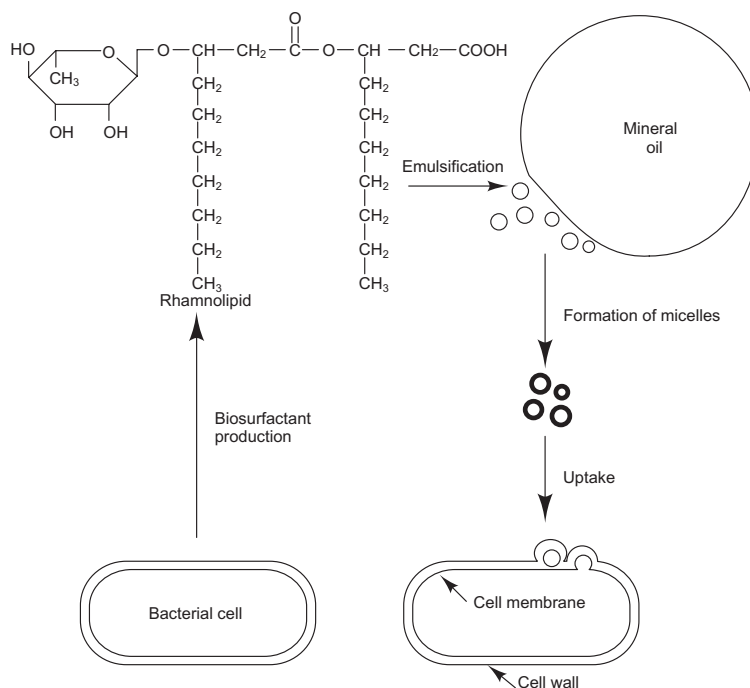


**Figure 14.3** - Alkane degradation in bacteria. Enzymes: (1) *n*-alkane monooxygenase; (2) alcohol dehydrogenase; (3) aldehyde dehydrogenase.



**Figure 14.4** - Metabolic pathways for degradation of cycloaliphatic compounds.

In general, benzene and related compounds are characterized by greater thermodynamic stability than aliphatics ones. The first step in benzene oxidation is a hydroxylation that is catalyzed by a dioxygenase (Fig. 14.2). The product, a diol, is then converted to catechol by a dehydrogenase. These initial reactions, hydroxylation and dehydrogenation, are also common to pathways of degradation of other aromatic hydrocarbons.

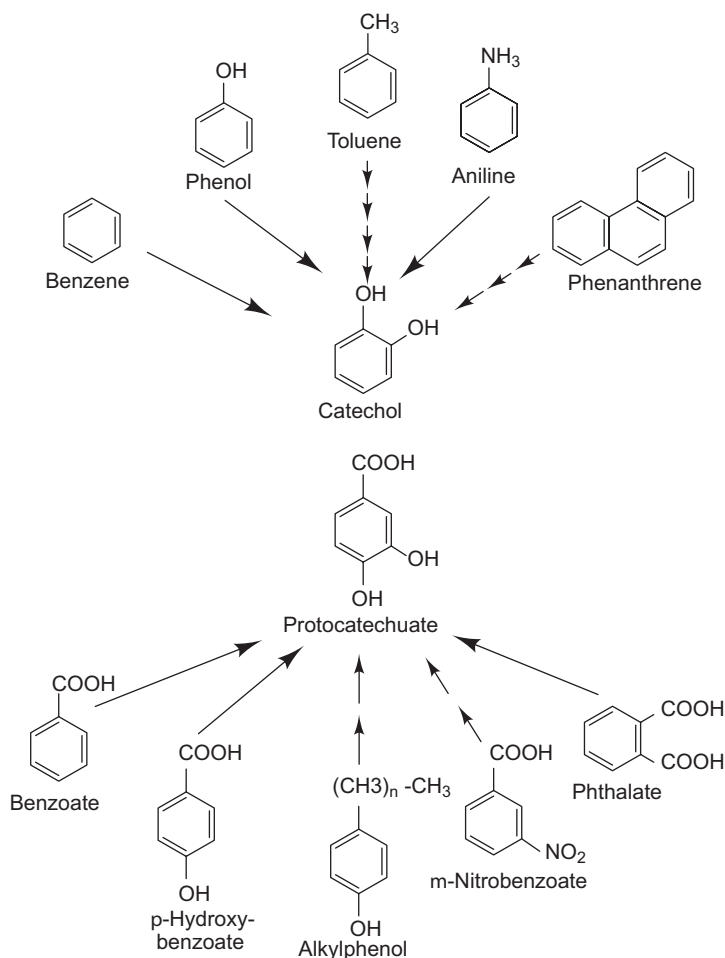


**Figure 14.5** - Degradation of rhamnolipids through biosurfactant production.

The introduction of a substituent onto the benzene ring makes it possible to attack sidechains or to oxidize the aromatic ring by alternative mechanisms. The versatility and adaptability of bacteria are based on the existence of catabolic plasmids. Catabolic plasmids have been found in bacteria, that encode enzymes for degrading such naturally occurring aromatics as camphor, naphthalene and salicylate. Most of the catabolic plasmids are self transmissible and have a broad host range. The majority of Gram-negative soil bacteria isolated from polluted areas possess degradative plasmids (TOL plasmids). The main reaction involved in the oxidation of toluene and related arenes is methyl-group hydroxylation. The methyl group of toluene is oxidized stepwise to the corresponding alcohol, aldehyde and carboxylic group. Benzoate formed or its alkylated derivatives are then oxidized by toluate dioxygenase and decarboxylated to catechol.

The oxygenolytic cleavage of the aromatic ring occurs via *ortho* or *meta* cleavage. Both pathways may be present in one bacterial species. The metabolism of a broad spectrum of aromatic compounds by one species requires the metabolic

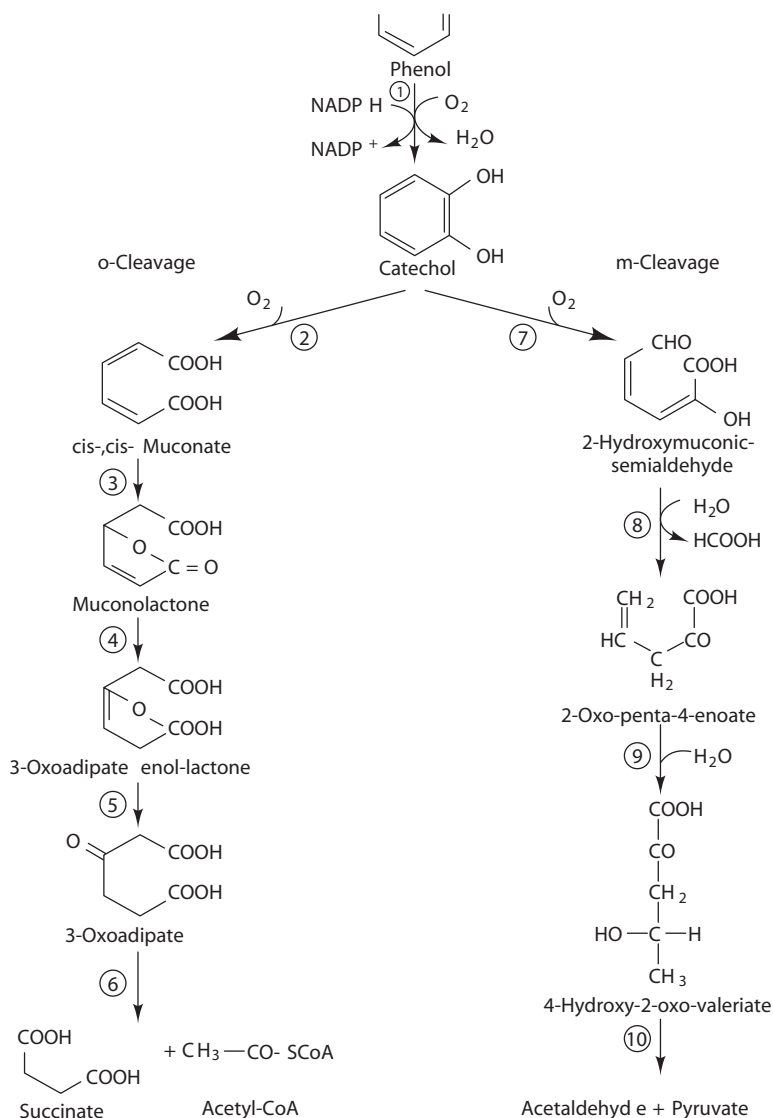
isolation of intermediates into distinct pathways. This kind of metabolic compartmentation seems to be implemented by metabolic regulation. The key enzymes in the degradation of aromatic substrates are induced and synthesized in appreciable amounts only when the substrate or structurally related compounds are present. The enzyme induction depends on the concentration of the inducing molecules. The substrate-specific concentrations represent the threshold of utilization and growth and are in the micromolar range.



**Figure 14.6** - The central metabolic path of degradation of aromatic hydrocarbons through the formation of catechol and protocatechuate intermediates.

In the pathways of oxygenolytic ring cleavage, catechol is oxidized by either intradiol *ortho* cleavage or extradiol *meta* cleavage (Fig. 14.7) catalyzed by specific dioxygenases. The product of the *ortho* cleavage—*cis, cis*-muconate—is transformed into an unstable enol-lactone, which is in turn hydrolyzed to oxoadipate. This dicarboxylic acid is activated by transfer to CoA, followed by thiolytic cleavage to acetyl-CoA and succinate. Protocatechuate is metabolized

by a homologous set of enzymes. The additional carboxylic group is decarboxylated and the double bond is simultaneously shifted to form oxiadipate enol-lactone. The oxygenolytic *meta* cleavage yields 2-hydroxymuconic semialdehyde, which is metabolized by hydrolytic enzymes to formate, acetaldehyde and pyruvate. These are then utilized in the central metabolism.



**Figure 14.7 -** The *ortho*- and *meta*-cleavage pathway of aerobic degradation of aromatic hydrocarbons. Enzymes: (1) phenol monooxygenase; (2) & (7) catechol dioxygenase; (3) muconate-lactonizing enzyme; (4) muconolactone isomerase; (5) oxiadipate enol-lactone hydrolase; (6) oxiadipate-succinyl CoA transferase; (8) 2-hydroxymuconic semialdehyde hydrolase; (9) oxopentaenoate hydrolase; (10) 4-hydroxy-2-oxovalerate aldolase.

## **14.4 ANAEROBIC DEGRADATION OF HYDROCARBONS**

### **14.4.1 Degradation of Saturated Aliphatics**

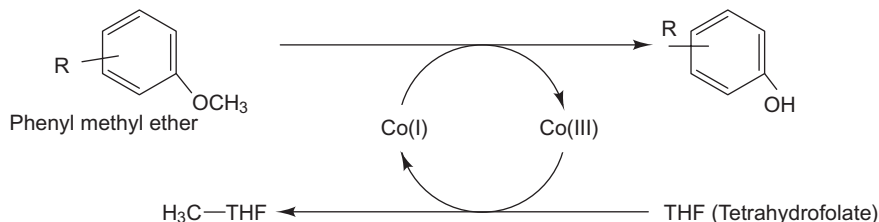
Saturated aliphatic hydrocarbons are attacked only slowly in the absence of oxygen mostly by nitrate-reducing and sulphate-reducing bacteria. Several strains of alkane-oxidizing anaerobes have been isolated, which are specialized for either long-chain ( $C_{12}$ – $C_{20}$ ) or medium-chain ( $C_6$ – $C_{16}$ ) alkanes and use either sulphate or nitrate as electron acceptor. A special example is the anaerobic degradation of methane, e.g., with sulphate as electron acceptor, a process that is of major importance in global carbon transformations. No bacterium that catalyzes this reaction has been isolated so far, although it is thermodynamically feasible. Recent evidence has shown that this reaction is most probably carried out by archaea similar to methanogens, which operate methane formation in the backward reaction, most often in syntrophic association with sulphate-reducing partner organisms.

Unsaturated long-chain hydrocarbons with terminal double bonds can be hydrated to the corresponding primary alcohols and completely degraded. A branched-chain unsaturated hydrocarbon such as squalene is degraded in methanogenic enrichment cultures but the degradation is usually incomplete, probably due to the formation of saturated branched derivatives. Other unsaturated isoprene derivatives such as terpenes have been shown recently to be completely degraded, with nitrate as electron acceptor. Acetylene, a highly unsaturated hydrocarbon, is fermented at a comparatively rapid rate to acetate and ethanol through acetaldehyde, which is formed by a hydratase enzyme. Acetylene hydratase is an iron-sulphur protein containing a tungsten cofactor and is active only in the reduced state, but the reaction mechanism is still unknown. The anaerobes producing this enzyme grow metabolically well in acetylene containing environment but toxicity is felt when the acetylene concentration in the growth medium goes up, indicating that for these microbes acetylene is not the most favoured substrate, when alternative substrates are available. However, the presence of an alternative growth substrate is not a necessity to stimulate the acetylene metabolism but syntrophism significantly facilitates microbial growth vis-à-vis acetylene degradation. No anaerobic degradation has been documented so far for ethylene, propylene, propine and higher homologs having up to six carbon atoms.

### **14.4.2 Degradation of Ether Compounds and Nonionic Surfactant**

Biological ether cleavage in the presence of oxygen employs oxygen as cosubstrate in an oxygenase reaction, which transforms the ether into an unstable hemiacetal. Thus, methyl groups of lignin monomers are released as formaldehyde, not as methanol. Anaerobic demethylation of lignin monomers by

the homoacetogen *Acetobacterium woodii*, *Holophaga foetida* and by several other homoacetogens showed that the methyl group was first transferred as a methyl cation to a fully reduced cob(I)alamin carrier, which later methylates the coenzyme tetrahydrofolate (Fig. 14.8).



**Figure 14.8** - Cobalamin catalyzed anaerobic demethylation of phenyl methyl ether by *Acetobacterium* spp.

A different type of anaerobic ether cleavage has been observed with polyethylene glycol (PEG). The first step in this degradation is a cobalamin-dependent shift of the terminal hydroxyl group to the subterminal C atom, analogous to a diol dehydratase reaction. This reaction again transforms the ether into a hemiacetal derivative that decomposes spontaneously. Since the ether-cleaving enzyme is located in the cytoplasmic space, the polymeric PEG has to cross the cell membrane(s) before it is cleaved inside and the same is true for PEG-containing nonionic surfactants. Since the bacterial strains studied so far are specialized only for the degradation of the PEG chain, the lipophilic residues of the surfactants have to cross the membrane(s) again on their way back. It is obvious that these transport steps considerably limit the applicability of such degradation capacities to treatment of wastewaters rich in such surfactants. Nonetheless, the applicability of anaerobic fixed-bed reactors for treatment of various types of nonionic surfactants to methane, carbon dioxide and mixtures of fatty acids has been demonstrated.

### 14.4.3 Degradation of N-alkyl Compounds and Nitrilotriacetate

The N-alkyl compounds includes several methylated amines such as trimethylamine and tetramethylamine. The former is formed during the initial decay of animal tissue through reduction of trimethylamine-N-oxide by several enterobacteria and many other species. Under strictly anoxic conditions, methanogenic archaea have been found to efficiently demethylate trimethylamine via dimethylamine to monomethylamine and to ferment the methyl moieties by dismutation to methane and carbon dioxide. The cleavage between the nitrogen and the neighbouring carbon atom is accomplished by a nucleophilic attack by a cob(I)alamin, analogous to the demethylation of phenyl methyl ethers by homoacetogenic bacteria. So far, only methanogens have been found to demethylate methylamines, whereas homoacetogens appear to be specialists for

demethylation of phenyl methyl ethers, which in turn are not attacked by methanogens.

N-alkyl compounds of technical and environmental concern include ethylene diamine tetraacetate (EDTA) and nitrilotriacetate (NTA). The latter has largely replaced polyphosphates as a calcium chelator in most commercial washing detergents. The main problem in degrading EDTA is the formation of strong complexes of EDTA with metal ions, which make this substrate very difficult to attack. Nonetheless, microbial EDTA degradation in the presence of oxygen has been documented, but no reports exist on possible anaerobic degradation of EDTA. NTA is degraded aerobically through oxygenase-dependent hydroxylation of one methylene carbon. The resulting hydroxy compound is unstable and releases glyoxylic acid. Removal of an additional carboxymethylene residue produces glycine as coproduct. Anaerobic degradation of NTA is possible with nitrate as electron acceptor. The first degradation step is dehydrogenation to an unsaturated iminium derivative, which upon hydration, can again release glyoxylic acid to form the iminodiacetate derivative. Since oxidation of NTA to the unsaturated derivative has a rather high redox potential ( $> +100$  mV), only nitrate reducers have been found to be able to degrade NTA. Under strictly reducing conditions, i.e., in deeper sediment layers, NTA is stable to microbial attack, because neither sulphate-reducing nor fermenting bacteria can release electrons arising at this high redox potential.

#### 14.4.4 Degradation of S-alkyl Compounds

Dimethylsulfoniopropionate is an osmoprotectant found in several green algae and seaweeds. Its cleavage by anaerobic bacteria leads to acrylate and dimethylsulphide, which can escape into the atmosphere. The thioether dimethylsulphide can also be degraded anaerobically by methanogenic archaea. The carbon-sulphur linkage is cleaved by methyl cation removal, analogous to the demethylation reactions described above (see section 14.4.3).

#### 14.4.5 Degradation of Ketones

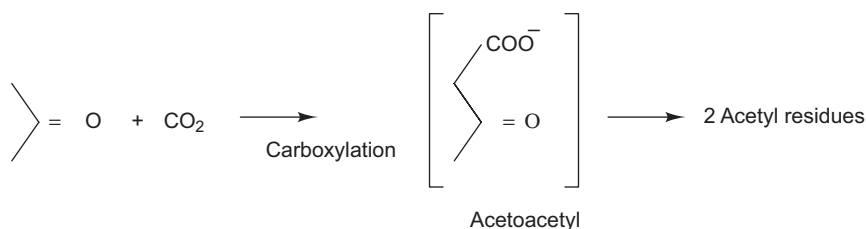
Anaerobic degradation of acetone by nitrate-reducing, sulphate-reducing, or fermenting bacteria cooperating with methanogenic partners uses a carboxylation reaction as primary step of activation, leading to an acetoacetyl derivative that undergoes subsequent cleavage to two acetate moieties (Fig. 14.9). An acetone-carboxylating enzyme complex of high activity has been found in an aerobic *Xanthobacter* strain but the activity in the phototrophic anaerobes like *Rhodobacter capsulatus* is very weak in vitro. The reaction requires one ATP as energy source (which is hydrolyzed to AMP plus two Pi) for the acetone carboxylation. Whether sulphate-reducing or fermenting bacteria provide the necessary energy for carboxylation in different ways has still to be examined. Especially the fermenting bacteria cannot afford to spend the equivalent of two ATP molecules on this carboxylation reaction resulting in very slow degradation



of ketones under anaerobic condition by some specialized strains of bacteria. Higher homologs of acetone appear to be attacked in a similar manner by carboxylations but cometabolic degradation of ketones is always much higher than the metabolic fermentations.

#### 14.4.6 Degradation of Aromatic Compounds

The benzoyl-CoA pathway appears to be the most important one in anaerobic degradation of aromatics, because a broad variety of compounds enter this path, including phenol, various hydroxybenzoates, phenylacetate, aniline, certain cresols and even the hydrocarbon, toluene (Fig. 14.10). Once benzoyl-CoA is formed, the stability of the aromatic ring structure is overcome by a reductive step that introduces two single electrons and protons, through radical intermediates, to form cyclohexadiene carboxyl-CoA as the first identifiable product. Because reduction of the benzene ring to a cyclohexadiene derivative, even with electrons at the ferredoxin level, is endergonic, it requires the investment of energy in the form of two ATP molecules. Nitrate-reducing bacteria can recover this energy investment through further breakdown of the C<sub>7</sub>-dicarboxylic acid derivative produced upon ring cleavage, via beta oxidation to three acetyl-CoA residues that are finally oxidized in the citric acid cycle. Fermenting bacteria and sulphate-reducing bacteria recover only a little energy in the further breakdown of the open-chain intermediate. They may use a different type of reaction for benzoyl-CoA dearomatization that leads to a cyclohexene carboxyl derivative or to a hydroxylated derivative through hydroxylations.



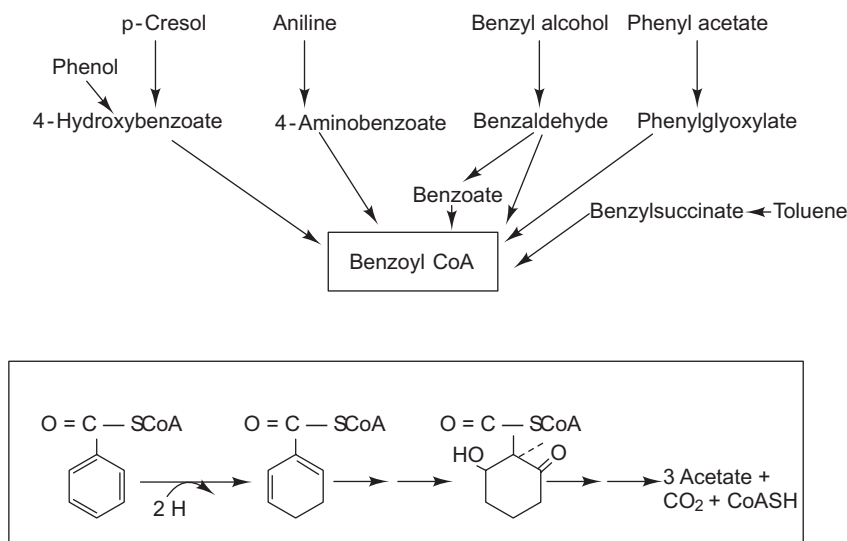
**Figure 14.9** - Anaerobic degradation of acetone by carboxylation reaction.

**(a) Phenol, Hydroxybenzoate and Aniline:** Aromatic compounds, phenol or aniline, are first carboxylated to a *p*-hydroxy or a *p*-amino benzoic acid residue, which is subsequently activated with coenzyme A (Fig. 14.11). The carboxylation of phenol by a nitrate-reducing *Thauera aromatica* strain can be followed *in vitro* with phenyl phosphate as substrate, which is carboxylated to 4-hydroxybenzoate and further degraded as such. The phosphate donor for phenol phosphorylation is still unknown. The overall energy budget of fermentative phenol degradation is very tight and hardly allows spending a full ATP equivalent or even more on this carboxylation reaction. The 4-hydroxybenzoate formed is activated through a ligase reaction, analogous to benzoyl-CoA formation from benzoate to form 4-hydroxybenzoyl-CoA, which is subsequently reductively dehydroxylated to

benzoyl-CoA. 4-Hydroxybenzoate may be degraded anaerobically through the same pathway (see box of Fig. 14.10).

Aniline is degraded anaerobically through a pathway analogous to that for phenol degradation. The initial activation is accomplished through carboxylation to 4-aminobenzoate, which is subsequently activated to 4-aminobenzoyl-CoA and undergoes reductive deamination to benzoyl-CoA. Aminobenzoates, diaminobenzenes and aminohydroxybenzenes are degraded very slowly in anaerobic enrichment cultures and nothing is known about the degradation pathways.

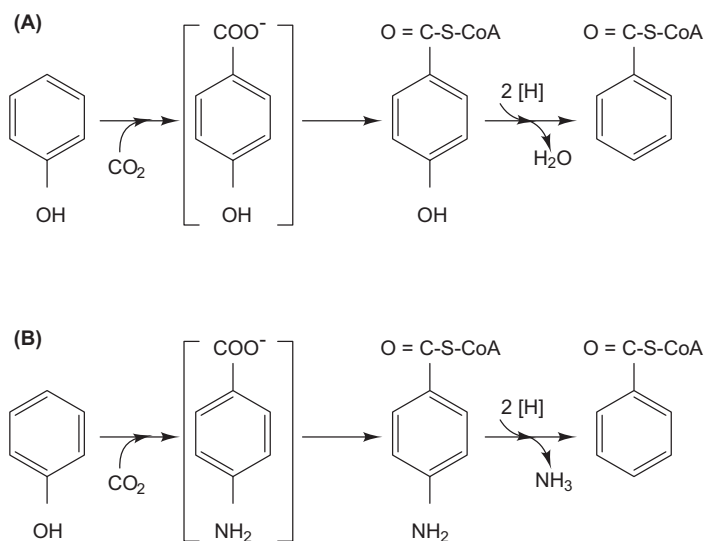
**(b) Cresols:** Cresols (methyl phenols) are anaerobically degraded through three different pathways, depending on the type of substitution. *p*-Cresol is hydroxylated at the methyl group by an oxygen-independent reaction, probably through a quinomethide intermediate as suggested earlier for an aerobic *Pseudomonas* strain (Fig. 14.12A). *o*-Cresol can be carboxylated to 3-methyl-4-hydroxybenzoate and further degraded as such (Fig. 14.12B). The pathway of anaerobic *m*-cresol degradation, elucidated recently in the sulphate-reducing bacterium *Desulfobacterium cetonicum*, follows a strategy analogous to anaerobic toluene degradation by nitrate-reducing bacteria. The methyl group of *m*-cresol adds to fumarate to form 3-hydroxybenzylsuccinate. Activation and  $\beta$ -oxidation lead to succinyl-CoA and benzoyl-CoA (Fig. 14.12C).



**Figure 14.10** - Benzyl CoA pathway of anaerobic degradation of aromatic compounds. The box shows anaerobic degradation of benzoate by the nitrate-reducing bacterium *Thauera aromatica* (Adopted from Schink, 2005).

**(c) Hydroquinone and Catechol:** Hydroquinone is degraded by sulphate-reducing and fermenting bacteria. The degradation pathway, studied in a sulphate-reducing *Desulfococcus* strain and a fermenting bacterium *Syntrophus*

*gentianae* showed that hydroquinone is carboxylated to gentisic acid. The later is activated to gentisyl-CoA through a CoA-ligase reaction and reductively dehydroxylated to benzoyl-CoA, which enters the modified benzoyl-CoA pathway. The dehydroxylation of both hydroxyl groups appears to proceed in a single step. Alternatively, gentisic acid is utilized by the fermenting bacterium.



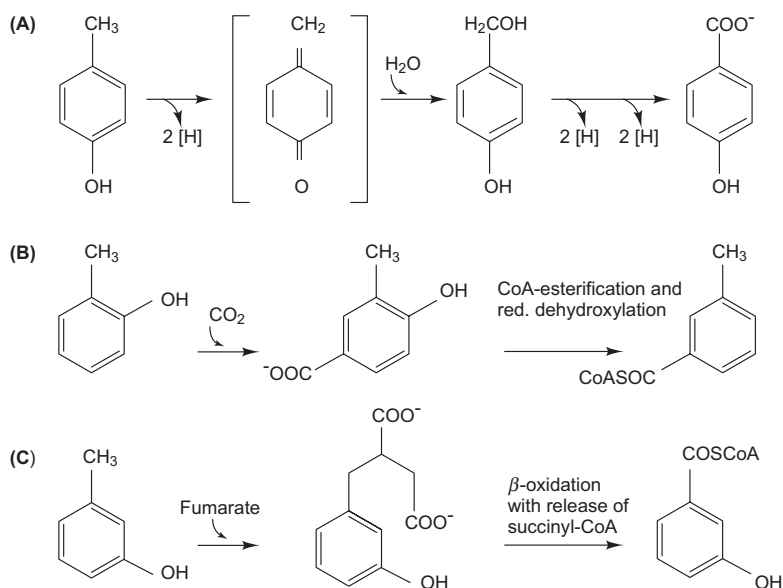
**Figure 14.11** - Anaerobic degradation of (A) phenol and (B) aniline by *Pseudomonads* (Adopted from Schink, 2005).

Catechol, the key intermediate in aerobic breakdown of aromatic compounds, is by far the slowest phenolic compound to be degraded under anoxic conditions. A *Desulfobacterium* strain carboxylates catechol to protocatechuate, which is activated to form protocatechuy-CoA and subsequently dehydroxylated to benzoyl-CoA. Efforts to isolate nitrate-reducing or fermenting bacteria with hydroquinone or catechol as substrate have failed so far.

**(d) Resorcinol:** An entirely different strategy is used in the anaerobic degradation of resorcinol and its derivatives. The two hydroxyl groups in resorcinol are in positions relative to each other that allow tautomerization to a cyclohexene dione derivative having three isolated double bonds. Cell-free extracts of a fermenting *Clostridium* strain convert resorcinol to 5-oxohexanoate, probably through a nucleophilic attack on one of the carbonyl carbon atoms. The resorcinol carboxylates  $\beta$ - and  $\gamma$ -resorcyrate are degraded by the same fermenting bacterium after decarboxylation to resorcinol.

The nitrate reducer *Azoarcus anaerobius* destabilizes the ring by introducing an additional hydroxyl group to form hydroxyhydroquinone. The enzyme involved is membrane-bound and the hydroxylation is coupled to reduction of nitrate to nitrite. In a later oxidation step, hydroxyhydroquinone is oxidized to hydroxybenzoquinone.

**(e) Trihydroxybenzenes and Trihydroxybenzoates:** Among the three trihydroxybenzene isomers, pyrogallol and phloroglucinol are degraded quickly by fermenting bacteria. Phloroglucinol degradation has been studied in detail with *Eubacterium oxidoreducens* and *Pelobacter acidigallici*. Phloroglucinol is reduced by an NADPH-dependent reductase to dihydrophloroglucinol and the same strategy is followed by *Holophaga foetida* strain TMBS4. Hydrolytic ring cleavage leads to 3-hydroxy-5-oxohexanoic acid, which is thiolitically cleaved to three acetate residues. Pyrogallol is hydrolyzed directly but is isomerized to phloroglucinol through a transhydroxylation reaction.



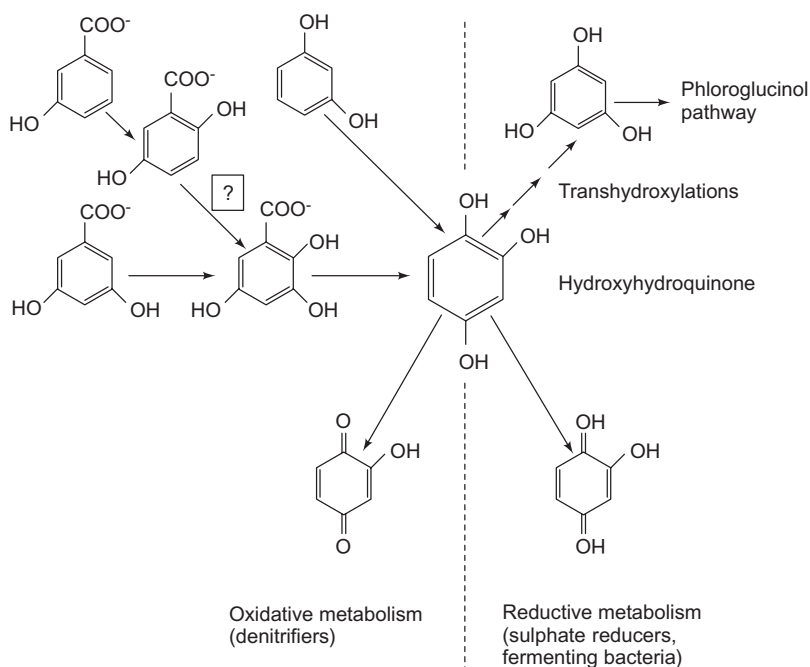
**Figure 14.12** - Anaerobic degradation of (A) *p*-cresol, (B) *o*-cresol and (C) *m*-cresol (Schink, 2005).

The third trihydroxybenzene isomer, hydroxyhydroquinone, is converted by the fermenting bacterium *Pelobacter massiliensis* to three acetates via three subsequent transhydroxylation reactions, analogous to the pyrogallol-phloroglucinol transhydroxylation (Fig. 14.13). Alternative pathways of degradation has been found in nitrate-reducing and sulphate-reducing bacteria. The sulphate-reducing bacterium *Desulfovibrio inopinatus* destabilizes hydroxyhydroquinone by reduction to dihydroxyhydroquinone, to form acetate and an unidentified four-carbon derivative. Since the bacterium is unable to oxidize acetate, the final products are two acetate and two carbon dioxide and one mole of sulphate is reduced concomitantly to sulphide.

#### 14.4.7 Degradation of Halogenated Organics

Halogenated organics are widespread in nature and are formed especially as secondary metabolites by plants, marine algae, fungi and certain bacteria at low

rates. They also include a majority of synthetic halogenated compounds. Dehalogenation can proceed basically through an oxidative, a hydrolytic, or a reductive reaction. Among anaerobic bacteria, the reductive elimination of halogen substituents is the most common type of reaction. Nearly all chlorinated organics can be dehalogenated and further degraded in strictly anaerobic microbial cultures. As a rule, reductive dechlorination is the preferred process for degradation of compounds with higher degrees of halogenation. The reaction is an excellent electron sink and highly halogenated compounds are much more amenable to a nucleophilic attack on the respective carbon atoms than to an oxidative reaction. The electrons derived from molecular hydrogen, formate, or more complex organic compounds are transferred to the halogenated substrate to release the organic residue in a reduced form, together with chloride. It has been shown in several instances that this redox process can yield metabolic energy through a respiratory mechanism, which implies that the process may establish a net translocation of protons across the cytoplasmic membrane and that the proton gradient drives ATP synthesis through a membrane-bound ATP synthase complex.



**Figure 14.13** - The hydroxyhydroquinone pathway for anaerobic microbial degradation of various aromatic hydrocarbons.

## 14.5 MICROBIAL DEGRADATION OF PESTICIDES

Microorganisms differ widely in their ability to metabolize synthetic pesticides. Whilst a large number of reactions of pesticides have been observed in the presence of microorganisms, many workers have failed to differentiate enzymatic

processes from those resulting from the extracellular or intracellular conditions of redox and pH, or from the presence of heavy metals, etc. Although details of many enzyme reactions with pesticides are still lacking, the volume of knowledge is steadily increasing. Studies of the oxygenases, for example, are well-advanced and many of these enzymes have been isolated, purified and prepared in crystalline form. Investigation of the requirements, and complex for enzyme activity have mostly progressed simultaneously. *Pseudomonas putida*, for example, can hydroxylate camphor in the presence of NADH, molecular oxygen, a flavoprotein and the iron sulphur protein putidaredoxin. Studies on the degradation of organophosphorus insecticides have progressed with the purification and characterization of phosphotriesterases. Many other enzymes are now known to cause degradation of xenobiotics, excepting limited progress on the metabolism of pyrethroid insecticides, a new entrant into the environment.

The principal reactions involved in pesticide metabolism, indeed in the metabolism of most organic compounds include  $\beta$ -oxidation, oxidative dealkylation, thioether oxidation, phosphorothionate oxidation, epoxidation of carbon-carbon double bonds, hydroxylation, aromatic ring cleavage, hydrolysis, dehalogenation, nitro-reduction, condensation and conjugate formation. The number of reactions is lesser compared to the number of compounds metabolized, reflecting the relatively small number of chemical types. It is no surprise that there have been many observed correlations between the chemical structure of different pesticides and their routes of metabolism. The numbers and types of substituents can all influence the rate of metabolism. The positions of linkage of side chains onto aromatic rings can also affect the reaction rate.

The pesticides serve as a carbon, nitrogen and/or energy source for the microorganism, or are used as a substrate for cometabolism. There is now considerable evidence that the microbial metabolism of some pesticides involves co-oxidation stages both under controlled and field conditions. The site (chromosomal or plasmid) of genetic information for coding the enzymes responsible for pesticide metabolism has received much attention in the recent years. The occurrence of bacterial drug resistance factors composed of extra-chromosomal infectious plasmids and non-infectious resistance determinants is well documented. The phosphotriesterase (PTE) gene (encoding the enzyme PTE) of *Pseudomonas diminutum* has been well-characterized, transformed and expressed in other bacteria. The PTE homologous sequences have been found in the genome of *Synechocystis* PCC 6803 and *Anabaena* PCC 7119. Herbicide resistance genes have been used for the production of transgenic crops. There are evidences for the presence of bacterial transfer factors able to confer specific pesticidal degradative ability upon a microbial recipient. These factors may carry genetic information controlling the enzymes that are responsible for the degradation of the concerned pesticides and may effectively preserve the ability of natural microbial populations to metabolize these chemicals. Extensive reports are also available on the enzymatic activities, structural complexity and variations of Lindane Induced Proteins (LIPs). The *lin* genes encoding dehydrochlorinase

enzymes have been well-characterized in *Sphingomonas paucimobilis* in the presence of different isomers of HCH and other OC insecticides.

### 14.5.1 Microbial Degradation of Organochlorines

Many organochlorine insecticides, including DDT, lindane ( $\gamma$ -BHC), endrin, endosulfan, aldrin, dieldrin, chlordane, heptachlor and methoxychlor have been frequently used in the past in different agricultural practices. Although the mode of degradation of these compounds in natural environments can be mediated through chemical or biological methods, the later processes perhaps play the most significant role in their dissipation. More often, microbial degradation of these chemicals is hindered by their low water solubility and thus is not readily accessible to microbial or enzymatic attack. Biological conversion of organochlorines via dechlorination, dehydrochlorination, oxidation and/or isomerization, are the common mechanisms of their degradation in natural habitats and many microbes have the ability to metabolise these insecticides (Table 14.2).

Microbial conversion of DDT, the most rampantly used OC in the past, occurs through (a) reductive dechlorination and (b) dehydrochlorination. The facultative anaerobes, *Aerobacter aerogenes*, *Escherichia coli*, *Proteus vulgaris*, *Clostridium* sp. and *Klebsiella pneumoniae* dechlorinate DDT to DDD. The cell-free extracts of *A. aerogenes*, *Pseudomonas fluorescens*, *E. coli* and *K. pneumoniae* dechlorinate *p,p'*-DDT to DDE, DDD, DDMU and DDNU exclusively under anaerobic conditions. Dechlorination of DDT is slightly accelerated in the presence of light-activated flavin mononucleotide (FMN). Six actinomycetes belonging to *Nocardia* and *Streptomyces* genera can metabolise DDT under natural condition. In contrast, *Mucor alternans* in shake cultures converts DDT to five unidentified metabolites. A baker's yeast can convert DDT to DDD through reductive dechlorination, without forming DDE as an intermediate. Many soil microorganisms, mainly bacteria, are able to degrade other OC insecticides like endrin, aldrin and dieldrin. Mixed bacterial cultures containing *Pseudomonas*, *Acetomonas* and *Acinetobacter* sp. are more efficient in degrading DDT than when act individually.

DDM, an analog of DDT, is cometabolized by a sewage sludge bacterium, *Hydrogenomonas*, under aerobic conditions. *Arthrobacter* sp. can convert *p*-chlorophenyl acetate to 4-chloro-3-hydroxyphenyl acetate and an unidentified metabolite. A *Pseudomonas* sp., which utilize diphenyl ethane as sole carbon source, also metabolizes several monochlorinated and nonchlorinated analogues of DDT affecting ring cleavage. *Pseudomonas putida* growing on diphenyl methane and benzhydrol, cometabolizes bis(*p*-chlorophenyl) methane producing several dehalogenated products. This bacterium also affected ring cleavage on certain products, but failed to cometabolize these in the presence of DPM. On formation by *P. putida* grown in DPM salts broth, BHE was again cometabolized by the resting cells of this bacterium to BH and BP. It has also been observed that a strain of *Sphingomonas paucimobilis* can effectively degrade  $\gamma$ -HCH as well as



lindane to harmless products. The exposure of the strain to elevated concentrations of the insecticides caused induction of synthesis of seven novel polypeptides and concomitantly equipped the strain to degrade the insecticides.

### 14.5.2 Microbes in Degradation of Organophosphates

Microbial rather than chemical processes are the principal means of detoxification of organophosphates observed in natural ecosystems. Microbial degradation through hydrolysis of P-O alkyl and P-O-aryl bonds are considered as the most significant steps in the detoxification of these insecticides. An *Arthrobacter* sp. and a *Streptomyces* sp. isolated through selective enrichment technique, synergistically utilize diazinon and release CO<sub>2</sub> from the pyrimidinyl ring. Similarly, an adapted species of *Flavobacterium* also hydrolyzes diazinon to IMHP in mineral solution. This bacterium also hydrolyzes parathion and dursban to more unstable products but can not metabolize malathion. *p*-Nitrophenol, the major hydrolytic product of parathion degradation, does not serve as energy source for this *Flavobacterium* sp. whereas IMHP is completely mineralized to CO<sub>2</sub>. *Flavobacterium* sp. also hydrolyzes methyl parathion and fenitrothion. The enzyme for this degradation has been purified and identified as phosphotriesterase. Pseudomonads also use diazinon or malathion as a sole source of phosphorus and even hydrolyse aspon, monocrotophos, fensulfothion, diazinon, malathion, acephate, parathion and trithion, but they fail to hydrolyze dimethoate, trichlorofon, methyl parathion and dichlorvos. *Pseudomonas stutzeri* and *P. aeruginosa* very effectively metabolise parathion and the degrading potential of the bacteria increases with prolonged exposure due to adaptive enzyme synthesis. Fenitrothion can be decomposed by a soil bacterium, *Bacillus subtilis*, via nitro group reduction. *Pseudomonas diminutum* and a *Flavobacterium* sp. are known to degrade many phosphorothioate and vinyl OP insecticides (dimethoate, monocrotophos, quinalphos, chlorfenvinphos, etc.). Dimethoate has been found to be degraded by the fungus *Aspergillus niger*. The insecticide-degrading enzyme has been purified from *A. niger* ZHY256 strain and has been found to be very similar to the bacterial enzyme (Table 14.3).

Biological hydrolysis of methyl parathion is the principal means of degradation observed in soil. Methyl parathion is cometabolically degraded by a mixed bacterial culture obtained from oligotrophic or eutrophic waters. *Pseudomonas* and *Acinetobacter* spp. can effectively degrade *p*-nitrophenol but not methyl parathion. A pesticide-degrading *Bacillus* sp. can cometabolize methyl parathion. Many *Pseudomonas* sp. have the ability to utilize parathion and methyl parathion in the presence of an organic carbon source.

Numerous bacterial isolates mainly belonging to *Bacillus* and *Arthrobacter* sp. have been isolated from parathion-treated Israeli (Gilat) soil. These species have exceptional ability to hydrolyze parathion to *p*-nitrophenol. Many microorganisms, including bacteria, algae fungi and yeasts can reduce parathion in pure culture. The soil fungi *Trichoderma viride* and *Penicillium waksmanii* can convert parathion to aminoparathion in pure cultures. The alga, *Chlorella pyrenoidosa*, was reported to

**Table 14.2:** Microorganisms known for the degradation of organochlorine insecticides in cultures and field conditions.

| <i>Name of the Insecticide</i> | <i>Bacteria</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <i>Fungi</i>                                                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Aldicarb                       | <i>Aerobacter</i> sp., <i>Bacillus</i> sp., <i>E. coli</i> , <i>Erwinia</i> sp., <i>Pseudomonas morsprunorum</i> , <i>P. putida</i> , <i>P. fluorescens</i> , <i>Pseudomonas</i> sp.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>Mucor</i> sp.<br><i>Fusarium</i> sp.                                                            |
| DDT                            | <i>Acetomonas</i> sp., <i>Acinetobacter</i> sp., <i>Actinomyces</i> sp., <i>Aerobacter aerogenes</i> , <i>Agrobacterium tumefaciens</i> , <i>Arthrobacter</i> sp., <i>Bacillus cereus</i> , <i>B. coagulans</i> , <i>B. megaterium</i> , <i>B. subtilis</i> , <i>Clostridium pasteurianum</i> , <i>C. michiganense</i> , <i>Enterobacter aerogenes</i> , <i>Erwinia</i> sp., <i>E. amylovora</i> , <i>E. ananas</i> , <i>E. carotovora</i> , <i>E. crysanthemii</i> , <i>Escherichia coli</i> , <i>Hydrogenomonas</i> sp., <i>Klebsiella pneumoniae</i> , <i>Klebsiella</i> sp., <i>Kurthia zopfii</i> , <i>Micrococcus</i> sp., <i>Nocardia</i> sp., <i>Proteus vulgaris</i> , <i>Pseudomonas</i> sp., <i>P. putida</i> , <i>P. fluorescens</i> , <i>P. glycinea</i> , <i>P. marginalis</i> , <i>P. morsprunorum</i> , <i>P. syringae</i> , <i>P. tabaci</i> , <i>Streptococcus</i> sp., <i>Rhizobium</i> spp., <i>Streptomyces annamoneus</i> , <i>S. aurofaciens</i> , <i>S. viridochromogenes</i> , <i>Xanthomonas pruni</i> , <i>X. stewartii</i> , <i>X. uredovorus</i> , <i>X. vesicatoria</i> | <i>Aspergillus niger</i> , <i>Mucor alternans</i> , <i>Mucor</i> sp., <i>Penicillium</i> sp.       |
| Endosulfan                     | <i>Acinetobacter</i> sp., <i>Aerobacter aerogenes</i> , <i>Agrobacterium</i> sp., <i>Bacillus subtilis</i> , <i>Bacillus</i> sp., <i>Clostridium</i> sp., <i>C. pasteurianum</i> , <i>Erwinia ananas</i> , <i>Rhizobium</i> spp., <i>Streptomyces</i> sp., <i>Streptomyces</i> sp. MT21, <i>S. aurofaciens</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Aspergillus</i> sp.<br><i>Mucor</i> sp.<br><i>Claviceps</i> sp.<br><i>Penicillium</i> sp.       |
| HCH                            | <i>Acetomonas</i> sp., <i>Aerobacter aerogenes</i> , <i>Arthrobacter</i> sp., <i>Bacillus cereus</i> , <i>B. megaterium</i> , <i>Citrobacter freundii</i> , <i>Clostridium</i> sp., <i>C. rectum</i> , <i>E. coli</i> , <i>P. fluorescens</i> , <i>P. putida</i> , <i>P. diminutum</i> , <i>Sphingomonas</i> sp., <i>S. paucimobilis</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>Aspergillus</i> sp.<br><i>Mucor alternans</i><br><i>Rhizopus</i> sp.,<br><i>Saprolegnia</i> sp. |
| Aldrin                         | <i>Acinetobacter</i> sp., <i>Arthrobacter</i> sp., <i>Bacillus</i> sp., <i>Clostridium</i> sp., <i>Corynebacterium</i> sp., <i>Micrococcus</i> sp., <i>Micromonospora</i> sp., <i>Erwinia ananas</i> , <i>Nocardia</i> sp., <i>Pseudomonas</i> sp., <i>P. putida</i> , <i>P. fluorescens</i> , <i>P. tabaci</i> , <i>Streptococcus</i> sp., <i>Streptomyces</i> sp., <i>Thermoactinomyces</i> sp.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Aspergillus niger</i> ,<br><i>Mucor</i> sp. MT4,<br><i>Penicillium</i> sp.                      |

Table 14.2: (Contd.)

|           |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Dieldrin  | <i>Actinomyces</i> sp., <i>Aerobacter aerogenes</i> ,<br><i>Aerobacter</i> sp., <i>Arthrobacter</i> sp., <i>Bacillus</i> sp.,<br><i>B. megaterium</i> , <i>Corynebacterium</i> sp.,<br><i>Micrococcus</i> sp., <i>Nocardia</i> sp., <i>Pseudomonas</i><br>sp., <i>P. melophthora</i> , <i>Sphingomonas</i> sp.                                                                                                          |                                                                                                                       |
| Endrin    | <i>A. aerogenes</i> , <i>Arthrobacter</i> sp. <i>Bacillus</i> sp.,<br><i>B. subtilis</i> , <i>B. cereus</i> , <i>B. megaterium</i> ,<br><i>Corynebacterium</i> sp., <i>Micrococcus</i> sp.,<br><i>Pseudomonas</i> sp., <i>P. putida</i> , <i>P. fluorescens</i> ,<br><i>Rhizobium</i> spp., <i>Streptococcus</i> sp.,<br><i>Streptomyces</i> sp., <i>Sphingomonas</i><br><i>paucimobilis</i> , <i>Xanthomonas pruni</i> | <i>Aspergillus</i> sp.<br><i>Penicillium</i> sp.<br><i>Rhizopus</i> spp.                                              |
| Toxaphene | <i>Aerobacter</i> sp., <i>A. aerogenes</i> , <i>Bacillus</i> sp., <i>B.</i><br><i>subtilis</i> , <i>B. cereus</i> , <i>B. megaterium</i> ,<br><i>Corynebacterium</i> sp., <i>Clastridium</i> sp., <i>E. coli</i> ,<br><i>Erwinia</i> spp., <i>Streptomyces</i> sp.,<br><i>Sphingomonas</i> sp., <i>Pseudomonas</i> sp.,<br><i>Xanthomonas</i> sp., <i>X. pruni</i>                                                      | <i>Albugo candida</i> ,<br><i>Alternaria</i> sp.,<br><i>Fusarium</i> sp.,<br><i>Mucor</i> sp.,<br><i>Rhizopus</i> sp. |

reduce parathion to its amino analog. *Chlorella vulgaris* also show tolerance to many OP insecticides (dimethoate, monocrotophos, quinalphos, parathion and chlorfenvinphos) through esterase mediated hydrolytic degradation of the chemicals. Very little information is available on the microbial degradation of chlorpyrophos in soil, water and sediments. Phosdrin and phorate degradation is believed to be microbial but proper identification of the species has not yet been made. Demeton has been found to be degraded by *Pseudomonas putida* culture isolated from soil and grown with disodium formate. A *Nocardia* sp. isolated from the same soil showed similar results.

Prolonged exposure of the cyanobacteria *Anabaena* sp. PCC 7119 and *Synechocystis* sp. PCC 6803 to different OP insecticides separately or in combination provides tolerance to the cyanobacteria against the chemicals (Biswal, 2007). Preexposed strains of *Synechocystis* showed an enhanced growth performance in the insecticide-containing medium. For example, at sub-lethal concentrations of dimethoate, the preexposed strains of *Anabaena* showed five times more photosynthetic activity than of the normal strain. Similar tolerance has also been reported in the cyanobacterium to other OP insecticides. However, none of the preexposed strains showed cross tolerance indicating that there is chemical specific tolerance development in cyanobacteria. A comparison of the degradation capacity of *Anabaena* sp. PCC 7119, grown in normal and sterile soils, showed that the cyanobacteria, soil bacteria assemblage is more effective in removing dimethoate from soil than the individual group. Multiple species in an association are more effective degraders of OP insecticides than of individual species (Sahu, 2007).

**Table 14.3:** Microorganisms known for metabolism of organophosphate pesticides in cultures and field conditions.

| Name of the Insecticides | Bacteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fungi                                                                                                                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diazinon                 | <i>Acetomonas</i> sp., <i>Arthrobacter</i> spp., <i>Flavobacterium</i> sp., <i>Hydrogenomonas</i> sp., <i>Erwinia</i> sp., <i>E. carotovora</i> , <i>E. ananas</i> , <i>Kurthia</i> sp., <i>Micrococcus</i> sp., <i>Pseudomonas</i> sp., <i>P. aeruginosa</i> , <i>P. diminutum</i> , <i>P. melophthora</i> , <i>P. glycinea</i> , <i>Streptomyces</i> sp., <i>S. griseus</i>                                                                                                                                                                                                | <i>Aspergillus oryzae</i> , <i>A. niger</i> , <i>Mucor</i> sp., <i>Fusarium</i> sp., <i>Penicillium</i> sp., <i>Trichoderma</i> sp.                                                                      |
| Chlorpyrifos             | <i>Arthrobacter</i> sp., <i>Corynebacterium</i> sp., <i>Flavobacterium</i> sp., <i>Erwinia carotovora</i> , <i>Kurthia</i> sp., <i>Nocardia</i> sp., <i>Pseudomonas</i> sp., <i>P. diminutum</i> , <i>Streptomyces</i> sp., <i>S. griseus</i> , <i>Streptococcus</i> sp., <i>Bacillus</i> sp., <i>B. subtilis</i>                                                                                                                                                                                                                                                            | <i>Aspergillus</i> sp., <i>Claviceps</i> sp., <i>Mucor</i> sp., <i>Rhizopus</i> sp., <i>Penicillium waksmanii</i> , <i>Fusarium</i> sp., <i>Rhizopus</i> sp.                                             |
| Dichlorvos               | <i>Arthrobacter</i> sp., <i>Bacillus</i> sp., <i>B. subtilis</i> , <i>B. coagulans</i> , <i>B. cereus</i> , <i>Erwinia</i> sp., <i>E. amylovora</i> , <i>Enterobacter</i> sp., <i>Pseudomonas</i> sp., <i>P. fluorescens</i> , <i>P. glycinea</i> , <i>P. tobaci</i> , <i>P. melophthora</i> , <i>P. diminutum</i> , <i>Flavobacterium</i> sp., <i>Streptomyces</i> sp.                                                                                                                                                                                                      | <i>A. niger</i> , <i>A. smithii</i> , <i>Mucor alternans</i> , <i>Mucor</i> sp., <i>Saprolegnia</i> sp., <i>Fusarium</i> sp., <i>Penicillium waksmanii</i> , <i>P. notatum</i> , <i>Phytophthora</i> sp. |
| Dimethoate               | <i>Clostridium</i> sp., <i>Flavobacterium</i> spp., <i>Kurthia</i> sp., <i>Micrococcus</i> sp., <i>Nocardia</i> sp., <i>Proteus vulgaris</i> , <i>Pseudomonas</i> sp., <i>P. diminutum</i> , <i>P. fluorescens</i> , <i>P. aeruginosa</i> , <i>P. melophthora</i> , <i>P. putida</i> , <i>Streptococcus</i> sp., <i>Xanthomonas</i> sp., <i>Rhizobium meliloti</i>                                                                                                                                                                                                           | <i>Aspergillus niger</i> , <i>Fusarium</i> sp., <i>Claviceps</i> sp., <i>Rhizopus</i> spp., <i>Mucor</i> sp., <i>Penicillium notatum</i>                                                                 |
| Malathion                | <i>Acinetobacter</i> sp., <i>Arthrobacter</i> sp., <i>Enterobacter</i> sp., <i>Erwinia</i> sp., <i>Klebsiella pneumoniae</i> , <i>Microsomonas</i> sp., <i>Micrococcus</i> sp., <i>Nocardia</i> sp., <i>Pseudomonas</i> sp., <i>P. aeruginosa</i> , <i>P. diminutum</i> , <i>P. fluorescens</i> , <i>P. melophthora</i> , <i>Proteus</i> sp., <i>Bacillus</i> sp., <i>B. cereus</i> , <i>B. subtilis</i> , <i>B. megaterium</i> , <i>Streptococcus</i> sp., <i>Streptomyces</i> sp., <i>S. aurofaciens</i> , <i>Rhizobium</i> spp., <i>R. japonicum</i> , <i>R. meliloti</i> | <i>Saprolegnia</i> sp., <i>Mucor</i> sp., <i>Rhizopus</i> sp., <i>Aspergillus</i> sp., <i>A. niger</i> , <i>Penicillium</i> sp., <i>P. waksmanii</i> , <i>Trichoderma viride</i>                         |

The metabolites of the OP degradation are actively used as the sole carbon source by many species of bacteria. A bacterial isolate, *Pseudomonas putida* can utilize alkylphosphonates as sole carbon, nitrogen and phosphorus source. Several bacterial species utilize the products of many organophosphorus pesticides containing ionic phosphorus groups in a metabolic way to support their growth even in the presence of oxidizable substrates. *Pseudomonas testoteroni*, use *o*-alkylphosphonates as sole source of phosphorus and produce more natural products such as alcohols, alkanes and inorganic phosphates. This bacterium cleaves the C-P bond of various phosphonates and forms methane or ethane from phosphorus-bound methyl or ethyl groups under aerobic conditions. *p*-Nitrophenol, the major hydrolytic product of parathion and methyl parathion, is metabolized by *Pseudomonas* spp. and *Corynebacterium* spp. Various strains of *Flavobacterium* spp. and *Nocardia* spp. utilize *o*- and *p*-nitrobenzoic acid and *o*- and *p*-nitrophenol as nutrient sources. *Pseudomonas fluorescens* can hydrolyze gusathion and cleave the heterocyclic ring forming anthranilic acid.

A strain of *Trichoderma viride* and a *Pseudomonas* sp. were isolated from soil and could hydrolyze malathion; both organisms possess soluble carboxyesterase activity, the most important mechanism involved in malathion hydrolysis. Another soil fungus, *Penicillium notatum* also metabolizes malathion and use it as a nutrient source. Two species of *Rhizobium*, *R. leguminosarum* and *R. trifolii*, isolated from an Egyptian soil, hydrolyzed malathion using a carboxyesterase and a demethylation process. *Aspergillus oryzae* and an *Arthrobacter* sp. can degrade malathion to  $\beta$ -malathion monoacid and malathion dicarboxylic acid products, similar to bacterial decompositions. The degradation of malathion by *A. oryzae* is slow, but the rate of malathion transformation is proportional to the increase in dry weight of fungal biomass, suggesting that the fungus utilizes malathion as a nutrient source.

### 14.5.3 Microbes in Metabolism of Carbamates

Many soil microorganisms metabolize carbamate pesticides to non-toxic metabolites (Table 14.4). *Pseudomonas melophthora*, a bacterial symbiont of the apple maggot, degrades carbaryl to support its growth in the culture and similar activity is also reported in natural environment. Several bacterial and fungal isolates have been obtained from carbaryl-treated soils and these organisms degrade the insecticide in a mineral medium, but at a slower rate. The metabolism of carbaryl is more in flooded rice field and is dependent on soil types, pH and redox potential. The bacterium, *Pseudomonas cepacia*, isolated from a carbofuran-amended flooded soil is known to cometabolize carbaryl in presence of yeast extract or soil extract. Different species of fungi, *Gliocladium roseum*, *Trichoderma viride* and the species of the genera *Aspergillus*, *Mucor*, *Penicillium* and *Fusarium* degrade carbaryl either by side chain or ring hydroxylations, producing water-soluble metabolites. An isolate of *Pseudomonas melophthora* degrades carbaryl to several metabolites. Several microorganisms isolated from marine environments are capable of using carbaryl as source of energy and

produce hydroxylated water-soluble products. In contrast, species of *Brevibacterium* and *Spirillum* degrade carbaryl and produce only ether-soluble metabolites. *Aspergillus terreus* degrades carbaryl at low concentrations. An *Achromobacter* sp. uses carbaryl as sole carbon source.

Many cyanobacteria are known to degrade carbaryl in the culture conditions, in the presence of growth-supporting nutrients. The cyanobacterium *Westiellopsis prolifica* metabolizes carbaryl more effectively than other carbamate insecticides. Carbaryl degradation has also been observed in flooded rice fields by the indigenous cyanobacterial flora. The cyanobacteria *Nostoc linckia*, *Aulosira fertilissima*, *Anabaena doliolum* and *Calothrix* sp. have been found as effective degraders of carbaryl.

The breakdown of carbofuran is much faster than carbaryl and occurs primarily either via hydrolysis of the carbamate ester, resulting in the formation of a phenol or by the oxidation of the methylene group of the furanyl ring forming alcohols and ketones. Due to the water-soluble nature of carbofuran and resulting metabolites, no serious bio-accumulation or other toxic problems of carbofuran appears in natural environments. Biological hydrolysis is the principal pathway of degradation of carbofuran in natural condition. *Pseudomonas cepacia*, from carbofuran-amended flooded soil, decomposes the insecticide very effectively by using it as a carbon source. Additions of rice-straw to flooded rice soils creates anaerobic conditions and helps to promote rapid decomposition of carbofuran under such conditions by the indigenous microflora.

Microorganisms are also responsible for the breakdown of 3-hydroxycarbofuran, a metabolic product of carbofuran. Three fungal isolates, *Helminthosporium* sp., *Aspergillus niger* and *Trichoderma viride* degrade carbofuran to form 3-hydroxycarbofuran under good supply of oxygen. Rapid loss of carbofuran in soils previously treated with the insecticide (Furadan 10 G) and phorate has been reported from contaminated soils by bacterial activity but

**Table 14.4:** Microorganisms known for metabolism of carbamate pesticides.

| Pesticides | Bacteria                                                                                                                                                                                                                                                                                                                                              | Fungi                                                                                                                                                                                                      |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Carbaryl   | <i>Arthrobactor</i> sp., <i>Corynebacterium</i> sp., <i>Erwinia</i> sp., <i>E. carotovora</i> , <i>E. coli</i> , <i>Hydrogenomonas</i> sp., <i>Micrococcus</i> sp., <i>Proteus</i> sp., <i>Pseudomonas cepacia</i> , <i>P. fluorescens</i> , <i>P. putida</i> , <i>P. syringae</i> , <i>P. melophthora</i> , <i>R. meliloti</i> , <i>R. japonicum</i> | <i>Saprolegnia</i> sp.<br><i>Neurospora</i> sp.<br><i>Mucor</i> sp.                                                                                                                                        |
| Carbofuran | <i>Acinetobactor</i> sp., <i>Corynebacter</i> sp., <i>E. coli</i> , <i>Flavobacterium</i> , <i>Thermophilus</i> sp., <i>Micrococcus</i> sp., <i>Pseudomonas cepacia</i> , <i>P. fluorescens</i> , <i>Xanthomonas pruni</i>                                                                                                                            | <i>Saprolegnia</i> sp.<br><i>Neurospora</i> sp.<br><i>Mucor</i> sp.<br><i>Penicillium notatum</i><br><i>Aspergillus</i> sp.<br><i>Fusarium</i> sp.<br><i>Phytophthora</i> sp.<br><i>Trichoderma viride</i> |



this capacity was completely lost when the soils were sterilized by autoclaving. Two bacteria, an *Achromobacter* sp. and a *Pseudomonas* sp. were isolated from carbofuran-treated soils and these bacteria degraded carbofuran in mineral medium and also when inoculated to sterile amended soils. However, removal of insecticide stress caused the loss of degradative potential of the bacteria isolates. The fungi *Fusarium solani*, *Rhizoctonia praticola*, *Rhizoctonia* sp. and the bacterium *Pseudomonas* sp. use 1-naphthol, a carbamate insecticide, as a nutrient source and degrade the compound in a mineral medium metabolically even on the absence of other growth supporting substrates. In many cases, the degradation of carbamates have been reported to be an esterase-mediated hydrolytic process.

#### 14.5.4 Microbes in Metabolism of Pyrethroids

Much information is not available on the metabolism of pyrethroids by microorganisms. However, it has been well-established that the pyrethroids are highly unstable chemicals susceptible to microbial as well as chemical hydrolysis. A direct continuous spectrophotometric assay to measure pyrethroid-cleaving enzymes has been developed using *cis*- and *trans*- $\alpha$ -naphthyl-(2,2-dichlorovinyl)-2,2-dimethylcyclopropane carboxylate. This substrate shows a structure very similar to permethrin and cyfluthrin. The assay has shown that the insecticides can be effectively degraded by the live microbial cells, crude cell extracts as well as by the purified enzymes of microbial origin. Elevated esterase activity is observed in the resistant strains of bacteria in response to the exposure to pyrethroid chemicals showing that esterase mediated degradation is an important process of detoxification of these chemicals. The pseudomonads, *Achromobacter*, *Nocardia*, *Rhizobium* and *Flavobacterium* and the fungi *Aspergillus niger*, *Fusarium*, *Rhizoctonia praticola* and *Rhizoctonia* sp. have all shown ability to degrade pyrethroids in pure culture as well as in natural environment. The cyanobacteria *Anabaena* sp. PCC 7119 and *Synechocystis* sp. PCC 6803 are effective degrader of the pyrethroid cypermethrin and deltamethrin. The tolerant strains of these cyanobacteria, developed through prolonged exposure to elevated concentrations of the insecticides, show higher esterase activity than the sensitive strains (Biswal, 2007).

### 14.6 COMETABOLIC DEGRADATION OF ORGANOPOLLUTANTS

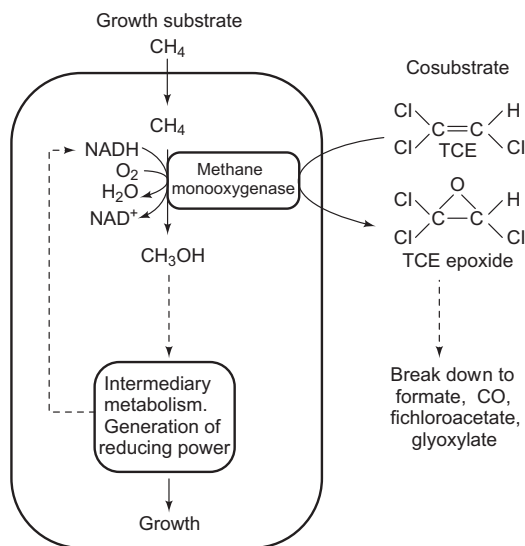
Cometabolism, the transformation of a substance without much nutritional benefit in the presence of a growth supporting substrate, is a common phenomenon in microorganisms. It is the basis of biotransformations (bioconversions) used in biotechnology to convert a substance to a chemically modified form. Microorganisms growing on a particular substrate can gratuitously oxidize a second substrate (cosubstrate). The cosubstrate is not assimilated, but the product can be available as a substrate for other organisms in a mixed culture.

The prerequisites for cometabolic transformations are the enzymes of the growing cells and the synthesis of cofactors necessary for enzymatic reactions, e.g.,



of hydrogen donors (reducing equivalents, NAD(P)H) for oxygenases. For example, methanotrophic bacteria can use methane or other  $C_1$  compounds as a sole source of carbon and energy. They oxidize methane to carbon dioxide via methanol, formaldehyde and formate. The assimilation requires special pathways and formaldehyde is the intermediate that is assimilated. The first step in methane oxidation is catalyzed by methane monooxygenase, which attacks the inert  $CH_4$  (Fig. 14.14). Methane monooxygenase is unspecific and also oxidizes various other compounds, e.g., alkanes, aromatic compounds and trichloroethylene (TCE). The latter is oxidized to an epoxide that is excreted from the cell. The unstable oxidation product breaks down to compounds that can be used by other microorganisms.

Cometabolism of chloroaromatics is a widespread activity of bacteria in mixtures of industrial pollutants. Cometabolic transformation of 2-chlorophenol gives rise to dead-end metabolites, e.g., 3-chlorocatechol. This reaction product can be autooxidized or polymerized in soils to humic-like structures. Irreversible binding of dead-end metabolites may fulfill the function of detoxification.



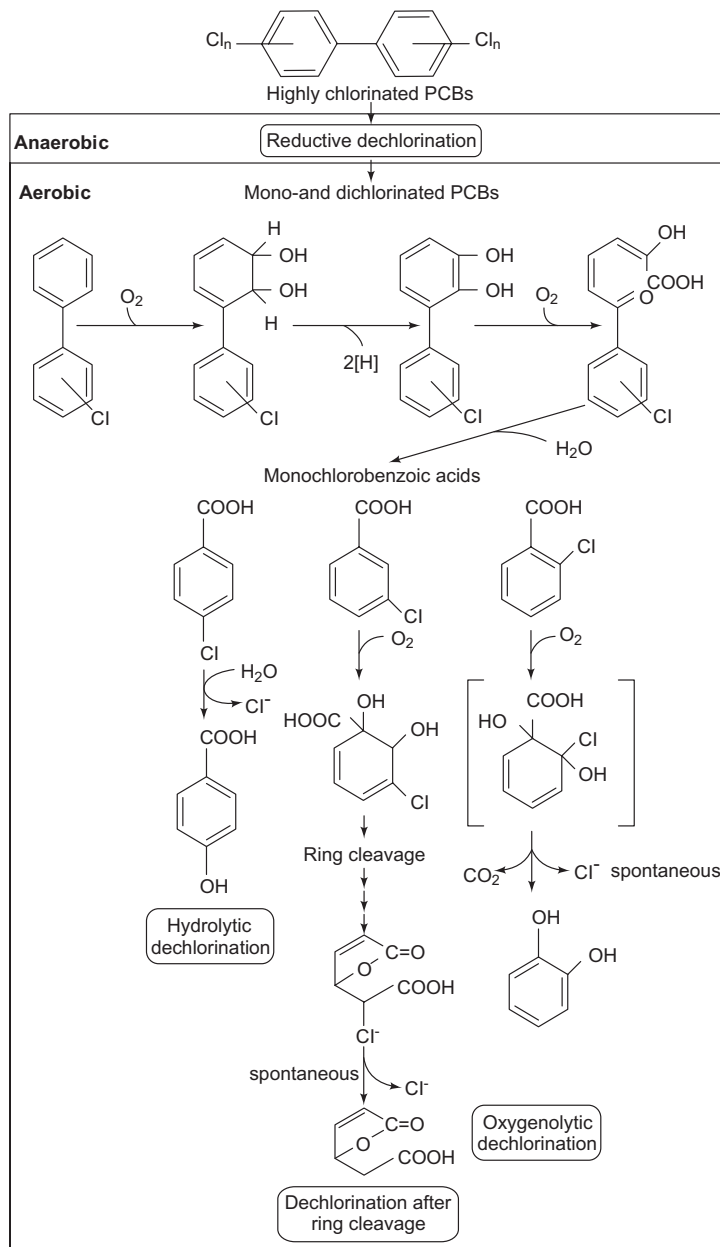
**Figure 14.14** - Cometabolic degradation of trichloroethylene by methanotrophs.

### 14.6.1 Concurrence Between Anaerobic and Aerobic Bacteria

To overcome the relatively high persistence of halogenated xenobiotics, reductive attack by anaerobic bacteria is important. Reductive dehalogenation, as the first step in the degradation of higher halogenated compounds by anaerobic bacteria, is a significant finding, because it is either a gratuitous reaction or a new type of anaerobic respiration. The process reduces the degree of chlorination and thus makes the products more accessible to mineralization by aerobic bacteria.

A sequence of anaerobic and aerobic bacterial activities for the mineralization of chlorinated xenobiotics is shown in Fig. 14.15. PCBs, which are selected as an

example for the degradation of halogenated compounds, are well-studied substances. The implementation of the reactions depends on the structure of the chemical compounds as well as on the microorganisms and conditions in a polluted ecosystem. We have to distinguish between the general degradation



**Figure 14.15** - Cooperation between aerobic and anaerobic microbes for degradation of PCB.

potential and the actual conditions necessary for it to occur. Reductive dehalogenation, the first step in PCBs degradation, requires anaerobic conditions and organic substrates acting as electron donors. The PCBs function as electron acceptors to allow the anaerobic bacteria to transfer electrons to these compounds. Anaerobic dechlorination is always incomplete; products are di- and monochlorinated biphenyls. These products can be metabolized further by aerobic microorganisms. Hydrolytic dechlorination has been elucidated by using 4-chlorobenzoate as the substrate for *Pseudomonas* and *Nocardia* spp. A halidohydrolase is capable of replacing the halogen substituent by a hydroxy group originating from water. This type of reaction seems to be restricted to halobenzoates substituted in the *para* position. Dechlorination after ring cleavage is a common reaction in the *ortho* pathway of chlorocatechols, catalyzed by catechol-1,2-dioxygenases to produce chloromuconates. Oxygenolytic dechlorination is a rare fortuitous reaction catalyzed by mono- and dioxygenases. During this reaction, the halogen substituent is replaced by an atom of O<sub>2</sub>.

Higher-chlorinated phenols, e.g., pentachlorophenol, have been widely used as biocides. Several aerobic bacteria that degrade chlorophenols have been isolated (*Flavobacterium*, *Rhodococcus*, etc.). *Rhodococcus chlorophenolicus* degrades pentachlorophenol through a hydrolytic dechlorination and three reductive dechlorinations, producing trihydroxybenzene. The potential of these bacteria is limited to some specialists and to specific conditions.

### 14.6.2 Concurrence Between Bacteria and Fungi

The syntrophic activity of fungi and bacteria has been found to be very efficient in degrading both natural and synthetic organic substances. Though the rate of degradation is highly influenced by the microbial density and diversity, the association, in general, more efficiently degrades complex organic substances than the association of any of the two groups of organisms. The recalcitrant molecules like lignin and hemicellulose are first enzymatically attacked by fungi to form products that become the substrates for bacterial metabolism. Many microfungi like yeasts and molds biodegrade aliphatic and aromatic hydrocarbons and chlorinated organic pollutants to form suitable substrates for bacteria.

## 14.7 FUNGAL DEGRADATION OF ORGANOPOLLUTANTS

### 14.7.1 Degradation of Organopollutants

The unique ligninolytic enzyme system of basidiomycetous fungi, being based on a highly reactive free-radical depolymerization mechanism, should be ideal for the biodegradation of organopollutants in the environment. Compared with other potential bioremediation systems, the extracellular, nonspecific, nonstereoselective lignin-degrading system of basidiomycetous fungi has the advantage of being applicable to a variety of recalcitrant and toxic chemicals. Examples of such

chemicals and selected fungi responsible for their biodegradation are given in Table 14.5. These substances include hazardous xenobiotic compounds such as polychlorinated dibenzodioxines and dibenzofurans, other chlorinated aromatics, nitroaromatic compounds (explosives) and carcinogenic organopollutants belonging to the polycyclic aromatic hydrocarbons.

**Table 14.5:** Lignolytic transformation of some selected organopollutants by basidiomycetes fungi.

| Name of the fungus                 | Organopollutants                                                                                              |
|------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <i>Bjerkandera adusta</i>          | Benzopyrene, other PAHs, TNT, dyes                                                                            |
| <i>Nematoloma frowardii</i>        | Benzopyrene, other PAHs, TNT, DCP, PCP, AsO                                                                   |
| <i>Phanerochaete chrysosporium</i> | Benzopyrene, other PAHs, BTX, DNT, TNT, DDT, DCP, PCP, PCBs, DCA, dyes, polystyrenes, KCN, aromatic sulphides |
| <i>Phanerochaete sordida</i>       | PAHs, polychlorinated DBDs, DBFs TNT, dyes                                                                    |
| <i>Phlebia radiata</i>             |                                                                                                               |
| <i>Pleurotus ostreatus</i>         | Benzopyrene, other PAHs, dibenzothiophene, TNT                                                                |
| <i>Stropharia rugosoannulata</i>   | DCP, PCP, TNT                                                                                                 |
| <i>Trametes versicolor</i>         | Benzopyrene, other PAHs, DCP, PCP, DCA                                                                        |

Over the last decades, efforts have been made to use ligninolytic basidiomycetes in bioremediation technologies. Laboratory experiments have shown that degradation of certain organopollutants (e.g., PAHs, PCP, TNT) is stimulated by wood-inhabiting white-rot fungi in contaminated soils (*Phanerochaete chrysosporium*, *Lentinus edodes*, *Kuehneromyces mutabilis*). However, a disadvantage of these fungi is their small competitive potential in soil and still lesser ability to grow in sewage sludge.

### 14.7.2 Degradation of Aliphatic Hydrocarbons

Biodegradation of aliphatic hydrocarbons occurring in crude oil and petroleum products has been investigated well, especially for yeasts. The n-alkanes are the most widely and readily utilized hydrocarbons, with those between C<sub>10</sub> and C<sub>20</sub> being most suitable as substrates for microfungi. However, the biodegradation of n-alkanes having chain lengths up to n-C<sub>24</sub> has also been demonstrated. Typical representatives of alkane-utilizing yeasts include *Candida lipolytica*, *C. tropicalis*, *Rhodoturula rubra* and *Aureobasidium (Trichosporon) pullulans*; examples of molds using n-alkanes as growth substrates are *Cunninghamella blakesleeana*, *Aspergillus niger* and *Penicillium frequentans*.

Yeasts and molds preferentially oxidize long-chain n-alkanes, because short-chain liquid alkanes (n = C<sub>5</sub> – C<sub>9</sub>) are toxic. The toxic effects of short-chain alkanes can be reversed by adding nontoxic long-chain hydrocarbons as metabolizable substrates. The soil fungus *Penicillium frequentans* is capable of utilizing monohalogenated n-alkanes (e.g., 1-fluorotetradecane) and dehalogenates them completely. Since all aliphatic hydrocarbons are nearly insoluble in water, fungi

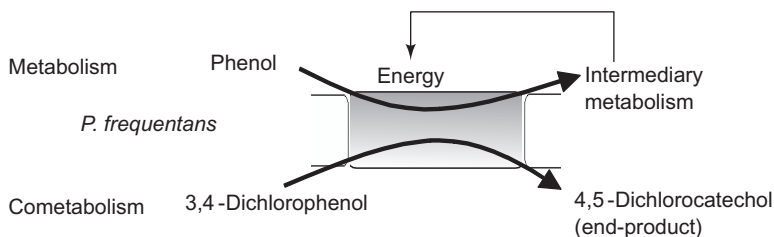
produce biosurfactants, which disperse the substrates into oil-in-water emulsions, increasing the interfacial area and thereby enhancing the bioavailability of hydrocarbons. In microfungi, the alkanes are mostly terminally oxidized to their corresponding primary alcohols (n-alkan-1-ols) by a monooxygenase enzyme complex containing cytochrome P450 and NAD(P)H-cytochrome P450 reductase:



Subterminal oxidation yielding various secondary alcohols is a rarer type of primary attack on n-alkanes and has been documented in certain molds (*Aspergillus* spp., *Fusarium* spp.). Peroxisomal enzymes carry out the remaining degradation of alkanols to intermediates that are transferable to the mitochondria. After terminal oxidation, the alcohol produced is normally oxidized to the corresponding aldehyde and fatty acid by means of pyridine nucleotide-linked dehydrogenases. In some *Candida* spp. and several molds, alcohol oxidases have been shown to be present instead of dehydrogenases. Secondary alcohols resulting from subterminal oxidation systems are oxidized to the corresponding ester and hydrolytically cleaved to acetic acid and an alcohol, which is subsequently also converted to a fatty acid. The fatty acids produced are always further metabolized by  $\beta$ -oxidation and finally to  $\text{CO}_2$  via the tricarboxylic acid cycle. However, fungi cannot utilize branched alkanes or cycloaliphatic compounds as sole sources of carbon and energy.

### 14.7.3 Degradation of Aromatic Compounds

Several yeasts and molds can utilize aromatic compounds as growth substrates (Table 14.6), but more important is their ability to convert aromatic substances cometabolically. The table is a partial list of microfungal species that have been shown to utilize aromatic compounds as sole sources of carbon and energy. For example, 3,4-dichlorophenol oxidation in the presence of phenol as the actual growth substrate can be caused by the soil-dwelling mold *Penicillium frequentans* whereas phenol is completely converted into biomass, carbon dioxide and water, the chlorinated phenol is only transformed to a chlorinated catechol that cannot be further degraded and remains as a dead-end product (Fig. 14.16).



**Figure 14.16** - Cometabolic degradation of phenol and dichlorophenol by *Penicillium frequentans*.

**Table 14.6:** The species of yeasts and molds and the organopollutants they use as growth substrates.

| Species                           | Organopollutants used as growth substrate                                                                          |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Yeasts</b>                     |                                                                                                                    |
| <i>Aureobasidium pullulans</i>    | Phenol, <i>o</i> -cresol, <i>p</i> -cresol, benzoate                                                               |
| <i>Candida maltosa</i>            | Phenol, catechol, benzoate                                                                                         |
| <i>Exophiala jeanselmei</i>       | Phenol, styrene, benzoate, acetophenone                                                                            |
| <i>Rhodotorula glutinis</i>       | Phenol, <i>p</i> -cresol, benzoate, salicylate                                                                     |
| <b>Molds</b>                      |                                                                                                                    |
| <i>Aspergillus niger</i>          | 2,4-dichlorophenoxy acetic acid, benzoate, salicylate, monochlorobenzoate                                          |
| <i>Aspergillus fumigatus</i>      | Phenol, 4-ethylphenol, phenyl acetate, <i>p</i> -cresol                                                            |
| <i>Fusarium flocciferum</i>       | Phenol, resorcinol                                                                                                 |
| <i>Penicillium frequentans</i>    | Phenol, <i>p</i> -cresol, resorcinol, phloroglucinol, anisole, benzyl alcohol, benzoate, salicylate, gallate, etc. |
| <i>Penicillium simplicissimum</i> | Phenol, phloroglucinol, monofluorophenols                                                                          |

Source: Fritsche and Hofrichter (2000).

Some species, such as the soil yeast *Trichosporon cutaneum*, possess specific energy-dependent uptake systems for aromatic substrates (e.g., phenol). They cleave aromatic rings exclusively via the *ortho* pathway analogous to many bacteria. To this end, they first insert activating hydroxyl groups into the aromatic ring. Phenol hydroxylase, benzoate-4-hydroxylase and 4 -hydroxybenzoate-3-hydroxylase are examples of enzymes hydroxylating the aromatic ring in the *ortho* or *para* position. They are NADPH-dependent monooxygenases and have been described in both yeasts and molds (e.g., *Trichosporon cutaneum*, *Penicillium frequentans*, *Aspergillus fumigatus*). After activation, dioxygenases cleave the aromatic ring to form *cis,cis*-muconic acids. The latter are lactonized, isomerized and hydrolyzed, resulting in the formation of  $\beta$ -ketoadipate, which is degraded to CO<sub>2</sub> via the tricarboxylic acid cycle.

Hydroxylating and ring-cleaving enzymes of yeasts and molds are relatively unspecific and usually convert related compounds, including halogenated and nitro aromatics. Thus, depending on the substitution pattern, fluoro- and chlorophenols are converted to the corresponding catechols, cleaved to halogenated muconic acids, or dehalogenated. The alkane- and phenol-utilizing mold *Penicillium frequentans* has been shown to degrade a mixture of phenol, *p*-cresol and two chlorophenols without additional growth substrate in soil. In addition to halogenated aromatic compounds, microfungi transform numerous other aromatic organopollutants cometabolically, including polycyclic aromatic hydrocarbons (PAHs) and biphenyls, dibenzofurans, nitro aromatics, various pesticides and plasticizers. Typical fungal transformations are glycosylations, hydroxylations and ring cleavage, methoxylations and the reduction of nitro

groups to amino groups (Table 14.7). Hydroxylation reactions are of particular importance for elimination of organopollutants in soils, because they increase the reactivity of the molecules and make their subsequent covalent coupling to humus possible. The introduction of hydroxyl groups is often catalyzed by relatively unspecific cytochrome P450-containing monooxygenases, for example, oxidize benzo(a)pyrene in *Aspergillus ochraceus* (Fitsche and Hofrichter, 2005).

**Table 14.7:** Selected organopollutants and their catabolites that are selectively formed by yeasts and molds.

| <i>Organopollutants</i>   | <i>Metabolites</i>                                                              | <i>Organisms</i>                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mono- and Dichlorophenols | Chlorocatechols                                                                 | <i>Rhodotorula</i> spp.,<br><i>Trichosporon</i> spp.,<br><i>Penicillium</i> spp.<br><i>Penicillium</i> spp.                                                |
| Pentachlorophenol         | Chloroguaiacols,<br>Chlorinated muconic acid,<br>Chloride                       | <i>Trichoderma</i> spp.<br><i>Penicillium</i> spp.                                                                                                         |
| Fluorophenols             | Pentachloroanisole<br>Fluorocatechol,<br>Fluorinated muconic<br>acids, Fluoride | <i>Fusarium</i> spp.<br><i>Aspergillus</i> spp.,<br><i>Fusarium</i> spp.,<br><i>Penicillium</i> spp.<br><i>Aspergillus</i> spp.<br><i>Penicillium</i> spp. |
| 2,4-Dinitrophenol         | Monoaminonitrophenols                                                           | <i>Aspergillus</i> spp.                                                                                                                                    |
| 2,4,6-TNT                 | Amino-DNT,<br>Hydroxylamino-DNT                                                 | <i>Aspergillus</i> spp.,<br><i>Penicillium</i> spp.                                                                                                        |
| Pyrene                    | 1-Pyrenol<br>1-Methoxypyrene<br>Dihydroxypyrenes<br>1-Pyrenol                   | <i>Aspergillus</i> spp.,<br><i>Fusarium</i> spp.,<br><i>Penicillium</i> spp.<br><i>Penicillium</i> spp.<br><i>Aspergillus</i> spp.                         |
| Benzo(a)pyrene            | Dihydroxypyrenes<br>1-Methoxypyrene                                             | <i>Aspergillus</i> spp.<br><i>Trichosporon</i> spp.                                                                                                        |
| Dibenzofurane             | Hydroxybenzopyrenes<br>Dihydroxybenzofuran,<br>ring cleavage products           | <i>Aspergillus</i> spp.<br><i>Aspergillus</i> spp.<br><i>Fusarium</i> spp.<br><i>Trichoderma</i> spp.<br><i>Penicillium</i> spp.                           |
| Biphenyl                  | Dihydroxybiphenyl                                                               |                                                                                                                                                            |
| Atrazine                  | N-Dealkylated products                                                          |                                                                                                                                                            |

## 14.8 BIOMANIPULATION FOR ACCELERATED DEGRADATION

In natural habitats, complete biodegradation of many synthetic organic molecules to inorganic end products is channeled with the participation of indigenous



microbial biomass. This process is achieved through the occurrence of stepwise pollutant transformations. A mixed microbial population offers many advantages in transformation of many toxic pollutants, due to the participation of multiple degrading enzymes. The nature of organic xenobiotic-microbial interaction can be studied in the laboratory under simulated conditions corresponding to natural environments. Isolation of microbial cultures, mixed or pure, from natural ecosystems is, therefore, instrumental to establish the pathways of degradation of many synthetic environmental contaminants. Among the several methods used, “enrichment” for specific microflora in the presence of toxic compounds has proved to be the best practical approach. Microbial cultures have been isolated using target chemical as the main source of carbon and energy, nitrogen or phosphorus. Isolation of toxicant-degrading microorganisms have been made after repeated applications of the pollutant to flooded paddy fields, soils, sediments and water samples. The biomanipulation of contaminated habitats with such enriched microorganisms have proved fruitful in accelerating the recovery of contaminated sites

Model ecosystems are now being extensively used for the evaluation of the disappearance of synthetic organic pollutants and to predict their physical and microbiological transformations under natural conditions. Microcosm or ecocores have been subsequently designed and natural microflora have been acclimated to many xenobiotics. Adaptation of microflora under these test conditions has resulted in the isolation of many microorganisms able to degrade some micropollutants, indicating the feasibility of these processes in isolating pollutant-degrading microorganisms.

### **14.8.1 Cometabolic Adaptation to Pesticides**

Many microorganisms do not use these toxic chemicals as the main source of carbon and energy, but possess enzyme systems to make structural modifications—a phenomenon known as cometabolism and can be defined as “a microbial process where an organism transforms organic compounds without using them as energy for their metabolism, resulting in product accumulation with little or no biomass production”. The resultant compounds are more vulnerable to microbial attack than the parent molecules and thus become sensitive to microbial degradation. For the cultivation of these cometabolic organisms in the laboratory, additional energy sources other than pesticide compounds must be provided. Many reactions in nature are now thought to be cometabolic including the degradation of hydrocarbon and the non-toxic lignocelluloses. Though cometabolism does not necessarily lead to complete detoxification of a compound, the initial resistance of pesticides to microbial attack is often overcome during this process. A new group of adapted microbes act on these modified organic molecules to degrade them to less toxic or non-toxic products. Although the feasibility of providing extra co-energy substrates in studying pesticide metabolism is debatable, this process can offer a possible alternative for biodegradation of many toxic xenobiotic compounds, even under

field conditions. However, the interactive effects of other physicochemical factors need to be considered.

### 14.8.2 Enzymatic Decontamination Option

Several disposal methods have been recognized for treating recalcitrant hydrocarbons and pesticide wastes. These include: (a) use as a diluent in spray tanks, (b) incineration at high temperatures (1000°C), (c) soil injection by ordinary farm equipment (plows, harrows, etc.), (d) photodecomposition by placement of the chemical on the soil surface or in a lagoon, (e) chemical degradation by chemical oxidation or reduction and alkaline or acid hydrolysis, (f) biodegradation under controlled microbial culture conditions, (g) disposal pits, (h) special landfills, (i) storage (which merely delays disposal) and (j) enzymatic treatment. The least expensive and most convenient methods are usually sought and the particular method should be chosen to minimize environmental damage. Biological systems are advantageous because they are capable of metabolizing a wide variety of pesticides under mild conditions and the resulting detoxification is generally more complete than physical or chemical procedures. Also, facilities that utilize biological methods for pesticide disposal are more easily constructed than pesticide incinerators or chemical treatment plants.

Enzymatic decontamination of recalcitrant organics from wastewater and highly contaminated soils has been conceptualized since 1980s but it has not yet emerged as a viable solution to the problem. Munnecke (1976) suggested detoxification of contaminated areas and pesticide-industrial effluents using soluble or immobilized enzymes. A crude cell extract from a mixed bacterial culture grown on parathion hydrolyzed this chemical 2450 times faster than chemical hydrolysis by 0.1 N sodium hydroxide. Still further refined enzyme preparations were reported by Munnecke (1977); an immobilized bacterial enzyme system was capable of hydrolyzing nine organophosphate insecticides. This enzyme was further used to decontaminate pesticide containers, suggesting its higher catalytic properties even at elevated concentrations. Detoxification of very high concentrations (up to 1%) of parathion and diazinon in soils by enzymatic treatment has become more or less successful. The insecticides are hydrolyzed to less toxic and unstable compounds within a short period of incubation. However, the presence of high concentrations of salts, solvents, other micropollutants and detergents (also present in the commercial formulations of pesticides) are the main negative forces in enzymatic pesticide hydrolysis when the inoculum is viable microbial cells. Microbial enzymes can be the only alternative to physical and chemical decontamination for detoxifying pesticides in these ecological areas but requires much refinement. More efforts are needed to determine the potential application for enzymatic detoxifications to clean up used pesticide containers and contaminated areas and to reduce pesticide concentrations in industrial waste disposals to acceptable levels. Enzymatic detoxification of pesticide residue treatment can be a viable alternative to the physical and the chemical methods recommended for the decontamination of toxic chemicals.

Different microbial enzymes have generated considerable interests for their application as microsomal vesicles to reclaim the pesticide contaminated habitats as well as to use them for detoxification in case of the acute toxicity to man. The crude enzyme extracts as well as the purified enzymes covalently bound to porous glass, silica beads or alumina are suggested as the immobilized catalytic surfaces to perform the degradation of pesticides. Talbot *et al.* (1982) have tried many surfaces to immobilize enzymes from crude cell extracts and have observed that the catalytic properties of the enzymes remain intact on such surfaces if the pH and temperature of the medium is not significantly changed. These surfaces can be considered as enzyme reactors capable of hydrolyzing toxic chemicals.

Detoxification by enzymes rather than by whole microbial cells is beneficial since enzymes can sometimes better tolerate environmental extremes as they have a lower physiological need than of an actively growing cell. Extremes of pH and temperature, as well as high salt and solvent concentrations, are often encountered in pesticide production wastewaters. These parameters could prevent the growth of the *Pseudomonas* sp. that produced the enzyme. The use of enzymes is also attractive because the transport of pesticides into whole microbial cells can be problematic. Such membrane transport problems could be avoided when soluble enzymes are employed in disposal processes. It seems quite feasible that the combination of whole cells and enzymes may be particularly beneficial for detoxifying spills such as those occurring in soils as the result of transportation or handling accidents.

The use of immobilized enzymes in a continuous flow bioreactor for detoxification of pesticide production wastewaters has also been tried. These systems have been examined to determine their ability to hydrolyze residual parathion in process wastewaters before discharge into waste treatment systems. The enzyme columns are effective in reducing parathion levels from 10 mg/l to 0.5 mg/l in approximately 1 min and the enzyme remains active under continuous use conditions for a considerably long period (up to 10 weeks).

Enzymes that show the greatest potential for pesticide detoxification are those that can function without co-factors or co-enzymes and can detoxify a pesticide molecule by a hydrolytic or other simple enzymatic reaction. Presently, there are some enzymes that would be useful for detoxification reactions but are not likely candidates for immediate development because they require co-factors or co-enzymes. These chemicals make such use economically prohibitive. For example, oxygenase enzymes (EC 1.14.12 and EC 1.14.13) require molecular oxygen and either NADPH or NADH to initiate aromatic ring fission. The development of oxygenase enzymes for large-scale applications will be prohibited until economical methods for co-factor regeneration are discovered. Carboxylase enzymes (EC 4.1.1) could also be important degradative enzymes for industrial pesticide detoxification, but these require thiamin-diphosphate or pyridoxal-phosphate and until an inexpensive source of these co-factors is available, carboxylase enzymes cannot be developed for treatment systems.

There are attempt to modify the catalytic properties of different pesticide-degrading enzymes to improve their catalytic efficiency. The procedures of the

enzyme preparations are now highly refined to produce purified enzymes with intact catalytic activity. PTEs have been successfully tried to hydrolyse the OP insecticides in the blood and tissues of rats artificially injected with OP chemicals. These enzymes are instrumental for organophosphate and carbamate insecticide degradation and are excellent candidates for industrial or large-scale use because they are stable over wide ranges of pH and temperature and require no co-factors. Hydrolysis generally results in a significant decrease in toxicity. In addition, the bioactivity of the parent insecticidal compound is destroyed and the metabolites of the hydrolytic reactions are most typically less stable in the environment than the original insecticidal molecule.

## 14.9 CONCLUSION

The microorganisms and their degradative capabilities are absolutely essential for the renewal of our environment and maintenance of the global carbon cycle. Among the substances that can be degraded or transformed by microorganisms are a huge number of synthetic compounds and other chemicals having environmental relevance (e.g., mineral oil components). However, it has to be considered that this statement concerns potential degradabilities which, in most instances, were estimated in the laboratory by using pure cultures and ideal growth conditions. Under natural conditions in soil and surface waters, the actual degradability of organopollutants is lower, due to a whole range of factors: competition with other microorganisms, insufficient supply with essential substrates (C, N, P, S sources), unfavourable external conditions ( $O_2$ ,  $H_2O$ , pH, temperature) and low bioavailability of the pollutant that is to be degraded. For this purpose, it is necessary to support the activities of the indigenous microorganisms in polluted soils and to enhance their degradative potential by bioaugmentation. The former measure particularly applies to bacteria, whereas bioaugmentation mainly concerns basidiomycetous fungi. In this context, we should point out that neither bacteria nor fungi are 'better' degraders. In nature, both groups of microorganisms work together and complement one another in their degradative capabilities.

Anaerobic degradation can be applied in technical devices for treatment of waste material, often leading to  $CH_4$  and  $CO_2$  as products, which can be exploited as energy source or as a basis for biosynthetic processes. Moreover, anaerobic degradation proceeds in many anoxic habitats such as the intestinal tracts of humans and animals, sediments and oxygen-deprived microenvironments in soil, sewage sludge, etc. Knowledge of the capacities, strategies and limits of anaerobic degradation processes is, therefore, needed to assess the potential risk of synthetic compounds to health or to the environment, no matter whether such synthetics are released intentionally (as with plant protection agents), inadvertently through wastewater treatment, or accidentally through spills.

This overview shows that the degradative potential of anaerobic microbial communities is much greater than assumed only a few years ago: a broad variety of compounds can be subject to anaerobic degradation, most often down to methane and carbon dioxide as final products. Aliphatic hydrocarbons are

degraded if they contain unsaturated bonds, preferentially if these are located terminally, but saturated long-chain aliphatic hydrocarbons are also anaerobically degradable. These processes are slow and can be applied only in long-term incubations. Ether compounds are degraded anaerobically if they are methyl ethers or if they can be transformed into hemiacetals through, e.g., hydroxyl shift reactions. In the anaerobic degradation of ketones, the primary activation reaction is a carboxylation rather than an oxidation step.

Basic research regarding pesticide transformation in natural environments and under laboratory conditions has established the importance of biological degradation as a major mechanism of pesticide detoxification. Microorganisms are now believed to be the major catalytic forces that degrade pesticides in soil, sediments and aquatic ecosystems. Once acclimated to these synthetic chemicals, their versatility in degradation of the xenobiotic compound is intensive. Biological hydrolysis is one of the major reactions to occur in the decontamination process of many pesticides. This is particularly true in the metabolism of organophosphates, dithioates, carbamates and most probably for the other pesticides like fungicides, herbicides and bactericides. However, many of these degradation products remain in the environment for a considerably long period as soil-bound residues. Our knowledge of the fate and the metabolism of bound residues is now extensive. Research is only needed in this area to develop applicable processes involving the usefulness of microbial cells or their enzymes for the detoxification of these residues.

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# BIOINDICATION IN THE ENVIRONMENT

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## 15.1 INTRODUCTION

During the past few decades significant improvements in the monitoring and quality determination of the aquatic environments have been made through the upgradation of the chemical as well as biological test parameters. Biological test systems are providing in-depth information on the effects of almost all anthropogenic substances added to the environment as residues of the human activities. The synergistic and antagonistic effects of environmental contaminants, that could not be assessed before by chemical analysis, are now being effectively determined by using biological test batteries. Because each organism, especially microorganism, has its characteristic sensitivity/resistance to an anthropogenic substance, it can be possible to assess the quality of the environment by monitoring the qualitative and the quantitative changes of such organisms in the presence of the concerned stress.

While some microorganisms show tolerance to a xenobiotic substance at a selected concentration (resistant microbes), others show sensitivity to the same concentration of the toxicant (sensitive microbes). Some organisms like to grow in a definite environment with a specific combination of abiotic parameters and are favoured or discouraged by the change of the physicochemistry of the environment. These are called indicator organisms. For example, coliform bacteria, which normally occur in the intestines of all warm-blooded animals, are excreted in great numbers in feces. In polluted water, coliform bacteria are found in densities roughly proportional to the degree of fecal pollution. Because coliform bacteria are generally harder than disease causing bacteria, their absence from water is an indication that the water is bacteriologically safe for human consumption. Conversely, the presence of the coliform group of bacteria is indicative that other kinds of microorganisms capable of causing disease may also be present and that the water is potentially unsafe to drink.

## 15.2 STANDARD CRITERIA FOR INDICATOR ORGANISMS

The coliform group has been adapted as an indicator of fecal contamination of drinking water. Many countries have adopted coliforms and other groups of bacteria as official standards for drinking water, recreational bathing waters, wastewater discharges and various foods. Indicator microorganisms have also been used to assess the efficacy of food processing and water and wastewater treatment processes. As an ideal assessor of fecal contamination, it has been suggested that they meet the criteria listed in Table 15.1. Unfortunately, not even one indicator meets all these criteria. Thus, various groups of microorganisms have been suggested and used as indicator organisms. Concentrations of indicator bacteria found in wastewater and feces are shown in Tables 15.2.

**Table 15.1:** Criteria for an ideal indicator organism.

| <i>Microbial group</i> | <i>Criteria</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bacteria               | <p>The organism should be useful for all types of water.</p> <p>In case of coliforms, the organism should be present whenever enteric pathogens are present.</p> <p>The organism should have a reasonably longer survival time than the hardiest enteric pathogen.</p> <p>The organism should not grow in water.</p> <p>The testing method should be easy to perform.</p> <p>The density of the indicator organism should have some direct relationship to the degree of fecal pollution.</p> <p>The organism should be a member of the intestinal microflora of warm-blooded animals.</p> |
| Algae                  | <p>The organism should have definite choice for a nutrient regime.</p> <p>The indicator alga should have preference for specific nutrients, for example, diatoms for silica.</p> <p>For eutrophic indicators, the growth should be very fast in comparison to other species.</p> <p>For oligotrophic indicators, the growth rate should be very sensitive to change of nutrient status.</p>                                                                                                                                                                                                |

In addition to coliform bacteria, many other microorganisms are also used as indicator species, especially in aquatic ecosystems. The diversity, density, distribution and ecological responses of these microorganisms also act as indicators of water contamination and sources of pollution of the surface waters. These microbes includes phytoplankton, protozoa, diatoms, zooplankton, etc. while protozoa and bacteria indicate the saprobic status of surface water bodies, cyanobacteria and other planktonic algae indicate the nutrient loading of the water bodies (trophic status indication). The species with high sensitivity to changes of the physicochemistry of waters are better indicators than those, which are not significantly changed with minor changes of the saprobic or trophic status



**Table 15.2:** The expected density of indicator organisms (bacteria and viruses) in raw sewage having high saprobic status.

| Organism                            | CFU per 100 ml   |
|-------------------------------------|------------------|
| Coliforms                           | $10^7 - 10^9$    |
| Fecal coliforms                     | $10^6 - 10^7$    |
| Fecal streptococci                  | $10^5 - 10^6$    |
| Enterococci                         | $10^4 - 10^5$    |
| Clostridium perfringens             | $10^4$           |
| Staphylococcus (coagulase positive) | $10^3$           |
| Pseudomonas aeruginosa              | $10^5$           |
| Acid-fast bacteria                  | $10^2$           |
| Coliphages                          | $10^2 - 10^3$    |
| Bacteroides                         | $10^7 - 10^{10}$ |

Modified from Gleeson and Gray (1997).

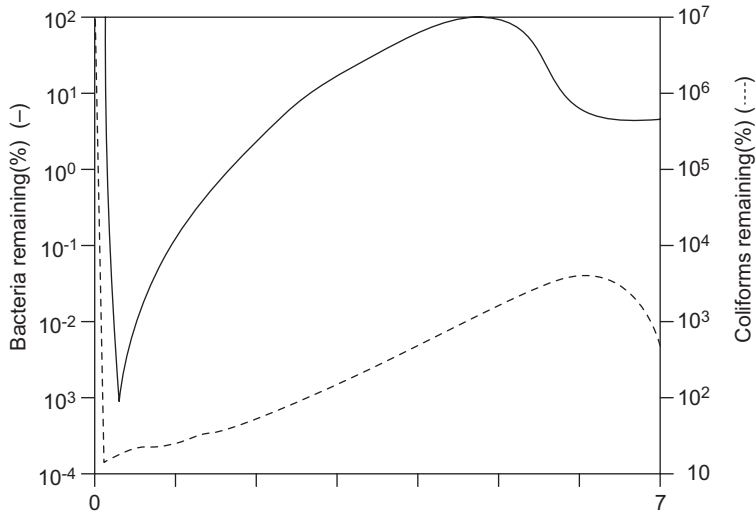
of a water body. For example, the diversity of protozoa and their population growth rate is significantly enhanced by organic loading of surface waters but the density and the diversity of diatoms and desmids are negatively affected by such changes in the water body. Desmids show high sensitivity to eutrophication and are thus good indicators of surface water pollution.

### 15.3 BACTERIA AS INDICATORS

#### 15.3.1 Coliform Group

The coliform group, which includes *Escherichia*, *Citrobacter*, *Enterobacter* and *Klebsiella* species, is relatively easy to detect. Specifically, this group includes all aerobic and facultatively anaerobic, Gram-negative, non-spore-forming, rod-shaped bacteria that produce gas upon lactose fermentation in prescribed culture media within 48 hours at 35°C. The coliform group has been used as the standard for assessing fecal contamination of recreational and drinking waters. The absence of this organism in 100 ml of drinking water ensures the prevention of bacterial waterborne disease outbreaks.

All members of the coliform group have been observed to regrow in natural surface and drinking water distribution systems. The die-off rate of coliform bacteria depends on the amount and the type of organic matter in water and its temperature. If the water contains significant concentrations of organic matter and is at an elevated temperature, the bacteria may increase in numbers. This phenomenon has been observed in eutrophic tropical waters, waters receiving pulp and paper mill effluents, wastewater, aquatic sediments and organically enriched soil (i.e., sewage sludge amended) after periods of heavy rainfall. Of greatest concern is the growth or the recovery of injured coliform bacteria in a distribution system because this may give a false indication of fecal contamination (Fig.15.1). Coliforms may colonize and grow in the biofilm found on the distribution system pipes, even in the presence of free chlorine. *Escherichia coli* is 2400 times more resistant to free chlorine when attached to a surface than as free cells in water.



**Figure 15.1** - The growth pattern of coliforms and *E. coli* during regrowth in sewage effluents after treatment with chlorine (5 mg/l) (Modified from Shuval *et al.*, 1973). The populations are in number per 100 ml of sample.

Because large numbers of heterotrophic bacteria in the water may mask the growth of coliform bacteria on selective media used for their isolation, true numbers of coliforms may be underestimated. This often becomes a problem when aerobic heterotrophic bacterial numbers exceed 500/ml. Finally, the longer survival and greater resistance to disinfectants of pathogenic enteric viruses and protozoan parasites limit the use of coliform bacteria as an indicator for these organisms. Still, the coliform group of bacteria has proved its merit in assessing the bacterial quality of water. Three methods are commonly used to identify coliforms in water. These are the most probable number (MPN), the membrane filter (MF) and the presence-absence (P-A) tests.

**(a) Most Probable Number (MPN) test:** The MPN test allows detection of the presence of coliforms in a sample and estimation of their numbers. This test consists of three steps: a presumptive test, a confirmed test and a completed test. In the presumptive test, lauryl sulphate-tryptose-lactose broth is placed in a set of test tubes with different dilutions of the water to be tested and incubated at 35°C for 24 to 48 hours. The tubes are then examined for the presence of coliforms, which is indicated by gas and acid production. Once the positive tubes have been identified and recorded, it is possible to estimate the total number of coliforms in the original sample by using an MPN table that gives numbers of coliforms per 100 ml. In the confirming test, the presence of coliforms is verified by inoculating selective bacteriological agars such as Levine's eosin-methylene blue (EMB) agar or Endo agar with a small amount of culture from the positive tubes. Lactose-fermenting bacteria are indicated on the medium by the production of colonies with a green sheen or colonies with a dark centre. In some cases, a

completed test is performed in which colonies from the agar are inoculated back into lauryl sulphate-tryptose-lactose broth to demonstrate the production of acid and gas.

**(b) Membrane Filter (MF) test:** In this technique, a measured amount of water (usually 100 ml for drinking water) is passed through a membrane filter (pore size 0.45  $\mu\text{m}$ ) that traps bacteria on its surface. This membrane is then placed on a thin absorbent pad that has been saturated with a specific medium designed to permit growth and differentiation of the organisms being sought. For example, if total coliform organisms are sought, a modified Endo medium is used. For coliform bacteria, the filter is incubated at 35°C for 18-24 hours. The success of the method depends on using effective differential or selective media that can facilitate identification of the bacterial colonies growing on the membrane filter surface. To determine the number of coliform bacteria in a water sample, the colonies having a green sheen are enumerated.

**(c) Presence-Absence (P-A) test:** The presence-absence tests (P-A tests) are not quantitative tests; instead, they answer the simple question of whether the target organism is present in a sample or not. A single tube of lauryl sulphate-tryptose-lactose broth as used in the MPN test, but without dilutions, would be used in a P-A test. In recent years, enzymatic assays have been developed that allow the simultaneous detection of total coliform bacteria and *E. coli* in drinking water. The assay can be a simple P-A test or an MPN assay.

### 15.3.2 Fecal Coliforms

Despite that total coliform group has served as the main indicator of water pollution, many of the organisms in this group are not limited to fecal sources. Therefore, it is necessary to enumerate the coliforms that are more clearly of fecal origin (the fecal coliforms). These organisms, which include the genera *Escherichia* and *Klebsiella*, are differentiated in the laboratory by their ability to ferment lactose with the production of acid and gas at 44.5°C within 24 hours. The frequent occurrence of fecal coliform bacteria in unpolluted tropical waters and their ability to survive for considerable periods of time outside the intestine in these waters, have made us to believe that these organisms occur naturally in tropical waters and that new indicators for these waters need to be developed.

Some have suggested the use of *E. coli* as an indicator, because it can be easily distinguished from other members of the fecal coliform group (e.g., absence of urease and presence of  $\beta$ -glucuronidase). Fecal coliforms also have some of the limitations in use as the coliform bacteria, i.e., regrowth and less resistant to water treatment than viruses and protozoa. They may be detected by methods similar to those used for coliform bacteria. For the MPN method, EC broth is used and for the membrane filter method m-FC agar is used for water analysis. The Colilert test has the advantage of detecting coliforms and *E. coli*, the principal fecal coliform, simultaneously within 24 hours.

### 15.3.3 Fecal Streptococci

The Gram-positive Lancefield group D streptococci are the fecal streptococci belonging to the genera *Enterococcus* and *Streptococcus*. The former includes all streptococci that share certain biochemical properties and have a wide range of tolerance to adverse growth conditions. They are differentiated from other streptococci by their ability to grow in 6.5% sodium chloride, at pH 9.6 and at 45°C and include *Enterococcus avium*, *E. faecium*, *E. durans*, *E. faecalis* and *E. gallinarum*. In the water industry, the genus is often given as *Streptococcus* for this group. Of the genus *Streptococcus*, only *S. bovis* and *S. equinus* are considered to be true fecal streptococci. These two species are predominately found in animals; *Enterococcus faecalis* and *E. faecium* are more specific to the human gut. It has been suggested that a fecal coliform/fecal streptococci (FC/FS) ratio of 4 or more indicates a contamination of human origin, whereas a ratio below 0.7 is indicative of animal pollution (Table 15.3). Both the membrane filtration method and MPN method are used for the isolation of fecal streptococci. Fecal streptococci are considered to have certain advantages over the coliform and fecal coliform bacteria as indicators, viz., (a) They rarely multiply in water, (b) They are more resistant to environmental stress and chlorination than coliforms and (c) They generally persist longer in the environment. The enterococci have been suggested as useful indicators of risk of gastroenteritis for recreational bathers and standards have been recommended. They have been suggested as useful indicators of the presence of enteric viruses in the environment.

### 15.3.4 *Clostridium perfringens*

It is a Gram-positive, rod-shaped sulphite-reducing anaerobic spore former exclusively of fecal origin. The spores are very heat resistant (75°C for 15 minutes), persist for long periods in the environment and are very resistant to disinfectants. The hardy spores of this organism limit its usefulness as an indicator. However, it has been suggested that it could be an indicator of past pollution, a tracer of less hardy indicators and an indicator of removal of protozoan parasites or viruses during drinking water and wastewater treatment.

**Table 15.3:** The FC/FS ratio in water bodies receiving pollutants from various sources.

| FC/FS ratio | Source of pollution                                                             |
|-------------|---------------------------------------------------------------------------------|
| > 4.0       | Strong evidence that pollution is of human origin.                              |
| 2.0–4.0     | Good evidence of the predominance of human wastes in mixed pollution.           |
| 0.7–2.0     | Good evidence of the predominance of domestic animal wastes in mixed pollution. |
| < 0.7       | Strong evidence that pollution is of animal origin.                             |

Source: Maier *et al.* (2000).

Other anaerobic bacteria such as *Bifidobacterium* and *Bacteroides* have been suggested as potential indicators. Because some of the *Bifidobacterium* species are primarily associated with humans, they could potentially help in distinguishing human contamination from that of animal origin. However, better and more standard methods are needed for detection of all of the anaerobic bacteria in the environment before they can be adequately monitored in a routine fashion.

### 15.3.5 Heterotrophic Plate Count

Heterotrophic plate count (HPC) is done to assess the numbers of aerobic and facultatively anaerobic bacteria in water that derive their carbon and energy from organic compounds. This group includes gram-negative bacteria belonging to the genera *Pseudomonas*, *Aeromonas*, *Klebsiella*, *Flavobacterium*, *Citrobacter*, *Serratia*, *Acinetobacter*, *Proteus*, *Alcaligenes*, *Enterobacter* and *Moraxella*. The heterotrophic plate counts of microorganisms found in untreated drinking water and chlorinated distribution water are shown in Table 15.4. These bacteria are commonly isolated from surface waters and groundwater and are widespread in soil and vegetation (including many vegetables eaten raw). In drinking water, the number of HPC bacteria may vary from less than 1 to more than  $10^4$  CFU/ml and they are influenced mainly by temperature, presence of residual chlorine and level of assimilable organic matter. However, the HPC is not a direct indicator of fecal contamination but it does indicate variation in water quality and potential for pathogen survival and regrowth. These bacteria may also interfere with coliform and fecal coliform detection when present in high numbers.

### 15.3.6 Surface Water Indication by Diversity Pattern of Bacteria

The bacterial flora of surface waters is dependent on the nutrient concentrations and the rate of production of organic matters. In an oligotrophic system, with limited availability of organic matter to bacteria, the flora is well-represented by a diverse type of bacteria that are well-demarcated by their role in remineralization process. A number of bacteria also make their regular entry from point and non-point sources through wastewater discharge or through surface run-off during rain. Majority of these bacteria are ubiquitous in their distribution, though some like the faecal bacteria make relatively a short appearance in surface waters to multiply. The standardization of individual and species or an assemblage of bacteria as indicator of water quality is practically difficult as the activity and the density of bacteria are controlled by a number of biotic and abiotic variables. In addition, the availability of surface and the area of biofilm plays a significant role in determining the bacterial activity. The aggregates and biofilm bacteria show different metabolic efficiencies making it difficult to quantify the bacterial activities from simple determination of their densities. There is also regular exchange of cells and spores of bacteria between the bulk liquid and the biofilm surface bringing a bacterial cell to different metabolic states at different times. However, the bacteria, that remain in

**Table 15.4:** Identification of HPC bacteria (measured as % of the total number of organisms encountered) in chlorinated distribution system and in the untreated drinking water bodies.

| Organism                              | Distribution water | Untreated water |
|---------------------------------------|--------------------|-----------------|
| <i>Actinomycetes</i>                  | 10.7               | 0               |
| <i>Arthrobacter</i> spp.              | 2.3                | 1.3             |
| <i>Bacillus</i> spp.                  | 4.9                | 0.6             |
| <i>Corynebacterium</i> spp.           | 8.9                | 1.9             |
| <i>Micrococcus luteus</i>             | 3.5                | 3.2             |
| <i>Staphylococcus aureus</i>          | 0.6                | 0               |
| <i>S. epidermidis</i>                 | 5.2                | 5.1             |
| <i>Acinetobacter</i> spp.             | 5.5                | 10.8            |
| <i>Alcaligenes</i> spp.               | 3.7                | 0.6             |
| <i>Flavobacterium meningosepticum</i> | 2.0                | 0               |
| <i>Moraxella</i> spp.                 | 0.3                | 0.6             |
| <i>Pseudomonas alcaligenes</i>        | 6.9                | 2.5             |
| <i>P. cepacia</i>                     | 1.2                | 0               |
| <i>P. fluorescens</i>                 | 0.6                | 0               |
| <i>P. mallei</i>                      | 1.4                | 0               |
| <i>P. maltophilia</i>                 | 1.2                | 5.7             |
| <i>Pseudomonas</i> spp.               | 2.9                | 0               |
| <i>Aeromonas</i> spp.                 | 9.5                | 15.9            |
| <i>Citrobacter freundii</i>           | 1.7                | 5.1             |
| <i>Enterobacter agglomerans</i>       | 1.2                | 11.5            |
| <i>Escherichia coli</i>               | 0.3                | 0               |
| <i>Yersinia enterocolitica</i>        | 0.9                | 6.4             |
| <i>Hafnia alvei</i>                   | 0                  | 5.7             |
| <i>Enterobacter aerogenes</i>         | 0                  | 0.6             |
| <i>Enterobacter cloacae</i>           | 0                  | 0.6             |
| <i>Klebsiella pneumoniae</i>          | 0                  | 0               |
| <i>Serratia liquefaciens</i>          | 0                  | 0.6             |
| Unidentified                          | 18.7               | 17.8            |

Modified from LeChevallier *et al.* (1980).

sessile forms and enter into aquatic system for a short period, can indicate the water quality better than other species; the faecal bacteria are thus the best for the purpose.

Competition, grazing, O<sub>2</sub> availability, pH and temperature are the important parameters influencing the bacterial activity. The saprophytic activity of bacteria influences the rate of change of organic matters and thereby determines the rate of remineralization. The self purification process in a water body is governed by the microbial activity in water column and sediment, which in turn maintains the water quality. The qualitative and the quantitative changes of bacterial flora and their fluctuation over time is proportional to the pace of self purification. Anthropogenic interference can encourage or retard the growth and the activities of the bacterial assemblage and regulate the bacterial succession in surface waters.

The direct microscopic determination of the total bacterial number (TBN) is a common method to determine the quality of surface waters. Though there is no direct relationship between the surface water quality and TBN, there is nevertheless an increase of TBN with increase in organic load of surface waters. In undisturbed surface waters having no external source of organic carbon, the TBN lies within a million range whereas in a highly polluted water body it can exceed 10 million per milliliter of sample. In addition, the TBN also changes according to the environmental fluctuations (variation in dissolved  $O_2$ , pH and temperature).

Basically, there is a great degree of morphological variation in the bacterial assemblage of a water body, which can be used as an index of water quality. The differentiation of bacteria, on the basis of their size and form as well as the proportion of different forms in the total population, can be used as criteria for water quality determination. In general, the oligotrophic waters with low organic loading has dominance of coccal bacterial forms, that are nutritionally fastidious and comparatively small in average size. The other extreme is the dominance of large filamentous and rod-shaped bacteria in eutrophic waters with high organic load. It is not yet clear whether the small bacterial forms grow as an indigenous bacterial flora in oligotrophic water and are succeeded by small rods or filaments with increase in the nutrient load of water or the small filaments are only a morphological modification of coccal bacterial forms during eutrophication.

### 15.3.7 Saprophytic Bacteria as Indicators

The total bacterial population in a water sample is determined relatively as the population density of saprophytic bacteria through inoculation in organically-enriched nutrient media. The saprophytic population is used as an indicator of water quality and of the concentration and fate of degradable organic matters in wastewaters. These bacteria have a high reproduction potential and are sensitive to the nutrient regime, to which they often show immediate reaction. The population of saprophytic bacteria (cells or colonies per milliliter) ranges from few hundred to over a million depending on the saprophytic status of the water body (Table 15.5). The quantitative determination of heterotrophic bacteria in agar or liquid broth show a lot of variation with handling of sample, growth conditions, medium type, incubation period, temperature and pH. This may cause a variation of saprophytic population to up to one order of magnitude. In addition, the quantitative determination of bacteria by incubation in a selective medium is considered incomplete as many species do not grow in artificial media despite manipulation of all growth conditions, for reasons not yet clearly understood though, however, some are believed to be estimated better after growth in nutrient-enriched media.

The bacterial population also show significant annual fluctuation making it necessary to have regular and repeated quantification to draw a conclusion on the microbiological status of a water body. There is a need to correlate bacterial density with the hydrological, physico-chemical and biological parameters of a water body for determining the saprobic status.



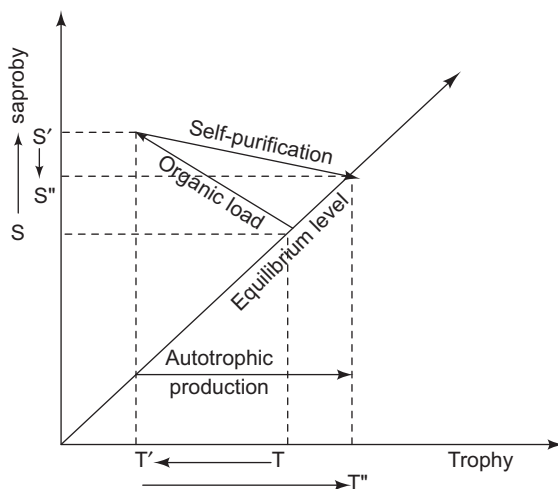
**Table 15.5:** The density of saprophytic bacteria in waters of varied saprobic status.

| <i>Saprobity level</i>           | <i>Saprophytic bacteria</i><br>( $\times 1000/\text{ml}$ ) |
|----------------------------------|------------------------------------------------------------|
| Catharobic                       | < 0.1                                                      |
| Xenosaprobic                     | < 0.5                                                      |
| Oligosaprobic                    | < 1.0                                                      |
| Oligo- $\beta$ -Mesosaprobic     | $\leq 10$                                                  |
| $\beta$ -Mesosaprobic            | < 20                                                       |
| $\beta$ - $\alpha$ -Mesosaprobic | < 50                                                       |
| $\alpha$ -Mesosaprobic           | < 100                                                      |
| $\alpha$ Meso-Polysaprobic       | < 400                                                      |
| Polysaprobic                     | > 400                                                      |

Source: Gunkel (1994).

In rivers receiving wastewaters, there is remarkable annual rhythm of saprophytic bacterial population showing distinct summer peaks and winter trough. The successional changes of bacterial population is more distinct in the early summer than in any other part of the year and it is much influenced by the types of organic matters entering into the surface water. In small flowing waters like low-order springs and streams, there are strong interactions of the water bodies with the surrounding terrestrial habitat, which facilitates the growth of an indigenous microflora. However, wastewater discharge into these water bodies significantly affect the water quality and thereby the microbial community. Such shallow water bodies have relatively high interaction with the sediment and experience a gradient of saprophytic density from the sediment surface to the water surface, the bottom water having high saprophytic population than surface waters. Extremely high saprophytic population is a marker of the poor management of water treatment systems and fish ponds.

The self-purification process in wastewater, through microbial degradation of organic matter, brings a dynamic change in the trophic status of a water body. As shown in Fig. 15.2, microbial metabolism of degradable organic matters reduces the saprobic status whereas the gain of organic matter in the water body increases the saprobity level (from  $S'$  to  $S''$ ). There is an inverse relationship between the saprobity and trophy of a water body, the saprobity being regulated by organic matter input while trophy being regulated by concentration of inorganic nutrients (especially of nitrogen and phosphorus). Self purification decreases the saprobic level ( $S'$  to  $S''$ ) but increases the trophic level ( $T'$  to  $T''$ ) whereas organic matter addition from allochthonous sources or by phytoplankton production by bloom formation causes the reverse process. The principal difference between the natural saprobic state (caused by gain of autochthonous organic matter) and the induced saprobic state (gain of allochthonous organic matter) is that in the former the self purification brings a stable equilibrium between the saprobity and trophy while in latter only a quasi-equilibrium is reached.



**Figure 15.2** - The relationship between saprobity and trophy in rivers (Adopted from Schwaerbel, 1993).

## 15.4 ALGAE AS INDICATOR

Algae are an ecologically important group of organisms in most aquatic ecosystems but often are ignored as indicators of disturbances and conditions of aquatic ecosystem. Because of their nutritional needs and their position at the base of the food web, algal indicators provide relatively unique information concerning ecosystem conditions in comparison to commonly used animal and bacterial indicators. Algae, in general, and phytoplankton as specific group, respond rapidly and predictably to a wide range of pollutants and thus provide potentially useful early warning signals of the deteriorating conditions and the possible causes. Algal assemblages provide one of the few benchmarks for establishing the required water quality conditions and for characterizing the minimally impacted biological conditions of many disturbing ecosystems.

The planktonic and periphytic algae react to the change in the characteristics of the aquatic environment through a succession of the assemblage. The stability of a changed environment is built only after 3–5 generations, with the planktonic species reacting faster than the other forms. The in place studies on the distribution of phytoplankton and macrophytes have shown that many algal groups (red algae, brown algae, xanthophytes and cyanobacteria) can indicate the degree of threat perceived in a water body. The red list of algal groups prepared from the pattern of change in their existing biodiversity in surface water contains five threatened classes: (a) extinct—the species that are not found in any of the water bodies, at least during last 10 years; (b) nearly extinct—that species that are now found in very few and restricted habitats but may come back to the vast ecosystem by improvement of water quality; (c) highly threatened species whose abundance has been substantially reduced in many places; (d) threatened—the

species whose distribution has been affected in some regions and there is possibility of their reduced distribution in other regions in near future; and (e) potentially threatened—the species that are hitherto not threatened but going by the trend they are expected to be affected in long term future. Not threatened are the species that are not included in the above five classes. Table 15.6 summarises the data of Blab *et al.* (1984) on the threatened algal groups in sub-tropical water bodies. This distribution indicates that red and brown algae are very much affected by eutrophication than other algal groups.

**Table 15.6:** The threatened class of different algal groups in sub-tropical water bodies.

| Algal groups          | No. of spp. studied | Threatened class (as mentioned in text) (do) |          |          |          |          |                |
|-----------------------|---------------------|----------------------------------------------|----------|----------|----------|----------|----------------|
|                       |                     | <i>a</i>                                     | <i>b</i> | <i>c</i> | <i>d</i> | <i>e</i> | Not threatened |
| Red algae             | 28                  | -                                            | 100      | -        | -        | -        | -              |
| Brown algae           | 04                  | -                                            | 100      | -        | -        | -        | -              |
| Green algae           | 49                  | 5                                            | 45       | 10       | 16       | 7        | 17             |
| Cyanobacteria         | 52                  | -                                            | 39       | -        | 33       | 9        | 19             |
| Xanthophytes          | 26                  | -                                            | 40       | 10       | 11       | 14       | 25             |
| Light pigmented forms | 34                  | 6                                            | 6        | 30       | 41       | -        | 13             |

Source: Blab *et al.*, (1984).

### 15.4.1 Phytoplankton as Indicators

The micronutrients (P, N, Si), micronutrients and Xenobiotic substances in freshwater bodies are well-indicated by the diversity and association patterns of planktonic algae. The disturbances in the phytoplankton assemblage by the top down zooplankton grazing or bottom up nutrient flow are indicated by the qualitative and the quantitative changes in phytoplankton composition. The diversity changes of phytoplankton with the change in trophic status of water body caused by the gain of population density as well as the blooming of few species (eutrophic indicators) at the expense of many less tolerant and affinity adapted species (oligotrophic indicators) results in increased accumulation of autochthonous organic matters. For example, increase in phosphorus concentration promotes the growth of cyanobacteria but adversely affect the desmids and diatoms. Nitrogen enrichment encourages the growth of green algae but not of desmids as the latter has less competitive ability to use as nutrients in an assemblage under rate-adapted conditions. High level of silicon in the surface waters facilitates the growth of diatoms produced, the concentration of other key nutrients, nitrogen and phosphorus, do not become limiting. Table 15.7 is a comprehensive list of indicator species of water bodies of different trophic status.

**Table 15.7:** The list of common planktonic algae indicating the trophic status of surface waters.

| <i>Algal group</i> | <i>Oligotrophic</i>                                                                                                                                                                                            | <i>Mesotrophic</i>                                                                                                                                                                                         | <i>Eutrophic</i>                                                                                                                                                                                               |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyanobacteria      | <i>Merismopedia</i> sp.<br><i>Gloeotrichia</i> sp.<br><i>Anabaena solitaria</i>                                                                                                                                | <i>Coelosphaerium</i> sp.<br><i>Planktothrix</i> sp.<br><i>Gomphosphaeria</i> sp.<br><i>Anabaena lemmermanni</i><br><i>Anabaena flos-aquae</i>                                                             | <i>Aphanizomenon</i> spp.<br><i>Microcystis</i> spp.<br><i>Anabaenopsis arnoldii</i><br><i>Anabaena planktonica</i>                                                                                            |
| Green algae        | <i>Ankistrodesmus</i> spp.<br><i>Chlamydomonas</i> sp.<br><i>Pandorina</i> spp.<br><i>Gonium</i> spp.<br><i>Sphaerocystis</i> sp.<br><i>Staurodesmus</i> sp.<br><i>Cosmarium</i> spp.<br><i>Desmidium</i> spp. | <i>Chlorella</i> spp.<br><i>Scenedesmus bijugatus</i><br><i>Eudorina</i> spp.<br><i>Pediastrum</i> spp.<br><i>Gemmellicystis</i> sp.<br><i>Staurastrum</i> sp.<br><i>Closterium</i> sp.                    | <i>Chlorella</i> spp.<br><i>Mougeotia</i> spp.                                                                                                                                                                 |
| Chrysophytes       | <i>Dinobryon</i> sp.<br><i>Dinobryon divergens</i><br><i>Uroglena</i> spp.<br><i>Mallomonas</i> spp.<br><i>Synura</i> spp.<br><i>Chrysosphaerella</i> spp.                                                     | —                                                                                                                                                                                                          | —                                                                                                                                                                                                              |
| Diatoms            | <i>Cyclotella glomerata</i><br><i>C.comensis</i><br><i>Asterionella formosa</i><br><i>Urosolenia</i> sp.<br><i>Aulacoseira distans</i>                                                                         | <i>Cyclotella meneghiniana</i><br><i>Stephanodiscus minutulus</i><br><i>S. neoastraea</i><br><i>Fragilaria</i> sp.<br><i>Asterionello</i> spp.<br><i>Melosira varians</i><br><i>Aulacoseira subarctica</i> | <i>Stephanodiscus rotula</i><br><i>S. hantzschii</i><br><i>S. astraea</i><br><i>Tabellaria fenestrata</i><br><i>Tabellaria</i> sp.<br><i>Diatoma</i> spp.<br><i>Aulacoseira ambigua</i><br><i>A. granulata</i> |
| Dinoflagellates    | —                                                                                                                                                                                                              | <i>Peridinium</i> spp.<br><i>Ceratium</i> spp.                                                                                                                                                             | —<br>—                                                                                                                                                                                                         |
| Euglenoids         | —                                                                                                                                                                                                              | —                                                                                                                                                                                                          | <i>Euglena</i> spp.<br><i>Phacus</i> spp.<br><i>Lepocinclis</i> spp.                                                                                                                                           |

Compiled from Gunkel (1994); Reynolds (1998).

Phytoplakton can well indicate the presence as well as the levels of toxic chemicals like pesticides and heavy metals. For example, presence of copper (algicide) at the concentration range 5–10 µg/l causes the elimination of cyanobacteria, thus facilitating the dominance of diatoms (*Nitzschia* and *Navicula*

spp.) and dinoflagellates. Atrazine acts selectively on planktonic algae causing elimination of *Oocystis*, *Chlorella* spp., *Ankistrodesmus* spp. and *Selenastrum* spp. but does not significantly change the distribution and the growth patterns of cyanobacteria. This indicates that cyanobacteria are not good indicators of herbicide pollution. At 120 µg/l atrazine facilitates the relative dominance of diatoms like *Nitzschia* spp., *Navicula* spp., *Fragilaria* and *Tabellaria* spp. Similarly, acidic pH (pH ≤ 4.0) selectively affects the specific groups of phytoplankton and the species showing acidity tolerance usually have the ability to grow heterotrophically when absorbable organic matters are present. The species of *Lobomonas*, *Cryptomonas* and *Moagoetia* (Chlorophyceae), *Synura* and *Dinobryon* (Chrysophyceae) and *Peridinium* (Dinophyceae) are well-known acidity indicators.

### 15.4.2 Periphytic Algae as Indicators

As sessile forms, the periphytic algae build a periphytic assemblage, which is very specific to the substratum, in which they grow. Many periphytic species, especially the diatoms, have been well-standardized as bioindicators. Different experimental protocols have been developed to use benthic and periphytic diatoms as indicators of water quality. Sladecek (1986) has described the saprobic activities of 323 diatom species for using them in water quality indices. The diatom index (CEC index) of European Union Environmental Protection Agency indicates to types of surface water types (ultraoligotrophy to polytropy) on the basis of the index value. Like planktonic algae, the periphytic algae are also highly sensitive to the changes in the physicochemistry of surface waters. Many periphytic diatoms are negatively affected by the enhanced nutrient load of water as the extensive growth of filamentous green algae and cyanobacteria restricts the habitat for diatoms. However, the diatoms also grow as epiphytes on the surface of other algal filaments when conditions favour them. In organically-loaded water bodies, which are moving from a trophic condition to a saprobic one, the periphytic growth becomes highly restricted due to enhanced turbidity of water (caused by increased bacterial density). The high sedimentation rate and the organic deposit on the substratum not only discourages the growth of periphytic algae but also affects their photosynthetic activities by light attenuation. However, certain diatoms and cyanobacteria are preferred in such water and lead mixotrophic mode of nutrition.

Uptake and accumulation of heavy metals also cause significant change in the periphytic species assemblage. Growth inhibiting levels of Cu (≥ 90 µg/l) retard the growth of many cyanobacteria but green algae show a comparatively less sensitivity to this metal concentration (Table 15.8). Periphytic diatoms also sensitively react to increase of acidity of surface waters. Acidophilic forms are encouraged to grow when pH falls below 5.0 resulting in simultaneous elimination of pH sensitive species. Green algal species and cyanobacteria prefer to grow at near neutral to alkaline pH. Species like *Aphanizomenon flos-aquae*, *Anabaena flos-aquae*, *Anabaenopsis* spp. and *Microcystis* spp. grow luxuriantly at pH ≥ 9.0. The acidophilic species like *Actinella punctata*, *Amphicampa hemicyclus*, *Anomoeoneis serians*, *Eunotia exigua*, etc., are preferred within the pH range of 6.0 – 8.0.

**Table 15.8:** The periphytic indicators showing different types of conditions in surface waters.

| Environmental condition | Indicator species                                                                                                                                                                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organic load            | <i>Calothrix ramenskii</i> , <i>Rivularia biasolettiana</i> , <i>Oscillatoria</i> sp., <i>Lyngbya</i> spp., <i>Schizothrix lacustris</i> , <i>Draparnaldia glomerata</i> .                                                                                                                               |
| Metal pollution         | <i>Chamaesiphon fuscus</i> , <i>Zygnema</i> spp., <i>Spirogyra</i> spp., <i>Clastidium setigerum</i> , <i>Mougeotia</i> a., <i>Ulothrix subtilis</i> , <i>Microspora palustris</i> , <i>Phormidium</i> sp., <i>Stigonema mamillosum</i> , <i>Cyanophanon mirabile</i> , <i>Tolypothrix penicillata</i> . |
| Acidic water            | <i>Anomoeoneis brachysira</i> , <i>Eunotia tenella</i> , <i>Eunotia veneris</i> , <i>Frustulia rhomboides</i> , <i>Asterionella rolfsii</i> , <i>Navicula</i> spp., <i>Tabellaria flocculosa</i> , <i>Tabellaria quadrisepata</i> .                                                                      |
| Alkaline water          | <i>Oscillatoria</i> spp., <i>Navicula subtilissima</i> , <i>Tabellaria binalis</i> .                                                                                                                                                                                                                     |

Compiled with modification from Lindstrom and Rorslett (1991).

## 15.5 BIOINDICATION BY BACTERIOPHAGES

Because of their constant presence in sewage and polluted waters, the use of bacteriophage as appropriate indicators of fecal pollution has been proposed. These organisms have also been suggested as indicators of viral pollution. This is because the structure, the morphology and the size, as well as the behaviour in the aquatic environment of many bacteriophages closely resemble those of enteric viruses. For these reasons, they have also been used extensively to evaluate virus resistance to disinfectants, to evaluate virus fate during water and wastewater treatment and as surface and groundwater tracers. The use of bacteriophages as indicators of fecal pollution is based on the assumption that their presence in water samples denotes the presence of bacteria capable of supporting the replication of the phage.

Two groups of phage in particular have been studied, viz., (a) the somatic coliphage, which infect *E. coli* host strains through cell wall receptors and (b) the F-specific RNA coliphage, which infect strains of *E. coli* and related bacteria through the F<sup>+</sup> or sex pili. A significant advantage of using coliphage is that they can be detected by simple and inexpensive techniques that yield results in 8–18 hours. Both a plating method (the agar overlay method) and the MPN method can be used to detect coliphage in volumes ranging from 1 ml. to 100 ml. The F-specific coliphage (male-specific phage) have received the greatest amount of attention because they are similar in size and shape to many of the pathogenic human enteric viruses. Coliphage f2, Φ174, MS-2 and PRD-1 are most commonly used as tracers and for evaluation of disinfectants. Because F-specific phage are infrequently detected in human fecal matter and show no direct relationship to the fecal pollution level, they cannot be considered indicators of fecal pollution. However, their presence in high numbers in wastewaters and their relatively high resistance to chlorination contribute to their consideration as an index of wastewater contamination and as potential indicators of enteric viruses.

Phage of *Bacteroides fragilis* have also been suggested as potential indicators of human viruses in the environment. *Bacteroides* spp. are strict anaerobes and are a major component of human feces. So bacteriophage active against these organisms have the potential to be suitable indicators of viral contamination. The phage that infect *B. fragilis* appear to be exclusively of human origin and appear to be present only in environmental samples contaminated with human fecal pollution.

## 15.6 OTHER INDICATOR ORGANISMS

A number of other organisms have also been considered to have potential as alternative indicator organisms or for use in certain applications (e.g., recreational waters). These include *Pseudomonas* spp., yeasts, acid-fast mycobacteria (*Mycobacterium fortuitum* and *M. phlei*), *Aeromonas* and *Staphylococcus*.

Within the genus *Pseudomonas*, the species of significant public health concern is *P. aeruginosa*. A characteristic of this species is that it can produce the blue pigment pyocyanin or the fluorescent pigment fluorescein or both. Numerous cases of folliculitis, dermatitis and ear (swimmer's ear) and urinary tract infections are due to *P. aeruginosa* associated with swimming in contaminated water or poorly-maintained swimming pools and hot tubs. Because of this association and its consistent presence in high numbers in sewage, *P. aeruginosa* has been suggested as a potential indicator for bad water quality in swimming pools, hot tubs and other recreational waters. However, as this organism is known to be ubiquitous in nature and can multiply under natural conditions (it can even grow in distilled water), it is believed to be of little value for fecal contamination studies.

*Staphylococcus aureus* and *Candida albicans* have been proposed as better indicators of body (ear nose and skin) infection associated with swimming. Recreational waters may serve as a vehicle for skin infections caused by *S. aureus* and some observers have recommended that this organism be used as an additional indicator of the sanitary quality of recreational waters, because its presence is associated with human activity in recreational waters.

The genus *Aeromonas* includes straight Gram-negative rods, facultatively anaerobic, that are included in the family Vibrionaceae. Only *Aeromonas hydrophila* has received attention as an organism of potential sanitary significance. *Aeromonas* occur in uncontaminated waters as well as in sewage and sewage-contaminated waters. The organism can be pathogenic for humans, other warm-blooded animals and cold-blooded animals including fish. Food borne outbreaks associated with *A. hydrophila* have been documented and it is considered an opportunistic pathogen in humans. Because of its association with nutrient-rich conditions, it has been suggested as an indicator of the nutrient status of natural waters. Table 15.9 summarizes potential applications of indicator organisms in the assessment of water quality.



**Table 15.9:** Bacterial indicators of water quality, their significant sources and potential uses.

| <i>Indicator</i>               | <i>Significant source<sup>a</sup></i> | <i>Potential use<sup>b</sup></i> |
|--------------------------------|---------------------------------------|----------------------------------|
| Coliforms                      | F S I R A                             | S                                |
| Fecal coliforms                | F S I R A                             | F S                              |
| Enterococcus                   | F S                                   | F S A D                          |
| <i>Clostridium perfringens</i> | F S                                   | F S D                            |
| <i>Candida albicans</i>        | F S                                   | P F S                            |
| Bifidobacteria                 | F S                                   | F S A D                          |
| Coliphage                      | S                                     | S                                |
| <i>Pseudomonas aeruginosa</i>  | S I R A                               | P S N                            |
| <i>Aeromonas hydrophila</i>    | S I R A                               | P S N                            |

Modified from Cabelli (1978).

<sup>a</sup> Relative to other sources: F, feces of warm-blooded animals; S, sewage; I, industrial wastes; R, runoff from uncontaminated soils; A, fresh and marine waters.

<sup>b</sup> Potential use: P, pathogen; F, fecal indicators; S, sewage indicator; A, separation of human from lower animal sources; D, proximity to fecal source; N, indicator of nutrient pollution.

## 15.7 BIOINDICATION OF SURFACE WATER QUALITY

In many of the limnetic systems, the productivity (primary as well as secondary productivity) is mostly determined by the growth and the photosynthetic carbon fixation rate of phytoplankton. In small and shallow water bodies, where the light can reach the bottom, a significant part of the primary production is attributed to benthic, submerged and floating macrophytes activities. In lake ecosystem, where the nutrient loading is significant from the catchment areas, the primary production is limited at least by the available inorganic phosphorus concentration of water. Anthropogenic input of phosphorus brings in qualitative and quantitative changes of the phytoplankton assemblage in a water body, thereby determining the lake productivity and consumer density. Plants that have a high phosphorus demand or have rate-limited adaptation for phosphorus uptake are encouraged by phosphorus input indicating an enhanced P-level of surface water.

The planktonic algae and macrophytes react differently to the nutrient variation of surface waters. The algae respond to the nutrient enrichment through an immediate increase of density resulting in formation of a bloom. Nutrient deficiency, thereafter, causes mass death of cells and enhances sinking flux. In addition, the density of phytoplankton, under nutrient limited conditions, is additionally stressed by an intense grazing pressure by a population of herbivores, that are encouraged in the preceeding nutrient-enriched condition. Variation of nutrient concentrations also change the composition of phytoplankton, certain species being favoured at a particular concentration range, while some others preferring another concentration range. As a result, the variations of nutrients,

especially nitrogen, phosphorus and silicon, facilitates regular succession of algal community. However, with respect to macrophytes, the nutrient variation can only limit the growth but rarely cause succession of species.

### 15.7.1 Indication of Trophy by Phytoplankton

The nutrient cycle begins with the standing phytoplankton biomass in surface-waters. The composition and the growth of grazers depend on the type and the population pattern of phytoplankton. In coastal oceans, the phytoplankton density and growth rate is regulated by the bioavailable forms of nutrients contributed by the terrestrial habitats, which in turn influence the coastal secondary productivity. The disturbances in phytoplankton growth and composition by the availability and remineralization rate of nutrients is reflected by the population fluctuation of the entire grazer group, which is known as the bottom-up reaction. Similarly, the influence of the grazers on the phytoplankton group through regulation of phytoplankton population by grazing is known as top-down disturbances. In a lake ecosystem, the eutrophication is caused by reduction of top-down disturbance.

It is a known fact that the phytoplankton productivity is strongly influenced by the available phosphorus load of the surface water. The change in primary production and the standing phytoplankton biomass regulate the net productivity of other higher trophic levels. In oligotrophic waters, the seasonal fluctuation of phytoplankton density lies at a minimum : maximum ratio of 1 : 10 while in a eutrophic water body the ratio can be as high as 1 : 250. Many local conditions, however, make temporary changes of such ratio, especially in small water bodies, through a sudden change in water physicochemistry, thus making the indication of trophy of such water bodies difficult. For example, Raeder (1990) observed that a manipulation of grazer population can convert a eutrophic ratio to an oligotrophic one within a fortnight in a small water body even though the physicochemistry of water is not appreciably changed. However, such problems are not encountered in large water bodies like lakes and rivers, in which the biota is at a stable homeostatic equilibrium with high elasticity of homeostatic plateau.

The growth strategies of phytoplankton in limnetic systems can be separated into two distinct categories:

1. *r* – Strategists: The relatively short-lived species with high growth rate, who prefer simple nutrients, which are easily available and readily taken up by cells. Such species react quickly to the changes in the concentration of bioavailable nutrients in surface waters.
2. *K*-Strategists: The relatively long-lived species that grow extremely slowly but are able to use nutrients and organic carbon forms that are usually not preferred by species showing *r*-strategy. These species are not much influenced by sudden change in the nutrient concentrations of the medium.

Sommer (1981) had observed an inverse correlation between the cell size and the net growth rate of phytoplankton. The phytoplankton composition having definite ratios of algal groups can be used as a parameter for bioindication of surface waters. Nutrient deficiency through an anthropogenic

manipulation of phosphate concentration can be marked by a change in the composition of phytoplankton assemblage. In relation to P-gradients, the growth rate of the assemblage lies between the groups showing r-strategy and K-strategy of growth. The assemblage experiences r-strategy of growth when the water body has a continuous growth supporting levels of nutrients, especially of nitrogen and phosphorus. Thus, the nutrient status of a water body is not only marked by phytoplankton biomass but also by the qualitative changes in the phytoplankton assemblage.

The PEG model (Plankton Ecology Group Model) of Sommer *et al.* (2006) states that not only the nutrient load of a water body but also the ratios of the growth limiting nutrients regulate the phytoplankton composition. For example, an increase in silicate : phosphate ratio stimulates diatom growth, which in turn improves the phytoplankton : zooplankton ratio as diatoms are slowly digested and, therefore, are not preferably grazed by zooplankton. By anthropogenic manipulation of phosphate level, the silicate : phosphate ratio can be changed to facilitate green algal growth. Under eutrophic situations, where P becomes limiting, the algal association is dominated by green algae provided nitrogen availability is not limited. When P becomes limiting (ratio  $\geq 8 : 1$ ), the dominance shifts to cyanobacteria (*Anabaena* spp., *Anabaenopsis* spp., *Aphanizomenon* spp., *Microcystis* spp.), majority of whom has a tendency to produce a bloom.

### 15.7.2 Trophy Indication by Periphyton

Limited information are available on the use of periphytic algae as indicator organisms to determine the surface water quality, though it is a known fact that in many systems the primary production by periphyton significantly contribute to the organic carbon demand of the surface water ecosystems. The periphytic algal assemblage acts as a good indicator of the hardness of surface water. The periphytic population increases with the progress of the trophy but decline is observed after eutrophic level to polytrophic level. This is due to increased turbidity of a polytrophic water body which discourages periphyton growth. A maximum possible periphyton biomass of 500 mg/m<sup>2</sup> has been predicted in surface waters with high eutrophic status.

Diatoms dominate in the density and the diversity of periphytic assemblage in almost all surface water bodies but the diatom dominance is comparatively less as compared to cyanobacteria in warm waters of tropics. Hofmann (1993) used the diatom biomass as the indicator of the trophic status of surface water ecosystem. By analyzing the diatom composition and determining the trophic spectrum of different diatom species, he assigned a location value to each diatom species and grouped them into different valence groups as given in Table 15.10. These values were used for determining the trophic status of a surface water by using the equation

$$T = \Sigma (n_i \cdot g_i \cdot l_i) / \Sigma n_i \cdot g_i$$

where T = Trophic index,  $l_i$  = Trophic location of the  $i$ th species,  $n_i$  = Abundance of the  $i$ th species and  $g_i$  = Biomass of the  $i$ th species.

**Table 15.10:** The location values of indicator species occurring in the water bodies of different trophy.

| <i>Trophic level (valence group)</i>   | <i>Local value (Rel. units)</i> |
|----------------------------------------|---------------------------------|
| Oligotrophic                           | 1.5                             |
| Oligo-to- $\beta$ -mesotrophic         | 2.0                             |
| $\beta$ -Meso-to $\alpha$ -mesotrophic | 2.5                             |
| $\alpha$ -Meso-to eutrophic            | 4.0                             |
| Eutrophic                              | 4.5                             |
| Eutrophic onwards                      | 5.0                             |

**Table 15.11:** The trophic index value of water bodies of different trophic status.

| <i>Trophic index (Rel. units)</i> | <i>Trophic state</i> |
|-----------------------------------|----------------------|
| 1.00 – 1.99                       | Oligotrophic         |
| 2.00 – 2.49                       | Oligo-to-mesotrophic |
| 2.50 – 3.49                       | Mesotrophic          |
| 3.50 – 3.99                       | Meso-to-eutrophic    |
| $\geq 4.0$                        | Eutrophic            |

Source: Gunkel (1994).

The location value of a species has been determined from the mean abundance of the species in the trophic spectrum of its occurrence. The calculated trophic indices remain between 1 and 5 and indicate different trophic states of surface-waters (Table 15.11).

### 15.7.3 Bioindication by Macrophytes

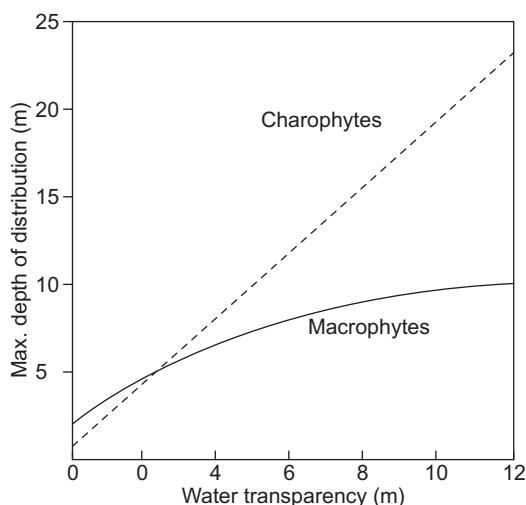
Not only the density and the diversity but also the distribution pattern of macroalgae and macrophytes have been used as reliable parameters for determining the surface water quality. Macrophytes, as a bioindicator parameter, were first developed in the Scandanavians and then were extremely used in Europe to determine the trophic states of surface waters. It has been observed that the rate of eutrophication of a water body shows an inverse relationship with the area of occupation by characean macrophytes (*Chara hispida*, *C. tomentosa*, *C. aspera*, *Nitella syncarpa*, *N. opaca*, etc.). For example, distribution of characean members in German–Switzerland Boden lake has been reduced from 4.35 km<sup>2</sup> in the 1990's to 1.65 km<sup>2</sup> in 2003, which has been well-correlated with the increased density of phytoplankton and reduced water transparency. On the other hand, the characean members are found well-distributed without any significant change during last 10 years in Deras Reservoir of Chandaka Elephant Sanctuary, Bhubaneswar. This show that there is a direct relation between the anthropogenic impact on the water body and distribution and abundance of characean macrophytes.

Blindow (1991) have observed that distribution of characean macrophytes is directly influenced by the phytoplankton density vis-à-vis water transparency. The macrophytes are restricted to shallow zones of water bodies with increase in the

eutrophic status. This indicates that under eutrophic condition, the distribution pattern of characean macrophytes is more influenced by light availability than by nutrient load. There are evidences that the growth of *Chara globularis* did not change appreciably between the P-concentration of 100–2000  $\mu\text{g/l}$ . Similarly, *C.tomentosa* and *C.hispida* growth remained unchanged up to 1000  $\mu\text{g/l}$  of phosphorus (Gunkel, 1994). However, other macrophytes like *Potamogeton*, *Zannichellia* and *Fragmitus* grow profusely and form a thick submerged vegetation with increase in the nutrient load, especially of phosphorus, of surface waters. The best example of growth of such weeds is seen in the Chilika Lake of Orissa and Dal Lake of Jammu and Kashmir. Fig. 15.3 shows the relationship between the water transparency and the macrophyte distribution in shallow lakes and reservoirs.

It is a known fact that macrophytes distribution is generally restricted up to 10 m depth in lakes and reservoirs. Within this depth, the hydrostatic pressure is so high that the plants require to fill the internal tissues with gas in order to keep themselves upright inside water. In rare cases, the macrophytes like *Chara* and *Utricularia* are present beyond 12 m depth in freshwater lakes. However, ultraoligotrophic and extremely transparent freshwater lakes can have distribution of macrophytes, especially charophytes, up to a depth as deep as 150 m. The interaction of a water body with the surrounding terrestrial systems and the biomass as well as nutrient contribution of the catchments to the water body strongly influences the macrophyte productivity and distribution.

The distribution pattern of macrophytes and the relationship between water transparency and macrophyte abundance can be used as a good parameter for bioindication. The total nutrient load, in general, and phosphorus load, in particular, are well-indicated by the macrophytes distribution. As phosphorus level in water significantly influences phytoplankton growth and carbon availability to bacterial



**Figure 15.3** - The relationship between the maximum depth of macrophyte distribution and the water transparency of water bodies with different trophic status.

activity, increased P-loading negatively affects the water transparency and limits macrophyte growth. At high P-load, extensive light attenuation eliminates sensitive macrophytes, that are not adapted to low light regime, from the benthic vegetation. Gradual increase in water turbidity brings in a succession of macrophytes facilitating the growth of floating members like *Pistia*, *Eichhornia*, *Lemna*, *Azolla*, *Salvinia*, etc. The rooted but emergent macrophytes such as *Trapa*, *Nymphaea*, etc. are, however, not adversely affected by the change in water transparency.

The trophic status vis-à-vis the pollution load of a water body can be determined by using the presence and the distribution of macrophytes, called macrophyte index. On the basis of their preference to occupy the depth in a water body, the macrophytes can be grouped into indicator levels (Table 15.12). The index values range between 1 (ultraoligotrophic and oligotrophic waters) to 5 (polytrophic waters). The species of *Chara* are found to be limited to ultraoligotrophic up to mesotrophic water bodies, however, in mesotrophic waters the depth, to which chara distributes itself, is substantially less than that of oligotrophic waters. On the other hand, there is high abundance of floating and emergent macrophytes in eutrophic and polytrophic waters with these species having high indication value. The macrophyte index value for a water body is calculated by taking the average of index values of the macrophytes present in the water body and assigned an indication value. Thus

**Table 15.12:** The macrophyte indication groups with their assigned indication value.

| Indication value<br>(Rel. units) | Name of the species                                                                                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0                              | <i>Chara hispida</i> , <i>C. polyacantha</i> , <i>C. strigosa</i> , <i>Potamogeton coloratus</i> <i>Utricularia achroleuca</i>                                                                                                      |
| 1.5                              | <i>Chara aspera</i> , <i>C. intermedia</i>                                                                                                                                                                                          |
| 2.0                              | <i>Chara delicatula</i> , <i>C. tomentosa</i> , <i>Tolypella glomerata</i> , <i>Potamogeton alpinus</i>                                                                                                                             |
| 2.5                              | <i>Chara contraria</i> , <i>C. fragilis</i> , <i>Nitella opaca</i> , <i>Nitellopsis obtusa</i> , <i>Potamogeton gramineus</i> , <i>P. natans</i> , <i>Potamogeton</i> sp.                                                           |
| 3.0                              | <i>Chara vulgaris</i> , <i>Myriophyllum spicatum</i> , <i>Potamogeton filiformis</i> , <i>P. perfoliatus</i>                                                                                                                        |
| 3.5                              | <i>Myriophyllum verticillatum</i> , <i>Potamogeton berchtoldii</i> , <i>P. lucens</i> , <i>P. praelongus</i> , <i>P. pusillus</i>                                                                                                   |
| 4.0                              | <i>Hippuris vulgaris</i> , <i>Lagarosiphon major</i> , <i>Potamogeton pectinatus</i> , <i>Azolla pinnata</i>                                                                                                                        |
| 4.5                              | <i>Callitriche cophocarpa</i> , <i>Elodea nuttallii</i> , <i>E. canadensis</i> , <i>Potamogeton crispus</i> , <i>P. obtusifolius</i> , <i>Ranunculus circinatus</i> , <i>R. trichophyllum</i>                                       |
| 5.0                              | <i>Eichhornia crassipes</i> , <i>Lemna minor</i> , <i>Ceratophyllum demersum</i> , <i>Potamogeton mucronatus</i> , <i>P. nodosus</i> , <i>Sagittaria sogittifolia</i> , <i>Spirodela polyrrhiza</i> , <i>Zannichellia palustris</i> |

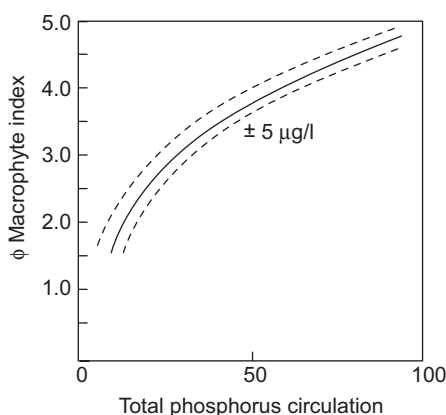
Modified from Gunkel (1994).

$$MI = \frac{\sum_{i=1}^n IV_i}{n}$$

where  $IV_i$  = Indication value of  $i$ th species

and  $n$  = No. of species with indication value taken into consideration.

Taking 154 water bodies Melzer (1992) has observed that the MI values could be well-correlated with total phosphorus content of a water body. The multiple regression analysis between MI and total phosphorus level and phosphorus remineralization rate showed a significant relationship, with MI increasing on enrichment of the water body, with phosphorus (Fig. 15.4). In the selected water bodies, the author observed the MI value between 2.43 and 4.88 with an average of 3.25 *i.e.*,  $\beta$ -mesotrophic condition. There is, however, no relationship between the MI value and nitrate or ammonia concentration of water as at higher trophic conditions, nitrogen no more becomes limiting for the phytoplankton population.



**Figure 15.4** - The values of average macrophyte index in relation to phosphorus concentration that undergo total cycling in a water body.

## 15.8 BIOINDICATION OF GROUNDWATER QUALITY

The groundwater, separated from the surface waters by numerous soil layers and stones, has a characteristic fauna and microbial diversity. Being kept away from light, the microbial population of this zone is highly deficient of producers, especially of those who perform oxygenic photosynthesis. In turn, there is abundance and dominance of protozoa and heterotrophic bacteria in the groundwater microbial population. The size, distribution and population of protozoa, bacteria and microscopic animals in groundwater is determined and influenced by the pore size of the soil layer separating the groundwater from surface waters. The quality of surface waters and the microbial activity in the sediment of surface water bodies also determine the microbial diversity of groundwater as they act as



the main sources of carbon and nutrients to the groundwater. The organic matter load and leaching of soluble and particulate organic matters from the surface and sediments to the groundwater zone is the determining force of groundwater microbial processes. Similarly, the size of the groundwater animals and their movement to and from the surface waters are dependent on the separating layer's permeability and food availability.

There are worldwide research to study the nature and the characteristics of groundwater microbes with an aim to maintain the quality of groundwater and to restrict the microbial population of groundwater below the acceptable level following appropriate measures. However, we are still short of knowledge on the nature, growth requirements, reproduction rates and control mechanisms of many groundwater microbes and small animals, making it difficult for us to take appropriate steps to maintain the groundwater quality. Majority of information available to us are from the natural systems, as the attempts to culture, maintain and study the characteristics of groundwater microbes have failed in many cases. However, there are successes in study and cultivation of indicator microorganisms, which are also found in surface waters. (e.g. coliforms, ammonifiers, sulphate reducers, etc).

The zooplankton and animals observed in groundwater can be basically categorised into three types viz;

1. The *ostigobionts*—The organisms that are true groundwater species living for their entire life cycle in groundwater;
2. The *stylophiles*—The organisms, who live in groundwater but a significant part of their life cycle is required to be completed in surface waters; and
3. The *stylogenesis*—The organisms, who live in groundwater for a significantly short period and always prefer to move to the surface water environment. The *stygobionts* live in complete darkness and at a relatively constant temperature without having capacity to show adjustments to fluctuations of physicochemical characteristics of groundwater. Fortunately, the groundwater physicochemistry is not changed as frequently as in the surface waters. Thus, the *stygobionts* (including both zooplankton and bacteria) act as good indicators of the groundwater quality.

The composition of groundwater adapted organisms is very heterogeneous. There is, however, many phytoplankton typically adapted to groundwater conditions, though in shallow aquifers, there is presence of some cyanobacteria and diatoms. These organisms are transported through groundwater recharge from surface and are indicators of groundwater contamination. While diatoms have a very short residence time in groundwater environment, cyanobacteria can prevail for a considerably long period by adopting heterotrophic mode of nutrition.

## 15.9 CONCLUSION

Bacterial indicators such as coliforms have been used for the development of water quality standards. For example, the US Environmental Protection Agency

(US EPA) has set a standard of no detectable coliforms per 100 ml of drinking water. Similarly, standards for recreational waters and wastewater reuse have also been determined by different countries. Criteria and guidelines are terms used to describe recommendations for acceptable levels of indicator microorganisms. They are not legally enforceable but serve as guidance indicating that a potential water quality problem exists. Ideally, all standards would indicate that an unacceptable public health threat exists or that some relationship exists between the amount of illness and the level of indicator organisms. Such information is difficult to acquire because of the involvement of costly epidemiological studies that are often difficult to interpret because of confounding factors.

The use of microbial standards also requires the development of standard methods and quality assurance or quality control plans for the laboratories that will do the monitoring. Knowledge of how to sample and how often to sample is also important. Proper and frequent sampling and microbiological evaluation of water quality prevent the deterioration of the water bodies and facilitates the planning for maintaining the water quality. The use of different microbial indicators like bacteria, algae, viruses and protozoa provide better information on the pollution status of a water body than that determined by a single indicator group. For determination of surface water quality, the macrophytes and their associated periphytic microbes have been found very useful.

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# SUBJECT INDEX

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## A

- Acetogens 276
- Acidophiles 186
- Acinetobacter* 408
- Actinomycetes 36
- Activated sludge systems 399
  - aeration 399
- Adenylate energy charge 230
- Aerobic anoxygenic phototrophs 109
  - metabolic efficiency 109
  - photosynthesis 109
  - pigments 109
- Aerobic degradation 385, 451
  - by bacteria 385
  - energy balance 385
  - of aliphatics 451
  - of carbohydrates 387
  - of hydrocarbon 451
  - of lipids 389
  - of neutral fats 389
  - of proteins 388
- Aerobic degrading bacteria 450
- Aerobic hydrolysis 383
  - biological aspects 383
  - kinetic aspects 383
  - of biopolymers 383
- Aerobic microbes in sediment 92
- Airborne pathogens 176
- Algae 45
  - Charophyta 48
  - Chlorophyta 48
  - Chrysophyta 48
  - Diatoms 48
  - Dinoflagellates 49
  - Euglenophyta 48
  - in metal contaminated sites 417
  - indicator species 496
  - Phaeophyta 49
  - Pyrrhophyta 49
  - Rhodophyta 50
  - symbiosis 302-306
  - threatened species 497
- Algal
  - classification 47
  - diversity 46
  - pigments 47
  - succession 73
  - symbiosis 312
    - in bryophytes 312
    - with angiosperms 314
- Alkaliphiles 5, 186
- AM Fungi 3, 160
  - classification 161
  - diversity 160, 163
  - growth 162
- Ammonia assimilation 346
- Ammonifex* 197
- Ammonification 347, 403
- Ammonium oxidation 348
- Anabaena azollae* 313
- Anabaena cycadoidae* 314
  - pigments 314
- Anabaena* in cycads 313
- Anaerobic degradation 383, 390, 457, 458
  - control 391
  - of acetone 460
  - of aliphatics 457
  - of amino acids 393
  - of aniline 460
  - of aromatic compounds 460
  - of biopolymers 383
  - of carbohydrates 392
  - of catechol 462
  - of cresol 461, 463
  - of EDTA 459
  - of ether compounds 457
  - of halogenated organics 464
  - of hydroquinone 462

- of hydroxybenzoate 460
  - of hydroxyhydroquinone 463
  - of ketones 459
  - of N-alkyl compounds 458
  - of nitrolotriacetate 458
  - of non-ionic surfactants 457
  - of phenol 460
  - of phenylmethyl ether 458
  - of polyethylene glycol 458
  - of proteins 393
  - of resorcinol 463
  - of S-alkyl compounds 459
  - of tetramethyl amines 458
  - of trihydroxybenzenes 463
  - of trimethyl amines 458
  - syntrophic interaction 390
  - through food chain 390
- Anaerobic degrading bacteria 448
- Anaerobic microbes
- in sediment 93,124
- Anaerobic respiration 233
- Anamox 406
- Animal symbiosis 325
- with algae 326
  - with chemolithotrophs 329
  - with cyanobacteria 328
  - with luminescent bacteria 330
  - with methanotrophs 329
  - with Prochlorophytes 327
- Animal-microbe interaction 292
- mutualism 292
  - predation 292
  - symbiosis 294, 324
- Anoxygenic photosynthesis 25
- Antagonism 280
- Antibiosis 280
- Antibiotic production 281
- Apocarotenoids 164
- Appresorium 162
- Aquatic fungi 85
- Aquefcae 25
- Aquifers 137
- Aquifex 25
- Archaea 19
- characteristics 23
  - chemolithotrophs 20
  - chromosome 20
  - genetics 20
  - genome 20
  - hyperthermophilic 195
  - mesophiles 19
  - metabolism 22
  - morphology 19
  - multiplication 20
  - of extreme habitats 20
  - organotrophs 20, 22
  - physiology 20
- Archaeoglobi 24
- Arsenate resistance 422
- Aspergillus* 5
- Athalassohaline environment 212
- microbes 212
  - adaptation 212
  - archaea 212
  - bacteria 212
  - cyanobacteria 212
- Atmosphere 165
- Atmospheric dispersal of microbes 167
- eddy layer 166
  - laminar boundary layers 166
  - turbulent boundary layer 166
  - viable particles 166
- Azolla-Anabaena* symbiosis 313
- Azorhizobium* 317
- BAC 258
- Bacillus* 35
- Bacteria
- bioindicator 488
  - diversity pattern 492
  - in polluted soil 449
  - lactic acid 36
  - low G+C 34
  - marine 105
  - of cold habitats 204
  - of hotspots 199
  - sulphate reducers 358
  - symbiosis 302, 305
  - vertical variation 91
- Bacterial
- abundance 19
  - diversity of soil 140
  - indicators 502
  - of surface water 502
- Bacterioplankton 81, 87,105
- biomass 87,105
  - distribution 87,106
  - grazing 81,122
  - in freshwater 87

- in littoral zones 94
  - in marine water 105
  - in rivers 88
  - mobility 89
  - of different ecosystems 89
  - population 88
  - spatial variability 89
- Bacteriophage 81
- Bacteroides* 293
  - pelagic 105
- Bacteroidetes 27
  - diseases 27
  - habitat 27
- Barophiles 186
- Bdellovibrio* 31, 282
  - replication pattern 282
- Beggiatoa* 356
- Benthic bacteria 92,123
- Benzoyl CoA pathway 461
- Bergey's manual 19
- Bioactive molecules 9
- Bioaerosols 167
  - composition 168
  - control 184
  - deposition 171
    - electrostatic 172
    - gravitational settling 171
    - molecular diffusion 172
    - surface impaction 172
  - diffusion 170
  - from agriculture 173
  - in residential buildings 175
  - in waste disposal 175
  - indoor 175
  - lunching 168
  - outdoor 173
  - sources 169
  - transport 170
- Biocatalysts 9
- Biochemical agents 12
- Biofilm 286, 375
  - colonization strategy 286
  - community 379
  - for wastewater treatment 379
  - formation 376
  - interaction of microbes 380
  - metabolic communication 378
  - microbes 376, 378
  - microbial
    - behaviour 378
    - competition 380
    - succession 380
    - regulation 378
    - spatial heterogeneity 380
- Biofilters 395
  - conventional 395
  - high rate 397
  - hydraulic loading 396, 398
  - packing media 396
  - percolating 397
  - rotating 398
- Biogeochemical cycles 333
  - carbon 335
  - interrelationship 365
  - nitrogen 343
  - phosphorus 359
  - sulphur 353
- Bioindication 486
  - by bacterial assemblage 494
  - by bacteriophages 500
  - by coliform bacteria 486
  - by heterotrophic plate count 492
  - by saprophytic bacteria 494
  - of ground water quality 508
  - of surface water 502
    - by microphytes 505
    - by periphyton 504
    - by phytoplankton 503
- Bioindication value 507
- Bioindicator list 498
- Bioindicator
  - of trophic status 498
  - organisms 487, 491
    - Aeromonas* 501
    - algae 496
    - faecal streptococci 491
    - mycobacteria 501
    - periphytic algae 499
    - potential uses 502
    - Pseudomonas* 501
    - saprophytic bacteria 494
    - Staphylococcus* 501
    - periphyton 504
- Bioleaching 13, 437
  - controlled 438
  - factors 439
  - heap leaching 438
  - microbes 437

- uncontrolled 439
- vat leaching 439
- Biological pest control 12
- Bio-manipulation 14, 436, 480
  - for metal removal 436
- Biopolymers 385
  - degradation 385
- Bioremediation 11
  - potential 11
- Biosorption 421, 440
  - by microbial cells 441
  - by particles 442
  - in biofilm 443
  - in immobilized cells 442
  - of metals 440
- Biosurfactant 14, 454, 426
- BOD measurement 231
- Borella* 26
- Bradyrhizobium* 317
- Brucella* 28
- cad* genes 431
- Cadmium
  - binding proteins 426, 434
  - efflux 431
  - resistance 421, 431, 435
  - toxicity 421
- Caminibacter* 197
- Campylobacter* 32
- Carbon
  - cycle 335
  - energy flow 337
  - fixation 229
  - respiration 338
  - transport 336
  - utilization 226, 232
- Carbonaceous oxidation 374
- Carboxidotrophs 108
- Catechol 454
  - aerobic degradation 456
  - metacleaveage 457
  - orthocleaveage 457
  - pathway 455
- Cation exchange capacity 132
- Cellobiose cleavage 384
- Cellulolytic organisms 384
- Cellulose degradation 383, 392
- Cellulosome 384
- Chlorobi 26
- Chloroflexi 26
- Clays 132
  - microbial attachment 132
- Clostridium* 34
- Clostridium perfringens* 491
- CO degradation 342
- CO oxidation 198
- CO oxidizers 108
- CO<sub>2</sub> stripping 382
- Coccolithophorids
  - marine 118
- Cold adapted enzymes 203
- Cold habitat 204
  - algal diversity 206
  - cyanobacteria 206
  - microbes 204
  - lipid composition 207
  - membrane structure 207
- Cold signal transduction 208
  - in cyanobacteria 208
- Coliform bacteria 488
  - faecal 490
  - in biofilm 489
  - MF test 490
  - MPN test 489
  - P-A test 490
- Colony hybridization 242
- Cometabolic adaptation 481
- Cometabolic degradation 272, 474
- Commensal relationship 270
  - metabolism 271
  - of fungi 271
- Commensalism 292
- Community bacterial DNA 238
  - extraction 238
  - separation 240
- Community genome 4
- Competition 277
  - in biofilm 380
- Competitive exclusive principle 278
- Competitive interactions 11
- Concurrence among microbes 474
- Conidial fungi 85
- Contaminated
  - ecosystems 7
  - sites 14
- Cooperative interactions 264
- Copper



- efflux 433
- MT 434
- resistance 421, 435
- transporting genes 433
- Coral reefs 121
  - microalgae 121
- Coxiella* 30
- Crenarchaeota 24
- Crenothrix* 27
- cta* genes 433
- CUP locus 435
- Cyanelles 308
- Cyanobacteria 25, 37
  - classification 38
  - cold signal transduction 208
  - diversity 39
  - fatty acid desaturation 207
  - in liverworts 312
  - in metal contaminated sites 416
  - marine 118
  - metal resistance 432
  - of cold habitats 206
  - of metal polluted habitat 217
  - pigments 37
  - symbiosis 40, 307
    - in bryophytes 311
    - with plant 311
- Cyanobiont 306
  - location in host 312
- Cytophaga* 27
- Deep sea
  - bacteria 200
  - conidial fungi 201
  - eukaryotes 201
  - floor 124
    - degradation processes 125
    - microbial diversity 126
    - organic matter 125
  - fungi 200
  - microorganisms 124, 199
    - adaptation 202
  - prokaryotes 200
  - protists 201
- Deferribacter* 196
- Dehydrogenase assay 230, 248
- Deinococci 25
- Denitrification 350, 405
- Denitrifying bacteria 351
  - diversity 352
  - metabolism 351
- Desulfocella* 211
- Desulfotomaculum* 34
- Desulfovibrio* 277
- Desulfurococcales 23
- Desulphurella* 195
- Desulphurobacterium* 196
- Desulphuromonas* 358
- Detrital dynamics 61
- Detrital particulate matter 95
- Detritosphere 103
- Detritus 61, 103
  - metabolism 62, 103
- Didinium* 284
- Dinitrogenase 315
  - activity 315
- Dinoflagellates 118, 308
  - marine 118
  - symbiosis 308
- Disposal of biosolids 175
- DOM plume 103
- Drainage basin 59
- Drug resistance 9
- EBPR 408
- Ecological relationship 262
  - of microbes 262
- Ecosystem function 4
  - stability 5
- Ectoendomycorrhizae 158
- Ectomycorrhizae 157, 159
- Ectomycorrhizal infection 159
- Ectoparasites 281
- Electrical conductivity estimation 232
- Elite organisms 7
- Endobiotic habitat 98
- Endomycorrhizae 157, 160
- Endoparasites 281
- Endophytic microorganisms 8
- Endosymbiotic
  - bacteria 306
  - methanogens 306
- Endozoic algae 326
  - in corals 327
- Environmental niche 4
- Enzymatic degradation 482
  - of hydrocarbons 482
  - of recalcitrant organics 482

- Enzyme assays 246
  - ATPase 252
  - dehydrogenase 248
  - esterase 250
  - phosphatase 247, 251
- Epibiosis 380
- Epibiotic habitat 98
- EPS for metal binding 420
- EPS of bacteria 420
- EPSS 322
- Epulopiscium* 331
- Estuarine habitat 97
- Euarchaeota 24
- Eutrophic indicators 497
- exo* genes 323
- Extreme environments 7, 187
  - acid and alkali stress 189
  - metal stress 188
  - salinity stress 188
  - temperature stress 187
  - toxic organic pollutants 189
- Extreme halophiles 24
- Extremophiles 186
- Extremophilic bacteria 4
- Faecal coliforms 490
  - enterococci 491
  - streptococci 491
  - Clostridium* spp. 491
- FC/FS ratio 491
- FDA hydrolysis 230
- Ferroplasma* 25
- Firmicutes 26
- fix* genes 323
- Flavonoids 318
- Flexibacter* 27
- Fluorescent immunolabeling 255
- Fluorescent pseudomonads 5
- Frankia* 289
- Freshwater
  - algal diversity 64
  - biological community 60
  - chrysophytes 67
  - ciliates 78
  - cyanobacteria 64
  - diatoms 68
  - dinoflagellates 68
  - ecology 58
  - environment 56
  - euglenoids 68
  - flagellates 77
  - green algae 65
  - microbes 55
  - phytoplankton productivity 69
  - phytoplankton size 67
  - protists 77
  - protozoa 77
  - sarcodines 78
  - trophic structure 60
  - zonation 57
  - zooplankton 76
- Fungal
  - classification 43
  - complexity 42
  - degradation 476
    - by bacidiomycetes 477
    - for pollutant degradation 481
    - in contaminated habitat 481
    - of aliphatic hydrocarbons 477
    - of aromatic compounds 478
    - of organopollutants 476
    - of phenol 479
    - through ring cleavage 479
    - to organopollutants 482
    - to pesticides 481
  - disease 41
  - nutrition 42
  - pathogens 174
  - remineralization 86
- Fungi 40
  - Acrasiomycota 45
  - aquatic 85
  - Ascomycota 43
  - Basidiomycota 44
  - Chytridiomycota 44
  - conidial 85
  - Deuteromycota 44
  - environmental importance 41
  - in marine environment 112
  - in phyllosphere 291
  - Myxomycota 44
  - of mangrove habitat 113
  - of metal polluted habitat 216
  - of soil 141
  - Oomycota 45
  - Slime molds 45
  - symbiosis 302

- Thermophilic 193
- Zygomycota 43
- Gaia hypothesis 334
- Gene database 240
- Gene probes 240
  - detection 241
- Genetic intelligence 7
- Geobacillus* 196
- Global carbon reservoir 335
- Glomus* 160
- Glycosyl hydrolase 384
- Graphium* 203
- Grassibacillus* 211
- Green sulphur bacteria 25
- Ground water organisms 508
- Growth associated degradation 451
  - of aliphatics 451
  - of aromatic compounds 452
  - of benzene 453
  - of cyclic alkanes 451
  - of rhamnolipids 454
- Growth of microbes 223
  - BOD 225
  - carbon utilization 226, 232
  - cell mass 225
  - nitrate reduction 225
  - nitrification 223
  - protein measurement 225
  - substrate utilization 223
  - Sulphur oxidation 224
  - terminal electron acceptance 224
- Gunnera* 291, 314
- Gut microflora 9
- Haloalkalithermophiles 191, 212
  - adaptation 215
  - electrochemical gradient 215
  - lateral proton transfer 215
- Halobacteria 24, 210
- Halococcus* 24
- Halogenated organics 370
  - mobilization 370
- Halomethanes 370
- Halomonas* 211
- Halonatronum* 191
- Halophiles 186
  - physiological properties 213
  - temperature tolerance 212
- Halophilic
  - aerobes 210
  - archaea 211
  - microbes 192, 209
  - proteins 214
- Halothermophiles 211, 215
- Heavy metal 411
  - binding 414
    - sites 413
  - bioavailability 414
  - cellular interaction 420
  - community resistance 416
  - complexation 424 - 427
  - by organic metabolites 424
  - extracellular 424
  - intracellular 426
  - enzymatic transformation 427
  - enzyme modification 419
  - intracellular accumulation 419, 420
  - methylation 423
    - in sulphate rich habitat 423, 425
  - microbial transformation 423
  - precipitation 425
  - resistance 421
    - adsorption 425
    - biosorption 421
    - genetic aspect 429
    - in bacteria 421, 425, 430
    - in cyanobacteria 432
    - in eukaryotes 435
    - in fungi 434
    - transport related 421
  - tolerance 418
  - toxicity 412
    - in aquatic sediment 417
    - in ecosystem 415
    - to algae 417
    - to cyanobacteria 416
    - to protozoa 417
  - transformation 419
  - uptake 414
  - volatilization 424
- Helicobacter* 32
- Heliobacterium* 34
- Heliophilum* 34
- Hemicellulose degradation 384, 392
- HEPA filters 184
- Heterotrophic
  - metabolism 88, 108
  - nitrifiers 405
  - plate count 492

- Hik33 208  
Hok 267  
Host-parasite relationship 281  
Hotspring thermophiles 198  
*Humicola* 193  
*hup* genes 324  
Hydrocarbon  
  contamination 11  
  decontamination 482  
    by immobilized enzymes 483  
    by microbial cells 483  
  degradation 451  
    by aerobic bacteria 451  
    by anaerobic bacteria 458  
  utilizing lithotrophs 197  
*Hydrogenobacter* 25  
Hydrothermal vents 25  
Hydroxyhydroquinone pathway 464  
Hyperparasitism 283  
Hypersaline environment 188, 209  
Hyperthermophiles 23  
Hyperthermophilic archaea 195  
Immunoassays 253  
Immunocytochemical assay 256  
Immunofluorescence 255  
Immunomagnetic separation 255  
Indigenous soil microbes 140  
Infection thread 321  
Intestinal symbionts 325  
*ips* genes 323  
Jasmonates 163  
*Lactobacillus* 35  
Lake  
  ecosystem 56  
    concept 59  
  living biota 61  
  phytoplankton 62  
    association 63  
  POC load 61  
Lectins 321  
*Legionella* 30  
Legume flavonoids 318  
Legume-*Rhizobium*  
  interaction 289  
Legume-*Rhizobium* symbiosis 315  
  biochemical 321  
  molecular 323  
Lentic ecosystem 56  
*Leptospira* 26  
Lichen 303  
  metabolic association 304  
  symbiosis 303  
  Littoral decomposition 94  
Lotic ecosystem 56  
LPS of bacteria 420  
Macrophyte  
  distribution 506  
  index 507  
  indicator 505, 507  
Marine bacteria 105  
  anoxygenic phototrophy 109  
  carbon fixation 107  
  CO oxidation 108  
  density 105  
  distribution 106  
  lithoheterotrophy 108  
  seasonal dynamics 107  
Marine ecosystems 99  
  bacterial activity 99  
  microbial groups 100  
  microbial structure 99  
Marine environment 97  
  characteristics 98  
  colloidal particles 101  
  detritus metabolism 103  
  DOM degradation 102  
  gel particles 105  
  macromolecules 102  
  microbial food web 111  
  microbial interaction 102  
    algae-bacteria 102  
    bacteria-protists 103  
    with detritus 103  
  mucous sheets 101  
  organic matter 101  
Marine fungi 112  
  colonization 113  
  diversity 112, 115  
Marine microbes 97  
  carbon assimilation 101  
  hydrolytic enzymes 101  
  metabolic interaction 101  
  spatial distribution 100  
Marine phytoplankton 114  
  abundance 119  
  Coccolithophorids 118

- Cyanobacteria 118
- dinoflagellates 118
- diversity 114-120
- picobiliphytes 118
- primary production 114
- Prochlorophytes 118
- regional pattern 120
- silicoflagellates 118
- size distribution 117
- Marine protozoa 122
  - in microbial food web 122
  - Tintinids 122
- Marine snow 104
  - formation 104
  - plumes 105
  - sinking rate 105
- Mass culture 9
- Membrane filter test 490
- mer* genes 428, 430
- Mercuric reductase 427
- Mercury methylation 429
- Mesorhizobium* 317
- Metabolic diversity 18
- Metaclevage of aromatics 456
- Metagenome 11
  - analysis 258
  - library 259
- Metal binding proteins 426
- Metal contaminated environment 216
  - algal diversity 218
  - bacteria 216
  - cyanobacteria 217
  - fungi 216
  - microbes 216
- Metal efflux 422, 431
- Metal pollution 411
  - control 411
- Metal removal 427, 437
  - by bioleaching 437
  - by biosorption 440
  - by volatilization 427
- Metal resistance 427
- Metallothionein 426
  - copper induced 434
  - in cyanobacteria 432
  - in fungi 434
- Metazoan hypothesis 84
- Methane oxidation 341
- Methanobacteria 24
  - Methanobacterium* 20, 277
  - Methanococcus* 24
  - Methanogenesis 276, 340
  - Methanogenium* 24
  - Methanogens 23, 276
  - Methanosalsum* 212
  - Methanosarcina* 277
  - Methanothermus* 24
  - Methylococcus* 30
  - Methylomonas* 30
  - Methylotroph 30, 342
  - Microaerophiles 25
  - Microarrays 242
  - Microbes
    - hydrocarbon interaction 11
    - barophilic 200
    - deep sea 124
    - freshwater 55
    - in cold habitats 204
    - in environment 2
    - in estuarine habitat 97
    - in human health 9
    - in human welfare 12
    - in marine habitat 97
    - of agricultural soil 143
    - of deep sub-surface soil 148
    - of thalassohaline environment 211
    - of upper sub-surface soil 146
    - psychrophilic 200
    - thermal adaptation 199
  - Microbial activity
    - in ocean floor 123
    - in running waters 95
    - in sediment 90, 95
  - Microbial
    - application 10
    - biochemical diversity 4
    - classification 19
    - community 3
      - analysis 7
    - ecology 4
    - competition 266
    - consortia 11
    - cooperation 263
    - corrosion 366
      - by biofilm 368
      - by SRB 369
    - inhibitors 369
    - of concrete 369

- of metal 367
- counts 149
  - of sub surface soil 149
  - of surface soil 149
- Microbial degradation 338, 381, 446, 465
  - anaerobic 340
  - of aromatic compounds 452
  - of C1 compounds 342
  - of carbamates 471
  - of carbaryl 472
  - of carbofuran 472
  - of cellulose 339
  - of CO 342
  - of DDT 466
  - of hemicellulose 339
  - of lignin 339
  - of methyl parathion 467
  - of organic carbon 339
  - of organic matters 381
  - of organochlorines 466
  - of organophosphates 467
  - of pesticides 464
  - of pyrethroids 473
  - of xenobiotics 446
- Microbial diversity 4, 16, 128, 199, 216, 221
  - in metal contaminated habitat 415
  - of air 165
  - of aquatic environment 55
  - of chemically perturbed soil 221
  - of deep sea 199
  - of extreme environment 186
  - of shallow aquifers 149
  - of sub-surface soil 146
  - of terrestrial environment 128
- Microbial domains 17
- Microbial enzymes 12
- Microbial food chain
  - in rhizosphere 156
- Microbial food web 82, 111, 123
- Microbial functional diversity 11
- Microbial gene pool 11
- Microbial genetic diversity 4
- Microbial growth 223
- Microbial immobilization 13
  - of metals 411
- Microbial interaction 286
  - in biofilm 286
  - pathogenic 294
- Microbial loop 82
- Microbial metabolism 18
- Microbial mobilization 333
  - of carbon 333
  - of haloorganics 370
  - of metals 437
- Microbial
  - morphology 24
  - phylogeny 17
  - physiology 24
  - population 11
  - predation 285
- Microbial relationship 262
  - amensalism 268
  - combined 264
  - commensalisms 269, 270
  - competition 266
  - ecological 262
  - interspecies 268
  - negative 264, 269
  - neutralism 269
  - parasitism 268
  - positive 264, 268
  - predation 268
  - protocooperation 268, 272
  - symbiosis 268
- Microbial remineralization 333
- Microbial respiration 227
  - in aquatic environment 231
  - in terrestrial samples 227
  - measurement 227
  - of radiolabeled substrates 228
- Microbial succession
  - in sediment 93
- Microbial survival 178
  - factors 178
    - atmospheric humidity 178
    - ions 179
    - oxygen 179
    - radiation 179
    - temperature 179
  - in air 178
- Microbial symbiosis 301
- Microbial taxonomy 19
- Microbial tolerance 219
  - to heavy metals 219
  - to organic contaminants 219
- Microbial transformation 411, 423, 464
  - of metal 423

- Microbial treatment 373
  - of wastewater 373
- Microbially enhanced oil recovery 13
- Microbiological methods 222
- Microhabitats 10
- Microplankton 69
- Microzooplankton 78
  - distribution 78
  - diversity 79
  - mixotrophy 79
  - seasonal dynamics 80
- Minimum infectious dose 264
- Modified microbes 8
- Mollicutes 32
- Most probable number 489
- mtn* genes 432
- Mutualism 277, 292
  - by cyanobacteria 291
  - in non-legumes 289
- Mycetomes 331
- Mycoplasma 33
- Mycorrhizae 156, 288, 310
  - arbuscular 157, 159
  - arbutoid 157
  - ectotrophic 157
  - ericoid 157
  - orchid 157
- Mycorrhizal association 157, 162, 288, 310
  - signaling 162
- Myxobacteria 31
- Myxococcus* 32
- Natural microbial community 11
- nif* genes 315
- Nitrate reduction 349
- Nitrate respiration 349
- Nitrification 348, 374, 404
  - autotrophic 404
  - heterotrophic 404
- Nitrifying bacteria 28, 348
- Nitrite oxidation 348
- Nitrococcus* 29
- Nitrogen cycle 275, 343
  - repression 353
- Nitrogen fixation 275, 344
  - by cyanobacteria 345
  - in phyllosphere 345
  - in rhizosphere 345
  - rate 345
- Nitrogen fixing bacteria 28
- Nitrogen removal 403
- Nitrogen reserves 343
- Nitrosococcus* 29
- Nitrosomonas* 29
- Nitrospira* 29
- Nod factors 321
- nod* gene induction 318
  - flavonoids 318
- nod* genes 319, 323
  - expression 323
- NodD 320
- Nodulation 319
  - biochemical signals 322
  - infection thread 321
- nol* genes 323
- Non-legume symbiosis 289
  - host plants 290
- Non-photosynthetic bacteria 30
- Non-proteobacteria 25
- Non-radioactive labeling 240
  - by digoxigenin 241
  - of probes 241
  - with biotin 241
  - with fluorescein 241
- Nucleic acid assay 238
- Nutrient cycle 333
- Obligate symbionts 3
- Ocean floor 123
  - metabolic processes 123
  - microbial activity 123
  - physicochemistry 123
- Oceanithermus* 196
- Oligotrophic indicators 497
- omph* 202
- Organic contaminants 11
- Organic osmolyte 214
  - compatible solutes 215
  - strategy 214
- Organic pollutants 446
- Organohalogenes 371
- Ostigobionts 509
- Oxygenases 382
- Oxygenolytic cleavage 454
- pac* genes 433
- PAH 221
  - degradation 221
- Paracoccus* sp. 351
- Paramecium* 278, 284



- Parasitism 281
- Particulate organic matters 95
  - colloidal 95
  - fine 95
- Pathogenic interactions 294
  - enzymes 295
  - in animals 296
  - morphological changes 295
  - of fungi 295, 298
  - phytoalexins 295
  - toxin production 297
  - with bacteria 295, 299
  - with plants 294
- PEG model 504
- Pelagic bacteroids 105
- Pelagic foodweb 82, 123
- Penicillium* 5
- Periphyton indicator 504
- Peroxidases 382
- Pesticide degradation 464
- PGPB 155, 365
- Phagotrophic flagellates 82
- Phenazines 3
- Phloroglucinols 3
- Phosphate accumulation 408
- Phosphate transport system 364
- Phosphorus cycle 359
  - biomanipulation 364
- Phosphorus
  - in ecosystem 359
  - minerals 359
  - oxidation 360
  - removal 407
  - solubilization 360
    - by 2-oxogluconate 360
    - microbial 360
  - solubilizing microbes 361
- Phosphotriesterase 464, 484
- Photobacterium* 202
- Photosynthetic bacteria 25
- Phylloplane 181
- Phyllosphere 179, 291
  - association 182
  - bacteria 180
  - fungi 180, 291
  - microbes 180
  - microbes 291
- Phytoalexins 295
- Phytoplankton
  - assemblage 70
  - association 73
  - growth strategies 503
  - indices 497
  - of coral reefs 121
  - of mangrove habitat 114
  - periodicity 72
  - pigments 66
  - productivity 77
  - seasonal succession 71
  - size distribution 67
- Phytoremediation 11
- Picobiliphytes 118
- Picoplankton 69
- Picrophilus* 25
- Planktonic bacteria 88
- Plant-cyanobacteria symbiosis 311
- Plant-microbe interactions 287
- Plasmid analysis 245
- Plasmid profile 245
- Polygalacturonase 321
- Porous media 129
- Predation 283, 292
  - by ciliates 286
  - by flagellates 285
  - microbial 285
- Predator- prey interaction 283
- Presence-absence test 490
- Pressure induced proteins 202
- Prochloron* 40
- Prochlorophytes 40
- Prokaryote diversity 4
- Proteobacteria 27
  - metabolic modes a 28
- Proteorhodopsin 110
- Protozoa 50
  - Apicomplexa 53
  - Asctospora 53
  - Ciliophora 54
  - classification 52
  - in ground water 508
  - in metal contaminated sites 417
  - in rumen 293
  - Labyrinthomorpha 53
  - marine 51
  - Microspora 53
  - morphology 51

- Myxozoa 54
  - of marine habitat 122
  - pathogenic 50
  - Sacromastigophora 53
  - symbiosis 305
- Pseudomonads 30
- PSMs 333
- pst* genes 364
- Psychrophiles 186
- Purple bacteria 25
- Purple non-sulphur bacteria 25, 28
- Pyrodictium* 24
- Pyrrolnitrin 3
- Qualitative detection method 11
- Radiolabeling 233
  - for membrane transport 235
  - of DNA 234
  - of protein 235
  - with 2-deoxyglucose 237
  - with leucine 235
- Raynold's number 171
- Recalcitrant organics 446
- Recombinant vaccines 9
- Reservoir ecosystem 57
- Respiration gases 228
- RFLP analysis 243
- Rhizobia 315
  - association with legumes 318
  - characteristics 316, 318
  - chemotaxis 318
  - classification 317
  - hosts 317
- Rhizobium* 310, 315
  - genome 323
  - in non-legumes 289
- Rhizodeposition 154
- Rhizoplane 149
- Rhizoseath 149
- Rhizosolenia* 120
- Rhizosphere 2, 149, 287
  - adaptation 155
  - allelopathic substances 153
  - and bulk soil diversity 154
  - bacteria 151
  - influence 150
  - interactions 288
  - microbes 149, 152, 287
  - microbial compounds 152
  - microbial food chain 156
  - microbial hormones 152
  - nutrient mobility 154
  - Pseudomonads 287
  - soil 150
  - succession 151
- Rhizospheric bacteria 309
- Rickettsia* 28
- Riverine system 58
- Rolling surface behaviour 378
- Root exudates 151
- Root influence 155
- Root-derived factors 155
- Rotating biological contactors 398, 443
- Rumen
  - ciliates 293
  - metabolites 293
  - microflora 293
  - protozoa 293
- Salt accumulation 213
  - chloride pump 213
  - halorhodopsin 213
  - ion gradient 213
  - osmotic adaptation 214
  - strategy 213
- Saprophytic bacteria
  - density in wastewater 495
- Saprosira* 27
- Secondary treatment 395
  - activated sludge systems 399
  - conventional biofilters 395
  - high-rate biofilters 397
  - of wastewater 395
  - RBC 399
  - SAF 401
  - UASB 402
- Sediment O<sub>2</sub> uptake 92
- Self purification of water 493, 495
- Shewanella* 203
- Siderophores 421
- Silaffin 104
- Silicoflagellates 118
- Silicon cycle 104
- Sinorhizobium* 317
- smt* genes 432
- Soda lakes 213
- Soil
  - actinomycetes 141
  - aggregates 131
  - air 135

- algae 143
- algal colonization 144
- amoebae 145
- autotrophic bacteria 143
- bacterial diversity 140
- biological constituents 131
- chemically perturbed 221
- components 130
- cyanobacteria 144
- environment 129
- flagellates 145
- fungi 141
- heterotrophic bacteria 143
- humus 133
  - degradation 134
- ion sorption 132
- layers 129
- microbes 147
  - distribution 137
- microbial activity 135
- microbial aggregation 138
- microbial attachment 138
- microbial colonization 138
- microbial consortia 147
- microbial film 134
- mineral exchange 135
- minerals 138
- O<sub>2</sub> diffusion 135
- organic matters 133, 136
- organically contaminated 220
  - microbes 220
- particles 130
- phases 130, 134
- pore space 131, 135
- protozoa 145
- S/V ratio 130
- solids 130
- solution 134
- sub-surface 136
- surface environment 136
- water 134
  - movement 135
  - potential 134
- Sok 267
- Solar salterns 211
- Spingobacterium* 27
- Spirochaete 26
- Sporocytophaga* 27
- Staphyllococcus* 35
- Stigmatella* 32
- Stromatolites 334
- Stygogenes 509
- Stygophiles 509
- Sub-marine hot vents 195
- Submerged aerated filters 401
- Sub-surface
  - culturable bacteria 147
  - eukaryotes 146
  - heterotrophic activity 147
  - soil microbes 140
  - soil microbial diversity 146
- Sulfolobales 23
- Sulfolobus* 24
- Sulphate reducers 25, 31
- Sulphate reduction 358
- Sulphur
  - assimilation 354
  - bacteria 356
  - cycle 275, 353
  - global reserve 354
  - mineralization 356
  - oxidation state 353
  - oxidizer 29
  - precipitation 358
  - reducing thermophiles 194
  - respiration 358
  - transformation 356
    - aerobic 357
    - anaerobic 357
    - oxidative 356
    - reductive 358
- Superbugs 128
- Superorganism 334
- sym* plasmids 324
- Symbiodinium* 121
- Symbiosis 301
  - algae-algae 306
  - algae-fungi 302
  - algae-protozoa 305
  - angiosperm-algae 314
  - animal-algae 326
  - animal-microbe 324
  - Azolla-Anabaena* 313
  - Azolla*-bacteria 309
  - bacteria-protozoa 305
  - between bacteria 302
  - by methanotrophs 329
  - cyanelles 308

- cyanobacteria 307
- dinoflagellates 308
- in diatoms 307
- in lichen 303
- legume-*Rhizobium* 315
- microbe-microbe 302
- mycobiont 303
- of algae 310
- phycobiont 303
- plant-bacteria 309
- plant-algae 310
- plant-cyanobacteria 311
- plant-fungi 310
- plant-microbes 309
- reconstitution 331
- Synergism 272-276
- Synergistic methane emission 276
- Synthetic organics 7
- Syntrophic degradation 273
- Syntrophism 273
- Syntrophobacter* 276
- Syntrophomonas* 276
- Syntrophy 274
  - in aggregates 276
  - in N<sub>2</sub> cycle 274
  - of acetogens 277
- Tetrazolium salts 249
- Thalassohaline microbes 211
- Thermoacidophiles 24
- Thermoanaerobacter* 198
- Thermococci 24
- Thermophiles 186, 190
  - hydrogen producing 198
  - iron reducing 197
  - nitrate reducing 196
  - sulphur reducing 194
- Thermophilic composting 193
  - temperature optima 193
- Thermophilic fungi 193
  - cellulolytic systems 193
- Thermophilic microbes 190
  - actinomycetes 191
  - aerobic 190
  - bacteria 190
  - diversity 190
  - Thermophilic protein sequence 199
  - Thermoplasma* 20, 24
  - Thermoproteales 195
  - Thermoproteus* 24
  - Thermos 25
  - Thermostable enzymes 194
  - Thermoterribacterium* 198
  - Thermothrix* 196
  - Thermotogae 25
  - Thermovenabulum* 198
  - Thermus* 196
  - Thiobacillus* 356
  - Thiobacillus denitrificans* 357
  - Thiobacillus ferrooxidans* 13
  - Thiobacillus thiooxidans* 13
  - Thiomicrospira* 29
  - Thymidine incorporation 229, 234
  - Tintinids 122
  - tnp* genes 430
  - Torula-Humicola* complex 194
  - Total bacterial number 494
  - Toxic contamination 4
  - Trepollemma* 26
  - Trichodesmium* 120
  - Trichormus azollae* 313
  - Trickling filters 386
  - Trophic index 504
  - Trophy indication 503
    - by periphyton 504
    - by phytoplankton 503
  - TTSS 322
  - UASB 402
  - Ultraphytoplankton 70
  - Unusual microbes 4
  - Uranium mining 14
  - UVGI 184
  - Vedose zone 136
  - Wastewater treatment 374
  - Xenobiotics degrading microbes 448
  - zia* genes 433
  - Zinc exporters 433
  - Zoogloea* 375
  - Zoogloeal slime 375
  - Zooplankton in ground water 509

# Textbook of Environmental Microbiology

The present book meets the requirement of the students of Microbiology, Biotechnology, Life Sciences and Environmental Sciences. It provides the fundamental knowledge on the microbial diversity, processes and applications in the environment. This can also be used as a source of basic knowledge for the researchers in the field of environmental microbiology.

## **This book contains**

- Microbial approaches to wastewater treatment
- Microbial diversity and processes in different habitats
- The microbiology of extreme environments
- Microbial remediation of wasteland and pesticide contaminated habitats
- Microbial interactions in the environment and their role in the environmental processes
- Symbiotic interaction of microbes with plants and animals and their role in ecosystem stability
- The rhizosphere microbiology and its role in plant productivity
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- Microbial mobilization of nutrients with special reference to the mobilization of nitrogen and phosphorus
- Advanced methodologies to study the microbial diversity and phylogenetic relationship in the environment
- Microbial remediation of the metal contaminated habitats
- Microbial mobilization and recovery of valuable metals from low grade ores and diluted solutions
- Microbial degradation of recalcitrant pollutants
- Determination of the environmental quality by microbial indication

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